UM1725
User Manual

Description of STM32F4xx HAL drivers

Introduction

STM32Cube™ is STMicroelectronics's original initiative to ease developers' life by reducing development efforts, time and cost. STM32Cube™ covers the STM32 portfolio.

STM32Cube™ Version 1.x includes:

- The STM32CubeMX, a graphical software configuration tool that allows generating C initialization code using graphical wizards.
- A comprehensive embedded software platform, delivered per series (such as STM32CubeF4 for STM32F4 series)
  - The STM32Cube HAL, an STM32 abstraction layer embedded software, ensuring maximized portability across the STM32 portfolio
  - A consistent set of middleware components such as RTOS, USB, TCP/IP, Graphics
  - All embedded software utilities coming with a full set of examples.

The HAL driver layer provides a generic multi instance simple set of APIs (application programming interfaces) to interact with the upper layer (application, libraries and stacks). It is composed of generic and extension APIs. It is directly built around a generic architecture and allows the built-upon layers, such as the middleware layer, to implement their functions without knowing in-depth how to use the MCU. This structure improves the library code reusability and guarantees easy portability to other devices.

The HAL drivers include a complete set of ready-to-use APIs which simplify the user application implementation. As an example, the communication peripherals contain APIs to initialize and configure the peripheral, to manage data transfers based on polling, to handle interrupts or DMA, and to manage communication errors.

The HAL driver APIs are split into two categories: generic APIs which provide common and generic functions for all the STM32 series and extension APIs which include specific and customized functions for a given family or part number.

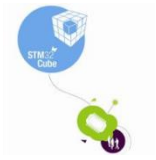
The HAL drivers are feature-oriented instead of IP-oriented. As an example, the timer APIs are split into several categories following the functions offered by the IP: basic timer, capture, pulse width modulation (PWM), etc..

The drivers source code is developed in Strict ANSI-C which makes it independent from the development tools. It is checked with CodeSonar™ static analysis tool. It is fully documented and is MISRA-C 2004 compliant.

The HAL driver layer implements run-time failure detection by checking the input values of all functions. Such dynamic checking contributes to enhance the firmware robustness. Run-time detection is also suitable for user application development and debugging.

This user manual is structured as follows:

- Overview of the HAL drivers
- Detailed description of each peripheral driver: configuration structures, functions, and how to use the given API to build your application.



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1 Acronyms and definitions

Table 1: Acronyms and definitions

Acronym	Definition
ADC	Analog-to-digital converter
ANSI	American National Standards Institute
API	Application Programming Interface
BSP	Board Support Package
CAN	Controller area network
CEC	Consumer Electronics Control
CMSIS	Cortex Microcontroller Software Interface Standard
CPU	Central Processing Unit
CRYP	Cryptographic processor unit
CRC	CRC calculation unit
DAC	Digital to analog converter
DCMI	Digital Camera Module Interface
DFSDM	Digital filter for sigma delta modulators
DMA	Direct Memory Access
DMA2D	Chrom-Art Accelerator™ controller
DSI	Display Serial Interface
ETH	Ethernet controller
EXTI	External interrupt/event controller
FLASH	Flash memory
FSMC	Flexible Static Memory controller
FMC	Flexible Memory controller
FMPI2C	Fast-mode Plus inter-integrated circuit
GPIO	General purpose I/Os
HAL	Hardware abstraction layer
HASH	Hash processor
HCD	USB Host Controller Driver
I2C	Inter-integrated circuit
I2S	Inter-integrated sound
IRDA	InfraRed Data Association
IWDG	Independent watchdog
LPTIM	Low-power Timer
LTDC	LCD TFT Display Controller
MSP	MCU Specific Package
NAND	NAND external Flash memory

Acronym	Definition
NOR	NOR external Flash memory
NVIC	Nested Vectored Interrupt Controller
PCCARD	PCCARD external memory
PCD	USB Peripheral Controller Driver
PWR	Power controller
QSPI	QuadSPI Flash memory Interface
RCC	Reset and clock controller
RNG	Random Number Generator
RTC	Real-time clock
SAI	Serial Audio Interface
SD	Secure Digital
SDRAM	SDRAM external memory
SRAM	SRAM external memory
SMARTCARD	Smartcard IC
SPDIFRX	SPDIF-RX Receiver Interface
SPI	Serial Peripheral interface
SysTick	System tick timer
TIM	Advanced-control, general-purpose or basic timer
UART	Universal asynchronous receiver/transmitter
USART	Universal synchronous receiver/transmitter
WWDG	Window watchdog
USB	Universal Serial Bus
PPP	STM32 peripheral or block

2 Overview of HAL drivers

The HAL drivers were designed to offer a rich set of APIs and to interact easily with the application upper layers.

Each driver consists of a set of functions covering the most common peripheral features. The development of each driver is driven by a common API which standardizes the driver structure, the functions and the parameter names.

The HAL drivers consist of a set of driver modules, each module being linked to standalone peripheral. However, in some cases, the module is linked to a peripheral functional mode. As an example, several modules exist for the USART peripheral: UART driver module, USART driver module, SMARTCARD driver module and IRDA driver module.

The HAL main features are the following:

- Cross-family portable set of APIs covering the common peripheral features as well as extension APIs in case of specific peripheral features.
- Three API programming models: polling, interrupt and DMA.
- APIs are RTOS compliant:
  - Fully reentrant APIs
  - Systematic usage of timeouts in polling mode.
- Peripheral multi-instance support allowing concurrent API calls for multiple instances of a given peripheral (USART1, USART2...)
- All HAL APIs implement user-callback functions mechanism:
  - Peripheral Init/DeInit HAL APIs can call user-callback functions to perform peripheral system level Initialization/De-Initialization (clock, GPIOs, interrupt, DMA)
  - Peripherals interrupt events
  - Error events.
- Object locking mechanism: safe hardware access to prevent multiple spurious accesses to shared resources.
- Timeout used for all blocking processes: the timeout can be a simple counter or a timebase.

2.1 HAL and user-application files

2.1.1 HAL driver files

HAL drivers are composed of the following set of files:

Table 2: HAL driver files

File	Description
stm32f4xx_hal_ppp.c	Main peripheral/module driver file. It includes the APIs that are common to all STM32 devices. *Example: stm32f4xx_hal_adc.c, stm32f4xx_hal_irda.c, …*
stm32f4xx_hal_ppp.h	Header file of the main driver C file It includes common data, handle and enumeration structures, define statements and macros, as well as the exported generic APIs. *Example: stm32f4xx_hal_adc.h, stm32f4xx_hal_irda.h, …*

File	Description
stm32f4xx_hal_ppp_ex.c	Extension file of a peripheral/module driver. It includes the specific APIs for a given part number or family, as well as the newly defined APIs that overwrite the default generic APIs if the internal process is implemented in different way. *Example: stm32f4xx_hal_adc_ex.c, stm32f4xx_hal_dma_ex.c, …*
stm32f4xx_hal_ppp_ex.h	Header file of the extension C file. It includes the specific data and enumeration structures, define statements and macros, as well as the exported device part number specific APIs *Example: stm32f4xx_hal_adc_ex.h, stm32f4xx_hal_dma_ex.h, …*
stm32f4xx_ll_ppp.c	Peripheral low layer driver that can be accessed from one or more HAL drivers. It offers a set of APIs and services used by the upper driver. From the user point of view, low-level drivers are not accessible directly. They are used only by the HAL drivers built upon them. *Example: stm32f4xx_ll_fmc.c offers a set of API used by stm32f4xx_hal_sdram.c, stm32f4xx_hal_sram.c, stm32f4xx_hal_nor.c, stm32f4xx_hal_nand.c,…*
stm32f4xx_ll_ppp.h	Header file of the low layer C file. It is included in the HAL driver header file, thus making the low-level driver an intrinsic add-on of the HAL driver that is not visible from the application. *Example: stm32f4xx_ll_fmc.h, stm32f4xx_ll_usb.h,...*
stm32f4xx_hal.c	This file is used for HAL initialization and contains DBGMCU, Remap and Time Delay based on systick APIs.
stm32f4xx_hal.h	stm32f4xx_hal.c header file
stm32f4xx_hal_msp_template.c	Template file to be copied to the user application folder. It contains the MSP initialization and de-initialization (main routine and callbacks) of the peripheral used in the user application.
stm32f4xx_hal_conf_template.h	Template file allowing to customize the drivers for a given application.
stm32f4xx_hal_def.h	Common HAL resources such as common define statements, enumerations, structures and macros.

Since the low-level drivers are only used by the HAL drivers built upon them, the APIs provided by these drivers will not be described in this document.

2.1.2 User-application files

The minimum files required to build an application using the HAL drivers are listed in the table below:

Table 3: User-application files

File	Description
system_stm32f4xx.c	This file contains SystemInit() which is called at startup just after reset and before branching to the main program. It does not configure the system clock at startup (contrary to the standard library). This must be performed using the HAL APIs in the user files. It allows to : • relocate the vector table in internal SRAM. • configure FSMC/FMC peripheral (when available) to use as data memory the external SRAM or SDRAM mounted on the evaluation board.
startup_stm32f4xx.s	Toolchain specific file that contains reset handler and exception vectors. For some toolchains, it allows adapting the stack/heap size to fit the application requirements.
stm32f4xx_flash.icf (optional)	Linker file for EWARM toolchain allowing mainly to adapt the stack/heap size to fit the application requirements.
stm32f4xx_hal_msp.c	This file contains the MSP initialization and de-initialization (main routine and callbacks) of the peripheral used in the user application.
stm32f4xx_hal_conf.h	This file allows the user to customize the HAL drivers for a specific application. It is not mandatory to modify this configuration. The application can use the default configuration without any modification.
stm32f4xx_it.c/.h	This file contains the exceptions handler and peripherals interrupt service routine, and calls HAL_IncTick() at regular time intervals to increment a local variable (declared in stm32*f4*xx_hal.c) used as HAL timebase. By default, this function is called each 1ms in Systick ISR. . The PPP_IRQHandler() routine must call HAL_PPP_IRQHandler() if an interrupt based process is used within the application.
main.c/.h	This file contains the main program routine, mainly: • the call to HAL_Init() • assert_failed() implementation • system clock configuration • peripheral HAL initialization and user application code.

The STM32Cube package comes with ready-to-use project templates, one for each supported board. Each project contains the files listed above and a preconfigured project for the supported toolchains.

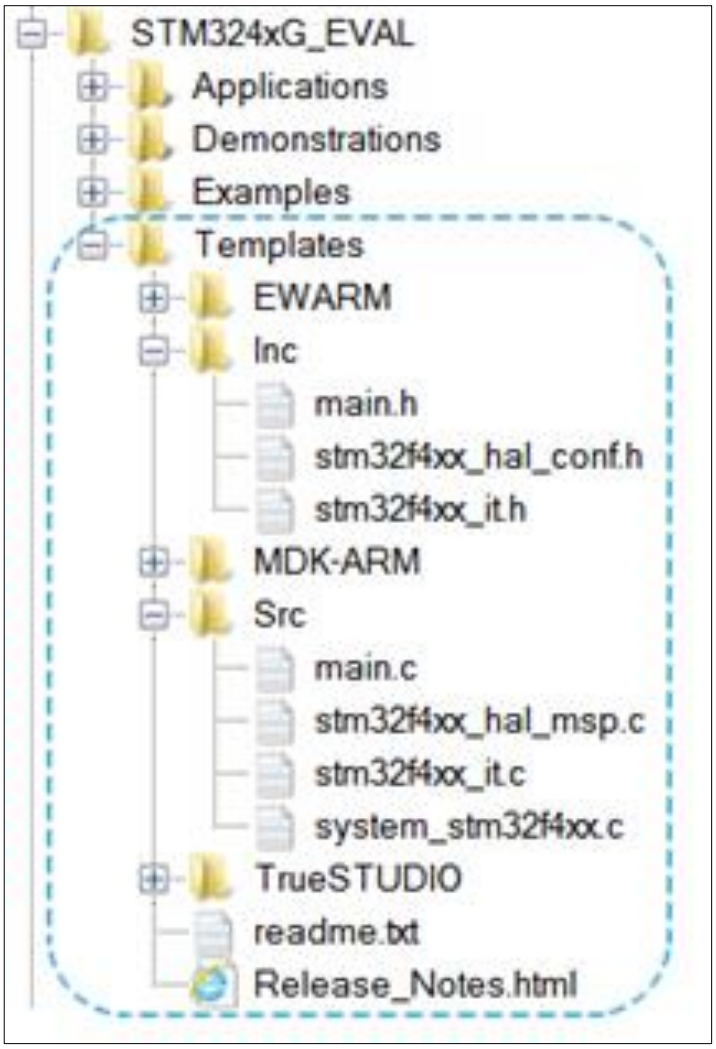
Each project template provides empty main loop function and can be used as a starting point to get familiar with project settings for STM32Cube. Their characteristics are the following:

- It contains sources of HAL, CMSIS and BSP drivers which are the minimal components to develop a code on a given board.
- It contains the include paths for all the firmware components.

- It defines the STM32 device supported, and allows to configure the CMSIS and HAL drivers accordingly.
- It provides ready to use user files preconfigured as defined below:
  - HAL is initialized
  - SysTick ISR implemented for HAL_Delay()
  - System clock configured with the maximum frequency of the device.

If an existing project is copied to another location, then include paths must be updated.

Figure 1: Example of project template



2.2 HAL data structures

Each HAL driver can contain the following data structures:

- Peripheral handle structures
- Initialization and configuration structures
- Specific process structures.

2.2.1 Peripheral handle structures

The APIs have a modular generic multi-instance architecture that allows working with several IP instances simultaneous.

PPP_HandleTypeDef *handle is the main structure that is implemented in the HAL drivers. It handles the peripheral/module configuration and registers and embeds all the structures and variables needed to follow the peripheral device flow.

The peripheral handle is used for the following purposes:

- Multi-instance support: each peripheral/module instance has its own handle. As a result instance resources are independent.
- Peripheral process intercommunication: the handle is used to manage shared data resources between the process routines.
  Example: global pointers, DMA handles, state machine.
- Storage : this handle is used also to manage global variables within a given HAL driver.

An example of peripheral structure is shown below:

```
typedef struct
{
 USART_TypeDef       *Instance; /* USART registers base address   */
 USART InitTypeDef     Init;       /* Usart communication parameters  */
 uint8 t           *pTxBuffPtr;/* Pointer to Usart Tx transfer Buffer */
 uint16_t           TxXferSize; /* Usart Tx Transfer size       */
 __IO uint16_t       TxXferCount;/* Usart Tx Transfer Counter      */
 uint8_t           *pRxBuffPtr;/* Pointer to Usart Rx transfer Buffer */
 uint16 t           RxXferSize;  /* Usart Rx Transfer size      */
   IO uint16 t       RxXferCount; /* Usart Rx Transfer Counter     */
 DMA_HandleTypeDef     *hdmatx;    /* Usart Tx DMA Handle parameters */
 DMA_HandleTypeDef     *hdmarx;    /* Usart Rx DMA Handle parameters */
 HAL_LockTypeDef      Lock;     /* Locking object       */
   IO HAL USART StateTypeDef  State;  /* Usart communication state     */
   IO HAL USART ErrorTypeDef  ErrorCode;/* USART Error code         */
 }USART_HandleTypeDef;
```

1) The multi-instance feature implies that all the APIs used in the application are re-entrant and avoid using global variables because a subroutine can fail to be re-entrant if they rely on a global variable to remain unchanged but that variable is modified when the subroutine is recursively invoked. For this reason, the following rules are respected:

- Re-entrant code does not hold any static (or global) non-constant data: re-entrant functions can work with global data. For example, a re-entrant interrupt service routine can grab a piece of hardware status to work with (e.g. serial port read buffer) which is not only global, but volatile. Still, typical use of static variables and global data is not advised, in the sense that only atomic read-modify-write instructions should be used in these variables. It should not be possible for an interrupt or signal to occur during the execution of such an instruction.
- Reentrant code does not modify its own code.

2) When a peripheral can manage several processes simultaneously using the DMA (full duplex case), the DMA interface handle for each process is added in the PPP_HandleTypeDef.

3) For the shared and system peripherals, no handle or instance object is used. The peripherals concerned by this exception are the following:

- GPIO
- SYSTICK
- NVIC
- PWR
- RCC
- FLASH.

2.2.2 Initialization and configuration structure

These structures are defined in the generic driver header file when it is common to all part numbers. When they can change from one part number to another, the structures are defined in the extension header file for each part number.

```
typedef struct
{
uint32_t BaudRate;   /*!< This member configures the UART communication baudrate.*/
uint32_t WordLength;  /*!< Specifies the number of data bits transmitted or received
in a frame.*/
uint32 t StopBits;   /*!< Specifies the number of stop bits transmitted.*/
uint32_t Parity;    /*!< Specifies the parity mode. */
uint32_t Mode;     /*!< Specifies wether the Receive or Transmit mode is enabled or
disabled.*/
uint32 t HwFlowCtl;  /*!< Specifies wether the hardware flow control mode is enabled
or disabled.*/
uint32 t OverSampling; /*!< Specifies wether the Over sampling 8 is enabled or
disabled,
to achieve higher speed (up to fPCLK/8).*/
}UART_InitTypeDef;
```

The config structure is used to initialize the sub-modules or sub-instances. See below example:

```
HAL ADC ConfigChannel (ADC HandleTypeDef* hadc, ADC ChannelConfTypeDef*
sConfig)
```

2.2.3 Specific process structures

The specific process structures are used for specific process (common APIs). They are defined in the generic driver header file.

Example:

```
HAL_PPP_Process (PPP_HandleTypeDef* hadc,PPP_ProcessConfig* sConfig)
```

2.3 API classification

The HAL APIs are classified into three categories:

- **Generic APIs:** common generic APIs applying to all STM32 devices. These APIs are consequently present in the generic HAL drivers files of all STM32 microcontrollers.

```
HAL StatusTypeDef HAL ADC Init(ADC HandleTypeDef* hadc); HAL StatusTypeDef
HAL ADC DeInit(ADC HandleTypeDef *hadc); HAL StatusTypeDef
HAL ADC Start(ADC HandleTypeDef* hadc); HAL StatusTypeDef
HAL_ADC_Stop(ADC_HandleTypeDef* hadc); HAL_StatusTypeDef
HAL_ADC_Start_IT(ADC_HandleTypeDef* hadc); HAL_StatusTypeDef
HAL ADC Stop IT(ADC HandleTypeDef* hadc); void HAL ADC IRQHandler(ADC HandleTypeDef*
hadc);
```

- **Extension APIs:** This set of API is divided into two sub-categories :
  - **Family specific APIs:** APIs applying to a given family. They are located in the extension HAL driver file (see example below related to the ADC).

```
HAL StatusTypeDef HAL ADCEx InjectedStop(ADC HandleTypeDef* hadc); HAL StatusTypeDef
HAL_ADCEx_InjectedStop_IT(ADC_HandleTypeDef* hadc); HAL_StatusTypeDef
HAL_ADCEx_InjectedStart(ADC_HandleTypeDef* hadc); HAL_StatusTypeDef
HAL_ADCEx_InjectedStart_IT(ADC_HandleTypeDef* hadc);
```

  - **Device part number specific APIs:** These APIs are implemented in the extension file and delimited by specific define statements relative to a given part number.

```
#if defined(STM32F427xx) || defined(STM32F437xx) || defined(STM32F429xx) ||
defined(STM32F439xx) HAL StatusTypeDef HAL FLASHEx OB SelectPCROP(void);
HAL StatusTypeDef HAL FLASHEx OB DeSelectPCROP(void);
#endif /* STM32F427xx || STM32F437xx || STM32F429xx || STM32F439xx || */
```

  - The data structure related to the specific APIs is delimited by the device part number define statement. It is located in the corresponding extension header C file.

The following table summarizes the location of the different categories of HAL APIs in the driver files.

Table 4: API classification

	Generic file	Extension file
Common APIs	X	X (1)
Family specific APIs		X
Device specific APIs		X

Notes:

(1)In some cases, the implementation for a specific device part number may change . In this case the generic API is declared as weak function in the extension file. The API is implemented again to overwrite the default function

Family specific APIs are only related to a given family. This means that if a specific API is implemented in another family, and the arguments of this latter family are different, additional structures and arguments might need to be added.

The IRQ handlers are used for common and family specific processes.

2.4 Devices supported by HAL drivers

Table 5: List of devices supported by HAL drivers

IP/Module	STM32F405xx	STM32F415xx	STM32F407xx	STM32F417xx	STM32F427xx	STM32F437xx	STM32F429xx	STM32F439xx	STM32F401xC	STM32F401xE	STM32F446xx	STM32F469xx	STM32F479xx	STM32F410xx	STM32F412xx
stm32f4xx_hal.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_adc.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_adc_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_can.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_cec.c	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No
stm32f4xx_hal_cortex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_crc.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_cryp.c	No	Yes	No	Yes	No	Yes	No	Yes	No	No	No	No	Yes	No	No
stm32f4xx_hal_cryp_ex.c	No	Yes	No	Yes	No	Yes	No	Yes	No	No	No	No	Yes	No	No
stm32f4xx_hal_dac.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No
stm32f4xx_hal_dac_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No
stm32f4xx_hal_dcmi.c	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_dcmi_ex.c	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_dfsdm.c	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
stm32f4xx_hal_dma.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_dma2d.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	No	No
stm32f4xx_hal_dma_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_dsi.c	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes	No	No
stm32f4xx_hal_eth.c	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	No	No
stm32f4xx_hal_flash.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_flash_ex.c	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_flash_ramfunc.c	No	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes	Yes
stm32f4xx_hal_fmpi2c.c	No	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes	Yes
stm32f4xx_hal_fmpi2c_ex.c	No	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes	Yes
stm32f4xx_hal_gpio.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_hash.c	No	Yes	No	Yes	No	Yes	No	Yes	No	No	No	No	Yes	No	No
stm32f4xx_hal_hash_ex.c	No	No	No	No	No	Yes	No	Yes	No	No	No	No	Yes	No	No
stm32f4xx_hal_hcd.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_i2c.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_i2c_ex.c	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_i2s.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

IP/Module	STM32F405xx	STM32F415xx	STM32F407xx	STM32F417xx	STM32F427xx	STM32F437xx	STM32F429xx	STM32F439xx	STM32F401xC	STM32F401xE	STM32F446xx	STM32F469xx	STM32F479xx	STM32F410xx	STM32F412xx
stm32f4xx_hal_i2s_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_irda.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_iwdg.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_lptim.c	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No
stm32f4xx_hal_ltdc.c	No	No	No	No	No	No	Yes	Yes	No	No	No	Yes	Yes	No	No
stm32f4xx_hal_ltdc_ex.c	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes	No	No
stm32f4xx_hal_msp_ template.c	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
stm32f4xx_hal_nand.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_nor.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_pccard.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No
stm32f4xx_hal_pcd.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_pcd_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_pwr.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_pwr_ex.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_qspi.c	No	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_rcc.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_rcc_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_rng.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
stm32f4xx_hal_rtc.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_rtc_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_sai.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_sai_ex.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_sd.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_sdram.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_hal_smartcard.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_spdifrx.c	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No
stm32f4xx_hal_spi.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_sram.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	Yes
stm32f4xx_hal_tim.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_tim_ex.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_uart.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_usart.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_hal_wwdg.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
stm32f4xx_ll_fmc.c	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
stm32f4xx_ll_fsmc.c	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	No	Yes
stm32f4xx_ll_sdmmc.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
stm32f4xx_ll_usb.c	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes

2.5 HAL driver rules

2.5.1 HAL API naming rules

The following naming rules are used in HAL drivers:

Table 6: HAL API naming rules

	Generic	Family specific	Device specific
File names	*stm32f4xx_hal_ppp (c/h)*	*stm32f4xx_hal_ppp_ex (c/h)*	*stm32f4xx_ hal_ppp_ex (c/h)*
Module name	*HAL_PPP_ MODULE*		
Function name	*HAL_PPP_Function HAL_PPP_FeatureFunction_ MODE*	*HAL_PPPEx_Function HAL_PPPEx_FeatureFunction_ MODE*	*HAL_PPPEx_Function HAL_PPPEx_FeatureFunction_ MODE*
Handle name	*PPP_HandleTypedef*	*NA*	*NA*
Init structure name	*PPP_InitTypeDef*	*NA*	*PPP_InitTypeDef*
Enum name	*HAL_PPP_StructnameTypeDef*	*NA*	*NA*

- The **PPP** prefix refers to the peripheral functional mode and not to the peripheral itself. For example, if the USART, PPP can be USART, IRDA, UART or SMARTCARD depending on the peripheral mode.
- The constants used in one file are defined within this file. A constant used in several files is defined in a header file. All constants are written in uppercase, except for peripheral driver function parameters.
- typedef variable names should be suffixed with _TypeDef.
- Registers are considered as constants. In most cases, their name is in uppercase and uses the same acronyms as in the STM32F4xx reference manuals.
- Peripheral registers are declared in the PPP_TypeDef structure (e.g. ADC_TypeDef) in stm32f4xxx.h header file.stm32f4xxx.h corresponds to stm32f401xc.h, stm32f401xe.h, stm32f405xx.h, stm32f415xx.h, stm32f407xx.h, stm32f417xx.h, stm32f427xx.h, stm32f437xx.h, stm32f429xx.h, stm32f439xx.h, stm32f446xx.h, stm32f469xx.h, stm32f479xx.h, stm32f410cx.h, stm32f410tx.h, stm32f410rx.h, stm32f412cx.h, stm32f412rx.h, stm32f412vx.h or stm32f412zx.h.
- Peripheral function names are prefixed by HAL_, then the corresponding peripheral acronym in uppercase followed by an underscore. The first letter of each word is in uppercase (e.g. HAL_UART_Transmit()). Only one underscore is allowed in a function name to separate the peripheral acronym from the rest of the function name.
- The structure containing the PPP peripheral initialization parameters are named PPP_InitTypeDef (e.g. ADC_InitTypeDef).
- The structure containing the Specific configuration parameters for the PPP peripheral are named PPP_xxxxConfTypeDef (e.g. ADC_ChannelConfTypeDef).
- Peripheral handle structures are named PPP_HandleTypedef (e.g DMA_HandleTypeDef)
- The functions used to initialize the PPP peripheral according to parameters specified in PPP_InitTypeDef are named HAL_PPP_Init (e.g. HAL_TIM_Init()).

- The functions used to reset the PPP peripheral registers to their default values are named PPP_DeInit, e.g. TIM_DeInit.
- The **MODE** suffix refers to the process mode, which can be polling, interrupt or DMA. As an example, when the DMA is used in addition to the native resources, the function should be called: *HAL_PPP_Function_DMA ().*
- The **Feature** prefix should refer to the new feature.
  Example: *HAL_ADC_InjectionStart*() refers to the injection mode

2.5.2 HAL general naming rules

- For the shared and system peripherals, no handle or instance object is used. This rule applies to the following peripherals:
  – GPIO
  – SYSTICK
  – NVIC
  – RCC
  – FLASH.

Example: The *HAL_GPIO_Init()* requires only the GPIO address and its configuration parameters.

```
HAL StatusTypeDef HAL GPIO Init (GPIO TypeDef* GPIOx, GPIO InitTypeDef *Init)
{
/*GPIO Initialization body */
}
```

- The macros that handle interrupts and specific clock configurations are defined in each peripheral/module driver. These macros are exported in the peripheral driver header files so that they can be used by the extension file. The list of these macros is defined below: This list is not exhaustive and other macros related to peripheral features can be added, so that they can be used in the user application.

Table 7: Macros handling interrupts and specific clock configurations

Macros	Description
__HAL_PPP_ENABLE_IT(__HANDLE__, __INTERRUPT__)	Enables a specific peripheral interrupt
__HAL_PPP_DISABLE_IT(__HANDLE__, __INTERRUPT__)	Disables a specific peripheral interrupt
__HAL_PPP_GET_IT (__HANDLE__, __ INTERRUPT __)	Gets a specific peripheral interrupt status
__HAL_PPP_CLEAR_IT (__HANDLE__, __ INTERRUPT __)	Clears a specific peripheral interrupt status
__HAL_PPP_GET_FLAG (__HANDLE__, __FLAG__)	Gets a specific peripheral flag status
__HAL_PPP_CLEAR_FLAG (__HANDLE__, __FLAG__)	Clears a specific peripheral flag status
__HAL_PPP_ENABLE(__HANDLE__)	Enables a peripheral
__HAL_PPP_DISABLE(__HANDLE__)	Disables a peripheral
__HAL_PPP_XXXX (__HANDLE__, __PARAM__)	Specific PPP HAL driver macro
__HAL_PPP_GET_ IT_SOURCE (__HANDLE__, __ INTERRUPT __)	Checks the source of specified interrupt

- NVIC and SYSTICK are two ARM Cortex core features. The APIs related to these features are located in the stm32f4xx_hal_cortex.c file.
- When a status bit or a flag is read from registers, it is composed of shifted values depending on the number of read values and of their size. In this case, the returned status width is 32 bits. Example : STATUS = XX | (YY << 16) or STATUS=".
- The PPP handles are valid before using the HAL_PPP_Init() API. The init function performs a check before modifying the handle fields.

```
HAL_PPP_Init(PPP_HandleTypeDef) if(hppp == NULL) { return HAL_ERROR; }
```

- The macros defined below are used:
  - Conditional macro:

```
#define ABS(x) (((x) > 0) ? (x) : -(x))
```

  - Pseudo-code macro (multiple instructions macro):

```
#define   HAL LINKDMA(  HANDLE  ,   PPP DMA FIELD ,   DMA HANDLE ) \ do{ \
(__HANDLE__)->__PPP_DMA_FIELD_ = &(__DMA_HANDLE_); \ (__DMA_HANDLE_).Parent =
(__HANDLE__); \ } while(0)
```

2.5.3 HAL interrupt handler and callback functions

Besides the APIs, HAL peripheral drivers include:

- HAL_PPP_IRQHandler() peripheral interrupt handler that should be called from stm32f4xx_it.c
- User callback functions.

The user callback functions are defined as empty functions with “weak” attribute. They have to be defined in the user code.

There are three types of user callbacks functions:

- Peripheral system level initialization/ de-Initialization callbacks: HAL_PPP_MspInit() and HAL_PPP_MspDeInit
- Process complete callbacks : HAL_PPP_ProcessCpltCallback
- Error callback: HAL_PPP_ErrorCallback.

Table 8: Callback functions

Callback functions	Example
HAL_PPP_MspInit() / _DeInit()	Ex: HAL_USART_MspInit() Called from HAL_PPP_Init() API function to perform peripheral system level initialization (GPIOs, clock, DMA, interrupt)
HAL_PPP_ProcessCpltCallback	Ex: HAL_USART_TxCpltCallback Called by peripheral or DMA interrupt handler when the process completes
HAL_PPP_ErrorCallback	Ex: HAL_USART_ErrorCallback Called by peripheral or DMA interrupt handler when an error occurs

2.6 HAL generic APIs

The generic APIs provide common generic functions applying to all STM32 devices. They are composed of four APIs groups:

- **Initialization and de-initialization functions:**HAL_PPP_Init(), HAL_PPP_DeInit()
- **IO operation functions:** HAL_PPP_Read(), HAL_PPP_Write(),HAL_PPP_Transmit(), HAL_PPP_Receive()
- **Control functions:**HAL_PPP_Set (), HAL_PPP_Get ().
- **State and Errors functions:** HAL_PPP_GetState (), HAL_PPP_GetError ().

For some peripheral/module drivers, these groups are modified depending on the peripheral/module implementation.

Example: in the timer driver, the API grouping is based on timer features (PWM, OC, IC...).

The initialization and de-initialization functions allow initializing a peripheral and configuring the low-level resources, mainly clocks, GPIO, alternate functions (AF) and possibly DMA and interrupts. The *HAL_DeInit()*function restores the peripheral default state, frees the low-level resources and removes any direct dependency with the hardware.

The IO operation functions perform a row access to the peripheral payload data in write and read modes.

The control functions are used to change dynamically the peripheral configuration and set another operating mode.

The peripheral state and errors functions allow retrieving in runtime the peripheral and data flow states, and identifying the type of errors that occurred. The example below is based on the ADC peripheral. The list of generic APIs is not exhaustive. It is only given as an example.

Table 9: HAL generic APIs

Function Group	Common API Name	Description
Initialization group	*HAL_ADC_Init()*	This function initializes the peripheral and configures the low -level resources (clocks, GPIO, AF..)
	HAL_ADC_DeInit()	This function restores the peripheral default state, frees the low-level resources and removes any direct dependency with the hardware.
IO operation group	*HAL_ADC_Start ()*	This function starts ADC conversions when the polling method is used
	HAL_ADC_Stop ()	This function stops ADC conversions when the polling method is used
	HAL_ADC_PollForConversion()	This function allows waiting for the end of conversions when the polling method is used. In this case, a timout value is specified by the user according to the application.
	HAL_ADC_Start_IT()	This function starts ADC conversions when the interrupt method is used
	HAL_ADC_Stop_IT()	This function stops ADC conversions when the interrupt method is used
	HAL_ADC_IRQHandler()	This function handles ADC interrupt requests

Function Group	Common API Name	Description
	HAL_ADC_ConvCpltCallback()	Callback function called in the IT subroutine to indicate the end of the current process or when a DMA transfer has completed
	HAL_ADC_ErrorCallback()	Callback function called in the IT subroutine if a peripheral error or a DMA transfer error occurred
Control group	*HAL_ADC_ConfigChannel()*	This function configures the selected ADC regular channel, the corresponding rank in the sequencer and the sample time
	HAL_ADC_AnalogWDGConfig	This function configures the analog watchdog for the selected ADC
State and Errors group	*HAL_ADC_GetState()*	This function allows getting in runtime the peripheral and the data flow states.
	HAL_ADC_GetError()	This fuction allows getting in runtime the error that occurred during IT routine

2.7 HAL extension APIs

2.7.1 HAL extension model overview

The extension APIs provide specific functions or overwrite modified APIs for a specific family (series) or specific part number within the same family.

The extension model consists of an additional file, stm32f4xx_hal_ppp_ex.c, that includes all the specific functions and define statements (stm32f4xx_hal_ppp_ex.h) for a given part number.

Below an example based on the ADC peripheral:

Table 10: HAL extension APIs

Function Group	Common API Name
HAL_ADCEx_InjectedStart()	This function starts injected channel ADC conversions when the polling method is used
HAL_ADCEx_InjectedStop()	This function stops injected channel ADC conversions when the polling method is used
HAL_ADCEx_InjectedStart_IT()	This function starts injected channel ADC conversions when the interrupt method is used
HAL_ADCEx_InjectedStop_IT()	This function stops injected channel ADC conversions when the interrupt method is used
HAL_ADCEx_InjectedConfigChannel()	This function configures the selected ADC Injected channel (corresponding rank in the sequencer and sample time)

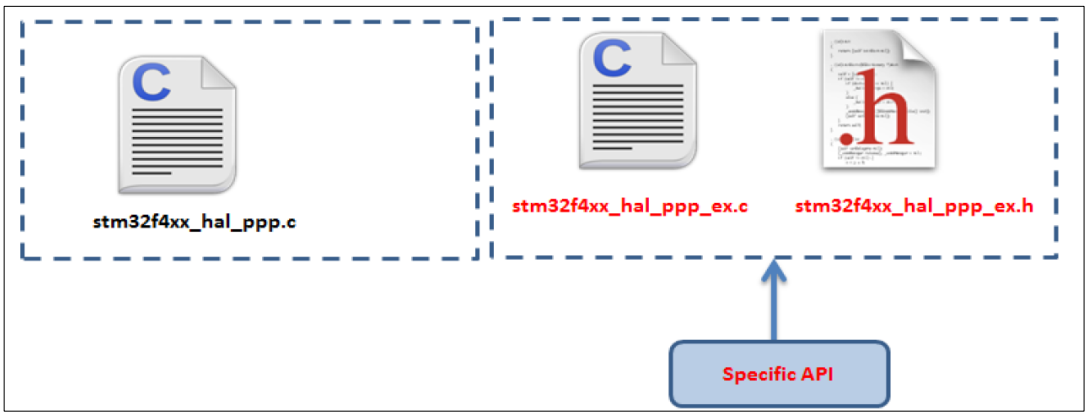
2.7.2 HAL extension model cases

The specific IP features can be handled by the HAL drivers in five different ways. They are described below.

Case1: Adding a part number-specific function

When a new feature specific to a given device is required, the new APIs are added in the stm32f4xx_hal_ppp_ex.c extension file. They are named HAL_PPPEx_Function().

Figure 2: Adding device-specific functions



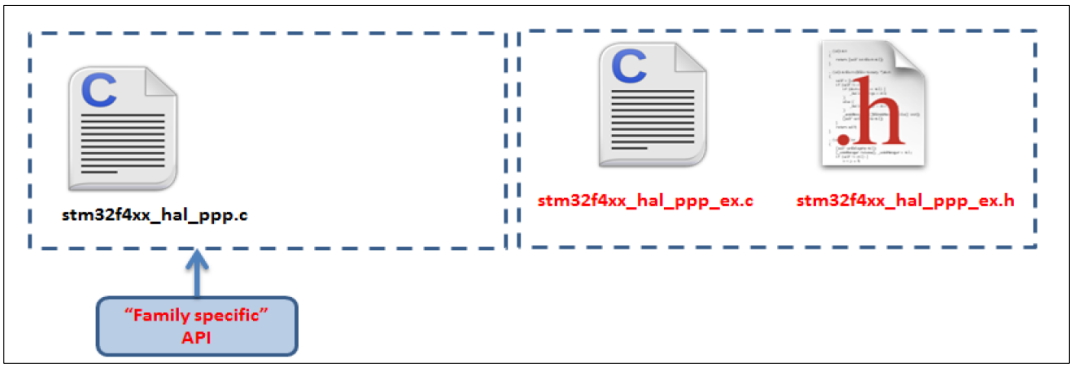
Example: stm32f4xx_hal_flash_ex.c/h

```
#if defined(STM32F427xx) || defined(STM32F437xx) || defined(STM32F429xx)||
defined(STM32F439xx)
HAL_StatusTypeDef HAL_FLASHEx_OB_SelectPCROP(void);
HAL_StatusTypeDef HAL_FLASHEx_OB_DeSelectPCROP(void);
#endif /* STM32F427xx ||STM32F437xx || STM32F429xx || STM32F439xx  */
```

Case2: Adding a family-specific function

In this case, the API is added in the extension driver C file and named HAL_PPPEx_Function ().

Figure 3: Adding family-specific functions



Example: stm32f4xx_hal_adc_ex.c/h

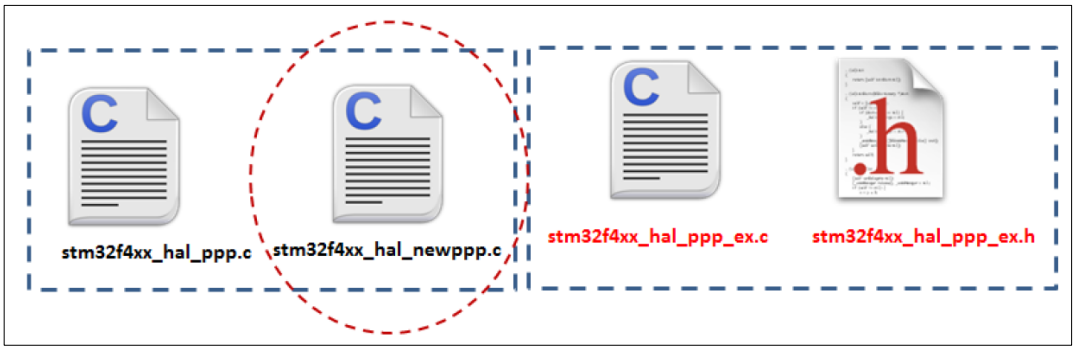
```
HAL_StatusTypeDef HAL_ADCEx_InjectedStop(ADC_HandleTypeDef* hadc);
HAL_StatusTypeDef HAL_ADCEx_InjectedStop_IT(ADC_HandleTypeDef* hadc);
HAL_StatusTypeDef HAL_ADCEx_InjectedStart(ADC_HandleTypeDef* hadc);
HAL_StatusTypeDef HAL_ADCEx_InjectedStart_IT(ADC_HandleTypeDef* hadc);
```

Case3 : Adding a new peripheral (specific to a device belonging to a given family)

When a peripheral which is available only in a specific device is required, the APIs corresponding to this new peripheral/module are added in stm32f4xx_hal_newppp.c. However the inclusion of this file is selected in the stm32fxx_hal_conf.h using the macro:

```
#define HAL_NEWPPP_MODULE_ENABLED
```

Figure 4: Adding new peripherals

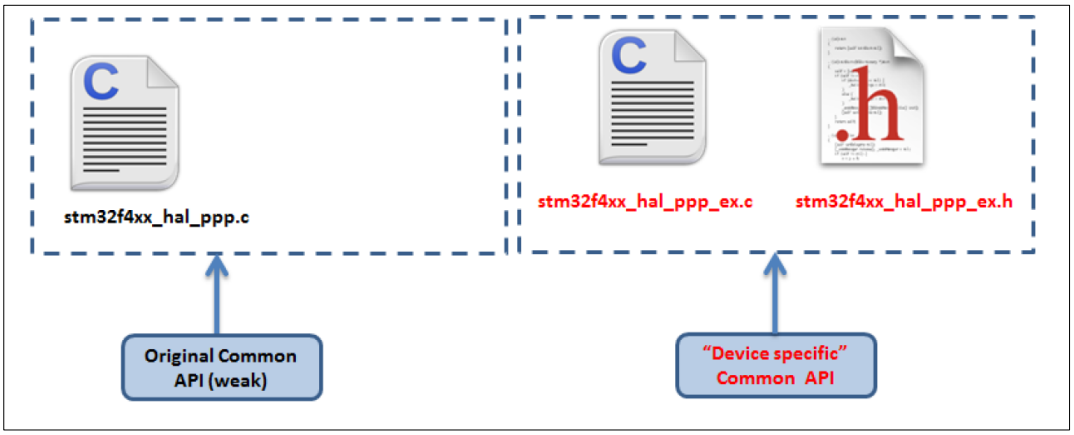


Example: stm32f4xx_hal_sai.c/h

Case4: Updating existing common APIs

In this case, the routines are defined with the same names in the stm32f4xx_hal_ppp_ex.c extension file, while the generic API is defined as *weak*, so that the compiler will overwrite the original routine by the new defined function.

Figure 5: Updating existing APIs



Case5 : Updating existing data structures

The data structure for a specific device part number (e.g. PPP_InitTypeDef) can have different fields. In this case, the data structure is defined in the extension header file and delimited by the specific part number define statement.

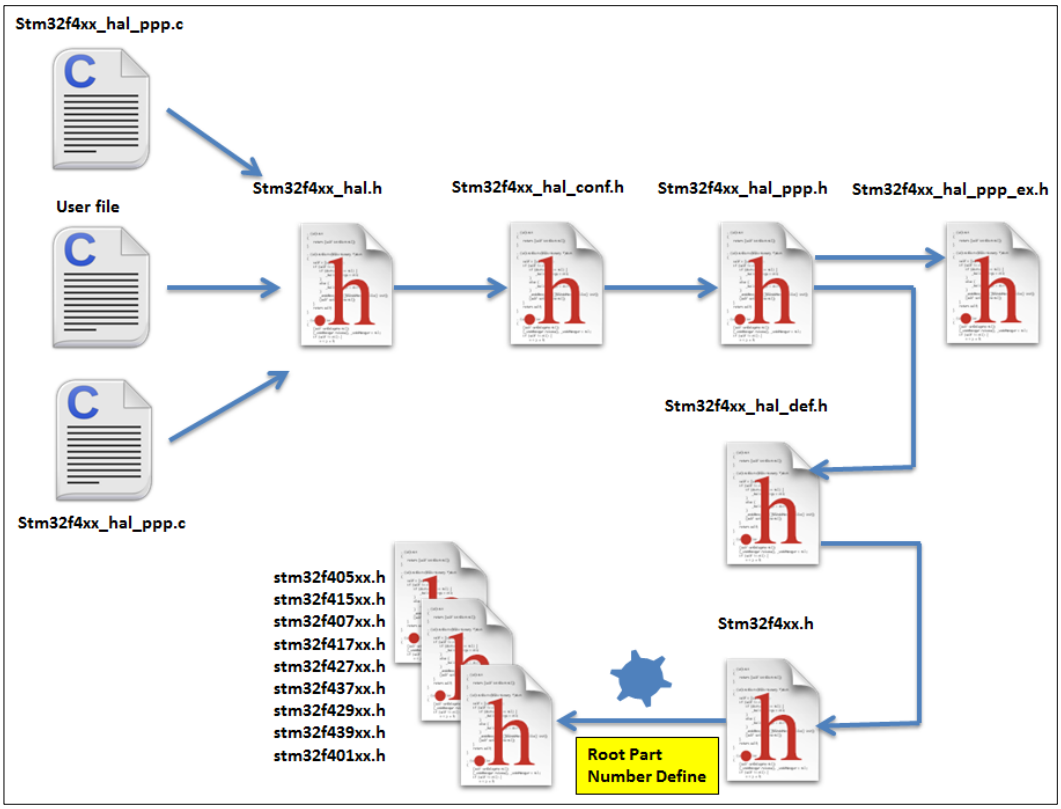
Example:

```
#if defined (STM32F401xx)
typedef struct
{
(…)
}PPP InitTypeDef;
#endif /* STM32F401xx */
```

2.8 File inclusion model

The header of the common HAL driver file (stm32f4xx_hal.h) includes the common configurations for the whole HAL library. It is the only header file that is included in the user sources and the HAL C sources files to be able to use the HAL resources.

Figure 6: File inclusion model



A PPP driver is a standalone module which is used in a project. The user must enable the corresponding USE_HAL_PPP_MODULE define statement in the configuration file.

```
/*****************************************************************
* @file stm32f4xx hal conf.h
* @author MCD Application Team
* @version VX.Y.Z * @date dd-mm-yyyy
* @brief This file contains the modules to be used
*****************************************************************
(…)
#define USE HAL USART MODULE
#define USE_HAL_IRDA_MODULE
#define USE_HAL_DMA_MODULE
#define USE HAL RCC MODULE
(…)
```

2.9 HAL common resources

The common HAL resources, such as common define enumerations, structures and macros, are defined in *stm32f4xx_hal_def.h.*The main common define enumeration is *HAL_StatusTypeDef*.

- **HAL Status:** The HAL status is used by almost all HAL APIs, except for boolean functions and IRQ handler. It returns the status of the current API operations. It has four possible values as described below:

```
Typedef enum
 { HAL OK = 0x00, HAL ERROR = 0x01, HAL BUSY = 0x02, HAL TIMEOUT = 0x03
 } HAL_StatusTypeDef;
```

- **HAL Locked:** The HAL lock is used by all HAL APIs to prevent accessing by accident shared resources.

```
typedef enum
 { HAL UNLOCKED = 0x00, /*!<Resources unlocked */ HAL LOCKED = 0x01 /*!< Resources
locked */
 } HAL_LockTypeDef;
```

- In addition to common resources, the stm32f4xx_hal_def.h file calls the stm32f4xx.h file in CMSIS library to get the data structures and the address mapping for all peripherals:
  - Declarations of peripheral registers and bits definition.
  - Macros to access peripheral registers hardware (Write register, Read register…etc.).
- **Common macros**
  - Macro defining HAL_MAX_DELAY

```
#define HAL_MAX_DELAY 0xFFFFFFFF
```

  - Macro linking a PPP peripheral to a DMA structure pointer: *__HAL_LINKDMA();*

```
#define __HAL_LINKDMA(__HANDLE__, __PPP_DMA_FIELD_, __DMA_HANDLE_) \
do{ \
 (__HANDLE__)->__PPP_DMA_FIELD_ = &(__DMA_HANDLE_); \
 (  DMA HANDLE ).Parent = (  HANDLE  ); \
 } while(0)
```

2.10 HAL configuration

The configuration file*, stm32f4xx_hal_conf.h,* allows customizing the drivers for the user application. Modifying this configuration is not mandatory: the application can use the default configuration without any modification.

To configure these parameters, the user should enable, disable or modify some options by uncommenting, commenting or modifying the values of the related define statements as described in the table below:

Table 11: Define statements used for HAL configuration

Configuration item	Description	Default Value
HSE_VALUE	Defines the value of the external oscillator (HSE) expressed in Hz. The user must adjust this define statement when using a different crystal value.	25 000 000 (Hz)
HSE_STARTUP_TIMEOUT	Timeout for HSE start up, expressed in ms	5000

Configuration item	Description	Default Value
HSI_VALUE	Defines the value of the internal oscillator (HSI) expressed in Hz.	16 000 000 (Hz)
EXTERNAL_CLOCK_VALUE	This value is used by the I2S/SAI HAL module to compute the I2S/SAI clock source frequency, this source is inserted directly through I2S_CKIN pad.	12288000 (Hz)
VDD_VALUE	VDD value	3300 (mV)
USE_RTOS	Enables the use of RTOS	FALSE (for future use)
PREFETCH_ENABLE	Enables prefetch feature	TRUE
INSTRUCTION_CACHE_ENABLE	Enables instruction cache	TRUE
DATA_CACHE_ENABLE	Enables data cache	TRUE
USE HAL_PPP_MODULE	Enables module to be used in the HAL driver	
MAC_ADDRx	Ethernet peripheral configuration : MAC address	
ETH_RX_BUF_SIZE	Ethernet buffer size for receive	ETH_MAX_PACKET_SIZE
ETH_TX_BUF_SIZE	Ethernet buffer size for trasmit	ETH_MAX_PACKET_SIZE
ETH_RXBUFNB	The number of Rx buffers of size ETH_RX_BUF_SIZE	4
ETH_TXBUFNB	The number of Tx buffers of size ETH_RX_BUF_SIZE	4
DP83848_PHY_ADDRESS	DB83848 Ethernet PHY Address	0x01
PHY_RESET_DELAY	PHY Reset delay these values are based on a 1 ms Systick interrupt	0x000000FF
PHY_CONFIG_DELAY	PHY Configuration delay	0x00000FFF
PHY_BCR PHY_BSR	Common PHY Registers	
PHY_SR PHY_MICR PHY_MISR	Extended PHY registers	

The stm32f4xx_hal_conf_template.h file is located in the HAL drivers *Inc* folder. It should be copied to the user folder, renamed and modified as described above.

By default, the values defined in the stm32f4xx_hal_conf_template.h file are the same as the ones used for the examples and demonstrations. All HAL include files are enabled so that they can be used in the user code without modifications.

2.11 HAL system peripheral handling

This chapter gives an overview of how the system peripherals are handled by the HAL drivers. The full API list is provided within each peripheral driver description section.

2.11.1 Clock

Two main functions can be used to configure the system clock:

- HAL_RCC_OscConfig (RCC_OscInitTypeDef *RCC_OscInitStruct). This function configures/enables multiple clock sources (HSE, HSI, LSE, LSI, PLL).
- HAL_RCC_ClockConfig (RCC_ClkInitTypeDef *RCC_ClkInitStruct, uint32_t FLatency). This function
  - Selects the system clock source
  - Configures AHB, APB1 and APB2 clock dividers
  - Configures the number od Flash memory wait states
  - Updates the SysTick configuration when HCLK clock changes.

Some peripheral clocks are not derived from the system clock (RTC, SDIO, I2S, SAI, Audio PLL…). In this case, the clock configuration is performed by an extended API defined in stm32f4xx_hal_ppp_ex.c: *HAL_RCCEx_PeriphCLKConfig(RCC_PeriphCLKInitTypeDef *PeriphClkInit)*.

Additional RCC HAL driver functions are available:

- HAL_RCC_DeInit() Clock de-init function that return clock configuration to reset state
- Get clock functions that allow retreiving various clock configurations (system clock, HCLK, PCLK1, PCLK2, …)
- MCO and CSS configuration functions

A set of macros are defined in stm32f4xx_hal_rcc.h. They allows executing elementary operations on RCC block registers, such as peripherals clock gating/reset control:

- __PPP_CLK_ENABLE/__PPP_CLK_DISABLE to enable/disable the peripheral clock
- __PPP_FORCE_RESET/__PPP_RELEASE_RESET to force/release peripheral reset
- __PPP_CLK_SLEEP_ENABLE/__PPP_CLK_SLEEP_DISABLE to enable/disable the peripheral clock during low power (Sleep) mode.

2.11.2 GPIOs

GPIO HAL APIs are the following:

- HAL_GPIO_Init() / HAL_GPIO_DeInit()
- HAL_GPIO_ReadPin() / HAL_GPIO_WritePin()
- HAL_GPIO_TogglePin ().

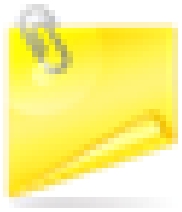
In addition to standard GPIO modes (input, output, analog), pin mode can be configured as EXTI with interrupt or event generation.

When selecting EXTI mode with interrupt generation, the user must call HAL_GPIO_EXTI_IRQHandler() from stm32f4xx_it.c and implement HAL_GPIO_EXTI_Callback()

The table below describes the GPIO_InitTypeDef structure field.

Table 12: Description of GPIO_InitTypeDef structure

Structure field	Description
Pin	Specifies the GPIO pins to be configured. Possible values: GPIO_PIN_x or GPIO_PIN_All, where x[0..15]
Mode	Specifies the operating mode for the selected pins: GPIO mode or EXTI mode. Possible values are: • GPIO mode – GPIO_MODE_INPUT : Input Floating – GPIO_MODE_OUTPUT_PP : Output Push Pull – GPIO_MODE_OUTPUT_OD : Output Open Drain – GPIO_MODE_AF_PP : Alternate Function Push Pull – GPIO_MODE_AF_OD : Alternate Function Open Drain – GPIO_MODE_ANALOG : Analog mode • External Interrupt Mode – GPIO_MODE_IT_RISING : Rising edge trigger detection – GPIO_MODE_IT_FALLING : Falling edge trigger detection – GPIO_MODE_IT_RISING_FALLING : Rising/Falling edge trigger detection • External Event Mode – GPIO_MODE_EVT_RISING : Rising edge trigger detection – GPIO_MODE_EVT_FALLING : Falling edge trigger detection – GPIO_MODE_EVT_RISING_FALLING: Rising/Falling edge trigger detection
Pull	Specifies the Pull-up or Pull-down activation for the selected pins. Possible values are: GPIO_NOPULL GPIO_PULLUP GPIO_PULLDOWN
Speed	Specifies the speed for the selected pins Possible values are: GPIO_SPEED_LOW GPIO_SPEED_MEDIUM GPIO_SPEED_FAST GPIO_SPEED_HIGH

Structure field	Description
Alternate	Peripheral to be connected to the selected pins. Possible values: GPIO_AFx_PPP, where AFx: is the alternate function index PPP: is the peripheral instance Example: use GPIO_AF1_TIM1 to connect TIM1 IOs on AF1. These values are defined in the GPIO extended driver, since the AF mapping may change between product lines.  Refer to the “Alternate function mapping” table in the datasheets for the detailed description of the system and peripheral I/O alternate functions.

Please find below typical GPIO configuration examples:

- Configuring GPIOs as output push-pull to drive external LEDs

```
GPIO InitStruct.Pin = GPIO PIN 12 | GPIO PIN 13 | GPIO PIN 14 | GPIO PIN 15;
GPIO InitStruct.Mode = GPIO MODE OUTPUT PP;
GPIO_InitStruct.Pull = GPIO_PULLUP;
GPIO_InitStruct.Speed = GPIO_SPEED_FAST;
HAL_GPIO_Init(GPIOD, &GPIO_InitStruct);
```

- Configuring PA0 as external interrupt with falling edge sensitivity:

```
GPIO_InitStructure.Mode = GPIO_MODE_IT_FALLING;
GPIO InitStructure.Pull = GPIO NOPULL;
GPIO InitStructure.Pin = GPIO PIN 0;
HAL_GPIO_Init(GPIOA, &GPIO_InitStructure);
```

- Configuring USART3 Tx (PC10, mapped on AF7) as alternate function:

```
GPIO InitStruct.Pin = GPIO PIN 10;
GPIO InitStruct.Mode = GPIO MODE AF PP;
GPIO InitStruct.Pull = GPIO PULLUP;
GPIO_InitStruct.Speed = GPIO_SPEED_FAST;
GPIO_InitStruct.Alternate = GPIO_AF7_USART3;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);
```

2.11.3 Cortex NVIC and SysTick timer

The Cortex HAL driver, stm32f4xx_hal_cortex.c, provides APIs to handle NVIC and Systick. The supported APIs include:

- HAL_NVIC_SetPriorityGrouping()
- HAL_NVIC_SetPriority()
- HAL_NVIC_EnableIRQ()/HAL_NVIC_DisableIRQ()
- HAL_NVIC_SystemReset()
- HAL_NVIC_GetPendingIRQ() / HAL_NVIC_SetPendingIRQ () / HAL_NVIC_ClearPendingIRQ()
- HAL_SYSTICK_Config()
- HAL_SYSTICK_CLKSourceConfig()

2.11.4 PWR

The PWR HAL driver handles power management. The features shared between all STM32 Series are listed below:

- PVD configuration, enabling/disabling and interrupt handling
  - HAL_PWR_PVDConfig()
  - HAL_PWR_EnablePVD() / HAL_PWR_DisablePVD()
  - HAL_PWR_PVD_IRQHandler()
  - HAL_PWR_PVDCallback()
- Wakeup pin configuration
  - HAL_PWR_EnableWakeUpPin() / HAL_PWR_DisableWakeUpPin()
- Low power mode entry
  - HAL_PWR_EnterSLEEPMode()
  - HAL_PWR_EnterSTOPMode()
  - HAL_PWR_EnterSTANDBYMode()

Depending on the STM32 Series, extension functions are available in stm32f4xx_hal_pwr_ex. Here are a few examples (the list is not exhaustive)

- Backup domain registers enable/disable
  - HAL_PWREx_EnableBkUpReg() / HAL_PWREx_DisableBkUpReg()
- Flash overdrive control and flash power-down, for STM32F429/F439xx only
  - HAL_PWREx_ActivateOverDrive()
  - HAL_PWREx_EnableFlashPowerDown().

2.11.5 EXTI

The EXTI is not considered as a standalone peripheral but rather as a service used by other peripheral. As a result there are no EXTI APIs but each peripheral HAL driver implements the associated EXTI configuration and EXTI function are implemented as macros in its header file.

The first 16 EXTI lines connected to the GPIOs are managed within the GPIO driver. The GPIO_InitTypeDef structure allows configuring an I/O as external interrupt or external event.

The EXTI lines connected internally to the PVD, RTC, USB, and COMP are configured within the HAL drivers of these peripheral through the macros given in the table below. The EXTI internal connections depend on the targeted STM32 microcontroller (refer to the product datasheet for more details):

Table 13: Description of EXTI configuration macros

Macros	Description
PPP_EXTI_LINE_FUNCTION	Defines the EXTI line connected to the internal peripheral. Example: #define PWR_EXTI_LINE_PVD ((uint32_t)0x00010000) /*!<External interrupt line 16 Connected to the PVD EXTI Line */
__HAL_PPP_EXTI_ENABLE_IT(__EXTI_LINE__)	Enables a given EXTI line Example: __HAL_PVD_EXTI_ENABLE_IT(PWR_EXTI_LINE_PVD)
__HAL_PPP_EXTI_DISABLE_IT(__EXTI_LINE__)	Disables a given EXTI line. Example: __HAL_PVD_EXTI_DISABLE_IT(PWR_EXTI_LINE_PVD)
__HAL_PPP_EXTI_GET_FLAG(__EXTI_LINE__)	Gets a given EXTI line interrupt flag pending bit status. Example: __HAL_PVD_EXTI_GET_FLAG(PWR_EXTI_LINE_PVD)
__HAL_PPP_EXTI_CLEAR_FLAG(__EXTI_LINE__)	Clears a given EXTI line interrupt flag pending bit. Example; __HAL_PVD_EXTI_CLEAR_FLAG(PWR_EXTI_LINE_PVD)
__HAL_PPP_EXTI_GENERATE_SWIT (__EXTI_LINE__)	Generates a software interrupt for a given EXTI line. Example: __HAL_PVD_EXTI_ GENERATE_SWIT (PWR_EXTI_LINE_PVD)

If the EXTI interrupt mode is selected, the user application must call HAL_PPP_FUNCTION_IRQHandler() (for example HAL_PWR_PVD_IRQHandler()), from stm32f4xx_it.c file, and implement HAL_PPP_FUNCTIONCallback() callback function (for example HAL_PWR_PVDCallback().

2.11.6 DMA

The DMA HAL driver allows enabling and configuring the peripheral to be connected to the DMA Stream (except for internal SRAM/FLASH memory which do not require any initialization). Refer to the product reference manual for details on the DMA request corresponding to each peripheral.

For a given stream, HAL_DMA_Init() API allows programming the required configuration through the following parameters:

- Transfer Direction
- Source and Destination data formats
- Circular, Normal or peripheral flow control mode
- Stream Priority level
- Source and Destination Increment mode
- FIFO mode and its Threshold (if needed)
- Burst mode for Source and/or Destination (if needed).

Two operating modes are available:

- Polling mode I/O operation
  a. Use HAL_DMA_Start() to start DMA transfer when the source and destination addresses and the Length of data to be transferred have been configured.
  b. Use HAL_DMA_PollForTransfer() to poll for the end of current transfer. In this case a fixed timeout can be configured depending on the user application.
- Interrupt mode I/O operation
  a. Configure the DMA interrupt priority using HAL_NVIC_SetPriority()
  b. Enable the DMA IRQ handler using HAL_NVIC_EnableIRQ()
  c. Use HAL_DMA_Start_IT() to start DMA transfer when the source and destination addresses and the length of data to be transferred have been confgured. In this case the DMA interrupt is configured.
  d. Use HAL_DMA_IRQHandler() called under DMA_IRQHandler() Interrupt subroutine
  e. When data transfer is complete, HAL_DMA_IRQHandler() function is executed and a user function can be called by customizing XferCpltCallback and XferErrorCallback function pointer (i.e. a member of DMA handle structure).

Additional functions and macros are available to ensure efficient DMA management:

- Use HAL_DMA_GetState() function to return the DMA state and HAL_DMA_GetError() in case of error detection.
- Use HAL_DMA_Abort() function to abort the current transfer

The most used DMA HAL driver macros are the following:

- __HAL_DMA_ENABLE: enablse the specified DMA Stream.
- __HAL_DMA_DISABLE: disables the specified DMA Stream.
- __HAL_DMA_GET_FS: returns the current DMA Stream FIFO filled level.
- __HAL_DMA_GET_FLAG: gets the DMA Stream pending flags.
- __HAL_DMA_CLEAR_FLAG: clears the DMA Stream pending flags.
- __HAL_DMA_ENABLE_IT: enables the specified DMA Stream interrupts.
- __HAL_DMA_DISABLE_IT: disables the specified DMA Stream interrupts.

- __HAL_DMA_GET_IT_SOURCE: checks whether the specified DMA stream interrupt has occurred or not.

When a peripheral is used in DMA mode, the DMA initialization should be done in the HAL_PPP_MspInit() callback. In addition, the user application should associate the DMA handle to the PPP handle (refer to section “HAL IO operation functions”).

DMA double-buffering feature is handled as an extension API.

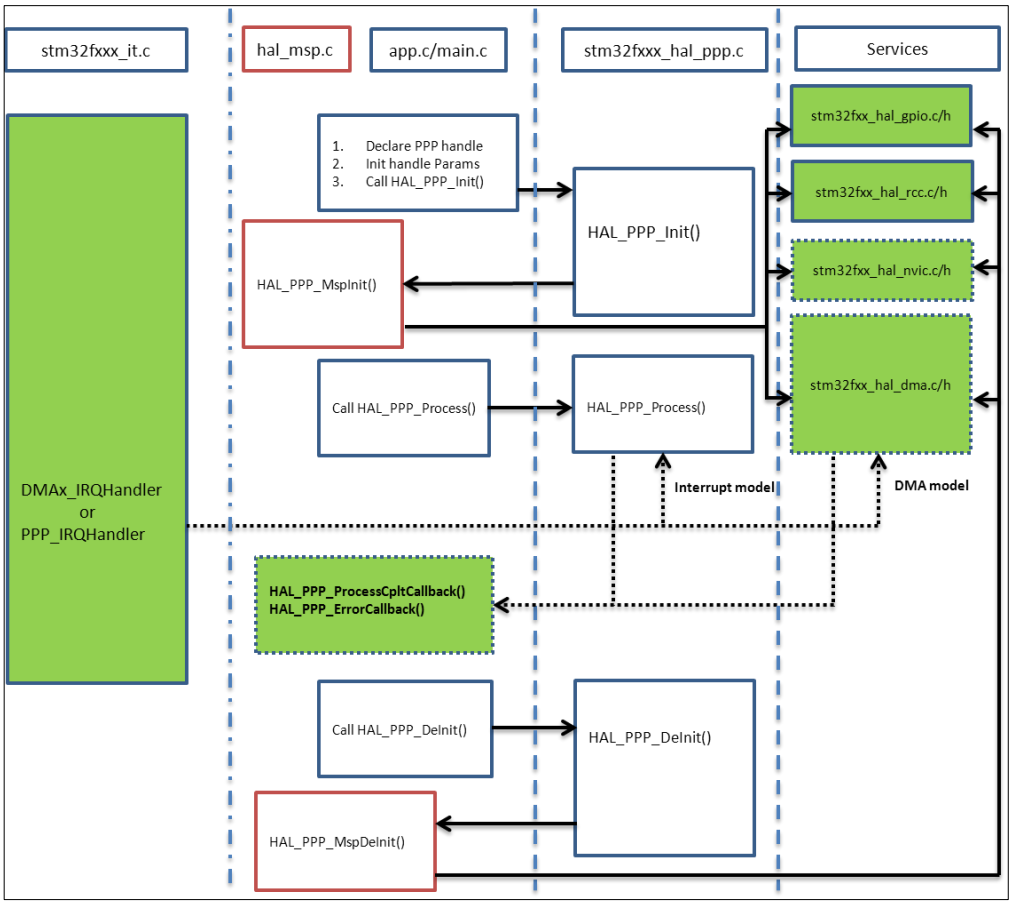
DMA channel callbacks need to be initialized by the user application only in case of memory-to-memory transfer. However when peripheral-to-memory transfers are used, these callbacks are automatically initialized by calling a process API function that uses the DMA.

2.12 How to use HAL drivers

2.12.1 HAL usage models

The following figure shows the typical use of the HAL driver and the interaction between the application user, the HAL driver and the interrupts.

Figure 7: HAL driver model



Basically, the HAL driver APIs are called from user files and optionally from interrupt handlers file when the APIs based on the DMA or the PPP peripheral dedicated interrupts are used.

When DMA or PPP peripheral interrupts are used, the PPP process complete callbacks are called to inform the user about the process completion in real-time event mode (interrupts). Note that the same process completion callbacks are used for DMA in interrupt mode.

2.12.2 HAL initialization

2.12.2.1 HAL global initialization

In addition to the peripheral initialization and de-initialization functions, a set of APIs are provided to initialize the HAL core implemented in file stm32f4xx_hal.c.

- HAL_Init(): this function must be called at application startup to
  - Initialize data/instruction cache and pre-fetch queue
  - Set Systick timer to generate an interrupt each 1ms (based on HSI clock) with the lowest priority
  - Set priority grouping to 4 preemption bits
  - Call HAL_MspInit() user callback function to perform system level initializations (Clock, GPIOs, DMA, interrupts). HAL_MspInit() is defined as “weak” empty function in the HAL drivers.
- HAL_DeInit()
  - Resets all peripherals
  - Calls function HAL_MspDeInit() which is a user callback function to do system level De-Initalizations.
- HAL_GetTick(): this function gets current SysTick counter value (incremented in SysTick interrupt) used by peripherals drivers to handle timeouts.
- HAL_Delay(). this function implements a delay (expressed in milliseconds) using the SysTick timer.
  Care must be taken when using HAL_Delay() since this function provides an accurate delay (expressed in milliseconds) based on a variable incremented in SysTick ISR. This means that if HAL_Delay() is called from a peripheral ISR, then the SysTick interrupt must have highest priority (numerically lower) than the peripheral interrupt, otherwise the caller ISR will be blocked.

In STM32Cube V1.0 implemented in STM32CubeF2 and STM32CubeF4 first versions, the SysTick timer is used as default timebase. This has been modified to allow implementing user-defined timebases (such as a general-purpose timer), keeping in mind that the timebase duration must be kept at 1 ms since all PPP_TIMEOUT_VALUEs are defined and handled in milliseconds. This enhancement is implemented in STM32Cube V1.1 that is deployed starting from STM32CubeL0/F0/F3 and later. This modification is backward compatible with STM32Cube V1.0 implementation. Functions affecting timebase configurations are declared as __Weak to allow different implementations in the user file.

2.12.2.2 System clock initialization

The clock configuration is done at the beginning of the user code. However the user can change the configuration of the clock in his own code. Please find below the typical Clock configuration sequence:

```
static void SystemClock Config(void)
{
RCC_ClkInitTypeDef RCC_ClkInitStruct;
RCC_OscInitTypeDef RCC_OscInitStruct;
/* Enable HSE Oscillator and activate PLL with HSE as source */
RCC OscInitStruct.OscillatorType = RCC OSCILLATORTYPE HSE;
RCC OscInitStruct.HSEState = RCC HSE ON;
RCC OscInitStruct.PLL.PLLState = RCC PLL ON;
RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSE;
RCC_OscInitStruct.PLL.PLLM = 25;
RCC_OscInitStruct.PLL.PLLN = 336;
RCC OscInitStruct.PLL.PLLP = RCC PLLP DIV2;
RCC OscInitStruct.PLL.PLLQ = 7;
HAL RCC OscConfig(&RCC OscInitStruct);
/* Select PLL as system clock source and configure the HCLK, PCLK1 and PCLK2 clocks
dividers */
RCC ClkInitStruct.ClockType = (RCC CLOCKTYPE SYSCLK | RCC CLOCKTYPE HCLK |
RCC CLOCKTYPE PCLK1 | RCC CLOCKTYPE PCLK2);
RCC ClkInitStruct.SYSCLKSource = RCC SYSCLKSOURCE PLLCLK;
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV4;
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV2;
HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_5); }
```

2.12.2.3 HAL MSP initialization process

The peripheral initialization is done through *HAL_PPP_Init()* while the hardware resources initialization used by a peripheral (PPP) is performed during this initialization by calling MSP callback function *HAL_PPP_MspInit().*

The MspInit callback performs the low level initialization related to the different additional hardware resources: RCC, GPIO, NVIC and DMA.

All the HAL drivers with handles include two MSP callbacks for initialization and de-initialization:

```
/**
* @brief Initializes the PPP MSP.
* @param hppp: PPP handle
* @retval None */
void   weak HAL PPP MspInit(PPP HandleTypeDef *hppp) {
/* NOTE : This function Should not be modified, when the callback is needed,
the HAL_PPP_MspInit could be implemented in the user file */
}
/**
* @brief DeInitializes PPP MSP.
* @param hppp: PPP handle
* @retval None */
void __weak HAL_PPP_MspDeInit(PPP_HandleTypeDef *hppp) {
/* NOTE : This function Should not be modified, when the callback is needed,
the HAL PPP MspDeInit could be implemented in the user file */
}
```

The MSP callbacks are declared empty as weak functions in each peripheral driver. The user can use them to set the low level initialization code or omit them and use his own initialization routine.

The HAL MSP callback is implemented inside the *stm32f4xx_hal_msp.c* file in the user folders. An *stm32f4xx_hal_msp.c* file template is located in the HAL folder and should be copied to the user folder. It can be generated automatically by STM32CubeMX tool and further modified. Note that all the routines are declared as weak functions and could be overwritten or removed to use user low level initialization code.

Stm32f4xx_hal_msp.c file contains the following functions:

Table 14: MSP functions

Routine	Description
void HAL_MspInit()	Global MSP initialization routine
void HAL_MspDeInit()	Global MSP de-initialization routine
void HAL_PPP_MspInit()	PPP MSP initialization routine
void HAL_PPP_MspDeInit()	PPP MSP de-initialization routine

By default, if no peripheral needs to be de-initialized during the program execution, the whole MSP initialization is done in *Hal_MspInit()* and MSP De-Initialization in the *Hal_MspDeInit().* In this case the *HAL_PPP_MspInit()* and *HAL_PPP_MspDeInit()* are not implemented.

When one or more peripherals needs to be de-initialized in run time and the low level resources of a given peripheral need to be released and used by another peripheral, HAL_PPP_MspDeInit() and *HAL_PPP_MspInit()* are implemented for the concerned peripheral and other peripherals initialization and de-Initialization are kept in the global *HAL_MspInit()* and the *HAL_MspDeInit().*

If there is nothing to be initialized by the global *HAL_MspInit()* and *HAL_MspDeInit()*, the two routines can simply be omitted.

2.12.3 HAL IO operation process

The HAL functions with internal data processing like Transmit, Receive, Write and Read are generally provided with three data processing modes as follows:

- Polling mode
- Interrupt mode
- DMA mode

2.12.3.1 Polling mode

In polling mode, the HAL functions return the process status when the data processing in blocking mode is complete. The operation is considered complete when the function returns the HAL_OK status, otherwise an error status is returned. The user can get more information through the *HAL_PPP_GetState()* function. The data processing is handled internally in a loop. A timeout (expressed in ms) is used to prevent process hanging.

The example below shows the typical polling mode processing sequence :

```
HAL_StatusTypeDef HAL_PPP_Transmit ( PPP_HandleTypeDef * phandle, uint8_t pData,
int16 tSize,uint32 tTimeout)
{
if((pData == NULL ) || (Size == 0))
{
return HAL_ERROR;
}
(…) while (data processing is running)
{
if( timeout reached )
```

```
{
return HAL_TIMEOUT;
}
}
 (…)
return HELIAC; }
```

2.12.3.2 Interrupt mode

In Interrupt mode, the HAL function returns the process status after starting the data processing and enabling the appropriate interruption. The end of the operation is indicated by a callback declared as a weak function. It can be customized by the user to be informed in real-time about the process completion. The user can also get the process status through the *HAL_PPP_GetState()* function.

In interrupt mode, four functions are declared in the driver:

- *HAL_PPP_Process_IT()*: launch the process
- *HAL_PPP_IRQHandler()*: the global PPP peripheral interruption
- *__weak HAL_PPP_ProcessCpltCallback ()*: the callback relative to the process completion.
- *__weak HAL_PPP_ProcessErrorCallback()*: the callback relative to the process Error.

To use a process in interrupt mode, *HAL_PPP_Process_IT()* is called in the user file and HAL_PPP_IRQHandler in *stm32f4xx_it.c.*

The *HAL_PPP_ProcessCpltCallback()* function is declared as weak function in the driver. This means that the user can declare it again in the application. The function in the driver is not modified.

An example of use is illustrated below:

main.c file:

```
UART HandleTypeDef UartHandle;
int main(void)
{
/* Set User Parameters */
UartHandle.Init.BaudRate = 9600;
UartHandle.Init.WordLength = UART DATABITS 8;
UartHandle.Init.StopBits = UART STOPBITS 1;
UartHandle.Init.Parity = UART PARITY NONE;
UartHandle.Init.HwFlowCtl = UART_HWCONTROL_NONE;
UartHandle.Init.Mode = UART_MODE_TX_RX;
UartHandle.Init.Instance = USART3;
HAL UART Init(&UartHandle);
HAL UART SendIT(&UartHandle, TxBuffer, sizeof(TxBuffer));
while (1);
}
void HAL_UART_TxCpltCallback(UART_HandleTypeDef *huart)
{
}
void HAL UART ErrorCallback(UART HandleTypeDef *huart)
{
}
```

*stm32f4xx_it.c*file:

```
extern UART_HandleTypeDef UartHandle;
void USART3 IRQHandler(void)
{
HAL UART IRQHandler(&UartHandle);
 }
```

2.12.3.3 DMA mode

In DMA mode, the HAL function returns the process status after starting the data processing through the DMA and after enabling the appropriate DMA interruption. The end of the operation is indicated by a callback declared as a weak function and can be customized by the user to be informed in real-time about the process completion. The user can also get the process status through the *HAL_PPP_GetState()* function. For the DMA mode, three functions are declared in the driver:

- *HAL_PPP_Process_DMA()*: launch the process
- *HAL_PPP_DMA_IRQHandler()*: the DMA interruption used by the PPP peripheral
- *__weak HAL_PPP_ProcessCpltCallback()*: the callback relative to the process completion.
- *__weak HAL_PPP_ErrorCpltCallback()*: the callback relative to the process Error.

To use a process in DMA mode, *HAL_PPP_Process_DMA()* is called in the user file and the *HAL_PPP_DMA_IRQHandler()* is placed in the *stm32f4xx_it.c*. When DMA mode is used, the DMA initialization is done in the *HAL_PPP_MspInit()* callback. The user should also associate the DMA handle to the PPP handle. For this purpose, the handles of all the peripheral drivers that use the DMA must be declared as follows:

```
typedef struct
{
PPP TypeDef *Instance; /* Register base address */
PPP_InitTypeDef Init; /* PPP communication parameters */
HAL_StateTypeDef State; /* PPP communication state */
(…)
DMA HandleTypeDef *hdma; /* associated DMA handle */
} PPP_HandleTypeDef;
```

The initialization is done as follows (UART example):

```
int main(void)
{
/* Set User Parameters */
UartHandle.Init.BaudRate = 9600;
UartHandle.Init.WordLength = UART_DATABITS_8;
UartHandle.Init.StopBits = UART_STOPBITS_1;
UartHandle.Init.Parity = UART PARITY NONE;
UartHandle.Init.HwFlowCtl = UART HWCONTROL NONE;
UartHandle.Init.Mode = UART_MODE_TX_RX;
UartHandle.Init.Instance = UART3;
HAL_UART_Init(&UartHandle);
(..)
}
void HAL USART MspInit (UART HandleTypeDef * huart)
{
static DMA_HandleTypeDef hdma_tx;
static DMA_HandleTypeDef hdma_rx;
(…)
   HAL LINKDMA(UartHandle, DMA Handle tx, hdma tx);
      HAL LINKDMA(UartHandle, DMA Handle rx, hdma rx);
(…)
}
```

The *HAL_PPP_ProcessCpltCallback()* function is declared as weak function in the driver that means, the user can declare it again in the application code. The function in the driver should not be modified.

An example of use is illustrated below:

main.c file:

```
UART HandleTypeDef UartHandle;
int main(void)
{
/* Set User Paramaters */
```

```
UartHandle.Init.BaudRate = 9600;
UartHandle.Init.WordLength = UART_DATABITS_8;
UartHandle.Init.StopBits = UART_STOPBITS_1;
UartHandle.Init.Parity = UART_PARITY_NONE;
UartHandle.Init.HwFlowCtl = UART HWCONTROL NONE;
UartHandle.Init.Mode = UART MODE TX RX;
UartHandle.Init.Instance = USART3;
HAL_UART_Init(&UartHandle);
HAL_UART_Send_DMA(&UartHandle, TxBuffer, sizeof(TxBuffer));
while (1);
}
void HAL_UART_TxCpltCallback(UART_HandleTypeDef *phuart)
{
}
void HAL_UART_TxErrorCallback(UART_HandleTypeDef *phuart)
{
}
```

stm32f4xx_it.c file:

```
extern UART HandleTypeDef UartHandle;
void DMAx_IRQHandler(void)
{
HAL_DMA_IRQHandler(&UartHandle.DMA_Handle_tx);
}
```

HAL_USART_TxCpltCallback() and *HAL_USART_ErrorCallback()* should be linked in the *HAL_PPP_Process_DMA()* function to the DMA transfer complete callback and the DMA transfer Error callback by using the following statement:

```
HAL PPP Process DMA (PPP HandleTypeDef *hppp, Params….)
{
(…)
hppp->DMA_Handle->XferCpltCallback = HAL_UART_TxCpltCallback ;
hppp->DMA Handle->XferErrorCallback = HAL UART ErrorCallback ;
(…)
}
```

2.12.4 Timeout and error management

2.12.4.1 Timeout management

The timeout is often used for the APIs that operate in polling mode. It defines the delay during which a blocking process should wait till an error is returned. An example is provided below:

```
HAL StatusTypeDef HAL DMA PollForTransfer(DMA HandleTypeDef *hdma, uint32 t
CompleteLevel, uint32_t Timeout)
```

The timeout possible value are the following:

Table 15: Timeout values

Timeout value	Description
0	No poll : Immediate process check and exit
1 ... (HAL_MAX_DELAY -1)(1)	Timeout in ms
HAL_MAX_DELAY	Infinite poll till process is successful

Notes:

(1)HAL_MAX_DELAY is defined in the stm32fxxx_hal_def.h as 0xFFFFFFFF

However, in some cases, a fixed timeout is used for system peripherals or internal HAL driver processes. In these cases, the timeout has the same meaning and is used in the same way, except when it is defined locally in the drivers and cannot be modified or introduced as an argument in the user application.

Example of fixed timeout:

```
#define LOCAL_PROCESS_TIMEOUT 100
HAL_StatusTypeDef HAL_PPP_Process(PPP_HandleTypeDef)
{
 (…)
 timeout = HAL GetTick() + LOCAL PROCESS TIMEOUT;
 (…)
 while(ProcessOngoing)
 {
 (…)
 if(HAL GetTick() >= timeout)
 {
  /* Process unlocked */
  __HAL_UNLOCK(hppp);
  hppp->State= HAL PPP STATE TIMEOUT;
  return HAL PPP STATE TIMEOUT;
 }
 }
 (…)
}
```

The following example shows how to use the timeout inside the polling functions:

```
HAL_PPP_StateTypeDef HAL_PPP_Poll (PPP_HandleTypeDef *hppp, uint32_t Timeout)
{
 (…)
 timeout = HAL GetTick() + Timeout;
 (…)
 while(ProcessOngoing)
 {
 (…)
 if(Timeout != HAL MAX DELAY)
 {
 if(HAL_GetTick() >= timeout)
 {
 /* Process unlocked */
 __HAL_UNLOCK(hppp);
 hppp->State= HAL PPP STATE TIMEOUT;
 return hppp->State;
 }
 }
 (…)
}
```

2.12.4.2 Error management

The HAL drivers implement a check for the following items:

- Valid parameters: for some process the used parameters should be valid and already defined, otherwise the system can crash or go into an undefined state. These critical parameters are checked before they are used (see example below).

```
HAL StatusTypeDef HAL PPP Process(PPP HandleTypeDef* hppp, uint32 t *pdata, uint32
Size)
 { if ((pData == NULL ) || (Size == 0))
 { return HAL ERROR;
 }
 }
```

- Valid handle: the PPP peripheral handle is the most important argument since it keeps the PPP driver vital parameters. It is always checked in the beginning of the *HAL_PPP_Init()* function.

```
HAL StatusTypeDef HAL PPP Init(PPP HandleTypeDef* hppp)
 { if (hppp == NULL) //the handle should be already allocated
 { return HAL_ERROR;
 }
 }
```

- Timeout error: the following statement is used when a timeout error occurs:

```
while (Process ongoing)
{
  timeout = HAL GetTick() + Timeout;
  while (data processing is running)
   {
    if(timeout)
   { return HAL_TIMEOUT;
 }
}
```

When an error occurs during a peripheral process, *HAL_PPP_Process ()* returns with a *HAL_ERROR* status. The HAL PPP driver implements the *HAL_PPP_GetError ()* to allow retrieving the origin of the error.

```
HAL_PPP_ErrorTypeDef HAL_PPP_GetError (PPP_HandleTypeDef *hppp);
```

In all peripheral handles, a *HAL_PPP_ErrorTypeDef* is defined and used to store the last error code.

```
typedef struct
{
 PPP TypeDef * Instance; /* PPP registers base address */
 PPP InitTypeDef Init; /* PPP initialization parameters */
 HAL_LockTypeDef Lock; /* PPP locking object */
 __IO HAL_PPP_StateTypeDef State; /* PPP state */
   IO HAL PPP ErrorTypeDef ErrorCode; /* PPP Error code */
 (…)
 /* PPP specific parameters */
}
PPP_HandleTypeDef;
```

The error state and the peripheral global state are always updated before returning an error:

```
PPP->State = HAL PPP READY;   /* Set the peripheral ready */
PP->ErrorCode = HAL ERRORCODE ; /* Set the error code */
_HAL_UNLOCK(PPP) ;       /* Unlock the PPP resources */
return HAL_ERROR;        /*return with HAL error */
```

HAL_PPP_GetError () must be used in interrupt mode in the error callback:

```
void HAL PPP ProcessCpltCallback(PPP HandleTypeDef *hspi)
{
 ErrorCode = HAL PPP GetError (hppp); /* retreive error code */
}
```

2.12.4.3 Run-time checking

The HAL implements run-time failure detection by checking the input values of all HAL driver functions. The run-time checking is achieved by using an *assert_param* macro. This macro is used in all the HAL drivers' functions which have an input parameter. It allows verifying that the input value lies within the parameter allowed values.

To enable the run-time checking, use the *assert_param* macro, and leave the define ***USE_FULL_ASSERT*** uncommented in *stm32f34xx_hal_conf.h* file.

```
void HAL_UART_Init(UART_HandleTypeDef *huart)
{
 (..) /* Check the parameters */
assert param(IS UART INSTANCE(huart->Instance));
assert param(IS UART BAUDRATE(huart->Init.BaudRate));
assert_param(IS_UART_WORD_LENGTH(huart->Init.WordLength));
assert_param(IS_UART_STOPBITS(huart->Init.StopBits));
assert param(IS UART PARITY(huart->Init.Parity));
assert param(IS UART MODE(huart->Init.Mode));
assert param(IS UART HARDWARE FLOW CONTROL(huart->Init.HwFlowCtl));
 (..)
```

```
/** @defgroup UART_Word_Length *
@{
*/
#define UART WORDLENGTH 8B ((uint32 t)0x00000000)
#define UART WORDLENGTH 9B ((uint32 t)USART CR1 M)
#define IS_UART_WORD_LENGTH(LENGTH) (((LENGTH) == UART_WORDLENGTH_8B) ||
\ ((LENGTH) == UART_WORDLENGTH_9B))
```

If the expression passed to the assert_param macro is false, the*assert_failed* function is called and returns the name of the source file and the source line number of the call that failed. If the expression is true, no value is returned.

The *assert_param* macro is implemented in stm32f4xx_hal_conf.h:

```
/* Exported macro ------------------------------------------------------------*/
#ifdef USE FULL ASSERT
/**
* @brief The assert param macro is used for function's parameters check.
* @param expr: If expr is false, it calls assert_failed function
* which reports the name of the source file and the source
* line number of the call that failed.
* If expr is true, it returns no value.
* @retval None */
#define assert param(expr) ((expr)?(void)0:assert failed((uint8 t *)  FILE  ,
__LINE__))
/* Exported functions --------------------------------------*/
void assert_failed(uint8_t* file, uint32_t line);
#else
#define assert param(expr)((void)0)
#endif /* USE_FULL_ASSERT */
```

The *assert_failed* function is implemented in the main.c file or in any other user C file:

```
#ifdef USE FULL ASSERT /**
* @brief Reports the name of the source file and the source line number
* where the assert_param error has occurred.
* @param file: pointer to the source file name
* @param line: assert param error line source number
* @retval None */
void assert failed(uint8 t* file, uint32 t line)
{
/* User can add his own implementation to report the file name and line number,
ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
/* Infinite loop */
while (1)
{
 }
}
```

Because of the overhead run-time checking introduces, it is recommended to use it during application code development and debugging, and to remove it from the final application to improve code size and speed.

3 HAL System Driver

3.1 HAL Firmware driver API description

3.1.1 How to use this driver

The common HAL driver contains a set of generic and common APIs that can be used by the PPP peripheral drivers and the user to start using the HAL.

The HAL contains two APIs' categories:

- Common HAL APIs
- Services HAL APIs

3.1.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initializes the Flash interface the NVIC allocation and initial clock configuration. It initializes the systick also when timeout is needed and the backup domain when enabled.
- de-Initializes common part of the HAL
- Configure The time base source to have 1ms time base with a dedicated Tick interrupt priority.
  - Systick timer is used by default as source of time base, but user can eventually implement his proper time base source (a general purpose timer for example or other time source), keeping in mind that Time base duration should be kept 1ms since PPP_TIMEOUT_VALUEs are defined and handled in milliseconds basis.
  - Time base configuration function (HAL_InitTick ()) is called automatically at the beginning of the program after reset by HAL_Init() or at any time when clock is configured, by HAL_RCC_ClockConfig().
  - Source of time base is configured to generate interrupts at regular time intervals. Care must be taken if HAL_Delay() is called from a peripheral ISR process, the Tick interrupt line must have higher priority (numerically lower) than the peripheral interrupt. Otherwise the caller ISR process will be blocked.
  - functions affecting time base configurations are declared as __weak to make override possible in case of other implementations in user file.

This section contains the following APIs:

- ***HAL_Init()***
- ***HAL_DeInit()***
- ***HAL_MspInit()***
- ***HAL_MspDeInit()***
- ***HAL_InitTick()***

3.1.3 HAL Control functions

This section provides functions allowing to:

- Provide a tick value in millisecond
- Provide a blocking delay in millisecond
- Suspend the time base source interrupt
- Resume the time base source interrupt
- Get the HAL API driver version
- Get the device identifier

- Get the device revision identifier
- Enable/Disable Debug module during SLEEP mode
- Enable/Disable Debug module during STOP mode
- Enable/Disable Debug module during STANDBY mode

This section contains the following APIs:

- ***HAL_IncTick()***
- ***HAL_GetTick()***
- ***HAL_Delay()***
- ***HAL_SuspendTick()***
- ***HAL_ResumeTick()***
- ***HAL_GetHalVersion()***
- ***HAL_GetREVID()***
- ***HAL_GetDEVID()***
- ***HAL_DBGMCU_EnableDBGSleepMode()***
- ***HAL_DBGMCU_DisableDBGSleepMode()***
- ***HAL_DBGMCU_EnableDBGStopMode()***
- ***HAL_DBGMCU_DisableDBGStopMode()***
- ***HAL_DBGMCU_EnableDBGStandbyMode()***
- ***HAL_DBGMCU_DisableDBGStandbyMode()***
- ***HAL_EnableCompensationCell()***
- ***HAL_DisableCompensationCell()***
- ***HAL_EnableMemorySwappingBank()***
- ***HAL_DisableMemorySwappingBank()***

3.1.4 Detailed description of functions

HAL_Init

Function Name	**HAL_StatusTypeDef HAL_Init (void)**
Function Description	This function is used to initialize the HAL Library; it must be the first instruction to be executed in the main program (before to call any other HAL function), it performs the following: Configure the Flash prefetch, instruction and Data caches.
Return values	• **HAL:** status
Notes	• SysTick is used as time base for the HAL_Delay() function, the application need to ensure that the SysTick time base is always set to 1 millisecond to have correct HAL operation.

HAL_DeInit

Function Name	**HAL_StatusTypeDef HAL_DeInit (void)**
Function Description	This function de-Initializes common part of the HAL and stops the systick.
Return values	• **HAL:** status

HAL_MspInit

Function Name	**void HAL_MspInit (void)**
Function Description	Initializes the MSP.

Return values
- **None**

HAL_MspDeInit

Function Name **void HAL_MspDeInit (void)**

Function Description DeInitializes the MSP.

Return values
- **None**

HAL_InitTick

Function Name **HAL_StatusTypeDef HAL_InitTick (uint32_t TickPriority)**

Function Description This function configures the source of the time base.

Parameters
- **TickPriority:** Tick interrupt priority.

Return values
- **HAL:** status

Notes
- This function is called automatically at the beginning of program after reset by HAL_Init() or at any time when clock is reconfigured by HAL_RCC_ClockConfig().
- In the default implementation, SysTick timer is the source of time base. It is used to generate interrupts at regular time intervals. Care must be taken if HAL_Delay() is called from a peripheral ISR process, The the SysTick interrupt must have higher priority (numerically lower) than the peripheral interrupt. Otherwise the caller ISR process will be blocked. The function is declared as __weak to be overwritten in case of other implementation in user file.

HAL_IncTick

Function Name **void HAL_IncTick (void)**

Function Description This function is called to increment a global variable "uwTick" used as application time base.

Return values
- **None**

Notes
- In the default implementation, this variable is incremented each 1ms in Systick ISR.
- This function is declared as __weak to be overwritten in case of other implementations in user file.

HAL_Delay

Function Name **void HAL_Delay (__IO uint32_t Delay)**

Function Description This function provides accurate delay (in milliseconds) based on variable incremented.

Parameters
- **Delay:** specifies the delay time length, in milliseconds.

Return values
- **None**

Notes
- In the default implementation , SysTick timer is the source of time base. It is used to generate interrupts at regular time intervals where uwTick is incremented.

- This function is declared as __weak to be overwritten in case of other implementations in user file.

HAL_GetTick

Function Name	**uint32_t HAL_GetTick (void)**
Function Description	Provides a tick value in millisecond.
Return values	• **tick:** value
Notes	• This function is declared as __weak to be overwritten in case of other implementations in user file.

HAL_SuspendTick

Function Name	**void HAL_SuspendTick (void)**
Function Description	Suspend Tick increment.
Return values	• **None**
Notes	• In the default implementation , SysTick timer is the source of time base. It is used to generate interrupts at regular time intervals. Once HAL_SuspendTick() is called, the SysTick interrupt will be disabled and so Tick increment is suspended. • This function is declared as __weak to be overwritten in case of other implementations in user file.

HAL_ResumeTick

Function Name	**void HAL_ResumeTick (void)**
Function Description	Resume Tick increment.
Return values	• **None**
Notes	• In the default implementation , SysTick timer is the source of time base. It is used to generate interrupts at regular time intervals. Once HAL_ResumeTick() is called, the SysTick interrupt will be enabled and so Tick increment is resumed. • This function is declared as __weak to be overwritten in case of other implementations in user file.

HAL_GetHalVersion

Function Name	**uint32_t HAL_GetHalVersion (void)**
Function Description	Returns the HAL revision.
Return values	• **version:** : 0xXYZR (8bits for each decimal, R for RC)

HAL_GetREVID

Function Name	**uint32_t HAL_GetREVID (void)**
Function Description	Returns the device revision identifier.
Return values	• **Device:** revision identifier

HAL_GetDEVID

Function Name	**uint32_t HAL_GetDEVID (void)**
Function Description	Returns the device identifier.
Return values	• **Device:** identifier

HAL_DBGMCU_EnableDBGSleepMode

Function Name	**void HAL_DBGMCU_EnableDBGSleepMode (void)**
Function Description	Enable the Debug Module during SLEEP mode.
Return values	• **None**

HAL_DBGMCU_DisableDBGSleepMode

Function Name	**void HAL_DBGMCU_DisableDBGSleepMode (void)**
Function Description	Disable the Debug Module during SLEEP mode.
Return values	• **None**

HAL_DBGMCU_EnableDBGStopMode

Function Name	**void HAL_DBGMCU_EnableDBGStopMode (void)**
Function Description	Enable the Debug Module during STOP mode.
Return values	• **None**

HAL_DBGMCU_DisableDBGStopMode

Function Name	**void HAL_DBGMCU_DisableDBGStopMode (void)**
Function Description	Disable the Debug Module during STOP mode.
Return values	• **None**

HAL_DBGMCU_EnableDBGStandbyMode

Function Name	**void HAL_DBGMCU_EnableDBGStandbyMode (void)**
Function Description	Enable the Debug Module during STANDBY mode.
Return values	• **None**

HAL_DBGMCU_DisableDBGStandbyMode

Function Name	**void HAL_DBGMCU_DisableDBGStandbyMode (void)**
Function Description	Disable the Debug Module during STANDBY mode.
Return values	• **None**

HAL_EnableCompensationCell

Function Name	**void HAL_EnableCompensationCell (void)**
Function Description	Enables the I/O Compensation Cell.

Return values	• **None**
Notes	• The I/O compensation cell can be used only when the device supply voltage ranges from 2.4 to 3.6 V.

HAL_DisableCompensationCell

Function Name	**void HAL_DisableCompensationCell (void)**
Function Description	Power-down the I/O Compensation Cell.
Return values	• **None**
Notes	• The I/O compensation cell can be used only when the device supply voltage ranges from 2.4 to 3.6 V.

HAL_EnableMemorySwappingBank

Function Name	**void HAL_EnableMemorySwappingBank (void)**
Function Description	Enables the Internal FLASH Bank Swapping.
Return values	• **None**
Notes	• This function can be used only for STM32F42xxx/43xxx devices. • Flash Bank2 mapped at 0x08000000 (and aliased @0x00000000) and Flash Bank1 mapped at 0x08100000 (and aliased at 0x00100000)

HAL_DisableMemorySwappingBank

Function Name	**void HAL_DisableMemorySwappingBank (void)**
Function Description	Disables the Internal FLASH Bank Swapping.
Return values	• **None**
Notes	• This function can be used only for STM32F42xxx/43xxx devices. • The default state : Flash Bank1 mapped at 0x08000000 (and aliased @0x00000000) and Flash Bank2 mapped at 0x08100000 (and aliased at 0x00100000)

3.2 HAL Firmware driver defines

3.2.1 HAL

HAL CAN Error Code

HAL_CAN_ERROR_NONE	No error
HAL_CAN_ERROR_EWG	EWG error
HAL_CAN_ERROR_EPV	EPV error
HAL_CAN_ERROR_BOF	BOF error
HAL_CAN_ERROR_STF	Stuff error
HAL_CAN_ERROR_FOR	Form error

HAL_CAN_ERROR_ACK	Acknowledgment error
HAL_CAN_ERROR_BR	Bit recessive
HAL_CAN_ERROR_BD	LEC dominant
HAL_CAN_ERROR_CRC	LEC transfer error

HAL Exported Macros

__HAL_DBGMCU_FREEZE_TIM2
__HAL_DBGMCU_FREEZE_TIM3
__HAL_DBGMCU_FREEZE_TIM4
__HAL_DBGMCU_FREEZE_TIM5
__HAL_DBGMCU_FREEZE_TIM6
__HAL_DBGMCU_FREEZE_TIM7
__HAL_DBGMCU_FREEZE_TIM12
__HAL_DBGMCU_FREEZE_TIM13
__HAL_DBGMCU_FREEZE_TIM14
__HAL_DBGMCU_FREEZE_RTC
__HAL_DBGMCU_FREEZE_WWDG
__HAL_DBGMCU_FREEZE_IWDG
__HAL_DBGMCU_FREEZE_I2C1_TIMEOUT
__HAL_DBGMCU_FREEZE_I2C2_TIMEOUT
__HAL_DBGMCU_FREEZE_I2C3_TIMEOUT
__HAL_DBGMCU_FREEZE_CAN1
__HAL_DBGMCU_FREEZE_CAN2
__HAL_DBGMCU_FREEZE_TIM1
__HAL_DBGMCU_FREEZE_TIM8
__HAL_DBGMCU_FREEZE_TIM9
__HAL_DBGMCU_FREEZE_TIM10
__HAL_DBGMCU_FREEZE_TIM11
__HAL_DBGMCU_UNFREEZE_TIM2
__HAL_DBGMCU_UNFREEZE_TIM3
__HAL_DBGMCU_UNFREEZE_TIM4
__HAL_DBGMCU_UNFREEZE_TIM5
__HAL_DBGMCU_UNFREEZE_TIM6
__HAL_DBGMCU_UNFREEZE_TIM7
__HAL_DBGMCU_UNFREEZE_TIM12
__HAL_DBGMCU_UNFREEZE_TIM13
__HAL_DBGMCU_UNFREEZE_TIM14

__HAL_DBGMCU_UNFREEZE_RTC
__HAL_DBGMCU_UNFREEZE_WWDG
__HAL_DBGMCU_UNFREEZE_IWDG
__HAL_DBGMCU_UNFREEZE_I2C1_TIMEOUT
__HAL_DBGMCU_UNFREEZE_I2C2_TIMEOUT
__HAL_DBGMCU_UNFREEZE_I2C3_TIMEOUT
__HAL_DBGMCU_UNFREEZE_CAN1
__HAL_DBGMCU_UNFREEZE_CAN2
__HAL_DBGMCU_UNFREEZE_TIM1
__HAL_DBGMCU_UNFREEZE_TIM8
__HAL_DBGMCU_UNFREEZE_TIM9
__HAL_DBGMCU_UNFREEZE_TIM10
__HAL_DBGMCU_UNFREEZE_TIM11
__HAL_SYSCFG_REMAPMEMORY_FLASH
__HAL_SYSCFG_REMAPMEMORY_SYSTEMFLASH
__HAL_SYSCFG_REMAPMEMORY_SRAM
__HAL_SYSCFG_REMAPMEMORY_FMC
__HAL_SYSCFG_REMAPMEMORY_FMC_SDRAM

4 HAL ADC Generic Driver

4.1 ADC Firmware driver registers structures

4.1.1 ADC_InitTypeDef

Data Fields

- ***uint32_t ClockPrescaler***
- ***uint32_t Resolution***
- ***uint32_t DataAlign***
- ***uint32_t ScanConvMode***
- ***uint32_t EOCSelection***
- ***uint32_t ContinuousConvMode***
- ***uint32_t NbrOfConversion***
- ***uint32_t DiscontinuousConvMode***
- ***uint32_t NbrOfDiscConversion***
- ***uint32_t ExternalTrigConv***
- ***uint32_t ExternalTrigConvEdge***
- ***uint32_t DMAContinuousRequests***

Field Documentation

- ***uint32_t ADC_InitTypeDef::ClockPrescaler***
  Select ADC clock prescaler. The clock is common for all the ADCs. This parameter can be a value of ***ADC_ClockPrescaler***
- ***uint32_t ADC_InitTypeDef::Resolution***
  Configures the ADC resolution. This parameter can be a value of ***ADC_Resolution***
- ***uint32_t ADC_InitTypeDef::DataAlign***
  Specifies ADC data alignment to right (MSB on register bit 11 and LSB on register bit 0) (default setting) or to left (if regular group: MSB on register bit 15 and LSB on register bit 4, if injected group (MSB kept as signed value due to potential negative value after offset application): MSB on register bit 14 and LSB on register bit 3). This parameter can be a value of ***ADC_Data_align***
- ***uint32_t ADC_InitTypeDef::ScanConvMode***
  Configures the sequencer of regular and injected groups. This parameter can be associated to parameter 'DiscontinuousConvMode' to have main sequence subdivided in successive parts. If disabled: Conversion is performed in single mode (one channel converted, the one defined in rank 1). Parameters 'NbrOfConversion' and 'InjectedNbrOfConversion' are discarded (equivalent to set to 1). If enabled: Conversions are performed in sequence mode (multiple ranks defined by 'NbrOfConversion'/'InjectedNbrOfConversion' and each channel rank). Scan direction is upward: from rank1 to rank 'n'. This parameter can be set to ENABLE or DISABLE
- ***uint32_t ADC_InitTypeDef::EOCSelection***
  Specifies what EOC (End Of Conversion) flag is used for conversion by polling and interruption: end of conversion of each rank or complete sequence. This parameter can be a value of ***ADC_EOCSelection***. Note: For injected group, end of conversion (flag&IT) is raised only at the end of the sequence. Therefore, if end of conversion is set to end of each conversion, injected group should not be used with interruption (HAL_ADCEx_InjectedStart_IT) or polling (HAL_ADCEx_InjectedStart and

HAL_ADCEx_InjectedPollForConversion). By the way, polling is still possible since driver will use an estimated timing for end of injected conversion. Note: If overrun feature is intended to be used, use ADC in mode 'interruption' (function **HAL_ADC_Start_IT()**) with parameter EOCSelection set to end of each conversion or in mode 'transfer by DMA' (function **HAL_ADC_Start_DMA()**). If overrun feature is intended to be bypassed, use ADC in mode 'polling' or 'interruption' with parameter EOCSelection must be set to end of sequence

- ***uint32_t ADC_InitTypeDef::ContinuousConvMode***
  Specifies whether the conversion is performed in single mode (one conversion) or continuous mode for regular group, after the selected trigger occurred (software start or external trigger). This parameter can be set to ENABLE or DISABLE.
- ***uint32_t ADC_InitTypeDef::NbrOfConversion***
  Specifies the number of ranks that will be converted within the regular group sequencer. To use regular group sequencer and convert several ranks, parameter 'ScanConvMode' must be enabled. This parameter must be a number between Min_Data = 1 and Max_Data = 16.
- ***uint32_t ADC_InitTypeDef::DiscontinuousConvMode***
  Specifies whether the conversions sequence of regular group is performed in Complete-sequence/Discontinuous-sequence (main sequence subdivided in successive parts). Discontinuous mode is used only if sequencer is enabled (parameter 'ScanConvMode'). If sequencer is disabled, this parameter is discarded. Discontinuous mode can be enabled only if continuous mode is disabled. If continuous mode is enabled, this parameter setting is discarded. This parameter can be set to ENABLE or DISABLE.
- ***uint32_t ADC_InitTypeDef::NbrOfDiscConversion***
  Specifies the number of discontinuous conversions in which the main sequence of regular group (parameter NbrOfConversion) will be subdivided. If parameter 'DiscontinuousConvMode' is disabled, this parameter is discarded. This parameter must be a number between Min_Data = 1 and Max_Data = 8.
- ***uint32_t ADC_InitTypeDef::ExternalTrigConv***
  Selects the external event used to trigger the conversion start of regular group. If set to ADC_SOFTWARE_START, external triggers are disabled. If set to external trigger source, triggering is on event rising edge by default. This parameter can be a value of ***ADC_External_trigger_Source_Regular***
- ***uint32_t ADC_InitTypeDef::ExternalTrigConvEdge***
  Selects the external trigger edge of regular group. If trigger is set to ADC_SOFTWARE_START, this parameter is discarded. This parameter can be a value of ***ADC_External_trigger_edge_Regular***
- ***uint32_t ADC_InitTypeDef::DMAContinuousRequests***
  Specifies whether the DMA requests are performed in one shot mode (DMA transfer stop when number of conversions is reached) or in Continuous mode (DMA transfer unlimited, whatever number of conversions). Note: In continuous mode, DMA must be configured in circular mode. Otherwise an overrun will be triggered when DMA buffer maximum pointer is reached. Note: This parameter must be modified when no conversion is on going on both regular and injected groups (ADC disabled, or ADC enabled without continuous mode or external trigger that could launch a conversion). This parameter can be set to ENABLE or DISABLE.

4.1.2 ADC_ChannelConfTypeDef

Data Fields

- ***uint32_t Channel***
- ***uint32_t Rank***
- ***uint32_t SamplingTime***
- ***uint32_t Offset***

Field Documentation

- ***uint32_t ADC_ChannelConfTypeDef::Channel***
  Specifies the channel to configure into ADC regular group. This parameter can be a value of ***ADC_channels***
- ***uint32_t ADC_ChannelConfTypeDef::Rank***
  Specifies the rank in the regular group sequencer. This parameter must be a number between Min_Data = 1 and Max_Data = 16
- ***uint32_t ADC_ChannelConfTypeDef::SamplingTime***
  Sampling time value to be set for the selected channel. Unit: ADC clock cycles Conversion time is the addition of sampling time and processing time (12 ADC clock cycles at ADC resolution 12 bits, 11 cycles at 10 bits, 9 cycles at 8 bits, 7 cycles at 6 bits). This parameter can be a value of ***ADC_sampling_times*** Caution: This parameter updates the parameter property of the channel, that can be used into regular and/or injected groups. If this same channel has been previously configured in the other group (regular/injected), it will be updated to last setting. Note: In case of usage of internal measurement channels (VrefInt/Vbat/TempSensor), sampling time constraints must be respected (sampling time can be adjusted in function of ADC clock frequency and sampling time setting) Refer to device datasheet for timings values, parameters TS_vrefint, TS_temp (values rough order: 4us min).
- ***uint32_t ADC_ChannelConfTypeDef::Offset***
  Reserved for future use, can be set to 0

4.1.3 ADC_AnalogWDGConfTypeDef

Data Fields

- ***uint32_t WatchdogMode***
- ***uint32_t HighThreshold***
- ***uint32_t LowThreshold***
- ***uint32_t Channel***
- ***uint32_t ITMode***
- ***uint32_t WatchdogNumber***

Field Documentation

- ***uint32_t ADC_AnalogWDGConfTypeDef::WatchdogMode***
  Configures the ADC analog watchdog mode. This parameter can be a value of ***ADC_analog_watchdog_selection***

- ***uint32_t ADC_AnalogWDGConfTypeDef::HighThreshold***
  Configures the ADC analog watchdog High threshold value. This parameter must be a 12-bit value.
- ***uint32_t ADC_AnalogWDGConfTypeDef::LowThreshold***
  Configures the ADC analog watchdog High threshold value. This parameter must be a 12-bit value.
- ***uint32_t ADC_AnalogWDGConfTypeDef::Channel***
  Configures ADC channel for the analog watchdog. This parameter has an effect only if watchdog mode is configured on single channel This parameter can be a value of ***ADC_channels***
- ***uint32_t ADC_AnalogWDGConfTypeDef::ITMode***
  Specifies whether the analog watchdog is configured is interrupt mode or in polling mode. This parameter can be set to ENABLE or DISABLE
- ***uint32_t ADC_AnalogWDGConfTypeDef::WatchdogNumber***
  Reserved for future use, can be set to 0

4.1.4 ADC_HandleTypeDef

Data Fields

- ***ADC_TypeDef * Instance***
- ***ADC_InitTypeDef Init***
- ***__IO uint32_t NbrOfCurrentConversionRank***
- ***DMA_HandleTypeDef * DMA_Handle***
- ***HAL_LockTypeDef Lock***
- ***__IO uint32_t State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***ADC_TypeDef* ADC_HandleTypeDef::Instance***
  Register base address
- ***ADC_InitTypeDef ADC_HandleTypeDef::Init***
  ADC required parameters
- ***__IO uint32_t ADC_HandleTypeDef::NbrOfCurrentConversionRank***
  ADC number of current conversion rank
- ***DMA_HandleTypeDef* ADC_HandleTypeDef::DMA_Handle***
  Pointer DMA Handler
- ***HAL_LockTypeDef ADC_HandleTypeDef::Lock***
  ADC locking object
- ***__IO uint32_t ADC_HandleTypeDef::State***
  ADC communication state
- ***__IO uint32_t ADC_HandleTypeDef::ErrorCode***
  ADC Error code

4.2 ADC Firmware driver API description

4.2.1 ADC Peripheral features

1. 12-bit, 10-bit, 8-bit or 6-bit configurable resolution.
2. Interrupt generation at the end of conversion, end of injected conversion, and in case of analog watchdog or overrun events
3. Single and continuous conversion modes.
4. Scan mode for automatic conversion of channel 0 to channel x.
5. Data alignment with in-built data coherency.
6. Channel-wise programmable sampling time.
7. External trigger option with configurable polarity for both regular and injected conversion.
8. Dual/Triple mode (on devices with 2 ADCs or more).
9. Configurable DMA data storage in Dual/Triple ADC mode.
10. Configurable delay between conversions in Dual/Triple interleaved mode.
11. ADC conversion type (refer to the datasheets).
12. ADC supply requirements: 2.4 V to 3.6 V at full speed and down to 1.8 V at slower speed.
13. ADC input range: VREF(minus) = VIN = VREF(plus).
14. DMA request generation during regular channel conversion.

4.2.2 How to use this driver

1. Initialize the ADC low level resources by implementing the HAL_ADC_MspInit():
   a. Enable the ADC interface clock using __HAL_RCC_ADC_CLK_ENABLE()
   b. ADC pins configuration
      – Enable the clock for the ADC GPIOs using the following function: __HAL_RCC_GPIOx_CLK_ENABLE()
      – Configure these ADC pins in analog mode using HAL_GPIO_Init()
   c. In case of using interrupts (e.g. HAL_ADC_Start_IT())
      – Configure the ADC interrupt priority using HAL_NVIC_SetPriority()
      – Enable the ADC IRQ handler using HAL_NVIC_EnableIRQ()
      – In ADC IRQ handler, call HAL_ADC_IRQHandler()
   d. In case of using DMA to control data transfer (e.g. HAL_ADC_Start_DMA())
      – Enable the DMAx interface clock using __HAL_RCC_DMAx_CLK_ENABLE()
      – Configure and enable two DMA streams stream for managing data transfer from peripheral to memory (output stream)
      – Associate the initialized DMA handle to the CRYP DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the two DMA Streams. The output stream should have higher priority than the input stream.

Configuration of ADC, groups regular/injected, channels parameters

1. Configure the ADC parameters (resolution, data alignment, ...) and regular group parameters (conversion trigger, sequencer, ...) using function HAL_ADC_Init().
2. Configure the channels for regular group parameters (channel number, channel rank into sequencer, ..., into regular group) using function HAL_ADC_ConfigChannel().
3. Optionally, configure the injected group parameters (conversion trigger, sequencer, ..., of injected group) and the channels for injected group parameters (channel number, channel rank into sequencer, ..., into injected group) using function HAL_ADCEx_InjectedConfigChannel().
4. Optionally, configure the analog watchdog parameters (channels monitored, thresholds, ...) using function HAL_ADC_AnalogWDGConfig().
5. Optionally, for devices with several ADC instances: configure the multimode parameters using function HAL_ADCEx_MultiModeConfigChannel().

Execution of ADC conversions

1. ADC driver can be used among three modes: polling, interruption, transfer by DMA.

Polling mode IO operation

- Start the ADC peripheral using HAL_ADC_Start()
- Wait for end of conversion using HAL_ADC_PollForConversion(), at this stage user can specify the value of timeout according to his end application
- To read the ADC converted values, use the HAL_ADC_GetValue() function.
- Stop the ADC peripheral using HAL_ADC_Stop()

Interrupt mode IO operation

- Start the ADC peripheral using HAL_ADC_Start_IT()
- Use HAL_ADC_IRQHandler() called under ADC_IRQHandler() Interrupt subroutine
- At ADC end of conversion HAL_ADC_ConvCpltCallback() function is executed and user can add his own code by customization of function pointer HAL_ADC_ConvCpltCallback
- In case of ADC Error, HAL_ADC_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_ADC_ErrorCallback
- Stop the ADC peripheral using HAL_ADC_Stop_IT()

DMA mode IO operation

- Start the ADC peripheral using HAL_ADC_Start_DMA(), at this stage the user specify the length of data to be transferred at each end of conversion
- At The end of data transfer by HAL_ADC_ConvCpltCallback() function is executed and user can add his own code by customization of function pointer HAL_ADC_ConvCpltCallback
- In case of transfer Error, HAL_ADC_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_ADC_ErrorCallback
- Stop the ADC peripheral using HAL_ADC_Stop_DMA()

ADC HAL driver macros list

Below the list of most used macros in ADC HAL driver.

- __HAL_ADC_ENABLE : Enable the ADC peripheral
- __HAL_ADC_DISABLE : Disable the ADC peripheral
- __HAL_ADC_ENABLE_IT: Enable the ADC end of conversion interrupt
- __HAL_ADC_DISABLE_IT: Disable the ADC end of conversion interrupt
- __HAL_ADC_GET_IT_SOURCE: Check if the specified ADC interrupt source is enabled or disabled
- __HAL_ADC_CLEAR_FLAG: Clear the ADC's pending flags
- __HAL_ADC_GET_FLAG: Get the selected ADC's flag status
- ADC_GET_RESOLUTION: Return resolution bits in CR1 register

You can refer to the ADC HAL driver header file for more useful macros

Deinitialization of ADC

1. Disable the ADC interface
   - ADC clock can be hard reset and disabled at RCC top level.
   - Hard reset of ADC peripherals using macro __HAL_RCC_ADC_FORCE_RESET(), __HAL_RCC_ADC_RELEASE_RESET().
   - ADC clock disable using the equivalent macro/functions as configuration step.
     - Example: Into HAL_ADC_MspDeInit() (recommended code location) or with other device clock parameters configuration:
     - HAL_RCC_GetOscConfig(&RCC_OscInitStructure);
     - RCC_OscInitStructure.OscillatorType = RCC_OSCILLATORTYPE_HSI;
     - RCC_OscInitStructure.HSIState = RCC_HSI_OFF; (if not used for system clock)
     - HAL_RCC_OscConfig(&RCC_OscInitStructure);
2. ADC pins configuration
   - Disable the clock for the ADC GPIOs using macro __HAL_RCC_GPIOx_CLK_DISABLE()
3. Optionally, in case of usage of ADC with interruptions:
   - Disable the NVIC for ADC using function HAL_NVIC_DisableIRQ(ADCx_IRQn)
4. Optionally, in case of usage of DMA:
   - Deinitialize the DMA using function HAL_DMA_DeInit().
   - Disable the NVIC for DMA using function HAL_NVIC_DisableIRQ(DMAx_Channelx_IRQn)

4.2.3 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize and configure the ADC.
- De-initialize the ADC.

This section contains the following APIs:

- ***HAL_ADC_Init()***
- ***HAL_ADC_DeInit()***
- ***HAL_ADC_MspInit()***
- ***HAL_ADC_MspDeInit()***

4.2.4 IO operation functions

This section provides functions allowing to:

- Start conversion of regular channel.
- Stop conversion of regular channel.
- Start conversion of regular channel and enable interrupt.
- Stop conversion of regular channel and disable interrupt.
- Start conversion of regular channel and enable DMA transfer.
- Stop conversion of regular channel and disable DMA transfer.
- Handle ADC interrupt request.

This section contains the following APIs:

- ***HAL_ADC_Start()***
- ***HAL_ADC_Stop()***
- ***HAL_ADC_PollForConversion()***
- ***HAL_ADC_PollForEvent()***
- ***HAL_ADC_Start_IT()***
- ***HAL_ADC_Stop_IT()***
- ***HAL_ADC_IRQHandler()***
- ***HAL_ADC_Start_DMA()***
- ***HAL_ADC_Stop_DMA()***
- ***HAL_ADC_GetValue()***
- ***HAL_ADC_ConvCpltCallback()***
- ***HAL_ADC_ConvHalfCpltCallback()***
- ***HAL_ADC_LevelOutOfWindowCallback()***
- ***HAL_ADC_ErrorCallback()***

4.2.5 Peripheral Control functions

This section provides functions allowing to:

- Configure regular channels.
- Configure injected channels.
- Configure multimode.
- Configure the analog watch dog.

This section contains the following APIs:

- ***HAL_ADC_ConfigChannel()***
- ***HAL_ADC_AnalogWDGConfig()***

4.2.6 Peripheral State and errors functions

This subsection provides functions allowing to

- Check the ADC state
- Check the ADC Error

This section contains the following APIs:

- ***HAL_ADC_GetState()***
- ***HAL_ADC_GetError()***

4.2.7 Detailed description of functions

HAL_ADC_Init

Function Name	**HAL_StatusTypeDef HAL_ADC_Init (ADC_HandleTypeDef * hadc)**
Function Description	Initializes the ADCx peripheral according to the specified parameters in the ADC_InitStruct and initializes the ADC MSP.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status
Notes	• This function is used to configure the global features of the ADC (ClockPrescaler, Resolution, Data Alignment and number of conversion), however, the rest of the configuration parameters are specific to the regular channels group (scan mode activation, continuous mode activation, External trigger source and edge, DMA continuous request after the last transfer and End of conversion selection).

HAL_ADC_DeInit

Function Name	**HAL_StatusTypeDef HAL_ADC_DeInit (ADC_HandleTypeDef * hadc)**
Function Description	Deinitializes the ADCx peripheral registers to their default reset values.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status

HAL_ADC_MspInit

Function Name	**void HAL_ADC_MspInit (ADC_HandleTypeDef * hadc)**
Function Description	Initializes the ADC MSP.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **None**

HAL_ADC_MspDeInit

Function Name	**void HAL_ADC_MspDeInit (ADC_HandleTypeDef * hadc)**
Function Description	DeInitializes the ADC MSP.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **None**

HAL_ADC_Start

Function Name	**HAL_StatusTypeDef HAL_ADC_Start (ADC_HandleTypeDef * hadc)**
Function Description	Enables ADC and starts conversion of the regular channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status

HAL_ADC_Stop

Function Name	**HAL_StatusTypeDef HAL_ADC_Stop (ADC_HandleTypeDef * hadc)**
Function Description	Disables ADC and stop conversion of regular channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status.
Notes	• Caution: This function will stop also injected channels.

HAL_ADC_PollForConversion

Function Name	**HAL_StatusTypeDef HAL_ADC_PollForConversion (ADC_HandleTypeDef * hadc, uint32_t Timeout)**
Function Description	Poll for regular conversion complete.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **Timeout:** Timeout value in millisecond.
Return values	• **HAL:** status
Notes	• ADC conversion flags EOS (end of sequence) and EOC (end of conversion) are cleared by this function. • This function cannot be used in a particular setup: ADC configured in DMA mode and polling for end of each conversion (ADC init parameter "EOCSelection" set to ADC_EOC_SINGLE_CONV). In this case, DMA resets the flag EOC and polling cannot be performed on each conversion. Nevertheless, polling can still be performed on the complete sequence.

HAL_ADC_PollForEvent

Function Name	**HAL_StatusTypeDef HAL_ADC_PollForEvent (ADC_HandleTypeDef * hadc, uint32_t EventType, uint32_t Timeout)**
Function Description	Poll for conversion event.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **EventType:** the ADC event type. This parameter can be one of the following values: – ADC_AWD_EVENT: ADC Analog watch Dog event. – ADC_OVR_EVENT: ADC Overrun event. • **Timeout:** Timeout value in millisecond.
Return values	• **HAL:** status

HAL_ADC_Start_IT

Function Name	**HAL_StatusTypeDef HAL_ADC_Start_IT (ADC_HandleTypeDef * hadc)**
Function Description	Enables the interrupt and starts ADC conversion of regular channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status.

HAL_ADC_Stop_IT

Function Name	**HAL_StatusTypeDef HAL_ADC_Stop_IT (ADC_HandleTypeDef * hadc)**
Function Description	Disables the interrupt and stop ADC conversion of regular channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status.
Notes	• Caution: This function will stop also injected channels.

HAL_ADC_IRQHandler

Function Name	**void HAL_ADC_IRQHandler (ADC_HandleTypeDef * hadc)**
Function Description	Handles ADC interrupt request.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **None**

HAL_ADC_Start_DMA

Function Name	**HAL_StatusTypeDef HAL_ADC_Start_DMA (ADC_HandleTypeDef * hadc, uint32_t * pData, uint32_t Length)**
Function Description	Enables ADC DMA request after last transfer (Single-ADC mode) and enables ADC peripheral.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **pData:** The destination Buffer address. • **Length:** The length of data to be transferred from ADC peripheral to memory.
Return values	• **HAL:** status

HAL_ADC_Stop_DMA

Function Name	**HAL_StatusTypeDef HAL_ADC_Stop_DMA (ADC_HandleTypeDef * hadc)**
Function Description	Disables ADC DMA (Single-ADC mode) and disables ADC peripheral.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status

HAL_ADC_GetValue

Function Name	**uint32_t HAL_ADC_GetValue (ADC_HandleTypeDef * hadc)**
Function Description	Gets the converted value from data register of regular channel.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **Converted:** value

HAL_ADC_ConvCpltCallback

Function Name	**void HAL_ADC_ConvCpltCallback (ADC_HandleTypeDef * hadc)**
Function Description	Regular conversion complete callback in non blocking mode.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **None**

HAL_ADC_ConvHalfCpltCallback

Function Name	**void HAL_ADC_ConvHalfCpltCallback (ADC_HandleTypeDef * hadc)**
Function Description	Regular conversion half DMA transfer callback in non blocking mode.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **None**

HAL_ADC_LevelOutOfWindowCallback

Function Name **void HAL_ADC_LevelOutOfWindowCallback (ADC_HandleTypeDef * hadc)**

Function Description Analog watchdog callback in non blocking mode.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **None**

HAL_ADC_ErrorCallback

Function Name **void HAL_ADC_ErrorCallback (ADC_HandleTypeDef * hadc)**

Function Description Error ADC callback.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **None**

Notes
- In case of error due to overrun when using ADC with DMA transfer (HAL ADC handle paramater "ErrorCode" to state "HAL_ADC_ERROR_OVR"): Reinitialize the DMA using function "HAL_ADC_Stop_DMA()".If needed, restart a new ADC conversion using function "HAL_ADC_Start_DMA()" (this function is also clearing overrun flag)

HAL_ADC_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_ADC_ConfigChannel (ADC_HandleTypeDef * hadc, ADC_ChannelConfTypeDef * sConfig)**

Function Description Configures for the selected ADC regular channel its corresponding rank in the sequencer and its sample time.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
- **sConfig:** ADC configuration structure.

Return values
- **HAL:** status

HAL_ADC_AnalogWDGConfig

Function Name **HAL_StatusTypeDef HAL_ADC_AnalogWDGConfig (ADC_HandleTypeDef * hadc, ADC_AnalogWDGConfTypeDef * AnalogWDGConfig)**

Function Description Configures the analog watchdog.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

- **AnalogWDGConfig:** : pointer to an ADC_AnalogWDGConfTypeDef structure that contains the configuration information of ADC analog watchdog.

Return values
- **HAL:** status

Notes
- Analog watchdog thresholds can be modified while ADC conversion is on going. In this case, some constraints must be taken into account: The programmed threshold values are effective from the next ADC EOC (end of unitary conversion). Considering that registers write delay may happen due to bus activity, this might cause an uncertainty on the effective timing of the new programmed threshold values.

HAL_ADC_GetState

Function Name **uint32_t HAL_ADC_GetState (ADC_HandleTypeDef * hadc)**

Function Description return the ADC state

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **HAL:** state

HAL_ADC_GetError

Function Name **uint32_t HAL_ADC_GetError (ADC_HandleTypeDef * hadc)**

Function Description Return the ADC error code.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **ADC:** Error Code

4.3 ADC Firmware driver defines

4.3.1 ADC

ADC Analog Watchdog Selection

ADC_ANALOGWATCHDOG_SINGLE_REG

ADC_ANALOGWATCHDOG_SINGLE_INJEC

ADC_ANALOGWATCHDOG_SINGLE_REGINJEC

ADC_ANALOGWATCHDOG_ALL_REG

ADC_ANALOGWATCHDOG_ALL_INJEC

ADC_ANALOGWATCHDOG_ALL_REGINJEC

ADC_ANALOGWATCHDOG_NONE

ADC Common Channels

ADC_CHANNEL_0

ADC_CHANNEL_1

ADC_CHANNEL_2

ADC_CHANNEL_3
ADC_CHANNEL_4
ADC_CHANNEL_5
ADC_CHANNEL_6
ADC_CHANNEL_7
ADC_CHANNEL_8
ADC_CHANNEL_9
ADC_CHANNEL_10
ADC_CHANNEL_11
ADC_CHANNEL_12
ADC_CHANNEL_13
ADC_CHANNEL_14
ADC_CHANNEL_15
ADC_CHANNEL_16
ADC_CHANNEL_17
ADC_CHANNEL_18
ADC_CHANNEL_VREFINT
ADC_CHANNEL_VBAT

ADC Channels Type

ADC_ALL_CHANNELS
ADC_REGULAR_CHANNELS reserved for future use
ADC_INJECTED_CHANNELS reserved for future use

ADC Clock Prescaler

ADC_CLOCK_SYNC_PCLK_DIV2
ADC_CLOCK_SYNC_PCLK_DIV4
ADC_CLOCK_SYNC_PCLK_DIV6
ADC_CLOCK_SYNC_PCLK_DIV8

ADC Data Align

ADC_DATAALIGN_RIGHT
ADC_DATAALIGN_LEFT

ADC Delay Between 2 Sampling Phases

ADC_TWOSAMPLINGDELAY_5CYCLES
ADC_TWOSAMPLINGDELAY_6CYCLES
ADC_TWOSAMPLINGDELAY_7CYCLES
ADC_TWOSAMPLINGDELAY_8CYCLES
ADC_TWOSAMPLINGDELAY_9CYCLES

ADC_TWOSAMPLINGDELAY_10CYCLES

ADC_TWOSAMPLINGDELAY_11CYCLES

ADC_TWOSAMPLINGDELAY_12CYCLES

ADC_TWOSAMPLINGDELAY_13CYCLES

ADC_TWOSAMPLINGDELAY_14CYCLES

ADC_TWOSAMPLINGDELAY_15CYCLES

ADC_TWOSAMPLINGDELAY_16CYCLES

ADC_TWOSAMPLINGDELAY_17CYCLES

ADC_TWOSAMPLINGDELAY_18CYCLES

ADC_TWOSAMPLINGDELAY_19CYCLES

ADC_TWOSAMPLINGDELAY_20CYCLES

ADC EOC Selection

ADC_EOC_SEQ_CONV

ADC_EOC_SINGLE_CONV

ADC_EOC_SINGLE_SEQ_CONV reserved for future use

ADC Error Code

HAL_ADC_ERROR_NONE	No error
HAL_ADC_ERROR_INTERNAL	ADC IP internal error: if problem of clocking, enable/disable, erroneous state
HAL_ADC_ERROR_OVR	Overrun error
HAL_ADC_ERROR_DMA	DMA transfer error

ADC Event Type

ADC_AWD_EVENT

ADC_OVR_EVENT

ADC Exported Macros

__HAL_ADC_RESET_HANDLE_STATE **Description:**

- Reset ADC handle state.

Parameters:

- __HANDLE__: ADC handle

Return value:

- None

__HAL_ADC_ENABLE **Description:**

- Enable the ADC peripheral.

Parameters:

- __HANDLE__: ADC handle

Return value:

- None

__HAL_ADC_DISABLE

Description:

- Disable the ADC peripheral.

Parameters:

- __HANDLE__: ADC handle

Return value:

- None

__HAL_ADC_ENABLE_IT

Description:

- Enable the ADC end of conversion interrupt.

Parameters:

- __HANDLE__: specifies the ADC Handle.
- __INTERRUPT__: ADC Interrupt.

Return value:

- None

__HAL_ADC_DISABLE_IT

Description:

- Disable the ADC end of conversion interrupt.

Parameters:

- __HANDLE__: specifies the ADC Handle.
- __INTERRUPT__: ADC interrupt.

Return value:

- None

__HAL_ADC_GET_IT_SOURCE

Description:

- Check if the specified ADC interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: specifies the ADC Handle.
- __INTERRUPT__: specifies the ADC interrupt source to check.

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_ADC_CLEAR_FLAG

Description:

- Clear the ADC's pending flags.

Parameters:

- __HANDLE__: specifies the ADC Handle.
- __FLAG__: ADC flag.

	Return value: • None
__HAL_ADC_GET_FLAG	**Description:** • Get the selected ADC's flag status. **Parameters:** • __HANDLE__: specifies the ADC Handle. • __FLAG__: ADC flag. **Return value:** • None

ADC Exported Types

HAL_ADC_STATE_RESET	ADC not yet initialized or disabled
HAL_ADC_STATE_READY	ADC peripheral ready for use
HAL_ADC_STATE_BUSY_INTERNAL	ADC is busy to internal process (initialization, calibration)
HAL_ADC_STATE_TIMEOUT	TimeOut occurrence
HAL_ADC_STATE_ERROR_INTERNAL	Internal error occurrence
HAL_ADC_STATE_ERROR_CONFIG	Configuration error occurrence
HAL_ADC_STATE_ERROR_DMA	DMA error occurrence
HAL_ADC_STATE_REG_BUSY	A conversion on group regular is ongoing or can occur (either by continuous mode, external trigger, low power auto power-on (if feature available), multimode ADC master control (if feature available))
HAL_ADC_STATE_REG_EOC	Conversion data available on group regular
HAL_ADC_STATE_REG_OVR	Overrun occurrence
HAL_ADC_STATE_INJ_BUSY	A conversion on group injected is ongoing or can occur (either by auto-injection mode, external trigger, low power auto power-on (if feature available), multimode ADC master control (if feature available))
HAL_ADC_STATE_INJ_EOC	Conversion data available on group injected
HAL_ADC_STATE_AWD1	Out-of-window occurrence of analog watchdog 1
HAL_ADC_STATE_AWD2	Not available on STM32F4 device: Out-of-window occurrence of analog watchdog 2
HAL_ADC_STATE_AWD3	Not available on STM32F4 device: Out-of-window occurrence of analog watchdog 3
HAL_ADC_STATE_MULTIMODE_SLAVE	Not available on STM32F4 device: ADC in multimode slave state, controlled by another ADC master (

ADC External Trigger Edge Regular

ADC_EXTERNALTRIGCONVEDGE_NONE

ADC_EXTERNALTRIGCONVEDGE_RISING

ADC_EXTERNALTRIGCONVEDGE_FALLING

ADC_EXTERNALTRIGCONVEDGE_RISINGFALLING

ADC External Trigger Source Regular

ADC_EXTERNALTRIGCONV_T1_CC1

ADC_EXTERNALTRIGCONV_T1_CC2

ADC_EXTERNALTRIGCONV_T1_CC3

ADC_EXTERNALTRIGCONV_T2_CC2

ADC_EXTERNALTRIGCONV_T2_CC3

ADC_EXTERNALTRIGCONV_T2_CC4

ADC_EXTERNALTRIGCONV_T2_TRGO

ADC_EXTERNALTRIGCONV_T3_CC1

ADC_EXTERNALTRIGCONV_T3_TRGO

ADC_EXTERNALTRIGCONV_T4_CC4

ADC_EXTERNALTRIGCONV_T5_CC1

ADC_EXTERNALTRIGCONV_T5_CC2

ADC_EXTERNALTRIGCONV_T5_CC3

ADC_EXTERNALTRIGCONV_T8_CC1

ADC_EXTERNALTRIGCONV_T8_TRGO

ADC_EXTERNALTRIGCONV_Ext_IT11

ADC_SOFTWARE_START

ADC Flags Definition

ADC_FLAG_AWD

ADC_FLAG_EOC

ADC_FLAG_JEOC

ADC_FLAG_JSTRT

ADC_FLAG_STRT

ADC_FLAG_OVR

ADC Interrupts Definition

ADC_IT_EOC

ADC_IT_AWD

ADC_IT_JEOC

ADC_IT_OVR

ADC Resolution

ADC_RESOLUTION_12B

ADC_RESOLUTION_10B

ADC_RESOLUTION_8B

ADC_RESOLUTION_6B

ADC Sampling Times

ADC_SAMPLETIME_3CYCLES

ADC_SAMPLETIME_15CYCLES

ADC_SAMPLETIME_28CYCLES

ADC_SAMPLETIME_56CYCLES

ADC_SAMPLETIME_84CYCLES

ADC_SAMPLETIME_112CYCLES

ADC_SAMPLETIME_144CYCLES

ADC_SAMPLETIME_480CYCLES

5 HAL ADC Extension Driver

5.1 ADCEx Firmware driver registers structures

5.1.1 ADC_InjectionConfTypeDef

Data Fields

- ***uint32_t InjectedChannel***
- ***uint32_t InjectedRank***
- ***uint32_t InjectedSamplingTime***
- ***uint32_t InjectedOffset***
- ***uint32_t InjectedNbrOfConversion***
- ***uint32_t InjectedDiscontinuousConvMode***
- ***uint32_t AutoInjectedConv***
- ***uint32_t ExternalTrigInjecConv***
- ***uint32_t ExternalTrigInjecConvEdge***

Field Documentation

- ***uint32_t ADC_InjectionConfTypeDef::InjectedChannel***
  Selection of ADC channel to configure This parameter can be a value of ***ADC_channels*** Note: Depending on devices, some channels may not be available on package pins. Refer to device datasheet for channels availability.
- ***uint32_t ADC_InjectionConfTypeDef::InjectedRank***
  Rank in the injected group sequencer This parameter must be a value of ***ADCEx_injected_rank*** Note: In case of need to disable a channel or change order of conversion sequencer, rank containing a previous channel setting can be overwritten by the new channel setting (or parameter number of conversions can be adjusted)
- ***uint32_t ADC_InjectionConfTypeDef::InjectedSamplingTime***
  Sampling time value to be set for the selected channel. Unit: ADC clock cycles Conversion time is the addition of sampling time and processing time (12 ADC clock cycles at ADC resolution 12 bits, 11 cycles at 10 bits, 9 cycles at 8 bits, 7 cycles at 6 bits). This parameter can be a value of ***ADC_sampling_times*** Caution: This parameter updates the parameter property of the channel, that can be used into regular and/or injected groups. If this same channel has been previously configured in the other group (regular/injected), it will be updated to last setting. Note: In case of usage of internal measurement channels (VrefInt/Vbat/TempSensor), sampling time constraints must be respected (sampling time can be adjusted in function of ADC clock frequency and sampling time setting) Refer to device datasheet for timings values, parameters TS_vrefint, TS_temp (values rough order: 4us min).
- ***uint32_t ADC_InjectionConfTypeDef::InjectedOffset***
  Defines the offset to be subtracted from the raw converted data (for channels set on injected group only). Offset value must be a positive number. Depending of ADC resolution selected (12, 10, 8 or 6 bits), this parameter must be a number between Min_Data = 0x000 and Max_Data = 0xFFF, 0x3FF, 0xFF or 0x3F respectively.
- ***uint32_t ADC_InjectionConfTypeDef::InjectedNbrOfConversion***
  Specifies the number of ranks that will be converted within the injected group sequencer. To use the injected group sequencer and convert several ranks, parameter 'ScanConvMode' must be enabled. This parameter must be a number

between Min_Data = 1 and Max_Data = 4. Caution: this setting impacts the entire injected group. Therefore, call of **HAL_ADCEx_InjectedConfigChannel()** to configure a channel on injected group can impact the configuration of other channels previously set.

- ***uint32_t ADC_InjectionConfTypeDef::InjectedDiscontinuousConvMode***
  Specifies whether the conversions sequence of injected group is performed in Complete-sequence/Discontinuous-sequence (main sequence subdivided in successive parts). Discontinuous mode is used only if sequencer is enabled (parameter 'ScanConvMode'). If sequencer is disabled, this parameter is discarded. Discontinuous mode can be enabled only if continuous mode is disabled. If continuous mode is enabled, this parameter setting is discarded. This parameter can be set to ENABLE or DISABLE. Note: For injected group, number of discontinuous ranks increment is fixed to one-by-one. Caution: this setting impacts the entire injected group. Therefore, call of **HAL_ADCEx_InjectedConfigChannel()** to configure a channel on injected group can impact the configuration of other channels previously set.
- ***uint32_t ADC_InjectionConfTypeDef::AutoInjectedConv***
  Enables or disables the selected ADC automatic injected group conversion after regular one This parameter can be set to ENABLE or DISABLE. Note: To use Automatic injected conversion, discontinuous mode must be disabled ('DiscontinuousConvMode' and 'InjectedDiscontinuousConvMode' set to DISABLE) Note: To use Automatic injected conversion, injected group external triggers must be disabled ('ExternalTrigInjecConv' set to ADC_SOFTWARE_START) Note: In case of DMA used with regular group: if DMA configured in normal mode (single shot) JAUTO will be stopped upon DMA transfer complete. To maintain JAUTO always enabled, DMA must be configured in circular mode. Caution: this setting impacts the entire injected group. Therefore, call of **HAL_ADCEx_InjectedConfigChannel()** to configure a channel on injected group can impact the configuration of other channels previously set.
- ***uint32_t ADC_InjectionConfTypeDef::ExternalTrigInjecConv***
  Selects the external event used to trigger the conversion start of injected group. If set to ADC_INJECTED_SOFTWARE_START, external triggers are disabled. If set to external trigger source, triggering is on event rising edge. This parameter can be a value of ***ADCEx_External_trigger_Source_Injected*** Note: This parameter must be modified when ADC is disabled (before ADC start conversion or after ADC stop conversion). If ADC is enabled, this parameter setting is bypassed without error reporting (as it can be the expected behaviour in case of another parameter update on the fly) Caution: this setting impacts the entire injected group. Therefore, call of **HAL_ADCEx_InjectedConfigChannel()** to configure a channel on injected group can impact the configuration of other channels previously set.
- ***uint32_t ADC_InjectionConfTypeDef::ExternalTrigInjecConvEdge***
  Selects the external trigger edge of injected group. This parameter can be a value of ***ADCEx_External_trigger_edge_Injected***. If trigger is set to ADC_INJECTED_SOFTWARE_START, this parameter is discarded. Caution: this setting impacts the entire injected group. Therefore, call of **HAL_ADCEx_InjectedConfigChannel()** to configure a channel on injected group can impact the configuration of other channels previously set.

5.1.2 ADC_MultiModeTypeDef

Data Fields

- ***uint32_t Mode***
- ***uint32_t DMAAccessMode***
- ***uint32_t TwoSamplingDelay***

Field Documentation

- ***uint32_t ADC_MultiModeTypeDef::Mode***
  Configures the ADC to operate in independent or multi mode. This parameter can be a value of ***ADCEx_Common_mode***
- ***uint32_t ADC_MultiModeTypeDef::DMAAccessMode***
  Configures the Direct memory access mode for multi ADC mode. This parameter can be a value of ***ADCEx_Direct_memory_access_mode_for_multi_mode***
- ***uint32_t ADC_MultiModeTypeDef::TwoSamplingDelay***
  Configures the Delay between 2 sampling phases. This parameter can be a value of ***ADC_delay_between_2_sampling_phases***

5.2 ADCEx Firmware driver API description

5.2.1 How to use this driver

1. Initialize the ADC low level resources by implementing the HAL_ADC_MspInit():
   a. Enable the ADC interface clock using __HAL_RCC_ADC_CLK_ENABLE()
   b. ADC pins configuration
      – Enable the clock for the ADC GPIOs using the following function: __HAL_RCC_GPIOx_CLK_ENABLE()
      – Configure these ADC pins in analog mode using HAL_GPIO_Init()
   c. In case of using interrupts (e.g. HAL_ADC_Start_IT())
      – Configure the ADC interrupt priority using HAL_NVIC_SetPriority()
      – Enable the ADC IRQ handler using HAL_NVIC_EnableIRQ()
      – In ADC IRQ handler, call HAL_ADC_IRQHandler()
   d. In case of using DMA to control data transfer (e.g. HAL_ADC_Start_DMA())
      – Enable the DMAx interface clock using __HAL_RCC_DMAx_CLK_ENABLE()
      – Configure and enable two DMA streams stream for managing data transfer from peripheral to memory (output stream)
      – Associate the initialized DMA handle to the ADC DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the two DMA Streams. The output stream should have higher priority than the input stream.
2. Configure the ADC Prescaler, conversion resolution and data alignment using the HAL_ADC_Init() function.
3. Configure the ADC Injected channels group features, use HAL_ADC_Init() and HAL_ADC_ConfigChannel() functions.

4. Three operation modes are available within this driver :

Polling mode IO operation

- Start the ADC peripheral using HAL_ADCEx_InjectedStart()
- Wait for end of conversion using HAL_ADC_PollForConversion(), at this stage user can specify the value of timeout according to his end application
- To read the ADC converted values, use the HAL_ADCEx_InjectedGetValue() function.
- Stop the ADC peripheral using HAL_ADCEx_InjectedStop()

Interrupt mode IO operation

- Start the ADC peripheral using HAL_ADCEx_InjectedStart_IT()
- Use HAL_ADC_IRQHandler() called under ADC_IRQHandler() Interrupt subroutine
- At ADC end of conversion HAL_ADCEx_InjectedConvCpltCallback() function is executed and user can add his own code by customization of function pointer HAL_ADCEx_InjectedConvCpltCallback
- In case of ADC Error, HAL_ADCEx_InjectedErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_ADCEx_InjectedErrorCallback
- Stop the ADC peripheral using HAL_ADCEx_InjectedStop_IT()

DMA mode IO operation

- Start the ADC peripheral using HAL_ADCEx_InjectedStart_DMA(), at this stage the user specify the length of data to be transferred at each end of conversion
- At The end of data transfer ba HAL_ADCEx_InjectedConvCpltCallback() function is executed and user can add his own code by customization of function pointer HAL_ADCEx_InjectedConvCpltCallback
- In case of transfer Error, HAL_ADCEx_InjectedErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_ADCEx_InjectedErrorCallback
- Stop the ADC peripheral using HAL_ADCEx_InjectedStop_DMA()

Multi mode ADCs Regular channels configuration

- Select the Multi mode ADC regular channels features (dual or triple mode) and configure the DMA mode using HAL_ADCEx_MultiModeConfigChannel() functions.
- Start the ADC peripheral using HAL_ADCEx_MultiModeStart_DMA(), at this stage the user specify the length of data to be transferred at each end of conversion
- Read the ADCs converted values using the HAL_ADCEx_MultiModeGetValue() function.

5.2.2 Extended features functions

This section provides functions allowing to:

- Start conversion of injected channel.
- Stop conversion of injected channel.
- Start multimode and enable DMA transfer.
- Stop multimode and disable DMA transfer.
- Get result of injected channel conversion.
- Get result of multimode conversion.
- Configure injected channels.
- Configure multimode.

This section contains the following APIs:

- ***HAL_ADCEx_InjectedStart()***
- ***HAL_ADCEx_InjectedStart_IT()***
- ***HAL_ADCEx_InjectedStop()***
- ***HAL_ADCEx_InjectedPollForConversion()***
- ***HAL_ADCEx_InjectedStop_IT()***
- ***HAL_ADCEx_InjectedGetValue()***
- ***HAL_ADCEx_MultiModeStart_DMA()***
- ***HAL_ADCEx_MultiModeStop_DMA()***
- ***HAL_ADCEx_MultiModeGetValue()***
- ***HAL_ADCEx_InjectedConvCpltCallback()***
- ***HAL_ADCEx_InjectedConfigChannel()***
- ***HAL_ADCEx_MultiModeConfigChannel()***

5.2.3 Detailed description of functions

HAL_ADCEx_InjectedStart

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedStart (ADC_HandleTypeDef * hadc)**
Function Description	Enables the selected ADC software start conversion of the injected channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status

HAL_ADCEx_InjectedStop

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedStop (ADC_HandleTypeDef * hadc)**
Function Description	Stop conversion of injected channels.
Parameters	• **hadc:** ADC handle
Return values	• **None**
Notes	• If ADC must be disabled and if conversion is on going on regular group, function HAL_ADC_Stop must be used to stop both injected and regular groups, and disable the ADC.

- If injected group mode auto-injection is enabled, function HAL_ADC_Stop must be used.
- In case of auto-injection mode, HAL_ADC_Stop must be used.

HAL_ADCEx_InjectedPollForConversion

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedPollForConversion (ADC_HandleTypeDef * hadc, uint32_t Timeout)**
Function Description	Poll for injected conversion complete.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **Timeout:** Timeout value in millisecond.
Return values	• **HAL:** status

HAL_ADCEx_InjectedStart_IT

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedStart_IT (ADC_HandleTypeDef * hadc)**
Function Description	Enables the interrupt and starts ADC conversion of injected channels.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **HAL:** status.

HAL_ADCEx_InjectedStop_IT

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedStop_IT (ADC_HandleTypeDef * hadc)**
Function Description	Stop conversion of injected channels, disable interruption of end-of-conversion.
Parameters	• **hadc:** ADC handle
Return values	• **None**
Notes	• If ADC must be disabled and if conversion is on going on regular group, function HAL_ADC_Stop must be used to stop both injected and regular groups, and disable the ADC. • If injected group mode auto-injection is enabled, function HAL_ADC_Stop must be used.

HAL_ADCEx_InjectedGetValue

Function Name	**uint32_t HAL_ADCEx_InjectedGetValue (ADC_HandleTypeDef * hadc, uint32_t InjectedRank)**
Function Description	Gets the converted value from data register of injected channel.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **InjectedRank:** the ADC injected rank. This parameter can be one of the following values:

- ADC_INJECTED_RANK_1: Injected Channel1 selected
- ADC_INJECTED_RANK_2: Injected Channel2 selected
- ADC_INJECTED_RANK_3: Injected Channel3 selected
- ADC_INJECTED_RANK_4: Injected Channel4 selected

Return values
- **None**

HAL_ADCEx_MultiModeStart_DMA

Function Name **HAL_StatusTypeDef HAL_ADCEx_MultiModeStart_DMA (ADC_HandleTypeDef * hadc, uint32_t * pData, uint32_t Length)**

Function Description Enables ADC DMA request after last transfer (Multi-ADC mode) and enables ADC peripheral.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
- **pData:** Pointer to buffer in which transferred from ADC peripheral to memory will be stored.
- **Length:** The length of data to be transferred from ADC peripheral to memory.

Return values
- **HAL:** status

Notes
- Caution: This function must be used only with the ADC master.

HAL_ADCEx_MultiModeStop_DMA

Function Name **HAL_StatusTypeDef HAL_ADCEx_MultiModeStop_DMA (ADC_HandleTypeDef * hadc)**

Function Description Disables ADC DMA (multi-ADC mode) and disables ADC peripheral.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **HAL:** status

HAL_ADCEx_MultiModeGetValue

Function Name **uint32_t HAL_ADCEx_MultiModeGetValue (ADC_HandleTypeDef * hadc)**

Function Description Returns the last ADC1, ADC2 and ADC3 regular conversions results data in the selected multi mode.

Parameters
- **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.

Return values
- **The:** converted data value.

HAL_ADCEx_InjectedConvCpltCallback

Function Name	**void HAL_ADCEx_InjectedConvCpltCallback (ADC_HandleTypeDef * hadc)**
Function Description	Injected conversion complete callback in non blocking mode.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC.
Return values	• **None**

HAL_ADCEx_InjectedConfigChannel

Function Name	**HAL_StatusTypeDef HAL_ADCEx_InjectedConfigChannel (ADC_HandleTypeDef * hadc, ADC_InjectionConfTypeDef * sConfigInjected)**
Function Description	Configures for the selected ADC injected channel its corresponding rank in the sequencer and its sample time.
Parameters	• **hadc:** pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **sConfigInjected:** ADC configuration structure for injected channel.
Return values	• **None**

HAL_ADCEx_MultiModeConfigChannel

Function Name	**HAL_StatusTypeDef HAL_ADCEx_MultiModeConfigChannel (ADC_HandleTypeDef * hadc, ADC_MultiModeTypeDef * multimode)**
Function Description	Configures the ADC multi-mode.
Parameters	• **hadc:** : pointer to a ADC_HandleTypeDef structure that contains the configuration information for the specified ADC. • **multimode:** : pointer to an ADC_MultiModeTypeDef structure that contains the configuration information for multimode.
Return values	• **HAL:** status

5.3 ADCEx Firmware driver defines

5.3.1 ADCEx

ADC Specific Channels

ADC_CHANNEL_DIFFERENCIATION_TEMPSENSOR_VBAT

ADC_CHANNEL_TEMPSENSOR

ADC Common Mode

ADC_MODE_INDEPENDENT

ADC_DUALMODE_REGSIMULT_INJECSIMULT

ADC_DUALMODE_REGSIMULT_ALTERTRIG

ADC_DUALMODE_INJECSIMULT

ADC_DUALMODE_REGSIMULT

ADC_DUALMODE_INTERL

ADC_DUALMODE_ALTERTRIG

ADC_TRIPLEMODE_REGSIMULT_INJECSIMULT

ADC_TRIPLEMODE_REGSIMULT_AlterTrig

ADC_TRIPLEMODE_INJECSIMULT

ADC_TRIPLEMODE_REGSIMULT

ADC_TRIPLEMODE_INTERL

ADC_TRIPLEMODE_ALTERTRIG

ADC Direct Memory Access Mode For Multi Mode

ADC_DMAACCESSMODE_DISABLED	DMA mode disabled
ADC_DMAACCESSMODE_1	DMA mode 1 enabled (2 / 3 half-words one by one - 1 then 2 then 3)
ADC_DMAACCESSMODE_2	DMA mode 2 enabled (2 / 3 half-words by pairs - 2&1 then 1&3 then 3&2)
ADC_DMAACCESSMODE_3	DMA mode 3 enabled (2 / 3 bytes by pairs - 2&1 then 1&3 then 3&2)

ADC External Trigger Edge Injected

ADC_EXTERNALTRIGINJECCONVEDGE_NONE

ADC_EXTERNALTRIGINJECCONVEDGE_RISING

ADC_EXTERNALTRIGINJECCONVEDGE_FALLING

ADC_EXTERNALTRIGINJECCONVEDGE_RISINGFALLING

ADC External Trigger Source Injected

ADC_EXTERNALTRIGINJECCONV_T1_CC4

ADC_EXTERNALTRIGINJECCONV_T1_TRGO

ADC_EXTERNALTRIGINJECCONV_T2_CC1

ADC_EXTERNALTRIGINJECCONV_T2_TRGO

ADC_EXTERNALTRIGINJECCONV_T3_CC2

ADC_EXTERNALTRIGINJECCONV_T3_CC4

ADC_EXTERNALTRIGINJECCONV_T4_CC1

ADC_EXTERNALTRIGINJECCONV_T4_CC2

ADC_EXTERNALTRIGINJECCONV_T4_CC3

ADC_EXTERNALTRIGINJECCONV_T4_TRGO

ADC_EXTERNALTRIGINJECCONV_T5_CC4

ADC_EXTERNALTRIGINJECCONV_T5_TRGO

ADC_EXTERNALTRIGINJECCONV_T8_CC2

ADC_EXTERNALTRIGINJECCONV_T8_CC3

ADC_EXTERNALTRIGINJECCONV_T8_CC4

ADC_EXTERNALTRIGINJECCONV_EXT_IT15

ADC_INJECTED_SOFTWARE_START

ADC Injected Rank

ADC_INJECTED_RANK_1

ADC_INJECTED_RANK_2

ADC_INJECTED_RANK_3

ADC_INJECTED_RANK_4

6 HAL CAN Generic Driver

6.1 CAN Firmware driver registers structures

6.1.1 CAN_InitTypeDef

Data Fields

- ***uint32_t Prescaler***
- ***uint32_t Mode***
- ***uint32_t SJW***
- ***uint32_t BS1***
- ***uint32_t BS2***
- ***uint32_t TTCM***
- ***uint32_t ABOM***
- ***uint32_t AWUM***
- ***uint32_t NART***
- ***uint32_t RFLM***
- ***uint32_t TXFP***

Field Documentation

- ***uint32_t CAN_InitTypeDef::Prescaler***
  Specifies the length of a time quantum. This parameter must be a number between Min_Data = 1 and Max_Data = 1024
- ***uint32_t CAN_InitTypeDef::Mode***
  Specifies the CAN operating mode. This parameter can be a value of ***CAN_operating_mode***
- ***uint32_t CAN_InitTypeDef::SJW***
  Specifies the maximum number of time quanta the CAN hardware is allowed to lengthen or shorten a bit to perform resynchronization. This parameter can be a value of ***CAN_synchronisation_jump_width***
- ***uint32_t CAN_InitTypeDef::BS1***
  Specifies the number of time quanta in Bit Segment 1. This parameter can be a value of ***CAN_time_quantum_in_bit_segment_1***
- ***uint32_t CAN_InitTypeDef::BS2***
  Specifies the number of time quanta in Bit Segment 2. This parameter can be a value of ***CAN_time_quantum_in_bit_segment_2***
- ***uint32_t CAN_InitTypeDef::TTCM***
  Enable or disable the time triggered communication mode. This parameter can be set to ENABLE or DISABLE.
- ***uint32_t CAN_InitTypeDef::ABOM***
  Enable or disable the automatic bus-off management. This parameter can be set to ENABLE or DISABLE
- ***uint32_t CAN_InitTypeDef::AWUM***
  Enable or disable the automatic wake-up mode. This parameter can be set to ENABLE or DISABLE
- ***uint32_t CAN_InitTypeDef::NART***
  Enable or disable the non-automatic retransmission mode. This parameter can be set to ENABLE or DISABLE

- ***uint32_t CAN_InitTypeDef::RFLM***
  Enable or disable the receive FIFO Locked mode. This parameter can be set to ENABLE or DISABLE
- ***uint32_t CAN_InitTypeDef::TXFP***
  Enable or disable the transmit FIFO priority. This parameter can be set to ENABLE or DISABLE

6.1.2 CAN_FilterConfTypeDef

Data Fields

- ***uint32_t FilterIdHigh***
- ***uint32_t FilterIdLow***
- ***uint32_t FilterMaskIdHigh***
- ***uint32_t FilterMaskIdLow***
- ***uint32_t FilterFIFOAssignment***
- ***uint32_t FilterNumber***
- ***uint32_t FilterMode***
- ***uint32_t FilterScale***
- ***uint32_t FilterActivation***
- ***uint32_t BankNumber***

Field Documentation

- ***uint32_t CAN_FilterConfTypeDef::FilterIdHigh***
  Specifies the filter identification number (MSBs for a 32-bit configuration, first one for a 16-bit configuration). This parameter must be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t CAN_FilterConfTypeDef::FilterIdLow***
  Specifies the filter identification number (LSBs for a 32-bit configuration, second one for a 16-bit configuration). This parameter must be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t CAN_FilterConfTypeDef::FilterMaskIdHigh***
  Specifies the filter mask number or identification number, according to the mode (MSBs for a 32-bit configuration, first one for a 16-bit configuration). This parameter must be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t CAN_FilterConfTypeDef::FilterMaskIdLow***
  Specifies the filter mask number or identification number, according to the mode (LSBs for a 32-bit configuration, second one for a 16-bit configuration). This parameter must be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t CAN_FilterConfTypeDef::FilterFIFOAssignment***
  Specifies the FIFO (0 or 1) which will be assigned to the filter. This parameter can be a value of ***CAN_filter_FIFO***
- ***uint32_t CAN_FilterConfTypeDef::FilterNumber***
  Specifies the filter which will be initialized. This parameter must be a number between Min_Data = 0 and Max_Data = 27
- ***uint32_t CAN_FilterConfTypeDef::FilterMode***
  Specifies the filter mode to be initialized. This parameter can be a value of ***CAN_filter_mode***
- ***uint32_t CAN_FilterConfTypeDef::FilterScale***
  Specifies the filter scale. This parameter can be a value of ***CAN_filter_scale***

- ***uint32_t CAN_FilterConfTypeDef::FilterActivation***
  Enable or disable the filter. This parameter can be set to ENABLE or DISABLE.
- ***uint32_t CAN_FilterConfTypeDef::BankNumber***
  Select the start slave bank filter. This parameter must be a number between Min_Data = 0 and Max_Data = 28

6.1.3 CanTxMsgTypeDef

Data Fields

- ***uint32_t StdId***
- ***uint32_t ExtId***
- ***uint32_t IDE***
- ***uint32_t RTR***
- ***uint32_t DLC***
- ***uint8_t Data***

Field Documentation

- ***uint32_t CanTxMsgTypeDef::StdId***
  Specifies the standard identifier. This parameter must be a number between Min_Data = 0 and Max_Data = 0x7FF
- ***uint32_t CanTxMsgTypeDef::ExtId***
  Specifies the extended identifier. This parameter must be a number between Min_Data = 0 and Max_Data = 0x1FFFFFFFU
- ***uint32_t CanTxMsgTypeDef::IDE***
  Specifies the type of identifier for the message that will be transmitted. This parameter can be a value of ***CAN_Identifier_Type***
- ***uint32_t CanTxMsgTypeDef::RTR***
  Specifies the type of frame for the message that will be transmitted. This parameter can be a value of ***CAN_remote_transmission_request***
- ***uint32_t CanTxMsgTypeDef::DLC***
  Specifies the length of the frame that will be transmitted. This parameter must be a number between Min_Data = 0 and Max_Data = 8
- ***uint8_t CanTxMsgTypeDef::Data[8]***
  Contains the data to be transmitted. This parameter must be a number between Min_Data = 0 and Max_Data = 0xFF

6.1.4 CanRxMsgTypeDef

Data Fields

- ***uint32_t StdId***
- ***uint32_t ExtId***
- ***uint32_t IDE***
- ***uint32_t RTR***
- ***uint32_t DLC***
- ***uint8_t Data***
- ***uint32_t FMI***

- ***uint32_t FIFONumber***

Field Documentation

- ***uint32_t CanRxMsgTypeDef::StdId***
  Specifies the standard identifier. This parameter must be a number between Min_Data = 0 and Max_Data = 0x7FF
- ***uint32_t CanRxMsgTypeDef::ExtId***
  Specifies the extended identifier. This parameter must be a number between Min_Data = 0 and Max_Data = 0x1FFFFFFFU
- ***uint32_t CanRxMsgTypeDef::IDE***
  Specifies the type of identifier for the message that will be received. This parameter can be a value of ***CAN_Identifier_Type***
- ***uint32_t CanRxMsgTypeDef::RTR***
  Specifies the type of frame for the received message. This parameter can be a value of ***CAN_remote_transmission_request***
- ***uint32_t CanRxMsgTypeDef::DLC***
  Specifies the length of the frame that will be received. This parameter must be a number between Min_Data = 0 and Max_Data = 8
- ***uint8_t CanRxMsgTypeDef::Data[8]***
  Contains the data to be received. This parameter must be a number between Min_Data = 0 and Max_Data = 0xFF
- ***uint32_t CanRxMsgTypeDef::FMI***
  Specifies the index of the filter the message stored in the mailbox passes through. This parameter must be a number between Min_Data = 0 and Max_Data = 0xFF
- ***uint32_t CanRxMsgTypeDef::FIFONumber***
  Specifies the receive FIFO number. This parameter can be CAN_FIFO0 or CAN_FIFO1

6.1.5 CAN_HandleTypeDef

Data Fields

- ***CAN_TypeDef * Instance***
- ***CAN_InitTypeDef Init***
- ***CanTxMsgTypeDef * pTxMsg***
- ***CanRxMsgTypeDef * pRxMsg***
- ***__IO HAL_CAN_StateTypeDef State***
- ***HAL_LockTypeDef Lock***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***CAN_TypeDef* CAN_HandleTypeDef::Instance***
  Register base address
- ***CAN_InitTypeDef CAN_HandleTypeDef::Init***
  CAN required parameters
- ***CanTxMsgTypeDef* CAN_HandleTypeDef::pTxMsg***
  Pointer to transmit structure
- ***CanRxMsgTypeDef* CAN_HandleTypeDef::pRxMsg***
  Pointer to reception structure

- ***__IO HAL_CAN_StateTypeDef CAN_HandleTypeDef::State***
  CAN communication state
- ***HAL_LockTypeDef CAN_HandleTypeDef::Lock***
  CAN locking object
- ***__IO uint32_t CAN_HandleTypeDef::ErrorCode***
  CAN Error code

6.2 CAN Firmware driver API description

6.2.1 How to use this driver

1. Enable the CAN controller interface clock using __HAL_RCC_CAN1_CLK_ENABLE() for CAN1 and __HAL_RCC_CAN2_CLK_ENABLE() for CAN2 In case you are using CAN2 only, you have to enable the CAN1 clock.
2. CAN pins configuration
   – Enable the clock for the CAN GPIOs using the following function: __GPIOx_CLK_ENABLE()
   – Connect and configure the involved CAN pins to AF9 using the following function HAL_GPIO_Init()
3. Initialize and configure the CAN using CAN_Init() function.
4. Transmit the desired CAN frame using HAL_CAN_Transmit() function.
5. Receive a CAN frame using HAL_CAN_Receive() function.

Polling mode IO operation

- Start the CAN peripheral transmission and wait the end of this operation using HAL_CAN_Transmit(), at this stage user can specify the value of timeout according to his end application
- Start the CAN peripheral reception and wait the end of this operation using HAL_CAN_Receive(), at this stage user can specify the value of timeout according to his end application

Interrupt mode IO operation

- Start the CAN peripheral transmission using HAL_CAN_Transmit_IT()
- Start the CAN peripheral reception using HAL_CAN_Receive_IT()
- Use HAL_CAN_IRQHandler() called under the used CAN Interrupt subroutine
- At CAN end of transmission HAL_CAN_TxCpltCallback() function is executed and user can add his own code by customization of function pointer HAL_CAN_TxCpltCallback
- In case of CAN Error, HAL_CAN_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_CAN_ErrorCallback

CAN HAL driver macros list

Below the list of most used macros in CAN HAL driver.

- __HAL_CAN_ENABLE_IT: Enable the specified CAN interrupts
- __HAL_CAN_DISABLE_IT: Disable the specified CAN interrupts

- __HAL_CAN_GET_IT_SOURCE: Check if the specified CAN interrupt source is enabled or disabled
- __HAL_CAN_CLEAR_FLAG: Clear the CAN's pending flags
- __HAL_CAN_GET_FLAG: Get the selected CAN's flag status

You can refer to the CAN HAL driver header file for more useful macros

6.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize and configure the CAN.
- De-initialize the CAN.

This section contains the following APIs:

- ***HAL_CAN_Init()***
- ***HAL_CAN_ConfigFilter()***
- ***HAL_CAN_DeInit()***
- ***HAL_CAN_MspInit()***
- ***HAL_CAN_MspDeInit()***

6.2.3 IO operation functions

This section provides functions allowing to:

- Transmit a CAN frame message.
- Receive a CAN frame message.
- Enter CAN peripheral in sleep mode.
- Wake up the CAN peripheral from sleep mode.

This section contains the following APIs:

- ***HAL_CAN_Transmit()***
- ***HAL_CAN_Transmit_IT()***
- ***HAL_CAN_Receive()***
- ***HAL_CAN_Receive_IT()***
- ***HAL_CAN_Sleep()***
- ***HAL_CAN_WakeUp()***
- ***HAL_CAN_IRQHandler()***
- ***HAL_CAN_TxCpltCallback()***
- ***HAL_CAN_RxCpltCallback()***
- ***HAL_CAN_ErrorCallback()***

6.2.4 Peripheral State and Error functions

This subsection provides functions allowing to :

- Check the CAN state.
- Check CAN Errors detected during interrupt process

This section contains the following APIs:

- ***HAL_CAN_GetState()***
- ***HAL_CAN_GetError()***

6.2.5 Detailed description of functions

HAL_CAN_Init

Function Name	**HAL_StatusTypeDef HAL_CAN_Init (CAN_HandleTypeDef * hcan)**
Function Description	Initializes the CAN peripheral according to the specified parameters in the CAN_InitStruct.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **HAL:** status

HAL_CAN_ConfigFilter

Function Name	**HAL_StatusTypeDef HAL_CAN_ConfigFilter (CAN_HandleTypeDef * hcan, CAN_FilterConfTypeDef * sFilterConfig)**
Function Description	Configures the CAN reception filter according to the specified parameters in the CAN_FilterInitStruct.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN. • **sFilterConfig:** pointer to a CAN_FilterConfTypeDef structure that contains the filter configuration information.
Return values	• **None**

HAL_CAN_DeInit

Function Name	**HAL_StatusTypeDef HAL_CAN_DeInit (CAN_HandleTypeDef * hcan)**
Function Description	Deinitializes the CANx peripheral registers to their default reset values.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **HAL:** status

HAL_CAN_MspInit

Function Name	**void HAL_CAN_MspInit (CAN_HandleTypeDef * hcan)**
Function Description	Initializes the CAN MSP.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **None**

HAL_CAN_MspDeInit

Function Name	**void HAL_CAN_MspDeInit (CAN_HandleTypeDef * hcan)**
Function Description	DeInitializes the CAN MSP.

Parameters
- **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.

Return values
- **None**

HAL_CAN_Transmit

Function Name **HAL_StatusTypeDef HAL_CAN_Transmit (CAN_HandleTypeDef * hcan, uint32_t Timeout)**

Function Description Initiates and transmits a CAN frame message.

Parameters
- **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CAN_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_CAN_Transmit_IT (CAN_HandleTypeDef * hcan)**

Function Description Initiates and transmits a CAN frame message.

Parameters
- **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.

Return values
- **HAL:** status

HAL_CAN_Receive

Function Name **HAL_StatusTypeDef HAL_CAN_Receive (CAN_HandleTypeDef * hcan, uint8_t FIFONumber, uint32_t Timeout)**

Function Description Receives a correct CAN frame.

Parameters
- **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
- **FIFONumber:** FIFO Number value
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CAN_Receive_IT

Function Name **HAL_StatusTypeDef HAL_CAN_Receive_IT (CAN_HandleTypeDef * hcan, uint8_t FIFONumber)**

Function Description Receives a correct CAN frame.

Parameters
- **hcan:** Pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
- **FIFONumber:** Specify the FIFO number

Return values
- **HAL:** status

HAL_CAN_Sleep

Function Name	**HAL_StatusTypeDef HAL_CAN_Sleep (CAN_HandleTypeDef * hcan)**
Function Description	Enters the Sleep (low power) mode.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **HAL:** status.

HAL_CAN_WakeUp

Function Name	**HAL_StatusTypeDef HAL_CAN_WakeUp (CAN_HandleTypeDef * hcan)**
Function Description	Wakes up the CAN peripheral from sleep mode, after that the CAN peripheral is in the normal mode.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **HAL:** status.

HAL_CAN_IRQHandler

Function Name	**void HAL_CAN_IRQHandler (CAN_HandleTypeDef * hcan)**
Function Description	Handles CAN interrupt request.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **None**

HAL_CAN_TxCpltCallback

Function Name	**void HAL_CAN_TxCpltCallback (CAN_HandleTypeDef * hcan)**
Function Description	Transmission complete callback in non blocking mode.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **None**

HAL_CAN_RxCpltCallback

Function Name	**void HAL_CAN_RxCpltCallback (CAN_HandleTypeDef * hcan)**
Function Description	Transmission complete callback in non blocking mode.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **None**

HAL_CAN_ErrorCallback

Function Name	**void HAL_CAN_ErrorCallback (CAN_HandleTypeDef * hcan)**
Function Description	Error CAN callback.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **None**

HAL_CAN_GetError

Function Name	**uint32_t HAL_CAN_GetError (CAN_HandleTypeDef * hcan)**
Function Description	Return the CAN error code.
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **CAN:** Error Code

HAL_CAN_GetState

Function Name	**HAL_CAN_StateTypeDef HAL_CAN_GetState (CAN_HandleTypeDef * hcan)**
Function Description	return the CAN state
Parameters	• **hcan:** pointer to a CAN_HandleTypeDef structure that contains the configuration information for the specified CAN.
Return values	• **HAL:** state

6.3 CAN Firmware driver defines

6.3.1 CAN

CAN Exported Macros

__HAL_CAN_RESET_HANDLE_STATE

Description:

- Reset CAN handle state.

Parameters:

- __HANDLE__: specifies the CAN Handle.

Return value:

- None

__HAL_CAN_ENABLE_IT

Description:

- Enable the specified CAN interrupts.

Parameters:

- __HANDLE__: CAN handle
- __INTERRUPT__: CAN Interrupt

Return value:

- None

__HAL_CAN_DISABLE_IT

Description:

- Disable the specified CAN interrupts.

Parameters:

- __HANDLE__: CAN handle
- __INTERRUPT__: CAN Interrupt

Return value:

- None

__HAL_CAN_MSG_PENDING

Description:

- Return the number of pending received messages.

Parameters:

- __HANDLE__: CAN handle
- __FIFONUMBER__: Receive FIFO number, CAN_FIFO0 or CAN_FIFO1.

Return value:

- The: number of pending message.

__HAL_CAN_GET_FLAG

Description:

- Check whether the specified CAN flag is set or not.

Parameters:

- __HANDLE__: CAN Handle
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - CAN_TSR_RQCP0: Request MailBox0 Flag
  - CAN_TSR_RQCP1: Request MailBox1 Flag
  - CAN_TSR_RQCP2: Request MailBox2 Flag
  - CAN_FLAG_TXOK0: Transmission OK MailBox0 Flag
  - CAN_FLAG_TXOK1: Transmission OK MailBox1 Flag
  - CAN_FLAG_TXOK2: Transmission OK MailBox2 Flag
  - CAN_FLAG_TME0: Transmit mailbox 0 empty Flag
  - CAN_FLAG_TME1: Transmit mailbox 1 empty Flag
  - CAN_FLAG_TME2: Transmit mailbox 2 empty Flag
  - CAN_FLAG_FMP0: FIFO 0 Message Pending Flag
  - CAN_FLAG_FF0: FIFO 0 Full Flag
  - CAN_FLAG_FOV0: FIFO 0 Overrun

Flag
- CAN_FLAG_FMP1: FIFO 1 Message Pending Flag
- CAN_FLAG_FF1: FIFO 1 Full Flag
- CAN_FLAG_FOV1: FIFO 1 Overrun Flag
- CAN_FLAG_WKU: Wake up Flag
- CAN_FLAG_SLAK: Sleep acknowledge Flag
- CAN_FLAG_SLAKI: Sleep acknowledge Flag
- CAN_FLAG_EWG: Error Warning Flag
- CAN_FLAG_EPV: Error Passive Flag
- CAN_FLAG_BOF: Bus-Off Flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_CAN_CLEAR_FLAG

Description:

- Clear the specified CAN pending flag.

Parameters:

- __HANDLE__: CAN Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - CAN_TSR_RQCP0: Request MailBox0 Flag
  - CAN_TSR_RQCP1: Request MailBox1 Flag
  - CAN_TSR_RQCP2: Request MailBox2 Flag
  - CAN_FLAG_TXOK0: Transmission OK MailBox0 Flag
  - CAN_FLAG_TXOK1: Transmission OK MailBox1 Flag
  - CAN_FLAG_TXOK2: Transmission OK MailBox2 Flag
  - CAN_FLAG_TME0: Transmit mailbox 0 empty Flag
  - CAN_FLAG_TME1: Transmit mailbox 1 empty Flag
  - CAN_FLAG_TME2: Transmit mailbox 2 empty Flag
  - CAN_FLAG_FMP0: FIFO 0 Message Pending Flag
  - CAN_FLAG_FF0: FIFO 0 Full Flag
  - CAN_FLAG_FOV0: FIFO 0 Overrun Flag
  - CAN_FLAG_FMP1: FIFO 1 Message Pending Flag
  - CAN_FLAG_FF1: FIFO 1 Full Flag

  - CAN_FLAG_FOV1: FIFO 1 Overrun Flag
  - CAN_FLAG_WKU: Wake up Flag
  - CAN_FLAG_SLAK: Sleep acknowledge Flag
  - CAN_FLAG_SLAKI: Sleep acknowledge Flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_CAN_GET_IT_SOURCE

Description:

- Check if the specified CAN interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: CAN Handle
- __INTERRUPT__: specifies the CAN interrupt source to check. This parameter can be one of the following values:
  - CAN_IT_TME: Transmit mailbox empty interrupt enable
  - CAN_IT_FMP0: FIFO0 message pending interrupt enable
  - CAN_IT_FMP1: FIFO1 message pending interrupt enable

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_CAN_TRANSMIT_STATUS

Description:

- Check the transmission status of a CAN Frame.

Parameters:

- __HANDLE__: CAN Handle
- __TRANSMITMAILBOX__: the number of the mailbox that is used for transmission.

Return value:

- The: new status of transmission (TRUE or FALSE).

__HAL_CAN_FIFO_RELEASE

Description:

- Release the specified receive FIFO.

Parameters:

- __HANDLE__: CAN handle
- __FIFONUMBER__: Receive FIFO number, CAN_FIFO0 or CAN_FIFO1.

Return value:

- None

__HAL_CAN_CANCEL_TRANSMIT

Description:

- Cancel a transmit request.

Parameters:

- __HANDLE__: CAN Handle
- __TRANSMITMAILBOX__: the number of the mailbox that is used for transmission.

Return value:

- None

__HAL_CAN_DBG_FREEZE

Description:

- Enable or disable the DBG Freeze for CAN.

Parameters:

- __HANDLE__: CAN Handle
- __NEWSTATE__: new state of the CAN peripheral. This parameter can be: ENABLE (CAN reception/transmission is frozen during debug. Reception FIFOs can still be accessed/controlled normally) or DISABLE (CAN is working during debug).

Return value:

- None

CAN Filter FIFO

CAN_FILTER_FIFO0 Filter FIFO 0 assignment for filter x

CAN_FILTER_FIFO1 Filter FIFO 1 assignment for filter x

CAN Filter Mode

CAN_FILTERMODE_IDMASK Identifier mask mode

CAN_FILTERMODE_IDLIST Identifier list mode

CAN Filter Scale

CAN_FILTERSCALE_16BIT Two 16-bit filters

CAN_FILTERSCALE_32BIT One 32-bit filter

CAN Flags

CAN_FLAG_RQCP0 Request MailBox0 flag

CAN_FLAG_RQCP1 Request MailBox1 flag

CAN_FLAG_RQCP2 Request MailBox2 flag

CAN_FLAG_TXOK0 Transmission OK MailBox0 flag

CAN_FLAG_TXOK1 Transmission OK MailBox1 flag

CAN_FLAG_TXOK2 Transmission OK MailBox2 flag

CAN_FLAG_TME0	Transmit mailbox 0 empty flag
CAN_FLAG_TME1	Transmit mailbox 0 empty flag
CAN_FLAG_TME2	Transmit mailbox 0 empty flag
CAN_FLAG_FF0	FIFO 0 Full flag
CAN_FLAG_FOV0	FIFO 0 Overrun flag
CAN_FLAG_FF1	FIFO 1 Full flag
CAN_FLAG_FOV1	FIFO 1 Overrun flag
CAN_FLAG_INAK	Initialization acknowledge flag
CAN_FLAG_SLAK	Sleep acknowledge flag
CAN_FLAG_ERRI	Error flag
CAN_FLAG_WKU	Wake up flag
CAN_FLAG_SLAKI	Sleep acknowledge flag
CAN_FLAG_EWG	Error warning flag
CAN_FLAG_EPV	Error passive flag
CAN_FLAG_BOF	Bus-Off flag

CAN Identifier Type

CAN_ID_STD	Standard Id
CAN_ID_EXT	Extended Id

CAN InitStatus

CAN_INITSTATUS_FAILED	CAN initialization failed
CAN_INITSTATUS_SUCCESS	CAN initialization OK

CAN Interrupts

CAN_IT_TME	Transmit mailbox empty interrupt
CAN_IT_FMP0	FIFO 0 message pending interrupt
CAN_IT_FF0	FIFO 0 full interrupt
CAN_IT_FOV0	FIFO 0 overrun interrupt
CAN_IT_FMP1	FIFO 1 message pending interrupt
CAN_IT_FF1	FIFO 1 full interrupt
CAN_IT_FOV1	FIFO 1 overrun interrupt
CAN_IT_WKU	Wake-up interrupt
CAN_IT_SLK	Sleep acknowledge interrupt
CAN_IT_EWG	Error warning interrupt
CAN_IT_EPV	Error passive interrupt
CAN_IT_BOF	Bus-off interrupt
CAN_IT_LEC	Last error code interrupt
CAN_IT_ERR	Error Interrupt

CAN Mailboxes Definition

CAN_TXMAILBOX_0

CAN_TXMAILBOX_1

CAN_TXMAILBOX_2

CAN Operating Mode

CAN_MODE_NORMAL	Normal mode
CAN_MODE_LOOPBACK	Loopback mode
CAN_MODE_SILENT	Silent mode
CAN_MODE_SILENT_LOOPBACK	Loopback combined with silent mode

CAN Receive FIFO Number Constants

CAN_FIFO0	CAN FIFO 0 used to receive
CAN_FIFO1	CAN FIFO 1 used to receive

CAN Remote Transmission Request

CAN_RTR_DATA	Data frame
CAN_RTR_REMOTE	Remote frame

CAN Synchronisation Jump Width

CAN_SJW_1TQ	1 time quantum
CAN_SJW_2TQ	2 time quantum
CAN_SJW_3TQ	3 time quantum
CAN_SJW_4TQ	4 time quantum

CAN Time Quantum in bit segment 1

CAN_BS1_1TQ	1 time quantum
CAN_BS1_2TQ	2 time quantum
CAN_BS1_3TQ	3 time quantum
CAN_BS1_4TQ	4 time quantum
CAN_BS1_5TQ	5 time quantum
CAN_BS1_6TQ	6 time quantum
CAN_BS1_7TQ	7 time quantum
CAN_BS1_8TQ	8 time quantum
CAN_BS1_9TQ	9 time quantum
CAN_BS1_10TQ	10 time quantum
CAN_BS1_11TQ	11 time quantum
CAN_BS1_12TQ	12 time quantum
CAN_BS1_13TQ	13 time quantum
CAN_BS1_14TQ	14 time quantum
CAN_BS1_15TQ	15 time quantum

CAN_BS1_16TQ	16 time quantum

CAN Time Quantum in bit segment 2

CAN_BS2_1TQ	1 time quantum
CAN_BS2_2TQ	2 time quantum
CAN_BS2_3TQ	3 time quantum
CAN_BS2_4TQ	4 time quantum
CAN_BS2_5TQ	5 time quantum
CAN_BS2_6TQ	6 time quantum
CAN_BS2_7TQ	7 time quantum
CAN_BS2_8TQ	8 time quantum

7 HAL CEC Generic Driver

7.1 CEC Firmware driver registers structures

7.1.1 CEC_InitTypeDef

Data Fields

- ***uint32_t SignalFreeTime***
- ***uint32_t Tolerance***
- ***uint32_t BRERxStop***
- ***uint32_t BREErrorBitGen***
- ***uint32_t LBPEErrorBitGen***
- ***uint32_t BroadcastMsgNoErrorBitGen***
- ***uint32_t SignalFreeTimeOption***
- ***uint32_t ListenMode***
- ***uint16_t OwnAddress***
- ***uint8_t * RxBuffer***

Field Documentation

- ***uint32_t CEC_InitTypeDef::SignalFreeTime***
  Set SFT field, specifies the Signal Free Time. It can be one of ***CEC_Signal_Free_Time*** and belongs to the set {0,...,7} where 0x0 is the default configuration else means 0.5 + (SignalFreeTime - 1) nominal data bit periods
- ***uint32_t CEC_InitTypeDef::Tolerance***
  Set RXTOL bit, specifies the tolerance accepted on the received waveforms, it can be a value of ***CEC_Tolerance*** : it is either CEC_STANDARD_TOLERANCE or CEC_EXTENDED_TOLERANCE
- ***uint32_t CEC_InitTypeDef::BRERxStop***
  Set BRESTP bit ***CEC_BRERxStop*** : specifies whether or not a Bit Rising Error stops the reception. CEC_NO_RX_STOP_ON_BRE: reception is not stopped. CEC_RX_STOP_ON_BRE: reception is stopped.
- ***uint32_t CEC_InitTypeDef::BREErrorBitGen***
  Set BREGEN bit ***CEC_BREErrorBitGen*** : specifies whether or not an Error-Bit is generated on the CEC line upon Bit Rising Error detection. CEC_BRE_ERRORBIT_NO_GENERATION: no error-bit generation. CEC_BRE_ERRORBIT_GENERATION: error-bit generation if BRESTP is set.
- ***uint32_t CEC_InitTypeDef::LBPEErrorBitGen***
  Set LBPEGEN bit ***CEC_LBPEErrorBitGen*** : specifies whether or not an Error-Bit is generated on the CEC line upon Long Bit Period Error detection. CEC_LBPE_ERRORBIT_NO_GENERATION: no error-bit generation. CEC_LBPE_ERRORBIT_GENERATION: error-bit generation.
- ***uint32_t CEC_InitTypeDef::BroadcastMsgNoErrorBitGen***
  Set BRDNOGEN bit ***CEC_BroadCastMsgErrorBitGen*** : allows to avoid an Error-Bit generation on the CEC line upon an error detected on a broadcast message.It supersedes BREGEN and LBPEGEN bits for a broadcast message error handling. It can take two values:1) CEC_BROADCASTERROR_ERRORBIT_GENERATION. a) BRE detection: error-bit generation on the CEC line if BRESTP=CEC_RX_STOP_ON_BRE and

BREGEN=CEC_BRE_ERRORBIT_NO_GENERATION. b) LBPE detection: error-bit generation on the CEC line if LBPGEN=CEC_LBPE_ERRORBIT_NO_GENERATION.2) CEC_BROADCASTERROR_NO_ERRORBIT_GENERATION. no error-bit generation in case neither a) nor b) are satisfied. Additionally, there is no error-bit generation in case of Short Bit Period Error detection in a broadcast message while LSTN bit is set.

- ***uint32_t CEC_InitTypeDef::SignalFreeTimeOption***
  Set SFTOP bit ***CEC_SFT_Option*** : specifies when SFT timer starts. CEC_SFT_START_ON_TXSOM SFT: timer starts when TXSOM is set by software. CEC_SFT_START_ON_TX_RX_END: SFT timer starts automatically at the end of message transmission/reception.
- ***uint32_t CEC_InitTypeDef::ListenMode***
  Set LSTN bit ***CEC_Listening_Mode*** : specifies device listening mode. It can take two values:CEC_REDUCED_LISTENING_MODE: CEC peripheral receives only message addressed to its own address (OAR). Messages addressed to different destination are ignored. Broadcast messages are always received.CEC_FULL_LISTENING_MODE: CEC peripheral receives messages addressed to its own address (OAR) with positive acknowledge. Messages addressed to different destination are received, but without interfering with the CEC bus: no acknowledge sent.
- ***uint16_t CEC_InitTypeDef::OwnAddress***
  Own addresses configuration This parameter can be a value of ***CEC_OWN_ADDRESS***
- ***uint8_t* CEC_InitTypeDef::RxBuffer***
  CEC Rx buffer pointeur

7.1.2 CEC_HandleTypeDef

Data Fields

- ***CEC_TypeDef * Instance***
- ***CEC_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferCount***
- ***uint16_t RxXferSize***
- ***HAL_LockTypeDef Lock***
- ***HAL_CEC_StateTypeDef gState***
- ***HAL_CEC_StateTypeDef RxState***
- ***uint32_t ErrorCode***

Field Documentation

- ***CEC_TypeDef* CEC_HandleTypeDef::Instance***
  CEC registers base address
- ***CEC_InitTypeDef CEC_HandleTypeDef::Init***
  CEC communication parameters
- ***uint8_t* CEC_HandleTypeDef::pTxBuffPtr***
  Pointer to CEC Tx transfer Buffer
- ***uint16_t CEC_HandleTypeDef::TxXferCount***
  CEC Tx Transfer Counter
- ***uint16_t CEC_HandleTypeDef::RxXferSize***
  CEC Rx Transfer size, 0: header received only

- ***HAL_LockTypeDef CEC_HandleTypeDef::Lock***
  Locking object
- ***HAL_CEC_StateTypeDef CEC_HandleTypeDef::gState***
  CEC state information related to global Handle management and also related to Tx operations. This parameter can be a value of **HAL_CEC_StateTypeDef**
- ***HAL_CEC_StateTypeDef CEC_HandleTypeDef::RxState***
  CEC state information related to Rx operations. This parameter can be a value of **HAL_CEC_StateTypeDef**
- ***uint32_t CEC_HandleTypeDef::ErrorCode***
  For errors handling purposes, copy of ISR register in case error is reported

7.2 CEC Firmware driver API description

7.2.1 How to use this driver

The CEC HAL driver can be used as follow:

1. Declare a CEC_HandleTypeDef handle structure.
2. Initialize the CEC low level resources by implementing the HAL_CEC_MspInit ()API:
   a. Enable the CEC interface clock.
   b. CEC pins configuration:
      – Enable the clock for the CEC GPIOs.
      – Configure these CEC pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_CEC_Transmit_IT() and HAL_CEC_Receive_IT() APIs):
      – Configure the CEC interrupt priority.
      – Enable the NVIC CEC IRQ handle.
      – The specific CEC interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_CEC_ENABLE_IT() and __HAL_CEC_DISABLE_IT() inside the transmit and receive process.
3. Program the Signal Free Time (SFT) and SFT option, Tolerance, reception stop in in case of Bit Rising Error, Error-Bit generation conditions, device logical address and Listen mode in the hcec Init structure.
4. Initialize the CEC registers by calling the HAL_CEC_Init() API.

This API (HAL_CEC_Init()) configures also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customed HAL_CEC_MspInit() API.

7.2.2 How to use this driver

The CEC HAL driver can be used as follow:

1. Declare a CEC_HandleTypeDef handle structure.
2. Initialize the CEC low level resources by implementing the HAL_CEC_MspInit ()API:
   a. Enable the CEC interface clock.
   b. CEC pins configuration:
      – Enable the clock for the CEC GPIOs.
      – Configure these CEC pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_CEC_Transmit_IT() and HAL_CEC_Receive_IT() APIs):
      – Configure the CEC interrupt priority.

  - Enable the NVIC CEC IRQ handle.
  - The specific CEC interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_CEC_ENABLE_IT() and __HAL_CEC_DISABLE_IT() inside the transmit and receive process.

3. Program the Signal Free Time (SFT) and SFT option, Tolerance, reception stop in in case of Bit Rising Error, Error-Bit generation conditions, device logical address and Listen mode in the hcec Init structure.
4. Initialize the CEC registers by calling the HAL_CEC_Init() API.

This API (HAL_CEC_Init()) configures also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customed HAL_CEC_MspInit() API.

7.2.3 Initialization and Configuration functions

This subsection provides a set of functions allowing to initialize the CEC

- The following parameters need to be configured:
  - SignalFreeTime
  - Tolerance
  - BRERxStop (RX stopped or not upon Bit Rising Error)
  - BREErrorBitGen (Error-Bit generation in case of Bit Rising Error)
  - LBPEErrorBitGen (Error-Bit generation in case of Long Bit Period Error)
  - BroadcastMsgNoErrorBitGen (Error-bit generation in case of broadcast message error)
  - SignalFreeTimeOption (SFT Timer start definition)
  - OwnAddress (CEC device address)
  - ListenMode

This section contains the following APIs:

- ***HAL_CEC_Init()***
- ***HAL_CEC_DeInit()***
- ***HAL_CEC_SetDeviceAddress()***
- ***HAL_CEC_MspInit()***
- ***HAL_CEC_MspDeInit()***

7.2.4 IO operation functions

This section contains the following APIs:

- ***HAL_CEC_Transmit_IT()***
- ***HAL_CEC_GetLastReceivedFrameSize()***
- ***HAL_CEC_ChangeRxBuffer()***
- ***HAL_CEC_IRQHandler()***
- ***HAL_CEC_TxCpltCallback()***
- ***HAL_CEC_RxCpltCallback()***
- ***HAL_CEC_ErrorCallback()***

7.2.5 Peripheral Control function

This subsection provides a set of functions allowing to control the CEC.

- HAL_CEC_GetState() API can be helpful to check in run-time the state of the CEC peripheral.
- HAL_CEC_GetError() API can be helpful to check in run-time the error of the CEC peripheral.

This section contains the following APIs:

- ***HAL_CEC_GetState()***
- ***HAL_CEC_GetError()***

7.2.6 Detailed description of functions

HAL_CEC_Init

Function Name	**HAL_StatusTypeDef HAL_CEC_Init (CEC_HandleTypeDef * hcec)**
Function Description	Initializes the CEC mode according to the specified parameters in the CEC_InitTypeDef and creates the associated handle .
Parameters	• **hcec:** CEC handle
Return values	• **HAL:** status

HAL_CEC_DeInit

Function Name	**HAL_StatusTypeDef HAL_CEC_DeInit (CEC_HandleTypeDef * hcec)**
Function Description	DeInitializes the CEC peripheral.
Parameters	• **hcec:** CEC handle
Return values	• **HAL:** status

HAL_CEC_SetDeviceAddress

Function Name	**HAL_StatusTypeDef HAL_CEC_SetDeviceAddress (CEC_HandleTypeDef * hcec, uint16_t CEC_OwnAddress)**
Function Description	Initializes the Own Address of the CEC device.
Parameters	• **hcec:** CEC handle • **CEC_OwnAddress:** The CEC own address.
Return values	• **HAL:** status

HAL_CEC_MspInit

Function Name	**void HAL_CEC_MspInit (CEC_HandleTypeDef * hcec)**
Function Description	CEC MSP Init.
Parameters	• **hcec:** CEC handle
Return values	• **None**

HAL_CEC_MspDeInit

Function Name	**void HAL_CEC_MspDeInit (CEC_HandleTypeDef * hcec)**
Function Description	CEC MSP DeInit.
Parameters	• **hcec:** CEC handle
Return values	• **None**

HAL_CEC_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_CEC_Transmit_IT (CEC_HandleTypeDef * hcec, uint8_t InitiatorAddress, uint8_t DestinationAddress, uint8_t * pData, uint32_t Size)**
Function Description	Send data in interrupt mode.
Parameters	• **hcec:** CEC handle • **InitiatorAddress:** Initiator logical address • **DestinationAddress:** destination logical address • **pData:** pointer to input byte data buffer • **Size:** amount of data to be sent in bytes (without counting the header). 0 means only the header is sent (ping operation). Maximum TX size is 15 bytes (1 opcode and up to 14 operands).
Return values	• **HAL:** status

HAL_CEC_GetLastReceivedFrameSize

Function Name	**uint32_t HAL_CEC_GetLastReceivedFrameSize (CEC_HandleTypeDef * hcec)**
Function Description	Get size of the received frame.
Parameters	• **hcec:** CEC handle
Return values	• **Frame:** size

HAL_CEC_ChangeRxBuffer

Function Name	**void HAL_CEC_ChangeRxBuffer (CEC_HandleTypeDef * hcec, uint8_t * Rxbuffer)**
Function Description	Change Rx Buffer.
Parameters	• **hcec:** CEC handle • **Rxbuffer:** Rx Buffer
Return values	• **Frame:** size
Notes	• This function can be called only inside the HAL_CEC_RxCpltCallback()

HAL_CEC_IRQHandler

Function Name	**void HAL_CEC_IRQHandler (CEC_HandleTypeDef * hcec)**
Function Description	This function handles CEC interrupt requests.

Parameters
- **hcec:** CEC handle

Return values
- **None**

HAL_CEC_TxCpltCallback

Function Name **void HAL_CEC_TxCpltCallback (CEC_HandleTypeDef * hcec)**

Function Description Tx Transfer completed callback.

Parameters
- **hcec:** CEC handle

Return values
- **None**

HAL_CEC_RxCpltCallback

Function Name **void HAL_CEC_RxCpltCallback (CEC_HandleTypeDef * hcec, uint32_t RxFrameSize)**

Function Description Rx Transfer completed callback.

Parameters
- **hcec:** CEC handle
- **RxFrameSize:** Size of frame

Return values
- **None**

HAL_CEC_ErrorCallback

Function Name **void HAL_CEC_ErrorCallback (CEC_HandleTypeDef * hcec)**

Function Description CEC error callbacks.

Parameters
- **hcec:** CEC handle

Return values
- **None**

HAL_CEC_GetState

Function Name **HAL_CEC_StateTypeDef HAL_CEC_GetState (CEC_HandleTypeDef * hcec)**

Function Description return the CEC state

Parameters
- **hcec:** pointer to a CEC_HandleTypeDef structure that contains the configuration information for the specified CEC module.

Return values
- **HAL:** state

HAL_CEC_GetError

Function Name **uint32_t HAL_CEC_GetError (CEC_HandleTypeDef * hcec)**

Function Description Return the CEC error code.

Parameters
- **hcec:** : pointer to a CEC_HandleTypeDef structure that contains the configuration information for the specified CEC.

Return values
- **CEC:** Error Code

7.3 CEC Firmware driver defines

7.3.1 CEC

CEC all RX or TX errors flags

CEC_ISR_ALL_ERROR

CEC Error Bit Generation if Bit Rise Error reported

CEC_BRE_ERRORBIT_NO_GENERATION

CEC_BRE_ERRORBIT_GENERATION

CEC Reception Stop on Error

CEC_NO_RX_STOP_ON_BRE

CEC_RX_STOP_ON_BRE

CEC Error Bit Generation on Broadcast message

CEC_BROADCASTERROR_ERRORBIT_GENERATION

CEC_BROADCASTERROR_NO_ERRORBIT_GENERATION

CEC Error Code

HAL_CEC_ERROR_NONE	no error
HAL_CEC_ERROR_RXOVR	CEC Rx-Overrun
HAL_CEC_ERROR_BRE	CEC Rx Bit Rising Error
HAL_CEC_ERROR_SBPE	CEC Rx Short Bit period Error
HAL_CEC_ERROR_LBPE	CEC Rx Long Bit period Error
HAL_CEC_ERROR_RXACKE	CEC Rx Missing Acknowledge
HAL_CEC_ERROR_ARBLST	CEC Arbitration Lost
HAL_CEC_ERROR_TXUDR	CEC Tx-Buffer Underrun
HAL_CEC_ERROR_TXERR	CEC Tx-Error
HAL_CEC_ERROR_TXACKE	CEC Tx Missing Acknowledge

CEC Exported Macros

__HAL_CEC_RESET_HANDLE_STATE

Description:

- Reset CEC handle gstate & RxState.

Parameters:

- __HANDLE__: CEC handle.

Return value:

- None

__HAL_CEC_GET_FLAG

Description:

- Checks whether or not the specified CEC interrupt flag is set.

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __FLAG__: specifies the flag to check.
  - CEC_FLAG_TXACKE: Tx Missing acknowledge Error
  - CEC_FLAG_TXERR: Tx Error.
  - CEC_FLAG_TXUDR: Tx-Buffer Underrun.
  - CEC_FLAG_TXEND: End of transmission (successful transmission of the last byte).
  - CEC_FLAG_TXBR: Tx-Byte Request.
  - CEC_FLAG_ARBLST: Arbitration Lost
  - CEC_FLAG_RXACKE: Rx-Missing Acknowledge
  - CEC_FLAG_LBPE: Rx Long period Error
  - CEC_FLAG_SBPE: Rx Short period Error
  - CEC_FLAG_BRE: Rx Bit Rising Error
  - CEC_FLAG_RXOVR: Rx Overrun.
  - CEC_FLAG_RXEND: End Of Reception.
  - CEC_FLAG_RXBR: Rx-Byte Received.

Return value:

- ITStatus

__HAL_CEC_CLEAR_FLAG

Description:

- Clears the interrupt or status flag when raised (write at 1)

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __FLAG__: specifies the interrupt/status flag to clear. This parameter can be one of the following values:
  - CEC_FLAG_TXACKE: Tx Missing acknowledge Error
  - CEC_FLAG_TXERR: Tx Error.
  - CEC_FLAG_TXUDR: Tx-Buffer Underrun.

- CEC_FLAG_TXEND: End of transmission (successful transmission of the last byte).
- CEC_FLAG_TXBR: Tx-Byte Request.
- CEC_FLAG_ARBLST: Arbitration Lost
- CEC_FLAG_RXACKE: Rx-Missing Acknowledge
- CEC_FLAG_LBPE: Rx Long period Error
- CEC_FLAG_SBPE: Rx Short period Error
- CEC_FLAG_BRE: Rx Bit Rising Error
- CEC_FLAG_RXOVR: Rx Overrun.
- CEC_FLAG_RXEND: End Of Reception.
- CEC_FLAG_RXBR: Rx-Byte Received.

Return value:

- none

__HAL_CEC_ENABLE_IT

Description:

- Enables the specified CEC interrupt.

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __INTERRUPT__: specifies the CEC interrupt to enable. This parameter can be one of the following values:
  - CEC_IT_TXACKE: Tx Missing acknowledge Error IT Enable
  - CEC_IT_TXERR: Tx Error IT Enable
  - CEC_IT_TXUDR: Tx-Buffer Underrun IT Enable
  - CEC_IT_TXEND: End of transmission IT Enable
  - CEC_IT_TXBR: Tx-Byte Request IT Enable
  - CEC_IT_ARBLST: Arbitration Lost IT Enable
  - CEC_IT_RXACKE: Rx-Missing Acknowledge IT Enable
  - CEC_IT_LBPE: Rx Long

period Error IT Enable
- CEC_IT_SBPE: Rx Short period Error IT Enable
- CEC_IT_BRE: Rx Bit Rising Error IT Enable
- CEC_IT_RXOVR: Rx Overrun IT Enable
- CEC_IT_RXEND: End Of Reception IT Enable
- CEC_IT_RXBR: Rx-Byte Received IT Enable

Return value:

- none

__HAL_CEC_DISABLE_IT

Description:

- Disables the specified CEC interrupt.

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __INTERRUPT__: specifies the CEC interrupt to disable. This parameter can be one of the following values:
  - CEC_IT_TXACKE: Tx Missing acknowledge Error IT Enable
  - CEC_IT_TXERR: Tx Error IT Enable
  - CEC_IT_TXUDR: Tx-Buffer Underrun IT Enable
  - CEC_IT_TXEND: End of transmission IT Enable
  - CEC_IT_TXBR: Tx-Byte Request IT Enable
  - CEC_IT_ARBLST: Arbitration Lost IT Enable
  - CEC_IT_RXACKE: Rx-Missing Acknowledge IT Enable
  - CEC_IT_LBPE: Rx Long period Error IT Enable
  - CEC_IT_SBPE: Rx Short period Error IT Enable
  - CEC_IT_BRE: Rx Bit Rising Error IT Enable
  - CEC_IT_RXOVR: Rx Overrun IT Enable
  - CEC_IT_RXEND: End Of Reception IT Enable
  - CEC_IT_RXBR: Rx-Byte Received IT Enable

Return value:

- none

__HAL_CEC_GET_IT_SOURCE

Description:

- Checks whether or not the specified CEC interrupt is enabled.

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __INTERRUPT__: specifies the CEC interrupt to check. This parameter can be one of the following values:
  - CEC_IT_TXACKE: Tx Missing acknowledge Error IT Enable
  - CEC_IT_TXERR: Tx Error IT Enable
  - CEC_IT_TXUDR: Tx-Buffer Underrun IT Enable
  - CEC_IT_TXEND: End of transmission IT Enable
  - CEC_IT_TXBR: Tx-Byte Request IT Enable
  - CEC_IT_ARBLST: Arbitration Lost IT Enable
  - CEC_IT_RXACKE: Rx-Missing Acknowledge IT Enable
  - CEC_IT_LBPE: Rx Long period Error IT Enable
  - CEC_IT_SBPE: Rx Short period Error IT Enable
  - CEC_IT_BRE: Rx Bit Rising Error IT Enable
  - CEC_IT_RXOVR: Rx Overrun IT Enable
  - CEC_IT_RXEND: End Of Reception IT Enable
  - CEC_IT_RXBR: Rx-Byte Received IT Enable

Return value:

- FlagStatus

__HAL_CEC_ENABLE

Description:

- Enables the CEC device.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- none

__HAL_CEC_DISABLE

Description:

- Disables the CEC device.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- none

__HAL_CEC_FIRST_BYTE_TX_SET

Description:

- Set Transmission Start flag.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- none

__HAL_CEC_LAST_BYTE_TX_SET

Description:

- Set Transmission End flag.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- none: If the CEC message consists of only one byte, TXEOM must be set before of TXSOM.

__HAL_CEC_GET_TRANSMISSION_START_FLAG

Description:

- Get Transmission Start flag.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- FlagStatus

__HAL_CEC_GET_TRANSMISSION_END_FLAG

Description:

- Get Transmission End flag.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- FlagStatus

__HAL_CEC_CLEAR_OAR

Description:

- Clear OAR register.

Parameters:

- __HANDLE__: specifies the CEC Handle.

Return value:

- none

__HAL_CEC_SET_OAR

Description:

- Set OAR register (without resetting previously set address in case of multi-address mode) To reset OAR,

Parameters:

- __HANDLE__: specifies the CEC Handle.
- __ADDRESS__: Own Address value (CEC logical address is identified by bit position)

Return value:

- none

CEC Flags definition

CEC_FLAG_TXACKE

CEC_FLAG_TXERR

CEC_FLAG_TXUDR

CEC_FLAG_TXEND

CEC_FLAG_TXBR

CEC_FLAG_ARBLST

CEC_FLAG_RXACKE

CEC_FLAG_LBPE

CEC_FLAG_SBPE

CEC_FLAG_BRE

CEC_FLAG_RXOVR

CEC_FLAG_RXEND

CEC_FLAG_RXBR

CEC all RX errors interrupts enabling flag

CEC_IER_RX_ALL_ERR

CEC all TX errors interrupts enabling flag

CEC_IER_TX_ALL_ERR

CEC Initiator logical address position in message header

CEC_INITIATOR_LSB_POS

CEC Interrupts definition

CEC_IT_TXACKE

CEC_IT_TXERR

CEC_IT_TXUDR

CEC_IT_TXEND

CEC_IT_TXBR

CEC_IT_ARBLST

CEC_IT_RXACKE

CEC_IT_LBPE

CEC_IT_SBPE

CEC_IT_BRE

CEC_IT_RXOVR

CEC_IT_RXEND

CEC_IT_RXBR

CEC Error Bit Generation if Long Bit Period Error reported

CEC_LBPE_ERRORBIT_NO_GENERATION

CEC_LBPE_ERRORBIT_GENERATION

CEC Listening mode option

CEC_REDUCED_LISTENING_MODE

CEC_FULL_LISTENING_MODE

CEC Device Own Address position in CEC CFGR register

CEC_CFGR_OAR_LSB_POS

CEC Own Address

CEC_OWN_ADDRESS_NONE

CEC_OWN_ADDRESS_0

CEC_OWN_ADDRESS_1

CEC_OWN_ADDRESS_2

CEC_OWN_ADDRESS_3

CEC_OWN_ADDRESS_4

CEC_OWN_ADDRESS_5

CEC_OWN_ADDRESS_6

CEC_OWN_ADDRESS_7

CEC_OWN_ADDRESS_8

CEC_OWN_ADDRESS_9

CEC_OWN_ADDRESS_10

CEC_OWN_ADDRESS_11

CEC_OWN_ADDRESS_12

CEC_OWN_ADDRESS_13

CEC_OWN_ADDRESS_14

CEC Signal Free Time start option

CEC_SFT_START_ON_TXSOM

CEC_SFT_START_ON_TX_RX_END

CEC Signal Free Time setting parameter

CEC_DEFAULT_SFT

CEC_0_5_BITPERIOD_SFT

CEC_1_5_BITPERIOD_SFT

CEC_2_5_BITPERIOD_SFT

CEC_3_5_BITPERIOD_SFT

CEC_4_5_BITPERIOD_SFT

CEC_5_5_BITPERIOD_SFT

CEC_6_5_BITPERIOD_SFT

CEC Receiver Tolerance

CEC_STANDARD_TOLERANCE

CEC_EXTENDED_TOLERANCE

8 HAL CORTEX Generic Driver

8.1 CORTEX Firmware driver registers structures

8.1.1 MPU_Region_InitTypeDef

Data Fields

- ***uint8_t Enable***
- ***uint8_t Number***
- ***uint32_t BaseAddress***
- ***uint8_t Size***
- ***uint8_t SubRegionDisable***
- ***uint8_t TypeExtField***
- ***uint8_t AccessPermission***
- ***uint8_t DisableExec***
- ***uint8_t IsShareable***
- ***uint8_t IsCacheable***
- ***uint8_t IsBufferable***

Field Documentation

- ***uint8_t MPU_Region_InitTypeDef::Enable***
  Specifies the status of the region. This parameter can be a value of ***CORTEX_MPU_Region_Enable***
- ***uint8_t MPU_Region_InitTypeDef::Number***
  Specifies the number of the region to protect. This parameter can be a value of ***CORTEX_MPU_Region_Number***
- ***uint32_t MPU_Region_InitTypeDef::BaseAddress***
  Specifies the base address of the region to protect.
- ***uint8_t MPU_Region_InitTypeDef::Size***
  Specifies the size of the region to protect. This parameter can be a value of ***CORTEX_MPU_Region_Size***
- ***uint8_t MPU_Region_InitTypeDef::SubRegionDisable***
  Specifies the number of the subregion protection to disable. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF
- ***uint8_t MPU_Region_InitTypeDef::TypeExtField***
  Specifies the TEX field level. This parameter can be a value of ***CORTEX_MPU_TEX_Levels***
- ***uint8_t MPU_Region_InitTypeDef::AccessPermission***
  Specifies the region access permission type. This parameter can be a value of ***CORTEX_MPU_Region_Permission_Attributes***
- ***uint8_t MPU_Region_InitTypeDef::DisableExec***
  Specifies the instruction access status. This parameter can be a value of ***CORTEX_MPU_Instruction_Access***
- ***uint8_t MPU_Region_InitTypeDef::IsShareable***
  Specifies the shareability status of the protected region. This parameter can be a value of ***CORTEX_MPU_Access_Shareable***

- ***uint8_t MPU_Region_InitTypeDef::IsCacheable***
  Specifies the cacheable status of the region protected. This parameter can be a value of ***CORTEX_MPU_Access_Cacheable***
- ***uint8_t MPU_Region_InitTypeDef::IsBufferable***
  Specifies the bufferable status of the protected region. This parameter can be a value of ***CORTEX_MPU_Access_Bufferable***

8.2 CORTEX Firmware driver API description

8.2.1 How to use this driver

How to configure Interrupts using CORTEX HAL driver

This section provides functions allowing to configure the NVIC interrupts (IRQ). The Cortex-M4 exceptions are managed by CMSIS functions.

1. Configure the NVIC Priority Grouping using HAL_NVIC_SetPriorityGrouping() function according to the following table.
2. Configure the priority of the selected IRQ Channels using HAL_NVIC_SetPriority().
3. Enable the selected IRQ Channels using HAL_NVIC_EnableIRQ().
4. please refer to programing manual for details in how to configure priority. When the NVIC_PRIORITYGROUP_0 is selected, IRQ preemption is no more possible. The pending IRQ priority will be managed only by the sub priority. IRQ priority order (sorted by highest to lowest priority): Lowest preemption priority Lowest sub priority Lowest hardware priority (IRQ number)

How to configure Systick using CORTEX HAL driver

Setup SysTick Timer for time base.

- The HAL_SYSTICK_Config() function calls the SysTick_Config() function which is a CMSIS function that:
  - Configures the SysTick Reload register with value passed as function parameter.
  - Configures the SysTick IRQ priority to the lowest value (0x0FU).
  - Resets the SysTick Counter register.
  - Configures the SysTick Counter clock source to be Core Clock Source (HCLK).
  - Enables the SysTick Interrupt.
  - Starts the SysTick Counter.
- You can change the SysTick Clock source to be HCLK_Div8 by calling the macro __HAL_CORTEX_SYSTICKCLK_CONFIG(SYSTICK_CLKSOURCE_HCLK_DIV8) just after the HAL_SYSTICK_Config() function call. The __HAL_CORTEX_SYSTICKCLK_CONFIG() macro is defined inside the stm32f4xx_hal_cortex.h file.
- You can change the SysTick IRQ priority by calling the HAL_NVIC_SetPriority(SysTick_IRQn,...) function just after the HAL_SYSTICK_Config() function call. The HAL_NVIC_SetPriority() call the NVIC_SetPriority() function which is a CMSIS function.
- To adjust the SysTick time base, use the following formula: Reload Value = SysTick Counter Clock (Hz) x Desired Time base (s)
  - Reload Value is the parameter to be passed for HAL_SYSTICK_Config() function

– Reload Value should not exceed 0xFFFFFF

8.2.2 Initialization and de-initialization functions

This section provides the CORTEX HAL driver functions allowing to configure Interrupts Systick functionalities

This section contains the following APIs:

- ***HAL_NVIC_SetPriorityGrouping()***
- ***HAL_NVIC_SetPriority()***
- ***HAL_NVIC_EnableIRQ()***
- ***HAL_NVIC_DisableIRQ()***
- ***HAL_NVIC_SystemReset()***
- ***HAL_SYSTICK_Config()***

8.2.3 Peripheral Control functions

This subsection provides a set of functions allowing to control the CORTEX (NVIC, SYSTICK, MPU) functionalities.

This section contains the following APIs:

- ***HAL_MPU_ConfigRegion()***
- ***HAL_NVIC_GetPriorityGrouping()***
- ***HAL_NVIC_GetPriority()***
- ***HAL_NVIC_SetPendingIRQ()***
- ***HAL_NVIC_GetPendingIRQ()***
- ***HAL_NVIC_ClearPendingIRQ()***
- ***HAL_NVIC_GetActive()***
- ***HAL_SYSTICK_CLKSourceConfig()***
- ***HAL_SYSTICK_IRQHandler()***
- ***HAL_SYSTICK_Callback()***

8.2.4 Detailed description of functions

HAL_NVIC_SetPriorityGrouping

Function Name **void HAL_NVIC_SetPriorityGrouping (uint32_t PriorityGroup)**

Function Description Sets the priority grouping field (preemption priority and subpriority) using the required unlock sequence.

Parameters

- **PriorityGroup:** The priority grouping bits length. This parameter can be one of the following values:
  - NVIC_PRIORITYGROUP_0: 0 bits for preemption priority 4 bits for subpriority
  - NVIC_PRIORITYGROUP_1: 1 bits for preemption priority 3 bits for subpriority
  - NVIC_PRIORITYGROUP_2: 2 bits for preemption priority 2 bits for subpriority
  - NVIC_PRIORITYGROUP_3: 3 bits for preemption priority 1 bits for subpriority
  - NVIC_PRIORITYGROUP_4: 4 bits for preemption priority 0 bits for subpriority

Return values

- **None**

Notes
- When the NVIC_PriorityGroup_0 is selected, IRQ preemption is no more possible. The pending IRQ priority will be managed only by the subpriority.

HAL_NVIC_SetPriority

Function Name **void HAL_NVIC_SetPriority (IRQn_Type IRQn, uint32_t PreemptPriority, uint32_t SubPriority)**

Function Description Sets the priority of an interrupt.

Parameters
- **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))
- **PreemptPriority:** The preemption priority for the IRQn channel. This parameter can be a value between 0 and 15 A lower priority value indicates a higher priority
- **SubPriority:** the subpriority level for the IRQ channel. This parameter can be a value between 0 and 15 A lower priority value indicates a higher priority.

Return values
- **None**

HAL_NVIC_EnableIRQ

Function Name **void HAL_NVIC_EnableIRQ (IRQn_Type IRQn)**

Function Description Enables a device specific interrupt in the NVIC interrupt controller.

Parameters
- **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **None**

Notes
- To configure interrupts priority correctly, the NVIC_PriorityGroupConfig() function should be called before.

HAL_NVIC_DisableIRQ

Function Name **void HAL_NVIC_DisableIRQ (IRQn_Type IRQn)**

Function Description Disables a device specific interrupt in the NVIC interrupt controller.

Parameters
- **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **None**

HAL_NVIC_SystemReset

Function Name	**void HAL_NVIC_SystemReset (void)**
Function Description	Initiates a system reset request to reset the MCU.
Return values	• **None**

HAL_SYSTICK_Config

Function Name	**uint32_t HAL_SYSTICK_Config (uint32_t TicksNumb)**
Function Description	Initializes the System Timer and its interrupt, and starts the System Tick Timer.
Parameters	• **TicksNumb:** Specifies the ticks Number of ticks between two interrupts.
Return values	• **status:** - 0 Function succeeded. – 1 Function failed.

HAL_MPU_ConfigRegion

Function Name	**void HAL_MPU_ConfigRegion (MPU_Region_InitTypeDef * MPU_Init)**
Function Description	Initializes and configures the Region and the memory to be protected.
Parameters	• **MPU_Init:** Pointer to a MPU_Region_InitTypeDef structure that contains the initialization and configuration information.
Return values	• **None**

HAL_NVIC_GetPriorityGrouping

Function Name	**uint32_t HAL_NVIC_GetPriorityGrouping (void)**
Function Description	Gets the priority grouping field from the NVIC Interrupt Controller.
Return values	• **Priority:** grouping field (SCB->AIRCR [10:8] PRIGROUP field)

HAL_NVIC_GetPriority

Function Name	**void HAL_NVIC_GetPriority (IRQn_Type IRQn, uint32_t PriorityGroup, uint32_t * pPreemptPriority, uint32_t * pSubPriority)**
Function Description	Gets the priority of an interrupt.
Parameters	• **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h)) • **PriorityGroup:** the priority grouping bits length. This parameter can be one of the following values: – NVIC_PRIORITYGROUP_0: 0 bits for preemption priority 4 bits for subpriority – NVIC_PRIORITYGROUP_1: 1 bits for preemption

priority 3 bits for subpriority
  - NVIC_PRIORITYGROUP_2: 2 bits for preemption priority 2 bits for subpriority
  - NVIC_PRIORITYGROUP_3: 3 bits for preemption priority 1 bits for subpriority
  - NVIC_PRIORITYGROUP_4: 4 bits for preemption priority 0 bits for subpriority
- **pPreemptPriority:** Pointer on the Preemptive priority value (starting from 0).
- **pSubPriority:** Pointer on the Subpriority value (starting from 0).

Return values
- **None**

HAL_NVIC_GetPendingIRQ

Function Name **uint32_t HAL_NVIC_GetPendingIRQ (IRQn_Type IRQn)**

Function Description Gets Pending Interrupt (reads the pending register in the NVIC and returns the pending bit for the specified interrupt).

Parameters
- **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **status:** - 0 Interrupt status is not pending.
  - 1 Interrupt status is pending.

HAL_NVIC_SetPendingIRQ

Function Name **void HAL_NVIC_SetPendingIRQ (IRQn_Type IRQn)**

Function Description Sets Pending bit of an external interrupt.

Parameters
- **IRQn:** External interrupt number This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **None**

HAL_NVIC_ClearPendingIRQ

Function Name **void HAL_NVIC_ClearPendingIRQ (IRQn_Type IRQn)**

Function Description Clears the pending bit of an external interrupt.

Parameters
- **IRQn:** External interrupt number. This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **None**

HAL_NVIC_GetActive

Function Name **uint32_t HAL_NVIC_GetActive (IRQn_Type IRQn)**

Function Description Gets active interrupt (reads the active register in NVIC and returns the active bit).

Parameters
- **IRQn:** External interrupt number This parameter can be an enumerator of IRQn_Type enumeration (For the complete STM32 Devices IRQ Channels list, please refer to the appropriate CMSIS device file (stm32f4xxxx.h))

Return values
- **status:** - 0 Interrupt status is not pending.
  - 1 Interrupt status is pending.

HAL_SYSTICK_CLKSourceConfig

Function Name **void HAL_SYSTICK_CLKSourceConfig (uint32_t CLKSource)**

Function Description Configures the SysTick clock source.

Parameters
- **CLKSource:** specifies the SysTick clock source. This parameter can be one of the following values:
  - SYSTICK_CLKSOURCE_HCLK_DIV8: AHB clock divided by 8 selected as SysTick clock source.
  - SYSTICK_CLKSOURCE_HCLK: AHB clock selected as SysTick clock source.

Return values
- **None**

HAL_SYSTICK_IRQHandler

Function Name **void HAL_SYSTICK_IRQHandler (void)**

Function Description This function handles SYSTICK interrupt request.

Return values
- **None**

HAL_SYSTICK_Callback

Function Name **void HAL_SYSTICK_Callback (void)**

Function Description SYSTICK callback.

Return values
- **None**

HAL_MPU_Disable

Function Name **__STATIC_INLINE void HAL_MPU_Disable (void)**

Function Description Disables the MPU.

Return values
- **None**

HAL_MPU_Enable

Function Name	**__STATIC_INLINE void HAL_MPU_Enable (uint32_t MPU_Control)**
Function Description	Enables the MPU.
Parameters	• **MPU_Control:** Specifies the control mode of the MPU during hard fault, NMI, FAULTMASK and privileged access to the default memory This parameter can be one of the following values: – MPU_HFNMI_PRIVDEF_NONE – MPU_HARDFAULT_NMI – MPU_PRIVILEGED_DEFAULT – MPU_HFNMI_PRIVDEF
Return values	• **None**

8.3 CORTEX Firmware driver defines

8.3.1 CORTEX

CORTEX MPU Instruction Access Bufferable

MPU_ACCESS_BUFFERABLE

MPU_ACCESS_NOT_BUFFERABLE

CORTEX MPU Instruction Access Cacheable

MPU_ACCESS_CACHEABLE

MPU_ACCESS_NOT_CACHEABLE

CORTEX MPU Instruction Access Shareable

MPU_ACCESS_SHAREABLE

MPU_ACCESS_NOT_SHAREABLE

MPU HFNMI and PRIVILEGED Access control

MPU_HFNMI_PRIVDEF_NONE

MPU_HARDFAULT_NMI

MPU_PRIVILEGED_DEFAULT

MPU_HFNMI_PRIVDEF

CORTEX MPU Instruction Access

MPU_INSTRUCTION_ACCESS_ENABLE

MPU_INSTRUCTION_ACCESS_DISABLE

CORTEX MPU Region Enable

MPU_REGION_ENABLE

MPU_REGION_DISABLE

CORTEX MPU Region Number

MPU_REGION_NUMBER0

MPU_REGION_NUMBER1

MPU_REGION_NUMBER2

MPU_REGION_NUMBER3

MPU_REGION_NUMBER4

MPU_REGION_NUMBER5

MPU_REGION_NUMBER6

MPU_REGION_NUMBER7

CORTEX MPU Region Permission Attributes

MPU_REGION_NO_ACCESS

MPU_REGION_PRIV_RW

MPU_REGION_PRIV_RW_URO

MPU_REGION_FULL_ACCESS

MPU_REGION_PRIV_RO

MPU_REGION_PRIV_RO_URO

CORTEX MPU Region Size

MPU_REGION_SIZE_32B

MPU_REGION_SIZE_64B

MPU_REGION_SIZE_128B

MPU_REGION_SIZE_256B

MPU_REGION_SIZE_512B

MPU_REGION_SIZE_1KB

MPU_REGION_SIZE_2KB

MPU_REGION_SIZE_4KB

MPU_REGION_SIZE_8KB

MPU_REGION_SIZE_16KB

MPU_REGION_SIZE_32KB

MPU_REGION_SIZE_64KB

MPU_REGION_SIZE_128KB

MPU_REGION_SIZE_256KB

MPU_REGION_SIZE_512KB

MPU_REGION_SIZE_1MB

MPU_REGION_SIZE_2MB

MPU_REGION_SIZE_4MB

MPU_REGION_SIZE_8MB

MPU_REGION_SIZE_16MB

MPU_REGION_SIZE_32MB

MPU_REGION_SIZE_64MB

MPU_REGION_SIZE_128MB

MPU_REGION_SIZE_256MB

MPU_REGION_SIZE_512MB

MPU_REGION_SIZE_1GB

MPU_REGION_SIZE_2GB

MPU_REGION_SIZE_4GB

MPU TEX Levels

MPU_TEX_LEVEL0

MPU_TEX_LEVEL1

MPU_TEX_LEVEL2

CORTEX Preemption Priority Group

NVIC_PRIORITYGROUP_0 0 bits for pre-emption priority 4 bits for subpriority

NVIC_PRIORITYGROUP_1 1 bits for pre-emption priority 3 bits for subpriority

NVIC_PRIORITYGROUP_2 2 bits for pre-emption priority 2 bits for subpriority

NVIC_PRIORITYGROUP_3 3 bits for pre-emption priority 1 bits for subpriority

NVIC_PRIORITYGROUP_4 4 bits for pre-emption priority 0 bits for subpriority

CORTEX _SysTick clock source

SYSTICK_CLKSOURCE_HCLK_DIV8

SYSTICK_CLKSOURCE_HCLK

9 HAL CRC Generic Driver

9.1 CRC Firmware driver registers structures

9.1.1 CRC_HandleTypeDef

Data Fields

- ***CRC_TypeDef * Instance***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_CRC_StateTypeDef State***

Field Documentation

- ***CRC_TypeDef* CRC_HandleTypeDef::Instance***
  Register base address
- ***HAL_LockTypeDef CRC_HandleTypeDef::Lock***
  CRC locking object
- ***__IO HAL_CRC_StateTypeDef CRC_HandleTypeDef::State***
  CRC communication state

9.2 CRC Firmware driver API description

9.2.1 How to use this driver

The CRC HAL driver can be used as follows:

1. Enable CRC AHB clock using __HAL_RCC_CRC_CLK_ENABLE();
2. Use HAL_CRC_Accumulate() function to compute the CRC value of a 32-bit data buffer using combination of the previous CRC value and the new one.
3. Use HAL_CRC_Calculate() function to compute the CRC Value of a new 32-bit data buffer. This function resets the CRC computation unit before starting the computation to avoid getting wrong CRC values.

9.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the CRC according to the specified parameters in the CRC_InitTypeDef and create the associated handle
- DeInitialize the CRC peripheral
- Initialize the CRC MSP
- DeInitialize CRC MSP

This section contains the following APIs:

- ***HAL_CRC_Init()***
- ***HAL_CRC_DeInit()***
- ***HAL_CRC_MspInit()***
- ***HAL_CRC_MspDeInit()***

9.2.3 Peripheral Control functions

This section provides functions allowing to:

- Compute the 32-bit CRC value of 32-bit data buffer, using combination of the previous CRC value and the new one.
- Compute the 32-bit CRC value of 32-bit data buffer, independently of the previous CRC value.

This section contains the following APIs:

- ***HAL_CRC_Accumulate()***
- ***HAL_CRC_Calculate()***

9.2.4 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_CRC_GetState()***

9.2.5 Detailed description of functions

HAL_CRC_Init

Function Name	**HAL_StatusTypeDef HAL_CRC_Init (CRC_HandleTypeDef * hcrc)**
Function Description	Initializes the CRC according to the specified parameters in the CRC_InitTypeDef and creates the associated handle.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC
Return values	• **HAL:** status

HAL_CRC_DeInit

Function Name	**HAL_StatusTypeDef HAL_CRC_DeInit (CRC_HandleTypeDef * hcrc)**
Function Description	DeInitializes the CRC peripheral.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC
Return values	• **HAL:** status

HAL_CRC_MspInit

Function Name	**void HAL_CRC_MspInit (CRC_HandleTypeDef * hcrc)**
Function Description	Initializes the CRC MSP.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC
Return values	• **None**

HAL_CRC_MspDeInit

Function Name	**void HAL_CRC_MspDeInit (CRC_HandleTypeDef * hcrc)**
Function Description	DeInitializes the CRC MSP.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC
Return values	• **None**

HAL_CRC_Accumulate

Function Name	**uint32_t HAL_CRC_Accumulate (CRC_HandleTypeDef * hcrc, uint32_t pBuffer, uint32_t BufferLength)**
Function Description	Computes the 32-bit CRC of 32-bit data buffer using combination of the previous CRC value and the new one.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC • **pBuffer:** pointer to the buffer containing the data to be computed • **BufferLength:** length of the buffer to be computed
Return values	• **32-bit:** CRC

HAL_CRC_Calculate

Function Name	**uint32_t HAL_CRC_Calculate (CRC_HandleTypeDef * hcrc, uint32_t pBuffer, uint32_t BufferLength)**
Function Description	Computes the 32-bit CRC of 32-bit data buffer independently of the previous CRC value.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC • **pBuffer:** Pointer to the buffer containing the data to be computed • **BufferLength:** Length of the buffer to be computed
Return values	• **32-bit:** CRC

HAL_CRC_GetState

Function Name	**HAL_CRC_StateTypeDef HAL_CRC_GetState (CRC_HandleTypeDef * hcrc)**
Function Description	Returns the CRC state.
Parameters	• **hcrc:** pointer to a CRC_HandleTypeDef structure that contains the configuration information for CRC
Return values	• **HAL:** state

9.3 CRC Firmware driver defines

9.3.1 CRC

CRC Exported Macros

__HAL_CRC_RESET_HANDLE_STATE

Description:

- Resets CRC handle state.

Parameters:

- __HANDLE__: CRC handle

Return value:

- None

__HAL_CRC_DR_RESET

Description:

- Resets CRC Data Register.

Parameters:

- __HANDLE__: CRC handle

Return value:

- None

__HAL_CRC_SET_IDR

Description:

- Stores a 8-bit data in the Independent Data(ID) register.

Parameters:

- __HANDLE__: CRC handle
- __VALUE__: 8-bit value to be stored in the ID register

Return value:

- None

__HAL_CRC_GET_IDR

Description:

- Returns the 8-bit data stored in the Independent Data(ID) register.

Parameters:

- __HANDLE__: CRC handle

Return value:

- 8-bit: value of the ID register

10 HAL CRYP Generic Driver

10.1 CRYP Firmware driver registers structures

10.1.1 CRYP_InitTypeDef

Data Fields

- ***uint32_t DataType***
- ***uint32_t KeySize***
- ***uint8_t * pKey***
- ***uint8_t * pInitVect***
- ***uint8_t IVSize***
- ***uint8_t TagSize***
- ***uint8_t * Header***
- ***uint32_t HeaderSize***
- ***uint8_t * pScratch***

Field Documentation

- ***uint32_t CRYP_InitTypeDef::DataType***
  32-bit data, 16-bit data, 8-bit data or 1-bit string. This parameter can be a value of ***CRYP_Data_Type***
- ***uint32_t CRYP_InitTypeDef::KeySize***
  Used only in AES mode only : 128, 192 or 256 bit key length. This parameter can be a value of ***CRYP_Key_Size***
- ***uint8_t* CRYP_InitTypeDef::pKey***
  The key used for encryption/decryption
- ***uint8_t* CRYP_InitTypeDef::pInitVect***
  The initialization vector used also as initialization counter in CTR mode
- ***uint8_t CRYP_InitTypeDef::IVSize***
  The size of initialization vector. This parameter (called nonce size in CCM) is used only in AES-128/192/256 encryption/decryption CCM mode
- ***uint8_t CRYP_InitTypeDef::TagSize***
  The size of returned authentication TAG. This parameter is used only in AES-128/192/256 encryption/decryption CCM mode
- ***uint8_t* CRYP_InitTypeDef::Header***
  The header used in GCM and CCM modes
- ***uint32_t CRYP_InitTypeDef::HeaderSize***
  The size of header buffer in bytes
- ***uint8_t* CRYP_InitTypeDef::pScratch***
  Scratch buffer used to append the header. It's size must be equal to header size + 21 bytes. This parameter is used only in AES-128/192/256 encryption/decryption CCM mode

10.1.2 CRYP_HandleTypeDef

Data Fields

- ***CRYP_TypeDef * Instance***
- ***CRYP_InitTypeDef Init***
- ***uint8_t * pCrypInBuffPtr***
- ***uint8_t * pCrypOutBuffPtr***
- ***__IO uint16_t CrypInCount***
- ***__IO uint16_t CrypOutCount***
- ***HAL_StatusTypeDef Status***
- ***HAL_PhaseTypeDef Phase***
- ***DMA_HandleTypeDef * hdmain***
- ***DMA_HandleTypeDef * hdmaout***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_CRYP_STATETypeDef State***

Field Documentation

- ***CRYP_TypeDef* CRYP_HandleTypeDef::Instance***
  CRYP registers base address
- ***CRYP_InitTypeDef CRYP_HandleTypeDef::Init***
  CRYP required parameters
- ***uint8_t* CRYP_HandleTypeDef::pCrypInBuffPtr***
  Pointer to CRYP processing (encryption, decryption,...) buffer
- ***uint8_t* CRYP_HandleTypeDef::pCrypOutBuffPtr***
  Pointer to CRYP processing (encryption, decryption,...) buffer
- ***__IO uint16_t CRYP_HandleTypeDef::CrypInCount***
  Counter of inputed data
- ***__IO uint16_t CRYP_HandleTypeDef::CrypOutCount***
  Counter of output data
- ***HAL_StatusTypeDef CRYP_HandleTypeDef::Status***
  CRYP peripheral status
- ***HAL_PhaseTypeDef CRYP_HandleTypeDef::Phase***
  CRYP peripheral phase
- ***DMA_HandleTypeDef* CRYP_HandleTypeDef::hdmain***
  CRYP In DMA handle parameters
- ***DMA_HandleTypeDef* CRYP_HandleTypeDef::hdmaout***
  CRYP Out DMA handle parameters
- ***HAL_LockTypeDef CRYP_HandleTypeDef::Lock***
  CRYP locking object
- ***__IO HAL_CRYP_STATETypeDef CRYP_HandleTypeDef::State***
  CRYP peripheral state

10.2 CRYP Firmware driver API description

10.2.1 How to use this driver

The CRYP HAL driver can be used as follows:

1. Initialize the CRYP low level resources by implementing the HAL_CRYP_MspInit():
   a. Enable the CRYP interface clock using __HAL_RCC_CRYP_CLK_ENABLE()
   b. In case of using interrupts (e.g. HAL_CRYP_AESECB_Encrypt_IT())
      – Configure the CRYP interrupt priority using HAL_NVIC_SetPriority()
      – Enable the CRYP IRQ handler using HAL_NVIC_EnableIRQ()
      – In CRYP IRQ handler, call HAL_CRYP_IRQHandler()
   c. In case of using DMA to control data transfer (e.g. HAL_CRYP_AESECB_Encrypt_DMA())
      – Enable the DMAx interface clock using __DMAx_CLK_ENABLE()
      – Configure and enable two DMA streams one for managing data transfer from memory to peripheral (input stream) and another stream for managing data transfer from peripheral to memory (output stream)
      – Associate the initialized DMA handle to the CRYP DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the two DMA Streams. The output stream should have higher priority than the input stream HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ()
2. Initialize the CRYP HAL using HAL_CRYP_Init(). This function configures mainly:
   a. The data type: 1-bit, 8-bit, 16-bit and 32-bit
   b. The key size: 128, 192 and 256. This parameter is relevant only for AES
   c. The encryption/decryption key. It's size depends on the algorithm used for encryption/decryption
   d. The initialization vector (counter). It is not used ECB mode.
3. Three processing (encryption/decryption) functions are available:
   a. Polling mode: encryption and decryption APIs are blocking functions i.e. they process the data and wait till the processing is finished, e.g. HAL_CRYP_AESCBC_Encrypt()
   b. Interrupt mode: encryption and decryption APIs are not blocking functions i.e. they process the data under interrupt, e.g. HAL_CRYP_AESCBC_Encrypt_IT()
   c. DMA mode: encryption and decryption APIs are not blocking functions i.e. the data transfer is ensured by DMA, e.g. HAL_CRYP_AESCBC_Encrypt_DMA()
4. When the processing function is called at first time after HAL_CRYP_Init() the CRYP peripheral is initialized and processes the buffer in input. At second call, the processing function performs an append of the already processed buffer. When a new data block is to be processed, call HAL_CRYP_Init() then the processing function.
5. Call HAL_CRYP_DeInit() to deinitialize the CRYP peripheral.

10.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the CRYP according to the specified parameters in the CRYP_InitTypeDef and creates the associated handle
- DeInitialize the CRYP peripheral
- Initialize the CRYP MSP
- DeInitialize CRYP MSP

This section contains the following APIs:

- ***HAL_CRYP_Init()***
- ***HAL_CRYP_DeInit()***
- ***HAL_CRYP_MspInit()***
- ***HAL_CRYP_MspDeInit()***

10.2.3 AES processing functions

This section provides functions allowing to:

- Encrypt plaintext using AES-128/192/256 using chaining modes
- Decrypt cyphertext using AES-128/192/256 using chaining modes

Three processing functions are available:

- Polling mode
- Interrupt mode
- DMA mode

This section contains the following APIs:

- ***HAL_CRYP_AESECB_Encrypt()***
- ***HAL_CRYP_AESCBC_Encrypt()***
- ***HAL_CRYP_AESCTR_Encrypt()***
- ***HAL_CRYP_AESECB_Decrypt()***
- ***HAL_CRYP_AESCBC_Decrypt()***
- ***HAL_CRYP_AESCTR_Decrypt()***
- ***HAL_CRYP_AESECB_Encrypt_IT()***
- ***HAL_CRYP_AESCBC_Encrypt_IT()***
- ***HAL_CRYP_AESCTR_Encrypt_IT()***
- ***HAL_CRYP_AESECB_Decrypt_IT()***
- ***HAL_CRYP_AESCBC_Decrypt_IT()***
- ***HAL_CRYP_AESCTR_Decrypt_IT()***
- ***HAL_CRYP_AESECB_Encrypt_DMA()***
- ***HAL_CRYP_AESCBC_Encrypt_DMA()***
- ***HAL_CRYP_AESCTR_Encrypt_DMA()***
- ***HAL_CRYP_AESECB_Decrypt_DMA()***
- ***HAL_CRYP_AESCBC_Decrypt_DMA()***
- ***HAL_CRYP_AESCTR_Decrypt_DMA()***

10.2.4 DES processing functions

This section provides functions allowing to:

- Encrypt plaintext using DES using ECB or CBC chaining modes
- Decrypt cyphertext using ECB or CBC chaining modes

Three processing functions are available:

- Polling mode
- Interrupt mode
- DMA mode

This section contains the following APIs:

- ***HAL_CRYP_DESECB_Encrypt()***
- ***HAL_CRYP_DESECB_Decrypt()***
- ***HAL_CRYP_DESCBC_Encrypt()***
- ***HAL_CRYP_DESCBC_Decrypt()***
- ***HAL_CRYP_DESECB_Encrypt_IT()***

- ***HAL_CRYP_DESCBC_Encrypt_IT()***
- ***HAL_CRYP_DESECB_Decrypt_IT()***
- ***HAL_CRYP_DESCBC_Decrypt_IT()***
- ***HAL_CRYP_DESECB_Encrypt_DMA()***
- ***HAL_CRYP_DESCBC_Encrypt_DMA()***
- ***HAL_CRYP_DESECB_Decrypt_DMA()***
- ***HAL_CRYP_DESCBC_Decrypt_DMA()***

10.2.5 TDES processing functions

This section provides functions allowing to:

- Encrypt plaintext using TDES based on ECB or CBC chaining modes
- Decrypt cyphertext using TDES based on ECB or CBC chaining modes

Three processing functions are available:

- Polling mode
- Interrupt mode
- DMA mode

This section contains the following APIs:

- ***HAL_CRYP_TDESECB_Encrypt()***
- ***HAL_CRYP_TDESECB_Decrypt()***
- ***HAL_CRYP_TDESCBC_Encrypt()***
- ***HAL_CRYP_TDESCBC_Decrypt()***
- ***HAL_CRYP_TDESECB_Encrypt_IT()***
- ***HAL_CRYP_TDESCBC_Encrypt_IT()***
- ***HAL_CRYP_TDESECB_Decrypt_IT()***
- ***HAL_CRYP_TDESCBC_Decrypt_IT()***
- ***HAL_CRYP_TDESECB_Encrypt_DMA()***
- ***HAL_CRYP_TDESCBC_Encrypt_DMA()***
- ***HAL_CRYP_TDESECB_Decrypt_DMA()***
- ***HAL_CRYP_TDESCBC_Decrypt_DMA()***

10.2.6 DMA callback functions

This section provides DMA callback functions:

- DMA Input data transfer complete
- DMA Output data transfer complete
- DMA error

This section contains the following APIs:

- ***HAL_CRYP_InCpltCallback()***
- ***HAL_CRYP_OutCpltCallback()***
- ***HAL_CRYP_ErrorCallback()***

10.2.7 CRYP IRQ handler management

This section provides CRYP IRQ handler function.

This section contains the following APIs:

- ***HAL_CRYP_IRQHandler()***

10.2.8 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral.

This section contains the following APIs:

- ***HAL_CRYP_GetState()***

10.2.9 Detailed description of functions

HAL_CRYP_Init

Function Name	**HAL_StatusTypeDef HAL_CRYP_Init (CRYP_HandleTypeDef * hcryp)**
Function Description	Initializes the CRYP according to the specified parameters in the CRYP_InitTypeDef and creates the associated handle.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
Return values	• **HAL:** status

HAL_CRYP_DeInit

Function Name	**HAL_StatusTypeDef HAL_CRYP_DeInit (CRYP_HandleTypeDef * hcryp)**
Function Description	DeInitializes the CRYP peripheral.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
Return values	• **HAL:** status

HAL_CRYP_MspInit

Function Name	**void HAL_CRYP_MspInit (CRYP_HandleTypeDef * hcryp)**
Function Description	Initializes the CRYP MSP.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
Return values	• **None**

HAL_CRYP_MspDeInit

Function Name	**void HAL_CRYP_MspDeInit (CRYP_HandleTypeDef * hcryp)**
Function Description	DeInitializes CRYP MSP.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
Return values	• **None**

HAL_CRYP_AESECB_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESECB_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES ECB encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESECB_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESECB_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES ECB decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pPlainData:** Pointer to the plaintext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESCBC_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCBC_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES CBC encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESCBC_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCBC_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES ECB decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pPlainData:** Pointer to the plaintext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESCTR_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCTR_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES CTR encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESCTR_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCTR_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES CTR decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pPlainData:** Pointer to the plaintext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_AESECB_Encrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYP_AESECB_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**
Function Description	Initializes the CRYP peripheral in AES ECB encryption mode using Interrupt.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes • **pCypherData:** Pointer to the cyphertext buffer
Return values	• **HAL:** status

HAL_CRYP_AESCBC_Encrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYP_AESCBC_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**
Function Description	Initializes the CRYP peripheral in AES CBC encryption mode using Interrupt.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes • **pCypherData:** Pointer to the cyphertext buffer
Return values	• **HAL:** status

HAL_CRYP_AESCTR_Encrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYP_AESCTR_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**
Function Description	Initializes the CRYP peripheral in AES CTR encryption mode using Interrupt.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes • **pCypherData:** Pointer to the cyphertext buffer
Return values	• **HAL:** status

HAL_CRYP_AESECB_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_AESECB_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES ECB decryption mode using Interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCTR_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCTR_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES CTR decryption mode using Interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCBC_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCBC_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES CBC decryption mode using IT.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_AESECB_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESECB_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES ECB encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_AESECB_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESECB_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES ECB decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCBC_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCBC_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES CBC encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCBC_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCBC_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES CBC encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16 bytes

- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCTR_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCTR_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES CTR encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16.
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_AESCTR_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_AESCTR_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES CTR decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_DESECB_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_DESECB_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in DES ECB encryption mode.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_DESCBC_Encrypt

Function Name	**HAL_StatusTypeDef HAL_CRYP_DESCBC_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**
Function Description	Initializes the CRYP peripheral in DES CBC encryption mode.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 8 • **pCypherData:** Pointer to the cyphertext buffer • **Timeout:** Specify Timeout value
Return values	• **HAL:** status

HAL_CRYP_DESECB_Decrypt

Function Name	**HAL_StatusTypeDef HAL_CRYP_DESECB_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**
Function Description	Initializes the CRYP peripheral in DES ECB decryption mode.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pCypherData:** Pointer to the cyphertext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 8 • **pPlainData:** Pointer to the plaintext buffer • **Timeout:** Specify Timeout value
Return values	• **HAL:** status

HAL_CRYP_DESCBC_Decrypt

Function Name	**HAL_StatusTypeDef HAL_CRYP_DESCBC_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**
Function Description	Initializes the CRYP peripheral in DES ECB decryption mode.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pCypherData:** Pointer to the cyphertext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 8 • **pPlainData:** Pointer to the plaintext buffer • **Timeout:** Specify Timeout value
Return values	• **HAL:** status

HAL_CRYP_DESECB_Encrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYP_DESECB_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**
Function Description	Initializes the CRYP peripheral in DES ECB encryption mode using IT.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESECB_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_DESECB_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in DES ECB decryption mode using IT.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESCBC_Encrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_DESCBC_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in DES CBC encryption mode using interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESCBC_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_DESCBC_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in DES ECB decryption mode using interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESECB_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_DESECB_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in DES ECB encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESECB_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_DESECB_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in DES ECB decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESCBC_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_DESCBC_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in DES CBC encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_DESCBC_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_DESCBC_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in DES ECB decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in TDES ECB encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in TDES CBC encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in TDES ECB decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in TDES CBC decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pPlainData:** Pointer to the plaintext buffer
- **Timeout:** Specify Timeout value

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Encrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in TDES ECB encryption mode using interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in TDES ECB decryption mode.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Encrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in TDES CBC encryption mode.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Decrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in TDES CBC decryption mode.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in TDES ECB encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESECB_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESECB_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in TDES ECB decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in TDES CBC encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYP_TDESCBC_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYP_TDESCBC_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in TDES CBC decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 8
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYP_InCpltCallback

Function Name **void HAL_CRYP_InCpltCallback (CRYP_HandleTypeDef * hcryp)**

Function Description Input FIFO transfer completed callbacks.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **None**

HAL_CRYP_OutCpltCallback

Function Name **void HAL_CRYP_OutCpltCallback (CRYP_HandleTypeDef * hcryp)**

Function Description Output FIFO transfer completed callbacks.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **None**

HAL_CRYP_ErrorCallback

Function Name **void HAL_CRYP_ErrorCallback (CRYP_HandleTypeDef * hcryp)**

Function Description CRYP error callbacks.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **None**

HAL_CRYP_IRQHandler

Function Name **void HAL_CRYP_IRQHandler (CRYP_HandleTypeDef * hcryp)**

Function Description This function handles CRYP interrupt request.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **None**

HAL_CRYP_GetState

Function Name **HAL_CRYP_STATETypeDef HAL_CRYP_GetState (CRYP_HandleTypeDef * hcryp)**

Function Description Returns the CRYP state.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **HAL:** state

10.3 CRYP Firmware driver defines

10.3.1 CRYP

CRYP Data Type

CRYP_DATATYPE_32B

CRYP_DATATYPE_16B

CRYP_DATATYPE_8B

CRYP_DATATYPE_1B

CRYP CRYP_AlgoModeDirection

CRYP_CR_ALGOMODE_DIRECTION

CRYP_CR_ALGOMODE_TDES_ECB_ENCRYPT

CRYP_CR_ALGOMODE_TDES_ECB_DECRYPT

CRYP_CR_ALGOMODE_TDES_CBC_ENCRYPT

CRYP_CR_ALGOMODE_TDES_CBC_DECRYPT

CRYP_CR_ALGOMODE_DES_ECB_ENCRYPT

CRYP_CR_ALGOMODE_DES_ECB_DECRYPT

CRYP_CR_ALGOMODE_DES_CBC_ENCRYPT

CRYP_CR_ALGOMODE_DES_CBC_DECRYPT

CRYP_CR_ALGOMODE_AES_ECB_ENCRYPT

CRYP_CR_ALGOMODE_AES_ECB_DECRYPT

CRYP_CR_ALGOMODE_AES_CBC_ENCRYPT

CRYP_CR_ALGOMODE_AES_CBC_DECRYPT

CRYP_CR_ALGOMODE_AES_CTR_ENCRYPT

CRYP_CR_ALGOMODE_AES_CTR_DECRYPT

CRYP CRYP_Interrupt

CRYP_IT_INI	Input FIFO Interrupt
CRYP_IT_OUTI	Output FIFO Interrupt

CRYP CRYP_Flags

CRYP_FLAG_BUSY	The CRYP core is currently processing a block of data or a key preparation (for AES decryption).
CRYP_FLAG_IFEM	Input FIFO is empty
CRYP_FLAG_IFNF	Input FIFO is not Full
CRYP_FLAG_OFNE	Output FIFO is not empty
CRYP_FLAG_OFFU	Output FIFO is Full
CRYP_FLAG_OUTRIS	Output FIFO service raw interrupt status
CRYP_FLAG_INRIS	Input FIFO service raw interrupt status

CRYP Exported Macros

__HAL_CRYP_RESET_HANDLE_STATE

Description:

- Reset CRYP handle state.

Parameters:

- __HANDLE__: specifies the CRYP handle.

Return value:

- None

__HAL_CRYP_ENABLE

Description:

- Enable/Disable the CRYP peripheral.

Parameters:

- __HANDLE__: specifies the CRYP handle.

Return value:

- None

__HAL_CRYP_DISABLE

__HAL_CRYP_FIFO_FLUSH

Description:

- Flush the data FIFO.

Parameters:

- __HANDLE__: specifies the CRYP handle.

Return value:

- None

__HAL_CRYP_SET_MODE

Description:

- Set the algorithm mode: AES-ECB, AES-CBC, AES-CTR, DES-ECB, DES-CBC.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- MODE: The algorithm mode.

Return value:

- None

__HAL_CRYP_GET_FLAG

Description:

- Check whether the specified CRYP flag is set or not.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - CRYP_FLAG_BUSY: The CRYP core is currently processing a block of data or a key preparation (for AES decryption).
  - CRYP_FLAG_IFEM: Input FIFO is empty
  - CRYP_FLAG_IFNF: Input FIFO is not full
  - CRYP_FLAG_INRIS: Input FIFO service raw interrupt is pending
  - CRYP_FLAG_OFNE: Output FIFO is not empty
  - CRYP_FLAG_OFFU: Output FIFO is full
  - CRYP_FLAG_OUTRIS: Input FIFO service raw interrupt is pending

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_CRYP_GET_IT

Description:

- Check whether the specified CRYP interrupt is set or not.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- __INTERRUPT__: specifies the interrupt to check. This parameter can be one of the following values:
  - CRYP_IT_INRIS: Input FIFO service raw interrupt is pending
  - CRYP_IT_OUTRIS: Output FIFO service raw interrupt is pending

Return value:

- The: new state of __INTERRUPT__ (TRUE or FALSE).

__HAL_CRYP_ENABLE_IT

Description:

- Enable the CRYP interrupt.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- __INTERRUPT__: CRYP Interrupt.

Return value:

- None

__HAL_CRYP_DISABLE_IT

Description:

- Disable the CRYP interrupt.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- __INTERRUPT__: CRYP interrupt.

Return value:

- None

CRYP Key Size

CRYP_KEYSIZE_128B

CRYP_KEYSIZE_192B

CRYP_KEYSIZE_256B

11 HAL CRYP Extension Driver

11.1 CRYPEx Firmware driver API description

11.1.1 How to use this driver

The CRYP Extension HAL driver can be used as follows:

1. Initialize the CRYP low level resources by implementing the HAL_CRYP_MspInit():
   a. Enable the CRYP interface clock using __HAL_RCC_CRYP_CLK_ENABLE()
   b. In case of using interrupts (e.g. HAL_CRYPEx_AESGCM_Encrypt_IT())
      – Configure the CRYP interrupt priority using HAL_NVIC_SetPriority()
      – Enable the CRYP IRQ handler using HAL_NVIC_EnableIRQ()
      – In CRYP IRQ handler, call HAL_CRYP_IRQHandler()
   c. In case of using DMA to control data transfer (e.g. HAL_AES_ECB_Encrypt_DMA())
      – Enable the DMAx interface clock using __DMAx_CLK_ENABLE()
      – Configure and enable two DMA streams one for managing data transfer from memory to peripheral (input stream) and another stream for managing data transfer from peripheral to memory (output stream)
      – Associate the initialized DMA handle to the CRYP DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the two DMA Streams. The output stream should have higher priority than the input stream HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ()
2. Initialize the CRYP HAL using HAL_CRYP_Init(). This function configures mainly:
   a. The data type: 1-bit, 8-bit, 16-bit and 32-bit
   b. The key size: 128, 192 and 256. This parameter is relevant only for AES
   c. The encryption/decryption key. Its size depends on the algorithm used for encryption/decryption
   d. The initialization vector (counter). It is not used ECB mode.
3. Three processing (encryption/decryption) functions are available:
   a. Polling mode: encryption and decryption APIs are blocking functions i.e. they process the data and wait till the processing is finished e.g. HAL_CRYPEx_AESGCM_Encrypt()
   b. Interrupt mode: encryption and decryption APIs are not blocking functions i.e. they process the data under interrupt e.g. HAL_CRYPEx_AESGCM_Encrypt_IT()
   c. DMA mode: encryption and decryption APIs are not blocking functions i.e. the data transfer is ensured by DMA e.g. HAL_CRYPEx_AESGCM_Encrypt_DMA()
4. When the processing function is called at first time after HAL_CRYP_Init() the CRYP peripheral is initialized and processes the buffer in input. At second call, the processing function performs an append of the already processed buffer. When a new data block is to be processed, call HAL_CRYP_Init() then the processing function.
5. In AES-GCM and AES-CCM modes are an authenticated encryption algorithms which provide authentication messages. HAL_AES_GCM_Finish() and HAL_AES_CCM_Finish() are used to provide those authentication messages. Call those functions after the processing ones (polling, interrupt or DMA). e.g. in AES-CCM mode call HAL_CRYPEx_AESCCM_Encrypt() to encrypt the plain data then call HAL_CRYPEx_AESCCM_Finish() to get the authentication message For CCM Encrypt/Decrypt API's, only DataType = 8-bit is supported by this version. The HAL_CRYPEx_AESGCM_xxxx() implementation is limited to 32bits inputs data length (Plain/Cyphertext, Header) compared with GCM standards specifications (800-38D).

6. Call HAL_CRYP_DeInit() to deinitialize the CRYP peripheral.

11.1.2 Extended AES processing functions

This section provides functions allowing to:

- Encrypt plaintext using AES-128/192/256 using GCM and CCM chaining modes
- Decrypt cyphertext using AES-128/192/256 using GCM and CCM chaining modes
- Finish the processing. This function is available only for GCM and CCM

Three processing methods are available:

- Polling mode
- Interrupt mode
- DMA mode

This section contains the following APIs:

- ***HAL_CRYPEx_AESCCM_Encrypt()***
- ***HAL_CRYPEx_AESGCM_Encrypt()***
- ***HAL_CRYPEx_AESGCM_Decrypt()***
- ***HAL_CRYPEx_AESGCM_Finish()***
- ***HAL_CRYPEx_AESCCM_Finish()***
- ***HAL_CRYPEx_AESCCM_Decrypt()***
- ***HAL_CRYPEx_AESGCM_Encrypt_IT()***
- ***HAL_CRYPEx_AESCCM_Encrypt_IT()***
- ***HAL_CRYPEx_AESGCM_Decrypt_IT()***
- ***HAL_CRYPEx_AESCCM_Decrypt_IT()***
- ***HAL_CRYPEx_AESGCM_Encrypt_DMA()***
- ***HAL_CRYPEx_AESCCM_Encrypt_DMA()***
- ***HAL_CRYPEx_AESGCM_Decrypt_DMA()***
- ***HAL_CRYPEx_AESCCM_Decrypt_DMA()***

11.1.3 CRYPEx IRQ handler management

This section provides CRYPEx IRQ handler function.

This section contains the following APIs:

- ***HAL_CRYPEx_GCMCCM_IRQHandler()***

11.1.4 Detailed description of functions

HAL_CRYPEx_AESGCM_Encrypt

Function Name	**HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**
Function Description	Initializes the CRYP peripheral in AES GCM encryption mode then encrypt pPlainData.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 • **pCypherData:** Pointer to the cyphertext buffer • **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_CRYPEx_AESGCM_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES GCM decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the cyphertext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_CRYPEx_AESGCM_Finish

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Finish (CRYP_HandleTypeDef * hcryp, uint32_t Size, uint8_t * AuthTag, uint32_t Timeout)**

Function Description Computes the authentication TAG.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **Size:** Total length of the plain/cyphertext buffer
- **AuthTag:** Pointer to the authentication buffer
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_CRYPEx_AESCCM_Encrypt

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Encrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES CCM encryption mode then encrypt pPlainData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_CRYPEx_AESCCM_Decrypt

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Decrypt (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData, uint32_t Timeout)**

Function Description Initializes the CRYP peripheral in AES CCM decryption mode then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pCypherData:** Pointer to the cyphertext buffer
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_CRYPEx_AESCCM_Finish

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Finish (CRYP_HandleTypeDef * hcryp, uint8_t * AuthTag, uint32_t Timeout)**

Function Description Computes the authentication TAG for AES CCM mode.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **AuthTag:** Pointer to the authentication buffer
- **Timeout:** Timeout duration

Return values
- **HAL:** status

Notes
- This API is called after HAL_AES_CCM_Encrypt()/HAL_AES_CCM_Decrypt()

HAL_CRYPEx_AESGCM_Encrypt_IT

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES GCM encryption mode using IT.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYPEx_AESGCM_Decrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**
Function Description	Initializes the CRYP peripheral in AES GCM decryption mode using IT.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pCypherData:** Pointer to the cyphertext buffer • **Size:** Length of the cyphertext buffer, must be a multiple of 16 • **pPlainData:** Pointer to the plaintext buffer
Return values	• **HAL:** status

HAL_CRYPEx_AESCCM_Encrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Encrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**
Function Description	Initializes the CRYP peripheral in AES CCM encryption mode using interrupt.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pPlainData:** Pointer to the plaintext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 • **pCypherData:** Pointer to the cyphertext buffer
Return values	• **HAL:** status

HAL_CRYPEx_AESCCM_Decrypt_IT

Function Name	**HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Decrypt_IT (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**
Function Description	Initializes the CRYP peripheral in AES CCM decryption mode using interrupt then decrypted pCypherData.
Parameters	• **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module • **pCypherData:** Pointer to the cyphertext buffer • **Size:** Length of the plaintext buffer, must be a multiple of 16 • **pPlainData:** Pointer to the plaintext buffer
Return values	• **HAL:** status

HAL_CRYPEx_AESGCM_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES GCM encryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYPEx_AESGCM_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESGCM_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES GCM decryption mode using DMA.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer.
- **Size:** Length of the cyphertext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYPEx_AESCCM_Encrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Encrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pPlainData, uint16_t Size, uint8_t * pCypherData)**

Function Description Initializes the CRYP peripheral in AES CCM encryption mode using interrupt.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pPlainData:** Pointer to the plaintext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pCypherData:** Pointer to the cyphertext buffer

Return values
- **HAL:** status

HAL_CRYPEx_AESCCM_Decrypt_DMA

Function Name **HAL_StatusTypeDef HAL_CRYPEx_AESCCM_Decrypt_DMA (CRYP_HandleTypeDef * hcryp, uint8_t * pCypherData, uint16_t Size, uint8_t * pPlainData)**

Function Description Initializes the CRYP peripheral in AES CCM decryption mode

using DMA then decrypted pCypherData.

Parameters
- **hcryp:** pointer to a CRYP_HandleTypeDef structure that contains the configuration information for CRYP module
- **pCypherData:** Pointer to the cyphertext buffer
- **Size:** Length of the plaintext buffer, must be a multiple of 16
- **pPlainData:** Pointer to the plaintext buffer

Return values
- **HAL:** status

HAL_CRYPEx_GCMCCM_IRQHandler

Function Name **void HAL_CRYPEx_GCMCCM_IRQHandler (CRYP_HandleTypeDef * hcryp)**

Function Description This function handles CRYPEx interrupt request.

Parameters
- **hcryp:** pointer to a CRYPEx_HandleTypeDef structure that contains the configuration information for CRYP module

Return values
- **None**

11.2 CRYPEx Firmware driver defines

11.2.1 CRYPEx

CRYP AlgoModeDirection

CRYP_CR_ALGOMODE_AES_GCM_ENCRYPT

CRYP_CR_ALGOMODE_AES_GCM_DECRYPT

CRYP_CR_ALGOMODE_AES_CCM_ENCRYPT

CRYP_CR_ALGOMODE_AES_CCM_DECRYPT

CRYP PhaseConfig

CRYP_PHASE_INIT

CRYP_PHASE_HEADER

CRYP_PHASE_PAYLOAD

CRYP_PHASE_FINAL

CRYP Exported Macros

__HAL_CRYP_SET_PHASE **Description:**

- Set the phase: Init, header, payload, final.

Parameters:

- __HANDLE__: specifies the CRYP handle.
- __PHASE__: The phase.

Return value:

- None

12 HAL DAC Generic Driver

12.1 DAC Firmware driver registers structures

12.1.1 DAC_HandleTypeDef

Data Fields

- ***DAC_TypeDef * Instance***
- ***__IO HAL_DAC_StateTypeDef State***
- ***HAL_LockTypeDef Lock***
- ***DMA_HandleTypeDef * DMA_Handle1***
- ***DMA_HandleTypeDef * DMA_Handle2***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***DAC_TypeDef* DAC_HandleTypeDef::Instance***
  Register base address
- ***__IO HAL_DAC_StateTypeDef DAC_HandleTypeDef::State***
  DAC communication state
- ***HAL_LockTypeDef DAC_HandleTypeDef::Lock***
  DAC locking object
- ***DMA_HandleTypeDef* DAC_HandleTypeDef::DMA_Handle1***
  Pointer DMA handler for channel 1
- ***DMA_HandleTypeDef* DAC_HandleTypeDef::DMA_Handle2***
  Pointer DMA handler for channel 2
- ***__IO uint32_t DAC_HandleTypeDef::ErrorCode***
  DAC Error code

12.1.2 DAC_ChannelConfTypeDef

Data Fields

- ***uint32_t DAC_Trigger***
- ***uint32_t DAC_OutputBuffer***

Field Documentation

- ***uint32_t DAC_ChannelConfTypeDef::DAC_Trigger***
  Specifies the external trigger for the selected DAC channel. This parameter can be a value of ***DAC_trigger_selection***
- ***uint32_t DAC_ChannelConfTypeDef::DAC_OutputBuffer***
  Specifies whether the DAC channel output buffer is enabled or disabled. This parameter can be a value of ***DAC_output_buffer***

12.2 DAC Firmware driver API description

12.2.1 DAC Peripheral features

DAC Channels

The device integrates two 12-bit Digital Analog Converters that can be used independently or simultaneously (dual mode):

1. DAC channel1 with DAC_OUT1 (PA4) as output
2. DAC channel2 with DAC_OUT2 (PA5) as output

DAC Triggers

Digital to Analog conversion can be non-triggered using DAC_TRIGGER_NONE and DAC_OUT1/DAC_OUT2 is available once writing to DHRx register.

Digital to Analog conversion can be triggered by:

1. External event: EXTI Line 9 (any GPIOx_Pin9) using DAC_TRIGGER_EXT_IT9. The used pin (GPIOx_Pin9) must be configured in input mode.
2. Timers TRGO: TIM2, TIM4, TIM5, TIM6, TIM7 and TIM8 (DAC_TRIGGER_T2_TRGO, DAC_TRIGGER_T4_TRGO...)
3. Software using DAC_TRIGGER_SOFTWARE

DAC Buffer mode feature

Each DAC channel integrates an output buffer that can be used to reduce the output impedance, and to drive external loads directly without having to add an external operational amplifier. To enable, the output buffer use sConfig.DAC_OutputBuffer = DAC_OUTPUTBUFFER_ENABLE;

Refer to the device datasheet for more details about output impedance value with and without output buffer.

DAC wave generation feature

Both DAC channels can be used to generate

1. Noise wave
2. Triangle wave

DAC data format

The DAC data format can be:

1. 8-bit right alignment using DAC_ALIGN_8B_R
2. 12-bit left alignment using DAC_ALIGN_12B_L
3. 12-bit right alignment using DAC_ALIGN_12B_R

DAC data value to voltage correspondence

The analog output voltage on each DAC channel pin is determined by the following equation: DAC_OUTx = VREF+ * DOR / 4095 with DOR is the Data Output Register VEF+ is the input voltage reference (refer to the device datasheet) e.g. To set DAC_OUT1 to 0.7V, use Assuming that VREF+ = 3.3V, DAC_OUT1 = (3.3 * 868) / 4095 = 0.7V

DMA requests

A DMA1 request can be generated when an external trigger (but not a software trigger) occurs if DMA1 requests are enabled using HAL_DAC_Start_DMA()

DMA1 requests are mapped as following:

1. DAC channel1 : mapped on DMA1 Stream5 channel7 which must be already configured
2. DAC channel2 : mapped on DMA1 Stream6 channel7 which must be already configured For Dual mode and specific signal (Triangle and noise) generation please refer to Extension Features Driver description

12.2.2 How to use this driver

- DAC APB clock must be enabled to get write access to DAC registers using HAL_DAC_Init()
- Configure DAC_OUTx (DAC_OUT1: PA4, DAC_OUT2: PA5) in analog mode.
- Configure the DAC channel using HAL_DAC_ConfigChannel() function.
- Enable the DAC channel using HAL_DAC_Start() or HAL_DAC_Start_DMA functions

Polling mode IO operation

- Start the DAC peripheral using HAL_DAC_Start()
- To read the DAC last data output value, use the HAL_DAC_GetValue() function.
- Stop the DAC peripheral using HAL_DAC_Stop()

DMA mode IO operation

- Start the DAC peripheral using HAL_DAC_Start_DMA(), at this stage the user specify the length of data to be transferred at each end of conversion
- At The end of data transfer HAL_DAC_ConvCpltCallbackCh1()or HAL_DAC_ConvCpltCallbackCh2() function is executed and user can add his own code by customization of function pointer HAL_DAC_ConvCpltCallbackCh1 or HAL_DAC_ConvCpltCallbackCh2
- In case of transfer Error, HAL_DAC_ErrorCallbackCh1() function is executed and user can add his own code by customization of function pointer HAL_DAC_ErrorCallbackCh1
- Stop the DAC peripheral using HAL_DAC_Stop_DMA()

DAC HAL driver macros list

Below the list of most used macros in DAC HAL driver.

- __HAL_DAC_ENABLE : Enable the DAC peripheral
- __HAL_DAC_DISABLE : Disable the DAC peripheral
- __HAL_DAC_CLEAR_FLAG: Clear the DAC's pending flags

- __HAL_DAC_GET_FLAG: Get the selected DAC's flag status

You can refer to the DAC HAL driver header file for more useful macros

12.2.3 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize and configure the DAC.
- De-initialize the DAC.

This section contains the following APIs:

- ***HAL_DAC_Init()***
- ***HAL_DAC_DeInit()***
- ***HAL_DAC_MspInit()***
- ***HAL_DAC_MspDeInit()***

12.2.4 IO operation functions

This section provides functions allowing to:

- Start conversion.
- Stop conversion.
- Start conversion and enable DMA transfer.
- Stop conversion and disable DMA transfer.
- Get result of conversion.

This section contains the following APIs:

- ***HAL_DAC_Start()***
- ***HAL_DAC_Stop()***
- ***HAL_DAC_Start_DMA()***
- ***HAL_DAC_Stop_DMA()***
- ***HAL_DAC_GetValue()***
- ***HAL_DAC_IRQHandler()***
- ***HAL_DAC_ConvCpltCallbackCh1()***
- ***HAL_DAC_ConvHalfCpltCallbackCh1()***
- ***HAL_DAC_ErrorCallbackCh1()***
- ***HAL_DAC_DMAUnderrunCallbackCh1()***

12.2.5 Peripheral Control functions

This section provides functions allowing to:

- Configure channels.
- Set the specified data holding register value for DAC channel.

This section contains the following APIs:

- ***HAL_DAC_ConfigChannel()***
- ***HAL_DAC_SetValue()***

12.2.6 Peripheral State and Errors functions

This subsection provides functions allowing to

- Check the DAC state.
- Check the DAC Errors.

This section contains the following APIs:

- ***HAL_DAC_GetState()***
- ***HAL_DAC_GetError()***
- ***HAL_DAC_IRQHandler()***
- ***HAL_DAC_ConvCpltCallbackCh1()***
- ***HAL_DAC_ConvHalfCpltCallbackCh1()***
- ***HAL_DAC_ErrorCallbackCh1()***
- ***HAL_DAC_DMAUnderrunCallbackCh1()***

12.2.7 Detailed description of functions

HAL_DAC_Init

Function Name	**HAL_StatusTypeDef HAL_DAC_Init (DAC_HandleTypeDef * hdac)**
Function Description	Initializes the DAC peripheral according to the specified parameters in the DAC_InitStruct.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **HAL:** status

HAL_DAC_DeInit

Function Name	**HAL_StatusTypeDef HAL_DAC_DeInit (DAC_HandleTypeDef * hdac)**
Function Description	Deinitializes the DAC peripheral registers to their default reset values.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **HAL:** status

HAL_DAC_MspInit

Function Name	**void HAL_DAC_MspInit (DAC_HandleTypeDef * hdac)**
Function Description	Initializes the DAC MSP.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DAC_MspDeInit

Function Name **void HAL_DAC_MspDeInit (DAC_HandleTypeDef * hdac)**

Function Description DeInitializes the DAC MSP.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.

Return values
- **None**

HAL_DAC_Start

Function Name **HAL_StatusTypeDef HAL_DAC_Start (DAC_HandleTypeDef * hdac, uint32_t Channel)**

Function Description Enables DAC and starts conversion of channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected

Return values
- **HAL:** status

HAL_DAC_Stop

Function Name **HAL_StatusTypeDef HAL_DAC_Stop (DAC_HandleTypeDef * hdac, uint32_t Channel)**

Function Description Disables DAC and stop conversion of channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected

Return values
- **HAL:** status

HAL_DAC_Start_DMA

Function Name **HAL_StatusTypeDef HAL_DAC_Start_DMA (DAC_HandleTypeDef * hdac, uint32_t Channel, uint32_t * pData, uint32_t Length, uint32_t Alignment)**

Function Description Enables DAC and starts conversion of channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected
- **pData:** The destination peripheral Buffer address.
- **Length:** The length of data to be transferred from memory to DAC peripheral

- **Alignment:** Specifies the data alignment for DAC channel. This parameter can be one of the following values:
  - DAC_ALIGN_8B_R: 8bit right data alignment selected
  - DAC_ALIGN_12B_L: 12bit left data alignment selected
  - DAC_ALIGN_12B_R: 12bit right data alignment selected

Return values
- **HAL:** status

HAL_DAC_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_DAC_Stop_DMA (DAC_HandleTypeDef * hdac, uint32_t Channel)**

Function Description Disables DAC and stop conversion of channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected

Return values
- **HAL:** status

HAL_DAC_GetValue

Function Name **uint32_t HAL_DAC_GetValue (DAC_HandleTypeDef * hdac, uint32_t Channel)**

Function Description Returns the last data output value of the selected DAC channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected

Return values
- **The:** selected DAC channel data output value.

HAL_DAC_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_DAC_ConfigChannel (DAC_HandleTypeDef * hdac, DAC_ChannelConfTypeDef * sConfig, uint32_t Channel)**

Function Description Configures the selected DAC channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **sConfig:** DAC configuration structure.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected

Return values
- **HAL:** status

HAL_DAC_SetValue

Function Name **HAL_StatusTypeDef HAL_DAC_SetValue (DAC_HandleTypeDef * hdac, uint32_t Channel, uint32_t Alignment, uint32_t Data)**

Function Description Set the specified data holding register value for DAC channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values:
  - DAC_CHANNEL_1: DAC Channel1 selected
  - DAC_CHANNEL_2: DAC Channel2 selected
- **Alignment:** Specifies the data alignment. This parameter can be one of the following values:
  - DAC_ALIGN_8B_R: 8bit right data alignment selected
  - DAC_ALIGN_12B_L: 12bit left data alignment selected
  - DAC_ALIGN_12B_R: 12bit right data alignment selected
- **Data:** Data to be loaded in the selected data holding register.

Return values
- **HAL:** status

HAL_DAC_GetState

Function Name **HAL_DAC_StateTypeDef HAL_DAC_GetState (DAC_HandleTypeDef * hdac)**

Function Description return the DAC state

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.

Return values
- **HAL:** state

HAL_DAC_IRQHandler

Function Name **void HAL_DAC_IRQHandler (DAC_HandleTypeDef * hdac)**

Function Description Handles DAC interrupt request.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.

Return values
- **None**

HAL_DAC_GetError

Function Name **uint32_t HAL_DAC_GetError (DAC_HandleTypeDef * hdac)**

Function Description Return the DAC error code.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.

Return values
- **DAC:** Error Code

HAL_DAC_ConvCpltCallbackCh1

Function Name	**void HAL_DAC_ConvCpltCallbackCh1 (DAC_HandleTypeDef * hdac)**
Function Description	Conversion complete callback in non blocking mode for Channel1.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DAC_ConvHalfCpltCallbackCh1

Function Name	**void HAL_DAC_ConvHalfCpltCallbackCh1 (DAC_HandleTypeDef * hdac)**
Function Description	Conversion half DMA transfer callback in non blocking mode for Channel1.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DAC_ErrorCallbackCh1

Function Name	**void HAL_DAC_ErrorCallbackCh1 (DAC_HandleTypeDef * hdac)**
Function Description	Error DAC callback for Channel1.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DAC_DMAUnderrunCallbackCh1

Function Name	**void HAL_DAC_DMAUnderrunCallbackCh1 (DAC_HandleTypeDef * hdac)**
Function Description	DMA underrun DAC callback for channel1.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

12.3 DAC Firmware driver defines

12.3.1 DAC

DAC Channel Selection

DAC_CHANNEL_1

DAC_CHANNEL_2

DAC Data Alignment

DAC_ALIGN_12B_R

DAC_ALIGN_12B_L

DAC_ALIGN_8B_R

DAC Error Code

HAL_DAC_ERROR_NONE	No error
HAL_DAC_ERROR_DMAUNDERRUNCH1	DAC channel1 DAM underrun error
HAL_DAC_ERROR_DMAUNDERRUNCH2	DAC channel2 DAM underrun error
HAL_DAC_ERROR_DMA	DMA error

DAC Exported Macros

__HAL_DAC_RESET_HANDLE_STATE

Description:

- Reset DAC handle state.

Parameters:

- __HANDLE__: specifies the DAC handle.

Return value:

- None

__HAL_DAC_ENABLE

Description:

- Enable the DAC channel.

Parameters:

- __HANDLE__: specifies the DAC handle.
- __DAC_Channel__: specifies the DAC channel

Return value:

- None

__HAL_DAC_DISABLE

Description:

- Disable the DAC channel.

Parameters:

- __HANDLE__: specifies the DAC handle
- __DAC_Channel__: specifies the DAC channel.

Return value:

- None

__HAL_DAC_ENABLE_IT

Description:

- Enable the DAC interrupt.

Parameters:

- __HANDLE__: specifies the DAC handle
- __INTERRUPT__: specifies the DAC

interrupt.

Return value:

- None

__HAL_DAC_DISABLE_IT

Description:

- Disable the DAC interrupt.

Parameters:

- __HANDLE__: specifies the DAC handle
- __INTERRUPT__: specifies the DAC interrupt.

Return value:

- None

__HAL_DAC_GET_IT_SOURCE

Description:

- Checks if the specified DAC interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: DAC handle
- __INTERRUPT__: DAC interrupt source to check This parameter can be any combination of the following values:
  - DAC_IT_DMAUDR1: DAC channel 1 DMA underrun interrupt
  - DAC_IT_DMAUDR2: DAC channel 2 DMA underrun interrupt

Return value:

- State: of interruption (SET or RESET)

__HAL_DAC_GET_FLAG

Description:

- Get the selected DAC's flag status.

Parameters:

- __HANDLE__: specifies the DAC handle.
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - DAC_FLAG_DMAUDR1: DMA underrun 1 flag
  - DAC_FLAG_DMAUDR2: DMA underrun 2 flag

Return value:

- None

__HAL_DAC_CLEAR_FLAG

Description:

- Clear the DAC's flag.

Parameters:

- __HANDLE__: specifies the DAC handle.

- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - DAC_FLAG_DMAUDR1: DMA underrun 1 flag
  - DAC_FLAG_DMAUDR2: DMA underrun 2 flag

Return value:

- None

DAC Flags Definition

DAC_FLAG_DMAUDR1

DAC_FLAG_DMAUDR2

DAC IT Definition

DAC_IT_DMAUDR1

DAC_IT_DMAUDR2

DAC Output Buffer

DAC_OUTPUTBUFFER_ENABLE

DAC_OUTPUTBUFFER_DISABLE

DAC Trigger Selection

DAC_TRIGGER_NONE	Conversion is automatic once the DAC1_DHRxxxx register has been loaded, and not by external trigger
DAC_TRIGGER_T2_TRGO	TIM2 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_T4_TRGO	TIM4 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_T5_TRGO	TIM5 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_T6_TRGO	TIM6 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_T7_TRGO	TIM7 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_T8_TRGO	TIM8 TRGO selected as external conversion trigger for DAC channel
DAC_TRIGGER_EXT_IT9	EXTI Line9 event selected as external conversion trigger for DAC channel
DAC_TRIGGER_SOFTWARE	Conversion started by software trigger for DAC channel

13 HAL DAC Extension Driver

13.1 DACEx Firmware driver API description

13.1.1 How to use this driver

- When Dual mode is enabled (i.e DAC Channel1 and Channel2 are used simultaneously) : Use HAL_DACEx_DualGetValue() to get digital data to be converted and use HAL_DACEx_DualSetValue() to set digital value to converted simultaneously in Channel 1 and Channel 2.
- Use HAL_DACEx_TriangleWaveGenerate() to generate Triangle signal.
- Use HAL_DACEx_NoiseWaveGenerate() to generate Noise signal.

13.1.2 Extended features functions

This section provides functions allowing to:

- Start conversion.
- Stop conversion.
- Start conversion and enable DMA transfer.
- Stop conversion and disable DMA transfer.
- Get result of conversion.
- Get result of dual mode conversion.

This section contains the following APIs:

- ***HAL_DACEx_DualGetValue()***
- ***HAL_DACEx_TriangleWaveGenerate()***
- ***HAL_DACEx_NoiseWaveGenerate()***
- ***HAL_DACEx_DualSetValue()***
- ***HAL_DACEx_ConvCpltCallbackCh2()***
- ***HAL_DACEx_ConvHalfCpltCallbackCh2()***
- ***HAL_DACEx_ErrorCallbackCh2()***
- ***HAL_DACEx_DMAUnderrunCallbackCh2()***

13.1.3 Detailed description of functions

HAL_DACEx_DualGetValue

Function Name	**uint32_t HAL_DACEx_DualGetValue (DAC_HandleTypeDef * hdac)**
Function Description	Returns the last data output value of the selected DAC channel.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **The:** selected DAC channel data output value.

HAL_DACEx_TriangleWaveGenerate

Function Name **HAL_StatusTypeDef HAL_DACEx_TriangleWaveGenerate (DAC_HandleTypeDef * hdac, uint32_t Channel, uint32_t Amplitude)**

Function Description Enables or disables the selected DAC channel wave generation.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values: DAC_CHANNEL_1 / DAC_CHANNEL_2
- **Amplitude:** Select max triangle amplitude. This parameter can be one of the following values:
  - DAC_TRIANGLEAMPLITUDE_1: Select max triangle amplitude of 1
  - DAC_TRIANGLEAMPLITUDE_3: Select max triangle amplitude of 3
  - DAC_TRIANGLEAMPLITUDE_7: Select max triangle amplitude of 7
  - DAC_TRIANGLEAMPLITUDE_15: Select max triangle amplitude of 15
  - DAC_TRIANGLEAMPLITUDE_31: Select max triangle amplitude of 31
  - DAC_TRIANGLEAMPLITUDE_63: Select max triangle amplitude of 63
  - DAC_TRIANGLEAMPLITUDE_127: Select max triangle amplitude of 127
  - DAC_TRIANGLEAMPLITUDE_255: Select max triangle amplitude of 255
  - DAC_TRIANGLEAMPLITUDE_511: Select max triangle amplitude of 511
  - DAC_TRIANGLEAMPLITUDE_1023: Select max triangle amplitude of 1023
  - DAC_TRIANGLEAMPLITUDE_2047: Select max triangle amplitude of 2047
  - DAC_TRIANGLEAMPLITUDE_4095: Select max triangle amplitude of 4095

Return values
- **HAL:** status

HAL_DACEx_NoiseWaveGenerate

Function Name **HAL_StatusTypeDef HAL_DACEx_NoiseWaveGenerate (DAC_HandleTypeDef * hdac, uint32_t Channel, uint32_t Amplitude)**

Function Description Enables or disables the selected DAC channel wave generation.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Channel:** The selected DAC channel. This parameter can be one of the following values: DAC_CHANNEL_1 / DAC_CHANNEL_2
- **Amplitude:** Unmask DAC channel LFSR for noise wave generation. This parameter can be one of the following

values:
- DAC_LFSRUNMASK_BIT0: Unmask DAC channel LFSR bit0 for noise wave generation
- DAC_LFSRUNMASK_BITS1_0: Unmask DAC channel LFSR bit[1:0] for noise wave generation
- DAC_LFSRUNMASK_BITS2_0: Unmask DAC channel LFSR bit[2:0] for noise wave generation
- DAC_LFSRUNMASK_BITS3_0: Unmask DAC channel LFSR bit[3:0] for noise wave generation
- DAC_LFSRUNMASK_BITS4_0: Unmask DAC channel LFSR bit[4:0] for noise wave generation
- DAC_LFSRUNMASK_BITS5_0: Unmask DAC channel LFSR bit[5:0] for noise wave generation
- DAC_LFSRUNMASK_BITS6_0: Unmask DAC channel LFSR bit[6:0] for noise wave generation
- DAC_LFSRUNMASK_BITS7_0: Unmask DAC channel LFSR bit[7:0] for noise wave generation
- DAC_LFSRUNMASK_BITS8_0: Unmask DAC channel LFSR bit[8:0] for noise wave generation
- DAC_LFSRUNMASK_BITS9_0: Unmask DAC channel LFSR bit[9:0] for noise wave generation
- DAC_LFSRUNMASK_BITS10_0: Unmask DAC channel LFSR bit[10:0] for noise wave generation
- DAC_LFSRUNMASK_BITS11_0: Unmask DAC channel LFSR bit[11:0] for noise wave generation

Return values
- **HAL:** status

HAL_DACEx_DualSetValue

Function Name **HAL_StatusTypeDef HAL_DACEx_DualSetValue (DAC_HandleTypeDef * hdac, uint32_t Alignment, uint32_t Data1, uint32_t Data2)**

Function Description Set the specified data holding register value for dual DAC channel.

Parameters
- **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
- **Alignment:** Specifies the data alignment for dual channel DAC. This parameter can be one of the following values: DAC_ALIGN_8B_R: 8bit right data alignment selected DAC_ALIGN_12B_L: 12bit left data alignment selected DAC_ALIGN_12B_R: 12bit right data alignment selected
- **Data1:** Data for DAC Channel2 to be loaded in the selected data holding register.
- **Data2:** Data for DAC Channel1 to be loaded in the selected data holding register.

Return values
- **HAL:** status

Notes
- In dual mode, a unique register access is required to write in both DAC channels at the same time.

HAL_DACEx_ConvCpltCallbackCh2

Function Name	**void HAL_DACEx_ConvCpltCallbackCh2 (DAC_HandleTypeDef * hdac)**
Function Description	Conversion complete callback in non blocking mode for Channel2.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DACEx_ConvHalfCpltCallbackCh2

Function Name	**void HAL_DACEx_ConvHalfCpltCallbackCh2 (DAC_HandleTypeDef * hdac)**
Function Description	Conversion half DMA transfer callback in non blocking mode for Channel2.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DACEx_ErrorCallbackCh2

Function Name	**void HAL_DACEx_ErrorCallbackCh2 (DAC_HandleTypeDef * hdac)**
Function Description	Error DAC callback for Channel2.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

HAL_DACEx_DMAUnderrunCallbackCh2

Function Name	**void HAL_DACEx_DMAUnderrunCallbackCh2 (DAC_HandleTypeDef * hdac)**
Function Description	DMA underrun DAC callback for channel2.
Parameters	• **hdac:** pointer to a DAC_HandleTypeDef structure that contains the configuration information for the specified DAC.
Return values	• **None**

DAC_DMAConvCpltCh2

Function Name	**void DAC_DMAConvCpltCh2 (DMA_HandleTypeDef * hdma)**
Function Description	DMA conversion complete callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

DAC_DMAErrorCh2

Function Name	**void DAC_DMAErrorCh2 (DMA_HandleTypeDef * hdma)**
Function Description	DMA error callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

DAC_DMAHalfConvCpltCh2

Function Name	**void DAC_DMAHalfConvCpltCh2 (DMA_HandleTypeDef * hdma)**
Function Description	DMA half transfer complete callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

13.2 DACEx Firmware driver defines

13.2.1 DACEx

DAC LFS Run Mask Triangle Amplitude

DAC_LFSRUNMASK_BIT0	Unmask DAC channel LFSR bit0 for noise wave generation
DAC_LFSRUNMASK_BITS1_0	Unmask DAC channel LFSR bit[1:0] for noise wave generation
DAC_LFSRUNMASK_BITS2_0	Unmask DAC channel LFSR bit[2:0] for noise wave generation
DAC_LFSRUNMASK_BITS3_0	Unmask DAC channel LFSR bit[3:0] for noise wave generation
DAC_LFSRUNMASK_BITS4_0	Unmask DAC channel LFSR bit[4:0] for noise wave generation
DAC_LFSRUNMASK_BITS5_0	Unmask DAC channel LFSR bit[5:0] for noise wave generation
DAC_LFSRUNMASK_BITS6_0	Unmask DAC channel LFSR bit[6:0] for noise wave generation
DAC_LFSRUNMASK_BITS7_0	Unmask DAC channel LFSR bit[7:0] for noise wave generation
DAC_LFSRUNMASK_BITS8_0	Unmask DAC channel LFSR bit[8:0] for noise wave generation
DAC_LFSRUNMASK_BITS9_0	Unmask DAC channel LFSR bit[9:0] for noise wave generation
DAC_LFSRUNMASK_BITS10_0	Unmask DAC channel LFSR bit[10:0] for noise wave generation

DAC_LFSRUNMASK_BITS11_0	Unmask DAC channel LFSR bit[11:0] for noise wave generation
DAC_TRIANGLEAMPLITUDE_1	Select max triangle amplitude of 1
DAC_TRIANGLEAMPLITUDE_3	Select max triangle amplitude of 3
DAC_TRIANGLEAMPLITUDE_7	Select max triangle amplitude of 7
DAC_TRIANGLEAMPLITUDE_15	Select max triangle amplitude of 15
DAC_TRIANGLEAMPLITUDE_31	Select max triangle amplitude of 31
DAC_TRIANGLEAMPLITUDE_63	Select max triangle amplitude of 63
DAC_TRIANGLEAMPLITUDE_127	Select max triangle amplitude of 127
DAC_TRIANGLEAMPLITUDE_255	Select max triangle amplitude of 255
DAC_TRIANGLEAMPLITUDE_511	Select max triangle amplitude of 511
DAC_TRIANGLEAMPLITUDE_1023	Select max triangle amplitude of 1023
DAC_TRIANGLEAMPLITUDE_2047	Select max triangle amplitude of 2047
DAC_TRIANGLEAMPLITUDE_4095	Select max triangle amplitude of 4095

14 HAL DCMI Generic Driver

14.1 DCMI Firmware driver registers structures

14.1.1 DCMI_HandleTypeDef

Data Fields

- ***DCMI_TypeDef * Instance***
- ***DCMI_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_DCMI_StateTypeDef State***
- ***__IO uint32_t XferCount***
- ***__IO uint32_t XferSize***
- ***uint32_t XferTransferNumber***
- ***uint32_t pBuffPtr***
- ***DMA_HandleTypeDef * DMA_Handle***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***DCMI_TypeDef* DCMI_HandleTypeDef::Instance***
  DCMI Register base address
- ***DCMI_InitTypeDef DCMI_HandleTypeDef::Init***
  DCMI parameters
- ***HAL_LockTypeDef DCMI_HandleTypeDef::Lock***
  DCMI locking object
- ***__IO HAL_DCMI_StateTypeDef DCMI_HandleTypeDef::State***
  DCMI state
- ***__IO uint32_t DCMI_HandleTypeDef::XferCount***
  DMA transfer counter
- ***__IO uint32_t DCMI_HandleTypeDef::XferSize***
  DMA transfer size
- ***uint32_t DCMI_HandleTypeDef::XferTransferNumber***
  DMA transfer number
- ***uint32_t DCMI_HandleTypeDef::pBuffPtr***
  Pointer to DMA output buffer
- ***DMA_HandleTypeDef* DCMI_HandleTypeDef::DMA_Handle***
  Pointer to the DMA handler
- ***__IO uint32_t DCMI_HandleTypeDef::ErrorCode***
  DCMI Error code

14.2 DCMI Firmware driver API description

14.2.1 How to use this driver

The sequence below describes how to use this driver to capture image from a camera module connected to the DCMI Interface. This sequence does not take into account the configuration of the camera module, which should be made before to configure and enable the DCMI to capture images.

1. Program the required configuration through following parameters: horizontal and vertical polarity, pixel clock polarity, Capture Rate, Synchronization Mode, code of the frame delimiter and data width using HAL_DCMI_Init() function.
2. Configure the DMA2_Stream1 channel1 to transfer Data from DCMI DR register to the destination memory buffer.
3. Program the required configuration through following parameters: DCMI mode, destination memory Buffer address and the data length and enable capture using HAL_DCMI_Start_DMA() function.
4. Optionally, configure and Enable the CROP feature to select a rectangular window from the received image using HAL_DCMI_ConfigCrop() and HAL_DCMI_EnableCROP() functions
5. The capture can be stopped using HAL_DCMI_Stop() function.
6. To control DCMI state you can use the function HAL_DCMI_GetState().

DCMI HAL driver macros list

Below the list of most used macros in DCMI HAL driver.

- __HAL_DCMI_ENABLE: Enable the DCMI peripheral.
- __HAL_DCMI_DISABLE: Disable the DCMI peripheral.
- __HAL_DCMI_GET_FLAG: Get the DCMI pending flags.
- __HAL_DCMI_CLEAR_FLAG: Clear the DCMI pending flags.
- __HAL_DCMI_ENABLE_IT: Enable the specified DCMI interrupts.
- __HAL_DCMI_DISABLE_IT: Disable the specified DCMI interrupts.
- __HAL_DCMI_GET_IT_SOURCE: Check whether the specified DCMI interrupt has occurred or not.

You can refer to the DCMI HAL driver header file for more useful macros

14.2.2 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and configure the DCMI
- De-initialize the DCMI

This section contains the following APIs:

- ***HAL_DCMI_Init()***
- ***HAL_DCMI_DeInit()***
- ***HAL_DCMI_MspInit()***
- ***HAL_DCMI_MspDeInit()***

14.2.3 IO operation functions

This section provides functions allowing to:

- Configure destination address and data length and Enables DCMI DMA request and enables DCMI capture
- Stop the DCMI capture.
- Handles DCMI interrupt request.

This section contains the following APIs:

- ***HAL_DCMI_Start_DMA()***
- ***HAL_DCMI_Stop()***
- ***HAL_DCMI_Suspend()***
- ***HAL_DCMI_Resume()***
- ***HAL_DCMI_IRQHandler()***
- ***HAL_DCMI_ErrorCallback()***
- ***HAL_DCMI_LineEventCallback()***
- ***HAL_DCMI_VsyncEventCallback()***
- ***HAL_DCMI_FrameEventCallback()***
- ***HAL_DCMI_VsyncCallback()***
- ***HAL_DCMI_HsyncCallback()***

14.2.4 Peripheral Control functions

This section provides functions allowing to:

- Configure the CROP feature.
- Enable/Disable the CROP feature.

This section contains the following APIs:

- ***HAL_DCMI_ConfigCrop()***
- ***HAL_DCMI_DisableCrop()***
- ***HAL_DCMI_EnableCrop()***

14.2.5 Peripheral State and Errors functions

This subsection provides functions allowing to

- Check the DCMI state.
- Get the specific DCMI error flag.

This section contains the following APIs:

- ***HAL_DCMI_GetState()***
- ***HAL_DCMI_GetError()***

14.2.6 Detailed description of functions

HAL_DCMI_Init

Function Name	**HAL_StatusTypeDef HAL_DCMI_Init (DCMI_HandleTypeDef * hdcmi)**
Function Description	Initializes the DCMI according to the specified parameters in the DCMI_InitTypeDef and create the associated handle.
Parameters	• **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **HAL:** status

HAL_DCMI_DeInit

Function Name **HAL_StatusTypeDef HAL_DCMI_DeInit (DCMI_HandleTypeDef * hdcmi)**

Function Description Deinitializes the DCMI peripheral registers to their default reset values.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **HAL:** status

HAL_DCMI_MspInit

Function Name **void HAL_DCMI_MspInit (DCMI_HandleTypeDef * hdcmi)**

Function Description Initializes the DCMI MSP.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_MspDeInit

Function Name **void HAL_DCMI_MspDeInit (DCMI_HandleTypeDef * hdcmi)**

Function Description DeInitializes the DCMI MSP.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_Start_DMA

Function Name **HAL_StatusTypeDef HAL_DCMI_Start_DMA (DCMI_HandleTypeDef * hdcmi, uint32_t DCMI_Mode, uint32_t pData, uint32_t Length)**

Function Description Enables DCMI DMA request and enables DCMI capture.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
- **DCMI_Mode:** DCMI capture mode snapshot or continuous grab.
- **pData:** The destination memory Buffer address (LCD Frame buffer).
- **Length:** The length of capture to be transferred.

Return values
- **HAL:** status

HAL_DCMI_Stop

Function Name **HAL_StatusTypeDef HAL_DCMI_Stop (DCMI_HandleTypeDef * hdcmi)**

Function Description Disable DCMI DMA request and Disable DCMI capture.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **HAL:** status

HAL_DCMI_Suspend

Function Name **HAL_StatusTypeDef HAL_DCMI_Suspend (DCMI_HandleTypeDef * hdcmi)**

Function Description

HAL_DCMI_Resume

Function Name **HAL_StatusTypeDef HAL_DCMI_Resume (DCMI_HandleTypeDef * hdcmi)**

Function Description Resume DCMI capture.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **HAL:** status

HAL_DCMI_ErrorCallback

Function Name **void HAL_DCMI_ErrorCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description Error DCMI callback.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_LineEventCallback

Function Name **void HAL_DCMI_LineEventCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description Line Event callback.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_FrameEventCallback

Function Name **void HAL_DCMI_FrameEventCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description Frame Event callback.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_VsyncEventCallback

Function Name **void HAL_DCMI_VsyncEventCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description VSYNC Event callback.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.

Return values
- **None**

HAL_DCMI_VsyncCallback

Function Name **void HAL_DCMI_VsyncCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description

HAL_DCMI_HsyncCallback

Function Name **void HAL_DCMI_HsyncCallback (DCMI_HandleTypeDef * hdcmi)**

Function Description

HAL_DCMI_IRQHandler

Function Name **void HAL_DCMI_IRQHandler (DCMI_HandleTypeDef * hdcmi)**

Function Description Handles DCMI interrupt request.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for the DCMI.

Return values
- **None**

HAL_DCMI_ConfigCrop

Function Name **HAL_StatusTypeDef HAL_DCMI_ConfigCrop (DCMI_HandleTypeDef * hdcmi, uint32_t X0, uint32_t Y0, uint32_t XSize, uint32_t YSize)**

Function Description Configure the DCMI CROP coordinate.

Parameters
- **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
- **X0:** DCMI window X offset
- **Y0:** DCMI window Y offset
- **XSize:** DCMI Pixel per line
- **YSize:** DCMI Line number

Return values
- **HAL:** status

HAL_DCMI_EnableCrop

Function Name	**HAL_StatusTypeDef HAL_DCMI_EnableCrop (DCMI_HandleTypeDef * hdcmi)**
Function Description	Enable the Crop feature.
Parameters	• **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
Return values	• **HAL:** status

HAL_DCMI_DisableCrop

Function Name	**HAL_StatusTypeDef HAL_DCMI_DisableCrop (DCMI_HandleTypeDef * hdcmi)**
Function Description	Disable the Crop feature.
Parameters	• **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
Return values	• **HAL:** status

HAL_DCMI_GetState

Function Name	**HAL_DCMI_StateTypeDef HAL_DCMI_GetState (DCMI_HandleTypeDef * hdcmi)**
Function Description	Return the DCMI state.
Parameters	• **hdcmi:** pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
Return values	• **HAL:** state

HAL_DCMI_GetError

Function Name	**uint32_t HAL_DCMI_GetError (DCMI_HandleTypeDef * hdcmi)**
Function Description	Return the DCMI error code.
Parameters	• **hdcmi:** : pointer to a DCMI_HandleTypeDef structure that contains the configuration information for DCMI.
Return values	• **DCMI:** Error Code

14.3 DCMI Firmware driver defines

14.3.1 DCMI

DCMI Capture Mode

DCMI_MODE_CONTINUOUS	The received data are transferred continuously into the destination memory through the DMA
DCMI_MODE_SNAPSHOT	Once activated, the interface waits for the start of frame and then transfers a single frame through the DMA

DCMI Capture Rate

DCMI_CR_ALL_FRAME	All frames are captured
DCMI_CR_ALTERNATE_2_FRAME	Every alternate frame captured
DCMI_CR_ALTERNATE_4_FRAME	One frame in 4 frames captured

DCMI Error Code

HAL_DCMI_ERROR_NONE	No error
HAL_DCMI_ERROR_OVR	Overrun error
HAL_DCMI_ERROR_SYNC	Synchronization error
HAL_DCMI_ERROR_TIMEOUT	Timeout error
HAL_DCMI_ERROR_DMA	DMA error

DCMI Exported Macros

__HAL_DCMI_RESET_HANDLE_STATE

Description:

- Reset DCMI handle state.

Parameters:

- __HANDLE__: specifies the DCMI handle.

Return value:

- None

__HAL_DCMI_ENABLE

Description:

- Enable the DCMI.

Parameters:

- __HANDLE__: DCMI handle

Return value:

- None

__HAL_DCMI_DISABLE

Description:

- Disable the DCMI.

Parameters:

- __HANDLE__: DCMI handle

Return value:

- None

__HAL_DCMI_GET_FLAG

Description:

- Get the DCMI pending flag.

Parameters:

- __HANDLE__: DCMI handle
- __FLAG__: Get the specified flag. This parameter can be one of the following values (no combination allowed)
  - DCMI_FLAG_HSYNC: HSYNC pin

state (active line / synchronization between lines)
  - DCMI_FLAG_VSYNC: VSYNC pin state (active frame / synchronization between frames)
  - DCMI_FLAG_FNE: FIFO empty flag
  - DCMI_FLAG_FRAMERI: Frame capture complete flag mask
  - DCMI_FLAG_OVRRI: Overrun flag mask
  - DCMI_FLAG_ERRRI: Synchronization error flag mask
  - DCMI_FLAG_VSYNCRI: VSYNC flag mask
  - DCMI_FLAG_LINERI: Line flag mask
  - DCMI_FLAG_FRAMEMI: DCMI Capture complete masked interrupt status
  - DCMI_FLAG_OVRMI: DCMI Overrun masked interrupt status
  - DCMI_FLAG_ERRMI: DCMI Synchronization error masked interrupt status
  - DCMI_FLAG_VSYNCMI: DCMI VSYNC masked interrupt status
  - DCMI_FLAG_LINEMI: DCMI Line masked interrupt status

Return value:

- The: state of FLAG.

__HAL_DCMI_CLEAR_FLAG

Description:

- Clear the DCMI pending flags.

Parameters:

- __HANDLE__: DCMI handle
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - DCMI_FLAG_FRAMERI: Frame capture complete flag mask
  - DCMI_FLAG_OVRRI: Overrun flag mask
  - DCMI_FLAG_ERRRI: Synchronization error flag mask
  - DCMI_FLAG_VSYNCRI: VSYNC flag mask
  - DCMI_FLAG_LINERI: Line flag mask

Return value:

- None

__HAL_DCMI_ENABLE_IT

Description:

- Enable the specified DCMI interrupts.

Parameters:

- __HANDLE__: DCMI handle
- __INTERRUPT__: specifies the DCMI interrupt sources to be enabled. This parameter can be any combination of the following values:
  - DCMI_IT_FRAME: Frame capture complete interrupt mask
  - DCMI_IT_OVR: Overrun interrupt mask
  - DCMI_IT_ERR: Synchronization error interrupt mask
  - DCMI_IT_VSYNC: VSYNC interrupt mask
  - DCMI_IT_LINE: Line interrupt mask

Return value:

- None

__HAL_DCMI_DISABLE_IT

Description:

- Disable the specified DCMI interrupts.

Parameters:

- __HANDLE__: DCMI handle
- __INTERRUPT__: specifies the DCMI interrupt sources to be enabled. This parameter can be any combination of the following values:
  - DCMI_IT_FRAME: Frame capture complete interrupt mask
  - DCMI_IT_OVR: Overrun interrupt mask
  - DCMI_IT_ERR: Synchronization error interrupt mask
  - DCMI_IT_VSYNC: VSYNC interrupt mask
  - DCMI_IT_LINE: Line interrupt mask

Return value:

- None

__HAL_DCMI_GET_IT_SOURCE

Description:

- Check whether the specified DCMI interrupt has occurred or not.

Parameters:

- __HANDLE__: DCMI handle
- __INTERRUPT__: specifies the DCMI interrupt source to check. This parameter can be one of the following values:

- DCMI_IT_FRAME: Frame capture complete interrupt mask
- DCMI_IT_OVR: Overrun interrupt mask
- DCMI_IT_ERR: Synchronization error interrupt mask
- DCMI_IT_VSYNC: VSYNC interrupt mask
- DCMI_IT_LINE: Line interrupt mask

Return value:

- The: state of INTERRUPT.

DCMI Extended Data Mode

DCMI_EXTEND_DATA_8B	Interface captures 8-bit data on every pixel clock
DCMI_EXTEND_DATA_10B	Interface captures 10-bit data on every pixel clock
DCMI_EXTEND_DATA_12B	Interface captures 12-bit data on every pixel clock
DCMI_EXTEND_DATA_14B	Interface captures 14-bit data on every pixel clock

DCMI Flags

DCMI_FLAG_HSYNC	HSYNC pin state (active line / synchronization between lines)
DCMI_FLAG_VSYNC	VSYNC pin state (active frame / synchronization between frames)
DCMI_FLAG_FNE	FIFO not empty flag
DCMI_FLAG_FRAMERI	Frame capture complete interrupt flag
DCMI_FLAG_OVRRI	Overrun interrupt flag
DCMI_FLAG_ERRRI	Synchronization error interrupt flag
DCMI_FLAG_VSYNCRI	VSYNC interrupt flag
DCMI_FLAG_LINERI	Line interrupt flag
DCMI_FLAG_FRAMEMI	DCMI Frame capture complete masked interrupt status
DCMI_FLAG_OVRMI	DCMI Overrun masked interrupt status
DCMI_FLAG_ERRMI	DCMI Synchronization error masked interrupt status
DCMI_FLAG_VSYNCMI	DCMI VSYNC masked interrupt status
DCMI_FLAG_LINEMI	DCMI Line masked interrupt status

DCMI HSYNC Polarity

DCMI_HSPOLARITY_LOW	Horizontal synchronization active Low
DCMI_HSPOLARITY_HIGH	Horizontal synchronization active High

DCMI interrupt sources

DCMI_IT_FRAME	Capture complete interrupt
DCMI_IT_OVR	Overrun interrupt
DCMI_IT_ERR	Synchronization error interrupt
DCMI_IT_VSYNC	VSYNC interrupt

DCMI_IT_LINE	Line interrupt

DCMI MODE JPEG

DCMI_JPEG_DISABLE	Mode JPEG Disabled
DCMI_JPEG_ENABLE	Mode JPEG Enabled

DCMI PIXCK Polarity

DCMI_PCKPOLARITY_FALLING	Pixel clock active on Falling edge
DCMI_PCKPOLARITY_RISING	Pixel clock active on Rising edge

DCMI Synchronization Mode

DCMI_SYNCHRO_HARDWARE	Hardware synchronization data capture (frame/line start/stop) is synchronized with the HSYNC/VSYNC signals
DCMI_SYNCHRO_EMBEDDED	Embedded synchronization data capture is synchronized with synchronization codes embedded in the data flow

DCMI VSYNC Polarity

DCMI_VSPOLARITY_LOW	Vertical synchronization active Low
DCMI_VSPOLARITY_HIGH	Vertical synchronization active High

DCMI Window Coordinate

DCMI_WINDOW_COORDINATE	Window coordinate

DCMI Window Height

DCMI_WINDOW_HEIGHT	Window Height

DCMI Window Vertical Line

DCMI_POSITION_CWSIZE_VLINE	Required left shift to set crop window vertical line count
DCMI_POSITION_CWSTRT_VST	Required left shift to set crop window vertical start line count

15 HAL DCMI Extension Driver

15.1 DCMIEx Firmware driver registers structures

15.1.1 DCMI_CodesInitTypeDef

Data Fields

- ***uint8_t FrameStartCode***
- ***uint8_t LineStartCode***
- ***uint8_t LineEndCode***
- ***uint8_t FrameEndCode***

Field Documentation

- ***uint8_t DCMI_CodesInitTypeDef::FrameStartCode***
  Specifies the code of the frame start delimiter.
- ***uint8_t DCMI_CodesInitTypeDef::LineStartCode***
  Specifies the code of the line start delimiter.
- ***uint8_t DCMI_CodesInitTypeDef::LineEndCode***
  Specifies the code of the line end delimiter.
- ***uint8_t DCMI_CodesInitTypeDef::FrameEndCode***
  Specifies the code of the frame end delimiter.

15.1.2 DCMI_InitTypeDef

Data Fields

- ***uint32_t SynchroMode***
- ***uint32_t PCKPolarity***
- ***uint32_t VSPolarity***
- ***uint32_t HSPolarity***
- ***uint32_t CaptureRate***
- ***uint32_t ExtendedDataMode***
- ***DCMI_CodesInitTypeDef SyncroCode***
- ***uint32_t JPEGMode***
- ***uint32_t ByteSelectMode***
- ***uint32_t ByteSelectStart***
- ***uint32_t LineSelectMode***
- ***uint32_t LineSelectStart***

Field Documentation

- ***uint32_t DCMI_InitTypeDef::SynchroMode***
  Specifies the Synchronization Mode: Hardware or Embedded. This parameter can be a value of ***DCMI_Synchronization_Mode***

- ***uint32_t DCMI_InitTypeDef::PCKPolarity***
  Specifies the Pixel clock polarity: Falling or Rising. This parameter can be a value of ***DCMI_PIXCK_Polarity***
- ***uint32_t DCMI_InitTypeDef::VSPolarity***
  Specifies the Vertical synchronization polarity: High or Low. This parameter can be a value of ***DCMI_VSYNC_Polarity***
- ***uint32_t DCMI_InitTypeDef::HSPolarity***
  Specifies the Horizontal synchronization polarity: High or Low. This parameter can be a value of ***DCMI_HSYNC_Polarity***
- ***uint32_t DCMI_InitTypeDef::CaptureRate***
  Specifies the frequency of frame capture: All, 1/2 or 1/4. This parameter can be a value of ***DCMI_Capture_Rate***
- ***uint32_t DCMI_InitTypeDef::ExtendedDataMode***
  Specifies the data width: 8-bit, 10-bit, 12-bit or 14-bit. This parameter can be a value of ***DCMI_Extended_Data_Mode***
- ***DCMI_CodesInitTypeDef DCMI_InitTypeDef::SyncroCode***
  Specifies the code of the frame start delimiter.
- ***uint32_t DCMI_InitTypeDef::JPEGMode***
  Enable or Disable the JPEG mode This parameter can be a value of ***DCMI_MODE_JPEG***
- ***uint32_t DCMI_InitTypeDef::ByteSelectMode***
  Specifies the data to be captured by the interface This parameter can be a value of ***DCMIEx_Byte_Select_Mode***
- ***uint32_t DCMI_InitTypeDef::ByteSelectStart***
  Specifies if the data to be captured by the interface is even or odd This parameter can be a value of ***DCMIEx_Byte_Select_Start***
- ***uint32_t DCMI_InitTypeDef::LineSelectMode***
  Specifies the line of data to be captured by the interface This parameter can be a value of ***DCMIEx_Line_Select_Mode***
- ***uint32_t DCMI_InitTypeDef::LineSelectStart***
  Specifies if the line of data to be captured by the interface is even or odd This parameter can be a value of ***DCMIEx_Line_Select_Start***

15.2 DCMIEx Firmware driver defines

15.2.1 DCMIEx

DCMI Byte Select Mode

DCMI_BSM_ALL	Interface captures all received data
DCMI_BSM_OTHER	Interface captures every other byte from the received data
DCMI_BSM_ALTERNATE_4	Interface captures one byte out of four
DCMI_BSM_ALTERNATE_2	Interface captures two bytes out of four

DCMI Byte Select Start

DCMI_OEBS_ODD	Interface captures first data from the frame/line start, second one being dropped
DCMI_OEBS_EVEN	Interface captures second data from the frame/line start, first one being dropped

DCMI Line Select Mode

DCMI_LSM_ALL	Interface captures all received lines
DCMI_LSM_ALTERNATE_2	Interface captures one line out of two

DCMI Line Select Start

DCMI_OELS_ODD	Interface captures first line from the frame start, second one being dropped
DCMI_OELS_EVEN	Interface captures second line from the frame start, first one being dropped

16 HAL DFSDM Generic Driver

16.1 DFSDM Firmware driver registers structures

16.1.1 DFSDM_Channel_OutputClockTypeDef

Data Fields

- ***FunctionalState Activation***
- ***uint32_t Selection***
- ***uint32_t Divider***

Field Documentation

- ***FunctionalState DFSDM_Channel_OutputClockTypeDef::Activation***
  Output clock enable/disable
- ***uint32_t DFSDM_Channel_OutputClockTypeDef::Selection***
  Output clock is system clock or audio clock. This parameter can be a value of ***DFSDM_Channel_OuputClock***
- ***uint32_t DFSDM_Channel_OutputClockTypeDef::Divider***
  Output clock divider. This parameter must be a number between Min_Data = 2 and Max_Data = 256

16.1.2 DFSDM_Channel_InputTypeDef

Data Fields

- ***uint32_t Multiplexer***
- ***uint32_t DataPacking***
- ***uint32_t Pins***

Field Documentation

- ***uint32_t DFSDM_Channel_InputTypeDef::Multiplexer***
  Input is external serial inputs or internal register. This parameter can be a value of ***DFSDM_Channel_InputMultiplexer***
- ***uint32_t DFSDM_Channel_InputTypeDef::DataPacking***
  Standard, interleaved or dual mode for internal register. This parameter can be a value of ***DFSDM_Channel_DataPacking***
- ***uint32_t DFSDM_Channel_InputTypeDef::Pins***
  Input pins are taken from same or following channel. This parameter can be a value of ***DFSDM_Channel_InputPins***

16.1.3 DFSDM_Channel_SerialInterfaceTypeDef

Data Fields

- ***uint32_t Type***
- ***uint32_t SpiClock***

Field Documentation

- ***uint32_t DFSDM_Channel_SerialInterfaceTypeDef::Type***
  SPI or Manchester modes. This parameter can be a value of ***DFSDM_Channel_SerialInterfaceType***
- ***uint32_t DFSDM_Channel_SerialInterfaceTypeDef::SpiClock***
  SPI clock select (external or internal with different sampling point). This parameter can be a value of ***DFSDM_Channel_SpiClock***

16.1.4 DFSDM_Channel_AwdTypeDef

Data Fields

- ***uint32_t FilterOrder***
- ***uint32_t Oversampling***

Field Documentation

- ***uint32_t DFSDM_Channel_AwdTypeDef::FilterOrder***
  Analog watchdog Sinc filter order. This parameter can be a value of ***DFSDM_Channel_AwdFilterOrder***
- ***uint32_t DFSDM_Channel_AwdTypeDef::Oversampling***
  Analog watchdog filter oversampling ratio. This parameter must be a number between Min_Data = 1 and Max_Data = 32

16.1.5 DFSDM_Channel_InitTypeDef

Data Fields

- ***DFSDM_Channel_OutputClockTypeDef OutputClock***
- ***DFSDM_Channel_InputTypeDef Input***
- ***DFSDM_Channel_SerialInterfaceTypeDef SerialInterface***
- ***DFSDM_Channel_AwdTypeDef Awd***
- ***int32_t Offset***
- ***uint32_t RightBitShift***

Field Documentation

- ***DFSDM_Channel_OutputClockTypeDef DFSDM_Channel_InitTypeDef::OutputClock***
  DFSDM channel output clock parameters
- ***DFSDM_Channel_InputTypeDef DFSDM_Channel_InitTypeDef::Input***
  DFSDM channel input parameters
- ***DFSDM_Channel_SerialInterfaceTypeDef DFSDM_Channel_InitTypeDef::SerialInterface***
  DFSDM channel serial interface parameters
- ***DFSDM_Channel_AwdTypeDef DFSDM_Channel_InitTypeDef::Awd***
  DFSDM channel analog watchdog parameters
- ***int32_t DFSDM_Channel_InitTypeDef::Offset***
  DFSDM channel offset. This parameter must be a number between Min_Data = -8388608 and Max_Data = 8388607
- ***uint32_t DFSDM_Channel_InitTypeDef::RightBitShift***
  DFSDM channel right bit shift. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0x1F

16.1.6 DFSDM_Channel_HandleTypeDef

Data Fields

- ***DFSDM_Channel_TypeDef * Instance***
- ***DFSDM_Channel_InitTypeDef Init***
- ***HAL_DFSDM_Channel_StateTypeDef State***

Field Documentation

- ***DFSDM_Channel_TypeDef* DFSDM_Channel_HandleTypeDef::Instance***
  DFSDM channel instance
- ***DFSDM_Channel_InitTypeDef DFSDM_Channel_HandleTypeDef::Init***
  DFSDM channel init parameters
- ***HAL_DFSDM_Channel_StateTypeDef DFSDM_Channel_HandleTypeDef::State***
  DFSDM channel state

16.1.7 DFSDM_Filter_RegularParamTypeDef

Data Fields

- ***uint32_t Trigger***
- ***FunctionalState FastMode***
- ***FunctionalState DmaMode***

Field Documentation

- ***uint32_t DFSDM_Filter_RegularParamTypeDef::Trigger***
  Trigger used to start regular conversion: software or synchronous. This parameter can be a value of ***DFSDM_Filter_Trigger***
- ***FunctionalState DFSDM_Filter_RegularParamTypeDef::FastMode***
  Enable/disable fast mode for regular conversion
- ***FunctionalState DFSDM_Filter_RegularParamTypeDef::DmaMode***
  Enable/disable DMA for regular conversion

16.1.8 DFSDM_Filter_InjectedParamTypeDef

Data Fields

- ***uint32_t Trigger***
- ***FunctionalState ScanMode***
- ***FunctionalState DmaMode***
- ***uint32_t ExtTrigger***
- ***uint32_t ExtTriggerEdge***

Field Documentation

- ***uint32_t DFSDM_Filter_InjectedParamTypeDef::Trigger***
  Trigger used to start injected conversion: software, external or synchronous. This parameter can be a value of ***DFSDM_Filter_Trigger***
- ***FunctionalState DFSDM_Filter_InjectedParamTypeDef::ScanMode***
  Enable/disable scanning mode for injected conversion
- ***FunctionalState DFSDM_Filter_InjectedParamTypeDef::DmaMode***
  Enable/disable DMA for injected conversion
- ***uint32_t DFSDM_Filter_InjectedParamTypeDef::ExtTrigger***
  External trigger. This parameter can be a value of ***DFSDM_Filter_ExtTrigger***
- ***uint32_t DFSDM_Filter_InjectedParamTypeDef::ExtTriggerEdge***
  External trigger edge: rising, falling or both. This parameter can be a value of ***DFSDM_Filter_ExtTriggerEdge***

16.1.9 DFSDM_Filter_FilterParamTypeDef

Data Fields

- ***uint32_t SincOrder***
- ***uint32_t Oversampling***
- ***uint32_t IntOversampling***

Field Documentation

- ***uint32_t DFSDM_Filter_FilterParamTypeDef::SincOrder***
  Sinc filter order. This parameter can be a value of ***DFSDM_Filter_SincOrder***
- ***uint32_t DFSDM_Filter_FilterParamTypeDef::Oversampling***
  Filter oversampling ratio. This parameter must be a number between Min_Data = 1 and Max_Data = 1024
- ***uint32_t DFSDM_Filter_FilterParamTypeDef::IntOversampling***
  Integrator oversampling ratio. This parameter must be a number between Min_Data = 1 and Max_Data = 256

16.1.10 DFSDM_Filter_InitTypeDef

Data Fields

- ***DFSDM_Filter_RegularParamTypeDef RegularParam***
- ***DFSDM_Filter_InjectedParamTypeDef InjectedParam***
- ***DFSDM_Filter_FilterParamTypeDef FilterParam***

Field Documentation

- ***DFSDM_Filter_RegularParamTypeDef DFSDM_Filter_InitTypeDef::RegularParam***
  DFSDM regular conversion parameters
- ***DFSDM_Filter_InjectedParamTypeDef DFSDM_Filter_InitTypeDef::InjectedParam***
  DFSDM injected conversion parameters
- ***DFSDM_Filter_FilterParamTypeDef DFSDM_Filter_InitTypeDef::FilterParam***
  DFSDM filter parameters

16.1.11 DFSDM_Filter_HandleTypeDef

Data Fields

- ***DFSDM_Filter_TypeDef * Instance***
- ***DFSDM_Filter_InitTypeDef Init***
- ***DMA_HandleTypeDef * hdmaReg***
- ***DMA_HandleTypeDef * hdmaInj***
- ***uint32_t RegularContMode***
- ***uint32_t RegularTrigger***
- ***uint32_t InjectedTrigger***
- ***uint32_t ExtTriggerEdge***
- ***FunctionalState InjectedScanMode***
- ***uint32_t InjectedChannelsNbr***
- ***uint32_t InjConvRemaining***
- ***HAL_DFSDM_Filter_StateTypeDef State***
- ***uint32_t ErrorCode***

Field Documentation

- ***DFSDM_Filter_TypeDef* DFSDM_Filter_HandleTypeDef::Instance***
  DFSDM filter instance
- ***DFSDM_Filter_InitTypeDef DFSDM_Filter_HandleTypeDef::Init***
  DFSDM filter init parameters
- ***DMA_HandleTypeDef* DFSDM_Filter_HandleTypeDef::hdmaReg***
  Pointer on DMA handler for regular conversions
- ***DMA_HandleTypeDef* DFSDM_Filter_HandleTypeDef::hdmaInj***
  Pointer on DMA handler for injected conversions
- ***uint32_t DFSDM_Filter_HandleTypeDef::RegularContMode***
  Regular conversion continuous mode
- ***uint32_t DFSDM_Filter_HandleTypeDef::RegularTrigger***
  Trigger used for regular conversion
- ***uint32_t DFSDM_Filter_HandleTypeDef::InjectedTrigger***
  Trigger used for injected conversion
- ***uint32_t DFSDM_Filter_HandleTypeDef::ExtTriggerEdge***
  Rising, falling or both edges selected
- ***FunctionalState DFSDM_Filter_HandleTypeDef::InjectedScanMode***
  Injected scanning mode
- ***uint32_t DFSDM_Filter_HandleTypeDef::InjectedChannelsNbr***
  Number of channels in injected sequence
- ***uint32_t DFSDM_Filter_HandleTypeDef::InjConvRemaining***
  Injected conversions remaining
- ***HAL_DFSDM_Filter_StateTypeDef DFSDM_Filter_HandleTypeDef::State***
  DFSDM filter state
- ***uint32_t DFSDM_Filter_HandleTypeDef::ErrorCode***
  DFSDM filter error code

16.1.12 DFSDM_Filter_AwdParamTypeDef

Data Fields

- ***uint32_t DataSource***
- ***uint32_t Channel***
- ***int32_t HighThreshold***
- ***int32_t LowThreshold***
- ***uint32_t HighBreakSignal***
- ***uint32_t LowBreakSignal***

Field Documentation

- ***uint32_t DFSDM_Filter_AwdParamTypeDef::DataSource***
  Values from digital filter or from channel watchdog filter. This parameter can be a value of ***DFSDM_Filter_AwdDataSource***
- ***uint32_t DFSDM_Filter_AwdParamTypeDef::Channel***
  Analog watchdog channel selection. This parameter can be a values combination of ***DFSDM_Channel_Selection***

- ***int32_t DFSDM_Filter_AwdParamTypeDef::HighThreshold***
  High threshold for the analog watchdog. This parameter must be a number between Min_Data = -8388608 and Max_Data = 8388607
- ***int32_t DFSDM_Filter_AwdParamTypeDef::LowThreshold***
  Low threshold for the analog watchdog. This parameter must be a number between Min_Data = -8388608 and Max_Data = 8388607
- ***uint32_t DFSDM_Filter_AwdParamTypeDef::HighBreakSignal***
  Break signal assigned to analog watchdog high threshold event. This parameter can be a values combination of ***DFSDM_BreakSignals***
- ***uint32_t DFSDM_Filter_AwdParamTypeDef::LowBreakSignal***
  Break signal assigned to analog watchdog low threshold event. This parameter can be a values combination of ***DFSDM_BreakSignals***

16.2 DFSDM Firmware driver API description

16.2.1 How to use this driver

Channel initialization

1. User has first to initialize channels (before filters initialization).
2. As prerequisite, fill in the HAL_DFSDM_ChannelMspInit() :
   – Enable DFSDMz clock interface with __HAL_RCC_DFSDMz_CLK_ENABLE().
   – Enable the clocks for the DFSDMz GPIOS with __HAL_RCC_GPIOx_CLK_ENABLE().
   – Configure these DFSDMz pins in alternate mode using HAL_GPIO_Init().
   – If interrupt mode is used, enable and configure DFSDMz_FLT0 global interrupt with HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ().
3. Configure the output clock, input, serial interface, analog watchdog, offset and data right bit shift parameters for this channel using the HAL_DFSDM_ChannelInit() function.

Channel clock absence detector

1. Start clock absence detector using HAL_DFSDM_ChannelCkabStart() or HAL_DFSDM_ChannelCkabStart_IT().
2. In polling mode, use HAL_DFSDM_ChannelPollForCkab() to detect the clock absence.
3. In interrupt mode, HAL_DFSDM_ChannelCkabCallback() will be called if clock absence is detected.
4. Stop clock absence detector using HAL_DFSDM_ChannelCkabStop() or HAL_DFSDM_ChannelCkabStop_IT().
5. Please note that the same mode (polling or interrupt) has to be used for all channels because the channels are sharing the same interrupt.
6. Please note also that in interrupt mode, if clock absence detector is stopped for one channel, interrupt will be disabled for all channels.

Channel short circuit detector

1. Start short circuit detector using HAL_DFSDM_ChannelScdStart() or or HAL_DFSDM_ChannelScdStart_IT().
2. In polling mode, use HAL_DFSDM_ChannelPollForScd() to detect short circuit.
3. In interrupt mode, HAL_DFSDM_ChannelScdCallback() will be called if short circuit is detected.
4. Stop short circuit detector using HAL_DFSDM_ChannelScdStop() or or HAL_DFSDM_ChannelScdStop_IT().
5. Please note that the same mode (polling or interrupt) has to be used for all channels because the channels are sharing the same interrupt.
6. Please note also that in interrupt mode, if short circuit detector is stopped for one channel, interrupt will be disabled for all channels.

Channel analog watchdog value

1. Get analog watchdog filter value of a channel using HAL_DFSDM_ChannelGetAwdValue().

Channel offset value

1. Modify offset value of a channel using HAL_DFSDM_ChannelModifyOffset().

Filter initialization

1. After channel initialization, user has to init filters.
2. As prerequisite, fill in the HAL_DFSDM_FilterMspInit() :
   – If interrupt mode is used , enable and configure DFSDMz_FLTx global interrupt with HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ(). Please note that DFSDMz_FLT0 global interrupt could be already enabled if interrupt is used for channel.
   – If DMA mode is used, configure DMA with HAL_DMA_Init() and link it with DFSDMz filter handle using __HAL_LINKDMA().
3. Configure the regular conversion, injected conversion and filter parameters for this filter using the HAL_DFSDM_FilterInit() function.

Filter regular channel conversion

1. Select regular channel and enable/disable continuous mode using HAL_DFSDM_FilterConfigRegChannel().
2. Start regular conversion using HAL_DFSDM_FilterRegularStart(), HAL_DFSDM_FilterRegularStart_IT(), HAL_DFSDM_FilterRegularStart_DMA() or HAL_DFSDM_FilterRegularMsbStart_DMA().
3. In polling mode, use HAL_DFSDM_FilterPollForRegConversion() to detect the end of regular conversion.
4. In interrupt mode, HAL_DFSDM_FilterRegConvCpltCallback() will be called at the end of regular conversion.

5. Get value of regular conversion and corresponding channel using HAL_DFSDM_FilterGetRegularValue().
6. In DMA mode, HAL_DFSDM_FilterRegConvHalfCpltCallback() and HAL_DFSDM_FilterRegConvCpltCallback() will be called respectively at the half transfer and at the transfer complete. Please note that HAL_DFSDM_FilterRegConvHalfCpltCallback() will be called only in DMA circular mode.
7. Stop regular conversion using HAL_DFSDM_FilterRegularStop(), HAL_DFSDM_FilterRegularStop_IT() or HAL_DFSDM_FilterRegularStop_DMA().

Filter injected channels conversion

1. Select injected channels using HAL_DFSDM_FilterConfigInjChannel().
2. Start injected conversion using HAL_DFSDM_FilterInjectedStart(), HAL_DFSDM_FilterInjectedStart_IT(), HAL_DFSDM_FilterInjectedStart_DMA() or HAL_DFSDM_FilterInjectedMsbStart_DMA().
3. In polling mode, use HAL_DFSDM_FilterPollForInjConversion() to detect the end of injected conversion.
4. In interrupt mode, HAL_DFSDM_FilterInjConvCpltCallback() will be called at the end of injected conversion.
5. Get value of injected conversion and corresponding channel using HAL_DFSDM_FilterGetInjectedValue().
6. In DMA mode, HAL_DFSDM_FilterInjConvHalfCpltCallback() and HAL_DFSDM_FilterInjConvCpltCallback() will be called respectively at the half transfer and at the transfer complete. Please note that HAL_DFSDM_FilterInjConvCpltCallback() will be called only in DMA circular mode.
7. Stop injected conversion using HAL_DFSDM_FilterInjectedStop(), HAL_DFSDM_FilterInjectedStop_IT() or HAL_DFSDM_FilterInjectedStop_DMA().

Filter analog watchdog

1. Start filter analog watchdog using HAL_DFSDM_FilterAwdStart_IT().
2. HAL_DFSDM_FilterAwdCallback() will be called if analog watchdog occurs.
3. Stop filter analog watchdog using HAL_DFSDM_FilterAwdStop_IT().

Filter extreme detector

1. Start filter extreme detector using HAL_DFSDM_FilterExdStart().
2. Get extreme detector maximum value using HAL_DFSDM_FilterGetExdMaxValue().
3. Get extreme detector minimum value using HAL_DFSDM_FilterGetExdMinValue().
4. Start filter extreme detector using HAL_DFSDM_FilterExdStop().

Filter conversion time

1. Get conversion time value using HAL_DFSDM_FilterGetConvTimeValue().

16.2.2 Channel initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the DFSDM channel.
- De-initialize the DFSDM channel.

This section contains the following APIs:

- ***HAL_DFSDM_ChannelInit()***
- ***HAL_DFSDM_ChannelDeInit()***
- ***HAL_DFSDM_ChannelMspInit()***
- ***HAL_DFSDM_ChannelMspDeInit()***

16.2.3 Channel operation functions

This section provides functions allowing to:

- Manage clock absence detector feature.
- Manage short circuit detector feature.
- Get analog watchdog value.
- Modify offset value.

This section contains the following APIs:

- ***HAL_DFSDM_ChannelCkabStart()***
- ***HAL_DFSDM_ChannelPollForCkab()***
- ***HAL_DFSDM_ChannelCkabStop()***
- ***HAL_DFSDM_ChannelCkabStart_IT()***
- ***HAL_DFSDM_ChannelCkabCallback()***
- ***HAL_DFSDM_ChannelCkabStop_IT()***
- ***HAL_DFSDM_ChannelScdStart()***
- ***HAL_DFSDM_ChannelPollForScd()***
- ***HAL_DFSDM_ChannelScdStop()***
- ***HAL_DFSDM_ChannelScdStart_IT()***
- ***HAL_DFSDM_ChannelScdCallback()***
- ***HAL_DFSDM_ChannelScdStop_IT()***
- ***HAL_DFSDM_ChannelGetAwdValue()***
- ***HAL_DFSDM_ChannelModifyOffset()***

16.2.4 Channel state function

This section provides function allowing to:

- Get channel handle state.

This section contains the following APIs:

- ***HAL_DFSDM_ChannelGetState()***

16.2.5 Filter initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the DFSDM filter.
- De-initialize the DFSDM filter.

This section contains the following APIs:

- ***HAL_DFSDM_FilterInit()***
- ***HAL_DFSDM_FilterDeInit()***
- ***HAL_DFSDM_FilterMspInit()***
- ***HAL_DFSDM_FilterMspDeInit()***

16.2.6 Filter control functions

This section provides functions allowing to:

- Select channel and enable/disable continuous mode for regular conversion.
- Select channels for injected conversion.

This section contains the following APIs:

- ***HAL_DFSDM_FilterConfigRegChannel()***
- ***HAL_DFSDM_FilterConfigInjChannel()***

16.2.7 Filter operation functions

This section provides functions allowing to:

- Start conversion of regular/injected channel.
- Poll for the end of regular/injected conversion.
- Stop conversion of regular/injected channel.
- Start conversion of regular/injected channel and enable interrupt.
- Call the callback functions at the end of regular/injected conversions.
- Stop conversion of regular/injected channel and disable interrupt.
- Start conversion of regular/injected channel and enable DMA transfer.
- Stop conversion of regular/injected channel and disable DMA transfer.
- Start analog watchdog and enable interrupt.
- Call the callback function when analog watchdog occurs.
- Stop analog watchdog and disable interrupt.
- Start extreme detector.
- Stop extreme detector.
- Get result of regular channel conversion.
- Get result of injected channel conversion.
- Get extreme detector maximum and minimum values.
- Get conversion time.
- Handle DFSDM interrupt request.

This section contains the following APIs:

- ***HAL_DFSDM_FilterRegularStart()***
- ***HAL_DFSDM_FilterPollForRegConversion()***
- ***HAL_DFSDM_FilterRegularStop()***
- ***HAL_DFSDM_FilterRegularStart_IT()***
- ***HAL_DFSDM_FilterRegularStop_IT()***
- ***HAL_DFSDM_FilterRegularStart_DMA()***
- ***HAL_DFSDM_FilterRegularMsbStart_DMA()***
- ***HAL_DFSDM_FilterRegularStop_DMA()***
- ***HAL_DFSDM_FilterGetRegularValue()***
- ***HAL_DFSDM_FilterInjectedStart()***
- ***HAL_DFSDM_FilterPollForInjConversion()***
- ***HAL_DFSDM_FilterInjectedStop()***
- ***HAL_DFSDM_FilterInjectedStart_IT()***
- ***HAL_DFSDM_FilterInjectedStop_IT()***
- ***HAL_DFSDM_FilterInjectedStart_DMA()***
- ***HAL_DFSDM_FilterInjectedMsbStart_DMA()***
- ***HAL_DFSDM_FilterInjectedStop_DMA()***
- ***HAL_DFSDM_FilterGetInjectedValue()***

- ***HAL_DFSDM_FilterAwdStart_IT()***
- ***HAL_DFSDM_FilterAwdStop_IT()***
- ***HAL_DFSDM_FilterExdStart()***
- ***HAL_DFSDM_FilterExdStop()***
- ***HAL_DFSDM_FilterGetExdMaxValue()***
- ***HAL_DFSDM_FilterGetExdMinValue()***
- ***HAL_DFSDM_FilterGetConvTimeValue()***
- ***HAL_DFSDM_IRQHandler()***
- ***HAL_DFSDM_FilterRegConvCpltCallback()***
- ***HAL_DFSDM_FilterRegConvHalfCpltCallback()***
- ***HAL_DFSDM_FilterInjConvCpltCallback()***
- ***HAL_DFSDM_FilterInjConvHalfCpltCallback()***
- ***HAL_DFSDM_FilterAwdCallback()***
- ***HAL_DFSDM_FilterErrorCallback()***

16.2.8 Filter state functions

This section provides functions allowing to:

- Get the DFSDM filter state.
- Get the DFSDM filter error.

This section contains the following APIs:

- ***HAL_DFSDM_FilterGetState()***
- ***HAL_DFSDM_FilterGetError()***

16.2.9 Detailed description of functions

HAL_DFSDM_ChannelInit

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelInit (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	Initialize the DFSDM channel according to the specified parameters in the DFSDM_ChannelInitTypeDef structure and initialize the associated handle.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **HAL:** status.

HAL_DFSDM_ChannelDeInit

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelDeInit (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	De-initialize the DFSDM channel.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **HAL:** status.

HAL_DFSDM_ChannelMspInit

Function Name	**void HAL_DFSDM_ChannelMspInit (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	Initialize the DFSDM channel MSP.

Parameters
- **hdfsdm_channel:** : DFSDM channel handle.

Return values
- **None**

HAL_DFSDM_ChannelMspDeInit

Function Name **void HAL_DFSDM_ChannelMspDeInit (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**

Function Description De-initialize the DFSDM channel MSP.

Parameters
- **hdfsdm_channel:** : DFSDM channel handle.

Return values
- **None**

HAL_DFSDM_ChannelCkabStart

Function Name **HAL_StatusTypeDef HAL_DFSDM_ChannelCkabStart (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**

Function Description This function allows to start clock absence detection in polling mode.

Parameters
- **hdfsdm_channel:** : DFSDM channel handle.

Return values
- **HAL:** status

Notes
- Same mode has to be used for all channels.
- If clock is not available on this channel during 5 seconds, clock absence detection will not be activated and function will return HAL_TIMEOUT error.

HAL_DFSDM_ChannelCkabStart_IT

Function Name **HAL_StatusTypeDef HAL_DFSDM_ChannelCkabStart_IT (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**

Function Description This function allows to start clock absence detection in interrupt mode.

Parameters
- **hdfsdm_channel:** : DFSDM channel handle.

Return values
- **HAL:** status

Notes
- Same mode has to be used for all channels.
- If clock is not available on this channel during 5 seconds, clock absence detection will not be activated and function will return HAL_TIMEOUT error.

HAL_DFSDM_ChannelCkabStop

Function Name **HAL_StatusTypeDef HAL_DFSDM_ChannelCkabStop (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**

Function Description This function allows to stop clock absence detection in polling mode.

Parameters
- **hdfsdm_channel:** : DFSDM channel handle.

Return values
- **HAL:** status

HAL_DFSDM_ChannelCkabStop_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelCkabStop_IT (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	This function allows to stop clock absence detection in interrupt mode.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **HAL:** status
Notes	• Interrupt will be disabled for all channels

HAL_DFSDM_ChannelScdStart

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelScdStart (DFSDM_Channel_HandleTypeDef * hdfsdm_channel, uint32_t Threshold, uint32_t BreakSignal)**
Function Description	This function allows to start short circuit detection in polling mode.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle. • **Threshold:** : Short circuit detector threshold. This parameter must be a number between Min_Data = 0 and Max_Data = 255. • **BreakSignal:** : Break signals assigned to short circuit event. This parameter can be a values combination of DFSDM break signals.
Return values	• **HAL:** status
Notes	• Same mode has to be used for all channels

HAL_DFSDM_ChannelScdStart_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelScdStart_IT (DFSDM_Channel_HandleTypeDef * hdfsdm_channel, uint32_t Threshold, uint32_t BreakSignal)**
Function Description	This function allows to start short circuit detection in interrupt mode.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle. • **Threshold:** : Short circuit detector threshold. This parameter must be a number between Min_Data = 0 and Max_Data = 255. • **BreakSignal:** : Break signals assigned to short circuit event. This parameter can be a values combination of DFSDM break signals.
Return values	• **HAL:** status
Notes	• Same mode has to be used for all channels

HAL_DFSDM_ChannelScdStop

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelScdStop (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**

Function Description	This function allows to stop short circuit detection in polling mode.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **HAL:** status

HAL_DFSDM_ChannelScdStop_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelScdStop_IT (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	This function allows to stop short circuit detection in interrupt mode.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **HAL:** status
Notes	• Interrupt will be disabled for all channels

HAL_DFSDM_ChannelGetAwdValue

Function Name	**int16_t HAL_DFSDM_ChannelGetAwdValue (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	This function allows to get channel analog watchdog value.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **Channel:** analog watchdog value.

HAL_DFSDM_ChannelModifyOffset

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelModifyOffset (DFSDM_Channel_HandleTypeDef * hdfsdm_channel, int32_t Offset)**
Function Description	This function allows to modify channel offset value.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle. • **Offset:** : DFSDM channel offset. This parameter must be a number between Min_Data = -8388608 and Max_Data = 8388607.
Return values	• **HAL:** status.

HAL_DFSDM_ChannelPollForCkab

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelPollForCkab (DFSDM_Channel_HandleTypeDef * hdfsdm_channel, uint32_t Timeout)**
Function Description	This function allows to poll for the clock absence detection.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle. • **Timeout:** : Timeout value in milliseconds.
Return values	• **HAL:** status

HAL_DFSDM_ChannelPollForScd

Function Name	**HAL_StatusTypeDef HAL_DFSDM_ChannelPollForScd (DFSDM_Channel_HandleTypeDef * hdfsdm_channel, uint32_t Timeout)**
Function Description	This function allows to poll for the short circuit detection.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle. • **Timeout:** : Timeout value in milliseconds.
Return values	• **HAL:** status

HAL_DFSDM_ChannelCkabCallback

Function Name	**void HAL_DFSDM_ChannelCkabCallback (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	Clock absence detection callback.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **None**

HAL_DFSDM_ChannelScdCallback

Function Name	**void HAL_DFSDM_ChannelScdCallback (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	Short circuit detection callback.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **None**

HAL_DFSDM_ChannelGetState

Function Name	**HAL_DFSDM_Channel_StateTypeDef HAL_DFSDM_ChannelGetState (DFSDM_Channel_HandleTypeDef * hdfsdm_channel)**
Function Description	This function allows to get the current DFSDM channel handle state.
Parameters	• **hdfsdm_channel:** : DFSDM channel handle.
Return values	• **DFSDM:** channel state.

HAL_DFSDM_FilterInit

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInit (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Initialize the DFSDM filter according to the specified parameters in the DFSDM_FilterInitTypeDef structure and initialize the associated handle.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status.

HAL_DFSDM_FilterDeInit

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterDeInit (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	De-initializes the DFSDM filter.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status.

HAL_DFSDM_FilterMspInit

Function Name	**void HAL_DFSDM_FilterMspInit (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Initializes the DFSDM filter MSP.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**

HAL_DFSDM_FilterMspDeInit

Function Name	**void HAL_DFSDM_FilterMspDeInit (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	De-initializes the DFSDM filter MSP.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**

HAL_DFSDM_FilterConfigRegChannel

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterConfigRegChannel (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t Channel, uint32_t ContinuousMode)**
Function Description	This function allows to select channel and to enable/disable continuous mode for regular conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Channel for regular conversion. This parameter can be a value of DFSDM Channel Selection. • **ContinuousMode:** : Enable/disable continuous mode for regular conversion. This parameter can be a value of DFSDM Continuous Mode.
Return values	• **HAL:** status

HAL_DFSDM_FilterConfigInjChannel

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterConfigInjChannel (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t Channel)**
Function Description	This function allows to select channels for injected conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Channels for injected conversion. This parameter

can be a values combination of DFSDM Channel Selection.

Return values
- **HAL:** status

HAL_DFSDM_FilterRegularStart

Function Name **HAL_StatusTypeDef HAL_DFSDM_FilterRegularStart (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**

Function Description This function allows to start regular conversion in polling mode.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.

Return values
- **HAL:** status

Notes
- This function should be called only when DFSDM filter instance is in idle state or if injected conversion is ongoing.

HAL_DFSDM_FilterRegularStart_IT

Function Name **HAL_StatusTypeDef HAL_DFSDM_FilterRegularStart_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**

Function Description This function allows to start regular conversion in interrupt mode.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.

Return values
- **HAL:** status

Notes
- This function should be called only when DFSDM filter instance is in idle state or if injected conversion is ongoing.

HAL_DFSDM_FilterRegularStart_DMA

Function Name **HAL_StatusTypeDef HAL_DFSDM_FilterRegularStart_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, int32_t * pData, uint32_t Length)**

Function Description This function allows to start regular conversion in DMA mode.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.
- **pData:** : The destination buffer address.
- **Length:** : The length of data to be transferred from DFSDM filter to memory.

Return values
- **HAL:** status

Notes
- This function should be called only when DFSDM filter instance is in idle state or if injected conversion is ongoing. Please note that data on buffer will contain signed regular conversion value on 24 most significant bits and corresponding channel on 3 least significant bits.

HAL_DFSDM_FilterRegularMsbStart_DMA

Function Name **HAL_StatusTypeDef HAL_DFSDM_FilterRegularMsbStart_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, int16_t * pData, uint32_t Length)**

Function Description	This function allows to start regular conversion in DMA mode and to get only the 16 most significant bits of conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **pData:** : The destination buffer address. • **Length:** : The length of data to be transferred from DFSDM filter to memory.
Return values	• **HAL:** status
Notes	• This function should be called only when DFSDM filter instance is in idle state or if injected conversion is ongoing. Please note that data on buffer will contain signed 16 most significant bits of regular conversion.

HAL_DFSDM_FilterRegularStop

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterRegularStop (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop regular conversion in polling mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if regular conversion is ongoing.

HAL_DFSDM_FilterRegularStop_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterRegularStop_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop regular conversion in interrupt mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if regular conversion is ongoing.

HAL_DFSDM_FilterRegularStop_DMA

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterRegularStop_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop regular conversion in DMA mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if regular conversion is ongoing.

HAL_DFSDM_FilterInjectedStart

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStart (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**

Function Description	This function allows to start injected conversion in polling mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only when DFSDM filter instance is in idle state or if regular conversion is ongoing.

HAL_DFSDM_FilterInjectedStart_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStart_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to start injected conversion in interrupt mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only when DFSDM filter instance is in idle state or if regular conversion is ongoing.

HAL_DFSDM_FilterInjectedStart_DMA

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStart_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, int32_t * pData, uint32_t Length)**
Function Description	This function allows to start injected conversion in DMA mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **pData:** : The destination buffer address. • **Length:** : The length of data to be transferred from DFSDM filter to memory.
Return values	• **HAL:** status
Notes	• This function should be called only when DFSDM filter instance is in idle state or if regular conversion is ongoing. Please note that data on buffer will contain signed injected conversion value on 24 most significant bits and corresponding channel on 3 least significant bits.

HAL_DFSDM_FilterInjectedMsbStart_DMA

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedMsbStart_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, int16_t * pData, uint32_t Length)**
Function Description	This function allows to start injected conversion in DMA mode and to get only the 16 most significant bits of conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **pData:** : The destination buffer address. • **Length:** : The length of data to be transferred from DFSDM filter to memory.
Return values	• **HAL:** status

Notes	• This function should be called only when DFSDM filter instance is in idle state or if regular conversion is ongoing. Please note that data on buffer will contain signed 16 most significant bits of injected conversion.

HAL_DFSDM_FilterInjectedStop

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStop (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop injected conversion in polling mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if injected conversion is ongoing.

HAL_DFSDM_FilterInjectedStop_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStop_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop injected conversion in interrupt mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if injected conversion is ongoing.

HAL_DFSDM_FilterInjectedStop_DMA

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterInjectedStop_DMA (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop injected conversion in DMA mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status
Notes	• This function should be called only if injected conversion is ongoing.

HAL_DFSDM_FilterAwdStart_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterAwdStart_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, DFSDM_Filter_AwdParamTypeDef * awdParam)**
Function Description	This function allows to start filter analog watchdog in interrupt mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **awdParam:** : DFSDM filter analog watchdog parameters.
Return values	• **HAL:** status

HAL_DFSDM_FilterAwdStop_IT

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterAwdStop_IT (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop filter analog watchdog in interrupt mode.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status

HAL_DFSDM_FilterExdStart

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterExdStart (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t Channel)**
Function Description	This function allows to start extreme detector feature.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Channels where extreme detector is enabled. This parameter can be a values combination of DFSDM Channel Selection.
Return values	• **HAL:** status

HAL_DFSDM_FilterExdStop

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterExdStop (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to stop extreme detector feature.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **HAL:** status

HAL_DFSDM_FilterGetRegularValue

Function Name	**int32_t HAL_DFSDM_FilterGetRegularValue (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t * Channel)**
Function Description	This function allows to get regular conversion value.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Corresponding channel of regular conversion.
Return values	• **Regular:** conversion value

HAL_DFSDM_FilterGetInjectedValue

Function Name	**int32_t HAL_DFSDM_FilterGetInjectedValue (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t * Channel)**
Function Description	This function allows to get injected conversion value.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Corresponding channel of injected conversion.

Return values
- **Injected:** conversion value

HAL_DFSDM_FilterGetExdMaxValue

Function Name **int32_t HAL_DFSDM_FilterGetExdMaxValue (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t * Channel)**

Function Description This function allows to get extreme detector maximum value.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.
- **Channel:** : Corresponding channel.

Return values
- **Extreme:** detector maximum value This value is between Min_Data = -8388608 and Max_Data = 8388607.

HAL_DFSDM_FilterGetExdMinValue

Function Name **int32_t HAL_DFSDM_FilterGetExdMinValue (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t * Channel)**

Function Description This function allows to get extreme detector minimum value.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.
- **Channel:** : Corresponding channel.

Return values
- **Extreme:** detector minimum value This value is between Min_Data = -8388608 and Max_Data = 8388607.

HAL_DFSDM_FilterGetConvTimeValue

Function Name **uint32_t HAL_DFSDM_FilterGetConvTimeValue (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**

Function Description This function allows to get conversion time value.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.

Return values
- **Conversion:** time value

Notes
- To get time in second, this value has to be divided by DFSDM clock frequency.

HAL_DFSDM_IRQHandler

Function Name **void HAL_DFSDM_IRQHandler (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**

Function Description This function handles the DFSDM interrupts.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.

Return values
- **None**

HAL_DFSDM_FilterPollForRegConversion

Function Name **HAL_StatusTypeDef HAL_DFSDM_FilterPollForRegConversion (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t**

Timeout)

Function Description	This function allows to poll for the end of regular conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Timeout:** : Timeout value in milliseconds.
Return values	• **HAL:** status
Notes	• This function should be called only if regular conversion is ongoing.

HAL_DFSDM_FilterPollForInjConversion

Function Name	**HAL_StatusTypeDef HAL_DFSDM_FilterPollForInjConversion (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t Timeout)**
Function Description	This function allows to poll for the end of injected conversion.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Timeout:** : Timeout value in milliseconds.
Return values	• **HAL:** status
Notes	• This function should be called only if injected conversion is ongoing.

HAL_DFSDM_FilterRegConvCpltCallback

Function Name	**void HAL_DFSDM_FilterRegConvCpltCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Regular conversion complete callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**
Notes	• In interrupt mode, user has to read conversion value in this function using HAL_DFSDM_FilterGetRegularValue.

HAL_DFSDM_FilterRegConvHalfCpltCallback

Function Name	**void HAL_DFSDM_FilterRegConvHalfCpltCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Half regular conversion complete callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**

HAL_DFSDM_FilterInjConvCpltCallback

Function Name	**void HAL_DFSDM_FilterInjConvCpltCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Injected conversion complete callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.

Return values	• **None**
Notes	• In interrupt mode, user has to read conversion value in this function using HAL_DFSDM_FilterGetInjectedValue.

HAL_DFSDM_FilterInjConvHalfCpltCallback

Function Name	**void HAL_DFSDM_FilterInjConvHalfCpltCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Half injected conversion complete callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**

HAL_DFSDM_FilterAwdCallback

Function Name	**void HAL_DFSDM_FilterAwdCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter, uint32_t Channel, uint32_t Threshold)**
Function Description	Filter analog watchdog callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle. • **Channel:** : Corresponding channel. • **Threshold:** : Low or high threshold has been reached.
Return values	• **None**

HAL_DFSDM_FilterErrorCallback

Function Name	**void HAL_DFSDM_FilterErrorCallback (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	Error callback.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **None**

HAL_DFSDM_FilterGetState

Function Name	**HAL_DFSDM_Filter_StateTypeDef HAL_DFSDM_FilterGetState (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to get the current DFSDM filter handle state.
Parameters	• **hdfsdm_filter:** : DFSDM filter handle.
Return values	• **DFSDM:** filter state.

HAL_DFSDM_FilterGetError

Function Name	**uint32_t HAL_DFSDM_FilterGetError (DFSDM_Filter_HandleTypeDef * hdfsdm_filter)**
Function Description	This function allows to get the current DFSDM filter error.

Parameters
- **hdfsdm_filter:** : DFSDM filter handle.

Return values
- **DFSDM:** filter error code.

16.3 DFSDM Firmware driver defines

16.3.1 DFSDM

DFSDM analog watchdog threshold

DFSDM_AWD_HIGH_THRESHOLD	Analog watchdog high threshold
DFSDM_AWD_LOW_THRESHOLD	Analog watchdog low threshold

DFSDM break signals

DFSDM_NO_BREAK_SIGNAL	No break signal
DFSDM_BREAK_SIGNAL_0	Break signal 0
DFSDM_BREAK_SIGNAL_1	Break signal 1
DFSDM_BREAK_SIGNAL_2	Break signal 2
DFSDM_BREAK_SIGNAL_3	Break signal 3

DFSDM channel analog watchdog filter order

DFSDM_CHANNEL_FASTSINC_ORDER	FastSinc filter type
DFSDM_CHANNEL_SINC1_ORDER	Sinc 1 filter type
DFSDM_CHANNEL_SINC2_ORDER	Sinc 2 filter type
DFSDM_CHANNEL_SINC3_ORDER	Sinc 3 filter type

DFSDM channel input data packing

DFSDM_CHANNEL_STANDARD_MODE	Standard data packing mode
DFSDM_CHANNEL_INTERLEAVED_MODE	Interleaved data packing mode
DFSDM_CHANNEL_DUAL_MODE	Dual data packing mode

DFSDM channel input multiplexer

DFSDM_CHANNEL_EXTERNAL_INPUTS	Data are taken from external inputs
DFSDM_CHANNEL_INTERNAL_REGISTER	Data are taken from internal register

DFSDM channel input pins

DFSDM_CHANNEL_SAME_CHANNEL_PINS	Input from pins on same channel
DFSDM_CHANNEL_FOLLOWING_CHANNEL_PINS	Input from pins on following channel

DFSDM channel output clock selection

DFSDM_CHANNEL_OUTPUT_CLOCK_SYSTEM	Source for ouput clock is system clock
DFSDM_CHANNEL_OUTPUT_CLOCK_AUDIO	Source for ouput clock is audio clock

DFSDM Channel Selection

DFSDM_CHANNEL_0

DFSDM_CHANNEL_1

DFSDM_CHANNEL_2

DFSDM_CHANNEL_3

DFSDM channel serial interface type

DFSDM_CHANNEL_SPI_RISING	SPI with rising edge
DFSDM_CHANNEL_SPI_FALLING	SPI with falling edge
DFSDM_CHANNEL_MANCHESTER_RISING	Manchester with rising edge
DFSDM_CHANNEL_MANCHESTER_FALLING	Manchester with falling edge

DFSDM channel SPI clock selection

DFSDM_CHANNEL_SPI_CLOCK_EXTERNAL	External SPI clock
DFSDM_CHANNEL_SPI_CLOCK_INTERNAL	Internal SPI clock
DFSDM_CHANNEL_SPI_CLOCK_INTERNAL_DIV2_FALLING	Internal SPI clock divided by 2, falling edge
DFSDM_CHANNEL_SPI_CLOCK_INTERNAL_DIV2_RISING	Internal SPI clock divided by 2, rising edge

DFSDM Continuous Mode

DFSDM_CONTINUOUS_CONV_OFF	Conversion are not continuous
DFSDM_CONTINUOUS_CONV_ON	Conversion are continuous

DFSDM Exported Macros

__HAL_DFSDM_CHANNEL_RESET_HANDLE_STATE

Description:

- Reset DFSDM channel handle state.

Parameters:

- __HANDLE__: DFSDM channel handle.

Return value:

- None

__HAL_DFSDM_FILTER_RESET_HANDLE_STATE

Description:

- Reset DFSDM filter handle state.

Parameters:

- __HANDLE__: DFSDM filter handle.

Return value:

- None

DFSDM filter analog watchdog data source

DFSDM_FILTER_AWD_FILTER_DATA	From digital filter
DFSDM_FILTER_AWD_CHANNEL_DATA	From analog watchdog channel

DFSDM filter error code

DFSDM_FILTER_ERROR_NONE	No error
DFSDM_FILTER_ERROR_REGULAR_OVERRUN	Overrun occurs during regular conversion
DFSDM_FILTER_ERROR_INJECTED_OVERRUN	Overrun occurs during injected conversion
DFSDM_FILTER_ERROR_DMA	DMA error occurs

DFSDM filter external trigger

DFSDM_FILTER_EXT_TRIG_TIM1_TRGO	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_TIM3_TRGO	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_TIM8_TRGO	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_TIM10_OC1	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_TIM4_TRGO	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_TIM6_TRGO	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_EXTI11	For DFSDM filter 0 and 1
DFSDM_FILTER_EXT_TRIG_EXTI15	For DFSDM filter 0 and 1

DFSDM filter external trigger edge

DFSDM_FILTER_EXT_TRIG_RISING_EDGE	External rising edge
DFSDM_FILTER_EXT_TRIG_FALLING_EDGE	External falling edge
DFSDM_FILTER_EXT_TRIG_BOTH_EDGES	External rising and falling edges

DFSDM filter sinc order

DFSDM_FILTER_FASTSINC_ORDER	FastSinc filter type
DFSDM_FILTER_SINC1_ORDER	Sinc 1 filter type
DFSDM_FILTER_SINC2_ORDER	Sinc 2 filter type
DFSDM_FILTER_SINC3_ORDER	Sinc 3 filter type
DFSDM_FILTER_SINC4_ORDER	Sinc 4 filter type
DFSDM_FILTER_SINC5_ORDER	Sinc 5 filter type

DFSDM filter conversion trigger

DFSDM_FILTER_SW_TRIGGER	Software trigger
DFSDM_FILTER_SYNC_TRIGGER	Synchronous with DFSDM_FLT0
DFSDM_FILTER_EXT_TRIGGER	External trigger (only for injected conversion)

17 HAL DMA2D Generic Driver

17.1 DMA2D Firmware driver registers structures

17.1.1 DMA2D_ColorTypeDef

Data Fields

- ***uint32_t Blue***
- ***uint32_t Green***
- ***uint32_t Red***

Field Documentation

- ***uint32_t DMA2D_ColorTypeDef::Blue***
  Configures the blue value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint32_t DMA2D_ColorTypeDef::Green***
  Configures the green value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint32_t DMA2D_ColorTypeDef::Red***
  Configures the red value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.

17.1.2 DMA2D_CLUTCfgTypeDef

Data Fields

- ***uint32_t * pCLUT***
- ***uint32_t CLUTColorMode***
- ***uint32_t Size***

Field Documentation

- ***uint32_t* DMA2D_CLUTCfgTypeDef::pCLUT***
  Configures the DMA2D CLUT memory address.
- ***uint32_t DMA2D_CLUTCfgTypeDef::CLUTColorMode***
  Configures the DMA2D CLUT color mode. This parameter can be one value of ***DMA2D_CLUT_CM***.
- ***uint32_t DMA2D_CLUTCfgTypeDef::Size***
  Configures the DMA2D CLUT size. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.

17.1.3 DMA2D_InitTypeDef

Data Fields

- ***uint32_t Mode***
- ***uint32_t ColorMode***
- ***uint32_t OutputOffset***

Field Documentation

- ***uint32_t DMA2D_InitTypeDef::Mode***
  Configures the DMA2D transfer mode. This parameter can be one value of ***DMA2D_Mode***.
- ***uint32_t DMA2D_InitTypeDef::ColorMode***
  Configures the color format of the output image. This parameter can be one value of ***DMA2D_Output_Color_Mode***.
- ***uint32_t DMA2D_InitTypeDef::OutputOffset***
  Specifies the Offset value. This parameter must be a number between Min_Data = 0x0000 and Max_Data = 0x3FFF.

17.1.4 DMA2D_LayerCfgTypeDef

Data Fields

- ***uint32_t InputOffset***
- ***uint32_t InputColorMode***
- ***uint32_t AlphaMode***
- ***uint32_t InputAlpha***

Field Documentation

- ***uint32_t DMA2D_LayerCfgTypeDef::InputOffset***
  Configures the DMA2D foreground or background offset. This parameter must be a number between Min_Data = 0x0000 and Max_Data = 0x3FFF.
- ***uint32_t DMA2D_LayerCfgTypeDef::InputColorMode***
  Configures the DMA2D foreground or background color mode. This parameter can be one value of ***DMA2D_Input_Color_Mode***.
- ***uint32_t DMA2D_LayerCfgTypeDef::AlphaMode***
  Configures the DMA2D foreground or background alpha mode. This parameter can be one value of ***DMA2D_Alpha_Mode***.
- ***uint32_t DMA2D_LayerCfgTypeDef::InputAlpha***
  Specifies the DMA2D foreground or background alpha value and color value in case of A8 or A4 color mode. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF except for the color modes detailed below.
  Note: In case of A8 or A4 color mode (ARGB), this parameter must be a number between Min_Data = 0x00000000 and Max_Data = 0xFFFFFFFF whereInputAlpha[24:31] is the alpha value ALPHA[0:7]InputAlpha[16:23] is the red value RED[0:7]InputAlpha[8:15] is the green value GREEN[0:7]InputAlpha[0:7] is the blue value BLUE[0:7].

17.1.5 __DMA2D_HandleTypeDef

Data Fields

- ***DMA2D_TypeDef * Instance***
- ***DMA2D_InitTypeDef Init***
- ***void(* XferCpltCallback***
- ***void(* XferErrorCallback***
- ***DMA2D_LayerCfgTypeDef LayerCfg***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_DMA2D_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***DMA2D_TypeDef* __DMA2D_HandleTypeDef::Instance***
  DMA2D register base address.
- ***DMA2D_InitTypeDef __DMA2D_HandleTypeDef::Init***
  DMA2D communication parameters.
- ***void(* __DMA2D_HandleTypeDef::XferCpltCallback)(struct __DMA2D_HandleTypeDef *hdma2d)***
  DMA2D transfer complete callback.
- ***void(* __DMA2D_HandleTypeDef::XferErrorCallback)(struct __DMA2D_HandleTypeDef *hdma2d)***
  DMA2D transfer error callback.
- ***DMA2D_LayerCfgTypeDef __DMA2D_HandleTypeDef::LayerCfg[MAX_DMA2D_LAYER]***
  DMA2D Layers parameters
- ***HAL_LockTypeDef __DMA2D_HandleTypeDef::Lock***
  DMA2D lock.
- ***__IO HAL_DMA2D_StateTypeDef __DMA2D_HandleTypeDef::State***
  DMA2D transfer state.
- ***__IO uint32_t __DMA2D_HandleTypeDef::ErrorCode***
  DMA2D error code.

17.2 DMA2D Firmware driver API description

17.2.1 How to use this driver

1. Program the required configuration through the following parameters: the transfer mode, the output color mode and the output offset using HAL_DMA2D_Init() function.
2. Program the required configuration through the following parameters: the input color mode, the input color, the input alpha value, the alpha mode, and the input offset using HAL_DMA2D_ConfigLayer() function for foreground or/and background layer.

Polling mode IO operation

1. Configure pdata parameter (explained hereafter), destination and data length and enable the transfer using HAL_DMA2D_Start().
2. Wait for end of transfer using HAL_DMA2D_PollForTransfer(), at this stage user can specify the value of timeout according to his end application.

Interrupt mode IO operation

1. Configure pdata parameter, destination and data length and enable the transfer using HAL_DMA2D_Start_IT().
2. Use HAL_DMA2D_IRQHandler() called under DMA2D_IRQHandler() interrupt subroutine
3. At the end of data transfer HAL_DMA2D_IRQHandler() function is executed and user can add his own function by customization of function pointer XferCpltCallback (member of DMA2D handle structure).
4. In case of error, the HAL_DMA2D_IRQHandler() function will call the callback XferErrorCallback. In Register-to-Memory transfer mode, pdata parameter is the register color, in Memory-to-memory or Memory-to-Memory with pixel format conversion pdata is the source address. Configure the foreground source address, the background source address, the destination and data length then Enable the transfer using HAL_DMA2D_BlendingStart() in polling mode and HAL_DMA2D_BlendingStart_IT() in interrupt mode HAL_DMA2D_BlendingStart() and HAL_DMA2D_BlendingStart_IT() functions are used if the memory to memory with blending transfer mode is selected.
5. Optionally, configure and enable the CLUT using HAL_DMA2D_CLUTLoad() in polling mode or HAL_DMA2D_CLUTLoad_IT() in interrupt mode.
6. Optionally, configure the line watermark in using the API HAL_DMA2D_ProgramLineEvent()
7. Optionally, configure the dead time value in the AHB clock cycle inserted between two consecutive accesses on the AHB master port in using the API HAL_DMA2D_ConfigDeadTime() and enable/disable the functionality with the APIs HAL_DMA2D_EnableDeadTime() or HAL_DMA2D_DisableDeadTime().
8. The transfer can be suspended, resumed and aborted using the following functions: HAL_DMA2D_Suspend(), HAL_DMA2D_Resume(), HAL_DMA2D_Abort().
9. The CLUT loading can be suspended, resumed and aborted using the following functions: HAL_DMA2D_CLUTLoading_Suspend(), HAL_DMA2D_CLUTLoading_Resume(), HAL_DMA2D_CLUTLoading_Abort().
10. To control the DMA2D state, use the following function: HAL_DMA2D_GetState().
11. To read the DMA2D error code, use the following function: HAL_DMA2D_GetError().

DMA2D HAL driver macros list

Below the list of most used macros in DMA2D HAL driver :

- __HAL_DMA2D_ENABLE: Enable the DMA2D peripheral.
- __HAL_DMA2D_GET_FLAG: Get the DMA2D pending flags.
- __HAL_DMA2D_CLEAR_FLAG: Clear the DMA2D pending flags.
- __HAL_DMA2D_ENABLE_IT: Enable the specified DMA2D interrupts.
- __HAL_DMA2D_DISABLE_IT: Disable the specified DMA2D interrupts.
- __HAL_DMA2D_GET_IT_SOURCE: Check whether the specified DMA2D interrupt is enabled or not

You can refer to the DMA2D HAL driver header file for more useful macros

17.2.2 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and configure the DMA2D
- De-initialize the DMA2D

This section contains the following APIs:

- ***HAL_DMA2D_Init()***
- ***HAL_DMA2D_DeInit()***
- ***HAL_DMA2D_MspInit()***
- ***HAL_DMA2D_MspDeInit()***

17.2.3 IO operation functions

This section provides functions allowing to:

- Configure the pdata, destination address and data size then start the DMA2D transfer.
- Configure the source for foreground and background, destination address and data size then start a MultiBuffer DMA2D transfer.
- Configure the pdata, destination address and data size then start the DMA2D transfer with interrupt.
- Configure the source for foreground and background, destination address and data size then start a MultiBuffer DMA2D transfer with interrupt.
- Abort DMA2D transfer.
- Suspend DMA2D transfer.
- Resume DMA2D transfer.
- Enable CLUT transfer.
- Configure CLUT loading then start transfer in polling mode.
- Configure CLUT loading then start transfer in interrupt mode.
- Abort DMA2D CLUT loading.
- Suspend DMA2D CLUT loading.
- Resume DMA2D CLUT loading.
- Poll for transfer complete.
- handle DMA2D interrupt request.
- Transfer watermark callback.
- CLUT Transfer Complete callback.

This section contains the following APIs:

- ***HAL_DMA2D_Start()***
- ***HAL_DMA2D_Start_IT()***
- ***HAL_DMA2D_BlendingStart()***
- ***HAL_DMA2D_BlendingStart_IT()***
- ***HAL_DMA2D_Abort()***
- ***HAL_DMA2D_Suspend()***
- ***HAL_DMA2D_Resume()***
- ***HAL_DMA2D_EnableCLUT()***
- ***HAL_DMA2D_CLUTLoad()***

- ***HAL_DMA2D_CLUTLoad_IT()***
- ***HAL_DMA2D_CLUTLoading_Abort()***
- ***HAL_DMA2D_CLUTLoading_Suspend()***
- ***HAL_DMA2D_CLUTLoading_Resume()***
- ***HAL_DMA2D_PollForTransfer()***
- ***HAL_DMA2D_IRQHandler()***
- ***HAL_DMA2D_LineEventCallback()***
- ***HAL_DMA2D_CLUTLoadingCpltCallback()***

17.2.4 Peripheral Control functions

This section provides functions allowing to:

- Configure the DMA2D foreground or background layer parameters.
- Configure the DMA2D CLUT transfer.
- Configure the line watermark
- Configure the dead time value.
- Enable or disable the dead time value functionality.

This section contains the following APIs:

- ***HAL_DMA2D_ConfigLayer()***
- ***HAL_DMA2D_ConfigCLUT()***
- ***HAL_DMA2D_ProgramLineEvent()***
- ***HAL_DMA2D_EnableDeadTime()***
- ***HAL_DMA2D_DisableDeadTime()***
- ***HAL_DMA2D_ConfigDeadTime()***

17.2.5 Peripheral State and Errors functions

This subsection provides functions allowing to :

- Get the DMA2D state
- Get the DMA2D error code

This section contains the following APIs:

- ***HAL_DMA2D_GetState()***
- ***HAL_DMA2D_GetError()***

17.2.6 Detailed description of functions

HAL_DMA2D_Init

Function Name	**HAL_StatusTypeDef HAL_DMA2D_Init (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Initialize the DMA2D according to the specified parameters in the DMA2D_InitTypeDef and create the associated handle.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **HAL:** status

HAL_DMA2D_DeInit

Function Name	**HAL_StatusTypeDef HAL_DMA2D_DeInit (DMA2D_HandleTypeDef * hdma2d)**

Function Description	Deinitializes the DMA2D peripheral registers to their default reset values.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **None**

HAL_DMA2D_MspInit

Function Name	**void HAL_DMA2D_MspInit (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Initializes the DMA2D MSP.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **None**

HAL_DMA2D_MspDeInit

Function Name	**void HAL_DMA2D_MspDeInit (DMA2D_HandleTypeDef * hdma2d)**
Function Description	DeInitializes the DMA2D MSP.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **None**

HAL_DMA2D_Start

Function Name	**HAL_StatusTypeDef HAL_DMA2D_Start (DMA2D_HandleTypeDef * hdma2d, uint32_t pdata, uint32_t DstAddress, uint32_t Width, uint32_t Height)**
Function Description	Start the DMA2D Transfer.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **pdata:** Configure the source memory Buffer address if Memory-to-Memory or Memory-to-Memory with pixel format conversion mode is selected, or configure the color value if Register-to-Memory mode is selected. • **DstAddress:** The destination memory Buffer address. • **Width:** The width of data to be transferred from source to destination (expressed in number of pixels per line). • **Height:** The height of data to be transferred from source to destination (expressed in number of lines).
Return values	• **HAL:** status

HAL_DMA2D_BlendingStart

Function Name **HAL_StatusTypeDef HAL_DMA2D_BlendingStart (DMA2D_HandleTypeDef * hdma2d, uint32_t SrcAddress1, uint32_t SrcAddress2, uint32_t DstAddress, uint32_t Width, uint32_t Height)**

Function Description Start the multi-source DMA2D Transfer.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **SrcAddress1:** The source memory Buffer address for the foreground layer.
- **SrcAddress2:** The source memory Buffer address for the background layer.
- **DstAddress:** The destination memory Buffer address.
- **Width:** The width of data to be transferred from source to destination (expressed in number of pixels per line).
- **Height:** The height of data to be transferred from source to destination (expressed in number of lines).

Return values
- **HAL:** status

HAL_DMA2D_Start_IT

Function Name **HAL_StatusTypeDef HAL_DMA2D_Start_IT (DMA2D_HandleTypeDef * hdma2d, uint32_t pdata, uint32_t DstAddress, uint32_t Width, uint32_t Height)**

Function Description Start the DMA2D Transfer with interrupt enabled.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **pdata:** Configure the source memory Buffer address if the Memory-to-Memory or Memory-to-Memory with pixel format conversion mode is selected, or configure the color value if Register-to-Memory mode is selected.
- **DstAddress:** The destination memory Buffer address.
- **Width:** The width of data to be transferred from source to destination (expressed in number of pixels per line).
- **Height:** The height of data to be transferred from source to destination (expressed in number of lines).

Return values
- **HAL:** status

HAL_DMA2D_BlendingStart_IT

Function Name **HAL_StatusTypeDef HAL_DMA2D_BlendingStart_IT (DMA2D_HandleTypeDef * hdma2d, uint32_t SrcAddress1, uint32_t SrcAddress2, uint32_t DstAddress, uint32_t Width, uint32_t Height)**

Function Description Start the multi-source DMA2D Transfer with interrupt enabled.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **SrcAddress1:** The source memory Buffer address for the foreground layer.

- **SrcAddress2:** The source memory Buffer address for the background layer.
- **DstAddress:** The destination memory Buffer address.
- **Width:** The width of data to be transferred from source to destination (expressed in number of pixels per line).
- **Height:** The height of data to be transferred from source to destination (expressed in number of lines).

Return values
- **HAL:** status

HAL_DMA2D_Suspend

Function Name **HAL_StatusTypeDef HAL_DMA2D_Suspend (DMA2D_HandleTypeDef * hdma2d)**

Function Description Suspend the DMA2D Transfer.

Parameters
- **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.

Return values
- **HAL:** status

HAL_DMA2D_Resume

Function Name **HAL_StatusTypeDef HAL_DMA2D_Resume (DMA2D_HandleTypeDef * hdma2d)**

Function Description Resume the DMA2D Transfer.

Parameters
- **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.

Return values
- **HAL:** status

HAL_DMA2D_Abort

Function Name **HAL_StatusTypeDef HAL_DMA2D_Abort (DMA2D_HandleTypeDef * hdma2d)**

Function Description Abort the DMA2D Transfer.

Parameters
- **hdma2d:** : pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.

Return values
- **HAL:** status

HAL_DMA2D_EnableCLUT

Function Name **HAL_StatusTypeDef HAL_DMA2D_EnableCLUT (DMA2D_HandleTypeDef * hdma2d, uint32_t LayerIdx)**

Function Description Enable the DMA2D CLUT Transfer.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)

Return values
- **HAL:** status

HAL_DMA2D_CLUTLoad

Function Name **HAL_StatusTypeDef HAL_DMA2D_CLUTLoad (DMA2D_HandleTypeDef * hdma2d, DMA2D_CLUTCfgTypeDef CLUTCfg, uint32_t LayerIdx)**

Function Description Start DMA2D CLUT Loading.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **CLUTCfg:** Pointer to a DMA2D_CLUTCfgTypeDef structure that contains the configuration information for the color look up table.
- **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)

Return values
- **HAL:** status

Notes
- Invoking this API is similar to calling HAL_DMA2D_ConfigCLUT() then HAL_DMA2D_EnableCLUT().

HAL_DMA2D_CLUTLoad_IT

Function Name **HAL_StatusTypeDef HAL_DMA2D_CLUTLoad_IT (DMA2D_HandleTypeDef * hdma2d, DMA2D_CLUTCfgTypeDef CLUTCfg, uint32_t LayerIdx)**

Function Description Start DMA2D CLUT Loading with interrupt enabled.

Parameters
- **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **CLUTCfg:** Pointer to a DMA2D_CLUTCfgTypeDef structure that contains the configuration information for the color look up table.
- **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)

Return values
- **HAL:** status

HAL_DMA2D_CLUTLoading_Abort

Function Name **HAL_StatusTypeDef HAL_DMA2D_CLUTLoading_Abort (DMA2D_HandleTypeDef * hdma2d, uint32_t LayerIdx)**

Function Description Abort the DMA2D CLUT loading.

Parameters
- **hdma2d:** : Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
- **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)

Return values
- **HAL:** status

HAL_DMA2D_CLUTLoading_Suspend

Function Name **HAL_StatusTypeDef HAL_DMA2D_CLUTLoading_Suspend (DMA2D_HandleTypeDef * hdma2d, uint32_t LayerIdx)**

Function Description	Suspend the DMA2D CLUT loading.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)
Return values	• **HAL:** status

HAL_DMA2D_CLUTLoading_Resume

Function Name	**HAL_StatusTypeDef HAL_DMA2D_CLUTLoading_Resume (DMA2D_HandleTypeDef * hdma2d, uint32_t LayerIdx)**
Function Description	Resume the DMA2D CLUT loading.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)
Return values	• **HAL:** status

HAL_DMA2D_PollForTransfer

Function Name	**HAL_StatusTypeDef HAL_DMA2D_PollForTransfer (DMA2D_HandleTypeDef * hdma2d, uint32_t Timeout)**
Function Description	Polling for transfer complete or CLUT loading.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_DMA2D_IRQHandler

Function Name	**void HAL_DMA2D_IRQHandler (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Handle DMA2D interrupt request.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **HAL:** status

HAL_DMA2D_LineEventCallback

Function Name	**void HAL_DMA2D_LineEventCallback (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Transfer watermark callback.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **None**

HAL_DMA2D_CLUTLoadingCpltCallback

Function Name	**void HAL_DMA2D_CLUTLoadingCpltCallback (DMA2D_HandleTypeDef * hdma2d)**
Function Description	CLUT Transfer Complete callback.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **None**

HAL_DMA2D_ConfigLayer

Function Name	**HAL_StatusTypeDef HAL_DMA2D_ConfigLayer (DMA2D_HandleTypeDef * hdma2d, uint32_t LayerIdx)**
Function Description	Configure the DMA2D Layer according to the specified parameters in the DMA2D_InitTypeDef and create the associated handle.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)
Return values	• **HAL:** status

HAL_DMA2D_ConfigCLUT

Function Name	**HAL_StatusTypeDef HAL_DMA2D_ConfigCLUT (DMA2D_HandleTypeDef * hdma2d, DMA2D_CLUTCfgTypeDef CLUTCfg, uint32_t LayerIdx)**
Function Description	Configure the DMA2D CLUT Transfer.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **CLUTCfg:** Pointer to a DMA2D_CLUTCfgTypeDef structure that contains the configuration information for the color look up table. • **LayerIdx:** DMA2D Layer index. This parameter can be one of the following values: 0(background) / 1(foreground)
Return values	• **HAL:** status

HAL_DMA2D_ProgramLineEvent

Function Name	**HAL_StatusTypeDef HAL_DMA2D_ProgramLineEvent (DMA2D_HandleTypeDef * hdma2d, uint32_t Line)**
Function Description	Configure the line watermark.
Parameters	• **hdma2d:** Pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D. • **Line:** Line Watermark configuration (maximum 16-bit long value expected).
Return values	• **HAL:** status
Notes	• HAL_DMA2D_ProgramLineEvent() API enables the transfer watermark interrupt.

- The transfer watermark interrupt is disabled once it has occurred.

HAL_DMA2D_EnableDeadTime

Function Name	**HAL_StatusTypeDef HAL_DMA2D_EnableDeadTime (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Enable DMA2D dead time feature.
Parameters	• **hdma2d:** DMA2D handle.
Return values	• **HAL:** status

HAL_DMA2D_DisableDeadTime

Function Name	**HAL_StatusTypeDef HAL_DMA2D_DisableDeadTime (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Disable DMA2D dead time feature.
Parameters	• **hdma2d:** DMA2D handle.
Return values	• **HAL:** status

HAL_DMA2D_ConfigDeadTime

Function Name	**HAL_StatusTypeDef HAL_DMA2D_ConfigDeadTime (DMA2D_HandleTypeDef * hdma2d, uint8_t DeadTime)**
Function Description	Configure dead time.
Parameters	• **hdma2d:** DMA2D handle. • **DeadTime:** dead time value.
Return values	• **HAL:** status
Notes	• The dead time value represents the guaranteed minimum number of cycles between two consecutive transactions on the AHB bus.

HAL_DMA2D_GetState

Function Name	**HAL_DMA2D_StateTypeDef HAL_DMA2D_GetState (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Return the DMA2D state.
Parameters	• **hdma2d:** pointer to a DMA2D_HandleTypeDef structure that contains the configuration information for the DMA2D.
Return values	• **HAL:** state

HAL_DMA2D_GetError

Function Name	**uint32_t HAL_DMA2D_GetError (DMA2D_HandleTypeDef * hdma2d)**
Function Description	Return the DMA2D error code.
Parameters	• **hdma2d:** : pointer to a DMA2D_HandleTypeDef structure

that contains the configuration information for DMA2D.

Return values
- **DMA2D:** Error Code

17.3 DMA2D Firmware driver defines

17.3.1 DMA2D

DMA2D API Aliases

HAL_DMA2D_DisableCLUT	Aliased to HAL_DMA2D_CLUTLoading_Abort for compatibility with legacy code

DMA2D Alpha Mode

DMA2D_NO_MODIF_ALPHA	No modification of the alpha channel value
DMA2D_REPLACE_ALPHA	Replace original alpha channel value by programmed alpha value
DMA2D_COMBINE_ALPHA	Replace original alpha channel value by programmed alpha value with original alpha channel value

DMA2D CLUT Color Mode

DMA2D_CCM_ARGB8888	ARGB8888 DMA2D CLUT color mode
DMA2D_CCM_RGB888	RGB888 DMA2D CLUT color mode

DMA2D CLUT Size

DMA2D_CLUT_SIZE	DMA2D CLUT size

DMA2D Color Value

DMA2D_COLOR_VALUE	Color value mask

DMA2D Error Code

HAL_DMA2D_ERROR_NONE	No error
HAL_DMA2D_ERROR_TE	Transfer error
HAL_DMA2D_ERROR_CE	Configuration error
HAL_DMA2D_ERROR_CAE	CLUT access error
HAL_DMA2D_ERROR_TIMEOUT	Timeout error

DMA2D Exported Macros

__HAL_DMA2D_RESET_HANDLE_STATE

Description:
- Reset DMA2D handle state.

Parameters:
- __HANDLE__: specifies the DMA2D handle.

Return value:
- None

__HAL_DMA2D_ENABLE

Description:
- Enable the DMA2D.

Parameters:

- __HANDLE__: DMA2D handle

Return value:

- None.

__HAL_DMA2D_GET_FLAG

Description:

- Get the DMA2D pending flags.

Parameters:

- __HANDLE__: DMA2D handle
- __FLAG__: flag to check. This parameter can be any combination of the following values:
  - DMA2D_FLAG_CE: Configuration error flag
  - DMA2D_FLAG_CTC: CLUT transfer complete flag
  - DMA2D_FLAG_CAE: CLUT access error flag
  - DMA2D_FLAG_TW: Transfer Watermark flag
  - DMA2D_FLAG_TC: Transfer complete flag
  - DMA2D_FLAG_TE: Transfer error flag

Return value:

- The: state of FLAG.

__HAL_DMA2D_CLEAR_FLAG

Description:

- Clear the DMA2D pending flags.

Parameters:

- __HANDLE__: DMA2D handle
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - DMA2D_FLAG_CE: Configuration error flag
  - DMA2D_FLAG_CTC: CLUT transfer complete flag
  - DMA2D_FLAG_CAE: CLUT access error flag
  - DMA2D_FLAG_TW: Transfer Watermark flag
  - DMA2D_FLAG_TC: Transfer complete flag
  - DMA2D_FLAG_TE: Transfer error flag

Return value:

- None

__HAL_DMA2D_ENABLE_IT

Description:

- Enable the specified DMA2D interrupts.

Parameters:

- __HANDLE__: DMA2D handle
- __INTERRUPT__: specifies the DMA2D interrupt sources to be enabled. This parameter can be any combination of the following values:
  - DMA2D_IT_CE: Configuration error interrupt mask
  - DMA2D_IT_CTC: CLUT transfer complete interrupt mask
  - DMA2D_IT_CAE: CLUT access error interrupt mask
  - DMA2D_IT_TW: Transfer Watermark interrupt mask
  - DMA2D_IT_TC: Transfer complete interrupt mask
  - DMA2D_IT_TE: Transfer error interrupt mask

Return value:

- None

__HAL_DMA2D_DISABLE_IT

Description:

- Disable the specified DMA2D interrupts.

Parameters:

- __HANDLE__: DMA2D handle
- __INTERRUPT__: specifies the DMA2D interrupt sources to be disabled. This parameter can be any combination of the following values:
  - DMA2D_IT_CE: Configuration error interrupt mask
  - DMA2D_IT_CTC: CLUT transfer complete interrupt mask
  - DMA2D_IT_CAE: CLUT access error interrupt mask
  - DMA2D_IT_TW: Transfer Watermark interrupt mask
  - DMA2D_IT_TC: Transfer complete interrupt mask
  - DMA2D_IT_TE: Transfer error interrupt mask

Return value:

- None

__HAL_DMA2D_GET_IT_SOURCE

Description:

- Check whether the specified DMA2D interrupt source is enabled or not.

Parameters:

- __HANDLE__: DMA2D handle
- __INTERRUPT__: specifies the DMA2D interrupt source to check. This parameter can be one of the following values:
  - DMA2D_IT_CE: Configuration error interrupt mask
  - DMA2D_IT_CTC: CLUT transfer complete interrupt mask
  - DMA2D_IT_CAE: CLUT access error interrupt mask
  - DMA2D_IT_TW: Transfer Watermark interrupt mask
  - DMA2D_IT_TC: Transfer complete interrupt mask
  - DMA2D_IT_TE: Transfer error interrupt mask

Return value:

- The: state of INTERRUPT source.

DMA2D Exported Types

MAX_DMA2D_LAYER

DMA2D Flags

DMA2D_FLAG_CE	Configuration Error Interrupt Flag
DMA2D_FLAG_CTC	CLUT Transfer Complete Interrupt Flag
DMA2D_FLAG_CAE	CLUT Access Error Interrupt Flag
DMA2D_FLAG_TW	Transfer Watermark Interrupt Flag
DMA2D_FLAG_TC	Transfer Complete Interrupt Flag
DMA2D_FLAG_TE	Transfer Error Interrupt Flag

DMA2D Input Color Mode

DMA2D_INPUT_ARGB8888	ARGB8888 color mode
DMA2D_INPUT_RGB888	RGB888 color mode
DMA2D_INPUT_RGB565	RGB565 color mode
DMA2D_INPUT_ARGB1555	ARGB1555 color mode
DMA2D_INPUT_ARGB4444	ARGB4444 color mode
DMA2D_INPUT_L8	L8 color mode
DMA2D_INPUT_AL44	AL44 color mode
DMA2D_INPUT_AL88	AL88 color mode
DMA2D_INPUT_L4	L4 color mode
DMA2D_INPUT_A8	A8 color mode
DMA2D_INPUT_A4	A4 color mode

DMA2D Interrupts

DMA2D_IT_CE	Configuration Error Interrupt
DMA2D_IT_CTC	CLUT Transfer Complete Interrupt
DMA2D_IT_CAE	CLUT Access Error Interrupt
DMA2D_IT_TW	Transfer Watermark Interrupt
DMA2D_IT_TC	Transfer Complete Interrupt
DMA2D_IT_TE	Transfer Error Interrupt

DMA2D Maximum Line Watermark

DMA2D_LINE_WATERMARK_MAX	DMA2D maximum line watermark

DMA2D Maximum Number of Layers

DMA2D_MAX_LAYER	DMA2D maximum number of layers

DMA2D Mode

DMA2D_M2M	DMA2D memory to memory transfer mode
DMA2D_M2M_PFC	DMA2D memory to memory with pixel format conversion transfer mode
DMA2D_M2M_BLEND	DMA2D memory to memory with blending transfer mode
DMA2D_R2M	DMA2D register to memory transfer mode

DMA2D Offset

DMA2D_OFFSET	Line Offset

DMA2D Output Color Mode

DMA2D_OUTPUT_ARGB8888	ARGB8888 DMA2D color mode
DMA2D_OUTPUT_RGB888	RGB888 DMA2D color mode
DMA2D_OUTPUT_RGB565	RGB565 DMA2D color mode
DMA2D_OUTPUT_ARGB1555	ARGB1555 DMA2D color mode
DMA2D_OUTPUT_ARGB4444	ARGB4444 DMA2D color mode

DMA2D Shifts

DMA2D_POSITION_FGPFCCR_CS	Required left shift to set foreground CLUT size
DMA2D_POSITION_BGPFCCR_CS	Required left shift to set background CLUT size
DMA2D_POSITION_FGPFCCR_CCM	Required left shift to set foreground CLUT color mode
DMA2D_POSITION_BGPFCCR_CCM	Required left shift to set background CLUT color mode
DMA2D_POSITION_AMTCR_DT	Required left shift to set deadtime value
DMA2D_POSITION_FGPFCCR_AM	Required left shift to set foreground alpha mode
DMA2D_POSITION_BGPFCCR_AM	Required left shift to set background alpha mode
DMA2D_POSITION_FGPFCCR_ALPHA	Required left shift to set foreground alpha value
DMA2D_POSITION_BGPFCCR_ALPHA	Required left shift to set background alpha

	value
DMA2D_POSITION_NLR_PL	Required left shift to set pixels per lines value

DMA2D Size

DMA2D_PIXEL	DMA2D number of pixels per line
DMA2D_LINE	DMA2D number of lines

DMA2D Time Out

DMA2D_TIMEOUT_ABORT	1s
DMA2D_TIMEOUT_SUSPEND	1s

18 HAL DMA Generic Driver

18.1 DMA Firmware driver registers structures

18.1.1 DMA_InitTypeDef

Data Fields

- ***uint32_t Channel***
- ***uint32_t Direction***
- ***uint32_t PeriphInc***
- ***uint32_t MemInc***
- ***uint32_t PeriphDataAlignment***
- ***uint32_t MemDataAlignment***
- ***uint32_t Mode***
- ***uint32_t Priority***
- ***uint32_t FIFOMode***
- ***uint32_t FIFOThreshold***
- ***uint32_t MemBurst***
- ***uint32_t PeriphBurst***

Field Documentation

- ***uint32_t DMA_InitTypeDef::Channel***
  Specifies the channel used for the specified stream. This parameter can be a value of ***DMA_Channel_selection***
- ***uint32_t DMA_InitTypeDef::Direction***
  Specifies if the data will be transferred from memory to peripheral, from memory to memory or from peripheral to memory. This parameter can be a value of ***DMA_Data_transfer_direction***
- ***uint32_t DMA_InitTypeDef::PeriphInc***
  Specifies whether the Peripheral address register should be incremented or not. This parameter can be a value of ***DMA_Peripheral_incremented_mode***
- ***uint32_t DMA_InitTypeDef::MemInc***
  Specifies whether the memory address register should be incremented or not. This parameter can be a value of ***DMA_Memory_incremented_mode***
- ***uint32_t DMA_InitTypeDef::PeriphDataAlignment***
  Specifies the Peripheral data width. This parameter can be a value of ***DMA_Peripheral_data_size***
- ***uint32_t DMA_InitTypeDef::MemDataAlignment***
  Specifies the Memory data width. This parameter can be a value of ***DMA_Memory_data_size***
- ***uint32_t DMA_InitTypeDef::Mode***
  Specifies the operation mode of the DMAy Streamx. This parameter can be a value of ***DMA_mode***
  Note: The circular buffer mode cannot be used if the memory-to-memory data transfer is configured on the selected Stream
- ***uint32_t DMA_InitTypeDef::Priority***
  Specifies the software priority for the DMAy Streamx. This parameter can be a value of ***DMA_Priority_level***

- ***uint32_t DMA_InitTypeDef::FIFOMode***
  Specifies if the FIFO mode or Direct mode will be used for the specified stream. This parameter can be a value of ***DMA_FIFO_direct_mode***
  Note: The Direct mode (FIFO mode disabled) cannot be used if the memory-to-memory data transfer is configured on the selected stream
- ***uint32_t DMA_InitTypeDef::FIFOThreshold***
  Specifies the FIFO threshold level. This parameter can be a value of ***DMA_FIFO_threshold_level***
- ***uint32_t DMA_InitTypeDef::MemBurst***
  Specifies the Burst transfer configuration for the memory transfers. It specifies the amount of data to be transferred in a single non interruptible transaction. This parameter can be a value of ***DMA_Memory_burst***
  Note: The burst mode is possible only if the address Increment mode is enabled.
- ***uint32_t DMA_InitTypeDef::PeriphBurst***
  Specifies the Burst transfer configuration for the peripheral transfers. It specifies the amount of data to be transferred in a single non interruptible transaction. This parameter can be a value of ***DMA_Peripheral_burst***
  Note: The burst mode is possible only if the address Increment mode is enabled.

18.1.2 __DMA_HandleTypeDef

Data Fields

- ***DMA_Stream_TypeDef * Instance***
- ***DMA_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_DMA_StateTypeDef State***
- ***void * Parent***
- ***void(* XferCpltCallback***
- ***void(* XferHalfCpltCallback***
- ***void(* XferM1CpltCallback***
- ***void(* XferM1HalfCpltCallback***
- ***void(* XferErrorCallback***
- ***void(* XferAbortCallback***
- ***__IO uint32_t ErrorCode***
- ***uint32_t StreamBaseAddress***
- ***uint32_t StreamIndex***

Field Documentation

- ***DMA_Stream_TypeDef* __DMA_HandleTypeDef::Instance***
  Register base address
- ***DMA_InitTypeDef __DMA_HandleTypeDef::Init***
  DMA communication parameters
- ***HAL_LockTypeDef __DMA_HandleTypeDef::Lock***
  DMA locking object
- ***__IO HAL_DMA_StateTypeDef __DMA_HandleTypeDef::State***
  DMA transfer state
- ***void* __DMA_HandleTypeDef::Parent***
  Parent object state

- ***void(* __DMA_HandleTypeDef::XferCpltCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA transfer complete callback
- ***void(* __DMA_HandleTypeDef::XferHalfCpltCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA Half transfer complete callback
- ***void(* __DMA_HandleTypeDef::XferM1CpltCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA transfer complete Memory1 callback
- ***void(* __DMA_HandleTypeDef::XferM1HalfCpltCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA transfer Half complete Memory1 callback
- ***void(* __DMA_HandleTypeDef::XferErrorCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA transfer error callback
- ***void(* __DMA_HandleTypeDef::XferAbortCallback)(struct __DMA_HandleTypeDef *hdma)***
  DMA transfer Abort callback
- ***__IO uint32_t __DMA_HandleTypeDef::ErrorCode***
  DMA Error code
- ***uint32_t __DMA_HandleTypeDef::StreamBaseAddress***
  DMA Stream Base Address
- ***uint32_t __DMA_HandleTypeDef::StreamIndex***
  DMA Stream Index

18.2 DMA Firmware driver API description

18.2.1 How to use this driver

1. Enable and configure the peripheral to be connected to the DMA Stream (except for internal SRAM/FLASH memories: no initialization is necessary) please refer to Reference manual for connection between peripherals and DMA requests.
2. For a given Stream, program the required configuration through the following parameters: Transfer Direction, Source and Destination data formats, Circular, Normal or peripheral flow control mode, Stream Priority level, Source and Destination Increment mode, FIFO mode and its Threshold (if needed), Burst mode for Source and/or Destination (if needed) using HAL_DMA_Init() function. Prior to HAL_DMA_Init() the clock must be enabled for DMA through the following macros: __HAL_RCC_DMA1_CLK_ENABLE() or __HAL_RCC_DMA2_CLK_ENABLE().

Polling mode IO operation

- Use HAL_DMA_Start() to start DMA transfer after the configuration of Source address and destination address and the Length of data to be transferred.
- Use HAL_DMA_PollForTransfer() to poll for the end of current transfer, in this case a fixed Timeout can be configured by User depending from his application.
- Use HAL_DMA_Abort() function to abort the current transfer.

Interrupt mode IO operation

- Configure the DMA interrupt priority using HAL_NVIC_SetPriority()
- Enable the DMA IRQ handler using HAL_NVIC_EnableIRQ()
- Use HAL_DMA_Start_IT() to start DMA transfer after the configuration of Source address and destination address and the Length of data to be transferred. In this case the DMA interrupt is configured
- Use HAL_DMA_IRQHandler() called under DMA_IRQHandler() Interrupt subroutine
- At the end of data transfer HAL_DMA_IRQHandler() function is executed and user can add his own function by customization of function pointer XferCpltCallback and XferErrorCallback (i.e a member of DMA handle structure).

1. Use HAL_DMA_GetState() function to return the DMA state and HAL_DMA_GetError() in case of error detection.
2. Use HAL_DMA_Abort_IT() function to abort the current transfer In Memory-to-Memory transfer mode, Circular mode is not allowed. The FIFO is used mainly to reduce bus usage and to allow data packing/unpacking: it is possible to set different Data Sizes for the Peripheral and the Memory (ie. you can set Half-Word data size for the peripheral to access its data register and set Word data size for the Memory to gain in access time. Each two half words will be packed and written in a single access to a Word in the Memory). When FIFO is disabled, it is not allowed to configure different Data Sizes for Source and Destination. In this case the Peripheral Data Size will be applied to both Source and Destination.

DMA HAL driver macros list

Below the list of most used macros in DMA HAL driver.

- __HAL_DMA_ENABLE: Enable the specified DMA Stream.
- __HAL_DMA_DISABLE: Disable the specified DMA Stream.
- __HAL_DMA_GET_IT_SOURCE: Check whether the specified DMA Stream interrupt has occurred or not.

You can refer to the DMA HAL driver header file for more useful macros

18.2.2 Initialization and de-initialization functions

This section provides functions allowing to initialize the DMA Stream source and destination addresses, incrementation and data sizes, transfer direction, circular/normal mode selection, memory-to-memory mode selection and Stream priority value.

The HAL_DMA_Init() function follows the DMA configuration procedures as described in reference manual.

This section contains the following APIs:

- ***HAL_DMA_Init()***
- ***HAL_DMA_DeInit()***

18.2.3 IO operation functions

This section provides functions allowing to:

- Configure the source, destination address and data length and Start DMA transfer
- Configure the source, destination address and data length and Start DMA transfer with interrupt
- Abort DMA transfer
- Poll for transfer complete
- Handle DMA interrupt request

This section contains the following APIs:

- ***HAL_DMA_Start()***
- ***HAL_DMA_Start_IT()***
- ***HAL_DMA_Abort()***
- ***HAL_DMA_Abort_IT()***
- ***HAL_DMA_PollForTransfer()***
- ***HAL_DMA_IRQHandler()***
- ***HAL_DMA_RegisterCallback()***
- ***HAL_DMA_UnRegisterCallback()***
- ***HAL_DMA_CleanCallbacks()***

18.2.4 State and Errors functions

This subsection provides functions allowing to

- Check the DMA state
- Get error code

This section contains the following APIs:

- ***HAL_DMA_GetState()***
- ***HAL_DMA_GetError()***

18.2.5 Detailed description of functions

HAL_DMA_Init

Function Name	**HAL_StatusTypeDef HAL_DMA_Init (DMA_HandleTypeDef * hdma)**
Function Description	Initializes the DMA according to the specified parameters in the DMA_InitTypeDef and create the associated handle.
Parameters	• **hdma:** Pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
Return values	• **HAL:** status

HAL_DMA_DeInit

Function Name	**HAL_StatusTypeDef HAL_DMA_DeInit (DMA_HandleTypeDef * hdma)**
Function Description	DeInitializes the DMA peripheral.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA

Stream.

Return values
- **HAL:** status

HAL_DMA_Start

Function Name **HAL_StatusTypeDef HAL_DMA_Start (DMA_HandleTypeDef * hdma, uint32_t SrcAddress, uint32_t DstAddress, uint32_t DataLength)**

Function Description Starts the DMA Transfer.

Parameters
- **hdma:** : pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
- **SrcAddress:** The source memory Buffer address
- **DstAddress:** The destination memory Buffer address
- **DataLength:** The length of data to be transferred from source to destination

Return values
- **HAL:** status

HAL_DMA_Start_IT

Function Name **HAL_StatusTypeDef HAL_DMA_Start_IT (DMA_HandleTypeDef * hdma, uint32_t SrcAddress, uint32_t DstAddress, uint32_t DataLength)**

Function Description Starts the DMA Transfer with interrupt enabled.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
- **SrcAddress:** The source memory Buffer address
- **DstAddress:** The destination memory Buffer address
- **DataLength:** The length of data to be transferred from source to destination

Return values
- **HAL:** status

HAL_DMA_Abort

Function Name **HAL_StatusTypeDef HAL_DMA_Abort (DMA_HandleTypeDef * hdma)**

Function Description Aborts the DMA Transfer.

Parameters
- **hdma:** : pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.

Return values
- **HAL:** status

Notes
- After disabling a DMA Stream, a check for wait until the DMA Stream is effectively disabled is added. If a Stream is disabled while a data transfer is ongoing, the current data will be transferred and the Stream will be effectively disabled only after the transfer of this single data is finished.

HAL_DMA_Abort_IT

Function Name **HAL_StatusTypeDef HAL_DMA_Abort_IT (DMA_HandleTypeDef * hdma)**

Function Description Aborts the DMA Transfer in Interrupt mode.

Parameters
- **hdma:** : pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.

Return values
- **HAL:** status

HAL_DMA_PollForTransfer

Function Name **HAL_StatusTypeDef HAL_DMA_PollForTransfer (DMA_HandleTypeDef * hdma, HAL_DMA_LevelCompleteTypeDef CompleteLevel, uint32_t Timeout)**

Function Description Polling for transfer complete.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
- **CompleteLevel:** Specifies the DMA level complete.
- **Timeout:** Timeout duration.

Return values
- **HAL:** status

Notes
- The polling mode is kept in this version for legacy. it is recommanded to use the IT model instead. This model could be used for debug purpose.
- The HAL_DMA_PollForTransfer API cannot be used in circular and double buffering mode (automatic circular mode).

HAL_DMA_IRQHandler

Function Name **void HAL_DMA_IRQHandler (DMA_HandleTypeDef * hdma)**

Function Description Handles DMA interrupt request.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.

Return values
- **None**

HAL_DMA_CleanCallbacks

Function Name **HAL_StatusTypeDef HAL_DMA_CleanCallbacks (DMA_HandleTypeDef * hdma)**

Function Description

HAL_DMA_RegisterCallback

Function Name **HAL_StatusTypeDef HAL_DMA_RegisterCallback (DMA_HandleTypeDef * hdma, HAL_DMA_CallbackIDTypeDef**

CallbackID, void(*)(DMA_HandleTypeDef *_hdma) pCallback)

Function Description	Register callbacks.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream. • **CallbackID:** User Callback identifer a DMA_HandleTypeDef structure as parameter. • **pCallback:** pointer to private callbacsk function which has pointer to a DMA_HandleTypeDef structure as parameter.
Return values	• **HAL:** status

HAL_DMA_UnRegisterCallback

Function Name	**HAL_StatusTypeDef HAL_DMA_UnRegisterCallback (DMA_HandleTypeDef * hdma, HAL_DMA_CallbackIDTypeDef CallbackID)**
Function Description	UnRegister callbacks.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream. • **CallbackID:** User Callback identifer a HAL_DMA_CallbackIDTypeDef ENUM as parameter.
Return values	• **HAL:** status

HAL_DMA_GetState

Function Name	**HAL_DMA_StateTypeDef HAL_DMA_GetState (DMA_HandleTypeDef * hdma)**
Function Description	Returns the DMA state.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
Return values	• **HAL:** state

HAL_DMA_GetError

Function Name	**uint32_t HAL_DMA_GetError (DMA_HandleTypeDef * hdma)**
Function Description	Return the DMA error code.
Parameters	• **hdma:** : pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream.
Return values	• **DMA:** Error Code

18.3 DMA Firmware driver defines

18.3.1 DMA

DMA Channel selection

DMA_CHANNEL_0	DMA Channel 0
DMA_CHANNEL_1	DMA Channel 1
DMA_CHANNEL_2	DMA Channel 2
DMA_CHANNEL_3	DMA Channel 3
DMA_CHANNEL_4	DMA Channel 4
DMA_CHANNEL_5	DMA Channel 5
DMA_CHANNEL_6	DMA Channel 6
DMA_CHANNEL_7	DMA Channel 7

DMA Data transfer direction

DMA_PERIPH_TO_MEMORY	Peripheral to memory direction
DMA_MEMORY_TO_PERIPH	Memory to peripheral direction
DMA_MEMORY_TO_MEMORY	Memory to memory direction

DMA Error Code

HAL_DMA_ERROR_NONE	No error
HAL_DMA_ERROR_TE	Transfer error
HAL_DMA_ERROR_FE	FIFO error
HAL_DMA_ERROR_DME	Direct Mode error
HAL_DMA_ERROR_TIMEOUT	Timeout error
HAL_DMA_ERROR_PARAM	Parameter error
HAL_DMA_ERROR_NO_XFER	Abort requested with no Xfer ongoing
HAL_DMA_ERROR_NOT_SUPPORTED	Not supported mode

DMA FIFO direct mode

DMA_FIFOMODE_DISABLE	FIFO mode disable
DMA_FIFOMODE_ENABLE	FIFO mode enable

DMA FIFO threshold level

DMA_FIFO_THRESHOLD_1QUARTERFULL	FIFO threshold 1 quart full configuration
DMA_FIFO_THRESHOLD_HALFFULL	FIFO threshold half full configuration
DMA_FIFO_THRESHOLD_3QUARTERSFULL	FIFO threshold 3 quarts full configuration
DMA_FIFO_THRESHOLD_FULL	FIFO threshold full configuration

DMA flag definitions

DMA_FLAG_FEIF0_4

DMA_FLAG_DMEIF0_4

DMA_FLAG_TEIF0_4

DMA_FLAG_HTIF0_4

DMA_FLAG_TCIF0_4

DMA_FLAG_FEIF1_5

DMA_FLAG_DMEIF1_5

DMA_FLAG_TEIF1_5

DMA_FLAG_HTIF1_5

DMA_FLAG_TCIF1_5

DMA_FLAG_FEIF2_6

DMA_FLAG_DMEIF2_6

DMA_FLAG_TEIF2_6

DMA_FLAG_HTIF2_6

DMA_FLAG_TCIF2_6

DMA_FLAG_FEIF3_7

DMA_FLAG_DMEIF3_7

DMA_FLAG_TEIF3_7

DMA_FLAG_HTIF3_7

DMA_FLAG_TCIF3_7

DMA Handle index

TIM_DMA_ID_UPDATE	Index of the DMA handle used for Update DMA requests
TIM_DMA_ID_CC1	Index of the DMA handle used for Capture/Compare 1 DMA requests
TIM_DMA_ID_CC2	Index of the DMA handle used for Capture/Compare 2 DMA requests
TIM_DMA_ID_CC3	Index of the DMA handle used for Capture/Compare 3 DMA requests
TIM_DMA_ID_CC4	Index of the DMA handle used for Capture/Compare 4 DMA requests
TIM_DMA_ID_COMMUTATION	Index of the DMA handle used for Commutation DMA requests
TIM_DMA_ID_TRIGGER	Index of the DMA handle used for Trigger DMA requests

DMA interrupt enable definitions

DMA_IT_TC

DMA_IT_HT

DMA_IT_TE

DMA_IT_DME

DMA_IT_FE

DMA Memory burst

DMA_MBURST_SINGLE

DMA_MBURST_INC4

DMA_MBURST_INC8

DMA_MBURST_INC16

DMA Memory data size

DMA_MDATAALIGN_BYTE	Memory data alignment: Byte
DMA_MDATAALIGN_HALFWORD	Memory data alignment: HalfWord
DMA_MDATAALIGN_WORD	Memory data alignment: Word

DMA Memory incremented mode

DMA_MINC_ENABLE	Memory increment mode enable
DMA_MINC_DISABLE	Memory increment mode disable

DMA mode

DMA_NORMAL	Normal mode
DMA_CIRCULAR	Circular mode
DMA_PFCTRL	Peripheral flow control mode

DMA Peripheral burst

DMA_PBURST_SINGLE

DMA_PBURST_INC4

DMA_PBURST_INC8

DMA_PBURST_INC16

DMA Peripheral data size

DMA_PDATAALIGN_BYTE	Peripheral data alignment: Byte
DMA_PDATAALIGN_HALFWORD	Peripheral data alignment: HalfWord
DMA_PDATAALIGN_WORD	Peripheral data alignment: Word

DMA Peripheral incremented mode

DMA_PINC_ENABLE	Peripheral increment mode enable
DMA_PINC_DISABLE	Peripheral increment mode disable

DMA Priority level

DMA_PRIORITY_LOW	Priority level: Low
DMA_PRIORITY_MEDIUM	Priority level: Medium
DMA_PRIORITY_HIGH	Priority level: High
DMA_PRIORITY_VERY_HIGH	Priority level: Very High

19 HAL DMA Extension Driver

19.1 DMAEx Firmware driver API description

19.1.1 How to use this driver

The DMA Extension HAL driver can be used as follows:

1. Start a multi buffer transfer using the HAL_DMA_MultiBufferStart() function for polling mode or HAL_DMA_MultiBufferStart_IT() for interrupt mode. In Memory-to-Memory transfer mode, Multi (Double) Buffer mode is not allowed. When Multi (Double) Buffer mode is enabled the, transfer is circular by default. In Multi (Double) buffer mode, it is possible to update the base address for the AHB memory port on the fly (DMA_SxM0AR or DMA_SxM1AR) when the stream is enabled.

19.1.2 Extended features functions

This section provides functions allowing to:

- Configure the source, destination address and data length and Start MultiBuffer DMA transfer
- Configure the source, destination address and data length and Start MultiBuffer DMA transfer with interrupt
- Change on the fly the memory0 or memory1 address.

This section contains the following APIs:

- ***HAL_DMAEx_MultiBufferStart()***
- ***HAL_DMAEx_MultiBufferStart_IT()***
- ***HAL_DMAEx_ChangeMemory()***

19.1.3 Detailed description of functions

HAL_DMAEx_MultiBufferStart

Function Name	**HAL_StatusTypeDef HAL_DMAEx_MultiBufferStart (DMA_HandleTypeDef * hdma, uint32_t SrcAddress, uint32_t DstAddress, uint32_t SecondMemAddress, uint32_t DataLength)**
Function Description	Starts the multi_buffer DMA Transfer.
Parameters	• **hdma:** : pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream. • **SrcAddress:** The source memory Buffer address • **DstAddress:** The destination memory Buffer address • **SecondMemAddress:** The second memory Buffer address in case of multi buffer Transfer • **DataLength:** The length of data to be transferred from source to destination
Return values	• **HAL:** status

HAL_DMAEx_MultiBufferStart_IT

Function Name	**HAL_StatusTypeDef HAL_DMAEx_MultiBufferStart_IT (DMA_HandleTypeDef * hdma, uint32_t SrcAddress, uint32_t DstAddress, uint32_t SecondMemAddress, uint32_t DataLength)**
Function Description	Starts the multi_buffer DMA Transfer with interrupt enabled.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream. • **SrcAddress:** The source memory Buffer address • **DstAddress:** The destination memory Buffer address • **SecondMemAddress:** The second memory Buffer address in case of multi buffer Transfer • **DataLength:** The length of data to be transferred from source to destination
Return values	• **HAL:** status

HAL_DMAEx_ChangeMemory

Function Name	**HAL_StatusTypeDef HAL_DMAEx_ChangeMemory (DMA_HandleTypeDef * hdma, uint32_t Address, HAL_DMA_MemoryTypeDef memory)**
Function Description	Change the memory0 or memory1 address on the fly.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA Stream. • **Address:** The new address • **memory:** the memory to be changed, This parameter can be one of the following values: MEMORY0 / MEMORY1
Return values	• **HAL:** status
Notes	• The MEMORY0 address can be changed only when the current transfer use MEMORY1 and the MEMORY1 address can be changed only when the current transfer use MEMORY0.

20 HAL DSI Generic Driver

20.1 DSI Firmware driver registers structures

20.1.1 DSI_InitTypeDef

Data Fields

- ***uint32_t AutomaticClockLaneControl***
- ***uint32_t TXEscapeCkdiv***
- ***uint32_t NumberOfLanes***

Field Documentation

- ***uint32_t DSI_InitTypeDef::AutomaticClockLaneControl***
  Automatic clock lane control This parameter can be any value of ***DSI_Automatic_Clk_Lane_Control***
- ***uint32_t DSI_InitTypeDef::TXEscapeCkdiv***
  TX Escape clock division The values 0 and 1 stop the TX_ESC clock generation
- ***uint32_t DSI_InitTypeDef::NumberOfLanes***
  Number of lanes This parameter can be any value of ***DSI_Number_Of_Lanes***

20.1.2 DSI_PLLInitTypeDef

Data Fields

- ***uint32_t PLLNDIV***
- ***uint32_t PLLIDF***
- ***uint32_t PLLODF***

Field Documentation

- ***uint32_t DSI_PLLInitTypeDef::PLLNDIV***
  PLL Loop Division Factor This parameter must be a value between 10 and 125
- ***uint32_t DSI_PLLInitTypeDef::PLLIDF***
  PLL Input Division Factor This parameter can be any value of ***DSI_PLL_IDF***
- ***uint32_t DSI_PLLInitTypeDef::PLLODF***
  PLL Output Division Factor This parameter can be any value of ***DSI_PLL_ODF***

20.1.3 DSI_VidCfgTypeDef

Data Fields

- ***uint32_t VirtualChannelID***
- ***uint32_t ColorCoding***
- ***uint32_t LooselyPacked***
- ***uint32_t Mode***
- ***uint32_t PacketSize***
- ***uint32_t NumberOfChunks***
- ***uint32_t NullPacketSize***
- ***uint32_t HSPolarity***
- ***uint32_t VSPolarity***
- ***uint32_t DEPolarity***
- ***uint32_t HorizontalSyncActive***
- ***uint32_t HorizontalBackPorch***
- ***uint32_t HorizontalLine***
- ***uint32_t VerticalSyncActive***
- ***uint32_t VerticalBackPorch***
- ***uint32_t VerticalFrontPorch***
- ***uint32_t VerticalActive***
- ***uint32_t LPCommandEnable***
- ***uint32_t LPLargestPacketSize***
- ***uint32_t LPVACTLargestPacketSize***
- ***uint32_t LPHorizontalFrontPorchEnable***
- ***uint32_t LPHorizontalBackPorchEnable***
- ***uint32_t LPVerticalActiveEnable***
- ***uint32_t LPVerticalFrontPorchEnable***
- ***uint32_t LPVerticalBackPorchEnable***
- ***uint32_t LPVerticalSyncActiveEnable***
- ***uint32_t FrameBTAAcknowledgeEnable***

Field Documentation

- ***uint32_t DSI_VidCfgTypeDef::VirtualChannelID***
  Virtual channel ID
- ***uint32_t DSI_VidCfgTypeDef::ColorCoding***
  Color coding for LTDC interface This parameter can be any value of ***DSI_Color_Coding***
- ***uint32_t DSI_VidCfgTypeDef::LooselyPacked***
  Enable or disable loosely packed stream (needed only when using 18-bit configuration). This parameter can be any value of ***DSI_LooselyPacked***
- ***uint32_t DSI_VidCfgTypeDef::Mode***
  Video mode type This parameter can be any value of ***DSI_Video_Mode_Type***
- ***uint32_t DSI_VidCfgTypeDef::PacketSize***
  Video packet size
- ***uint32_t DSI_VidCfgTypeDef::NumberOfChunks***
  Number of chunks
- ***uint32_t DSI_VidCfgTypeDef::NullPacketSize***
  Null packet size

- ***uint32_t DSI_VidCfgTypeDef::HSPolarity***
  HSYNC pin polarity This parameter can be any value of ***DSI_HSYNC_Polarity***
- ***uint32_t DSI_VidCfgTypeDef::VSPolarity***
  VSYNC pin polarity This parameter can be any value of ***DSI_VSYNC_Active_Polarity***
- ***uint32_t DSI_VidCfgTypeDef::DEPolarity***
  Data Enable pin polarity This parameter can be any value of ***DSI_DATA_ENABLE_Polarity***
- ***uint32_t DSI_VidCfgTypeDef::HorizontalSyncActive***
  Horizontal synchronism active duration (in lane byte clock cycles)
- ***uint32_t DSI_VidCfgTypeDef::HorizontalBackPorch***
  Horizontal back-porch duration (in lane byte clock cycles)
- ***uint32_t DSI_VidCfgTypeDef::HorizontalLine***
  Horizontal line duration (in lane byte clock cycles)
- ***uint32_t DSI_VidCfgTypeDef::VerticalSyncActive***
  Vertical synchronism active duration
- ***uint32_t DSI_VidCfgTypeDef::VerticalBackPorch***
  Vertical back-porch duration
- ***uint32_t DSI_VidCfgTypeDef::VerticalFrontPorch***
  Vertical front-porch duration
- ***uint32_t DSI_VidCfgTypeDef::VerticalActive***
  Vertical active duration
- ***uint32_t DSI_VidCfgTypeDef::LPCommandEnable***
  Low-power command enable This parameter can be any value of ***DSI_LP_Command***
- ***uint32_t DSI_VidCfgTypeDef::LPLargestPacketSize***
  The size, in bytes, of the low power largest packet that can fit in a line during VSA, VBP and VFP regions
- ***uint32_t DSI_VidCfgTypeDef::LPVACTLargestPacketSize***
  The size, in bytes, of the low power largest packet that can fit in a line during VACT region
- ***uint32_t DSI_VidCfgTypeDef::LPHorizontalFrontPorchEnable***
  Low-power horizontal front-porch enable This parameter can be any value of ***DSI_LP_HFP***
- ***uint32_t DSI_VidCfgTypeDef::LPHorizontalBackPorchEnable***
  Low-power horizontal back-porch enable This parameter can be any value of ***DSI_LP_HBP***
- ***uint32_t DSI_VidCfgTypeDef::LPVerticalActiveEnable***
  Low-power vertical active enable This parameter can be any value of ***DSI_LP_VACT***
- ***uint32_t DSI_VidCfgTypeDef::LPVerticalFrontPorchEnable***
  Low-power vertical front-porch enable This parameter can be any value of ***DSI_LP_VFP***
- ***uint32_t DSI_VidCfgTypeDef::LPVerticalBackPorchEnable***
  Low-power vertical back-porch enable This parameter can be any value of ***DSI_LP_VBP***
- ***uint32_t DSI_VidCfgTypeDef::LPVerticalSyncActiveEnable***
  Low-power vertical sync active enable This parameter can be any value of ***DSI_LP_VSYNC***
- ***uint32_t DSI_VidCfgTypeDef::FrameBTAAcknowledgeEnable***
  Frame bus-turn-around acknowledge enable This parameter can be any value of ***DSI_FBTA_acknowledge***

20.1.4 DSI_CmdCfgTypeDef

Data Fields

- ***uint32_t VirtualChannelID***
- ***uint32_t ColorCoding***
- ***uint32_t CommandSize***
- ***uint32_t TearingEffectSource***
- ***uint32_t TearingEffectPolarity***
- ***uint32_t HSPolarity***
- ***uint32_t VSPolarity***
- ***uint32_t DEPolarity***
- ***uint32_t VSyncPol***
- ***uint32_t AutomaticRefresh***
- ***uint32_t TEAcknowledgeRequest***

Field Documentation

- ***uint32_t DSI_CmdCfgTypeDef::VirtualChannelID***
  Virtual channel ID
- ***uint32_t DSI_CmdCfgTypeDef::ColorCoding***
  Color coding for LTDC interface This parameter can be any value of ***DSI_Color_Coding***
- ***uint32_t DSI_CmdCfgTypeDef::CommandSize***
  Maximum allowed size for an LTDC write memory command, measured in pixels. This parameter can be any value between 0x00 and 0xFFFFU
- ***uint32_t DSI_CmdCfgTypeDef::TearingEffectSource***
  Tearing effect source This parameter can be any value of ***DSI_TearingEffectSource***
- ***uint32_t DSI_CmdCfgTypeDef::TearingEffectPolarity***
  Tearing effect pin polarity This parameter can be any value of ***DSI_TearingEffectPolarity***
- ***uint32_t DSI_CmdCfgTypeDef::HSPolarity***
  HSYNC pin polarity This parameter can be any value of ***DSI_HSYNC_Polarity***
- ***uint32_t DSI_CmdCfgTypeDef::VSPolarity***
  VSYNC pin polarity This parameter can be any value of ***DSI_VSYNC_Active_Polarity***
- ***uint32_t DSI_CmdCfgTypeDef::DEPolarity***
  Data Enable pin polarity This parameter can be any value of ***DSI_DATA_ENABLE_Polarity***
- ***uint32_t DSI_CmdCfgTypeDef::VSyncPol***
  VSync edge on which the LTDC is halted This parameter can be any value of ***DSI_Vsync_Polarity***
- ***uint32_t DSI_CmdCfgTypeDef::AutomaticRefresh***
  Automatic refresh mode This parameter can be any value of ***DSI_AutomaticRefresh***
- ***uint32_t DSI_CmdCfgTypeDef::TEAcknowledgeRequest***
  Tearing Effect Acknowledge Request Enable This parameter can be any value of ***DSI_TE_AcknowledgeRequest***

20.1.5 DSI_LPCmdTypeDef

Data Fields

- ***uint32_t LPGenShortWriteNoP***
- ***uint32_t LPGenShortWriteOneP***
- ***uint32_t LPGenShortWriteTwoP***
- ***uint32_t LPGenShortReadNoP***
- ***uint32_t LPGenShortReadOneP***
- ***uint32_t LPGenShortReadTwoP***
- ***uint32_t LPGenLongWrite***
- ***uint32_t LPDcsShortWriteNoP***
- ***uint32_t LPDcsShortWriteOneP***
- ***uint32_t LPDcsShortReadNoP***
- ***uint32_t LPDcsLongWrite***
- ***uint32_t LPMaxReadPacket***
- ***uint32_t AcknowledgeRequest***

Field Documentation

- ***uint32_t DSI_LPCmdTypeDef::LPGenShortWriteNoP***
  Generic Short Write Zero parameters Transmission This parameter can be any value of ***DSI_LP_LPGenShortWriteNoP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenShortWriteOneP***
  Generic Short Write One parameter Transmission This parameter can be any value of ***DSI_LP_LPGenShortWriteOneP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenShortWriteTwoP***
  Generic Short Write Two parameters Transmission This parameter can be any value of ***DSI_LP_LPGenShortWriteTwoP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenShortReadNoP***
  Generic Short Read Zero parameters Transmission This parameter can be any value of ***DSI_LP_LPGenShortReadNoP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenShortReadOneP***
  Generic Short Read One parameter Transmission This parameter can be any value of ***DSI_LP_LPGenShortReadOneP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenShortReadTwoP***
  Generic Short Read Two parameters Transmission This parameter can be any value of ***DSI_LP_LPGenShortReadTwoP***
- ***uint32_t DSI_LPCmdTypeDef::LPGenLongWrite***
  Generic Long Write Transmission This parameter can be any value of ***DSI_LP_LPGenLongWrite***
- ***uint32_t DSI_LPCmdTypeDef::LPDcsShortWriteNoP***
  DCS Short Write Zero parameters Transmission This parameter can be any value of ***DSI_LP_LPDcsShortWriteNoP***
- ***uint32_t DSI_LPCmdTypeDef::LPDcsShortWriteOneP***
  DCS Short Write One parameter Transmission This parameter can be any value of ***DSI_LP_LPDcsShortWriteOneP***
- ***uint32_t DSI_LPCmdTypeDef::LPDcsShortReadNoP***
  DCS Short Read Zero parameters Transmission This parameter can be any value of ***DSI_LP_LPDcsShortReadNoP***

- ***uint32_t DSI_LPCmdTypeDef::LPDcsLongWrite***
  DCS Long Write Transmission This parameter can be any value of ***DSI_LP_LPDcsLongWrite***
- ***uint32_t DSI_LPCmdTypeDef::LPMaxReadPacket***
  Maximum Read Packet Size Transmission This parameter can be any value of ***DSI_LP_LPMaxReadPacket***
- ***uint32_t DSI_LPCmdTypeDef::AcknowledgeRequest***
  Acknowledge Request Enable This parameter can be any value of ***DSI_AcknowledgeRequest***

20.1.6 DSI_PHY_TimerTypeDef

Data Fields

- ***uint32_t ClockLaneHS2LPTime***
- ***uint32_t ClockLaneLP2HSTime***
- ***uint32_t DataLaneHS2LPTime***
- ***uint32_t DataLaneLP2HSTime***
- ***uint32_t DataLaneMaxReadTime***
- ***uint32_t StopWaitTime***

Field Documentation

- ***uint32_t DSI_PHY_TimerTypeDef::ClockLaneHS2LPTime***
  The maximum time that the D-PHY clock lane takes to go from high-speed to low-power transmission
- ***uint32_t DSI_PHY_TimerTypeDef::ClockLaneLP2HSTime***
  The maximum time that the D-PHY clock lane takes to go from low-power to high-speed transmission
- ***uint32_t DSI_PHY_TimerTypeDef::DataLaneHS2LPTime***
  The maximum time that the D-PHY data lanes takes to go from high-speed to low-power transmission
- ***uint32_t DSI_PHY_TimerTypeDef::DataLaneLP2HSTime***
  The maximum time that the D-PHY data lanes takes to go from low-power to high-speed transmission
- ***uint32_t DSI_PHY_TimerTypeDef::DataLaneMaxReadTime***
  The maximum time required to perform a read command
- ***uint32_t DSI_PHY_TimerTypeDef::StopWaitTime***
  The minimum wait period to request a High-Speed transmission after the Stop state

20.1.7 DSI_HOST_TimeoutTypeDef

Data Fields

- ***uint32_t TimeoutCkdiv***
- ***uint32_t HighSpeedTransmissionTimeout***
- ***uint32_t LowPowerReceptionTimeout***
- ***uint32_t HighSpeedReadTimeout***
- ***uint32_t LowPowerReadTimeout***

- ***uint32_t HighSpeedWriteTimeout***
- ***uint32_t HighSpeedWritePrespMode***
- ***uint32_t LowPowerWriteTimeout***
- ***uint32_t BTATimeout***

Field Documentation

- ***uint32_t DSI_HOST_TimeoutTypeDef::TimeoutCkdiv***
  Time-out clock division
- ***uint32_t DSI_HOST_TimeoutTypeDef::HighSpeedTransmissionTimeout***
  High-speed transmission time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::LowPowerReceptionTimeout***
  Low-power reception time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::HighSpeedReadTimeout***
  High-speed read time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::LowPowerReadTimeout***
  Low-power read time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::HighSpeedWriteTimeout***
  High-speed write time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::HighSpeedWritePrespMode***
  High-speed write presp mode This parameter can be any value of ***DSI_HS_PrespMode***
- ***uint32_t DSI_HOST_TimeoutTypeDef::LowPowerWriteTimeout***
  Low-speed write time-out
- ***uint32_t DSI_HOST_TimeoutTypeDef::BTATimeout***
  BTA time-out

20.1.8 DSI_HandleTypeDef

Data Fields

- ***DSI_TypeDef * Instance***
- ***DSI_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_DSI_StateTypeDef State***
- ***__IO uint32_t ErrorCode***
- ***uint32_t ErrorMsk***

Field Documentation

- ***DSI_TypeDef* DSI_HandleTypeDef::Instance***
  Register base address
- ***DSI_InitTypeDef DSI_HandleTypeDef::Init***
  DSI required parameters
- ***HAL_LockTypeDef DSI_HandleTypeDef::Lock***
  DSI peripheral status
- ***__IO HAL_DSI_StateTypeDef DSI_HandleTypeDef::State***
  DSI communication state
- ***__IO uint32_t DSI_HandleTypeDef::ErrorCode***
  DSI Error code

- ***uint32_t DSI_HandleTypeDef::ErrorMsk***
  DSI Error monitoring mask

20.2 DSI Firmware driver API description

20.2.1 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and configure the DSI
- De-initialize the DSI

This section contains the following APIs:

- ***HAL_DSI_Init()***
- ***HAL_DSI_DeInit()***
- ***HAL_DSI_GetError()***
- ***HAL_DSI_ConfigErrorMonitor()***
- ***HAL_DSI_MspInit()***
- ***HAL_DSI_MspDeInit()***

20.2.2 IO operation functions

This section provides function allowing to:

- Handle DSI interrupt request

This section contains the following APIs:

- ***HAL_DSI_IRQHandler()***
- ***HAL_DSI_TearingEffectCallback()***
- ***HAL_DSI_EndOfRefreshCallback()***
- ***HAL_DSI_ErrorCallback()***

20.2.3 Peripheral Control functions

This section contains the following APIs:

- ***HAL_DSI_SetGenericVCID()***
- ***HAL_DSI_ConfigVideoMode()***
- ***HAL_DSI_ConfigAdaptedCommandMode()***
- ***HAL_DSI_ConfigCommand()***
- ***HAL_DSI_ConfigFlowControl()***
- ***HAL_DSI_ConfigPhyTimer()***
- ***HAL_DSI_ConfigHostTimeouts()***
- ***HAL_DSI_Start()***
- ***HAL_DSI_Stop()***
- ***HAL_DSI_Refresh()***
- ***HAL_DSI_ColorMode()***
- ***HAL_DSI_Shutdown()***
- ***HAL_DSI_ShortWrite()***
- ***HAL_DSI_LongWrite()***
- ***HAL_DSI_Read()***
- ***HAL_DSI_EnterULPMData()***
- ***HAL_DSI_ExitULPMData()***
- ***HAL_DSI_EnterULPM()***

- ***HAL_DSI_ExitULPM()***
- ***HAL_DSI_PatternGeneratorStart()***
- ***HAL_DSI_PatternGeneratorStop()***
- ***HAL_DSI_SetSlewRateAndDelayTuning()***
- ***HAL_DSI_SetLowPowerRXFilter()***
- ***HAL_DSI_SetSDD()***
- ***HAL_DSI_SetLanePinsConfiguration()***
- ***HAL_DSI_SetPHYTimings()***
- ***HAL_DSI_ForceTXStopMode()***
- ***HAL_DSI_ForceRXLowPower()***
- ***HAL_DSI_ForceDataLanesInRX()***
- ***HAL_DSI_SetPullDown()***
- ***HAL_DSI_SetContentionDetectionOff()***

20.2.4 Peripheral State and Errors functions

This subsection provides functions allowing to

- Check the DSI state.
- Get error code.

This section contains the following APIs:

- ***HAL_DSI_GetState()***

20.2.5 Detailed description of functions

HAL_DSI_Init

Function Name	**HAL_StatusTypeDef HAL_DSI_Init (DSI_HandleTypeDef * hdsi, DSI_PLLInitTypeDef * PLLInit)**
Function Description	Initializes the DSI according to the specified parameters in the DSI_InitTypeDef and create the associated handle.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI. • **PLLInit:** pointer to a DSI_PLLInitTypeDef structure that contains the PLL Clock structure definition for the DSI.
Return values	• **HAL:** status

HAL_DSI_DeInit

Function Name	**HAL_StatusTypeDef HAL_DSI_DeInit (DSI_HandleTypeDef * hdsi)**
Function Description	De-initializes the DSI peripheral registers to their default reset values.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **HAL:** status

HAL_DSI_MspInit

Function Name	**void HAL_DSI_MspInit (DSI_HandleTypeDef * hdsi)**

Function Description	Initializes the DSI MSP.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **None**

HAL_DSI_MspDeInit

Function Name	**void HAL_DSI_MspDeInit (DSI_HandleTypeDef * hdsi)**
Function Description	De-initializes the DSI MSP.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **None**

HAL_DSI_IRQHandler

Function Name	**void HAL_DSI_IRQHandler (DSI_HandleTypeDef * hdsi)**
Function Description	Handles DSI interrupt request.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **HAL:** status

HAL_DSI_TearingEffectCallback

Function Name	**void HAL_DSI_TearingEffectCallback (DSI_HandleTypeDef * hdsi)**
Function Description	Tearing Effect DSI callback.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **None**

HAL_DSI_EndOfRefreshCallback

Function Name	**void HAL_DSI_EndOfRefreshCallback (DSI_HandleTypeDef * hdsi)**
Function Description	End of Refresh DSI callback.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **None**

HAL_DSI_ErrorCallback

Function Name	**void HAL_DSI_ErrorCallback (DSI_HandleTypeDef * hdsi)**
Function Description	Operation Error DSI callback.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values
- **None**

HAL_DSI_SetGenericVCID

Function Name **HAL_StatusTypeDef HAL_DSI_SetGenericVCID (DSI_HandleTypeDef * hdsi, uint32_t VirtualChannelID)**

Function Description Configure the Generic interface read-back Virtual Channel ID.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **VirtualChannelID:** Virtual channel ID

Return values
- **HAL:** status

HAL_DSI_ConfigVideoMode

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigVideoMode (DSI_HandleTypeDef * hdsi, DSI_VidCfgTypeDef * VidCfg)**

Function Description Select video mode and configure the corresponding parameters.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **VidCfg:** pointer to a DSI_VidCfgTypeDef structure that contains the DSI video mode configuration parameters

Return values
- **HAL:** status

HAL_DSI_ConfigAdaptedCommandMode

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigAdaptedCommandMode (DSI_HandleTypeDef * hdsi, DSI_CmdCfgTypeDef * CmdCfg)**

Function Description Select adapted command mode and configure the corresponding parameters.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **CmdCfg:** pointer to a DSI_CmdCfgTypeDef structure that contains the DSI command mode configuration parameters

Return values
- **HAL:** status

HAL_DSI_ConfigCommand

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigCommand (DSI_HandleTypeDef * hdsi, DSI_LPCmdTypeDef * LPCmd)**

Function Description Configure command transmission mode: High-speed or Low-power and enable/disable acknowledge request after packet transmission.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **LPCmd:** pointer to a DSI_LPCmdTypeDef structure that contains the DSI command transmission mode configuration parameters

Return values
- **HAL:** status

HAL_DSI_ConfigFlowControl

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigFlowControl (DSI_HandleTypeDef * hdsi, uint32_t FlowControl)**

Function Description Configure the flow control parameters.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **FlowControl:** flow control feature(s) to be enabled. This parameter can be any combination of DSI Flow Control.

Return values
- **HAL:** status

HAL_DSI_ConfigPhyTimer

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigPhyTimer (DSI_HandleTypeDef * hdsi, DSI_PHY_TimerTypeDef * PhyTimers)**

Function Description Configure the DSI PHY timer parameters.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **PhyTimers:** DSI_PHY_TimerTypeDef structure that contains the DSI PHY timing parameters

Return values
- **HAL:** status

HAL_DSI_ConfigHostTimeouts

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigHostTimeouts (DSI_HandleTypeDef * hdsi, DSI_HOST_TimeoutTypeDef * HostTimeouts)**

Function Description Configure the DSI HOST timeout parameters.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **HostTimeouts:** DSI_HOST_TimeoutTypeDef structure that contains the DSI host timeout parameters

Return values
- **HAL:** status

HAL_DSI_Start

Function Name **HAL_StatusTypeDef HAL_DSI_Start (DSI_HandleTypeDef * hdsi)**

Function Description Start the DSI module.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values
- **HAL:** status

HAL_DSI_Stop

Function Name	**HAL_StatusTypeDef HAL_DSI_Stop (DSI_HandleTypeDef * hdsi)**
Function Description	Stop the DSI module.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **HAL:** status

HAL_DSI_Refresh

Function Name	**HAL_StatusTypeDef HAL_DSI_Refresh (DSI_HandleTypeDef * hdsi)**
Function Description	Refresh the display in command mode.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
Return values	• **HAL:** status

HAL_DSI_ColorMode

Function Name	**HAL_StatusTypeDef HAL_DSI_ColorMode (DSI_HandleTypeDef * hdsi, uint32_t ColorMode)**
Function Description	Controls the display color mode in Video mode.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI. • **ColorMode:** Color mode (full or 8-colors). This parameter can be any value of DSI Color Mode
Return values	• **HAL:** status

HAL_DSI_Shutdown

Function Name	**HAL_StatusTypeDef HAL_DSI_Shutdown (DSI_HandleTypeDef * hdsi, uint32_t Shutdown)**
Function Description	Control the display shutdown in Video mode.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI. • **Shutdown:** Shut-down (Display-ON or Display-OFF). This parameter can be any value of DSI ShutDown
Return values	• **HAL:** status

HAL_DSI_ShortWrite

Function Name	**HAL_StatusTypeDef HAL_DSI_ShortWrite (DSI_HandleTypeDef * hdsi, uint32_t ChannelID, uint32_t Mode, uint32_t Param1, uint32_t Param2)**
Function Description	DCS or Generic short write command.
Parameters	• **hdsi:** pointer to a DSI_HandleTypeDef structure that

contains the configuration information for the DSI.
- **ChannelID:** Virtual channel ID.
- **Mode:** DSI short packet data type. This parameter can be any value of DSI SHORT WRITE PKT Data Type.
- **Param1:** DSC command or first generic parameter. This parameter can be any value of DSI DCS Command or a generic command code.
- **Param2:** DSC parameter or second generic parameter.

Return values
- **HAL:** status

HAL_DSI_LongWrite

Function Name **HAL_StatusTypeDef HAL_DSI_LongWrite (DSI_HandleTypeDef * hdsi, uint32_t ChannelID, uint32_t Mode, uint32_t NbParams, uint32_t Param1, uint8_t * ParametersTable)**

Function Description DCS or Generic long write command.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **ChannelID:** Virtual channel ID.
- **Mode:** DSI long packet data type. This parameter can be any value of DSI LONG WRITE PKT Data Type.
- **NbParams:** Number of parameters.
- **Param1:** DSC command or first generic parameter. This parameter can be any value of DSI DCS Command or a generic command code
- **ParametersTable:** Pointer to parameter values table.

Return values
- **HAL:** status

HAL_DSI_Read

Function Name **HAL_StatusTypeDef HAL_DSI_Read (DSI_HandleTypeDef * hdsi, uint32_t ChannelNbr, uint8_t * Array, uint32_t Size, uint32_t Mode, uint32_t DCSCmd, uint8_t * ParametersTable)**

Function Description Read command (DCS or generic)

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **ChannelNbr:** Virtual channel ID
- **Array:** pointer to a buffer to store the payload of a read back operation.
- **Size:** Data size to be read (in byte).
- **Mode:** DSI read packet data type. This parameter can be any value of DSI SHORT READ PKT Data Type.
- **DCSCmd:** DCS get/read command.
- **ParametersTable:** Pointer to parameter values table.

Return values
- **HAL:** status

HAL_DSI_EnterULPMData

Function Name **HAL_StatusTypeDef HAL_DSI_EnterULPMData**

(DSI_HandleTypeDef * hdsi)

Function Description: Enter the ULPM (Ultra Low Power Mode) with the D-PHY PLL running (only data lanes are in ULPM)

Parameters:
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values:
- **HAL:** status

HAL_DSI_ExitULPMData

Function Name: **HAL_StatusTypeDef HAL_DSI_ExitULPMData (DSI_HandleTypeDef * hdsi)**

Function Description: Exit the ULPM (Ultra Low Power Mode) with the D-PHY PLL running (only data lanes are in ULPM)

Parameters:
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values:
- **HAL:** status

HAL_DSI_EnterULPM

Function Name: **HAL_StatusTypeDef HAL_DSI_EnterULPM (DSI_HandleTypeDef * hdsi)**

Function Description: Enter the ULPM (Ultra Low Power Mode) with the D-PHY PLL turned off (both data and clock lanes are in ULPM)

Parameters:
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values:
- **HAL:** status

HAL_DSI_ExitULPM

Function Name: **HAL_StatusTypeDef HAL_DSI_ExitULPM (DSI_HandleTypeDef * hdsi)**

Function Description: Exit the ULPM (Ultra Low Power Mode) with the D-PHY PLL turned off (both data and clock lanes are in ULPM)

Parameters:
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values:
- **HAL:** status

HAL_DSI_PatternGeneratorStart

Function Name: **HAL_StatusTypeDef HAL_DSI_PatternGeneratorStart (DSI_HandleTypeDef * hdsi, uint32_t Mode, uint32_t Orientation)**

Function Description: Start test pattern generation.

Parameters:
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **Mode:** Pattern generator mode This parameter can be one of the following values: 0 : Color bars (horizontal or vertical) 1 :

BER pattern (vertical only)
- **Orientation:** Pattern generator orientation This parameter can be one of the following values: 0 : Vertical color bars 1 : Horizontal color bars

Return values
- **HAL:** status

HAL_DSI_PatternGeneratorStop

Function Name **HAL_StatusTypeDef HAL_DSI_PatternGeneratorStop (DSI_HandleTypeDef * hdsi)**

Function Description Stop test pattern generation.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values
- **HAL:** status

HAL_DSI_SetSlewRateAndDelayTuning

Function Name **HAL_StatusTypeDef HAL_DSI_SetSlewRateAndDelayTuning (DSI_HandleTypeDef * hdsi, uint32_t CommDelay, uint32_t Lane, uint32_t Value)**

Function Description Set Slew-Rate And Delay Tuning.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **CommDelay:** Communication delay to be adjusted. This parameter can be any value of DSI Communication Delay
- **Lane:** select between clock or data lanes. This parameter can be any value of DSI Lane Group
- **Value:** Custom value of the slew-rate or delay

Return values
- **HAL:** status

HAL_DSI_SetLowPowerRXFilter

Function Name **HAL_StatusTypeDef HAL_DSI_SetLowPowerRXFilter (DSI_HandleTypeDef * hdsi, uint32_t Frequency)**

Function Description Low-Power Reception Filter Tuning.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **Frequency:** cutoff frequency of low-pass filter at the input of LPRX

Return values
- **HAL:** status

HAL_DSI_SetSDD

Function Name **HAL_StatusTypeDef HAL_DSI_SetSDD (DSI_HandleTypeDef * hdsi, FunctionalState State)**

Function Description Activate an additional current path on all lanes to meet the SDDTx parameter defined in the MIPI D-PHY specification.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_SetLanePinsConfiguration

Function Name **HAL_StatusTypeDef HAL_DSI_SetLanePinsConfiguration (DSI_HandleTypeDef * hdsi, uint32_t CustomLane, uint32_t Lane, FunctionalState State)**

Function Description Custom lane pins configuration.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **CustomLane:** Function to be applyed on selected lane. This parameter can be any value of DSI CustomLane
- **Lane:** select between clock or data lane 0 or data lane 1. This parameter can be any value of DSI Lane Select
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_SetPHYTimings

Function Name **HAL_StatusTypeDef HAL_DSI_SetPHYTimings (DSI_HandleTypeDef * hdsi, uint32_t Timing, FunctionalState State, uint32_t Value)**

Function Description Set custom timing for the PHY.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **Timing:** PHY timing to be adjusted. This parameter can be any value of DSI PHY Timing
- **State:** ENABLE or DISABLE
- **Value:** Custom value of the timing

Return values
- **HAL:** status

HAL_DSI_ForceTXStopMode

Function Name **HAL_StatusTypeDef HAL_DSI_ForceTXStopMode (DSI_HandleTypeDef * hdsi, uint32_t Lane, FunctionalState State)**

Function Description Force the Clock/Data Lane in TX Stop Mode.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **Lane:** select between clock or data lanes. This parameter can be any value of DSI Lane Group
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_ForceRXLowPower

Function Name **HAL_StatusTypeDef HAL_DSI_ForceRXLowPower (DSI_HandleTypeDef * hdsi, FunctionalState State)**

Function Description Forces LP Receiver in Low-Power Mode.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_ForceDataLanesInRX

Function Name **HAL_StatusTypeDef HAL_DSI_ForceDataLanesInRX (DSI_HandleTypeDef * hdsi, FunctionalState State)**

Function Description Force Data Lanes in RX Mode after a BTA.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_SetPullDown

Function Name **HAL_StatusTypeDef HAL_DSI_SetPullDown (DSI_HandleTypeDef * hdsi, FunctionalState State)**

Function Description Enable a pull-down on the lanes to prevent from floating states when unused.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_SetContentionDetectionOff

Function Name **HAL_StatusTypeDef HAL_DSI_SetContentionDetectionOff (DSI_HandleTypeDef * hdsi, FunctionalState State)**

Function Description Switch off the contention detection on data lanes.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **State:** ENABLE or DISABLE

Return values
- **HAL:** status

HAL_DSI_GetError

Function Name **uint32_t HAL_DSI_GetError (DSI_HandleTypeDef * hdsi)**

Function Description Return the DSI error code.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values
- **DSI:** Error Code

HAL_DSI_ConfigErrorMonitor

Function Name **HAL_StatusTypeDef HAL_DSI_ConfigErrorMonitor (DSI_HandleTypeDef * hdsi, uint32_t ActiveErrors)**

Function Description Enable the error monitor flags.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.
- **ActiveErrors:** indicates which error interrupts will be enabled. This parameter can be any combination of DSI Error Data Type.

Return values
- **HAL:** status

HAL_DSI_GetState

Function Name **HAL_DSI_StateTypeDef HAL_DSI_GetState (DSI_HandleTypeDef * hdsi)**

Function Description Return the DSI state.

Parameters
- **hdsi:** pointer to a DSI_HandleTypeDef structure that contains the configuration information for the DSI.

Return values
- **HAL:** state

20.3 DSI Firmware driver defines

20.3.1 DSI

DSI Acknowledge Request

DSI_ACKNOWLEDGE_DISABLE

DSI_ACKNOWLEDGE_ENABLE

DSI Automatic Refresh

DSI_AR_DISABLE

DSI_AR_ENABLE

DSI Automatic Clk Lane Control

DSI_AUTO_CLK_LANE_CTRL_DISABLE

DSI_AUTO_CLK_LANE_CTRL_ENABLE

DSI Color Coding

DSI_RGB565 The values 0x00000001U and 0x00000002U can also be used for the RGB565 color mode configuration

DSI_RGB666 The value 0x00000004U can also be used for the RGB666 color mode configuration

DSI_RGB888

DSI Color Mode

DSI_COLOR_MODE_FULL

DSI_COLOR_MODE_EIGHT

DSI Communication Delay

DSI_SLEW_RATE_HSTX

DSI_SLEW_RATE_LPTX

DSI_HS_DELAY

DSI CustomLane

DSI_SWAP_LANE_PINS

DSI_INVERT_HS_SIGNAL

DSI DATA ENABLE Polarity

DSI_DATA_ENABLE_ACTIVE_HIGH

DSI_DATA_ENABLE_ACTIVE_LOW

DSI DCS Command

DSI_ENTER_IDLE_MODE

DSI_ENTER_INVERT_MODE

DSI_ENTER_NORMAL_MODE

DSI_ENTER_PARTIAL_MODE

DSI_ENTER_SLEEP_MODE

DSI_EXIT_IDLE_MODE

DSI_EXIT_INVERT_MODE

DSI_EXIT_SLEEP_MODE

DSI_GET_3D_CONTROL

DSI_GET_ADDRESS_MODE

DSI_GET_BLUE_CHANNEL

DSI_GET_DIAGNOSTIC_RESULT

DSI_GET_DISPLAY_MODE

DSI_GET_GREEN_CHANNEL

DSI_GET_PIXEL_FORMAT

DSI_GET_POWER_MODE

DSI_GET_RED_CHANNEL

DSI_GET_SCANLINE

DSI_GET_SIGNAL_MODE

DSI_NOP

DSI_READ_DDB_CONTINUE

DSI_READ_DDB_START

DSI_READ_MEMORY_CONTINUE

DSI_READ_MEMORY_START
DSI_SET_3D_CONTROL
DSI_SET_ADDRESS_MODE
DSI_SET_COLUMN_ADDRESS
DSI_SET_DISPLAY_OFF
DSI_SET_DISPLAY_ON
DSI_SET_GAMMA_CURVE
DSI_SET_PAGE_ADDRESS
DSI_SET_PARTIAL_COLUMNS
DSI_SET_PARTIAL_ROWS
DSI_SET_PIXEL_FORMAT
DSI_SET_SCROLL_AREA
DSI_SET_SCROLL_START
DSI_SET_TEAR_OFF
DSI_SET_TEAR_ON
DSI_SET_TEAR_SCANLINE
DSI_SET_VSYNC_TIMING
DSI_SOFT_RESET
DSI_WRITE_LUT
DSI_WRITE_MEMORY_CONTINUE
DSI_WRITE_MEMORY_START

DSI Error Data Type

HAL_DSI_ERROR_NONE
HAL_DSI_ERROR_ACK acknowledge errors
HAL_DSI_ERROR_PHY PHY related errors
HAL_DSI_ERROR_TX transmission error
HAL_DSI_ERROR_RX reception error
HAL_DSI_ERROR_ECC ECC errors
HAL_DSI_ERROR_CRC CRC error
HAL_DSI_ERROR_PSE Packet Size error
HAL_DSI_ERROR_EOT End Of Transmission error
HAL_DSI_ERROR_OVF FIFO overflow error
HAL_DSI_ERROR_GEN Generic FIFO related errors

DSI FBTA Acknowledge

DSI_FBTAA_DISABLE
DSI_FBTAA_ENABLE

DSI Flags

DSI_FLAG_TE

DSI_FLAG_ER

DSI_FLAG_BUSY

DSI_FLAG_PLLLS

DSI_FLAG_PLLL

DSI_FLAG_PLLU

DSI_FLAG_RRS

DSI_FLAG_RR

DSI Flow Control

DSI_FLOW_CONTROL_CRC_RX

DSI_FLOW_CONTROL_ECC_RX

DSI_FLOW_CONTROL_BTA

DSI_FLOW_CONTROL_EOTP_RX

DSI_FLOW_CONTROL_EOTP_TX

DSI_FLOW_CONTROL_ALL

DSI HSYNC Polarity

DSI_HSYNC_ACTIVE_HIGH

DSI_HSYNC_ACTIVE_LOW

DSI HS Presp Mode

DSI_HS_PM_DISABLE

DSI_HS_PM_ENABLE

DSI Interrupts

DSI_IT_TE

DSI_IT_ER

DSI_IT_PLLL

DSI_IT_PLLU

DSI_IT_RR

DSI Lane Group

DSI_CLOCK_LANE

DSI_CLOCK_LANE

DSI_DATA_LANES

DSI Lane Select

DSI_DATA_LANE0

DSI_DATA_LANE1

DSI LONG WRITE PKT Data Type

DSI_DCS_LONG_PKT_WRITE DCS long write

DSI_GEN_LONG_PKT_WRITE Generic long write

DSI Loosely Packed

DSI_LOOSELY_PACKED_ENABLE

DSI_LOOSELY_PACKED_DISABLE

DSI LP Command

DSI_LP_COMMAND_DISABLE

DSI_LP_COMMAND_ENABLE

DSI LP HBP

DSI_LP_HBP_DISABLE

DSI_LP_HBP_ENABLE

DSI LP HFP

DSI_LP_HFP_DISABLE

DSI_LP_HFP_ENABLE

DSI LP LPDcs Long Write

DSI_LP_DLW_DISABLE

DSI_LP_DLW_ENABLE

DSI LP LPDcs Short Read NoP

DSI_LP_DSR0P_DISABLE

DSI_LP_DSR0P_ENABLE

DSI LP LPDcs Short Write NoP

DSI_LP_DSW0P_DISABLE

DSI_LP_DSW0P_ENABLE

DSI LP LPDcs Short Write OneP

DSI_LP_DSW1P_DISABLE

DSI_LP_DSW1P_ENABLE

DSI LP LPGen LongWrite

DSI_LP_GLW_DISABLE

DSI_LP_GLW_ENABLE

DSI LP LPGen Short Read NoP

DSI_LP_GSR0P_DISABLE

DSI_LP_GSR0P_ENABLE

DSI LP LPGen Short Read OneP

DSI_LP_GSR1P_DISABLE

DSI_LP_GSR1P_ENABLE

DSI LP LPGen Short Read TwoP

DSI_LP_GSR2P_DISABLE

DSI_LP_GSR2P_ENABLE

DSI LP LPGen Short Write NoP

DSI_LP_GSW0P_DISABLE

DSI_LP_GSW0P_ENABLE

DSI LP LPGen Short Write OneP

DSI_LP_GSW1P_DISABLE

DSI_LP_GSW1P_ENABLE

DSI LP LPGen Short Write TwoP

DSI_LP_GSW2P_DISABLE

DSI_LP_GSW2P_ENABLE

DSI LP LPMax Read Packet

DSI_LP_MRDP_DISABLE

DSI_LP_MRDP_ENABLE

DSI LP VACT

DSI_LP_VACT_DISABLE

DSI_LP_VACT_ENABLE

DSI LP VBP

DSI_LP_VBP_DISABLE

DSI_LP_VBP_ENABLE

DSI LP VFP

DSI_LP_VFP_DISABLE

DSI_LP_VFP_ENABLE

DSI LP VSYNC

DSI_LP_VSYNC_DISABLE

DSI_LP_VSYNC_ENABLE

DSI Number Of Lanes

DSI_ONE_DATA_LANE

DSI_TWO_DATA_LANES

DSI PHY Timing

DSI_TCLK_POST

DSI_TLPX_CLK

DSI_THS_EXIT

DSI_TLPX_DATA

DSI_THS_ZERO

DSI_THS_TRAIL

DSI_THS_PREPARE

DSI_TCLK_ZERO

DSI_TCLK_PREPARE

DSI PLL IDF

DSI_PLL_IN_DIV1

DSI_PLL_IN_DIV2

DSI_PLL_IN_DIV3

DSI_PLL_IN_DIV4

DSI_PLL_IN_DIV5

DSI_PLL_IN_DIV6

DSI_PLL_IN_DIV7

DSI PLL ODF

DSI_PLL_OUT_DIV1

DSI_PLL_OUT_DIV2

DSI_PLL_OUT_DIV4

DSI_PLL_OUT_DIV8

DSI SHORT READ PKT Data Type

DSI_DCS_SHORT_PKT_READ	DCS short read
DSI_GEN_SHORT_PKT_READ_P0	Generic short read, no parameters
DSI_GEN_SHORT_PKT_READ_P1	Generic short read, one parameter
DSI_GEN_SHORT_PKT_READ_P2	Generic short read, two parameters

DSI SHORT WRITE PKT Data Type

DSI_DCS_SHORT_PKT_WRITE_P0	DCS short write, no parameters
DSI_DCS_SHORT_PKT_WRITE_P1	DCS short write, one parameter
DSI_GEN_SHORT_PKT_WRITE_P0	Generic short write, no parameters
DSI_GEN_SHORT_PKT_WRITE_P1	Generic short write, one parameter
DSI_GEN_SHORT_PKT_WRITE_P2	Generic short write, two parameters

DSI ShutDown

DSI_DISPLAY_ON

DSI_DISPLAY_OFF

DSI Tearing Effect Polarity

DSI_TE_RISING_EDGE

DSI_TE_FALLING_EDGE

DSI Tearing Effect Source

DSI_TE_DSILINK

DSI_TE_EXTERNAL

DSI TE Acknowledge Request

DSI_TE_ACKNOWLEDGE_DISABLE

DSI_TE_ACKNOWLEDGE_ENABLE

DSI Video Mode Type

DSI_VID_MODE_NB_PULSES

DSI_VID_MODE_NB_EVENTS

DSI_VID_MODE_BURST

DSI VSYNC Active Polarity

DSI_VSYNC_ACTIVE_HIGH

DSI_VSYNC_ACTIVE_LOW

DSI Vsync Polarity

DSI_VSYNC_FALLING

DSI_VSYNC_RISING

21 HAL ETH Generic Driver

21.1 ETH Firmware driver registers structures

21.1.1 ETH_InitTypeDef

Data Fields

- ***uint32_t AutoNegotiation***
- ***uint32_t Speed***
- ***uint32_t DuplexMode***
- ***uint16_t PhyAddress***
- ***uint8_t * MACAddr***
- ***uint32_t RxMode***
- ***uint32_t ChecksumMode***
- ***uint32_t MediaInterface***

Field Documentation

- ***uint32_t ETH_InitTypeDef::AutoNegotiation***
  Selects or not the AutoNegotiation mode for the external PHY The AutoNegotiation allows an automatic setting of the Speed (10/100Mbps) and the mode (half/full-duplex). This parameter can be a value of ***ETH_AutoNegotiation***
- ***uint32_t ETH_InitTypeDef::Speed***
  Sets the Ethernet speed: 10/100 Mbps. This parameter can be a value of ***ETH_Speed***
- ***uint32_t ETH_InitTypeDef::DuplexMode***
  Selects the MAC duplex mode: Half-Duplex or Full-Duplex mode This parameter can be a value of ***ETH_Duplex_Mode***
- ***uint16_t ETH_InitTypeDef::PhyAddress***
  Ethernet PHY address. This parameter must be a number between Min_Data = 0 and Max_Data = 32
- ***uint8_t* ETH_InitTypeDef::MACAddr***
  MAC Address of used Hardware: must be pointer on an array of 6 bytes
- ***uint32_t ETH_InitTypeDef::RxMode***
  Selects the Ethernet Rx mode: Polling mode, Interrupt mode. This parameter can be a value of ***ETH_Rx_Mode***
- ***uint32_t ETH_InitTypeDef::ChecksumMode***
  Selects if the checksum is check by hardware or by software. This parameter can be a value of ***ETH_Checksum_Mode***
- ***uint32_t ETH_InitTypeDef::MediaInterface***
  Selects the media-independent interface or the reduced media-independent interface. This parameter can be a value of ***ETH_Media_Interface***

21.1.2 ETH_MACInitTypeDef

Data Fields

- ***uint32_t Watchdog***
- ***uint32_t Jabber***
- ***uint32_t InterFrameGap***
- ***uint32_t CarrierSense***
- ***uint32_t ReceiveOwn***
- ***uint32_t LoopbackMode***
- ***uint32_t ChecksumOffload***
- ***uint32_t RetryTransmission***
- ***uint32_t AutomaticPadCRCStrip***
- ***uint32_t BackOffLimit***
- ***uint32_t DeferralCheck***
- ***uint32_t ReceiveAll***
- ***uint32_t SourceAddrFilter***
- ***uint32_t PassControlFrames***
- ***uint32_t BroadcastFramesReception***
- ***uint32_t DestinationAddrFilter***
- ***uint32_t PromiscuousMode***
- ***uint32_t MulticastFramesFilter***
- ***uint32_t UnicastFramesFilter***
- ***uint32_t HashTableHigh***
- ***uint32_t HashTableLow***
- ***uint32_t PauseTime***
- ***uint32_t ZeroQuantaPause***
- ***uint32_t PauseLowThreshold***
- ***uint32_t UnicastPauseFrameDetect***
- ***uint32_t ReceiveFlowControl***
- ***uint32_t TransmitFlowControl***
- ***uint32_t VLANTagComparison***
- ***uint32_t VLANTagIdentifier***

Field Documentation

- ***uint32_t ETH_MACInitTypeDef::Watchdog***
  Selects or not the Watchdog timer When enabled, the MAC allows no more then 2048 bytes to be received. When disabled, the MAC can receive up to 16384 bytes. This parameter can be a value of ***ETH_Watchdog***
- ***uint32_t ETH_MACInitTypeDef::Jabber***
  Selects or not Jabber timer When enabled, the MAC allows no more then 2048 bytes to be sent. When disabled, the MAC can send up to 16384 bytes. This parameter can be a value of ***ETH_Jabber***
- ***uint32_t ETH_MACInitTypeDef::InterFrameGap***
  Selects the minimum IFG between frames during transmission. This parameter can be a value of ***ETH_Inter_Frame_Gap***
- ***uint32_t ETH_MACInitTypeDef::CarrierSense***
  Selects or not the Carrier Sense. This parameter can be a value of ***ETH_Carrier_Sense***

- ***uint32_t ETH_MACInitTypeDef::ReceiveOwn***
  Selects or not the ReceiveOwn, ReceiveOwn allows the reception of frames when the TX_EN signal is asserted in Half-Duplex mode. This parameter can be a value of ***ETH_Receive_Own***
- ***uint32_t ETH_MACInitTypeDef::LoopbackMode***
  Selects or not the internal MAC MII Loopback mode. This parameter can be a value of ***ETH_Loop_Back_Mode***
- ***uint32_t ETH_MACInitTypeDef::ChecksumOffload***
  Selects or not the IPv4 checksum checking for received frame payloads' TCP/UDP/ICMP headers. This parameter can be a value of ***ETH_Checksum_Offload***
- ***uint32_t ETH_MACInitTypeDef::RetryTransmission***
  Selects or not the MAC attempt retries transmission, based on the settings of BL, when a collision occurs (Half-Duplex mode). This parameter can be a value of ***ETH_Retry_Transmission***
- ***uint32_t ETH_MACInitTypeDef::AutomaticPadCRCStrip***
  Selects or not the Automatic MAC Pad/CRC Stripping. This parameter can be a value of ***ETH_Automatic_Pad_CRC_Strip***
- ***uint32_t ETH_MACInitTypeDef::BackOffLimit***
  Selects the BackOff limit value. This parameter can be a value of ***ETH_Back_Off_Limit***
- ***uint32_t ETH_MACInitTypeDef::DeferralCheck***
  Selects or not the deferral check function (Half-Duplex mode). This parameter can be a value of ***ETH_Deferral_Check***
- ***uint32_t ETH_MACInitTypeDef::ReceiveAll***
  Selects or not all frames reception by the MAC (No filtering). This parameter can be a value of ***ETH_Receive_All***
- ***uint32_t ETH_MACInitTypeDef::SourceAddrFilter***
  Selects the Source Address Filter mode. This parameter can be a value of ***ETH_Source_Addr_Filter***
- ***uint32_t ETH_MACInitTypeDef::PassControlFrames***
  Sets the forwarding mode of the control frames (including unicast and multicast PAUSE frames) This parameter can be a value of ***ETH_Pass_Control_Frames***
- ***uint32_t ETH_MACInitTypeDef::BroadcastFramesReception***
  Selects or not the reception of Broadcast Frames. This parameter can be a value of ***ETH_Broadcast_Frames_Reception***
- ***uint32_t ETH_MACInitTypeDef::DestinationAddrFilter***
  Sets the destination filter mode for both unicast and multicast frames. This parameter can be a value of ***ETH_Destination_Addr_Filter***
- ***uint32_t ETH_MACInitTypeDef::PromiscuousMode***
  Selects or not the Promiscuous Mode This parameter can be a value of ***ETH_Promiscuous_Mode***
- ***uint32_t ETH_MACInitTypeDef::MulticastFramesFilter***
  Selects the Multicast Frames filter mode: None/HashTableFilter/PerfectFilter/PerfectHashTableFilter. This parameter can be a value of ***ETH_Multicast_Frames_Filter***
- ***uint32_t ETH_MACInitTypeDef::UnicastFramesFilter***
  Selects the Unicast Frames filter mode: HashTableFilter/PerfectFilter/PerfectHashTableFilter. This parameter can be a value of ***ETH_Unicast_Frames_Filter***
- ***uint32_t ETH_MACInitTypeDef::HashTableHigh***
  This field holds the higher 32 bits of Hash table. This parameter must be a number between Min_Data = 0x0 and Max_Data = 0xFFFFFFFFU

- ***uint32_t ETH_MACInitTypeDef::HashTableLow***
  This field holds the lower 32 bits of Hash table. This parameter must be a number between Min_Data = 0x0 and Max_Data = 0xFFFFFFFFU
- ***uint32_t ETH_MACInitTypeDef::PauseTime***
  This field holds the value to be used in the Pause Time field in the transmit control frame. This parameter must be a number between Min_Data = 0x0 and Max_Data = 0xFFFFU
- ***uint32_t ETH_MACInitTypeDef::ZeroQuantaPause***
  Selects or not the automatic generation of Zero-Quanta Pause Control frames. This parameter can be a value of ***ETH_Zero_Quanta_Pause***
- ***uint32_t ETH_MACInitTypeDef::PauseLowThreshold***
  This field configures the threshold of the PAUSE to be checked for automatic retransmission of PAUSE Frame. This parameter can be a value of ***ETH_Pause_Low_Threshold***
- ***uint32_t ETH_MACInitTypeDef::UnicastPauseFrameDetect***
  Selects or not the MAC detection of the Pause frames (with MAC Address0 unicast address and unique multicast address). This parameter can be a value of ***ETH_Unicast_Pause_Frame_Detect***
- ***uint32_t ETH_MACInitTypeDef::ReceiveFlowControl***
  Enables or disables the MAC to decode the received Pause frame and disable its transmitter for a specified time (Pause Time) This parameter can be a value of ***ETH_Receive_Flow_Control***
- ***uint32_t ETH_MACInitTypeDef::TransmitFlowControl***
  Enables or disables the MAC to transmit Pause frames (Full-Duplex mode) or the MAC back-pressure operation (Half-Duplex mode) This parameter can be a value of ***ETH_Transmit_Flow_Control***
- ***uint32_t ETH_MACInitTypeDef::VLANTagComparison***
  Selects the 12-bit VLAN identifier or the complete 16-bit VLAN tag for comparison and filtering. This parameter can be a value of ***ETH_VLAN_Tag_Comparison***
- ***uint32_t ETH_MACInitTypeDef::VLANTagIdentifier***
  Holds the VLAN tag identifier for receive frames

21.1.3 ETH_DMAInitTypeDef

Data Fields

- ***uint32_t DropTCPIPChecksumErrorFrame***
- ***uint32_t ReceiveStoreForward***
- ***uint32_t FlushReceivedFrame***
- ***uint32_t TransmitStoreForward***
- ***uint32_t TransmitThresholdControl***
- ***uint32_t ForwardErrorFrames***
- ***uint32_t ForwardUndersizedGoodFrames***
- ***uint32_t ReceiveThresholdControl***
- ***uint32_t SecondFrameOperate***
- ***uint32_t AddressAlignedBeats***
- ***uint32_t FixedBurst***
- ***uint32_t RxDMABurstLength***
- ***uint32_t TxDMABurstLength***
- ***uint32_t EnhancedDescriptorFormat***
- ***uint32_t DescriptorSkipLength***
- ***uint32_t DMAArbitration***

Field Documentation

- ***uint32_t ETH_DMAInitTypeDef::DropTCPIPChecksumErrorFrame***
  Selects or not the Dropping of TCP/IP Checksum Error Frames. This parameter can be a value of ***ETH_Drop_TCP_IP_Checksum_Error_Frame***
- ***uint32_t ETH_DMAInitTypeDef::ReceiveStoreForward***
  Enables or disables the Receive store and forward mode. This parameter can be a value of ***ETH_Receive_Store_Forward***
- ***uint32_t ETH_DMAInitTypeDef::FlushReceivedFrame***
  Enables or disables the flushing of received frames. This parameter can be a value of ***ETH_Flush_Received_Frame***
- ***uint32_t ETH_DMAInitTypeDef::TransmitStoreForward***
  Enables or disables Transmit store and forward mode. This parameter can be a value of ***ETH_Transmit_Store_Forward***
- ***uint32_t ETH_DMAInitTypeDef::TransmitThresholdControl***
  Selects or not the Transmit Threshold Control. This parameter can be a value of ***ETH_Transmit_Threshold_Control***
- ***uint32_t ETH_DMAInitTypeDef::ForwardErrorFrames***
  Selects or not the forward to the DMA of erroneous frames. This parameter can be a value of ***ETH_Forward_Error_Frames***
- ***uint32_t ETH_DMAInitTypeDef::ForwardUndersizedGoodFrames***
  Enables or disables the Rx FIFO to forward Undersized frames (frames with no Error and length less than 64 bytes) including pad-bytes and CRC) This parameter can be a value of ***ETH_Forward_Undersized_Good_Frames***
- ***uint32_t ETH_DMAInitTypeDef::ReceiveThresholdControl***
  Selects the threshold level of the Receive FIFO. This parameter can be a value of ***ETH_Receive_Threshold_Control***
- ***uint32_t ETH_DMAInitTypeDef::SecondFrameOperate***
  Selects or not the Operate on second frame mode, which allows the DMA to process a second frame of Transmit data even before obtaining the status for the first frame. This parameter can be a value of ***ETH_Second_Frame_Operate***
- ***uint32_t ETH_DMAInitTypeDef::AddressAlignedBeats***
  Enables or disables the Address Aligned Beats. This parameter can be a value of ***ETH_Address_Aligned_Beats***
- ***uint32_t ETH_DMAInitTypeDef::FixedBurst***
  Enables or disables the AHB Master interface fixed burst transfers. This parameter can be a value of ***ETH_Fixed_Burst***
- ***uint32_t ETH_DMAInitTypeDef::RxDMABurstLength***
  Indicates the maximum number of beats to be transferred in one Rx DMA transaction. This parameter can be a value of ***ETH_Rx_DMA_Burst_Length***
- ***uint32_t ETH_DMAInitTypeDef::TxDMABurstLength***
  Indicates the maximum number of beats to be transferred in one Tx DMA transaction. This parameter can be a value of ***ETH_Tx_DMA_Burst_Length***
- ***uint32_t ETH_DMAInitTypeDef::EnhancedDescriptorFormat***
  Enables the enhanced descriptor format. This parameter can be a value of ***ETH_DMA_Enhanced_descriptor_format***
- ***uint32_t ETH_DMAInitTypeDef::DescriptorSkipLength***
  Specifies the number of word to skip between two unchained descriptors (Ring mode) This parameter must be a number between Min_Data = 0 and Max_Data = 32
- ***uint32_t ETH_DMAInitTypeDef::DMAArbitration***
  Selects the DMA Tx/Rx arbitration. This parameter can be a value of ***ETH_DMA_Arbitration***

21.1.4 ETH_DMADescTypeDef

Data Fields

- ***__IO uint32_t Status***
- ***uint32_t ControlBufferSize***
- ***uint32_t Buffer1Addr***
- ***uint32_t Buffer2NextDescAddr***
- ***uint32_t ExtendedStatus***
- ***uint32_t Reserved1***
- ***uint32_t TimeStampLow***
- ***uint32_t TimeStampHigh***

Field Documentation

- ***__IO uint32_t ETH_DMADescTypeDef::Status***
  Status
- ***uint32_t ETH_DMADescTypeDef::ControlBufferSize***
  Control and Buffer1, Buffer2 lengths
- ***uint32_t ETH_DMADescTypeDef::Buffer1Addr***
  Buffer1 address pointer
- ***uint32_t ETH_DMADescTypeDef::Buffer2NextDescAddr***
  Buffer2 or next descriptor address pointer Enhanced ETHERNET DMA PTP Descriptors
- ***uint32_t ETH_DMADescTypeDef::ExtendedStatus***
  Extended status for PTP receive descriptor
- ***uint32_t ETH_DMADescTypeDef::Reserved1***
  Reserved
- ***uint32_t ETH_DMADescTypeDef::TimeStampLow***
  Time Stamp Low value for transmit and receive
- ***uint32_t ETH_DMADescTypeDef::TimeStampHigh***
  Time Stamp High value for transmit and receive

21.1.5 ETH_DMARxFrameInfos

Data Fields

- ***ETH_DMADescTypeDef * FSRxDesc***
- ***ETH_DMADescTypeDef * LSRxDesc***
- ***uint32_t SegCount***
- ***uint32_t length***
- ***uint32_t buffer***

Field Documentation

- ***ETH_DMADescTypeDef* ETH_DMARxFrameInfos::FSRxDesc***
  First Segment Rx Desc
- ***ETH_DMADescTypeDef* ETH_DMARxFrameInfos::LSRxDesc***
  Last Segment Rx Desc

- ***uint32_t ETH_DMARxFrameInfos::SegCount***
  Segment count
- ***uint32_t ETH_DMARxFrameInfos::length***
  Frame length
- ***uint32_t ETH_DMARxFrameInfos::buffer***
  Frame buffer

21.1.6 ETH_HandleTypeDef

Data Fields

- ***ETH_TypeDef * Instance***
- ***ETH_InitTypeDef Init***
- ***uint32_t LinkStatus***
- ***ETH_DMADescTypeDef * RxDesc***
- ***ETH_DMADescTypeDef * TxDesc***
- ***ETH_DMARxFrameInfos RxFrameInfos***
- ***__IO HAL_ETH_StateTypeDef State***
- ***HAL_LockTypeDef Lock***

Field Documentation

- ***ETH_TypeDef* ETH_HandleTypeDef::Instance***
  Register base address
- ***ETH_InitTypeDef ETH_HandleTypeDef::Init***
  Ethernet Init Configuration
- ***uint32_t ETH_HandleTypeDef::LinkStatus***
  Ethernet link status
- ***ETH_DMADescTypeDef* ETH_HandleTypeDef::RxDesc***
  Rx descriptor to Get
- ***ETH_DMADescTypeDef* ETH_HandleTypeDef::TxDesc***
  Tx descriptor to Set
- ***ETH_DMARxFrameInfos ETH_HandleTypeDef::RxFrameInfos***
  last Rx frame infos
- ***__IO HAL_ETH_StateTypeDef ETH_HandleTypeDef::State***
  ETH communication state
- ***HAL_LockTypeDef ETH_HandleTypeDef::Lock***
  ETH Lock

21.2 ETH Firmware driver API description

21.2.1 How to use this driver

1. Declare a ETH_HandleTypeDef handle structure, for example: ETH_HandleTypeDef heth;
2. Fill parameters of Init structure in heth handle
3. Call HAL_ETH_Init() API to initialize the Ethernet peripheral (MAC, DMA, ...)
4. Initialize the ETH low level resources through the HAL_ETH_MspInit() API:
   a. Enable the Ethernet interface clock using

- __HAL_RCC_ETHMAC_CLK_ENABLE();
- __HAL_RCC_ETHMACTX_CLK_ENABLE();
- __HAL_RCC_ETHMACRX_CLK_ENABLE();

b. Initialize the related GPIO clocks
c. Configure Ethernet pin-out
d. Configure Ethernet NVIC interrupt (IT mode)

5. Initialize Ethernet DMA Descriptors in chain mode and point to allocated buffers:
   a. HAL_ETH_DMATxDescListInit(); for Transmission process
   b. HAL_ETH_DMARxDescListInit(); for Reception process
6. Enable MAC and DMA transmission and reception:
   a. HAL_ETH_Start();
7. Prepare ETH DMA TX Descriptors and give the hand to ETH DMA to transfer the frame to MAC TX FIFO:
   a. HAL_ETH_TransmitFrame();
8. Poll for a received frame in ETH RX DMA Descriptors and get received frame parameters
   a. HAL_ETH_GetReceivedFrame(); (should be called into an infinite loop)
9. Get a received frame when an ETH RX interrupt occurs:
   a. HAL_ETH_GetReceivedFrame_IT(); (called in IT mode only)
10. Communicate with external PHY device:
    a. Read a specific register from the PHY HAL_ETH_ReadPHYRegister();
    b. Write data to a specific RHY register: HAL_ETH_WritePHYRegister();
11. Configure the Ethernet MAC after ETH peripheral initialization HAL_ETH_ConfigMAC(); all MAC parameters should be filled.
12. Configure the Ethernet DMA after ETH peripheral initialization HAL_ETH_ConfigDMA(); all DMA parameters should be filled. The PTP protocol and the DMA descriptors ring mode are not supported in this driver

21.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize and configure the Ethernet peripheral
- De-initialize the Ethernet peripheral

This section contains the following APIs:

- ***HAL_ETH_Init()***
- ***HAL_ETH_DeInit()***
- ***HAL_ETH_DMATxDescListInit()***
- ***HAL_ETH_DMARxDescListInit()***
- ***HAL_ETH_MspInit()***
- ***HAL_ETH_MspDeInit()***

21.2.3 IO operation functions

This section provides functions allowing to:

- Transmit a frame HAL_ETH_TransmitFrame();
- Receive a frame HAL_ETH_GetReceivedFrame(); HAL_ETH_GetReceivedFrame_IT();
- Read from an External PHY register HAL_ETH_ReadPHYRegister();
- Write to an External PHY register HAL_ETH_WritePHYRegister();

This section contains the following APIs:

- ***HAL_ETH_TransmitFrame()***
- ***HAL_ETH_GetReceivedFrame()***

- *HAL_ETH_GetReceivedFrame_IT()*
- *HAL_ETH_IRQHandler()*
- *HAL_ETH_TxCpltCallback()*
- *HAL_ETH_RxCpltCallback()*
- *HAL_ETH_ErrorCallback()*
- *HAL_ETH_ReadPHYRegister()*
- *HAL_ETH_WritePHYRegister()*

21.2.4 Peripheral Control functions

This section provides functions allowing to:

- Enable MAC and DMA transmission and reception. HAL_ETH_Start();
- Disable MAC and DMA transmission and reception. HAL_ETH_Stop();
- Set the MAC configuration in runtime mode HAL_ETH_ConfigMAC();
- Set the DMA configuration in runtime mode HAL_ETH_ConfigDMA();

This section contains the following APIs:

- *HAL_ETH_Start()*
- *HAL_ETH_Stop()*
- *HAL_ETH_ConfigMAC()*
- *HAL_ETH_ConfigDMA()*

21.2.5 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

- Get the ETH handle state: HAL_ETH_GetState();

This section contains the following APIs:

- *HAL_ETH_GetState()*

21.2.6 Detailed description of functions

HAL_ETH_Init

Function Name	**HAL_StatusTypeDef HAL_ETH_Init (ETH_HandleTypeDef * heth)**
Function Description	Initializes the Ethernet MAC and DMA according to default parameters.
Parameters	• **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
Return values	• **HAL:** status

HAL_ETH_DeInit

Function Name	**HAL_StatusTypeDef HAL_ETH_DeInit (ETH_HandleTypeDef * heth)**
Function Description	De-Initializes the ETH peripheral.
Parameters	• **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
Return values	• **HAL:** status

HAL_ETH_MspInit

Function Name **void HAL_ETH_MspInit (ETH_HandleTypeDef * heth)**

Function Description Initializes the ETH MSP.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **None**

HAL_ETH_MspDeInit

Function Name **void HAL_ETH_MspDeInit (ETH_HandleTypeDef * heth)**

Function Description DeInitializes ETH MSP.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **None**

HAL_ETH_DMATxDescListInit

Function Name **HAL_StatusTypeDef HAL_ETH_DMATxDescListInit (ETH_HandleTypeDef * heth, ETH_DMADescTypeDef * DMATxDescTab, uint8_t * TxBuff, uint32_t TxBuffCount)**

Function Description Initializes the DMA Tx descriptors in chain mode.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **DMATxDescTab:** Pointer to the first Tx desc list
- **TxBuff:** Pointer to the first TxBuffer list
- **TxBuffCount:** Number of the used Tx desc in the list

Return values
- **HAL:** status

HAL_ETH_DMARxDescListInit

Function Name **HAL_StatusTypeDef HAL_ETH_DMARxDescListInit (ETH_HandleTypeDef * heth, ETH_DMADescTypeDef * DMARxDescTab, uint8_t * RxBuff, uint32_t RxBuffCount)**

Function Description Initializes the DMA Rx descriptors in chain mode.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **DMARxDescTab:** Pointer to the first Rx desc list
- **RxBuff:** Pointer to the first RxBuffer list
- **RxBuffCount:** Number of the used Rx desc in the list

Return values
- **HAL:** status

HAL_ETH_TransmitFrame

Function Name **HAL_StatusTypeDef HAL_ETH_TransmitFrame (ETH_HandleTypeDef * heth, uint32_t FrameLength)**

Function Description Sends an Ethernet frame.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **FrameLength:** Amount of data to be sent

Return values
- **HAL:** status

HAL_ETH_GetReceivedFrame

Function Name **HAL_StatusTypeDef HAL_ETH_GetReceivedFrame (ETH_HandleTypeDef * heth)**

Function Description Checks for received frames.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **HAL:** status

HAL_ETH_ReadPHYRegister

Function Name **HAL_StatusTypeDef HAL_ETH_ReadPHYRegister (ETH_HandleTypeDef * heth, uint16_t PHYReg, uint32_t * RegValue)**

Function Description Reads a PHY register.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **PHYReg:** PHY register address, is the index of one of the 32 PHY register. This parameter can be one of the following values: PHY_BCR: Transceiver Basic Control Register, PHY_BSR: Transceiver Basic Status Register. More PHY register could be read depending on the used PHY
- **RegValue:** PHY register value

Return values
- **HAL:** status

HAL_ETH_WritePHYRegister

Function Name **HAL_StatusTypeDef HAL_ETH_WritePHYRegister (ETH_HandleTypeDef * heth, uint16_t PHYReg, uint32_t RegValue)**

Function Description Writes to a PHY register.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **PHYReg:** PHY register address, is the index of one of the 32 PHY register. This parameter can be one of the following values: PHY_BCR: Transceiver Control Register. More PHY register could be written depending on the used PHY
- **RegValue:** the value to write

Return values
- **HAL:** status

HAL_ETH_GetReceivedFrame_IT

Function Name **HAL_StatusTypeDef HAL_ETH_GetReceivedFrame_IT**

(ETH_HandleTypeDef * heth)

Function Description: Gets the Received frame in interrupt mode.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values:
- **HAL:** status

HAL_ETH_IRQHandler

Function Name: **void HAL_ETH_IRQHandler (ETH_HandleTypeDef * heth)**

Function Description: This function handles ETH interrupt request.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values:
- **HAL:** status

HAL_ETH_TxCpltCallback

Function Name: **void HAL_ETH_TxCpltCallback (ETH_HandleTypeDef * heth)**

Function Description: Tx Transfer completed callbacks.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values:
- **None**

HAL_ETH_RxCpltCallback

Function Name: **void HAL_ETH_RxCpltCallback (ETH_HandleTypeDef * heth)**

Function Description: Rx Transfer completed callbacks.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values:
- **None**

HAL_ETH_ErrorCallback

Function Name: **void HAL_ETH_ErrorCallback (ETH_HandleTypeDef * heth)**

Function Description: Ethernet transfer error callbacks.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values:
- **None**

HAL_ETH_Start

Function Name: **HAL_StatusTypeDef HAL_ETH_Start (ETH_HandleTypeDef * heth)**

Function Description: Enables Ethernet MAC and DMA reception/transmission.

Parameters:
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **HAL:** status

HAL_ETH_Stop

Function Name **HAL_StatusTypeDef HAL_ETH_Stop (ETH_HandleTypeDef * heth)**

Function Description Stop Ethernet MAC and DMA reception/transmission.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **HAL:** status

HAL_ETH_ConfigMAC

Function Name **HAL_StatusTypeDef HAL_ETH_ConfigMAC (ETH_HandleTypeDef * heth, ETH_MACInitTypeDef * macconf)**

Function Description Set ETH MAC Configuration.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **macconf:** MAC Configuration structure

Return values
- **HAL:** status

HAL_ETH_ConfigDMA

Function Name **HAL_StatusTypeDef HAL_ETH_ConfigDMA (ETH_HandleTypeDef * heth, ETH_DMAInitTypeDef * dmaconf)**

Function Description Sets ETH DMA Configuration.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module
- **dmaconf:** DMA Configuration structure

Return values
- **HAL:** status

HAL_ETH_GetState

Function Name **HAL_ETH_StateTypeDef HAL_ETH_GetState (ETH_HandleTypeDef * heth)**

Function Description Return the ETH HAL state.

Parameters
- **heth:** pointer to a ETH_HandleTypeDef structure that contains the configuration information for ETHERNET module

Return values
- **HAL:** state

21.3 ETH Firmware driver defines

21.3.1 ETH

ETH Address Aligned Beats

ETH_ADDRESSALIGNEDBEATS_ENABLE

ETH_ADDRESSALIGNEDBEATS_DISABLE

ETH Automatic Pad CRC Strip

ETH_AUTOMATICPADCRCSTRIP_ENABLE

ETH_AUTOMATICPADCRCSTRIP_DISABLE

ETH AutoNegotiation

ETH_AUTONEGOTIATION_ENABLE

ETH_AUTONEGOTIATION_DISABLE

ETH Back Off Limit

ETH_BACKOFFLIMIT_10

ETH_BACKOFFLIMIT_8

ETH_BACKOFFLIMIT_4

ETH_BACKOFFLIMIT_1

ETH Broadcast Frames Reception

ETH_BROADCASTFRAMESRECEPTION_ENABLE

ETH_BROADCASTFRAMESRECEPTION_DISABLE

ETH Buffers setting

ETH_MAX_PACKET_SIZE	ETH_HEADER + ETH_EXTRA + ETH_VLAN_TAG + ETH_MAX_ETH_PAYLOAD + ETH_CRC
ETH_HEADER	6 byte Dest addr, 6 byte Src addr, 2 byte length/type
ETH_CRC	Ethernet CRC
ETH_EXTRA	Extra bytes in some cases
ETH_VLAN_TAG	optional 802.1q VLAN Tag
ETH_MIN_ETH_PAYLOAD	Minimum Ethernet payload size
ETH_MAX_ETH_PAYLOAD	Maximum Ethernet payload size
ETH_JUMBO_FRAME_PAYLOAD	Jumbo frame payload size

ETH_RX_BUF_SIZE

ETH_RXBUFNB

ETH_TX_BUF_SIZE

ETH_TXBUFNB

ETH Carrier Sense

ETH_CARRIERSENCE_ENABLE

ETH_CARRIERSENCE_DISABLE

ETH Checksum Mode

ETH_CHECKSUM_BY_HARDWARE

ETH_CHECKSUM_BY_SOFTWARE

ETH Checksum Offload

ETH_CHECKSUMOFFLAOD_ENABLE

ETH_CHECKSUMOFFLAOD_DISABLE

ETH Deferral Check

ETH_DEFFERRALCHECK_ENABLE

ETH_DEFFERRALCHECK_DISABLE

ETH Destination Addr Filter

ETH_DESTINATIONADDRFILTER_NORMAL

ETH_DESTINATIONADDRFILTER_INVERSE

ETH DMA Arbitration

ETH_DMAARBITRATION_ROUNDROBIN_RXTX_1_1

ETH_DMAARBITRATION_ROUNDROBIN_RXTX_2_1

ETH_DMAARBITRATION_ROUNDROBIN_RXTX_3_1

ETH_DMAARBITRATION_ROUNDROBIN_RXTX_4_1

ETH_DMAARBITRATION_RXPRIORTX

ETH DMA Enhanced descriptor format

ETH_DMAENHANCEDDESCRIPTOR_ENABLE

ETH_DMAENHANCEDDESCRIPTOR_DISABLE

ETH DMA Flags

ETH_DMA_FLAG_TST	Time-stamp trigger interrupt (on DMA)
ETH_DMA_FLAG_PMT	PMT interrupt (on DMA)
ETH_DMA_FLAG_MMC	MMC interrupt (on DMA)
ETH_DMA_FLAG_DATATRANSFERERROR	Error bits 0-Rx DMA, 1-Tx DMA
ETH_DMA_FLAG_READWRITEERROR	Error bits 0-write transfer, 1-read transfer
ETH_DMA_FLAG_ACCESSERROR	Error bits 0-data buffer, 1-desc. access
ETH_DMA_FLAG_NIS	Normal interrupt summary flag
ETH_DMA_FLAG_AIS	Abnormal interrupt summary flag
ETH_DMA_FLAG_ER	Early receive flag
ETH_DMA_FLAG_FBE	Fatal bus error flag
ETH_DMA_FLAG_ET	Early transmit flag
ETH_DMA_FLAG_RWT	Receive watchdog timeout flag
ETH_DMA_FLAG_RPS	Receive process stopped flag
ETH_DMA_FLAG_RBU	Receive buffer unavailable flag
ETH_DMA_FLAG_R	Receive flag
ETH_DMA_FLAG_TU	Underflow flag
ETH_DMA_FLAG_RO	Overflow flag
ETH_DMA_FLAG_TJT	Transmit jabber timeout flag
ETH_DMA_FLAG_TBU	Transmit buffer unavailable flag

ETH_DMA_FLAG_TPS	Transmit process stopped flag
ETH_DMA_FLAG_T	Transmit flag

ETH DMA Interrupts

ETH_DMA_IT_TST	Time-stamp trigger interrupt (on DMA)
ETH_DMA_IT_PMT	PMT interrupt (on DMA)
ETH_DMA_IT_MMC	MMC interrupt (on DMA)
ETH_DMA_IT_NIS	Normal interrupt summary
ETH_DMA_IT_AIS	Abnormal interrupt summary
ETH_DMA_IT_ER	Early receive interrupt
ETH_DMA_IT_FBE	Fatal bus error interrupt
ETH_DMA_IT_ET	Early transmit interrupt
ETH_DMA_IT_RWT	Receive watchdog timeout interrupt
ETH_DMA_IT_RPS	Receive process stopped interrupt
ETH_DMA_IT_RBU	Receive buffer unavailable interrupt
ETH_DMA_IT_R	Receive interrupt
ETH_DMA_IT_TU	Underflow interrupt
ETH_DMA_IT_RO	Overflow interrupt
ETH_DMA_IT_TJT	Transmit jabber timeout interrupt
ETH_DMA_IT_TBU	Transmit buffer unavailable interrupt
ETH_DMA_IT_TPS	Transmit process stopped interrupt
ETH_DMA_IT_T	Transmit interrupt

ETH DMA overflow

ETH_DMA_OVERFLOW_RXFIFOCOUNTER	Overflow bit for FIFO overflow counter
ETH_DMA_OVERFLOW_MISSEDFRAMECOUNTER	Overflow bit for missed frame counter

ETH DMA receive process state

ETH_DMA_RECEIVEPROCESS_STOPPED	Stopped - Reset or Stop Rx Command issued
ETH_DMA_RECEIVEPROCESS_FETCHING	Running - fetching the Rx descriptor
ETH_DMA_RECEIVEPROCESS_WAITING	Running - waiting for packet
ETH_DMA_RECEIVEPROCESS_SUSPENDED	Suspended - Rx Descriptor unavailable
ETH_DMA_RECEIVEPROCESS_CLOSING	Running - closing descriptor
ETH_DMA_RECEIVEPROCESS_QUEUING	Running - queuing the receive frame into host memory

ETH DMA RX Descriptor

ETH_DMARXDESC_OWN	OWN bit: descriptor is owned by DMA engine

ETH_DMARXDESC_AFM	DA Filter Fail for the rx frame
ETH_DMARXDESC_FL	Receive descriptor frame length
ETH_DMARXDESC_ES	Error summary: OR of the following bits: DE || OE || IPC || LC || RWT || RE || CE
ETH_DMARXDESC_DE	Descriptor error: no more descriptors for receive frame
ETH_DMARXDESC_SAF	SA Filter Fail for the received frame
ETH_DMARXDESC_LE	Frame size not matching with length field
ETH_DMARXDESC_OE	Overflow Error: Frame was damaged due to buffer overflow
ETH_DMARXDESC_VLAN	VLAN Tag: received frame is a VLAN frame
ETH_DMARXDESC_FS	First descriptor of the frame
ETH_DMARXDESC_LS	Last descriptor of the frame
ETH_DMARXDESC_IPV4HCE	IPC Checksum Error: Rx Ipv4 header checksum error
ETH_DMARXDESC_LC	Late collision occurred during reception
ETH_DMARXDESC_FT	Frame type - Ethernet, otherwise 802.3
ETH_DMARXDESC_RWT	Receive Watchdog Timeout: watchdog timer expired during reception
ETH_DMARXDESC_RE	Receive error: error reported by MII interface
ETH_DMARXDESC_DBE	Dribble bit error: frame contains non int multiple of 8 bits
ETH_DMARXDESC_CE	CRC error
ETH_DMARXDESC_MAMPCE	Rx MAC Address/Payload Checksum Error: Rx MAC address matched/ Rx Payload Checksum Error
ETH_DMARXDESC_DIC	Disable Interrupt on Completion
ETH_DMARXDESC_RBS2	Receive Buffer2 Size
ETH_DMARXDESC_RER	Receive End of Ring
ETH_DMARXDESC_RCH	Second Address Chained
ETH_DMARXDESC_RBS1	Receive Buffer1 Size
ETH_DMARXDESC_B1AP	Buffer1 Address Pointer
ETH_DMARXDESC_B2AP	Buffer2 Address Pointer
ETH_DMAPTPRXDESC_PTPV	

ETH_DMAPTPRXDESC_PTPFT

ETH_DMAPTPRXDESC_PTPMT

ETH_DMAPTPRXDESC_PTPMT_SYNC

ETH_DMAPTPRXDESC_PTPMT_FOLLOWUP

ETH_DMAPTPRXDESC_PTPMT_DELAYREQ

ETH_DMAPTPRXDESC_PTPMT_DELAYRESP

ETH_DMAPTPRXDESC_PTPMT_PDELAYREQ_ANNOUNCE

ETH_DMAPTPRXDESC_PTPMT_PDELAYRESP_MANAG

ETH_DMAPTPRXDESC_PTPMT_PDELAYRESPFOLLOWUP_SIGNAL

ETH_DMAPTPRXDESC_IPV6PR

ETH_DMAPTPRXDESC_IPV4PR

ETH_DMAPTPRXDESC_IPCB

ETH_DMAPTPRXDESC_IPPE

ETH_DMAPTPRXDESC_IPHE

ETH_DMAPTPRXDESC_IPPT

ETH_DMAPTPRXDESC_IPPT_UDP

ETH_DMAPTPRXDESC_IPPT_TCP

ETH_DMAPTPRXDESC_IPPT_ICMP

ETH_DMAPTPRXDESC_RTSL

ETH_DMAPTPRXDESC_RTSH

ETH DMA Rx descriptor buffers

ETH_DMARXDESC_BUFFER1	DMA Rx Desc Buffer1
ETH_DMARXDESC_BUFFER2	DMA Rx Desc Buffer2

ETH DMA transmit process state

ETH_DMA_TRANSMITPROCESS_STOPPED	Stopped - Reset or Stop Tx Command issued
ETH_DMA_TRANSMITPROCESS_FETCHING	Running - fetching the Tx descriptor
ETH_DMA_TRANSMITPROCESS_WAITING	Running - waiting for status
ETH_DMA_TRANSMITPROCESS_READING	Running - reading the data from host memory
ETH_DMA_TRANSMITPROCESS_SUSPENDED	Suspended - Tx Descriptor unavailable
ETH_DMA_TRANSMITPROCESS_CLOSING	Running - closing Rx descriptor

ETH DMA TX Descriptor

ETH_DMATXDESC_OWN	OWN bit: descriptor is owned by DMA engine
ETH_DMATXDESC_IC	Interrupt on Completion
ETH_DMATXDESC_LS	Last Segment

ETH_DMATXDESC_FS	First Segment
ETH_DMATXDESC_DC	Disable CRC
ETH_DMATXDESC_DP	Disable Padding
ETH_DMATXDESC_TTSE	Transmit Time Stamp Enable
ETH_DMATXDESC_CIC	Checksum Insertion Control: 4 cases
ETH_DMATXDESC_CIC_BYPASS	Do Nothing: Checksum Engine is bypassed
ETH_DMATXDESC_CIC_IPV4HEADER	IPV4 header Checksum Insertion
ETH_DMATXDESC_CIC_TCPUDPICMP_SEGMENT	TCP/UDP/ICMP Checksum Insertion calculated over segment only
ETH_DMATXDESC_CIC_TCPUDPICMP_FULL	TCP/UDP/ICMP Checksum Insertion fully calculated
ETH_DMATXDESC_TER	Transmit End of Ring
ETH_DMATXDESC_TCH	Second Address Chained
ETH_DMATXDESC_TTSS	Tx Time Stamp Status
ETH_DMATXDESC_IHE	IP Header Error
ETH_DMATXDESC_ES	Error summary: OR of the following bits: UE || ED || EC || LCO || NC || LCA || FF || JT
ETH_DMATXDESC_JT	Jabber Timeout
ETH_DMATXDESC_FF	Frame Flushed: DMA/MTL flushed the frame due to SW flush
ETH_DMATXDESC_PCE	Payload Checksum Error
ETH_DMATXDESC_LCA	Loss of Carrier: carrier lost during transmission
ETH_DMATXDESC_NC	No Carrier: no carrier signal from the transceiver
ETH_DMATXDESC_LCO	Late Collision: transmission aborted due to collision
ETH_DMATXDESC_EC	Excessive Collision: transmission aborted after 16 collisions
ETH_DMATXDESC_VF	VLAN Frame
ETH_DMATXDESC_CC	Collision Count
ETH_DMATXDESC_ED	Excessive Deferral
ETH_DMATXDESC_UF	Underflow Error: late data arrival from the memory
ETH_DMATXDESC_DB	Deferred Bit
ETH_DMATXDESC_TBS2	Transmit Buffer2 Size
ETH_DMATXDESC_TBS1	Transmit Buffer1 Size

ETH_DMATXDESC_B1AP	Buffer1 Address Pointer
ETH_DMATXDESC_B2AP	Buffer2 Address Pointer
ETH_DMAPTPTXDESC_TTSL	
ETH_DMAPTPTXDESC_TTSH	

ETH DMA Tx descriptor Checksum Insertion Control

ETH_DMATXDESC_CHECKSUMBYPASS	Checksum engine bypass
ETH_DMATXDESC_CHECKSUMIPV4HEADER	IPv4 header checksum insertion
ETH_DMATXDESC_CHECKSUMTCPUDPICMPSEGMENT	TCP/UDP/ICMP checksum insertion. Pseudo header checksum is assumed to be present
ETH_DMATXDESC_CHECKSUMTCPUDPICMPFULL	TCP/UDP/ICMP checksum fully in hardware including pseudo header

ETH DMA Tx descriptor segment

ETH_DMATXDESC_LASTSEGMENTS	Last Segment
ETH_DMATXDESC_FIRSTSEGMENT	First Segment

ETH Drop TCP IP Checksum Error Frame

ETH_DROPTCPIPCHECKSUMERRORFRAME_ENABLE

ETH_DROPTCPIPCHECKSUMERRORFRAME_DISABLE

ETH Duplex Mode

ETH_MODE_FULLDUPLEX

ETH_MODE_HALFDUPLEX

ETH Exported Macros

__HAL_ETH_RESET_HANDLE_STATE

Description:

- Reset ETH handle state.

Parameters:

- __HANDLE__: specifies the ETH handle.

Return value:

- None

__HAL_ETH_DMATXDESC_GET_FLAG

Description:

- Checks whether the specified ETHERNET DMA Tx Desc flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __FLAG__: specifies the flag of TDES0 to check.

Return value:

- the: ETH_DMATxDescFlag (SET or RESET).

__HAL_ETH_DMARXDESC_GET_FLAG

Description:

- Checks whether the specified ETHERNET DMA Rx Desc flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __FLAG__: specifies the flag of RDES0 to check.

Return value:

- the: ETH_DMATxDescFlag (SET or RESET).

__HAL_ETH_DMARXDESC_ENABLE_IT

Description:

- Enables the specified DMA Rx Desc receive interrupt.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMARXDESC_DISABLE_IT

Description:

- Disables the specified DMA Rx Desc receive interrupt.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMARXDESC_SET_OWN_BIT

Description:

- Set the specified DMA Rx Desc Own bit.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMATXDESC_GET_COLLISION_COUNT

Description:

- Returns the specified ETHERNET DMA Tx Desc collision count.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- The: Transmit descriptor collision counter value.

__HAL_ETH_DMATXDESC_SET_OWN_BIT

Description:

- Set the specified DMA Tx Desc Own bit.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMATXDESC_ENABLE_IT

Description:

- Enables the specified DMA Tx Desc Transmit interrupt.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMATXDESC_DISABLE_IT

Description:

- Disables the specified DMA Tx Desc Transmit interrupt.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_DMATXDESC_CHECKSUM_INSERTION

Description:

- Selects the specified ETHERNET DMA Tx Desc Checksum Insertion.

Parameters:

- __HANDLE__: ETH Handle
- __CHECKSUM__: specifies is the DMA Tx desc checksum insertion. This parameter can be one of the following values:
  - ETH_DMATXDESC_CHECKSUMBYPASS : Checksum bypass
  - ETH_DMATXDESC_CHECKSUMIPV4HEADER : IPv4 header checksum
  - ETH_DMATXDESC_CHECKSUMTCPUDPICMPSEGMENT : TCP/UDP/ICMP checksum. Pseudo header checksum is assumed to be present

	– ETH_DMATXDESC_CHECKSUMTCPUDPICMPFULL : TCP/UDP/ICMP checksum fully in hardware including pseudo header **Return value:** • None
__HAL_ETH_DMATXDESC_CRC_ENABLE	**Description:** • Enables the DMA Tx Desc CRC. **Parameters:** • __HANDLE__: ETH Handle **Return value:** • None
__HAL_ETH_DMATXDESC_CRC_DISABLE	**Description:** • Disables the DMA Tx Desc CRC. **Parameters:** • __HANDLE__: ETH Handle **Return value:** • None
__HAL_ETH_DMATXDESC_SHORT_FRAME_PADDING_ENABLE	**Description:** • Enables the DMA Tx Desc padding for frame shorter than 64 bytes. **Parameters:** • __HANDLE__: ETH Handle **Return value:** • None
__HAL_ETH_DMATXDESC_SHORT_FRAME_PADDING_DISABLE	**Description:** • Disables the DMA Tx Desc padding for frame shorter than 64 bytes. **Parameters:** • __HANDLE__: ETH Handle **Return value:** • None
__HAL_ETH_MAC_ENABLE_IT	**Description:** • Enables the specified ETHERNET MAC interrupts. **Parameters:** • __HANDLE__: : ETH Handle • __INTERRUPT__: specifies the ETHERNET MAC interrupt sources to

be enabled or disabled. This parameter can be any combination of the following values:
  - ETH_MAC_IT_TST : Time stamp trigger interrupt
  - ETH_MAC_IT_PMT : PMT interrupt

Return value:

- None

__HAL_ETH_MAC_DISABLE_IT

Description:

- Disables the specified ETHERNET MAC interrupts.

Parameters:

- __HANDLE__: : ETH Handle
- __INTERRUPT__: specifies the ETHERNET MAC interrupt sources to be enabled or disabled. This parameter can be any combination of the following values:
  - ETH_MAC_IT_TST : Time stamp trigger interrupt
  - ETH_MAC_IT_PMT : PMT interrupt

Return value:

- None

__HAL_ETH_INITIATE_PAUSE_CONTROL_FRAME

Description:

- Initiate a Pause Control Frame (Full-duplex only).

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_GET_FLOW_CONTROL_BUSY_STATUS

Description:

- Checks whether the ETHERNET flow control busy bit is set or not.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- The: new state of flow control busy status bit (SET or RESET).

__HAL_ETH_BACK_PRESSURE_ACTIVATION_ENABLE

Description:

- Enables the MAC Back Pressure operation activation (Half-duplex only).

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_BACK_PRESSURE_ACTIVATION_DISABLE

Description:

- Disables the MAC BackPressure operation activation (Half-duplex only).

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_MAC_GET_FLAG

Description:

- Checks whether the specified ETHERNET MAC flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - ETH_MAC_FLAG_TST : Time stamp trigger flag
  - ETH_MAC_FLAG_MMCT : MMC transmit flag
  - ETH_MAC_FLAG_MMCR : MMC receive flag
  - ETH_MAC_FLAG_MMC : MMC flag
  - ETH_MAC_FLAG_PMT : PMT flag

Return value:

- The: state of ETHERNET MAC flag.

__HAL_ETH_DMA_ENABLE_IT

Description:

- Enables the specified ETHERNET DMA interrupts.

Parameters:

- __HANDLE__: : ETH Handle
- __INTERRUPT__: specifies the ETHERNET DMA interrupt sources to be enabled

Return value:

- None

__HAL_ETH_DMA_DISABLE_IT

Description:

- Disables the specified ETHERNET DMA interrupts.

Parameters:

- __HANDLE__: : ETH Handle
- __INTERRUPT__: specifies the ETHERNET DMA interrupt sources to be disabled.

Return value:

- None

__HAL_ETH_DMA_CLEAR_IT

Description:

- Clears the ETHERNET DMA IT pending bit.

Parameters:

- __HANDLE__: : ETH Handle
- __INTERRUPT__: specifies the interrupt pending bit to clear.

Return value:

- None

__HAL_ETH_DMA_GET_FLAG

Description:

- Checks whether the specified ETHERNET DMA flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __FLAG__: specifies the flag to check.

Return value:

- The: new state of ETH_DMA_FLAG (SET or RESET).

__HAL_ETH_DMA_CLEAR_FLAG

Description:

- Checks whether the specified ETHERNET DMA flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __FLAG__: specifies the flag to clear.

Return value:

- The: new state of ETH_DMA_FLAG (SET or RESET).

__HAL_ETH_GET_DMA_OVERFLOW_STATUS

Description:

- Checks whether the specified ETHERNET DMA overflow flag is set or not.

Parameters:

- __HANDLE__: ETH Handle
- __OVERFLOW__: specifies the DMA

overflow flag to check. This parameter can be one of the following values:
- ETH_DMA_OVERFLOW_RXFIFOCOUNTER : Overflow for FIFO Overflows Counter
- ETH_DMA_OVERFLOW_MISSEDFRAMECOUNTER : Overflow for Buffer Unavailable Missed Frame Counter

Return value:

- The: state of ETHERNET DMA overflow Flag (SET or RESET).

__HAL_ETH_SET_RECEIVE_WATCHDOG_TIMER

Description:

- Set the DMA Receive status watchdog timer register value.

Parameters:

- __HANDLE__: ETH Handle
- __VALUE__: DMA Receive status watchdog timer register value

Return value:

- None

__HAL_ETH_GLOBAL_UNICAST_WAKEUP_ENABLE

Description:

- Enables any unicast packet filtered by the MAC address recognition to be a wake-up frame.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_GLOBAL_UNICAST_WAKEUP_DISABLE

Description:

- Disables any unicast packet filtered by the MAC address recognition to be a wake-up frame.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_WAKEUP_FRAME_DETECTION_ENABLE

Description:

- Enables the MAC Wake-Up Frame Detection.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_WAKEUP_FRAME_DETECTION_DISABLE

Description:

- Disables the MAC Wake-Up Frame Detection.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_MAGIC_PACKET_DETECTION_ENABLE

Description:

- Enables the MAC Magic Packet Detection.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_MAGIC_PACKET_DETECTION_DISABLE

Description:

- Disables the MAC Magic Packet Detection.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_POWER_DOWN_ENABLE

Description:

- Enables the MAC Power Down.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_POWER_DOWN_DISABLE

Description:

- Disables the MAC Power Down.

Parameters:

- __HANDLE__: ETH Handle

Return value:

- None

__HAL_ETH_GET_PMT_FLAG_STATUS

Description:

- Checks whether the specified

ETHERNET PMT flag is set or not.

Parameters:

- __HANDLE__: ETH Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - ETH_PMT_FLAG_WUFFRPR : Wake-Up Frame Filter Register Pointer Reset
  - ETH_PMT_FLAG_WUFR : Wake-Up Frame Received
  - ETH_PMT_FLAG_MPR : Magic Packet Received

Return value:

- The: new state of ETHERNET PMT Flag (SET or RESET).

__HAL_ETH_MMC_COUNTER_FULL_PRESET

Description:

- Preset and Initialize the MMC counters to almost-full value: 0xFFFF_FFF0 (full - 16)

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_MMC_COUNTER_HALF_PRESET

Description:

- Preset and Initialize the MMC counters to almost-half value: 0x7FFF_FFF0 (half - 16)

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_MMC_COUNTER_FREEZE_ENABLE

Description:

- Enables the MMC Counter Freeze.

Parameters:

- __HANDLE__: ETH Handle.

Return value:

- None

__HAL_ETH_MMC_COUNTER_FREEZE_DISABLE

Description:

- Disables the MMC Counter Freeze.

Parameters:

	• __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_ETH_MMC_RESET_ONREAD_ENABLE	**Description:** • Enables the MMC Reset On Read. **Parameters:** • __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_ETH_MMC_RESET_ONREAD_DISABLE	**Description:** • Disables the MMC Reset On Read. **Parameters:** • __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_ETH_MMC_COUNTER_ROLLOVER_ENABLE	**Description:** • Enables the MMC Counter Stop Rollover. **Parameters:** • __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_ETH_MMC_COUNTER_ROLLOVER_DISABLE	**Description:** • Disables the MMC Counter Stop Rollover. **Parameters:** • __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_MMC_COUNTERS_RESET	**Description:** • Resets the MMC Counters. **Parameters:** • __HANDLE__: ETH Handle. **Return value:** • None
__HAL_ETH_MMC_RX_IT_ENABLE	**Description:** • Enables the specified ETHERNET MMC

Rx interrupts.

Parameters:

- __HANDLE__: ETH Handle.
- __INTERRUPT__: specifies the ETHERNET MMC interrupt sources to be enabled or disabled. This parameter can be one of the following values:
  - ETH_MMC_IT_RGUF : When Rx good unicast frames counter reaches half the maximum value
  - ETH_MMC_IT_RFAE : When Rx alignment error counter reaches half the maximum value
  - ETH_MMC_IT_RFCE : When Rx crc error counter reaches half the maximum value

Return value:

- None

__HAL_ETH_MMC_RX_IT_DISABLE

Description:

- Disables the specified ETHERNET MMC Rx interrupts.

Parameters:

- __HANDLE__: ETH Handle.
- __INTERRUPT__: specifies the ETHERNET MMC interrupt sources to be enabled or disabled. This parameter can be one of the following values:
  - ETH_MMC_IT_RGUF : When Rx good unicast frames counter reaches half the maximum value
  - ETH_MMC_IT_RFAE : When Rx alignment error counter reaches half the maximum value
  - ETH_MMC_IT_RFCE : When Rx crc error counter reaches half the maximum value

Return value:

- None

__HAL_ETH_MMC_TX_IT_ENABLE

Description:

- Enables the specified ETHERNET MMC Tx interrupts.

Parameters:

- __HANDLE__: ETH Handle.
- __INTERRUPT__: specifies the ETHERNET MMC interrupt sources to be enabled or disabled. This parameter can be one of the following values:

– ETH_MMC_IT_TGF : When Tx good frame counter reaches half the maximum value
– ETH_MMC_IT_TGFMSC: When Tx good multi col counter reaches half the maximum value
– ETH_MMC_IT_TGFSC : When Tx good single col counter reaches half the maximum value

Return value:

- None

__HAL_ETH_MMC_TX_IT_DISABLE

Description:

- Disables the specified ETHERNET MMC Tx interrupts.

Parameters:

- __HANDLE__: ETH Handle.
- __INTERRUPT__: specifies the ETHERNET MMC interrupt sources to be enabled or disabled. This parameter can be one of the following values:
  – ETH_MMC_IT_TGF : When Tx good frame counter reaches half the maximum value
  – ETH_MMC_IT_TGFMSC: When Tx good multi col counter reaches half the maximum value
  – ETH_MMC_IT_TGFSC : When Tx good single col counter reaches half the maximum value

Return value:

- None

__HAL_ETH_WAKEUP_EXTI_ENABLE_IT

Description:

- Enables the ETH External interrupt line.

Return value:

- None

__HAL_ETH_WAKEUP_EXTI_DISABLE_IT

Description:

- Disables the ETH External interrupt line.

Return value:

- None

__HAL_ETH_WAKEUP_EXTI_ENABLE_EVENT

Description:

- Enable event on ETH External event line.

Return value:

- None.

__HAL_ETH_WAKEUP_EXTI_DISABLE_EVENT	**Description:** • Disable event on ETH External event line. **Return value:** • None.
__HAL_ETH_WAKEUP_EXTI_GET_FLAG	**Description:** • Get flag of the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_CLEAR_FLAG	**Description:** • Clear flag of the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_ENABLE_RISING_EDGE_TRIGGER	**Description:** • Enables rising edge trigger to the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_DISABLE_RISING_EDGE_TRIGGER	**Description:** • Disables the rising edge trigger to the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_ENABLE_FALLING_EDGE_TRIGGER	**Description:** • Enables falling edge trigger to the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_DISABLE_FALLING_EDGE_TRIGGER	**Description:** • Disables falling edge trigger to the ETH External interrupt line. **Return value:** • None
__HAL_ETH_WAKEUP_EXTI_ENABLE_FALLINGRISING_TRIGGER	**Description:** • Enables rising/falling edge trigger to the ETH External interrupt line.

Return value:

- None

__HAL_ETH_WAKEUP_EXTI_DISABLE_FALLINGRISING_TRIGGER

Description:

- Disables rising/falling edge trigger to the ETH External interrupt line.

Return value:

- None

__HAL_ETH_WAKEUP_EXTI_GENERATE_SWIT

Description:

- Generate a Software interrupt on selected EXTI line.

Return value:

- None.

ETH EXTI LINE WAKEUP

ETH_EXTI_LINE_WAKEUP External interrupt line 19 Connected to the ETH EXTI Line

ETH Fixed Burst

ETH_FIXEDBURST_ENABLE

ETH_FIXEDBURST_DISABLE

ETH Flush Received Frame

ETH_FLUSHRECEIVEDFRAME_ENABLE

ETH_FLUSHRECEIVEDFRAME_DISABLE

ETH Forward Error Frames

ETH_FORWARDERRORFRAMES_ENABLE

ETH_FORWARDERRORFRAMES_DISABLE

ETH Forward Undersized Good Frames

ETH_FORWARDUNDERSIZEDGOODFRAMES_ENABLE

ETH_FORWARDUNDERSIZEDGOODFRAMES_DISABLE

ETH Inter Frame Gap

ETH_INTERFRAMEGAP_96BIT	minimum IFG between frames during transmission is 96Bit
ETH_INTERFRAMEGAP_88BIT	minimum IFG between frames during transmission is 88Bit
ETH_INTERFRAMEGAP_80BIT	minimum IFG between frames during transmission is 80Bit
ETH_INTERFRAMEGAP_72BIT	minimum IFG between frames during transmission is 72Bit
ETH_INTERFRAMEGAP_64BIT	minimum IFG between frames during transmission is 64Bit
ETH_INTERFRAMEGAP_56BIT	minimum IFG between frames during transmission is 56Bit

ETH_INTERFRAMEGAP_48BIT	minimum IFG between frames during transmission is 48Bit
ETH_INTERFRAMEGAP_40BIT	minimum IFG between frames during transmission is 40Bit

ETH Jabber

ETH_JABBER_ENABLE

ETH_JABBER_DISABLE

ETH Loop Back Mode

ETH_LOOPBACKMODE_ENABLE

ETH_LOOPBACKMODE_DISABLE

ETH MAC addresses

ETH_MAC_ADDRESS0

ETH_MAC_ADDRESS1

ETH_MAC_ADDRESS2

ETH_MAC_ADDRESS3

ETH MAC addresses filter Mask bytes

ETH_MAC_ADDRESSMASK_BYTE6	Mask MAC Address high reg bits [15:8]
ETH_MAC_ADDRESSMASK_BYTE5	Mask MAC Address high reg bits [7:0]
ETH_MAC_ADDRESSMASK_BYTE4	Mask MAC Address low reg bits [31:24]
ETH_MAC_ADDRESSMASK_BYTE3	Mask MAC Address low reg bits [23:16]
ETH_MAC_ADDRESSMASK_BYTE2	Mask MAC Address low reg bits [15:8]
ETH_MAC_ADDRESSMASK_BYTE1	Mask MAC Address low reg bits [70]

ETH MAC addresses filter SA DA

ETH_MAC_ADDRESSFILTER_SA

ETH_MAC_ADDRESSFILTER_DA

ETH MAC Flags

ETH_MAC_FLAG_TST	Time stamp trigger flag (on MAC)
ETH_MAC_FLAG_MMCT	MMC transmit flag
ETH_MAC_FLAG_MMCR	MMC receive flag
ETH_MAC_FLAG_MMC	MMC flag (on MAC)
ETH_MAC_FLAG_PMT	PMT flag (on MAC)

ETH MAC Interrupts

ETH_MAC_IT_TST	Time stamp trigger interrupt (on MAC)
ETH_MAC_IT_MMCT	MMC transmit interrupt
ETH_MAC_IT_MMCR	MMC receive interrupt
ETH_MAC_IT_MMC	MMC interrupt (on MAC)
ETH_MAC_IT_PMT	PMT interrupt (on MAC)

ETH Media Interface

ETH_MEDIA_INTERFACE_MII

ETH_MEDIA_INTERFACE_RMII

ETH MMC Rx Interrupts

ETH_MMC_IT_RGUF	When Rx good unicast frames counter reaches half the maximum value
ETH_MMC_IT_RFAE	When Rx alignment error counter reaches half the maximum value
ETH_MMC_IT_RFCE	When Rx crc error counter reaches half the maximum value

ETH MMC Tx Interrupts

ETH_MMC_IT_TGF	When Tx good frame counter reaches half the maximum value
ETH_MMC_IT_TGFMSC	When Tx good multi col counter reaches half the maximum value
ETH_MMC_IT_TGFSC	When Tx good single col counter reaches half the maximum value

ETH Multicast Frames Filter

ETH_MULTICASTFRAMESFILTER_PERFECTHASHTABLE

ETH_MULTICASTFRAMESFILTER_HASHTABLE

ETH_MULTICASTFRAMESFILTER_PERFECT

ETH_MULTICASTFRAMESFILTER_NONE

ETH Pass Control Frames

ETH_PASSCONTROLFRAMES_BLOCKALL	MAC filters all control frames from reaching the application
ETH_PASSCONTROLFRAMES_FORWARDALL	MAC forwards all control frames to application even if they fail the Address Filter
ETH_PASSCONTROLFRAMES_FORWARDPASSEDADDRFILTER	MAC forwards control frames that pass the Address Filter.

ETH Pause Low Threshold

ETH_PAUSELOWTHRESHOLD_MINUS4	Pause time minus 4 slot times
ETH_PAUSELOWTHRESHOLD_MINUS28	Pause time minus 28 slot times
ETH_PAUSELOWTHRESHOLD_MINUS144	Pause time minus 144 slot times
ETH_PAUSELOWTHRESHOLD_MINUS256	Pause time minus 256 slot times

ETH PMT Flags

ETH_PMT_FLAG_WUFFRPR	Wake-Up Frame Filter Register Pointer Reset
ETH_PMT_FLAG_WUFR	Wake-Up Frame Received

ETH_PMT_FLAG_MPR	Magic Packet Received

ETH Promiscuous Mode

ETH_PROMISCUOUS_MODE_ENABLE

ETH_PROMISCUOUS_MODE_DISABLE

ETH Receive All

ETH_RECEIVEALL_ENABLE

ETH_RECEIVEAll_DISABLE

ETH Receive Flow Control

ETH_RECEIVEFLOWCONTROL_ENABLE

ETH_RECEIVEFLOWCONTROL_DISABLE

ETH Receive Own

ETH_RECEIVEOWN_ENABLE

ETH_RECEIVEOWN_DISABLE

ETH Receive Store Forward

ETH_RECEIVESTOREFORWARD_ENABLE

ETH_RECEIVESTOREFORWARD_DISABLE

ETH Receive Threshold Control

ETH_RECEIVEDTHRESHOLDCONTROL_64BYTES	threshold level of the MTL Receive FIFO is 64 Bytes
ETH_RECEIVEDTHRESHOLDCONTROL_32BYTES	threshold level of the MTL Receive FIFO is 32 Bytes
ETH_RECEIVEDTHRESHOLDCONTROL_96BYTES	threshold level of the MTL Receive FIFO is 96 Bytes
ETH_RECEIVEDTHRESHOLDCONTROL_128BYTES	threshold level of the MTL Receive FIFO is 128 Bytes

ETH Retry Transmission

ETH_RETRYTRANSMISSION_ENABLE

ETH_RETRYTRANSMISSION_DISABLE

ETH Rx DMA Burst Length

ETH_RXDMABURSTLENGTH_1BEAT	maximum number of beats to be transferred in one RxDMA transaction is 1
ETH_RXDMABURSTLENGTH_2BEAT	maximum number of beats to be transferred in one RxDMA transaction is 2
ETH_RXDMABURSTLENGTH_4BEAT	maximum number of beats to be transferred in one RxDMA transaction is 4
ETH_RXDMABURSTLENGTH_8BEAT	maximum number of beats to be transferred in one RxDMA transaction

	is 8
ETH_RXDMABURSTLENGTH_16BEAT	maximum number of beats to be transferred in one RxDMA transaction is 16
ETH_RXDMABURSTLENGTH_32BEAT	maximum number of beats to be transferred in one RxDMA transaction is 32
ETH_RXDMABURSTLENGTH_4XPBL_4BEAT	maximum number of beats to be transferred in one RxDMA transaction is 4
ETH_RXDMABURSTLENGTH_4XPBL_8BEAT	maximum number of beats to be transferred in one RxDMA transaction is 8
ETH_RXDMABURSTLENGTH_4XPBL_16BEAT	maximum number of beats to be transferred in one RxDMA transaction is 16
ETH_RXDMABURSTLENGTH_4XPBL_32BEAT	maximum number of beats to be transferred in one RxDMA transaction is 32
ETH_RXDMABURSTLENGTH_4XPBL_64BEAT	maximum number of beats to be transferred in one RxDMA transaction is 64
ETH_RXDMABURSTLENGTH_4XPBL_128BEAT	maximum number of beats to be transferred in one RxDMA transaction is 128

ETH Rx Mode

ETH_RXPOLLING_MODE

ETH_RXINTERRUPT_MODE

ETH Second Frame Operate

ETH_SECONDFRAMEOPERARTE_ENABLE

ETH_SECONDFRAMEOPERARTE_DISABLE

ETH Source Addr Filter

ETH_SOURCEADDRFILTER_NORMAL_ENABLE

ETH_SOURCEADDRFILTER_INVERSE_ENABLE

ETH_SOURCEADDRFILTER_DISABLE

ETH Speed

ETH_SPEED_10M

ETH_SPEED_100M

ETH Transmit Flow Control

ETH_TRANSMITFLOWCONTROL_ENABLE

ETH_TRANSMITFLOWCONTROL_DISABLE

ETH Transmit Store Forward

ETH_TRANSMITSTOREFORWARD_ENABLE

ETH_TRANSMITSTOREFORWARD_DISABLE

ETH Transmit Threshold Control

ETH_TRANSMITTHRESHOLDCONTROL_64BYTES	threshold level of the MTL Transmit FIFO is 64 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_128BYTES	threshold level of the MTL Transmit FIFO is 128 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_192BYTES	threshold level of the MTL Transmit FIFO is 192 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_256BYTES	threshold level of the MTL Transmit FIFO is 256 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_40BYTES	threshold level of the MTL Transmit FIFO is 40 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_32BYTES	threshold level of the MTL Transmit FIFO is 32 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_24BYTES	threshold level of the MTL Transmit FIFO is 24 Bytes
ETH_TRANSMITTHRESHOLDCONTROL_16BYTES	threshold level of the MTL Transmit FIFO is 16 Bytes

ETH Tx DMA Burst Length

ETH_TXDMABURSTLENGTH_1BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 1
ETH_TXDMABURSTLENGTH_2BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 2
ETH_TXDMABURSTLENGTH_4BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 4
ETH_TXDMABURSTLENGTH_8BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 8
ETH_TXDMABURSTLENGTH_16BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 16
ETH_TXDMABURSTLENGTH_32BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 32
ETH_TXDMABURSTLENGTH_4XPBL_4BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 4
ETH_TXDMABURSTLENGTH_4XPBL_8BEAT	maximum number of beats to be transferred in one TxDMA (or both)

	transaction is 8
ETH_TXDMABURSTLENGTH_4XPBL_16BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 16
ETH_TXDMABURSTLENGTH_4XPBL_32BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 32
ETH_TXDMABURSTLENGTH_4XPBL_64BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 64
ETH_TXDMABURSTLENGTH_4XPBL_128BEAT	maximum number of beats to be transferred in one TxDMA (or both) transaction is 128

ETH Unicast Frames Filter

ETH_UNICASTFRAMESFILTER_PERFECTHASHTABLE

ETH_UNICASTFRAMESFILTER_HASHTABLE

ETH_UNICASTFRAMESFILTER_PERFECT

ETH Unicast Pause Frame Detect

ETH_UNICASTPAUSEFRAMEDETECT_ENABLE

ETH_UNICASTPAUSEFRAMEDETECT_DISABLE

ETH VLAN Tag Comparison

ETH_VLANTAGCOMPARISON_12BIT

ETH_VLANTAGCOMPARISON_16BIT

ETH Watchdog

ETH_WATCHDOG_ENABLE

ETH_WATCHDOG_DISABLE

ETH Zero Quanta Pause

ETH_ZEROQUANTAPAUSE_ENABLE

ETH_ZEROQUANTAPAUSE_DISABLE

22 HAL FLASH Generic Driver

22.1 FLASH Firmware driver registers structures

22.1.1 FLASH_ProcessTypeDef

Data Fields

- ***__IO FLASH_ProcedureTypeDef ProcedureOnGoing***
- ***__IO uint32_t NbSectorsToErase***
- ***__IO uint8_t VoltageForErase***
- ***__IO uint32_t Sector***
- ***__IO uint32_t Bank***
- ***__IO uint32_t Address***
- ***HAL_LockTypeDef Lock***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***__IO FLASH_ProcedureTypeDef FLASH_ProcessTypeDef::ProcedureOnGoing***
- ***__IO uint32_t FLASH_ProcessTypeDef::NbSectorsToErase***
- ***__IO uint8_t FLASH_ProcessTypeDef::VoltageForErase***
- ***__IO uint32_t FLASH_ProcessTypeDef::Sector***
- ***__IO uint32_t FLASH_ProcessTypeDef::Bank***
- ***__IO uint32_t FLASH_ProcessTypeDef::Address***
- ***HAL_LockTypeDef FLASH_ProcessTypeDef::Lock***
- ***__IO uint32_t FLASH_ProcessTypeDef::ErrorCode***

22.2 FLASH Firmware driver API description

22.2.1 FLASH peripheral features

The Flash memory interface manages CPU AHB I-Code and D-Code accesses to the Flash memory. It implements the erase and program Flash memory operations and the read and write protection mechanisms.

The Flash memory interface accelerates code execution with a system of instruction prefetch and cache lines.

The FLASH main features are:

- Flash memory read operations
- Flash memory program/erase operations
- Read / write protections
- Prefetch on I-Code
- 64 cache lines of 128 bits on I-Code
- 8 cache lines of 128 bits on D-Code

22.2.2 How to use this driver

This driver provides functions and macros to configure and program the FLASH memory of all STM32F4xx devices.

1. FLASH Memory IO Programming functions:
   - Lock and Unlock the FLASH interface using HAL_FLASH_Unlock() and HAL_FLASH_Lock() functions
   - Program functions: byte, half word, word and double word
   - There Two modes of programming :
     - Polling mode using HAL_FLASH_Program() function
     - Interrupt mode using HAL_FLASH_Program_IT() function
2. Interrupts and flags management functions :
   - Handle FLASH interrupts by calling HAL_FLASH_IRQHandler()
   - Wait for last FLASH operation according to its status
   - Get error flag status by calling HAL_SetErrorCode()

In addition to these functions, this driver includes a set of macros allowing to handle the following operations:

- Set the latency
- Enable/Disable the prefetch buffer
- Enable/Disable the Instruction cache and the Data cache
- Reset the Instruction cache and the Data cache
- Enable/Disable the FLASH interrupts
- Monitor the FLASH flags status

22.2.3 Programming operation functions

This subsection provides a set of functions allowing to manage the FLASH program operations.

This section contains the following APIs:

- ***HAL_FLASH_Program()***
- ***HAL_FLASH_Program_IT()***
- ***HAL_FLASH_IRQHandler()***
- ***HAL_FLASH_EndOfOperationCallback()***
- ***HAL_FLASH_OperationErrorCallback()***

22.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the FLASH memory operations.

This section contains the following APIs:

- ***HAL_FLASH_Unlock()***
- ***HAL_FLASH_Lock()***
- ***HAL_FLASH_OB_Unlock()***
- ***HAL_FLASH_OB_Lock()***
- ***HAL_FLASH_OB_Launch()***

22.2.5 Peripheral Errors functions

This subsection permits to get in run-time Errors of the FLASH peripheral.

This section contains the following APIs:

- ***HAL_FLASH_GetError()***

- *FLASH_WaitForLastOperation()*

22.2.6 Detailed description of functions

HAL_FLASH_Program

Function Name	**HAL_StatusTypeDef HAL_FLASH_Program (uint32_t TypeProgram, uint32_t Address, uint64_t Data)**
Function Description	Program byte, halfword, word or double word at a specified address.
Parameters	• **TypeProgram:** Indicate the way to program at a specified address. This parameter can be a value of FLASH Type Program • **Address:** specifies the address to be programmed. • **Data:** specifies the data to be programmed
Return values	• **HAL_StatusTypeDef:** HAL Status

HAL_FLASH_Program_IT

Function Name	**HAL_StatusTypeDef HAL_FLASH_Program_IT (uint32_t TypeProgram, uint32_t Address, uint64_t Data)**
Function Description	Program byte, halfword, word or double word at a specified address with interrupt enabled.
Parameters	• **TypeProgram:** Indicate the way to program at a specified address. This parameter can be a value of FLASH Type Program • **Address:** specifies the address to be programmed. • **Data:** specifies the data to be programmed
Return values	• **HAL:** Status

HAL_FLASH_IRQHandler

Function Name	**void HAL_FLASH_IRQHandler (void)**
Function Description	This function handles FLASH interrupt request.
Return values	• **None**

HAL_FLASH_EndOfOperationCallback

Function Name	**void HAL_FLASH_EndOfOperationCallback (uint32_t ReturnValue)**
Function Description	FLASH end of operation interrupt callback.
Parameters	• **ReturnValue:** The value saved in this parameter depends on the ongoing procedure Mass Erase: Bank number which has been requested to erase Sectors Erase: Sector which has been erased (if 0xFFFFFFFFU, it means that all the selected sectors have been erased) Program: Address which was selected for data program
Return values	• **None**

HAL_FLASH_OperationErrorCallback

Function Name	**void HAL_FLASH_OperationErrorCallback (uint32_t ReturnValue)**
Function Description	FLASH operation error interrupt callback.
Parameters	• **ReturnValue:** The value saved in this parameter depends on the ongoing procedure Mass Erase: Bank number which has been requested to erase Sectors Erase: Sector number which returned an error Program: Address which was selected for data program
Return values	• **None**

HAL_FLASH_Unlock

Function Name	**HAL_StatusTypeDef HAL_FLASH_Unlock (void)**
Function Description	Unlock the FLASH control register access.
Return values	• **HAL:** Status

HAL_FLASH_Lock

Function Name	**HAL_StatusTypeDef HAL_FLASH_Lock (void)**
Function Description	Locks the FLASH control register access.
Return values	• **HAL:** Status

HAL_FLASH_OB_Unlock

Function Name	**HAL_StatusTypeDef HAL_FLASH_OB_Unlock (void)**
Function Description	Unlock the FLASH Option Control Registers access.
Return values	• **HAL:** Status

HAL_FLASH_OB_Lock

Function Name	**HAL_StatusTypeDef HAL_FLASH_OB_Lock (void)**
Function Description	Lock the FLASH Option Control Registers access.
Return values	• **HAL:** Status

HAL_FLASH_OB_Launch

Function Name	**HAL_StatusTypeDef HAL_FLASH_OB_Launch (void)**
Function Description	Launch the option byte loading.
Return values	• **HAL:** Status

HAL_FLASH_GetError

Function Name	**uint32_t HAL_FLASH_GetError (void)**
Function Description	Get the specific FLASH error flag.

Return values
- **FLASH_ErrorCode:** The returned value can be a combination of:
  - HAL_FLASH_ERROR_RD: FLASH Read Protection error flag (PCROP)
  - HAL_FLASH_ERROR_PGS: FLASH Programming Sequence error flag
  - HAL_FLASH_ERROR_PGP: FLASH Programming Parallelism error flag
  - HAL_FLASH_ERROR_PGA: FLASH Programming Alignment error flag
  - HAL_FLASH_ERROR_WRP: FLASH Write protected error flag
  - HAL_FLASH_ERROR_OPERATION: FLASH operation Error flag

FLASH_WaitForLastOperation

Function Name **HAL_StatusTypeDef FLASH_WaitForLastOperation (uint32_t Timeout)**

Function Description Wait for a FLASH operation to complete.

Parameters
- **Timeout:** maximum flash operationtimeout

Return values
- **HAL:** Status

22.3 FLASH Firmware driver defines

22.3.1 FLASH

FLASH Error Code

HAL_FLASH_ERROR_NONE	No error
HAL_FLASH_ERROR_RD	Read Protection error
HAL_FLASH_ERROR_PGS	Programming Sequence error
HAL_FLASH_ERROR_PGP	Programming Parallelism error
HAL_FLASH_ERROR_PGA	Programming Alignment error
HAL_FLASH_ERROR_WRP	Write protection error
HAL_FLASH_ERROR_OPERATION	Operation Error

FLASH Exported Macros

__HAL_FLASH_SET_LATENCY

Description:
- Set the FLASH Latency.

Parameters:
- __LATENCY__: FLASH Latency The value of this parameter depend on device used within the same series

Return value:
- none

__HAL_FLASH_GET_LATENCY	**Description:** • Get the FLASH Latency. **Return value:** • FLASH: Latency The value of this parameter depend on device used within the same series
__HAL_FLASH_PREFETCH_BUFFER_ENABLE	**Description:** • Enable the FLASH prefetch buffer. **Return value:** • none
__HAL_FLASH_PREFETCH_BUFFER_DISABLE	**Description:** • Disable the FLASH prefetch buffer. **Return value:** • none
__HAL_FLASH_INSTRUCTION_CACHE_ENABLE	**Description:** • Enable the FLASH instruction cache. **Return value:** • none
__HAL_FLASH_INSTRUCTION_CACHE_DISABLE	**Description:** • Disable the FLASH instruction cache. **Return value:** • none
__HAL_FLASH_DATA_CACHE_ENABLE	**Description:** • Enable the FLASH data cache. **Return value:** • none
__HAL_FLASH_DATA_CACHE_DISABLE	**Description:** • Disable the FLASH data cache. **Return value:** • none
__HAL_FLASH_INSTRUCTION_CACHE_RESET	**Description:** • Resets the FLASH instruction Cache. **Return value:**

- None

Notes:

- This function must be used only when the Instruction Cache is disabled.

__HAL_FLASH_DATA_CACHE_RESET

Description:

- Resets the FLASH data Cache.

Return value:

- None

Notes:

- This function must be used only when the data Cache is disabled.

__HAL_FLASH_ENABLE_IT

Description:

- Enable the specified FLASH interrupt.

Parameters:

- __INTERRUPT__: : FLASH interrupt This parameter can be any combination of the following values:
  - FLASH_IT_EOP: End of FLASH Operation Interrupt
  - FLASH_IT_ERR: Error Interrupt

Return value:

- none

__HAL_FLASH_DISABLE_IT

Description:

- Disable the specified FLASH interrupt.

Parameters:

- __INTERRUPT__: : FLASH interrupt This parameter can be any combination of the following values:
  - FLASH_IT_EOP: End of FLASH Operation Interrupt
  - FLASH_IT_ERR: Error Interrupt

Return value:

- none

__HAL_FLASH_GET_FLAG

Description:

- Get the specified FLASH flag status.

Parameters:

- __FLAG__: specifies the FLASH flag to check. This parameter can be one of the following values:
  - FLASH_FLAG_EOP : FLASH End of Operation flag
  - FLASH_FLAG_OPERR : FLASH operation Error flag
  - FLASH_FLAG_WRPERR: FLASH Write protected error flag
  - FLASH_FLAG_PGAERR: FLASH Programming Alignment error flag
  - FLASH_FLAG_PGPERR: FLASH Programming Parallelism error flag
  - FLASH_FLAG_PGSERR: FLASH Programming Sequence error flag
  - FLASH_FLAG_RDERR : FLASH Read Protection error flag (PCROP)
  - FLASH_FLAG_BSY : FLASH Busy flag

Return value:

- The: new state of __FLAG__ (SET or RESET).

__HAL_FLASH_CLEAR_FLAG

Description:

- Clear the specified FLASH flag.

Parameters:

- __FLAG__: specifies the FLASH flags to clear. This parameter can be any combination of the following values:
  - FLASH_FLAG_EOP : FLASH End of Operation flag
  - FLASH_FLAG_OPERR : FLASH operation Error flag
  - FLASH_FLAG_WRPERR: FLASH Write protected error flag
  - FLASH_FLAG_PGAERR: FLASH Programming Alignment error flag
  - FLASH_FLAG_PGPERR: FLASH Programming Parallelism error flag
  - FLASH_FLAG_PGSERR: FLASH Programming Sequence error flag

– FLASH_FLAG_RDERR : FLASH Read Protection error flag (PCROP)

Return value:

- none

FLASH Flag definition

FLASH_FLAG_EOP	FLASH End of Operation flag
FLASH_FLAG_OPERR	FLASH operation Error flag
FLASH_FLAG_WRPERR	FLASH Write protected error flag
FLASH_FLAG_PGAERR	FLASH Programming Alignment error flag
FLASH_FLAG_PGPERR	FLASH Programming Parallelism error flag
FLASH_FLAG_PGSERR	FLASH Programming Sequence error flag
FLASH_FLAG_RDERR	Read Protection error flag (PCROP)
FLASH_FLAG_BSY	FLASH Busy flag

FLASH Interrupt definition

FLASH_IT_EOP	End of FLASH Operation Interrupt source
FLASH_IT_ERR	Error Interrupt source

FLASH Private macros to check input parameters

IS_FLASH_TYPEPROGRAM

FLASH Keys

RDP_KEY

FLASH_KEY1

FLASH_KEY2

FLASH_OPT_KEY1

FLASH_OPT_KEY2

FLASH Latency

FLASH_LATENCY_0	FLASH Zero Latency cycle
FLASH_LATENCY_1	FLASH One Latency cycle
FLASH_LATENCY_2	FLASH Two Latency cycles
FLASH_LATENCY_3	FLASH Three Latency cycles
FLASH_LATENCY_4	FLASH Four Latency cycles
FLASH_LATENCY_5	FLASH Five Latency cycles
FLASH_LATENCY_6	FLASH Six Latency cycles
FLASH_LATENCY_7	FLASH Seven Latency cycles
FLASH_LATENCY_8	FLASH Eight Latency cycles
FLASH_LATENCY_9	FLASH Nine Latency cycles
FLASH_LATENCY_10	FLASH Ten Latency cycles

FLASH_LATENCY_11	FLASH Eleven Latency cycles
FLASH_LATENCY_12	FLASH Twelve Latency cycles
FLASH_LATENCY_13	FLASH Thirteen Latency cycles
FLASH_LATENCY_14	FLASH Fourteen Latency cycles
FLASH_LATENCY_15	FLASH Fifteen Latency cycles

FLASH Program Parallelism

FLASH_PSIZE_BYTE

FLASH_PSIZE_HALF_WORD

FLASH_PSIZE_WORD

FLASH_PSIZE_DOUBLE_WORD

CR_PSIZE_MASK

FLASH Type Program

FLASH_TYPEPROGRAM_BYTE	Program byte (8-bit) at a specified address
FLASH_TYPEPROGRAM_HALFWORD	Program a half-word (16-bit) at a specified address
FLASH_TYPEPROGRAM_WORD	Program a word (32-bit) at a specified address
FLASH_TYPEPROGRAM_DOUBLEWORD	Program a double word (64-bit) at a specified address

23 HAL FLASH Extension Driver

23.1 FLASHEx Firmware driver registers structures

23.1.1 FLASH_EraseInitTypeDef

Data Fields

- ***uint32_t TypeErase***
- ***uint32_t Banks***
- ***uint32_t Sector***
- ***uint32_t NbSectors***
- ***uint32_t VoltageRange***

Field Documentation

- ***uint32_t FLASH_EraseInitTypeDef::TypeErase***
  Mass erase or sector Erase. This parameter can be a value of ***FLASHEx_Type_Erase***
- ***uint32_t FLASH_EraseInitTypeDef::Banks***
  Select banks to erase when Mass erase is enabled. This parameter must be a value of ***FLASHEx_Banks***
- ***uint32_t FLASH_EraseInitTypeDef::Sector***
  Initial FLASH sector to erase when Mass erase is disabled This parameter must be a value of ***FLASHEx_Sectors***
- ***uint32_t FLASH_EraseInitTypeDef::NbSectors***
  Number of sectors to be erased. This parameter must be a value between 1 and (max number of sectors - value of Initial sector)
- ***uint32_t FLASH_EraseInitTypeDef::VoltageRange***
  The device voltage range which defines the erase parallelism This parameter must be a value of ***FLASHEx_Voltage_Range***

23.1.2 FLASH_OBProgramInitTypeDef

Data Fields

- ***uint32_t OptionType***
- ***uint32_t WRPState***
- ***uint32_t WRPSector***
- ***uint32_t Banks***
- ***uint32_t RDPLevel***
- ***uint32_t BORLevel***
- ***uint8_t USERConfig***

Field Documentation

- ***uint32_t FLASH_OBProgramInitTypeDef::OptionType***
  Option byte to be configured. This parameter can be a value of ***FLASHEx_Option_Type***
- ***uint32_t FLASH_OBProgramInitTypeDef::WRPState***
  Write protection activation or deactivation. This parameter can be a value of ***FLASHEx_WRP_State***
- ***uint32_t FLASH_OBProgramInitTypeDef::WRPSector***
  Specifies the sector(s) to be write protected. The value of this parameter depend on device used within the same series
- ***uint32_t FLASH_OBProgramInitTypeDef::Banks***
  Select banks for WRP activation/deactivation of all sectors. This parameter must be a value of ***FLASHEx_Banks***
- ***uint32_t FLASH_OBProgramInitTypeDef::RDPLevel***
  Set the read protection level. This parameter can be a value of ***FLASHEx_Option_Bytes_Read_Protection***
- ***uint32_t FLASH_OBProgramInitTypeDef::BORLevel***
  Set the BOR Level. This parameter can be a value of ***FLASHEx_BOR_Reset_Level***
- ***uint8_t FLASH_OBProgramInitTypeDef::USERConfig***
  Program the FLASH User Option Byte: IWDG_SW / RST_STOP / RST_STDBY.

23.1.3 FLASH_AdvOBProgramInitTypeDef

Data Fields

- ***uint32_t OptionType***
- ***uint32_t PCROPState***
- ***uint32_t Banks***
- ***uint16_t SectorsBank1***
- ***uint16_t SectorsBank2***
- ***uint8_t BootConfig***

Field Documentation

- ***uint32_t FLASH_AdvOBProgramInitTypeDef::OptionType***
  Option byte to be configured for extension. This parameter can be a value of ***FLASHEx_Advanced_Option_Type***
- ***uint32_t FLASH_AdvOBProgramInitTypeDef::PCROPState***
  PCROP activation or deactivation. This parameter can be a value of ***FLASHEx_PCROP_State***
- ***uint32_t FLASH_AdvOBProgramInitTypeDef::Banks***
  Select banks for PCROP activation/deactivation of all sectors. This parameter must be a value of ***FLASHEx_Banks***
- ***uint16_t FLASH_AdvOBProgramInitTypeDef::SectorsBank1***
  Specifies the sector(s) set for PCROP for Bank1. This parameter can be a value of ***FLASHEx_Option_Bytes_PC_ReadWrite_Protection***
- ***uint16_t FLASH_AdvOBProgramInitTypeDef::SectorsBank2***
  Specifies the sector(s) set for PCROP for Bank2. This parameter can be a value of ***FLASHEx_Option_Bytes_PC_ReadWrite_Protection***
- ***uint8_t FLASH_AdvOBProgramInitTypeDef::BootConfig***
  Specifies Option bytes for boot config. This parameter can be a value of ***FLASHEx_Dual_Boot***

23.2 FLASHEx Firmware driver API description

23.2.1 Flash Extension features

Comparing to other previous devices, the FLASH interface for STM32F427xx/437xx and STM32F429xx/439xx devices contains the following additional features

- Capacity up to 2 Mbyte with dual bank architecture supporting read-while-write capability (RWW)
- Dual bank memory organization
- PCROP protection for all banks

23.2.2 How to use this driver

This driver provides functions to configure and program the FLASH memory of all STM32F427xx/437xx, STM32F429xx/439xx, STM32F469xx/479xx and STM32F446xx devices. It includes

1. FLASH Memory Erase functions:
   - Lock and Unlock the FLASH interface using HAL_FLASH_Unlock() and HAL_FLASH_Lock() functions
   - Erase function: Erase sector, erase all sectors
   - There are two modes of erase :
     - Polling Mode using HAL_FLASHEx_Erase()
     - Interrupt Mode using HAL_FLASHEx_Erase_IT()
2. Option Bytes Programming functions: Use HAL_FLASHEx_OBProgram() to :
   - Set/Reset the write protection
   - Set the Read protection Level
   - Set the BOR level
   - Program the user Option Bytes
3. Advanced Option Bytes Programming functions: Use HAL_FLASHEx_AdvOBProgram() to :
   - Extended space (bank 2) erase function
   - Full FLASH space (2 Mo) erase (bank 1 and bank 2)
   - Dual Boot activation
   - Write protection configuration for bank 2
   - PCROP protection configuration and control for both banks

23.2.3 Extended programming operation functions

This subsection provides a set of functions allowing to manage the Extension FLASH programming operations.

This section contains the following APIs:

- ***HAL_FLASHEx_Erase()***
- ***HAL_FLASHEx_Erase_IT()***
- ***HAL_FLASHEx_OBProgram()***
- ***HAL_FLASHEx_OBGetConfig()***
- ***HAL_FLASHEx_AdvOBProgram()***
- ***HAL_FLASHEx_AdvOBGetConfig()***
- ***HAL_FLASHEx_OB_SelectPCROP()***
- ***HAL_FLASHEx_OB_DeSelectPCROP()***
- ***HAL_FLASHEx_OB_GetBank2WRP()***

23.2.4 Detailed description of functions

HAL_FLASHEx_Erase

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_Erase (FLASH_EraseInitTypeDef * pEraseInit, uint32_t * SectorError)**
Function Description	Perform a mass erase or erase the specified FLASH memory sectors.
Parameters	• **pEraseInit:** pointer to an FLASH_EraseInitTypeDef structure that contains the configuration information for the erasing. • **SectorError:** pointer to variable that contains the configuration information on faulty sector in case of error (0xFFFFFFFFU means that all the sectors have been correctly erased)
Return values	• **HAL:** Status

HAL_FLASHEx_Erase_IT

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_Erase_IT (FLASH_EraseInitTypeDef * pEraseInit)**
Function Description	Perform a mass erase or erase the specified FLASH memory sectors with interrupt enabled.
Parameters	• **pEraseInit:** pointer to an FLASH_EraseInitTypeDef structure that contains the configuration information for the erasing.
Return values	• **HAL:** Status

HAL_FLASHEx_OBProgram

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_OBProgram (FLASH_OBProgramInitTypeDef * pOBInit)**
Function Description	Program option bytes.
Parameters	• **pOBInit:** pointer to an FLASH_OBInitStruct structure that contains the configuration information for the programming.
Return values	• **HAL:** Status

HAL_FLASHEx_OBGetConfig

Function Name	**void HAL_FLASHEx_OBGetConfig (FLASH_OBProgramInitTypeDef * pOBInit)**
Function Description	Get the Option byte configuration.
Parameters	• **pOBInit:** pointer to an FLASH_OBInitStruct structure that contains the configuration information for the programming.
Return values	• **None**

HAL_FLASHEx_AdvOBProgram

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_AdvOBProgram (FLASH_AdvOBProgramInitTypeDef * pAdvOBInit)**
Function Description	Program option bytes.
Parameters	• **pAdvOBInit:** pointer to an FLASH_AdvOBProgramInitTypeDef structure that contains the configuration information for the programming.
Return values	• **HAL:** Status

HAL_FLASHEx_AdvOBGetConfig

Function Name	**void HAL_FLASHEx_AdvOBGetConfig (FLASH_AdvOBProgramInitTypeDef * pAdvOBInit)**
Function Description	Get the OBEX byte configuration.
Parameters	• **pAdvOBInit:** pointer to an FLASH_AdvOBProgramInitTypeDef structure that contains the configuration information for the programming.
Return values	• **None**

HAL_FLASHEx_OB_SelectPCROP

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_OB_SelectPCROP (void)**
Function Description	Select the Protection Mode.
Return values	• **HAL:** Status
Notes	• After PCROP activated Option Byte modification NOT POSSIBLE! excepted Global Read Out Protection modification (from level1 to level0) • Once SPRMOD bit is active unprotection of a protected sector is not possible • Read a protected sector will set RDERR Flag and write a protected sector will set WRPERR Flag • This function can be used only for STM32F42xxx/STM32F43xxx/STM32F401xx/STM32F411xx/STM32F446xx/ STM32F469xx/STM32F479xx/STM32F412Zx/STM32F412Vx/STM32F412Rx/STM32F412Cx devices.

HAL_FLASHEx_OB_DeSelectPCROP

Function Name	**HAL_StatusTypeDef HAL_FLASHEx_OB_DeSelectPCROP (void)**
Function Description	Deselect the Protection Mode.
Return	• **HAL:** Status

values

Notes
- After PCROP activated Option Byte modification NOT POSSIBLE! excepted Global Read Out Protection modification (from level1 to level0)
- Once SPRMOD bit is active unprotection of a protected sector is not possible
- Read a protected sector will set RDERR Flag and write a protected sector will set WRPERR Flag
- This function can be used only for STM32F42xxx/STM32F43xxx/STM32F401xx/STM32F411xx/STM32F446xx/ STM32F469xx/STM32F479xx/STM32F412Zx/STM32F412Vx/STM32F412Rx/STM32F412Cx devices.

HAL_FLASHEx_OB_GetBank2WRP

Function Name **uint16_t HAL_FLASHEx_OB_GetBank2WRP (void)**

Function Description Returns the FLASH Write Protection Option Bytes value for Bank 2.

Return values
- **The:** FLASH Write Protection Option Bytes value

Notes
- This function can be used only for STM32F42xxx/STM32F43xxx/STM32F469xx/STM32F479xx devices.

FLASH_Erase_Sector

Function Name **void FLASH_Erase_Sector (uint32_t Sector, uint8_t VoltageRange)**

Function Description Erase the specified FLASH memory sector.

Parameters
- **Sector:** FLASH sector to erase The value of this parameter depend on device used within the same series
- **VoltageRange:** The device voltage range which defines the erase parallelism. This parameter can be one of the following values:
  - FLASH_VOLTAGE_RANGE_1: when the device voltage range is 1.8V to 2.1V, the operation will be done by byte (8-bit)
  - FLASH_VOLTAGE_RANGE_2: when the device voltage range is 2.1V to 2.7V, the operation will be done by half word (16-bit)
  - FLASH_VOLTAGE_RANGE_3: when the device voltage range is 2.7V to 3.6V, the operation will be done by word (32-bit)
  - FLASH_VOLTAGE_RANGE_4: when the device voltage range is 2.7V to 3.6V + External Vpp, the operation will be done by double word (64-bit)

Return values
- **None**

FLASH_FlushCaches

Function Name **void FLASH_FlushCaches (void)**

Function Description Flush the instruction and data caches.

Return values
- **None**

23.3 FLASHEx Firmware driver defines

23.3.1 FLASHEx

FLASH Advanced Option Type

OPTIONBYTE_PCROP PCROP option byte configuration

OPTIONBYTE_BOOTCONFIG BOOTConfig option byte configuration

FLASH Banks

FLASH_BANK_1 Bank 1

FLASH_BANK_2 Bank 2

FLASH_BANK_BOTH Bank1 and Bank2

FLASH BOR Reset Level

OB_BOR_LEVEL3 Supply voltage ranges from 2.70 to 3.60 V

OB_BOR_LEVEL2 Supply voltage ranges from 2.40 to 2.70 V

OB_BOR_LEVEL1 Supply voltage ranges from 2.10 to 2.40 V

OB_BOR_OFF Supply voltage ranges from 1.62 to 2.10 V

FLASH Dual Boot

OB_DUAL_BOOT_ENABLE Dual Bank Boot Enable

OB_DUAL_BOOT_DISABLE Dual Bank Boot Disable, always boot on User Flash

FLASH Private macros to check input parameters

IS_FLASH_TYPEERASE

IS_VOLTAGERANGE

IS_WRPSTATE

IS_OPTIONBYTE

IS_OB_RDP_LEVEL

IS_OB_IWDG_SOURCE

IS_OB_STOP_SOURCE

IS_OB_STDBY_SOURCE

IS_OB_BOR_LEVEL

IS_PCROPSTATE

IS_OBEX

IS_FLASH_LATENCY

IS_FLASH_BANK

IS_FLASH_SECTOR

IS_FLASH_ADDRESS

IS_FLASH_NBSECTORS

IS_OB_WRP_SECTOR

IS_OB_PCROP

IS_OB_BOOT

IS_OB_PCROP_SELECT

FLASH Mass Erase bit

FLASH_MER_BIT 2 MER bits here to clear

FLASH Option Bytes IWatchdog

OB_IWDG_SW Software IWDG selected

OB_IWDG_HW Hardware IWDG selected

FLASH Option Bytes nRST_STDBY

OB_STDBY_NO_RST No reset generated when entering in STANDBY

OB_STDBY_RST Reset generated when entering in STANDBY

FLASH Option Bytes nRST_STOP

OB_STOP_NO_RST No reset generated when entering in STOP

OB_STOP_RST Reset generated when entering in STOP

FLASH Option Bytes PC ReadWrite Protection

OB_PCROP_SECTOR_0 PC Read/Write protection of Sector0

OB_PCROP_SECTOR_1 PC Read/Write protection of Sector1

OB_PCROP_SECTOR_2 PC Read/Write protection of Sector2

OB_PCROP_SECTOR_3 PC Read/Write protection of Sector3

OB_PCROP_SECTOR_4 PC Read/Write protection of Sector4

OB_PCROP_SECTOR_5 PC Read/Write protection of Sector5

OB_PCROP_SECTOR_6 PC Read/Write protection of Sector6

OB_PCROP_SECTOR_7 PC Read/Write protection of Sector7

OB_PCROP_SECTOR_8 PC Read/Write protection of Sector8

OB_PCROP_SECTOR_9 PC Read/Write protection of Sector9

OB_PCROP_SECTOR_10 PC Read/Write protection of Sector10

OB_PCROP_SECTOR_11 PC Read/Write protection of Sector11

OB_PCROP_SECTOR_12 PC Read/Write protection of Sector12

OB_PCROP_SECTOR_13 PC Read/Write protection of Sector13

OB_PCROP_SECTOR_14 PC Read/Write protection of Sector14

OB_PCROP_SECTOR_15 PC Read/Write protection of Sector15

OB_PCROP_SECTOR_16 PC Read/Write protection of Sector16

OB_PCROP_SECTOR_17 PC Read/Write protection of Sector17

OB_PCROP_SECTOR_18 PC Read/Write protection of Sector18

OB_PCROP_SECTOR_19	PC Read/Write protection of Sector19
OB_PCROP_SECTOR_20	PC Read/Write protection of Sector20
OB_PCROP_SECTOR_21	PC Read/Write protection of Sector21
OB_PCROP_SECTOR_22	PC Read/Write protection of Sector22
OB_PCROP_SECTOR_23	PC Read/Write protection of Sector23
OB_PCROP_SECTOR_All	PC Read/Write protection of all Sectors

FLASH Option Bytes Read Protection

OB_RDP_LEVEL_0	
OB_RDP_LEVEL_1	
OB_RDP_LEVEL_2	Warning: When enabling read protection level 2 it s no more possible to go back to level 1 or 0

FLASH Option Bytes Write Protection

OB_WRP_SECTOR_0	Write protection of Sector0
OB_WRP_SECTOR_1	Write protection of Sector1
OB_WRP_SECTOR_2	Write protection of Sector2
OB_WRP_SECTOR_3	Write protection of Sector3
OB_WRP_SECTOR_4	Write protection of Sector4
OB_WRP_SECTOR_5	Write protection of Sector5
OB_WRP_SECTOR_6	Write protection of Sector6
OB_WRP_SECTOR_7	Write protection of Sector7
OB_WRP_SECTOR_8	Write protection of Sector8
OB_WRP_SECTOR_9	Write protection of Sector9
OB_WRP_SECTOR_10	Write protection of Sector10
OB_WRP_SECTOR_11	Write protection of Sector11
OB_WRP_SECTOR_12	Write protection of Sector12
OB_WRP_SECTOR_13	Write protection of Sector13
OB_WRP_SECTOR_14	Write protection of Sector14
OB_WRP_SECTOR_15	Write protection of Sector15
OB_WRP_SECTOR_16	Write protection of Sector16
OB_WRP_SECTOR_17	Write protection of Sector17
OB_WRP_SECTOR_18	Write protection of Sector18
OB_WRP_SECTOR_19	Write protection of Sector19
OB_WRP_SECTOR_20	Write protection of Sector20
OB_WRP_SECTOR_21	Write protection of Sector21
OB_WRP_SECTOR_22	Write protection of Sector22
OB_WRP_SECTOR_23	Write protection of Sector23
OB_WRP_SECTOR_All	Write protection of all Sectors

FLASH Option Type

OPTIONBYTE_WRP	WRP option byte configuration
OPTIONBYTE_RDP	RDP option byte configuration
OPTIONBYTE_USER	USER option byte configuration
OPTIONBYTE_BOR	BOR option byte configuration

FLASH PCROP State

OB_PCROP_STATE_DISABLE	Disable PCROP
OB_PCROP_STATE_ENABLE	Enable PCROP

FLASH Sectors

FLASH_SECTOR_0	Sector Number 0
FLASH_SECTOR_1	Sector Number 1
FLASH_SECTOR_2	Sector Number 2
FLASH_SECTOR_3	Sector Number 3
FLASH_SECTOR_4	Sector Number 4
FLASH_SECTOR_5	Sector Number 5
FLASH_SECTOR_6	Sector Number 6
FLASH_SECTOR_7	Sector Number 7
FLASH_SECTOR_8	Sector Number 8
FLASH_SECTOR_9	Sector Number 9
FLASH_SECTOR_10	Sector Number 10
FLASH_SECTOR_11	Sector Number 11
FLASH_SECTOR_12	Sector Number 12
FLASH_SECTOR_13	Sector Number 13
FLASH_SECTOR_14	Sector Number 14
FLASH_SECTOR_15	Sector Number 15
FLASH_SECTOR_16	Sector Number 16
FLASH_SECTOR_17	Sector Number 17
FLASH_SECTOR_18	Sector Number 18
FLASH_SECTOR_19	Sector Number 19
FLASH_SECTOR_20	Sector Number 20
FLASH_SECTOR_21	Sector Number 21
FLASH_SECTOR_22	Sector Number 22
FLASH_SECTOR_23	Sector Number 23

FLASH Selection Protection Mode

OB_PCROP_DESELECTED	Disabled PcROP, nWPRi bits used for Write Protection on sector i
OB_PCROP_SELECTED	Enable PcROP, nWPRi bits used for PCRoP Protection on

sector i

FLASH Type Erase

FLASH_TYPEERASE_SECTORS	Sectors erase only
FLASH_TYPEERASE_MASSERASE	Flash Mass erase activation

FLASH Voltage Range

FLASH_VOLTAGE_RANGE_1	Device operating range: 1.8V to 2.1V
FLASH_VOLTAGE_RANGE_2	Device operating range: 2.1V to 2.7V
FLASH_VOLTAGE_RANGE_3	Device operating range: 2.7V to 3.6V
FLASH_VOLTAGE_RANGE_4	Device operating range: 2.7V to 3.6V + External Vpp

FLASH WRP State

OB_WRPSTATE_DISABLE	Disable the write protection of the desired bank 1 sectors
OB_WRPSTATE_ENABLE	Enable the write protection of the desired bank 1 sectors

24 HAL FLASH__RAMFUNC Generic Driver

24.1 FLASH__RAMFUNC Firmware driver API description

24.1.1 APIs executed from Internal RAM

ARM Compiler

RAM functions are defined using the toolchain options. Functions that are be executed in RAM should reside in a separate source module. Using the 'Options for File' dialog you can simply change the 'Code / Const' area of a module to a memory space in physical RAM. Available memory areas are declared in the 'Target' tab of the Options for Target' dialog.

ICCARM Compiler

RAM functions are defined using a specific toolchain keyword "__ramfunc".

GNU Compiler

RAM functions are defined using a specific toolchain attribute "__attribute__((section(".RamFunc")))".

24.1.2 ramfunc functions

This subsection provides a set of functions that should be executed from RAM transfers.

This section contains the following APIs:

- ***HAL_FLASHEx_StopFlashInterfaceClk()***
- ***HAL_FLASHEx_StartFlashInterfaceClk()***
- ***HAL_FLASHEx_EnableFlashSleepMode()***
- ***HAL_FLASHEx_DisableFlashSleepMode()***

24.1.3 Detailed description of functions

HAL_FLASHEx_StopFlashInterfaceClk

Function Name	**__RAM_FUNC HAL_FLASHEx_StopFlashInterfaceClk (void)**
Function Description	Stop the flash interface while System Run.
Return values	• **None**
Notes	• This mode is only available for STM32F41xxx/STM32F446xx devices. • This mode couldn't be set while executing with the flash itself. It should be done with specific routine executed from RAM.

HAL_FLASHEx_StartFlashInterfaceClk

Function Name	**__RAM_FUNC HAL_FLASHEx_StartFlashInterfaceClk (void)**
Function Description	Start the flash interface while System Run.
Return values	• **None**

Notes	• This mode is only available for STM32F411xx/STM32F446xx devices. • This mode couldn't be set while executing with the flash itself. It should be done with specific routine executed from RAM.

HAL_FLASHEx_EnableFlashSleepMode

Function Name	**__RAM_FUNC HAL_FLASHEx_EnableFlashSleepMode (void)**
Function Description	Enable the flash sleep while System Run.
Return values	• **None**
Notes	• This mode is only available for STM32F41xxx/STM32F446xx devices. • This mode could n't be set while executing with the flash itself. It should be done with specific routine executed from RAM.

HAL_FLASHEx_DisableFlashSleepMode

Function Name	**__RAM_FUNC HAL_FLASHEx_DisableFlashSleepMode (void)**
Function Description	Disable the flash sleep while System Run.
Return values	• **None**
Notes	• This mode is only available for STM32F41xxx/STM32F446xx devices. • This mode couldn't be set while executing with the flash itself. It should be done with specific routine executed from RAM.

25 HAL FMPI2C Generic Driver

25.1 FMPI2C Firmware driver registers structures

25.1.1 FMPI2C_InitTypeDef

Data Fields

- ***uint32_t Timing***
- ***uint32_t OwnAddress1***
- ***uint32_t AddressingMode***
- ***uint32_t DualAddressMode***
- ***uint32_t OwnAddress2***
- ***uint32_t OwnAddress2Masks***
- ***uint32_t GeneralCallMode***
- ***uint32_t NoStretchMode***

Field Documentation

- ***uint32_t FMPI2C_InitTypeDef::Timing***
  Specifies the FMPI2C_TIMINGR_register value. This parameter calculated by referring to FMPI2C initialization section in Reference manual
- ***uint32_t FMPI2C_InitTypeDef::OwnAddress1***
  Specifies the first device own address. This parameter can be a 7-bit or 10-bit address.
- ***uint32_t FMPI2C_InitTypeDef::AddressingMode***
  Specifies if 7-bit or 10-bit addressing mode is selected. This parameter can be a value of ***FMPI2C_ADDRESSING_MODE***
- ***uint32_t FMPI2C_InitTypeDef::DualAddressMode***
  Specifies if dual addressing mode is selected. This parameter can be a value of ***FMPI2C_DUAL_ADDRESSING_MODE***
- ***uint32_t FMPI2C_InitTypeDef::OwnAddress2***
  Specifies the second device own address if dual addressing mode is selected This parameter can be a 7-bit address.
- ***uint32_t FMPI2C_InitTypeDef::OwnAddress2Masks***
  Specifies the acknowledge mask address second device own address if dual addressing mode is selected This parameter can be a value of ***FMPI2C_OWN_ADDRESS2_MASKS***
- ***uint32_t FMPI2C_InitTypeDef::GeneralCallMode***
  Specifies if general call mode is selected. This parameter can be a value of ***FMPI2C_GENERAL_CALL_ADDRESSING_MODE***
- ***uint32_t FMPI2C_InitTypeDef::NoStretchMode***
  Specifies if nostretch mode is selected. This parameter can be a value of ***FMPI2C_NOSTRETCH_MODE***

25.1.2 __FMPI2C_HandleTypeDef

Data Fields

- ***FMPI2C_TypeDef * Instance***
- ***FMPI2C_InitTypeDef Init***
- ***uint8_t * pBuffPtr***
- ***uint16_t XferSize***
- ***__IO uint16_t XferCount***
- ***__IO uint32_t XferOptions***
- ***__IO uint32_t PreviousState***
- ***HAL_StatusTypeDef(* XferISR***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_FMPI2C_StateTypeDef State***
- ***__IO HAL_FMPI2C_ModeTypeDef Mode***
- ***__IO uint32_t ErrorCode***
- ***__IO uint32_t AddrEventCount***

Field Documentation

- ***FMPI2C_TypeDef* __FMPI2C_HandleTypeDef::Instance***
  FMPI2C registers base address
- ***FMPI2C_InitTypeDef __FMPI2C_HandleTypeDef::Init***
  FMPI2C communication parameters
- ***uint8_t* __FMPI2C_HandleTypeDef::pBuffPtr***
  Pointer to FMPI2C transfer buffer
- ***uint16_t __FMPI2C_HandleTypeDef::XferSize***
  FMPI2C transfer size
- ***__IO uint16_t __FMPI2C_HandleTypeDef::XferCount***
  FMPI2C transfer counter
- ***__IO uint32_t __FMPI2C_HandleTypeDef::XferOptions***
  FMPI2C sequantial transfer options, this parameter can be a value of ***FMPI2C_XFEROPTIONS***
- ***__IO uint32_t __FMPI2C_HandleTypeDef::PreviousState***
  FMPI2C communication Previous state
- ***HAL_StatusTypeDef(* __FMPI2C_HandleTypeDef::XferISR)(struct __FMPI2C_HandleTypeDef *hfmpi2c, uint32_t ITFlags, uint32_t ITSources)***
  FMPI2C transfer IRQ handler function pointer
- ***DMA_HandleTypeDef* __FMPI2C_HandleTypeDef::hdmatx***
  FMPI2C Tx DMA handle parameters
- ***DMA_HandleTypeDef* __FMPI2C_HandleTypeDef::hdmarx***
  FMPI2C Rx DMA handle parameters
- ***HAL_LockTypeDef __FMPI2C_HandleTypeDef::Lock***
  FMPI2C locking object
- ***__IO HAL_FMPI2C_StateTypeDef __FMPI2C_HandleTypeDef::State***
  FMPI2C communication state
- ***__IO HAL_FMPI2C_ModeTypeDef __FMPI2C_HandleTypeDef::Mode***
  FMPI2C communication mode
- ***__IO uint32_t __FMPI2C_HandleTypeDef::ErrorCode***
  FMPI2C Error code
- ***__IO uint32_t __FMPI2C_HandleTypeDef::AddrEventCount***
  FMPI2C Address Event counter

25.2 FMPI2C Firmware driver API description

25.2.1 How to use this driver

The FMPI2C HAL driver can be used as follows:

1. Declare a FMPI2C_HandleTypeDef handle structure, for example: FMPI2C_HandleTypeDef hfmpi2c;
2. Initialize the FMPI2C low level resources by implementing the HAL_FMPI2C_MspInit() API:
   a. Enable the FMPI2Cx interface clock
   b. FMPI2C pins configuration
      – Enable the clock for the FMPI2C GPIOs
      – Configure FMPI2C pins as alternate function open-drain
   c. NVIC configuration if you need to use interrupt process
      – Configure the FMPI2Cx interrupt priority
      – Enable the NVIC FMPI2C IRQ Channel
   d. DMA Configuration if you need to use DMA process
      – Declare a DMA_HandleTypeDef handle structure for the transmit or receive channel
      – Enable the DMAx interface clock using
      – Configure the DMA handle parameters
      – Configure the DMA Tx or Rx channel
      – Associate the initialized DMA handle to the hfmpi2c DMA Tx or Rx handle
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx or Rx channel
3. Configure the Communication Clock Timing, Own Address1, Master Addressing mode, Dual Addressing mode, Own Address2, Own Address2 Mask, General call and Nostretch mode in the hfmpi2c Init structure.
4. Initialize the FMPI2C registers by calling the HAL_FMPI2C_Init(), configures also the low level Hardware (GPIO, CLOCK, NVIC...etc) by calling the customized HAL_FMPI2C_MspInit(&hfmpi2c) API.
5. To check if target device is ready for communication, use the function HAL_FMPI2C_IsDeviceReady()
6. For FMPI2C IO and IO MEM operations, three operation modes are available within this driver :

Polling mode IO operation

- Transmit in master mode an amount of data in blocking mode using HAL_FMPI2C_Master_Transmit()
- Receive in master mode an amount of data in blocking mode using HAL_FMPI2C_Master_Receive()
- Transmit in slave mode an amount of data in blocking mode using HAL_FMPI2C_Slave_Transmit()
- Receive in slave mode an amount of data in blocking mode using HAL_FMPI2C_Slave_Receive()

Polling mode IO MEM operation

- Write an amount of data in blocking mode to a specific memory address using HAL_FMPI2C_Mem_Write()

- Read an amount of data in blocking mode from a specific memory address using HAL_FMPI2C_Mem_Read()

Interrupt mode IO operation

- Transmit in master mode an amount of data in non-blocking mode using HAL_FMPI2C_Master_Transmit_IT()
- At transmission end of transfer, HAL_FMPI2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterTxCpltCallback()
- Receive in master mode an amount of data in non-blocking mode using HAL_FMPI2C_Master_Receive_IT()
- At reception end of transfer, HAL_FMPI2C_MasterRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterRxCpltCallback()
- Transmit in slave mode an amount of data in non-blocking mode using HAL_FMPI2C_Slave_Transmit_IT()
- At transmission end of transfer, HAL_FMPI2C_SlaveTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveTxCpltCallback()
- Receive in slave mode an amount of data in non-blocking mode using HAL_FMPI2C_Slave_Receive_IT()
- At reception end of transfer, HAL_FMPI2C_SlaveRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveRxCpltCallback()
- In case of transfer Error, HAL_FMPI2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ErrorCallback()
- Abort a master FMPI2C process communication with Interrupt using HAL_FMPI2C_Master_Abort_IT()
- End of abort process, HAL_FMPI2C_MasterRxCpltCallback() or HAL_FMPI2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterRxCpltCallback() or HAL_FMPI2C_MasterTxCpltCallback()
- Discard a slave FMPI2C process communication using __HAL_FMPI2C_GENERATE_NACK() macro. This action will inform Master to generate a Stop condition to discard the communication.

Interrupt mode IO sequential operation

These interfaces allow to manage a sequential transfer with a repeated start condition when a direction change during transfer

- A specific option field manage the different steps of a sequential transfer
- Option field values are defined through FMPI2C_XFEROPTIONS and are listed below:
  - FMPI2C_FIRST_AND_LAST_FRAME: No sequential usage, functionnal is same as associated interfaces in no sequential mode
  - FMPI2C_FIRST_FRAME: Sequential usage, this option allow to manage a sequence with start condition, address and data to transfer without a final stop condition

- FMPI2C_NEXT_FRAME: Sequential usage, this option allow to manage a sequence with a restart condition, address and with new data to transfer if the direction change or manage only the new data to transfer if no direction change and without a final stop condition in both cases
- FMPI2C_LAST_FRAME: Sequential usage, this option allow to manage a sequance with a restart condition, address and with new data to transfer if the direction change or manage only the new data to transfer if no direction change and with a final stop condition in both cases

- Differents sequential FMPI2C interfaces are listed below:
  - Sequential transmit in master FMPI2C mode an amount of data in non-blocking mode using HAL_FMPI2C_Master_Sequential_Transmit_IT()
    - At transmission end of current frame transfer, HAL_FMPI2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterTxCpltCallback()
  - Sequential receive in master FMPI2C mode an amount of data in non-blocking mode using HAL_FMPI2C_Master_Sequential_Receive_IT()
    - At reception end of current frame transfer, HAL_FMPI2C_MasterRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterRxCpltCallback()
  - Abort a master FMPI2C process communication with Interrupt using HAL_FMPI2C_Master_Abort_IT()
    - End of abort process, HAL_FMPI2C_AbortCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_AbortCpltCallback()
    - mean HAL_FMPI2C_MasterTxCpltCallback() in case of previous state was master transmit
    - mean HAL_FMPI2C_MasterRxCpltCallback() in case of previous state was master receive
  - Enable/disable the Address listen mode in slave FMPI2C mode using HAL_FMPI2C_EnableListen_IT() HAL_FMPI2C_DisableListen_IT()
    - When address slave FMPI2C match, HAL_FMPI2C_AddrCallback() is executed and user can add his own code to check the Address Match Code and the transmission direction request by master (Write/Read).
    - At Listen mode end HAL_FMPI2C_ListenCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ListenCpltCallback()
  - Sequential transmit in slave FMPI2C mode an amount of data in non-blocking mode using HAL_FMPI2C_Slave_Sequential_Transmit_IT()
    - At transmission end of current frame transfer, HAL_FMPI2C_SlaveTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveTxCpltCallback()
  - Sequential receive in slave FMPI2C mode an amount of data in non-blocking mode using HAL_FMPI2C_Slave_Sequential_Receive_IT()
    - At reception end of current frame transfer, HAL_FMPI2C_SlaveRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveRxCpltCallback()
  - In case of transfer Error, HAL_FMPI2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ErrorCallback()

- Abort a master FMPI2C process communication with Interrupt using HAL_FMPI2C_Master_Abort_IT()
- End of abort process, HAL_FMPI2C_AbortCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_AbortCpltCallback()
- Discard a slave FMPI2C process communication using __HAL_FMPI2C_GENERATE_NACK() macro. This action will inform Master to generate a Stop condition to discard the communication.

Interrupt mode IO MEM operation

- Write an amount of data in non-blocking mode with Interrupt to a specific memory address using HAL_FMPI2C_Mem_Write_IT()
- At Memory end of write transfer, HAL_FMPI2C_MemTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MemTxCpltCallback()
- Read an amount of data in non-blocking mode with Interrupt from a specific memory address using HAL_FMPI2C_Mem_Read_IT()
- At Memory end of read transfer, HAL_FMPI2C_MemRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MemRxCpltCallback()
- In case of transfer Error, HAL_FMPI2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ErrorCallback()

DMA mode IO operation

- Transmit in master mode an amount of data in non-blocking mode (DMA) using HAL_FMPI2C_Master_Transmit_DMA()
- At transmission end of transfer, HAL_FMPI2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterTxCpltCallback()
- Receive in master mode an amount of data in non-blocking mode (DMA) using HAL_FMPI2C_Master_Receive_DMA()
- At reception end of transfer, HAL_FMPI2C_MasterRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterRxCpltCallback()
- Transmit in slave mode an amount of data in non-blocking mode (DMA) using HAL_FMPI2C_Slave_Transmit_DMA()
- At transmission end of transfer, HAL_FMPI2C_SlaveTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveTxCpltCallback()
- Receive in slave mode an amount of data in non-blocking mode (DMA) using HAL_FMPI2C_Slave_Receive_DMA()
- At reception end of transfer, HAL_FMPI2C_SlaveRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_SlaveRxCpltCallback()
- In case of transfer Error, HAL_FMPI2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ErrorCallback()
- Abort a master FMPI2C process communication with Interrupt using HAL_FMPI2C_Master_Abort_IT()

- End of abort process, HAL_FMPI2C_MasterRxCpltCallback() or HAL_FMPI2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MasterRxCpltCallback() or HAL_FMPI2C_MasterTxCpltCallback()
- Discard a slave FMPI2C process communication using __HAL_FMPI2C_GENERATE_NACK() macro. This action will inform Master to generate a Stop condition to discard the communication.

DMA mode IO MEM operation

- Write an amount of data in non-blocking mode with DMA to a specific memory address using HAL_FMPI2C_Mem_Write_DMA()
- At Memory end of write transfer, HAL_FMPI2C_MemTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MemTxCpltCallback()
- Read an amount of data in non-blocking mode with DMA from a specific memory address using HAL_FMPI2C_Mem_Read_DMA()
- At Memory end of read transfer, HAL_FMPI2C_MemRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_FMPI2C_MemRxCpltCallback()
- In case of transfer Error, HAL_FMPI2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_FMPI2C_ErrorCallback()

FMPI2C HAL driver macros list

Below the list of most used macros in FMPI2C HAL driver.

- __HAL_FMPI2C_ENABLE: Enable the FMPI2C peripheral
- __HAL_FMPI2C_DISABLE: Disable the FMPI2C peripheral
- __HAL_FMPI2C_GENERATE_NACK: Generate a Non-Acknowledge FMPI2C peripheral in Slave mode
- __HAL_FMPI2C_GET_FLAG: Check whether the specified FMPI2C flag is set or not
- __HAL_FMPI2C_CLEAR_FLAG: Clear the specified FMPI2C pending flag
- __HAL_FMPI2C_ENABLE_IT: Enable the specified FMPI2C interrupt
- __HAL_FMPI2C_DISABLE_IT: Disable the specified FMPI2C interrupt

You can refer to the FMPI2C HAL driver header file for more useful macros

25.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and deinitialize the FMPI2Cx peripheral:

- User must Implement HAL_FMPI2C_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).
- Call the function HAL_FMPI2C_Init() to configure the selected device with the selected configuration:
  - Clock Timing
  - Own Address 1
  - Addressing mode (Master, Slave)

  - Dual Addressing mode
  - Own Address 2
  - Own Address 2 Mask
  - General call mode
  - Nostretch mode
- Call the function HAL_FMPI2C_DeInit() to restore the default configuration of the selected FMPI2Cx peripheral.

This section contains the following APIs:

- ***HAL_FMPI2C_Init()***
- ***HAL_FMPI2C_DeInit()***
- ***HAL_FMPI2C_MspInit()***
- ***HAL_FMPI2C_MspDeInit()***

25.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the FMPI2C data transfers.

1. There are two modes of transfer:
   - Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer startup. The end of the data processing will be indicated through the dedicated FMPI2C IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
2. Blocking mode functions are :
   - HAL_FMPI2C_Master_Transmit()
   - HAL_FMPI2C_Master_Receive()
   - HAL_FMPI2C_Slave_Transmit()
   - HAL_FMPI2C_Slave_Receive()
   - HAL_FMPI2C_Mem_Write()
   - HAL_FMPI2C_Mem_Read()
   - HAL_FMPI2C_IsDeviceReady()
3. No-Blocking mode functions with Interrupt are :
   - HAL_FMPI2C_Master_Transmit_IT()
   - HAL_FMPI2C_Master_Receive_IT()
   - HAL_FMPI2C_Slave_Transmit_IT()
   - HAL_FMPI2C_Slave_Receive_IT()
   - HAL_FMPI2C_Master_Sequential_Transmit_IT()
   - HAL_FMPI2C_Master_Sequential_Receive_IT()
   - HAL_FMPI2C_Slave_Sequential_Transmit_IT()
   - HAL_FMPI2C_Slave_Sequential_Receive_IT()
   - HAL_FMPI2C_Mem_Write_IT()
   - HAL_FMPI2C_Mem_Read_IT()
4. No-Blocking mode functions with DMA are :
   - HAL_FMPI2C_Master_Transmit_DMA()
   - HAL_FMPI2C_Master_Receive_DMA()
   - HAL_FMPI2C_Slave_Transmit_DMA()
   - HAL_FMPI2C_Slave_Receive_DMA()
   - HAL_FMPI2C_Mem_Write_DMA()
   - HAL_FMPI2C_Mem_Read_DMA()
5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   - HAL_FMPI2C_MemTxCpltCallback()

- HAL_FMPI2C_MemRxCpltCallback()
- HAL_FMPI2C_MasterTxCpltCallback()
- HAL_FMPI2C_MasterRxCpltCallback()
- HAL_FMPI2C_SlaveTxCpltCallback()
- HAL_FMPI2C_SlaveRxCpltCallback()
- HAL_FMPI2C_ErrorCallback()
- HAL_FMPI2C_AbortCpltCallback()

This section contains the following APIs:

- ***HAL_FMPI2C_Master_Transmit()***
- ***HAL_FMPI2C_Master_Receive()***
- ***HAL_FMPI2C_Slave_Transmit()***
- ***HAL_FMPI2C_Slave_Receive()***
- ***HAL_FMPI2C_Master_Transmit_IT()***
- ***HAL_FMPI2C_Master_Receive_IT()***
- ***HAL_FMPI2C_Slave_Transmit_IT()***
- ***HAL_FMPI2C_Slave_Receive_IT()***
- ***HAL_FMPI2C_Master_Transmit_DMA()***
- ***HAL_FMPI2C_Master_Receive_DMA()***
- ***HAL_FMPI2C_Slave_Transmit_DMA()***
- ***HAL_FMPI2C_Slave_Receive_DMA()***
- ***HAL_FMPI2C_Mem_Write()***
- ***HAL_FMPI2C_Mem_Read()***
- ***HAL_FMPI2C_Mem_Write_IT()***
- ***HAL_FMPI2C_Mem_Read_IT()***
- ***HAL_FMPI2C_Mem_Write_DMA()***
- ***HAL_FMPI2C_Mem_Read_DMA()***
- ***HAL_FMPI2C_IsDeviceReady()***
- ***HAL_FMPI2C_Master_Sequential_Transmit_IT()***
- ***HAL_FMPI2C_Master_Sequential_Receive_IT()***
- ***HAL_FMPI2C_Slave_Sequential_Transmit_IT()***
- ***HAL_FMPI2C_Slave_Sequential_Receive_IT()***
- ***HAL_FMPI2C_EnableListen_IT()***
- ***HAL_FMPI2C_DisableListen_IT()***
- ***HAL_FMPI2C_Master_Abort_IT()***

25.2.4 Peripheral State, Mode and Error functions

This subsection permit to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_FMPI2C_GetState()***
- ***HAL_FMPI2C_GetMode()***
- ***HAL_FMPI2C_GetError()***

25.2.5 Detailed description of functions

HAL_FMPI2C_Init

Function Name	**HAL_StatusTypeDef HAL_FMPI2C_Init (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Initializes the FMPI2C according to the specified parameters in the FMPI2C_InitTypeDef and initialize the associated handle.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **HAL:** status

HAL_FMPI2C_DeInit

Function Name **HAL_StatusTypeDef HAL_FMPI2C_DeInit (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description DeInitialize the FMPI2C peripheral.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **HAL:** status

HAL_FMPI2C_MspInit

Function Name **void HAL_FMPI2C_MspInit (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description Initialize the FMPI2C MSP.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_MspDeInit

Function Name **void HAL_FMPI2C_MspDeInit (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description DeInitialize the FMPI2C MSP.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_Master_Transmit

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Transmit (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Transmits in master mode an amount of data in blocking mode.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer

- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_Master_Receive

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Receive (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receives in master mode an amount of data in blocking mode.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Transmit

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Transmit (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Transmits in slave mode an amount of data in blocking mode.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Receive

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Receive (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receive in slave mode an amount of data in blocking mode.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Write

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Write (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Write an amount of data in blocking mode to a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Read

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Read (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Read an amount of data in blocking mode from a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_FMPI2C_IsDeviceReady

Function Name **HAL_StatusTypeDef HAL_FMPI2C_IsDeviceReady (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint32_t Trials, uint32_t Timeout)**

Function Description Checks if target device is ready for communication.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **Trials:** Number of trials
- **Timeout:** Timeout duration

Return values
- **HAL:** status

Notes
- This function is used with Memory devices

HAL_FMPI2C_Master_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Transmit_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Transmit in master mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Master_Receive_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Receive_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Receive in master mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Transmit_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size)**

Function Description Transmit in slave mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Receive_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Receive_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size)**

Function Description Receive in slave mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Write_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Write_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Write an amount of data in non-blocking mode with Interrupt to a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Read_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Read_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Read an amount of data in non-blocking mode with Interrupt from a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer

- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Master_Sequential_Transmit_IT

Function Name
HAL_StatusTypeDef HAL_FMPI2C_Master_Sequential_Transmit_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t XferOptions)

Function Description
Sequential transmit in master FMPI2C mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **XferOptions:** Options of Transfer, value of FMPI2C Sequential Transfer Options

Return values
- **HAL:** status

Notes
- This interface allow to manage repeated start condition when a direction change during transfer

HAL_FMPI2C_Master_Sequential_Receive_IT

Function Name
HAL_StatusTypeDef HAL_FMPI2C_Master_Sequential_Receive_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t XferOptions)

Function Description
Sequential receive in master FMPI2C mode an amount of data in non-blocking mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **XferOptions:** Options of Transfer, value of FMPI2C Sequential Transfer Options

Return values
- **HAL:** status

Notes
- This interface allow to manage repeated start condition when a direction change during transfer

HAL_FMPI2C_Slave_Sequential_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_FMPI2C_Slave_Sequential_Transmit_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size, uint32_t XferOptions)**
Function Description	Sequential transmit in slave/device FMPI2C mode an amount of data in non-blocking mode with Interrupt.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **XferOptions:** Options of Transfer, value of FMPI2C Sequential Transfer Options
Return values	• **HAL:** status
Notes	• This interface allow to manage repeated start condition when a direction change during transfer

HAL_FMPI2C_Slave_Sequential_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_FMPI2C_Slave_Sequential_Receive_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size, uint32_t XferOptions)**
Function Description	Sequential receive in slave/device FMPI2C mode an amount of data in non-blocking mode with Interrupt.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **XferOptions:** Options of Transfer, value of FMPI2C Sequential Transfer Options
Return values	• **HAL:** status
Notes	• This interface allow to manage repeated start condition when a direction change during transfer

HAL_FMPI2C_EnableListen_IT

Function Name	**HAL_StatusTypeDef HAL_FMPI2C_EnableListen_IT (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Enable the Address listen mode with Interrupt.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **HAL:** status

HAL_FMPI2C_DisableListen_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_DisableListen_IT (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description Disable the Address listen mode with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C

Return values
- **HAL:** status

HAL_FMPI2C_Master_Abort_IT

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Abort_IT (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress)**

Function Description Abort a master FMPI2C IT or DMA process communication with Interrupt.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface

Return values
- **HAL:** status

HAL_FMPI2C_Master_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Transmit_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Transmit in master mode an amount of data in non-blocking mode with DMA.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Master_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Master_Receive_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Receive in master mode an amount of data in non-blocking mode with DMA.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Transmit_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size)**

Function Description Transmit in slave mode an amount of data in non-blocking mode with DMA.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Slave_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Slave_Receive_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t * pData, uint16_t Size)**

Function Description Receive in slave mode an amount of data in non-blocking mode with DMA.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Write_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Write_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Write an amount of data in non-blocking mode with DMA to a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_FMPI2C_Mem_Read_DMA

Function Name **HAL_StatusTypeDef HAL_FMPI2C_Mem_Read_DMA (FMPI2C_HandleTypeDef * hfmpi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Reads an amount of data in non-blocking mode with DMA from a specific memory address.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be read

Return values
- **HAL:** status

HAL_FMPI2C_EV_IRQHandler

Function Name **void HAL_FMPI2C_EV_IRQHandler (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description This function handles FMPI2C event interrupt request.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_ER_IRQHandler

Function Name **void HAL_FMPI2C_ER_IRQHandler (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description This function handles FMPI2C error interrupt request.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_MasterTxCpltCallback

Function Name	**void HAL_FMPI2C_MasterTxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Master Tx Transfer completed callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **None**

HAL_FMPI2C_MasterRxCpltCallback

Function Name	**void HAL_FMPI2C_MasterRxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Master Rx Transfer completed callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **None**

HAL_FMPI2C_SlaveTxCpltCallback

Function Name	**void HAL_FMPI2C_SlaveTxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Slave Tx Transfer completed callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **None**

HAL_FMPI2C_SlaveRxCpltCallback

Function Name	**void HAL_FMPI2C_SlaveRxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Slave Rx Transfer completed callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **None**

HAL_FMPI2C_AddrCallback

Function Name	**void HAL_FMPI2C_AddrCallback (FMPI2C_HandleTypeDef * hfmpi2c, uint8_t TransferDirection, uint16_t AddrMatchCode)**
Function Description	Slave Address Match callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified

FMPI2C.
- **TransferDirection:** Master request Transfer Direction (Write/Read), value of FMPI2C Sequential Transfer Options
- **AddrMatchCode:** Address Match Code

Return values
- **None**

HAL_FMPI2C_ListenCpltCallback

Function Name **void HAL_FMPI2C_ListenCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description Listen Complete callback.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_MemTxCpltCallback

Function Name **void HAL_FMPI2C_MemTxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description Memory Tx Transfer completed callback.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_MemRxCpltCallback

Function Name **void HAL_FMPI2C_MemRxCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description Memory Rx Transfer completed callback.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_ErrorCallback

Function Name **void HAL_FMPI2C_ErrorCallback (FMPI2C_HandleTypeDef * hfmpi2c)**

Function Description FMPI2C error callback.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.

Return values
- **None**

HAL_FMPI2C_AbortCpltCallback

Function Name	**void HAL_FMPI2C_AbortCpltCallback (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	FMPI2C abort callback.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **None**

HAL_FMPI2C_GetState

Function Name	**HAL_FMPI2C_StateTypeDef HAL_FMPI2C_GetState (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Return the FMPI2C handle state.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **HAL:** state

HAL_FMPI2C_GetMode

Function Name	**HAL_FMPI2C_ModeTypeDef HAL_FMPI2C_GetMode (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Returns the FMPI2C Master, Slave, Memory or no mode.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for FMPI2C module
Return values	• **HAL:** mode

HAL_FMPI2C_GetError

Function Name	**uint32_t HAL_FMPI2C_GetError (FMPI2C_HandleTypeDef * hfmpi2c)**
Function Description	Return the FMPI2C error code.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2C.
Return values	• **FMPI2C:** Error Code

25.3 FMPI2C Firmware driver defines

25.3.1 FMPI2C

FMPI2C Addressing Mode

FMPI2C_ADDRESSINGMODE_7BIT

FMPI2C_ADDRESSINGMODE_10BIT

FMPI2C Dual Addressing Mode

FMPI2C_DUALADDRESS_DISABLE

FMPI2C_DUALADDRESS_ENABLE

FMPI2C Error Code definition

HAL_FMPI2C_ERROR_NONE	No error
HAL_FMPI2C_ERROR_BERR	BERR error
HAL_FMPI2C_ERROR_ARLO	ARLO error
HAL_FMPI2C_ERROR_AF	ACKF error
HAL_FMPI2C_ERROR_OVR	OVR error
HAL_FMPI2C_ERROR_DMA	DMA transfer error
HAL_FMPI2C_ERROR_TIMEOUT	Timeout error
HAL_FMPI2C_ERROR_SIZE	Size Management error

FMPI2C Exported Macros

__HAL_FMPI2C_RESET_HANDLE_STATE

Description:

- Reset FMPI2C handle state.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.

Return value:

- None

__HAL_FMPI2C_ENABLE_IT

Description:

- Enable the specified FMPI2C interrupt.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.
- __INTERRUPT__: specifies the interrupt source to enable. This parameter can be one of the following values:
  - FMPI2C_IT_ERRI Errors interrupt enable
  - FMPI2C_IT_TCI Transfer complete interrupt enable
  - FMPI2C_IT_STOPI STOP detection interrupt enable
  - FMPI2C_IT_NACKI NACK received interrupt enable
  - FMPI2C_IT_ADDRI Address match interrupt enable
  - FMPI2C_IT_RXI RX interrupt enable
  - FMPI2C_IT_TXI TX interrupt enable

Return value:

- None

__HAL_FMPI2C_DISABLE_IT

Description:

- Disable the specified FMPI2C interrupt.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.
- __INTERRUPT__: specifies the interrupt source to disable. This parameter can be one of the following values:
  - FMPI2C_IT_ERRI Errors interrupt enable
  - FMPI2C_IT_TCI Transfer complete interrupt enable
  - FMPI2C_IT_STOPI STOP detection interrupt enable
  - FMPI2C_IT_NACKI NACK received interrupt enable
  - FMPI2C_IT_ADDRI Address match interrupt enable
  - FMPI2C_IT_RXI RX interrupt enable
  - FMPI2C_IT_TXI TX interrupt enable

Return value:

- None

__HAL_FMPI2C_GET_IT_SOURCE

Description:

- Check whether the specified FMPI2C interrupt source is enabled or not.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.
- __INTERRUPT__: specifies the FMPI2C interrupt source to check. This parameter can be one of the following values:
  - FMPI2C_IT_ERRI Errors interrupt enable
  - FMPI2C_IT_TCI Transfer complete interrupt enable
  - FMPI2C_IT_STOPI STOP detection interrupt enable
  - FMPI2C_IT_NACKI NACK received interrupt enable
  - FMPI2C_IT_ADDRI Address match interrupt enable
  - FMPI2C_IT_RXI RX interrupt

enable
- FMPI2C_IT_TXI TX interrupt enable

Return value:

- The: new state of __INTERRUPT__ (SET or RESET).

__HAL_FMPI2C_GET_FLAG

Description:

- Check whether the specified FMPI2C flag is set or not.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - FMPI2C_FLAG_TXE Transmit data register empty
  - FMPI2C_FLAG_TXIS Transmit interrupt status
  - FMPI2C_FLAG_RXNE Receive data register not empty
  - FMPI2C_FLAG_ADDR Address matched (slave mode)
  - FMPI2C_FLAG_AF Acknowledge failure received flag
  - FMPI2C_FLAG_STOPF STOP detection flag
  - FMPI2C_FLAG_TC Transfer complete (master mode)
  - FMPI2C_FLAG_TCR Transfer complete reload
  - FMPI2C_FLAG_BERR Bus error
  - FMPI2C_FLAG_ARLO Arbitration lost
  - FMPI2C_FLAG_OVR Overrun/Underrun
  - FMPI2C_FLAG_PECERR PEC error in reception
  - FMPI2C_FLAG_TIMEOUT Timeout or Tlow detection flag
  - FMPI2C_FLAG_ALERT SMBus alert
  - FMPI2C_FLAG_BUSY Bus busy
  - FMPI2C_FLAG_DIR Transfer direction (slave mode)

Return value:

- The: new state of __FLAG__ (SET or RESET).

__HAL_FMPI2C_CLEAR_FLAG

Description:

- Clear the FMPI2C pending flags which are cleared by writing 1 in a specific bit.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - FMPI2C_FLAG_TXE Transmit data register empty
  - FMPI2C_FLAG_ADDR Address matched (slave mode)
  - FMPI2C_FLAG_AF Acknowledge failure received flag
  - FMPI2C_FLAG_STOPF STOP detection flag
  - FMPI2C_FLAG_BERR Bus error
  - FMPI2C_FLAG_ARLO Arbitration lost
  - FMPI2C_FLAG_OVR Overrun/Underrun
  - FMPI2C_FLAG_PECERR PEC error in reception
  - FMPI2C_FLAG_TIMEOUT Timeout or Tlow detection flag
  - FMPI2C_FLAG_ALERT SMBus alert

Return value:

- None

__HAL_FMPI2C_ENABLE

Description:

- Enable the specified FMPI2C peripheral.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.

Return value:

- None

__HAL_FMPI2C_DISABLE

Description:

- Disable the specified FMPI2C peripheral.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.

Return value:

- None

__HAL_FMPI2C_GENERATE_NACK

Description:

- Generate a Non-Acknowledge FMPI2C peripheral in Slave mode.

Parameters:

- __HANDLE__: specifies the FMPI2C Handle.

Return value:

- None

FMPI2C Flag definition

FMPI2C_FLAG_TXE

FMPI2C_FLAG_TXIS

FMPI2C_FLAG_RXNE

FMPI2C_FLAG_ADDR

FMPI2C_FLAG_AF

FMPI2C_FLAG_STOPF

FMPI2C_FLAG_TC

FMPI2C_FLAG_TCR

FMPI2C_FLAG_BERR

FMPI2C_FLAG_ARLO

FMPI2C_FLAG_OVR

FMPI2C_FLAG_PECERR

FMPI2C_FLAG_TIMEOUT

FMPI2C_FLAG_ALERT

FMPI2C_FLAG_BUSY

FMPI2C_FLAG_DIR

FMPI2C General Call Addressing Mode

FMPI2C_GENERALCALL_DISABLE

FMPI2C_GENERALCALL_ENABLE

FMPI2C Interrupt configuration definition

FMPI2C_IT_ERRI

FMPI2C_IT_TCI

FMPI2C_IT_STOPI

FMPI2C_IT_NACKI

FMPI2C_IT_ADDRI

FMPI2C_IT_RXI

FMPI2C_IT_TXI

FMPI2C Memory Address Size

FMPI2C_MEMADD_SIZE_8BIT

FMPI2C_MEMADD_SIZE_16BIT

FMPI2C No-Stretch Mode

FMPI2C_NOSTRETCH_DISABLE

FMPI2C_NOSTRETCH_ENABLE

FMPI2C Own Address2 Masks

FMPI2C_OA2_NOMASK

FMPI2C_OA2_MASK01

FMPI2C_OA2_MASK02

FMPI2C_OA2_MASK03

FMPI2C_OA2_MASK04

FMPI2C_OA2_MASK05

FMPI2C_OA2_MASK06

FMPI2C_OA2_MASK07

FMPI2C Reload End Mode

FMPI2C_RELOAD_MODE

FMPI2C_AUTOEND_MODE

FMPI2C_SOFTEND_MODE

FMPI2C Start or Stop Mode

FMPI2C_NO_STARTSTOP

FMPI2C_GENERATE_STOP

FMPI2C_GENERATE_START_READ

FMPI2C_GENERATE_START_WRITE

FMPI2C Transfer Direction

FMPI2C_DIRECTION_RECEIVE

FMPI2C_DIRECTION_TRANSMIT

FMPI2C Sequential Transfer Options

FMPI2C_NO_OPTION_FRAME

FMPI2C_FIRST_FRAME

FMPI2C_NEXT_FRAME

FMPI2C_FIRST_AND_LAST_FRAME

FMPI2C_LAST_FRAME

26 HAL FMPI2C Extension Driver

26.1 FMPI2CEx Firmware driver API description

26.1.1 FMPI2C peripheral Extended features

Comparing to other previous devices, the FMPI2C interface for STM32F4xx devices contains the following additional features

- Possibility to disable or enable Analog Noise Filter
- Use of a configured Digital Noise Filter
- Disable or enable wakeup from Stop mode

26.1.2 How to use this driver

This driver provides functions to configure Noise Filter and Wake Up Feature

1. Configure FMPI2C Analog noise filter using the function HAL_FMPI2CEx_ConfigAnalogFilter()
2. Configure FMPI2C Digital noise filter using the function HAL_FMPI2CEx_ConfigDigitalFilter()
3. Configure the enable or disable of FMPI2C Wake Up Mode using the functions :
   – HAL_FMPI2CEx_EnableWakeUp()
   – HAL_FMPI2CEx_DisableWakeUp()
4. Configure the enable or disable of fast mode plus driving capability using the functions :
   – HAL_FMPI2CEx_EnableFastModePlus()
   – HAL_FMPI2CEx_DisbleFastModePlus()

26.1.3 Extended features functions

This section provides functions allowing to:

- Configure Noise Filters
- Configure Wake Up Feature

This section contains the following APIs:

- ***HAL_FMPI2CEx_ConfigAnalogFilter()***
- ***HAL_FMPI2CEx_ConfigDigitalFilter()***
- ***HAL_FMPI2CEx_EnableFastModePlus()***
- ***HAL_FMPI2CEx_DisableFastModePlus()***

26.1.4 Detailed description of functions

HAL_FMPI2CEx_ConfigAnalogFilter

Function Name	**HAL_StatusTypeDef HAL_FMPI2CEx_ConfigAnalogFilter (FMPI2C_HandleTypeDef * hfmpi2c, uint32_t AnalogFilter)**
Function Description	Configure FMPI2C Analog noise filter.
Parameters	• **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2Cx peripheral. • **AnalogFilter:** New state of the Analog filter.

Return values
- **HAL:** status

HAL_FMPI2CEx_ConfigDigitalFilter

Function Name **HAL_StatusTypeDef HAL_FMPI2CEx_ConfigDigitalFilter (FMPI2C_HandleTypeDef * hfmpi2c, uint32_t DigitalFilter)**

Function Description Configure FMPI2C Digital noise filter.

Parameters
- **hfmpi2c:** Pointer to a FMPI2C_HandleTypeDef structure that contains the configuration information for the specified FMPI2Cx peripheral.
- **DigitalFilter:** Coefficient of digital noise filter between 0x00 and 0x0F.

Return values
- **HAL:** status

HAL_FMPI2CEx_EnableFastModePlus

Function Name **void HAL_FMPI2CEx_EnableFastModePlus (uint32_t ConfigFastModePlus)**

Function Description Enable the FMPI2C fast mode plus driving capability.

Parameters
- **ConfigFastModePlus:** Selects the pin. This parameter can be one of the FMPI2C Extended Fast Mode Plus values

Return values
- **None**

HAL_FMPI2CEx_DisableFastModePlus

Function Name **void HAL_FMPI2CEx_DisableFastModePlus (uint32_t ConfigFastModePlus)**

Function Description Disable the FMPI2C fast mode plus driving capability.

Parameters
- **ConfigFastModePlus:** Selects the pin. This parameter can be one of the FMPI2C Extended Fast Mode Plus values

Return values
- **None**

26.2 FMPI2CEx Firmware driver defines

26.2.1 FMPI2CEx

FMPI2C Extended Analog Filter

FMPI2C_ANALOGFILTER_ENABLE

FMPI2C_ANALOGFILTER_DISABLE

FMPI2C Extended Fast Mode Plus

FMPI2C_FASTMODEPLUS_SCL Enable Fast Mode Plus on FMPI2C1 SCL pins

FMPI2C_FASTMODEPLUS_SDA Enable Fast Mode Plus on FMPI2C1 SDA pins

27 HAL GPIO Generic Driver

27.1 GPIO Firmware driver registers structures

27.1.1 GPIO_InitTypeDef

Data Fields

- ***uint32_t Pin***
- ***uint32_t Mode***
- ***uint32_t Pull***
- ***uint32_t Speed***
- ***uint32_t Alternate***

Field Documentation

- ***uint32_t GPIO_InitTypeDef::Pin***
  Specifies the GPIO pins to be configured. This parameter can be any value of ***GPIO_pins_define***
- ***uint32_t GPIO_InitTypeDef::Mode***
  Specifies the operating mode for the selected pins. This parameter can be a value of ***GPIO_mode_define***
- ***uint32_t GPIO_InitTypeDef::Pull***
  Specifies the Pull-up or Pull-Down activation for the selected pins. This parameter can be a value of ***GPIO_pull_define***
- ***uint32_t GPIO_InitTypeDef::Speed***
  Specifies the speed for the selected pins. This parameter can be a value of ***GPIO_speed_define***
- ***uint32_t GPIO_InitTypeDef::Alternate***
  Peripheral to be connected to the selected pins. This parameter can be a value of ***GPIO_Alternate_function_selection***

27.2 GPIO Firmware driver API description

27.2.1 GPIO Peripheral features

Subject to the specific hardware characteristics of each I/O port listed in the datasheet, each port bit of the General Purpose IO (GPIO) Ports, can be individually configured by software in several modes:

- Input mode
- Analog mode
- Output mode
- Alternate function mode
- External interrupt/event lines

During and just after reset, the alternate functions and external interrupt lines are not active and the I/O ports are configured in input floating mode.

All GPIO pins have weak internal pull-up and pull-down resistors, which can be activated or not.

In Output or Alternate mode, each IO can be configured on open-drain or push-pull type and the IO speed can be selected depending on the VDD value.

All ports have external interrupt/event capability. To use external interrupt lines, the port must be configured in input mode. All available GPIO pins are connected to the 16 external interrupt/event lines from EXTI0 to EXTI15.

The external interrupt/event controller consists of up to 23 edge detectors (16 lines are connected to GPIO) for generating event/interrupt requests (each input line can be independently configured to select the type (interrupt or event) and the corresponding trigger event (rising or falling or both). Each line can also be masked independently.

27.2.2 How to use this driver

1. Enable the GPIO AHB clock using the following function: __HAL_RCC_GPIOx_CLK_ENABLE().
2. Configure the GPIO pin(s) using HAL_GPIO_Init().
   - Configure the IO mode using "Mode" member from GPIO_InitTypeDef structure
   - Activate Pull-up, Pull-down resistor using "Pull" member from GPIO_InitTypeDef structure.
   - In case of Output or alternate function mode selection: the speed is configured through "Speed" member from GPIO_InitTypeDef structure.
   - In alternate mode is selection, the alternate function connected to the IO is configured through "Alternate" member from GPIO_InitTypeDef structure.
   - Analog mode is required when a pin is to be used as ADC channel or DAC output.
   - In case of external interrupt/event selection the "Mode" member from GPIO_InitTypeDef structure select the type (interrupt or event) and the corresponding trigger event (rising or falling or both).
3. In case of external interrupt/event mode selection, configure NVIC IRQ priority mapped to the EXTI line using HAL_NVIC_SetPriority() and enable it using HAL_NVIC_EnableIRQ().
4. To get the level of a pin configured in input mode use HAL_GPIO_ReadPin().
5. To set/reset the level of a pin configured in output mode use HAL_GPIO_WritePin()/HAL_GPIO_TogglePin().
6. To lock pin configuration until next reset use HAL_GPIO_LockPin().
7. During and just after reset, the alternate functions are not active and the GPIO pins are configured in input floating mode (except JTAG pins).
8. The LSE oscillator pins OSC32_IN and OSC32_OUT can be used as general purpose (PC14 and PC15, respectively) when the LSE oscillator is off. The LSE has priority over the GPIO function.
9. The HSE oscillator pins OSC_IN/OSC_OUT can be used as general purpose PH0 and PH1, respectively, when the HSE oscillator is off. The HSE has priority over the GPIO function.

27.2.3 Initialization and de-initialization functions

This section provides functions allowing to initialize and de-initialize the GPIOs to be ready for use.

This section contains the following APIs:

- ***HAL_GPIO_Init()***
- ***HAL_GPIO_DeInit()***

27.2.4 IO operation functions

This section contains the following APIs:

- ***HAL_GPIO_ReadPin()***
- ***HAL_GPIO_WritePin()***
- ***HAL_GPIO_TogglePin()***
- ***HAL_GPIO_LockPin()***
- ***HAL_GPIO_EXTI_IRQHandler()***
- ***HAL_GPIO_EXTI_Callback()***

27.2.5 Detailed description of functions

HAL_GPIO_Init

Function Name	**void HAL_GPIO_Init (GPIO_TypeDef * GPIOx, GPIO_InitTypeDef * GPIO_Init)**
Function Description	Initializes the GPIOx peripheral according to the specified parameters in the GPIO_Init.
Parameters	• **GPIOx:** where x can be (A..K) to select the GPIO peripheral for STM32F429X device or x can be (A..I) to select the GPIO peripheral for STM32F40XX and STM32F427X devices. • **GPIO_Init:** pointer to a GPIO_InitTypeDef structure that contains the configuration information for the specified GPIO peripheral.
Return values	• **None**

HAL_GPIO_DeInit

Function Name	**void HAL_GPIO_DeInit (GPIO_TypeDef * GPIOx, uint32_t GPIO_Pin)**
Function Description	De-initializes the GPIOx peripheral registers to their default reset values.
Parameters	• **GPIOx:** where x can be (A..K) to select the GPIO peripheral for STM32F429X device or x can be (A..I) to select the GPIO peripheral for STM32F40XX and STM32F427X devices. • **GPIO_Pin:** specifies the port bit to be written. This parameter can be one of GPIO_PIN_x where x can be (0..15).
Return values	• **None**

HAL_GPIO_ReadPin

Function Name	**GPIO_PinState HAL_GPIO_ReadPin (GPIO_TypeDef * GPIOx, uint16_t GPIO_Pin)**
Function Description	Reads the specified input port pin.
Parameters	• **GPIOx:** where x can be (A..K) to select the GPIO peripheral for STM32F429X device or x can be (A..I) to select the GPIO peripheral for STM32F40XX and STM32F427X devices. • **GPIO_Pin:** specifies the port bit to read. This parameter can be GPIO_PIN_x where x can be (0..15).

Return values
- **The:** input port pin value.

HAL_GPIO_WritePin

Function Name **void HAL_GPIO_WritePin (GPIO_TypeDef * GPIOx, uint16_t GPIO_Pin, GPIO_PinState PinState)**

Function Description Sets or clears the selected data port bit.

Parameters
- **GPIOx:** where x can be (A..K) to select the GPIO peripheral for STM32F429X device or x can be (A..I) to select the GPIO peripheral for STM32F40XX and STM32F427X devices.
- **GPIO_Pin:** specifies the port bit to be written. This parameter can be one of GPIO_PIN_x where x can be (0..15).
- **PinState:** specifies the value to be written to the selected bit. This parameter can be one of the GPIO_PinState enum values:
  – GPIO_PIN_RESET: to clear the port pin
  – GPIO_PIN_SET: to set the port pin

Return values
- **None**

Notes
- This function uses GPIOx_BSRR register to allow atomic read/modify accesses. In this way, there is no risk of an IRQ occurring between the read and the modify access.

HAL_GPIO_TogglePin

Function Name **void HAL_GPIO_TogglePin (GPIO_TypeDef * GPIOx, uint16_t GPIO_Pin)**

Function Description Toggles the specified GPIO pins.

Parameters
- **GPIOx:** Where x can be (A..K) to select the GPIO peripheral for STM32F429X device or x can be (A..I) to select the GPIO peripheral for STM32F40XX and STM32F427X devices.
- **GPIO_Pin:** Specifies the pins to be toggled.

Return values
- **None**

HAL_GPIO_LockPin

Function Name **HAL_StatusTypeDef HAL_GPIO_LockPin (GPIO_TypeDef * GPIOx, uint16_t GPIO_Pin)**

Function Description Locks GPIO Pins configuration registers.

Parameters
- **GPIOx:** where x can be (A..F) to select the GPIO peripheral for STM32F4 family
- **GPIO_Pin:** specifies the port bit to be locked. This parameter can be any combination of GPIO_PIN_x where x can be (0..15).

Return values
- **None**

Notes
- The locked registers are GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR, GPIOx_AFRL and GPIOx_AFRH.
- The configuration of the locked GPIO pins can no longer be

modified until the next reset.

HAL_GPIO_EXTI_IRQHandler

Function Name	**void HAL_GPIO_EXTI_IRQHandler (uint16_t GPIO_Pin)**
Function Description	This function handles EXTI interrupt request.
Parameters	• **GPIO_Pin:** Specifies the pins connected EXTI line
Return values	• **None**

HAL_GPIO_EXTI_Callback

Function Name	**void HAL_GPIO_EXTI_Callback (uint16_t GPIO_Pin)**
Function Description	EXTI line detection callbacks.
Parameters	• **GPIO_Pin:** Specifies the pins connected EXTI line
Return values	• **None**

27.3 GPIO Firmware driver defines

27.3.1 GPIO

GPIO Alternate Function Selection

GPIO_AF0_RTC_50Hz

GPIO_AF0_MCO

GPIO_AF0_TAMPER

GPIO_AF0_SWJ

GPIO_AF0_TRACE

GPIO_AF1_TIM1

GPIO_AF1_TIM2

GPIO_AF2_TIM3

GPIO_AF2_TIM4

GPIO_AF2_TIM5

GPIO_AF3_TIM8

GPIO_AF3_TIM9

GPIO_AF3_TIM10

GPIO_AF3_TIM11

GPIO_AF4_I2C1

GPIO_AF4_I2C2

GPIO_AF4_I2C3

GPIO_AF5_SPI1

GPIO_AF5_SPI2

GPIO_AF5_SPI3
GPIO_AF5_SPI4
GPIO_AF5_SPI5
GPIO_AF5_SPI6
GPIO_AF5_I2S3ext
GPIO_AF6_SPI3
GPIO_AF6_I2S2ext
GPIO_AF6_SAI1
GPIO_AF7_USART1
GPIO_AF7_USART2
GPIO_AF7_USART3
GPIO_AF7_I2S3ext
GPIO_AF8_UART4
GPIO_AF8_UART5
GPIO_AF8_USART6
GPIO_AF8_UART7
GPIO_AF8_UART8
GPIO_AF9_CAN1
GPIO_AF9_CAN2
GPIO_AF9_TIM12
GPIO_AF9_TIM13
GPIO_AF9_TIM14
GPIO_AF9_LTDC
GPIO_AF9_QSPI
GPIO_AF10_OTG_FS
GPIO_AF10_OTG_HS
GPIO_AF10_QSPI
GPIO_AF11_ETH
GPIO_AF12_FMC
GPIO_AF12_OTG_HS_FS
GPIO_AF12_SDIO
GPIO_AF13_DCMI
GPIO_AF13_DSI
GPIO_AF14_LTDC
GPIO_AF15_EVENTOUT

GPIO Exported Macros

__HAL_GPIO_EXTI_GET_FLAG

Description:

- Checks whether the specified EXTI line flag is set or not.

Parameters:

- __EXTI_LINE__: specifies the EXTI line flag to check. This parameter can be GPIO_PIN_x where x can be(0..15)

Return value:

- The: new state of __EXTI_LINE__ (SET or RESET).

__HAL_GPIO_EXTI_CLEAR_FLAG

Description:

- Clears the EXTI's line pending flags.

Parameters:

- __EXTI_LINE__: specifies the EXTI lines flags to clear. This parameter can be any combination of GPIO_PIN_x where x can be (0..15)

Return value:

- None

__HAL_GPIO_EXTI_GET_IT

Description:

- Checks whether the specified EXTI line is asserted or not.

Parameters:

- __EXTI_LINE__: specifies the EXTI line to check. This parameter can be GPIO_PIN_x where x can be(0..15)

Return value:

- The: new state of __EXTI_LINE__ (SET or RESET).

__HAL_GPIO_EXTI_CLEAR_IT

Description:

- Clears the EXTI's line pending bits.

Parameters:

- __EXTI_LINE__: specifies the EXTI lines to clear. This parameter can be any combination of GPIO_PIN_x where x can be (0..15)

Return value:

- None

__HAL_GPIO_EXTI_GENERATE_SWIT

Description:

- Generates a Software interrupt on selected

EXTI line.

Parameters:

- __EXTI_LINE__: specifies the EXTI line to check. This parameter can be GPIO_PIN_x where x can be(0..15)

Return value:

- None

GPIO mode define

GPIO_MODE_INPUT	Input Floating Mode
GPIO_MODE_OUTPUT_PP	Output Push Pull Mode
GPIO_MODE_OUTPUT_OD	Output Open Drain Mode
GPIO_MODE_AF_PP	Alternate Function Push Pull Mode
GPIO_MODE_AF_OD	Alternate Function Open Drain Mode
GPIO_MODE_ANALOG	Analog Mode
GPIO_MODE_IT_RISING	External Interrupt Mode with Rising edge trigger detection
GPIO_MODE_IT_FALLING	External Interrupt Mode with Falling edge trigger detection
GPIO_MODE_IT_RISING_FALLING	External Interrupt Mode with Rising/Falling edge trigger detection
GPIO_MODE_EVT_RISING	External Event Mode with Rising edge trigger detection
GPIO_MODE_EVT_FALLING	External Event Mode with Falling edge trigger detection
GPIO_MODE_EVT_RISING_FALLING	External Event Mode with Rising/Falling edge trigger detection

GPIO pins define

GPIO_PIN_0

GPIO_PIN_1

GPIO_PIN_2

GPIO_PIN_3

GPIO_PIN_4

GPIO_PIN_5

GPIO_PIN_6

GPIO_PIN_7

GPIO_PIN_8

GPIO_PIN_9

GPIO_PIN_10

GPIO_PIN_11

GPIO_PIN_12

GPIO_PIN_13

GPIO_PIN_14

GPIO_PIN_15

GPIO_PIN_All

GPIO_PIN_MASK

GPIO pull define

GPIO_NOPULL	No Pull-up or Pull-down activation
GPIO_PULLUP	Pull-up activation
GPIO_PULLDOWN	Pull-down activation

GPIO speed define

GPIO_SPEED_FREQ_LOW	IO works at 2 MHz, please refer to the product datasheet
GPIO_SPEED_FREQ_MEDIUM	range 12,5 MHz to 50 MHz, please refer to the product datasheet
GPIO_SPEED_FREQ_HIGH	range 25 MHz to 100 MHz, please refer to the product datasheet
GPIO_SPEED_FREQ_VERY_HIGH	range 50 MHz to 200 MHz, please refer to the product datasheet

28 HAL GPIO Extension Driver

28.1 GPIOEx Firmware driver defines

28.1.1 GPIOEx

GPIO Get Port Index

GPIO_GET_INDEX

GPIO Check Alternate Function

IS_GPIO_AF

29 HAL HASH Generic Driver

29.1 HASH Firmware driver registers structures

29.1.1 HASH_InitTypeDef

Data Fields

- ***uint32_t DataType***
- ***uint32_t KeySize***
- ***uint8_t * pKey***

Field Documentation

- ***uint32_t HASH_InitTypeDef::DataType***
  32-bit data, 16-bit data, 8-bit data or 1-bit string. This parameter can be a value of ***HASH_Data_Type***
- ***uint32_t HASH_InitTypeDef::KeySize***
  The key size is used only in HMAC operation
- ***uint8_t* HASH_InitTypeDef::pKey***
  The key is used only in HMAC operation

29.1.2 HASH_HandleTypeDef

Data Fields

- ***HASH_InitTypeDef Init***
- ***uint8_t * pHashInBuffPtr***
- ***uint8_t * pHashOutBuffPtr***
- ***__IO uint32_t HashBuffSize***
- ***__IO uint32_t HashInCount***
- ***__IO uint32_t HashITCounter***
- ***HAL_StatusTypeDef Status***
- ***HAL_HASH_PhaseTypeDef Phase***
- ***DMA_HandleTypeDef * hdmain***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_HASH_StateTypeDef State***

Field Documentation

- ***HASH_InitTypeDef HASH_HandleTypeDef::Init***
  HASH required parameters
- ***uint8_t* HASH_HandleTypeDef::pHashInBuffPtr***
  Pointer to input buffer
- ***uint8_t* HASH_HandleTypeDef::pHashOutBuffPtr***
  Pointer to input buffer

- ***__IO uint32_t HASH_HandleTypeDef::HashBuffSize***
  Size of buffer to be processed
- ***__IO uint32_t HASH_HandleTypeDef::HashInCount***
  Counter of inputed data
- ***__IO uint32_t HASH_HandleTypeDef::HashITCounter***
  Counter of issued interrupts
- ***HAL_StatusTypeDef HASH_HandleTypeDef::Status***
  HASH peripheral status
- ***HAL_HASH_PhaseTypeDef HASH_HandleTypeDef::Phase***
  HASH peripheral phase
- ***DMA_HandleTypeDef* HASH_HandleTypeDef::hdmain***
  HASH In DMA handle parameters
- ***HAL_LockTypeDef HASH_HandleTypeDef::Lock***
  HASH locking object
- ***__IO HAL_HASH_StateTypeDef HASH_HandleTypeDef::State***
  HASH peripheral state

29.2 HASH Firmware driver API description

29.2.1 How to use this driver

The HASH HAL driver can be used as follows:

1. Initialize the HASH low level resources by implementing the HAL_HASH_MspInit():
   a. Enable the HASH interface clock using __HAL_RCC_HASH_CLK_ENABLE()
   b. In case of using processing APIs based on interrupts (e.g. HAL_HMAC_SHA1_Start_IT())
      – Configure the HASH interrupt priority using HAL_NVIC_SetPriority()
      – Enable the HASH IRQ handler using HAL_NVIC_EnableIRQ()
      – In HASH IRQ handler, call HAL_HASH_IRQHandler()
   c. In case of using DMA to control data transfer (e.g. HAL_HMAC_SHA1_Start_DMA())
      – Enable the DMAx interface clock using __DMAx_CLK_ENABLE()
      – Configure and enable one DMA stream one for managing data transfer from memory to peripheral (input stream). Managing data transfer from peripheral to memory can be performed only using CPU
      – Associate the initialized DMA handle to the HASH DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Stream using HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ()
2. Initialize the HASH HAL using HAL_HASH_Init(). This function configures mainly:
   a. The data type: 1-bit, 8-bit, 16-bit and 32-bit.
   b. For HMAC, the encryption key.
   c. For HMAC, the key size used for encryption.
3. Three processing functions are available:
   a. Polling mode: processing APIs are blocking functions i.e. they process the data and wait till the digest computation is finished e.g. HAL_HASH_SHA1_Start()
   b. Interrupt mode: encryption and decryption APIs are not blocking functions i.e. they process the data under interrupt e.g. HAL_HASH_SHA1_Start_IT()
   c. DMA mode: processing APIs are not blocking functions and the CPU is not used for data transfer i.e. the data transfer is ensured by DMA e.g. HAL_HASH_SHA1_Start_DMA()

4. When the processing function is called at first time after HAL_HASH_Init() the HASH peripheral is initialized and processes the buffer in input. After that, the digest computation is started. When processing multi-buffer use the accumulate function to write the data in the peripheral without starting the digest computation. In last buffer use the start function to input the last buffer ans start the digest computation.
   a. e.g. HAL_HASH_SHA1_Accumulate() : write 1st data buffer in the peripheral without starting the digest computation
   b. write (n-1)th data buffer in the peripheral without starting the digest computation
   c. HAL_HASH_SHA1_Start() : write (n)th data buffer in the peripheral and start the digest computation
5. In HMAC mode, there is no Accumulate API. Only Start API is available.
6. In case of using DMA, call the DMA start processing e.g. HAL_HASH_SHA1_Start_DMA(). After that, call the finish function in order to get the digest value e.g. HAL_HASH_SHA1_Finish()
7. Call HAL_HASH_DeInit() to deinitialize the HASH peripheral.

29.2.2 HASH processing using polling mode functions

This section provides functions allowing to calculate in polling mode the hash value using one of the following algorithms:

- MD5
- SHA1

This section contains the following APIs:

- ***HAL_HASH_MD5_Start()***
- ***HAL_HASH_MD5_Accumulate()***
- ***HAL_HASH_SHA1_Start()***
- ***HAL_HASH_SHA1_Accumulate()***

29.2.3 HASH processing using interrupt mode functions

This section provides functions allowing to calculate in interrupt mode the hash value using one of the following algorithms:

- MD5
- SHA1

This section contains the following APIs:

- ***HAL_HASH_MD5_Start_IT()***
- ***HAL_HASH_SHA1_Start_IT()***
- ***HAL_HASH_IRQHandler()***
- ***HAL_HMAC_SHA1_Start()***
- ***HAL_HMAC_MD5_Start()***

29.2.4 HASH processing using DMA mode functions

This section provides functions allowing to calculate in DMA mode the hash value using one of the following algorithms:

- MD5
- SHA1

This section contains the following APIs:

- ***HAL_HASH_MD5_Start_DMA()***
- ***HAL_HASH_MD5_Finish()***
- ***HAL_HASH_SHA1_Start_DMA()***

- ***HAL_HASH_SHA1_Finish()***
- ***HAL_HASH_SHA1_Start_IT()***
- ***HAL_HASH_MD5_Start_IT()***

29.2.5 HMAC processing using polling mode functions

This section provides functions allowing to calculate in polling mode the HMAC value using one of the following algorithms:

- MD5
- SHA1

This section contains the following APIs:

- ***HAL_HMAC_MD5_Start()***
- ***HAL_HMAC_SHA1_Start()***
- ***HAL_HASH_SHA1_Start_DMA()***
- ***HAL_HASH_SHA1_Finish()***
- ***HAL_HASH_MD5_Start_DMA()***
- ***HAL_HASH_MD5_Finish()***

29.2.6 HMAC processing using DMA mode functions

This section provides functions allowing to calculate in DMA mode the HMAC value using one of the following algorithms:

- MD5
- SHA1

This section contains the following APIs:

- ***HAL_HMAC_MD5_Start_DMA()***
- ***HAL_HMAC_SHA1_Start_DMA()***

29.2.7 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral.

This section contains the following APIs:

- ***HAL_HASH_GetState()***
- ***HAL_HASH_IRQHandler()***

29.2.8 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the HASH according to the specified parameters in the HASH_InitTypeDef and creates the associated handle.
- DeInitialize the HASH peripheral.
- Initialize the HASH MSP.
- DeInitialize HASH MSP.

This section contains the following APIs:

- ***HAL_HASH_Init()***
- ***HAL_HASH_DeInit()***
- ***HAL_HASH_MspInit()***
- ***HAL_HASH_MspDeInit()***
- ***HAL_HASH_InCpltCallback()***
- ***HAL_HASH_ErrorCallback()***

- ***HAL_HASH_DgstCpltCallback()***

29.2.9 Detailed description of functions

HAL_HASH_Init

Function Name	**HAL_StatusTypeDef HAL_HASH_Init (HASH_HandleTypeDef * hhash)**
Function Description	Initializes the HASH according to the specified parameters in the HASH_HandleTypeDef and creates the associated handle.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **HAL:** status

HAL_HASH_DeInit

Function Name	**HAL_StatusTypeDef HAL_HASH_DeInit (HASH_HandleTypeDef * hhash)**
Function Description	DeInitializes the HASH peripheral.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **HAL:** status
Notes	• This API must be called before starting a new processing.

HAL_HASH_SHA1_Start

Function Name	**HAL_StatusTypeDef HAL_HASH_SHA1_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in SHA1 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes. • **Timeout:** Timeout value
Return values	• **HAL:** status

HAL_HASH_MD5_Start

Function Name	**HAL_StatusTypeDef HAL_HASH_MD5_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in MD5 mode then processes pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is multiple of 64 bytes, appending the input buffer is possible. If the Size is not multiple of 64 bytes, the padding is managed by hardware and appending the input buffer is no more possible.
- **pOutBuffer:** Pointer to the computed digest. Its size must be 16 bytes.
- **Timeout:** Timeout value

Return values
- **HAL:** status

HAL_HASH_MD5_Accumulate

Function Name **HAL_StatusTypeDef HAL_HASH_MD5_Accumulate (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in MD5 mode then writes the pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is multiple of 64 bytes, appending the input buffer is possible. If the Size is not multiple of 64 bytes, the padding is managed by hardware and appending the input buffer is no more possible.

Return values
- **HAL:** status

HAL_HASH_SHA1_Accumulate

Function Name **HAL_StatusTypeDef HAL_HASH_SHA1_Accumulate (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in SHA1 mode then processes pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

Notes
- Input buffer size in bytes must be a multiple of 4 otherwise the digest computation is corrupted.

HAL_HMAC_SHA1_Start

Function Name **HAL_StatusTypeDef HAL_HMAC_SHA1_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**

Function Description — Initializes the HASH peripheral in HMAC SHA1 mode then processes pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
- **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.
- **Timeout:** Timeout value

Return values
- **HAL:** status

HAL_HMAC_MD5_Start

Function Name — **HAL_StatusTypeDef HAL_HMAC_MD5_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**

Function Description — Initializes the HASH peripheral in HMAC MD5 mode then processes pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
- **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.
- **Timeout:** Timeout value

Return values
- **HAL:** status

HAL_HASH_SHA1_Start_IT

Function Name — **HAL_StatusTypeDef HAL_HASH_SHA1_Start_IT (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer)**

Function Description — Initializes the HASH peripheral in SHA1 mode then processes pInBuffer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
- **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.

Return values
- **HAL:** status

HAL_HASH_MD5_Start_IT

Function Name — **HAL_StatusTypeDef HAL_HASH_MD5_Start_IT (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t**

Size, uint8_t * pOutBuffer)

Function Description: Initializes the HASH peripheral in MD5 mode then processes pInBuffer.

Parameters:
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
- **pOutBuffer:** Pointer to the computed digest. Its size must be 16 bytes.

Return values:
- **HAL:** status

HAL_HASH_SHA1_Start_DMA

Function Name: **HAL_StatusTypeDef HAL_HASH_SHA1_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description: Initializes the HASH peripheral in SHA1 mode then enables DMA to control data transfer.

Parameters:
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values:
- **HAL:** status

HAL_HASH_SHA1_Finish

Function Name: **HAL_StatusTypeDef HAL_HASH_SHA1_Finish (HASH_HandleTypeDef * hhash, uint8_t * pOutBuffer, uint32_t Timeout)**

Function Description: Returns the computed digest in SHA1 mode.

Parameters:
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.
- **Timeout:** Timeout value

Return values:
- **HAL:** status

HAL_HASH_MD5_Start_DMA

Function Name: **HAL_StatusTypeDef HAL_HASH_MD5_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description: Initializes the HASH peripheral in MD5 mode then enables DMA to control data transfer.

Parameters:
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module

- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

HAL_HASH_MD5_Finish

Function Name **HAL_StatusTypeDef HAL_HASH_MD5_Finish (HASH_HandleTypeDef * hhash, uint8_t * pOutBuffer, uint32_t Timeout)**

Function Description Returns the computed digest in MD5 mode.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pOutBuffer:** Pointer to the computed digest. Its size must be 16 bytes.
- **Timeout:** Timeout value

Return values
- **HAL:** status

HAL_HMAC_SHA1_Start_DMA

Function Name **HAL_StatusTypeDef HAL_HMAC_SHA1_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in HMAC SHA1 mode then enables DMA to control data transfer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

HAL_HMAC_MD5_Start_DMA

Function Name **HAL_StatusTypeDef HAL_HMAC_MD5_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in HMAC MD5 mode then enables DMA to control data transfer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

HAL_HASH_IRQHandler

Function Name	**void HAL_HASH_IRQHandler (HASH_HandleTypeDef * hhash)**
Function Description	This function handles HASH interrupt request.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**

HAL_HASH_GetState

Function Name	**HAL_HASH_StateTypeDef HAL_HASH_GetState (HASH_HandleTypeDef * hhash)**
Function Description	return the HASH state
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **HAL:** state

HAL_HASH_MspInit

Function Name	**void HAL_HASH_MspInit (HASH_HandleTypeDef * hhash)**
Function Description	Initializes the HASH MSP.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**

HAL_HASH_MspDeInit

Function Name	**void HAL_HASH_MspDeInit (HASH_HandleTypeDef * hhash)**
Function Description	DeInitializes HASH MSP.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**

HAL_HASH_InCpltCallback

Function Name	**void HAL_HASH_InCpltCallback (HASH_HandleTypeDef * hhash)**
Function Description	Input data transfer complete callback.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**

HAL_HASH_DgstCpltCallback

Function Name	**void HAL_HASH_DgstCpltCallback (HASH_HandleTypeDef * hhash)**
Function Description	Digest computation complete callback.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**
Notes	• This callback is not relevant with DMA.

HAL_HASH_ErrorCallback

Function Name	**void HAL_HASH_ErrorCallback (HASH_HandleTypeDef * hhash)**
Function Description	Data transfer Error callback.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
Return values	• **None**

29.3 HASH Firmware driver defines

29.3.1 HASH

HASH Data Type

HASH_DATATYPE_32B	32-bit data. No swapping
HASH_DATATYPE_16B	16-bit data. Each half word is swapped
HASH_DATATYPE_8B	8-bit data. All bytes are swapped
HASH_DATATYPE_1B	1-bit data. In the word all bits are swapped

HASH Algorithm Selection

HASH_ALGOSELECTION_SHA1	HASH function is SHA1
HASH_ALGOSELECTION_SHA224	HASH function is SHA224
HASH_ALGOSELECTION_SHA256	HASH function is SHA256
HASH_ALGOSELECTION_MD5	HASH function is MD5

HASH Algorithm Mode

HASH_ALGOMODE_HASH	Algorithm is HASH
HASH_ALGOMODE_HMAC	Algorithm is HMAC

HASH HMAC Long key

HASH_HMAC_KEYTYPE_SHORTKEY	HMAC Key is <= 64 bytes
HASH_HMAC_KEYTYPE_LONGKEY	HMAC Key is> 64 bytes

HASH Flags definition

HASH_FLAG_DINIS	16 locations are free in the DIN : A new block can be entered into the input buffer
HASH_FLAG_DCIS	Digest calculation complete
HASH_FLAG_DMAS	DMA interface is enabled (DMAE=1) or a transfer is ongoing
HASH_FLAG_BUSY	The hash core is Busy : processing a block of data
HASH_FLAG_DINNE	DIN not empty : The input buffer contains at least one word of data

HASH Interrupts definition

HASH_IT_DINI	A new block can be entered into the input buffer (DIN)
HASH_IT_DCI	Digest calculation complete

HASH Exported Macros

__HAL_HASH_RESET_HANDLE_STATE

Description:

- Reset HASH handle state.

Parameters:

- __HANDLE__: specifies the HASH handle.

Return value:

- None

__HAL_HASH_GET_FLAG

Description:

- Check whether the specified HASH flag is set or not.

Parameters:

- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - HASH_FLAG_DINIS: A new block can be entered into the input buffer.
  - HASH_FLAG_DCIS: Digest calculation complete
  - HASH_FLAG_DMAS: DMA interface is enabled (DMAE=1) or a transfer is ongoing
  - HASH_FLAG_BUSY: The hash core is Busy : processing a block of data
  - HASH_FLAG_DINNE: DIN not empty : The input buffer contains at least one word of data

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_HASH_SET_MDMAT

Description:

- Enable the multiple DMA mode.

Return value:

- None

__HAL_HASH_RESET_MDMAT

Description:

- Disable the multiple DMA mode.

Return value:

- None

__HAL_HASH_START_DIGEST

Description:

- Start the digest computation.

Return value:

- None

__HAL_HASH_SET_NBVALIDBITS

Description:

- Set the number of valid bits in last word written in Data register.

Parameters:

- SIZE: size in byte of last data written in Data register.

Return value:

- None

30 HAL HASH Extension Driver

30.1 HASHEx Firmware driver API description

30.1.1 How to use this driver

The HASH HAL driver can be used as follows:

1. Initialize the HASH low level resources by implementing the HAL_HASH_MspInit():
   a. Enable the HASH interface clock using __HAL_RCC_HASH_CLK_ENABLE()
   b. In case of using processing APIs based on interrupts (e.g. HAL_HMACEx_SHA224_Start())
      – Configure the HASH interrupt priority using HAL_NVIC_SetPriority()
      – Enable the HASH IRQ handler using HAL_NVIC_EnableIRQ()
      – In HASH IRQ handler, call HAL_HASH_IRQHandler()
   c. In case of using DMA to control data transfer (e.g. HAL_HMACEx_SH224_Start_DMA())
      – Enable the DMAx interface clock using __DMAx_CLK_ENABLE()
      – Configure and enable one DMA stream one for managing data transfer from memory to peripheral (input stream). Managing data transfer from peripheral to memory can be performed only using CPU
      – Associate the initialized DMA handle to the HASH DMA handle using __HAL_LINKDMA()
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Stream: HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ()
2. Initialize the HASH HAL using HAL_HASH_Init(). This function configures mainly:
   a. The data type: 1-bit, 8-bit, 16-bit and 32-bit.
   b. For HMAC, the encryption key.
   c. For HMAC, the key size used for encryption.
3. Three processing functions are available:
   a. Polling mode: processing APIs are blocking functions i.e. they process the data and wait till the digest computation is finished e.g. HAL_HASHEx_SHA224_Start()
   b. Interrupt mode: encryption and decryption APIs are not blocking functions i.e. they process the data under interrupt e.g. HAL_HASHEx_SHA224_Start_IT()
   c. DMA mode: processing APIs are not blocking functions and the CPU is not used for data transfer i.e. the data transfer is ensured by DMA e.g. HAL_HASHEx_SHA224_Start_DMA()
4. When the processing function is called at first time after HAL_HASH_Init() the HASH peripheral is initialized and processes the buffer in input. After that, the digest computation is started. When processing multi-buffer use the accumulate function to write the data in the peripheral without starting the digest computation. In last buffer use the start function to input the last buffer ans start the digest computation.
   a. e.g. HAL_HASHEx_SHA224_Accumulate() : write 1st data buffer in the peripheral without starting the digest computation
   b. write (n-1)th data buffer in the peripheral without starting the digest computation
   c. HAL_HASHEx_SHA224_Start() : write (n)th data buffer in the peripheral and start the digest computation
5. In HMAC mode, there is no Accumulate API. Only Start API is available.
6. In case of using DMA, call the DMA start processing e.g. HAL_HASHEx_SHA224_Start_DMA(). After that, call the finish function in order to get the digest value e.g. HAL_HASHEx_SHA224_Finish()

7. Call HAL_HASH_DeInit() to deinitialize the HASH peripheral.

30.1.2 HASH processing using polling mode functions

This section provides functions allowing to calculate in polling mode the hash value using one of the following algorithms:

- SHA224
- SHA256

This section contains the following APIs:

- ***HAL_HASHEx_SHA224_Start()***
- ***HAL_HASHEx_SHA256_Start()***
- ***HAL_HASHEx_SHA224_Accumulate()***
- ***HAL_HASHEx_SHA256_Accumulate()***

30.1.3 HMAC processing using polling mode functions

This section provides functions allowing to calculate in polling mode the HMAC value using one of the following algorithms:

- SHA224
- SHA256

This section contains the following APIs:

- ***HAL_HMACEx_SHA224_Start()***
- ***HAL_HMACEx_SHA256_Start()***

30.1.4 HASH processing using interrupt functions

This section provides functions allowing to calculate in interrupt mode the hash value using one of the following algorithms:

- SHA224
- SHA256

This section contains the following APIs:

- ***HAL_HASHEx_SHA224_Start_IT()***
- ***HAL_HASHEx_SHA256_Start_IT()***
- ***HAL_HASHEx_IRQHandler()***

30.1.5 HASH processing using DMA functions

This section provides functions allowing to calculate in DMA mode the hash value using one of the following algorithms:

- SHA224
- SHA256

This section contains the following APIs:

- ***HAL_HASHEx_SHA224_Start_DMA()***
- ***HAL_HASHEx_SHA224_Finish()***
- ***HAL_HASHEx_SHA256_Start_DMA()***
- ***HAL_HASHEx_SHA256_Finish()***

30.1.6 HMAC processing using DMA functions

This section provides functions allowing to calculate in DMA mode the HMAC value using one of the following algorithms:

- SHA224
- SHA256

This section contains the following APIs:

- ***HAL_HMACEx_SHA224_Start_DMA()***
- ***HAL_HMACEx_SHA256_Start_DMA()***

30.1.7 Detailed description of functions

HAL_HASHEx_SHA224_Start

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA224_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in SHA224 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 28 bytes. • **Timeout:** Specify Timeout value
Return values	• **HAL:** status

HAL_HASHEx_SHA256_Start

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA256_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in SHA256 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 32 bytes. • **Timeout:** Specify Timeout value
Return values	• **HAL:** status

HAL_HASHEx_SHA224_Accumulate

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA224_Accumulate (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**
Function Description	Initializes the HASH peripheral in SHA224 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
Return values	• **HAL:** status

HAL_HASHEx_SHA256_Accumulate

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA256_Accumulate (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**
Function Description	Initializes the HASH peripheral in SHA256 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
Return values	• **HAL:** status

HAL_HMACEx_SHA224_Start

Function Name	**HAL_StatusTypeDef HAL_HMACEx_SHA224_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in HMAC SHA224 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes. • **Timeout:** Timeout value
Return values	• **HAL:** status

HAL_HMACEx_SHA256_Start

Function Name	**HAL_StatusTypeDef HAL_HMACEx_SHA256_Start (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Initializes the HASH peripheral in HMAC SHA256 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes. • **Timeout:** Timeout value
Return values	• **HAL:** status

HAL_HASHEx_SHA224_Start_IT

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA224_Start_IT (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer)**
Function Description	Initializes the HASH peripheral in SHA224 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.
Return values	• **HAL:** status

HAL_HASHEx_SHA256_Start_IT

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA256_Start_IT (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size, uint8_t * pOutBuffer)**
Function Description	Initializes the HASH peripheral in SHA256 mode then processes pInBuffer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware. • **pOutBuffer:** Pointer to the computed digest. Its size must be 20 bytes.
Return values	• **HAL:** status

HAL_HASHEx_SHA224_Start_DMA

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA224_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**
Function Description	Initializes the HASH peripheral in SHA224 mode then enables DMA to control data transfer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
Return values	• **HAL:** status

HAL_HASHEx_SHA224_Finish

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA224_Finish (HASH_HandleTypeDef * hhash, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Returns the computed digest in SHA224.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pOutBuffer:** Pointer to the computed digest. Its size must be 28 bytes. • **Timeout:** Timeout value
Return values	• **HAL:** status

HAL_HASHEx_SHA256_Start_DMA

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA256_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**
Function Description	Initializes the HASH peripheral in SHA256 mode then enables DMA to control data transfer.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module • **pInBuffer:** Pointer to the input buffer (buffer to be hashed). • **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.
Return values	• **HAL:** status

HAL_HASHEx_SHA256_Finish

Function Name	**HAL_StatusTypeDef HAL_HASHEx_SHA256_Finish (HASH_HandleTypeDef * hhash, uint8_t * pOutBuffer, uint32_t Timeout)**
Function Description	Returns the computed digest in SHA256.
Parameters	• **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module

- **pOutBuffer:** Pointer to the computed digest. Its size must be 32 bytes.
- **Timeout:** Timeout value

Return values
- **HAL:** status

HAL_HMACEx_SHA224_Start_DMA

Function Name **HAL_StatusTypeDef HAL_HMACEx_SHA224_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in HMAC SHA224 mode then enables DMA to control data transfer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

HAL_HMACEx_SHA256_Start_DMA

Function Name **HAL_StatusTypeDef HAL_HMACEx_SHA256_Start_DMA (HASH_HandleTypeDef * hhash, uint8_t * pInBuffer, uint32_t Size)**

Function Description Initializes the HASH peripheral in HMAC SHA256 mode then enables DMA to control data transfer.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module
- **pInBuffer:** Pointer to the input buffer (buffer to be hashed).
- **Size:** Length of the input buffer in bytes. If the Size is not multiple of 64 bytes, the padding is managed by hardware.

Return values
- **HAL:** status

HAL_HASHEx_IRQHandler

Function Name **void HAL_HASHEx_IRQHandler (HASH_HandleTypeDef * hhash)**

Function Description This function handles HASH interrupt request.

Parameters
- **hhash:** pointer to a HASH_HandleTypeDef structure that contains the configuration information for HASH module

Return values
- **None**

31 HAL HCD Generic Driver

31.1 HCD Firmware driver registers structures

31.1.1 HCD_HandleTypeDef

Data Fields

- ***HCD_TypeDef * Instance***
- ***HCD_InitTypeDef Init***
- ***HCD_HCTypeDef hc***
- ***HAL_LockTypeDef Lock***
- ***__IO HCD_StateTypeDef State***
- ***void * pData***

Field Documentation

- ***HCD_TypeDef* HCD_HandleTypeDef::Instance***
  Register base address
- ***HCD_InitTypeDef HCD_HandleTypeDef::Init***
  HCD required parameters
- ***HCD_HCTypeDef HCD_HandleTypeDef::hc[15]***
  Host channels parameters
- ***HAL_LockTypeDef HCD_HandleTypeDef::Lock***
  HCD peripheral status
- ***__IO HCD_StateTypeDef HCD_HandleTypeDef::State***
  HCD communication state
- ***void* HCD_HandleTypeDef::pData***
  Pointer Stack Handler

31.2 HCD Firmware driver API description

31.2.1 How to use this driver

1. Declare a HCD_HandleTypeDef handle structure, for example: HCD_HandleTypeDef hhcd;
2. Fill parameters of Init structure in HCD handle
3. Call HAL_HCD_Init() API to initialize the HCD peripheral (Core, Host core, ...)
4. Initialize the HCD low level resources through the HAL_HCD_MspInit() API:
   a. Enable the HCD/USB Low Level interface clock using the following macros
      – __HAL_RCC_USB_OTG_FS_CLK_ENABLE();
      – __HAL_RCC_USB_OTG_HS_CLK_ENABLE(); (For High Speed Mode)
      – __HAL_RCC_USB_OTG_HS_ULPI_CLK_ENABLE(); (For High Speed Mode)
   b. Initialize the related GPIO clocks
   c. Configure HCD pin-out
   d. Configure HCD NVIC interrupt

5. Associate the Upper USB Host stack to the HAL HCD Driver:
   a. hhcd.pData = phost;
6. Enable HCD transmission and reception:
   a. HAL_HCD_Start();

31.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

This section contains the following APIs:

- ***HAL_HCD_Init()***
- ***HAL_HCD_HC_Init()***
- ***HAL_HCD_HC_Halt()***
- ***HAL_HCD_DeInit()***
- ***HAL_HCD_MspInit()***
- ***HAL_HCD_MspDeInit()***

31.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the USB Host Data Transfer

This section contains the following APIs:

- ***HAL_HCD_HC_SubmitRequest()***
- ***HAL_HCD_IRQHandler()***
- ***HAL_HCD_SOF_Callback()***
- ***HAL_HCD_Connect_Callback()***
- ***HAL_HCD_Disconnect_Callback()***
- ***HAL_HCD_HC_NotifyURBChange_Callback()***

31.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the HCD data transfers.

This section contains the following APIs:

- ***HAL_HCD_Start()***
- ***HAL_HCD_Stop()***
- ***HAL_HCD_ResetPort()***

31.2.5 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_HCD_GetState()***
- ***HAL_HCD_HC_GetURBState()***
- ***HAL_HCD_HC_GetXferCount()***
- ***HAL_HCD_HC_GetState()***
- ***HAL_HCD_GetCurrentFrame()***
- ***HAL_HCD_GetCurrentSpeed()***

31.2.6 Detailed description of functions

HAL_HCD_Init

Function Name **HAL_StatusTypeDef HAL_HCD_Init (HCD_HandleTypeDef * hhcd)**

Function Description Initialize the host driver.

Parameters
- **hhcd:** HCD handle

Return values
- **HAL:** status

HAL_HCD_DeInit

Function Name **HAL_StatusTypeDef HAL_HCD_DeInit (HCD_HandleTypeDef * hhcd)**

Function Description DeInitialize the host driver.

Parameters
- **hhcd:** HCD handle

Return values
- **HAL:** status

HAL_HCD_HC_Init

Function Name **HAL_StatusTypeDef HAL_HCD_HC_Init (HCD_HandleTypeDef * hhcd, uint8_t ch_num, uint8_t epnum, uint8_t dev_address, uint8_t speed, uint8_t ep_type, uint16_t mps)**

Function Description Initialize a host channel.

Parameters
- **hhcd:** HCD handle
- **ch_num:** Channel number. This parameter can be a value from 1 to 15
- **epnum:** Endpoint number. This parameter can be a value from 1 to 15
- **dev_address:** : Current device address This parameter can be a value from 0 to 255
- **speed:** Current device speed. This parameter can be one of these values: HCD_SPEED_HIGH: High speed mode, HCD_SPEED_FULL: Full speed mode, HCD_SPEED_LOW: Low speed mode
- **ep_type:** Endpoint Type. This parameter can be one of these values: EP_TYPE_CTRL: Control type, EP_TYPE_ISOC: Isochronous type, EP_TYPE_BULK: Bulk type, EP_TYPE_INTR: Interrupt type
- **mps:** Max Packet Size. This parameter can be a value from 0 to32K

Return values
- **HAL:** status

HAL_HCD_HC_Halt

Function Name **HAL_StatusTypeDef HAL_HCD_HC_Halt (HCD_HandleTypeDef * hhcd, uint8_t ch_num)**

Function Description Halt a host channel.

Parameters	• **hhcd:** HCD handle • **ch_num:** Channel number. This parameter can be a value from 1 to 15
Return values	• **HAL:** status

HAL_HCD_MspInit

Function Name	**void HAL_HCD_MspInit (HCD_HandleTypeDef * hhcd)**
Function Description	Initialize the HCD MSP.
Parameters	• **hhcd:** HCD handle
Return values	• **None**

HAL_HCD_MspDeInit

Function Name	**void HAL_HCD_MspDeInit (HCD_HandleTypeDef * hhcd)**
Function Description	DeInitialize the HCD MSP.
Parameters	• **hhcd:** HCD handle
Return values	• **None**

HAL_HCD_HC_SubmitRequest

Function Name	**HAL_StatusTypeDef HAL_HCD_HC_SubmitRequest (HCD_HandleTypeDef * hhcd, uint8_t ch_num, uint8_t direction, uint8_t ep_type, uint8_t token, uint8_t * pbuff, uint16_t length, uint8_t do_ping)**
Function Description	Submit a new URB for processing.
Parameters	• **hhcd:** HCD handle • **ch_num:** Channel number. This parameter can be a value from 1 to 15 • **direction:** Channel number. This parameter can be one of these values: 0 : Output / 1 : Input • **ep_type:** Endpoint Type. This parameter can be one of these values: EP_TYPE_CTRL: Control type/ EP_TYPE_ISOC: Isochronous type/ EP_TYPE_BULK: Bulk type/ EP_TYPE_INTR: Interrupt type/ • **token:** Endpoint Type. This parameter can be one of these values: 0: HC_PID_SETUP / 1: HC_PID_DATA1 • **pbuff:** pointer to URB data • **length:** Length of URB data • **do_ping:** activate do ping protocol (for high speed only). This parameter can be one of these values: 0 : do ping inactive / 1 : do ping active
Return values	• **HAL:** status

HAL_HCD_IRQHandler

Function Name	**void HAL_HCD_IRQHandler (HCD_HandleTypeDef * hhcd)**
Function Description	Handle HCD interrupt request.

Parameters
- **hhcd:** HCD handle

Return values
- **None**

HAL_HCD_SOF_Callback

Function Name **void HAL_HCD_SOF_Callback (HCD_HandleTypeDef * hhcd)**

Function Description SOF callback.

Parameters
- **hhcd:** HCD handle

Return values
- **None**

HAL_HCD_Connect_Callback

Function Name **void HAL_HCD_Connect_Callback (HCD_HandleTypeDef * hhcd)**

Function Description Connection Event callback.

Parameters
- **hhcd:** HCD handle

Return values
- **None**

HAL_HCD_Disconnect_Callback

Function Name **void HAL_HCD_Disconnect_Callback (HCD_HandleTypeDef * hhcd)**

Function Description Disconnection Event callback.

Parameters
- **hhcd:** HCD handle

Return values
- **None**

HAL_HCD_HC_NotifyURBChange_Callback

Function Name **void HAL_HCD_HC_NotifyURBChange_Callback (HCD_HandleTypeDef * hhcd, uint8_t chnum, HCD_URBStateTypeDef urb_state)**

Function Description Notify URB state change callback.

Parameters
- **hhcd:** HCD handle
- **chnum:** Channel number. This parameter can be a value from 1 to 15
- **urb_state:** This parameter can be one of these values: URB_IDLE/ URB_DONE/ URB_NOTREADY/ URB_NYET/ URB_ERROR/ URB_STALL/

Return values
- **None**

HAL_HCD_ResetPort

Function Name **HAL_StatusTypeDef HAL_HCD_ResetPort (HCD_HandleTypeDef * hhcd)**

Function Description Reset the host port.

Parameters	• **hhcd:** HCD handle
Return values	• **HAL:** status

HAL_HCD_Start

Function Name	**HAL_StatusTypeDef HAL_HCD_Start (HCD_HandleTypeDef * hhcd)**
Function Description	Start the host driver.
Parameters	• **hhcd:** HCD handle
Return values	• **HAL:** status

HAL_HCD_Stop

Function Name	**HAL_StatusTypeDef HAL_HCD_Stop (HCD_HandleTypeDef * hhcd)**
Function Description	Stop the host driver.
Parameters	• **hhcd:** HCD handle
Return values	• **HAL:** status

HAL_HCD_GetState

Function Name	**HCD_StateTypeDef HAL_HCD_GetState (HCD_HandleTypeDef * hhcd)**
Function Description	Return the HCD handle state.
Parameters	• **hhcd:** HCD handle
Return values	• **HAL:** state

HAL_HCD_HC_GetURBState

Function Name	**HCD_URBStateTypeDef HAL_HCD_HC_GetURBState (HCD_HandleTypeDef * hhcd, uint8_t chnum)**
Function Description	Return URB state for a channel.
Parameters	• **hhcd:** HCD handle • **chnum:** Channel number. This parameter can be a value from 1 to 15
Return values	• **URB:** state. This parameter can be one of these values: URB_IDLE/ URB_DONE/ URB_NOTREADY/ URB_NYET/ URB_ERROR/ URB_STALL

HAL_HCD_HC_GetXferCount

Function Name	**uint32_t HAL_HCD_HC_GetXferCount (HCD_HandleTypeDef * hhcd, uint8_t chnum)**
Function Description	Return the last host transfer size.
Parameters	• **hhcd:** HCD handle • **chnum:** Channel number. This parameter can be a value

from 1 to 15

Return values
- **last:** transfer size in byte

HAL_HCD_HC_GetState

Function Name **HCD_HCStateTypeDef HAL_HCD_HC_GetState (HCD_HandleTypeDef * hhcd, uint8_t chnum)**

Function Description Return the Host Channel state.

Parameters
- **hhcd:** HCD handle
- **chnum:** Channel number. This parameter can be a value from 1 to 15

Return values
- **Host:** channel state This parameter can be one of these values: HC_IDLE/ HC_XFRC/ HC_HALTED/ HC_NYET/ HC_NAK/ HC_STALL/ HC_XACTERR/ HC_BBLERR/ HC_DATATGLERR

HAL_HCD_GetCurrentFrame

Function Name **uint32_t HAL_HCD_GetCurrentFrame (HCD_HandleTypeDef * hhcd)**

Function Description Return the current Host frame number.

Parameters
- **hhcd:** HCD handle

Return values
- **Current:** Host frame number

HAL_HCD_GetCurrentSpeed

Function Name **uint32_t HAL_HCD_GetCurrentSpeed (HCD_HandleTypeDef * hhcd)**

Function Description Return the Host enumeration speed.

Parameters
- **hhcd:** HCD handle

Return values
- **Enumeration:** speed

31.3 HCD Firmware driver defines

31.3.1 HCD

HCD Exported Macros

__HAL_HCD_ENABLE

__HAL_HCD_DISABLE

__HAL_HCD_GET_FLAG

__HAL_HCD_CLEAR_FLAG

__HAL_HCD_IS_INVALID_INTERRUPT

__HAL_HCD_CLEAR_HC_INT

__HAL_HCD_MASK_HALT_HC_INT

__HAL_HCD_UNMASK_HALT_HC_INT

__HAL_HCD_MASK_ACK_HC_INT

__HAL_HCD_UNMASK_ACK_HC_INT

HCD PHY Module

HCD_PHY_ULPI

HCD_PHY_EMBEDDED

HCD Speed

HCD_SPEED_HIGH

HCD_SPEED_LOW

HCD_SPEED_FULL

32 HAL I2C Generic Driver

32.1 I2C Firmware driver registers structures

32.1.1 I2C_InitTypeDef

Data Fields

- ***uint32_t ClockSpeed***
- ***uint32_t DutyCycle***
- ***uint32_t OwnAddress1***
- ***uint32_t AddressingMode***
- ***uint32_t DualAddressMode***
- ***uint32_t OwnAddress2***
- ***uint32_t GeneralCallMode***
- ***uint32_t NoStretchMode***

Field Documentation

- ***uint32_t I2C_InitTypeDef::ClockSpeed***
  Specifies the clock frequency. This parameter must be set to a value lower than 400kHz
- ***uint32_t I2C_InitTypeDef::DutyCycle***
  Specifies the I2C fast mode duty cycle. This parameter can be a value of ***I2C_duty_cycle_in_fast_mode***
- ***uint32_t I2C_InitTypeDef::OwnAddress1***
  Specifies the first device own address. This parameter can be a 7-bit or 10-bit address.
- ***uint32_t I2C_InitTypeDef::AddressingMode***
  Specifies if 7-bit or 10-bit addressing mode is selected. This parameter can be a value of ***I2C_addressing_mode***
- ***uint32_t I2C_InitTypeDef::DualAddressMode***
  Specifies if dual addressing mode is selected. This parameter can be a value of ***I2C_dual_addressing_mode***
- ***uint32_t I2C_InitTypeDef::OwnAddress2***
  Specifies the second device own address if dual addressing mode is selected This parameter can be a 7-bit address.
- ***uint32_t I2C_InitTypeDef::GeneralCallMode***
  Specifies if general call mode is selected. This parameter can be a value of ***I2C_general_call_addressing_mode***
- ***uint32_t I2C_InitTypeDef::NoStretchMode***
  Specifies if nostretch mode is selected. This parameter can be a value of ***I2C_nostretch_mode***

32.1.2 I2C_HandleTypeDef

Data Fields

- ***I2C_TypeDef * Instance***
- ***I2C_InitTypeDef Init***
- ***uint8_t * pBuffPtr***
- ***uint16_t XferSize***
- ***__IO uint16_t XferCount***
- ***__IO uint32_t XferOptions***
- ***__IO uint32_t PreviousState***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_I2C_StateTypeDef State***
- ***__IO HAL_I2C_ModeTypeDef Mode***
- ***__IO uint32_t ErrorCode***
- ***__IO uint32_t Devaddress***
- ***__IO uint32_t Memaddress***
- ***__IO uint32_t MemaddSize***
- ***__IO uint32_t EventCount***

Field Documentation

- ***I2C_TypeDef* I2C_HandleTypeDef::Instance***
  I2C registers base address
- ***I2C_InitTypeDef I2C_HandleTypeDef::Init***
  I2C communication parameters
- ***uint8_t* I2C_HandleTypeDef::pBuffPtr***
  Pointer to I2C transfer buffer
- ***uint16_t I2C_HandleTypeDef::XferSize***
  I2C transfer size
- ***__IO uint16_t I2C_HandleTypeDef::XferCount***
  I2C transfer counter
- ***__IO uint32_t I2C_HandleTypeDef::XferOptions***
  I2C transfer options
- ***__IO uint32_t I2C_HandleTypeDef::PreviousState***
  I2C communication Previous state and mode context for internal usage
- ***DMA_HandleTypeDef* I2C_HandleTypeDef::hdmatx***
  I2C Tx DMA handle parameters
- ***DMA_HandleTypeDef* I2C_HandleTypeDef::hdmarx***
  I2C Rx DMA handle parameters
- ***HAL_LockTypeDef I2C_HandleTypeDef::Lock***
  I2C locking object
- ***__IO HAL_I2C_StateTypeDef I2C_HandleTypeDef::State***
  I2C communication state
- ***__IO HAL_I2C_ModeTypeDef I2C_HandleTypeDef::Mode***
  I2C communication mode
- ***__IO uint32_t I2C_HandleTypeDef::ErrorCode***
  I2C Error code

- ***__IO uint32_t I2C_HandleTypeDef::Devaddress***
  I2C Target device address
- ***__IO uint32_t I2C_HandleTypeDef::Memaddress***
  I2C Target memory address
- ***__IO uint32_t I2C_HandleTypeDef::MemaddSize***
  I2C Target memory address size
- ***__IO uint32_t I2C_HandleTypeDef::EventCount***
  I2C Event counter

32.2 I2C Firmware driver API description

32.2.1 How to use this driver

The I2C HAL driver can be used as follows:

1. Declare a I2C_HandleTypeDef handle structure, for example: I2C_HandleTypeDef hi2c;
2. Initialize the I2C low level resources by implement the HAL_I2C_MspInit() API:
   a. Enable the I2Cx interface clock
   b. I2C pins configuration
      – Enable the clock for the I2C GPIOs
      – Configure I2C pins as alternate function open-drain
   c. NVIC configuration if you need to use interrupt process
      – Configure the I2Cx interrupt priority
      – Enable the NVIC I2C IRQ Channel
   d. DMA Configuration if you need to use DMA process
      – Declare a DMA_HandleTypeDef handle structure for the transmit or receive stream
      – Enable the DMAx interface clock using
      – Configure the DMA handle parameters
      – Configure the DMA Tx or Rx Stream
      – Associate the initialized DMA handle to the hi2c DMA Tx or Rx handle
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx or Rx Stream
3. Configure the Communication Speed, Duty cycle, Addressing mode, Own Address1, Dual Addressing mode, Own Address2, General call and Nostretch mode in the hi2c Init structure.
4. Initialize the I2C registers by calling the HAL_I2C_Init(), configures also the low level Hardware (GPIO, CLOCK, NVIC...etc) by calling the customized HAL_I2C_MspInit(&hi2c) API.
5. To check if target device is ready for communication, use the function HAL_I2C_IsDeviceReady()
6. For I2C IO and IO MEM operations, three operation modes are available within this driver :

Polling mode IO operation

- Transmit in master mode an amount of data in blocking mode using HAL_I2C_Master_Transmit()
- Receive in master mode an amount of data in blocking mode using HAL_I2C_Master_Receive()
- Transmit in slave mode an amount of data in blocking mode using HAL_I2C_Slave_Transmit()

- Receive in slave mode an amount of data in blocking mode using HAL_I2C_Slave_Receive()

Polling mode IO MEM operation

- Write an amount of data in blocking mode to a specific memory address using HAL_I2C_Mem_Write()
- Read an amount of data in blocking mode from a specific memory address using HAL_I2C_Mem_Read()

Interrupt mode IO operation

- Transmit in master mode an amount of data in non blocking mode using HAL_I2C_Master_Transmit_IT()
- At transmission end of transfer HAL_I2C_MasterTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MasterTxCpltCallback
- Receive in master mode an amount of data in non blocking mode using HAL_I2C_Master_Receive_IT()
- At reception end of transfer HAL_I2C_MasterRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MasterRxCpltCallback
- Transmit in slave mode an amount of data in non blocking mode using HAL_I2C_Slave_Transmit_IT()
- At transmission end of transfer HAL_I2C_SlaveTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveTxCpltCallback
- Receive in slave mode an amount of data in non blocking mode using HAL_I2C_Slave_Receive_IT()
- At reception end of transfer HAL_I2C_SlaveRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveRxCpltCallback
- In case of transfer Error, HAL_I2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2C_ErrorCallback

Interrupt mode IO sequential operation

These interfaces allow to manage a sequential transfer with a repeated start condition when a direction change during transfer

- A specific option manage the different steps of a sequential transfer
- Differents steps option I2C_XferOptions_definition are listed below :
  - I2C_FIRST_AND_LAST_FRAME: No sequential usage, functionnal is same as associated interfaces in no sequential mode
  - I2C_FIRST_FRAME: Sequential usage, this option allow to manage a start condition with data to transfer without a final stop condition
  - I2C_NEXT_FRAME: Sequential usage, this option allow to manage a restart condition with new data to transfer if the direction change or manage only the new data to transfer if no direction change and without a final stop condition in both cases
  - I2C_LAST_FRAME: Sequential usage, this option allow to manage a restart condition with new data to transfer if the direction change or manage only the new data to transfer if no direction change and with a final stop condition in both cases

- Sequential transmit in master I2C mode an amount of data in non-blocking mode using HAL_I2C_Master_Sequential_Transmit_IT()
  - At transmission end of current frame transfer, HAL_I2C_MasterTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_I2C_MasterTxCpltCallback()
- Sequential receive in master I2C mode an amount of data in non-blocking mode using HAL_I2C_Master_Sequential_Receive_IT()
  - At reception end of current frame transfer, HAL_I2C_MasterRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_I2C_MasterRxCpltCallback()
- Abort a master I2C process communication with Interrupt using HAL_I2C_Master_Abort_IT()
  - The associated previous transfer callback is called at the end of abort process
  - mean HAL_I2C_MasterTxCpltCallback() in case of previous state was master transmit
  - mean HAL_I2c_MasterRxCpltCallback() in case of previous state was master receive
- Enable/disable the Address listen mode in slave I2C mode using HAL_I2C_EnableListen_IT() HAL_I2C_DisableListen_IT()
  - When address slave I2C match, HAL_I2C_AddrCallback() is executed and user can add his own code to check the Address Match Code and the transmission direction request by master (Write/Read).
  - At Listen mode end HAL_I2C_ListenCpltCallback() is executed and user can add his own code by customization of function pointer HAL_I2C_ListenCpltCallback()
- Sequential transmit in slave I2C mode an amount of data in non-blocking mode using HAL_I2C_Slave_Sequential_Transmit_IT()
  - At transmission end of current frame transfer, HAL_I2C_SlaveTxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveTxCpltCallback()
- Sequential receive in slave I2C mode an amount of data in non-blocking mode using HAL_I2C_Slave_Sequential_Receive_IT()
  - At reception end of current frame transfer, HAL_I2C_SlaveRxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveRxCpltCallback()

Interrupt mode IO MEM operation

- Write an amount of data in no-blocking mode with Interrupt to a specific memory address using HAL_I2C_Mem_Write_IT()
- At MEM end of write transfer HAL_I2C_MemTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MemTxCpltCallback
- Read an amount of data in no-blocking mode with Interrupt from a specific memory address using HAL_I2C_Mem_Read_IT()
- At MEM end of read transfer HAL_I2C_MemRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MemRxCpltCallback
- In case of transfer Error, HAL_I2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2C_ErrorCallback

DMA mode IO operation

- Transmit in master mode an amount of data in non blocking mode (DMA) using HAL_I2C_Master_Transmit_DMA()
- At transmission end of transfer HAL_I2C_MasterTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MasterTxCpltCallback

- Receive in master mode an amount of data in non blocking mode (DMA) using HAL_I2C_Master_Receive_DMA()
- At reception end of transfer HAL_I2C_MasterRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MasterRxCpltCallback
- Transmit in slave mode an amount of data in non blocking mode (DMA) using HAL_I2C_Slave_Transmit_DMA()
- At transmission end of transfer HAL_I2C_SlaveTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveTxCpltCallback
- Receive in slave mode an amount of data in non blocking mode (DMA) using HAL_I2C_Slave_Receive_DMA()
- At reception end of transfer HAL_I2C_SlaveRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_SlaveRxCpltCallback
- In case of transfer Error, HAL_I2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2C_ErrorCallback

DMA mode IO MEM operation

- Write an amount of data in no-blocking mode with DMA to a specific memory address using HAL_I2C_Mem_Write_DMA()
- At MEM end of write transfer HAL_I2C_MemTxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MemTxCpltCallback
- Read an amount of data in no-blocking mode with DMA from a specific memory address using HAL_I2C_Mem_Read_DMA()
- At MEM end of read transfer HAL_I2C_MemRxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2C_MemRxCpltCallback
- In case of transfer Error, HAL_I2C_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2C_ErrorCallback

I2C HAL driver macros list

Below the list of most used macros in I2C HAL driver.

- __HAL_I2C_ENABLE: Enable the I2C peripheral
- __HAL_I2C_DISABLE: Disable the I2C peripheral
- __HAL_I2C_GET_FLAG: Checks whether the specified I2C flag is set or not
- __HAL_I2C_CLEAR_FLAG: Clear the specified I2C pending flag
- __HAL_I2C_ENABLE_IT: Enable the specified I2C interrupt
- __HAL_I2C_DISABLE_IT: Disable the specified I2C interrupt

You can refer to the I2C HAL driver header file for more useful macros

32.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and de-initialize the I2Cx peripheral:

- User must Implement HAL_I2C_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).

- Call the function HAL_I2C_Init() to configure the selected device with the selected configuration:
  - Communication Speed
  - Duty cycle
  - Addressing mode
  - Own Address 1
  - Dual Addressing mode
  - Own Address 2
  - General call mode
  - Nostretch mode
- Call the function HAL_I2C_DeInit() to restore the default configuration of the selected I2Cx peripheral.

This section contains the following APIs:

- ***HAL_I2C_Init()***
- ***HAL_I2C_DeInit()***
- ***HAL_I2C_MspInit()***
- ***HAL_I2C_MspDeInit()***

32.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the I2C data transfers.

1. There are two modes of transfer:
   - Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer startup. The end of the data processing will be indicated through the dedicated I2C IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
2. Blocking mode functions are :
   - HAL_I2C_Master_Transmit()
   - HAL_I2C_Master_Receive()
   - HAL_I2C_Slave_Transmit()
   - HAL_I2C_Slave_Receive()
   - HAL_I2C_Mem_Write()
   - HAL_I2C_Mem_Read()
   - HAL_I2C_IsDeviceReady()
3. No-Blocking mode functions with Interrupt are :
   - HAL_I2C_Master_Transmit_IT()
   - HAL_I2C_Master_Receive_IT()
   - HAL_I2C_Slave_Transmit_IT()
   - HAL_I2C_Slave_Receive_IT()
   - HAL_I2C_Master_Sequential_Transmit_IT()
   - HAL_I2C_Master_Sequential_Receive_IT()
   - HAL_I2C_Slave_Sequential_Transmit_IT()
   - HAL_I2C_Slave_Sequential_Receive_IT()
   - HAL_I2C_Mem_Write_IT()
   - HAL_I2C_Mem_Read_IT()
4. No-Blocking mode functions with DMA are :
   - HAL_I2C_Master_Transmit_DMA()
   - HAL_I2C_Master_Receive_DMA()
   - HAL_I2C_Slave_Transmit_DMA()

- HAL_I2C_Slave_Receive_DMA()
- HAL_I2C_Mem_Write_DMA()
- HAL_I2C_Mem_Read_DMA()

5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   - HAL_I2C_MemTxCpltCallback()
   - HAL_I2C_MemRxCpltCallback()
   - HAL_I2C_MasterTxCpltCallback()
   - HAL_I2C_MasterRxCpltCallback()
   - HAL_I2C_SlaveTxCpltCallback()
   - HAL_I2C_SlaveRxCpltCallback()
   - HAL_I2C_ErrorCallback()
   - HAL_I2C_AbortCpltCallback()

This section contains the following APIs:

- ***HAL_I2C_Master_Transmit()***
- ***HAL_I2C_Master_Receive()***
- ***HAL_I2C_Slave_Transmit()***
- ***HAL_I2C_Slave_Receive()***
- ***HAL_I2C_Master_Transmit_IT()***
- ***HAL_I2C_Master_Receive_IT()***
- ***HAL_I2C_Master_Sequential_Transmit_IT()***
- ***HAL_I2C_Master_Sequential_Receive_IT()***
- ***HAL_I2C_Slave_Transmit_IT()***
- ***HAL_I2C_Slave_Receive_IT()***
- ***HAL_I2C_Slave_Sequential_Transmit_IT()***
- ***HAL_I2C_Slave_Sequential_Receive_IT()***
- ***HAL_I2C_EnableListen_IT()***
- ***HAL_I2C_DisableListen_IT()***
- ***HAL_I2C_Master_Transmit_DMA()***
- ***HAL_I2C_Master_Receive_DMA()***
- ***HAL_I2C_Master_Abort_IT()***
- ***HAL_I2C_Slave_Transmit_DMA()***
- ***HAL_I2C_Slave_Receive_DMA()***
- ***HAL_I2C_Mem_Write()***
- ***HAL_I2C_Mem_Read()***
- ***HAL_I2C_Mem_Write_IT()***
- ***HAL_I2C_Mem_Read_IT()***
- ***HAL_I2C_Mem_Write_DMA()***
- ***HAL_I2C_Mem_Read_DMA()***
- ***HAL_I2C_IsDeviceReady()***
- ***HAL_I2C_EV_IRQHandler()***
- ***HAL_I2C_ER_IRQHandler()***
- ***HAL_I2C_MasterTxCpltCallback()***
- ***HAL_I2C_MasterRxCpltCallback()***
- ***HAL_I2C_SlaveTxCpltCallback()***
- ***HAL_I2C_SlaveRxCpltCallback()***
- ***HAL_I2C_AddrCallback()***
- ***HAL_I2C_ListenCpltCallback()***
- ***HAL_I2C_MemTxCpltCallback()***
- ***HAL_I2C_MemRxCpltCallback()***
- ***HAL_I2C_ErrorCallback()***
- ***HAL_I2C_AbortCpltCallback()***

32.2.4 Peripheral State, Mode and Errors functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_I2C_GetState()***
- ***HAL_I2C_GetMode()***
- ***HAL_I2C_GetError()***

32.2.5 Detailed description of functions

HAL_I2C_Init

Function Name	**HAL_StatusTypeDef HAL_I2C_Init (I2C_HandleTypeDef * hi2c)**
Function Description	Initializes the I2C according to the specified parameters in the I2C_InitTypeDef and create the associated handle.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** status

HAL_I2C_DeInit

Function Name	**HAL_StatusTypeDef HAL_I2C_DeInit (I2C_HandleTypeDef * hi2c)**
Function Description	DeInitializes the I2C peripheral.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** status

HAL_I2C_MspInit

Function Name	**void HAL_I2C_MspInit (I2C_HandleTypeDef * hi2c)**
Function Description	I2C MSP Init.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_MspDeInit

Function Name	**void HAL_I2C_MspDeInit (I2C_HandleTypeDef * hi2c)**
Function Description	I2C MSP DeInit.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_Master_Transmit

Function Name	**HAL_StatusTypeDef HAL_I2C_Master_Transmit (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Transmits in master mode an amount of data in blocking mode.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_Master_Receive

Function Name	**HAL_StatusTypeDef HAL_I2C_Master_Receive (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Receives in master mode an amount of data in blocking mode.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_Slave_Transmit

Function Name	**HAL_StatusTypeDef HAL_I2C_Slave_Transmit (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Transmits in slave mode an amount of data in blocking mode.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_Slave_Receive

Function Name	**HAL_StatusTypeDef HAL_I2C_Slave_Receive (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Receive in slave mode an amount of data in blocking mode.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_Mem_Write

Function Name	**HAL_StatusTypeDef HAL_I2C_Mem_Write (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Write an amount of data in blocking mode to a specific memory address.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address • **MemAddress:** Internal memory address • **MemAddSize:** Size of internal memory address • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_Mem_Read

Function Name	**HAL_StatusTypeDef HAL_I2C_Mem_Read (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Read an amount of data in blocking mode from a specific memory address.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address • **MemAddress:** Internal memory address • **MemAddSize:** Size of internal memory address • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_I2C_IsDeviceReady

Function Name	**HAL_StatusTypeDef HAL_I2C_IsDeviceReady (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint32_t Trials, uint32_t Timeout)**
Function Description	Checks if target device is ready for communication.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address • **Trials:** Number of trials • **Timeout:** Timeout duration
Return values	• **HAL:** status
Notes	• This function is used with Memory devices

HAL_I2C_Master_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_I2C_Master_Transmit_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**
Function Description	Transmit in master mode an amount of data in no-blocking mode with Interrupt.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_I2C_Master_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_I2C_Master_Receive_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**
Function Description	Receive in master mode an amount of data in no-blocking mode with Interrupt.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_I2C_Slave_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_I2C_Slave_Transmit_IT (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size)**

Function Description Transmit in slave mode an amount of data in no-blocking mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Slave_Receive_IT

Function Name **HAL_StatusTypeDef HAL_I2C_Slave_Receive_IT (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size)**

Function Description Receive in slave mode an amount of data in no-blocking mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Mem_Write_IT

Function Name **HAL_StatusTypeDef HAL_I2C_Mem_Write_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Write an amount of data in no-blocking mode with Interrupt to a specific memory address.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Mem_Read_IT

Function Name **HAL_StatusTypeDef HAL_I2C_Mem_Read_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Read an amount of data in no-blocking mode with Interrupt from a specific memory address.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Master_Sequential_Transmit_IT

Function Name
HAL_StatusTypeDef HAL_I2C_Master_Sequential_Transmit_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t XferOptions)

Function Description
Sequential transmit in master mode an amount of data in no-blocking mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **XferOptions:** Options of Transfer, value of I2C XferOptions definition

Return values
- **HAL:** status

Notes
- This interface allow to manage repeated start condition when a direction change during transfer

HAL_I2C_Master_Sequential_Receive_IT

Function Name
HAL_StatusTypeDef HAL_I2C_Master_Sequential_Receive_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size, uint32_t XferOptions)

Function Description
Sequential receive in master mode an amount of data in no-blocking mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **XferOptions:** Options of Transfer, value of I2C XferOptions definition

Return values
- **HAL:** status

Notes
- This interface allow to manage repeated start condition when a direction change during transfer

HAL_I2C_Slave_Sequential_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_I2C_Slave_Sequential_Transmit_IT (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size, uint32_t XferOptions)**
Function Description	Sequential transmit in slave mode an amount of data in no-blocking mode with Interrupt.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **XferOptions:** Options of Transfer, value of I2C XferOptions definition
Return values	• **HAL:** status
Notes	• This interface allow to manage repeated start condition when a direction change during transfer

HAL_I2C_Slave_Sequential_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_I2C_Slave_Sequential_Receive_IT (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size, uint32_t XferOptions)**
Function Description	Sequential receive in slave mode an amount of data in no-blocking mode with Interrupt.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **XferOptions:** Options of Transfer, value of I2C XferOptions definition
Return values	• **HAL:** status
Notes	• This interface allow to manage repeated start condition when a direction change during transfer

HAL_I2C_Master_Abort_IT

Function Name	**HAL_StatusTypeDef HAL_I2C_Master_Abort_IT (I2C_HandleTypeDef * hi2c, uint16_t DevAddress)**
Function Description	Abort a master I2C process communication with Interrupt.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C. • **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
Return values	• **HAL:** status
Notes	• This abort can be called only if state is ready

HAL_I2C_EnableListen_IT

Function Name **HAL_StatusTypeDef HAL_I2C_EnableListen_IT (I2C_HandleTypeDef * hi2c)**

Function Description Enable the Address listen mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C.

Return values
- **HAL:** status

HAL_I2C_DisableListen_IT

Function Name **HAL_StatusTypeDef HAL_I2C_DisableListen_IT (I2C_HandleTypeDef * hi2c)**

Function Description Disable the Address listen mode with Interrupt.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C.

Return values
- **HAL:** status

HAL_I2C_Master_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Master_Transmit_DMA (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Transmit in master mode an amount of data in no-blocking mode with DMA.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Master_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Master_Receive_DMA (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint8_t * pData, uint16_t Size)**

Function Description Receive in master mode an amount of data in no-blocking mode with DMA.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address: The device 7 bits address value in datasheet must be shift at right before call interface
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Slave_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Slave_Transmit_DMA (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size)**

Function Description Transmit in slave mode an amount of data in no-blocking mode with DMA.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Slave_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Slave_Receive_DMA (I2C_HandleTypeDef * hi2c, uint8_t * pData, uint16_t Size)**

Function Description Receive in slave mode an amount of data in no-blocking mode with DMA.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Mem_Write_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Mem_Write_DMA (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description Write an amount of data in no-blocking mode with DMA to a specific memory address.

Parameters
- **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
- **DevAddress:** Target device address
- **MemAddress:** Internal memory address
- **MemAddSize:** Size of internal memory address
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_I2C_Mem_Read_DMA

Function Name **HAL_StatusTypeDef HAL_I2C_Mem_Read_DMA (I2C_HandleTypeDef * hi2c, uint16_t DevAddress, uint16_t MemAddress, uint16_t MemAddSize, uint8_t * pData, uint16_t Size)**

Function Description	Reads an amount of data in no-blocking mode with DMA from a specific memory address.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module • **DevAddress:** Target device address • **MemAddress:** Internal memory address • **MemAddSize:** Size of internal memory address • **pData:** Pointer to data buffer • **Size:** Amount of data to be read
Return values	• **HAL:** status

HAL_I2C_EV_IRQHandler

Function Name	**void HAL_I2C_EV_IRQHandler (I2C_HandleTypeDef * hi2c)**
Function Description	This function handles I2C event interrupt request.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** status

HAL_I2C_ER_IRQHandler

Function Name	**void HAL_I2C_ER_IRQHandler (I2C_HandleTypeDef * hi2c)**
Function Description	This function handles I2C error interrupt request.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** status

HAL_I2C_MasterTxCpltCallback

Function Name	**void HAL_I2C_MasterTxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Master Tx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_MasterRxCpltCallback

Function Name	**void HAL_I2C_MasterRxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Master Rx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_SlaveTxCpltCallback

Function Name	**void HAL_I2C_SlaveTxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Slave Tx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_SlaveRxCpltCallback

Function Name	**void HAL_I2C_SlaveRxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Slave Rx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_AddrCallback

Function Name	**void HAL_I2C_AddrCallback (I2C_HandleTypeDef * hi2c, uint8_t TransferDirection, uint16_t AddrMatchCode)**
Function Description	Slave Address Match callback.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C. • **TransferDirection:** Master request Transfer Direction (Write/Read), value of I2C XferOptions definition • **AddrMatchCode:** Address Match Code
Return values	• **None**

HAL_I2C_ListenCpltCallback

Function Name	**void HAL_I2C_ListenCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Listen Complete callback.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C.
Return values	• **None**

HAL_I2C_MemTxCpltCallback

Function Name	**void HAL_I2C_MemTxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Memory Tx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_MemRxCpltCallback

Function Name	**void HAL_I2C_MemRxCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	Memory Rx Transfer completed callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_ErrorCallback

Function Name	**void HAL_I2C_ErrorCallback (I2C_HandleTypeDef * hi2c)**
Function Description	I2C error callbacks.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **None**

HAL_I2C_AbortCpltCallback

Function Name	**void HAL_I2C_AbortCpltCallback (I2C_HandleTypeDef * hi2c)**
Function Description	I2C abort callback.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C.
Return values	• **None**

HAL_I2C_GetState

Function Name	**HAL_I2C_StateTypeDef HAL_I2C_GetState (I2C_HandleTypeDef * hi2c)**
Function Description	Returns the I2C state.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** state

HAL_I2C_GetMode

Function Name	**HAL_I2C_ModeTypeDef HAL_I2C_GetMode (I2C_HandleTypeDef * hi2c)**
Function Description	Returns the I2C Master, Slave, Memory or no mode.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for I2C module
Return values	• **HAL:** mode

HAL_I2C_GetError

Function Name	**uint32_t HAL_I2C_GetError (I2C_HandleTypeDef * hi2c)**
Function Description	Return the I2C error code.
Parameters	• **hi2c:** Pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2C.
Return values	• **I2C:** Error Code

32.3 I2C Firmware driver defines

32.3.1 I2C

I2C addressing mode

I2C_ADDRESSINGMODE_7BIT

I2C_ADDRESSINGMODE_10BIT

I2C dual addressing mode

I2C_DUALADDRESS_DISABLE

I2C_DUALADDRESS_ENABLE

I2C duty cycle in fast mode

I2C_DUTYCYCLE_2

I2C_DUTYCYCLE_16_9

I2C Error Code

HAL_I2C_ERROR_NONE	No error
HAL_I2C_ERROR_BERR	BERR error
HAL_I2C_ERROR_ARLO	ARLO error
HAL_I2C_ERROR_AF	AF error
HAL_I2C_ERROR_OVR	OVR error
HAL_I2C_ERROR_DMA	DMA transfer error
HAL_I2C_ERROR_TIMEOUT	Timeout Error

I2C Exported Macros

__HAL_I2C_RESET_HANDLE_STATE

Description:

- Reset I2C handle state.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.

Return value:

- None

__HAL_I2C_ENABLE_IT

Description:

- Enable or disable the specified I2C

interrupts.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.
- __INTERRUPT__: specifies the interrupt source to enable or disable. This parameter can be one of the following values:
  - I2C_IT_BUF: Buffer interrupt enable
  - I2C_IT_EVT: Event interrupt enable
  - I2C_IT_ERR: Error interrupt enable

Return value:

- None

__HAL_I2C_DISABLE_IT

__HAL_I2C_GET_IT_SOURCE

Description:

- Checks if the specified I2C interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.
- __INTERRUPT__: specifies the I2C interrupt source to check. This parameter can be one of the following values:
  - I2C_IT_BUF: Buffer interrupt enable
  - I2C_IT_EVT: Event interrupt enable
  - I2C_IT_ERR: Error interrupt enable

Return value:

- The: new state of __INTERRUPT__ (TRUE or FALSE).

__HAL_I2C_GET_FLAG

Description:

- Checks whether the specified I2C flag is set or not.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - I2C_FLAG_SMBALERT: SMBus Alert flag
  - I2C_FLAG_TIMEOUT: Timeout or Tlow error flag
  - I2C_FLAG_PECERR: PEC error in reception flag

- I2C_FLAG_OVR: Overrun/Underrun flag
- I2C_FLAG_AF: Acknowledge failure flag
- I2C_FLAG_ARLO: Arbitration lost flag
- I2C_FLAG_BERR: Bus error flag
- I2C_FLAG_TXE: Data register empty flag
- I2C_FLAG_RXNE: Data register not empty flag
- I2C_FLAG_STOPF: Stop detection flag
- I2C_FLAG_ADD10: 10-bit header sent flag
- I2C_FLAG_BTF: Byte transfer finished flag
- I2C_FLAG_ADDR: Address sent flag Address matched flag
- I2C_FLAG_SB: Start bit flag
- I2C_FLAG_DUALF: Dual flag
- I2C_FLAG_SMBHOST: SMBus host header
- I2C_FLAG_SMBDEFAULT: SMBus default header
- I2C_FLAG_GENCALL: General call header flag
- I2C_FLAG_TRA: Transmitter/Receiver flag
- I2C_FLAG_BUSY: Bus busy flag
- I2C_FLAG_MSL: Master/Slave flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_I2C_CLEAR_FLAG

Description:

- Clears the I2C pending flags which are cleared by writing 0 in a specific bit.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - I2C_FLAG_SMBALERT: SMBus Alert flag
  - I2C_FLAG_TIMEOUT: Timeout or Tlow error flag
  - I2C_FLAG_PECERR: PEC error in reception flag
  - I2C_FLAG_OVR: Overrun/Underrun flag (Slave mode)
  - I2C_FLAG_AF: Acknowledge failure

flag
  - I2C_FLAG_ARLO: Arbitration lost flag (Master mode)
  - I2C_FLAG_BERR: Bus error flag

Return value:

- None

__HAL_I2C_CLEAR_ADDRFLAG

Description:

- Clears the I2C ADDR pending flag.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.

Return value:

- None

__HAL_I2C_CLEAR_STOPFLAG

Description:

- Clears the I2C STOPF pending flag.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2C where x: 1, 2, or 3 to select the I2C peripheral.

Return value:

- None

__HAL_I2C_ENABLE

Description:

- Enable the I2C peripheral.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2Cx where x: 1 or 2 to select the I2C peripheral.

Return value:

- None

__HAL_I2C_DISABLE

Description:

- Disable the I2C peripheral.

Parameters:

- __HANDLE__: specifies the I2C Handle. This parameter can be I2Cx where x: 1 or 2 to select the I2C peripheral.

Return value:

- None

I2C Flag definition

I2C_FLAG_SMBALERT

I2C_FLAG_TIMEOUT

I2C_FLAG_PECERR

I2C_FLAG_OVR

I2C_FLAG_AF

I2C_FLAG_ARLO

I2C_FLAG_BERR

I2C_FLAG_TXE

I2C_FLAG_RXNE

I2C_FLAG_STOPF

I2C_FLAG_ADD10

I2C_FLAG_BTF

I2C_FLAG_ADDR

I2C_FLAG_SB

I2C_FLAG_DUALF

I2C_FLAG_SMBHOST

I2C_FLAG_SMBDEFAULT

I2C_FLAG_GENCALL

I2C_FLAG_TRA

I2C_FLAG_BUSY

I2C_FLAG_MSL

I2C general call addressing mode

I2C_GENERALCALL_DISABLE

I2C_GENERALCALL_ENABLE

I2C Interrupt configuration definition

I2C_IT_BUF

I2C_IT_EVT

I2C_IT_ERR

I2C Private macros to check input parameters

IS_I2C_DUTY_CYCLE

IS_I2C_ADDRESSING_MODE

IS_I2C_DUAL_ADDRESS

IS_I2C_GENERAL_CALL

IS_I2C_NO_STRETCH

IS_I2C_MEMADD_SIZE

IS_I2C_CLOCK_SPEED

IS_I2C_OWN_ADDRESS1

IS_I2C_OWN_ADDRESS2

IS_I2C_TRANSFER_OPTIONS_REQUEST

I2C Memory Address Size

I2C_MEMADD_SIZE_8BIT

I2C_MEMADD_SIZE_16BIT

I2C nostretch mode

I2C_NOSTRETCH_DISABLE

I2C_NOSTRETCH_ENABLE

I2C XferDirection definition

I2C_DIRECTION_RECEIVE

I2C_DIRECTION_TRANSMIT

I2C XferOptions definition

I2C_FIRST_FRAME

I2C_NEXT_FRAME

I2C_FIRST_AND_LAST_FRAME

I2C_LAST_FRAME

33 HAL I2C Extension Driver

33.1 I2CEx Firmware driver API description

33.1.1 I2C peripheral extension features

Comparing to other previous devices, the I2C interface for STM32F427xx/437xx/ 429xx/439xx devices contains the following additional features :

- Possibility to disable or enable Analog Noise Filter
- Use of a configured Digital Noise Filter

33.1.2 How to use this driver

This driver provides functions to configure Noise Filter

1. Configure I2C Analog noise filter using the function HAL_I2C_AnalogFilter_Config()
2. Configure I2C Digital noise filter using the function HAL_I2C_DigitalFilter_Config()

33.1.3 Extension features functions

This section provides functions allowing to:

- Configure Noise Filters

This section contains the following APIs:

- ***HAL_I2CEx_ConfigAnalogFilter()***
- ***HAL_I2CEx_ConfigDigitalFilter()***

33.1.4 Detailed description of functions

HAL_I2CEx_ConfigAnalogFilter

Function Name	**HAL_StatusTypeDef HAL_I2CEx_ConfigAnalogFilter (I2C_HandleTypeDef * hi2c, uint32_t AnalogFilter)**
Function Description	Configures I2C Analog noise filter.
Parameters	• **hi2c:** pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2Cx peripheral. • **AnalogFilter:** new state of the Analog filter.
Return values	• **HAL:** status

HAL_I2CEx_ConfigDigitalFilter

Function Name	**HAL_StatusTypeDef HAL_I2CEx_ConfigDigitalFilter (I2C_HandleTypeDef * hi2c, uint32_t DigitalFilter)**
Function Description	Configures I2C Digital noise filter.
Parameters	• **hi2c:** pointer to a I2C_HandleTypeDef structure that contains the configuration information for the specified I2Cx peripheral. • **DigitalFilter:** Coefficient of digital noise filter between 0x00 and 0x0F.
Return values	• **HAL:** status

33.2 I2CEx Firmware driver defines

33.2.1 I2CEx

I2C Analog Filter

I2C_ANALOGFILTER_ENABLE

I2C_ANALOGFILTER_DISABLE

34 HAL I2S Generic Driver

34.1 I2S Firmware driver registers structures

34.1.1 I2S_InitTypeDef

Data Fields

- ***uint32_t Mode***
- ***uint32_t Standard***
- ***uint32_t DataFormat***
- ***uint32_t MCLKOutput***
- ***uint32_t AudioFreq***
- ***uint32_t CPOL***
- ***uint32_t ClockSource***
- ***uint32_t FullDuplexMode***

Field Documentation

- ***uint32_t I2S_InitTypeDef::Mode***
  Specifies the I2S operating mode. This parameter can be a value of ***I2S_Mode***
- ***uint32_t I2S_InitTypeDef::Standard***
  Specifies the standard used for the I2S communication. This parameter can be a value of ***I2S_Standard***
- ***uint32_t I2S_InitTypeDef::DataFormat***
  Specifies the data format for the I2S communication. This parameter can be a value of ***I2S_Data_Format***
- ***uint32_t I2S_InitTypeDef::MCLKOutput***
  Specifies whether the I2S MCLK output is enabled or not. This parameter can be a value of ***I2S_MCLK_Output***
- ***uint32_t I2S_InitTypeDef::AudioFreq***
  Specifies the frequency selected for the I2S communication. This parameter can be a value of ***I2S_Audio_Frequency***
- ***uint32_t I2S_InitTypeDef::CPOL***
  Specifies the idle state of the I2S clock. This parameter can be a value of ***I2S_Clock_Polarity***
- ***uint32_t I2S_InitTypeDef::ClockSource***
  Specifies the I2S Clock Source. This parameter can be a value of ***I2S_Clock_Source***
- ***uint32_t I2S_InitTypeDef::FullDuplexMode***
  Specifies the I2S FullDuplex mode. This parameter can be a value of ***I2S_FullDuplex_Mode***

34.1.2 I2S_HandleTypeDef

Data Fields

- ***SPI_TypeDef * Instance***
- ***I2S_InitTypeDef Init***

- ***uint16_t * pTxBuffPtr***
- ***__IO uint16_t TxXferSize***
- ***__IO uint16_t TxXferCount***
- ***uint16_t * pRxBuffPtr***
- ***__IO uint16_t RxXferSize***
- ***__IO uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***__IO HAL_LockTypeDef Lock***
- ***__IO HAL_I2S_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***SPI_TypeDef* I2S_HandleTypeDef::Instance***
- ***I2S_InitTypeDef I2S_HandleTypeDef::Init***
- ***uint16_t* I2S_HandleTypeDef::pTxBuffPtr***
- ***__IO uint16_t I2S_HandleTypeDef::TxXferSize***
- ***__IO uint16_t I2S_HandleTypeDef::TxXferCount***
- ***uint16_t* I2S_HandleTypeDef::pRxBuffPtr***
- ***__IO uint16_t I2S_HandleTypeDef::RxXferSize***
- ***__IO uint16_t I2S_HandleTypeDef::RxXferCount***
- ***DMA_HandleTypeDef* I2S_HandleTypeDef::hdmatx***
- ***DMA_HandleTypeDef* I2S_HandleTypeDef::hdmarx***
- ***__IO HAL_LockTypeDef I2S_HandleTypeDef::Lock***
- ***__IO HAL_I2S_StateTypeDef I2S_HandleTypeDef::State***
- ***__IO uint32_t I2S_HandleTypeDef::ErrorCode***

34.2 I2S Firmware driver API description

34.2.1 How to use this driver

The I2S HAL driver can be used as follow:

1. Declare a I2S_HandleTypeDef handle structure.
2. Initialize the I2S low level resources by implement the HAL_I2S_MspInit() API:
   a. Enable the SPIx interface clock.
   b. I2S pins configuration:
      – Enable the clock for the I2S GPIOs.
      – Configure these I2S pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_I2S_Transmit_IT() and HAL_I2S_Receive_IT() APIs).
      – Configure the I2Sx interrupt priority.
      – Enable the NVIC I2S IRQ handle.
   d. DMA Configuration if you need to use DMA process (HAL_I2S_Transmit_DMA() and HAL_I2S_Receive_DMA() APIs:
      – Declare a DMA handle structure for the Tx/Rx stream.
      – Enable the DMAx interface clock.
      – Configure the declared DMA handle structure with the required Tx/Rx parameters.
      – Configure the DMA Tx/Rx Stream.
      – Associate the initialized DMA handle to the I2S DMA Tx/Rx handle.

– Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.

3. Program the Mode, Standard, Data Format, MCLK Output, Audio frequency and Polarity using HAL_I2S_Init() function. The specific I2S interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __I2S_ENABLE_IT() and __I2S_DISABLE_IT() inside the transmit and receive process. Make sure that either: I2S PLL is configured or External clock source is configured after setting correctly the define constant EXTERNAL_CLOCK_VALUE in the stm32f4xx_hal_conf.h file.
4. Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_I2S_Transmit()
- Receive an amount of data in blocking mode using HAL_I2S_Receive()

Interrupt mode IO operation

- Send an amount of data in non blocking mode using HAL_I2S_Transmit_IT()
- At transmission end of half transfer HAL_I2S_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxHalfCpltCallback
- At transmission end of transfer HAL_I2S_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxCpltCallback
- Receive an amount of data in non blocking mode using HAL_I2S_Receive_IT()
- At reception end of half transfer HAL_I2S_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxHalfCpltCallback
- At reception end of transfer HAL_I2S_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxCpltCallback
- In case of transfer Error, HAL_I2S_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2S_ErrorCallback

DMA mode IO operation

- Send an amount of data in non blocking mode (DMA) using HAL_I2S_Transmit_DMA()
- At transmission end of half transfer HAL_I2S_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxHalfCpltCallback
- At transmission end of transfer HAL_I2S_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxCpltCallback
- Receive an amount of data in non blocking mode (DMA) using HAL_I2S_Receive_DMA()
- At reception end of half transfer HAL_I2S_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxHalfCpltCallback
- At reception end of transfer HAL_I2S_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxCpltCallback
- In case of transfer Error, HAL_I2S_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2S_ErrorCallback

- Pause the DMA Transfer using HAL_I2S_DMAPause()
- Resume the DMA Transfer using HAL_I2S_DMAResume()
- Stop the DMA Transfer using HAL_I2S_DMAStop()

I2S HAL driver macros list

Below the list of most used macros in USART HAL driver.

- __HAL_I2S_ENABLE: Enable the specified SPI peripheral (in I2S mode)
- __HAL_I2S_DISABLE: Disable the specified SPI peripheral (in I2S mode)
- __HAL_I2S_ENABLE_IT : Enable the specified I2S interrupts
- __HAL_I2S_DISABLE_IT : Disable the specified I2S interrupts
- __HAL_I2S_GET_FLAG: Check whether the specified I2S flag is set or not

You can refer to the I2S HAL driver header file for more useful macros

34.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and de-initialize the I2Sx peripheral in simplex mode:

- User must Implement HAL_I2S_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).
- Call the function HAL_I2S_Init() to configure the selected device with the selected configuration:
  - Mode
  - Standard
  - Data Format
  - MCLK Output
  - Audio frequency
  - Polarity
- Call the function HAL_I2S_DeInit() to restore the default configuration of the selected I2Sx peripheral.

This section contains the following APIs:

- ***HAL_I2S_Init()***
- ***HAL_I2S_DeInit()***
- ***HAL_I2S_MspInit()***
- ***HAL_I2S_MspDeInit()***

34.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the I2S data transfers.

1. There are two modes of transfer:
   - Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer startup. The end of the data processing will be indicated through the dedicated I2S IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
2. Blocking mode functions are :

- HAL_I2S_Transmit()
- HAL_I2S_Receive()

3. No-Blocking mode functions with Interrupt are :
   - HAL_I2S_Transmit_IT()
   - HAL_I2S_Receive_IT()
4. No-Blocking mode functions with DMA are :
   - HAL_I2S_Transmit_DMA()
   - HAL_I2S_Receive_DMA()
5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   - HAL_I2S_TxCpltCallback()
   - HAL_I2S_RxCpltCallback()
   - HAL_I2S_ErrorCallback()

This section contains the following APIs:

- ***HAL_I2S_Transmit()***
- ***HAL_I2S_Receive()***
- ***HAL_I2S_Transmit_IT()***
- ***HAL_I2S_Receive_IT()***
- ***HAL_I2S_Transmit_DMA()***
- ***HAL_I2S_Receive_DMA()***
- ***HAL_I2S_DMAPause()***
- ***HAL_I2S_DMAResume()***
- ***HAL_I2S_DMAStop()***
- ***HAL_I2S_IRQHandler()***
- ***HAL_I2S_TxHalfCpltCallback()***
- ***HAL_I2S_TxCpltCallback()***
- ***HAL_I2S_RxHalfCpltCallback()***
- ***HAL_I2S_RxCpltCallback()***
- ***HAL_I2S_ErrorCallback()***
- ***HAL_I2S_GetState()***
- ***HAL_I2S_GetError()***

34.2.4 Peripheral State and Errors functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_I2S_GetState()***
- ***HAL_I2S_GetError()***

34.2.5 Detailed description of functions

HAL_I2S_Init

Function Name	**HAL_StatusTypeDef HAL_I2S_Init (I2S_HandleTypeDef * hi2s)**
Function Description	Initializes the I2S according to the specified parameters in the I2S_InitTypeDef and create the associated handle.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

HAL_I2S_DeInit

Function Name	**HAL_StatusTypeDef HAL_I2S_DeInit (I2S_HandleTypeDef * hi2s)**
Function Description	DeInitializes the I2S peripheral.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

HAL_I2S_MspInit

Function Name	**void HAL_I2S_MspInit (I2S_HandleTypeDef * hi2s)**
Function Description	I2S MSP Init.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **None**

HAL_I2S_MspDeInit

Function Name	**void HAL_I2S_MspDeInit (I2S_HandleTypeDef * hi2s)**
Function Description	I2S MSP DeInit.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **None**

HAL_I2S_Transmit

Function Name	**HAL_StatusTypeDef HAL_I2S_Transmit (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Transmit an amount of data in blocking mode.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pData:** a 16-bit pointer to data buffer. • **Size:** number of data sample to be sent: • **Timeout:** Timeout duration
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2S_Receive

Function Name	**HAL_StatusTypeDef HAL_I2S_Receive (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Receive an amount of data in blocking mode.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pData:** a 16-bit pointer to data buffer. • **Size:** number of data sample to be sent: • **Timeout:** Timeout duration
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming). • In I2S Master Receiver mode, just after enabling the peripheral the clock will be generate in continuous way and as the I2S is not disabled at the end of the I2S transaction.

HAL_I2S_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_I2S_Transmit_IT (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size)**
Function Description	Transmit an amount of data in non-blocking mode with Interrupt.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pData:** a 16-bit pointer to data buffer. • **Size:** number of data sample to be sent:
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2S_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_I2S_Receive_IT (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size)**
Function Description	Receive an amount of data in non-blocking mode with Interrupt.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
- **pData:** a 16-bit pointer to the Receive data buffer.
- **Size:** number of data sample to be sent:

Return values
- **HAL:** status

Notes
- When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length.
- The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).
- It is recommended to use DMA for the I2S receiver to avoid de-synchronisation between Master and Slave otherwise the I2S interrupt should be optimized.

HAL_I2S_IRQHandler

Function Name **void HAL_I2S_IRQHandler (I2S_HandleTypeDef * hi2s)**

Function Description This function handles I2S interrupt request.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module

Return values
- **None**

HAL_I2S_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_I2S_Transmit_DMA (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size)**

Function Description Transmit an amount of data in non-blocking mode with DMA.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
- **pData:** a 16-bit pointer to the Transmit data buffer.
- **Size:** number of data sample to be sent:

Return values
- **HAL:** status

Notes
- When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length.
- The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2S_Receive_DMA

Function Name	**HAL_StatusTypeDef HAL_I2S_Receive_DMA (I2S_HandleTypeDef * hi2s, uint16_t * pData, uint16_t Size)**
Function Description	Receive an amount of data in non-blocking mode with DMA.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pData:** a 16-bit pointer to the Receive data buffer. • **Size:** number of data sample to be sent:
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2S_DMAPause

Function Name	**HAL_StatusTypeDef HAL_I2S_DMAPause (I2S_HandleTypeDef * hi2s)**
Function Description	Pauses the audio stream playing from the Media.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

HAL_I2S_DMAResume

Function Name	**HAL_StatusTypeDef HAL_I2S_DMAResume (I2S_HandleTypeDef * hi2s)**
Function Description	Resumes the audio stream playing from the Media.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

HAL_I2S_DMAStop

Function Name	**HAL_StatusTypeDef HAL_I2S_DMAStop (I2S_HandleTypeDef * hi2s)**
Function Description	Resumes the audio stream playing from the Media.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

HAL_I2S_GetState

Function Name	**HAL_I2S_StateTypeDef HAL_I2S_GetState (I2S_HandleTypeDef * hi2s)**
Function Description	Return the I2S state.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** state

HAL_I2S_GetError

Function Name	**uint32_t HAL_I2S_GetError (I2S_HandleTypeDef * hi2s)**
Function Description	Return the I2S error code.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **I2S:** Error Code

HAL_I2S_TxHalfCpltCallback

Function Name	**void HAL_I2S_TxHalfCpltCallback (I2S_HandleTypeDef * hi2s)**
Function Description	Tx Transfer Half completed callbacks.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **None**

HAL_I2S_TxCpltCallback

Function Name	**void HAL_I2S_TxCpltCallback (I2S_HandleTypeDef * hi2s)**
Function Description	Tx Transfer completed callbacks.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **None**

HAL_I2S_RxHalfCpltCallback

Function Name	**void HAL_I2S_RxHalfCpltCallback (I2S_HandleTypeDef * hi2s)**
Function Description	Rx Transfer half completed callbacks.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **None**

HAL_I2S_RxCpltCallback

Function Name	**void HAL_I2S_RxCpltCallback (I2S_HandleTypeDef * hi2s)**
Function Description	Rx Transfer completed callbacks.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module

Return values
- **None**

HAL_I2S_ErrorCallback

Function Name **void HAL_I2S_ErrorCallback (I2S_HandleTypeDef * hi2s)**

Function Description I2S error callbacks.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module

Return values
- **None**

I2S_DMATxCplt

Function Name **void I2S_DMATxCplt (DMA_HandleTypeDef * hdma)**

Function Description DMA I2S transmit process complete callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

I2S_DMATxHalfCplt

Function Name **void I2S_DMATxHalfCplt (DMA_HandleTypeDef * hdma)**

Function Description DMA I2S transmit process half complete callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

I2S_DMARxCplt

Function Name **void I2S_DMARxCplt (DMA_HandleTypeDef * hdma)**

Function Description DMA I2S receive process complete callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

I2S_DMARxHalfCplt

Function Name **void I2S_DMARxHalfCplt (DMA_HandleTypeDef * hdma)**

Function Description DMA I2S receive process half complete callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

I2S_DMAError

Function Name **void I2S_DMAError (DMA_HandleTypeDef * hdma)**

Function Description DMA I2S communication error callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

I2S_WaitFlagStateUntilTimeout

Function Name **HAL_StatusTypeDef I2S_WaitFlagStateUntilTimeout (I2S_HandleTypeDef * hi2s, uint32_t Flag, uint32_t Status, uint32_t Timeout)**

Function Description This function handles I2S Communication Timeout.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
- **Flag:** Flag checked
- **Status:** Value of the flag expected
- **Timeout:** Duration of the timeout

Return values
- **HAL:** status

I2S_Transmit_IT

Function Name **HAL_StatusTypeDef I2S_Transmit_IT (I2S_HandleTypeDef * hi2s)**

Function Description Transmit an amount of data in non-blocking mode with Interrupt.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module

Return values
- **HAL:** status

I2S_Receive_IT

Function Name **HAL_StatusTypeDef I2S_Receive_IT (I2S_HandleTypeDef * hi2s)**

Function Description Receive an amount of data in non-blocking mode with Interrupt.

Parameters
- **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module

Return values
- **HAL:** status

34.3 I2S Firmware driver defines

34.3.1 I2S

I2S Audio Frequency

I2S_AUDIOFREQ_192K

I2S_AUDIOFREQ_96K

I2S_AUDIOFREQ_48K

I2S_AUDIOFREQ_44K

I2S_AUDIOFREQ_32K

I2S_AUDIOFREQ_22K

I2S_AUDIOFREQ_16K

I2S_AUDIOFREQ_11K

I2S_AUDIOFREQ_8K

I2S_AUDIOFREQ_DEFAULT

I2S Clock Polarity

I2S_CPOL_LOW

I2S_CPOL_HIGH

I2S Clock Source

I2S_CLOCK_PLL

I2S_CLOCK_EXTERNAL

I2S Data Format

I2S_DATAFORMAT_16B

I2S_DATAFORMAT_16B_EXTENDED

I2S_DATAFORMAT_24B

I2S_DATAFORMAT_32B

I2S Error Code

HAL_I2S_ERROR_NONE	No error
HAL_I2S_ERROR_UDR	I2S Underrun error
HAL_I2S_ERROR_OVR	I2S Overrun error
HAL_I2SEX_ERROR_UDR	I2S extended Underrun error
HAL_I2SEX_ERROR_OVR	I2S extended Overrun error
HAL_I2S_ERROR_FRE	I2S Frame format error
HAL_I2S_ERROR_DMA	DMA transfer error

I2S Exported Macros

__HAL_I2S_RESET_HANDLE_STATE

Description:

- Reset I2S handle state.

Parameters:

- __HANDLE__: specifies the I2S Handle.

Return value:

- None

__HAL_I2S_ENABLE

Description:

- Enable or disable the specified SPI peripheral (in I2S mode).

Parameters:

- __HANDLE__: specifies the I2S Handle.

Return value:

- None

__HAL_I2S_DISABLE

__HAL_I2S_ENABLE_IT

Description:

- Enable or disable the specified I2S interrupts.

Parameters:

- __HANDLE__: specifies the I2S Handle.
- __INTERRUPT__: specifies the interrupt source to enable or disable. This parameter can be one of the following values:
  - I2S_IT_TXE: Tx buffer empty interrupt enable
  - I2S_IT_RXNE: RX buffer not empty interrupt enable
  - I2S_IT_ERR: Error interrupt enable

Return value:

- None

__HAL_I2S_DISABLE_IT

__HAL_I2S_GET_IT_SOURCE

Description:

- Checks if the specified I2S interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: specifies the I2S Handle. This parameter can be I2S where x: 1, 2, or 3 to select the I2S peripheral.
- __INTERRUPT__: specifies the I2S interrupt source to check. This parameter can be one of the following values:
  - I2S_IT_TXE: Tx buffer empty interrupt enable
  - I2S_IT_RXNE: RX buffer not empty interrupt enable
  - I2S_IT_ERR: Error interrupt enable

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_I2S_GET_FLAG

Description:

- Checks whether the specified I2S flag is set

or not.

Parameters:

- __HANDLE__: specifies the I2S Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - I2S_FLAG_RXNE: Receive buffer not empty flag
  - I2S_FLAG_TXE: Transmit buffer empty flag
  - I2S_FLAG_UDR: Underrun flag
  - I2S_FLAG_OVR: Overrun flag
  - I2S_FLAG_FRE: Frame error flag
  - I2S_FLAG_CHSIDE: Channel Side flag
  - I2S_FLAG_BSY: Busy flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_I2S_CLEAR_OVRFLAG

Description:

- Clears the I2S OVR pending flag.

Parameters:

- __HANDLE__: specifies the I2S Handle.

Return value:

- None

__HAL_I2S_CLEAR_UDRFLAG

Description:

- Clears the I2S UDR pending flag.

Parameters:

- __HANDLE__: specifies the I2S Handle.

Return value:

- None

I2S Flags Definition

I2S_FLAG_TXE

I2S_FLAG_RXNE

I2S_FLAG_UDR

I2S_FLAG_OVR

I2S_FLAG_FRE

I2S_FLAG_CHSIDE

I2S_FLAG_BSY

I2S FullDuplex Mode

I2S_FULLDUPLEXMODE_DISABLE

I2S_FULLDUPLEXMODE_ENABLE

I2S Interrupts Definition

I2S_IT_TXE

I2S_IT_RXNE

I2S_IT_ERR

I2S Mclk Output

I2S_MCLKOUTPUT_ENABLE

I2S_MCLKOUTPUT_DISABLE

I2S Mode

I2S_MODE_SLAVE_TX

I2S_MODE_SLAVE_RX

I2S_MODE_MASTER_TX

I2S_MODE_MASTER_RX

I2S Standard

I2S_STANDARD_PHILIPS

I2S_STANDARD_MSB

I2S_STANDARD_LSB

I2S_STANDARD_PCM_SHORT

I2S_STANDARD_PCM_LONG

35 HAL I2S Extension Driver

35.1 I2SEx Firmware driver API description

35.1.1 I2S Extension features

1. In I2S full duplex mode, each SPI peripheral is able to manage sending and receiving data simultaneously using two data lines. Each SPI peripheral has an extended block called I2Sxext (i.e I2S2ext for SPI2 and I2S3ext for SPI3).
2. The extension block is not a full SPI IP, it is used only as I2S slave to implement full duplex mode. The extension block uses the same clock sources as its master.
3. Both I2Sx and I2Sx_ext can be configured as transmitters or receivers.

Only I2Sx can deliver SCK and WS to I2Sx_ext in full duplex mode, where I2Sx can be I2S2 or I2S3.

35.1.2 How to use this driver

Three operation modes are available within this driver :

Polling mode IO operation

- Send and receive in the same time an amount of data in blocking mode using HAL_I2S_TransmitReceive()

Interrupt mode IO operation

- Send and receive in the same time an amount of data in non blocking mode using HAL_I2S_TransmitReceive_IT()
- At transmission end of half transfer HAL_I2S_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxHalfCpltCallback
- At transmission end of transfer HAL_I2S_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxCpltCallback
- At reception end of half transfer HAL_I2S_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxHalfCpltCallback
- At reception end of transfer HAL_I2S_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxCpltCallback
- In case of transfer Error, HAL_I2S_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2S_ErrorCallback

DMA mode IO operation

- Send and receive an amount of data in non blocking mode (DMA) using HAL_I2S_TransmitReceive_DMA()
- At transmission end of half transfer HAL_I2S_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxHalfCpltCallback
- At transmission end of transfer HAL_I2S_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_TxCpltCallback
- At reception end of half transfer HAL_I2S_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxHalfCpltCallback
- At reception end of transfer HAL_I2S_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_I2S_RxCpltCallback
- In case of transfer Error, HAL_I2S_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_I2S_ErrorCallback
- Pause the DMA Transfer using HAL_I2S_DMAPause()
- Resume the DMA Transfer using HAL_I2S_DMAResume()
- Stop the DMA Transfer using HAL_I2S_DMAStop()

35.1.3 Extension features Functions

This subsection provides a set of functions allowing to manage the I2S data transfers.

1. There are two modes of transfer:
   – Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
   – No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer startup. The end of the data processing will be indicated through the dedicated I2S IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
2. Blocking mode functions are :
   – HAL_I2S_TransmitReceive()
3. No-Blocking mode functions with Interrupt are :
   – HAL_I2S_TransmitReceive_IT()
4. No-Blocking mode functions with DMA are :
   – HAL_I2S_TransmitReceive_DMA()
5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   – HAL_I2S_TxCpltCallback()
   – HAL_I2S_RxCpltCallback()
   – HAL_I2S_ErrorCallback()

This section contains the following APIs:

- ***HAL_I2S_Init()***
- ***HAL_I2SEx_TransmitReceive()***
- ***HAL_I2SEx_TransmitReceive_IT()***
- ***HAL_I2SEx_TransmitReceive_DMA()***
- ***HAL_I2S_DMAPause()***
- ***HAL_I2S_DMAResume()***
- ***HAL_I2S_DMAStop()***
- ***HAL_I2S_IRQHandler()***

35.1.4 Detailed description of functions

HAL_I2SEx_TransmitReceive

Function Name	**HAL_StatusTypeDef HAL_I2SEx_TransmitReceive (I2S_HandleTypeDef * hi2s, uint16_t * pTxData, uint16_t * pRxData, uint16_t Size, uint32_t Timeout)**
Function Description	Full-Duplex Transmit/Receive data in blocking mode.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pTxData:** a 16-bit pointer to the Transmit data buffer. • **pRxData:** a 16-bit pointer to the Receive data buffer. • **Size:** number of data sample to be sent: • **Timeout:** Timeout duration
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2SEx_TransmitReceive_IT

Function Name	**HAL_StatusTypeDef HAL_I2SEx_TransmitReceive_IT (I2S_HandleTypeDef * hi2s, uint16_t * pTxData, uint16_t * pRxData, uint16_t Size)**
Function Description	Full-Duplex Transmit/Receive data in non-blocking mode using Interrupt.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pTxData:** a 16-bit pointer to the Transmit data buffer. • **pRxData:** a 16-bit pointer to the Receive data buffer. • **Size:** number of data sample to be sent:
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

HAL_I2SEx_TransmitReceive_DMA

Function Name	**HAL_StatusTypeDef HAL_I2SEx_TransmitReceive_DMA (I2S_HandleTypeDef * hi2s, uint16_t * pTxData, uint16_t * pRxData, uint16_t Size)**
Function Description	Full-Duplex Transmit/Receive data in non-blocking mode using DMA.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module • **pTxData:** a 16-bit pointer to the Transmit data buffer. • **pRxData:** a 16-bit pointer to the Receive data buffer. • **Size:** number of data sample to be sent:
Return values	• **HAL:** status
Notes	• When a 16-bit data frame or a 16-bit data frame extended is selected during the I2S configuration phase, the Size parameter means the number of 16-bit data length in the transaction and when a 24-bit data frame or a 32-bit data frame is selected the Size parameter means the number of 16-bit data length. • The I2S is kept enabled at the end of transaction to avoid the clock de-synchronization between Master and Slave(example: audio streaming).

I2SEx_TransmitReceive_IT

Function Name	**HAL_StatusTypeDef I2SEx_TransmitReceive_IT (I2S_HandleTypeDef * hi2s)**
Function Description	Full-Duplex Transmit/Receive data in non-blocking mode using Interrupt.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **HAL:** status

I2S_GetInputClock

Function Name	**uint32_t I2S_GetInputClock (I2S_HandleTypeDef * hi2s)**
Function Description	Get I2S clock Input based on Source clock selection in RCC.
Parameters	• **hi2s:** pointer to a I2S_HandleTypeDef structure that contains the configuration information for I2S module
Return values	• **I2S:** Clock Input

36 HAL IRDA Generic Driver

36.1 IRDA Firmware driver registers structures

36.1.1 IRDA_InitTypeDef

Data Fields

- ***uint32_t BaudRate***
- ***uint32_t WordLength***
- ***uint32_t Parity***
- ***uint32_t Mode***
- ***uint8_t Prescaler***
- ***uint32_t IrDAMode***

Field Documentation

- ***uint32_t IRDA_InitTypeDef::BaudRate***
  This member configures the IRDA communication baud rate. The baud rate is computed using the following formula:IntegerDivider = ((PCLKx) / (8 * (hirda->Init.BaudRate)))FractionalDivider = ((IntegerDivider - ((uint32_t) IntegerDivider)) * 8) + 0.5
- ***uint32_t IRDA_InitTypeDef::WordLength***
  Specifies the number of data bits transmitted or received in a frame. This parameter can be a value of ***IRDA_Word_Length***
- ***uint32_t IRDA_InitTypeDef::Parity***
  Specifies the parity mode. This parameter can be a value of ***IRDA_Parity***
  Note: When parity is enabled, the computed parity is inserted at the MSB position of the transmitted data (9th bit when the word length is set to 9 data bits; 8th bit when the word length is set to 8 data bits).
- ***uint32_t IRDA_InitTypeDef::Mode***
  Specifies wether the Receive or Transmit mode is enabled or disabled. This parameter can be a value of ***IRDA_Mode***
- ***uint8_t IRDA_InitTypeDef::Prescaler***
  Specifies the Prescaler
- ***uint32_t IRDA_InitTypeDef::IrDAMode***
  Specifies the IrDA mode This parameter can be a value of ***IRDA_Low_Power***

36.1.2 IRDA_HandleTypeDef

Data Fields

- ***USART_TypeDef * Instance***
- ***IRDA_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferSize***
- ***uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***uint16_t RxXferSize***

- ***uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_IRDA_StateTypeDef gState***
- ***__IO HAL_IRDA_StateTypeDef RxState***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***USART_TypeDef* IRDA_HandleTypeDef::Instance***
- ***IRDA_InitTypeDef IRDA_HandleTypeDef::Init***
- ***uint8_t* IRDA_HandleTypeDef::pTxBuffPtr***
- ***uint16_t IRDA_HandleTypeDef::TxXferSize***
- ***uint16_t IRDA_HandleTypeDef::TxXferCount***
- ***uint8_t* IRDA_HandleTypeDef::pRxBuffPtr***
- ***uint16_t IRDA_HandleTypeDef::RxXferSize***
- ***uint16_t IRDA_HandleTypeDef::RxXferCount***
- ***DMA_HandleTypeDef* IRDA_HandleTypeDef::hdmatx***
- ***DMA_HandleTypeDef* IRDA_HandleTypeDef::hdmarx***
- ***HAL_LockTypeDef IRDA_HandleTypeDef::Lock***
- ***__IO HAL_IRDA_StateTypeDef IRDA_HandleTypeDef::gState***
- ***__IO HAL_IRDA_StateTypeDef IRDA_HandleTypeDef::RxState***
- ***__IO uint32_t IRDA_HandleTypeDef::ErrorCode***

36.2 IRDA Firmware driver API description

36.2.1 How to use this driver

The IRDA HAL driver can be used as follows:

1. Declare a IRDA_HandleTypeDef handle structure.
2. Initialize the IRDA low level resources by implementing the HAL_IRDA_MspInit() API:
   a. Enable the USARTx interface clock.
   b. IRDA pins configuration:
      – Enable the clock for the IRDA GPIOs.
      – Configure these IRDA pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_IRDA_Transmit_IT() and HAL_IRDA_Receive_IT() APIs):
      – Configure the USARTx interrupt priority.
      – Enable the NVIC USART IRQ handle.
   d. DMA Configuration if you need to use DMA process (HAL_IRDA_Transmit_DMA() and HAL_IRDA_Receive_DMA() APIs):
      – Declare a DMA handle structure for the Tx/Rx stream.
      – Enable the DMAx interface clock.
      – Configure the declared DMA handle structure with the required Tx/Rx parameters.
      – Configure the DMA Tx/Rx Stream.
      – Associate the initialized DMA handle to the IRDA DMA Tx/Rx handle.
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.

3. Program the Baud Rate, Word Length, Parity, IrDA Mode, Prescaler and Mode(Receiver/Transmitter) in the hirda Init structure.
4. Initialize the IRDA registers by calling the HAL_IRDA_Init() API:
   – This API configures also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customized HAL_IRDA_MspInit() API. The specific IRDA interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_IRDA_ENABLE_IT() and __HAL_IRDA_DISABLE_IT() inside the transmit and receive process.
5. Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_IRDA_Transmit()
- Receive an amount of data in blocking mode using HAL_IRDA_Receive()

Interrupt mode IO operation

- Send an amount of data in non blocking mode using HAL_IRDA_Transmit_IT()
- At transmission end of transfer HAL_IRDA_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_IRDA_TxCpltCallback
- Receive an amount of data in non blocking mode using HAL_IRDA_Receive_IT()
- At reception end of transfer HAL_IRDA_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_IRDA_RxCpltCallback
- In case of transfer Error, HAL_IRDA_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_IRDA_ErrorCallback

DMA mode IO operation

- Send an amount of data in non blocking mode (DMA) using HAL_IRDA_Transmit_DMA()
- At transmission end of transfer HAL_IRDA_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_IRDA_TxCpltCallback
- Receive an amount of data in non blocking mode (DMA) using HAL_IRDA_Receive_DMA()
- At reception end of transfer HAL_IRDA_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_IRDA_RxCpltCallback
- In case of transfer Error, HAL_IRDA_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_IRDA_ErrorCallback

IRDA HAL driver macros list

You can refer to the IRDA HAL driver header file for more useful macros

36.2.2 Initialization and Configuration functions

This subsection provides a set of functions allowing to initialize the USARTx or the UARTy in IrDA mode.

- For the asynchronous mode only these parameters can be configured:
  - BaudRate
  - WordLength
  - Parity: If the parity is enabled, then the MSB bit of the data written in the data register is transmitted but is changed by the parity bit. Depending on the frame length defined by the M bit (8-bits or 9-bits), please refer to Reference manual for possible IRDA frame formats.
  - Prescaler: A pulse of width less than two and greater than one PSC period(s) may or may not be rejected. The receiver set up time should be managed by software. The IrDA physical layer specification specifies a minimum of 10 ms delay between transmission and reception (IrDA is a half duplex protocol).
  - Mode: Receiver/transmitter modes
  - IrDAMode: the IrDA can operate in the Normal mode or in the Low power mode.

The HAL_IRDA_Init() API follows IRDA configuration procedures (details for the procedures are available in reference manual).

This section contains the following APIs:

- ***HAL_IRDA_Init()***
- ***HAL_IRDA_DeInit()***
- ***HAL_IRDA_MspInit()***
- ***HAL_IRDA_MspDeInit()***

36.2.3 IO operation functions

IrDA is a half duplex communication protocol. If the Transmitter is busy, any data on the IrDA receive line will be ignored by the IrDA decoder and if the Receiver is busy, data on the TX from the USART to IrDA will not be encoded by IrDA. While receiving data, transmission should be avoided as the data to be transmitted could be corrupted.

1. There are two modes of transfer:
   - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode: The communication is performed using Interrupts or DMA, These APIs return the HAL status. The end of the data processing will be indicated through the dedicated IRDA IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_IRDA_TxCpltCallback(), HAL_IRDA_RxCpltCallback() user callbacks will be executed respectively at the end of the transmit or Receive process The HAL_IRDA_ErrorCallback() user callback will be executed when a communication error is detected
2. Blocking mode API's are :
   - HAL_IRDA_Transmit()
   - HAL_IRDA_Receive()
3. Non Blocking mode APIs with Interrupt are :
   - HAL_IRDA_Transmit_IT()
   - HAL_IRDA_Receive_IT()
   - HAL_IRDA_IRQHandler()
4. Non Blocking mode functions with DMA are :
   - HAL_IRDA_Transmit_DMA()

– HAL_IRDA_Receive_DMA()

5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   – HAL_IRDA_TxCpltCallback()
   – HAL_IRDA_RxCpltCallback()
   – HAL_IRDA_ErrorCallback()

This section contains the following APIs:

- ***HAL_IRDA_Transmit()***
- ***HAL_IRDA_Receive()***
- ***HAL_IRDA_Transmit_IT()***
- ***HAL_IRDA_Receive_IT()***
- ***HAL_IRDA_Transmit_DMA()***
- ***HAL_IRDA_Receive_DMA()***
- ***HAL_IRDA_DMAPause()***
- ***HAL_IRDA_DMAResume()***
- ***HAL_IRDA_DMAStop()***
- ***HAL_IRDA_IRQHandler()***
- ***HAL_IRDA_TxCpltCallback()***
- ***HAL_IRDA_TxHalfCpltCallback()***
- ***HAL_IRDA_RxCpltCallback()***
- ***HAL_IRDA_RxHalfCpltCallback()***
- ***HAL_IRDA_ErrorCallback()***

36.2.4 Peripheral State and Errors functions

This subsection provides a set of functions allowing to return the State of IrDA communication process and also return Peripheral Errors occurred during communication process

- HAL_IRDA_GetState() API can be helpful to check in run-time the state of the IrDA peripheral.
- HAL_IRDA_GetError() check in run-time errors that could be occurred during communication.

This section contains the following APIs:

- ***HAL_IRDA_GetState()***
- ***HAL_IRDA_GetError()***

36.2.5 Detailed description of functions

HAL_IRDA_Init

Function Name	**HAL_StatusTypeDef HAL_IRDA_Init (IRDA_HandleTypeDef * hirda)**
Function Description	Initializes the IRDA mode according to the specified parameters in the IRDA_InitTypeDef and create the associated handle.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.
Return values	• **HAL:** status

HAL_IRDA_DeInit

Function Name **HAL_StatusTypeDef HAL_IRDA_DeInit (IRDA_HandleTypeDef * hirda)**

Function Description DeInitializes the IRDA peripheral.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **HAL:** status

HAL_IRDA_MspInit

Function Name **void HAL_IRDA_MspInit (IRDA_HandleTypeDef * hirda)**

Function Description IRDA MSP Init.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_MspDeInit

Function Name **void HAL_IRDA_MspDeInit (IRDA_HandleTypeDef * hirda)**

Function Description IRDA MSP DeInit.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_Transmit

Function Name **HAL_StatusTypeDef HAL_IRDA_Transmit (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Sends an amount of data in blocking mode.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Specify timeout value

Return values
- **HAL:** status

HAL_IRDA_Receive

Function Name **HAL_StatusTypeDef HAL_IRDA_Receive (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description	Receive an amount of data in blocking mode.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received • **Timeout:** Specify timeout value
Return values	• **HAL:** status

HAL_IRDA_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_IRDA_Transmit_IT (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size)**
Function Description	Send an amount of data in non blocking mode.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_IRDA_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_IRDA_Receive_IT (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size)**
Function Description	Receives an amount of data in non blocking mode.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status

HAL_IRDA_Transmit_DMA

Function Name	**HAL_StatusTypeDef HAL_IRDA_Transmit_DMA (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size)**
Function Description	Sends an amount of data in non blocking mode.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_IRDA_Receive_DMA

Function Name	**HAL_StatusTypeDef HAL_IRDA_Receive_DMA (IRDA_HandleTypeDef * hirda, uint8_t * pData, uint16_t Size)**
Function Description	Receives an amount of data in non blocking mode.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status
Notes	• When the IRDA parity is enabled (PCE = 1) the data received contain the parity bit.

HAL_IRDA_DMAPause

Function Name	**HAL_StatusTypeDef HAL_IRDA_DMAPause (IRDA_HandleTypeDef * hirda)**
Function Description	Pauses the DMA Transfer.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.
Return values	• **HAL:** status

HAL_IRDA_DMAResume

Function Name	**HAL_StatusTypeDef HAL_IRDA_DMAResume (IRDA_HandleTypeDef * hirda)**
Function Description	Resumes the DMA Transfer.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.
Return values	• **HAL:** status

HAL_IRDA_DMAStop

Function Name	**HAL_StatusTypeDef HAL_IRDA_DMAStop (IRDA_HandleTypeDef * hirda)**
Function Description	Stops the DMA Transfer.
Parameters	• **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.
Return values	• **HAL:** status

HAL_IRDA_IRQHandler

Function Name **void HAL_IRDA_IRQHandler (IRDA_HandleTypeDef * hirda)**

Function Description This function handles IRDA interrupt request.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_TxCpltCallback

Function Name **void HAL_IRDA_TxCpltCallback (IRDA_HandleTypeDef * hirda)**

Function Description Tx Transfer complete callbacks.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_RxCpltCallback

Function Name **void HAL_IRDA_RxCpltCallback (IRDA_HandleTypeDef * hirda)**

Function Description Rx Transfer complete callbacks.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_TxHalfCpltCallback

Function Name **void HAL_IRDA_TxHalfCpltCallback (IRDA_HandleTypeDef * hirda)**

Function Description Tx Half Transfer completed callbacks.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **None**

HAL_IRDA_RxHalfCpltCallback

Function Name **void HAL_IRDA_RxHalfCpltCallback (IRDA_HandleTypeDef * hirda)**

Function Description Rx Half Transfer complete callbacks.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_ErrorCallback

Function Name **void HAL_IRDA_ErrorCallback (IRDA_HandleTypeDef * hirda)**

Function Description IRDA error callbacks.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **None**

HAL_IRDA_GetState

Function Name **HAL_IRDA_StateTypeDef HAL_IRDA_GetState (IRDA_HandleTypeDef * hirda)**

Function Description Returns the IRDA state.

Parameters
- **hirda:** pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA module.

Return values
- **HAL:** state

HAL_IRDA_GetError

Function Name **uint32_t HAL_IRDA_GetError (IRDA_HandleTypeDef * hirda)**

Function Description Return the IARDA error code.

Parameters
- **hirda:** : pointer to a IRDA_HandleTypeDef structure that contains the configuration information for the specified IRDA.

Return values
- **IRDA:** Error Code

36.3 IRDA Firmware driver defines

36.3.1 IRDA

IRDA Error Code

HAL_IRDA_ERROR_NONE	No error
HAL_IRDA_ERROR_PE	Parity error
HAL_IRDA_ERROR_NE	Noise error
HAL_IRDA_ERROR_FE	Frame error
HAL_IRDA_ERROR_ORE	Overrun error
HAL_IRDA_ERROR_DMA	DMA transfer error

IRDA Exported Macros

__HAL_IRDA_RESET_HANDLE_STATE

Description:

- Reset IRDA handle gstate & RxState.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_FLUSH_DRREGISTER

Description:

- Flushs the IRDA DR register.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

__HAL_IRDA_GET_FLAG

Description:

- Checks whether the specified IRDA flag is set or not.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - IRDA_FLAG_TXE: Transmit data register empty flag
  - IRDA_FLAG_TC: Transmission Complete flag
  - IRDA_FLAG_RXNE: Receive data register not empty flag
  - IRDA_FLAG_IDLE: Idle Line detection flag
  - IRDA_FLAG_ORE: OverRun Error flag
  - IRDA_FLAG_NE: Noise Error flag
  - IRDA_FLAG_FE: Framing Error flag
  - IRDA_FLAG_PE: Parity Error flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_IRDA_CLEAR_FLAG

Description:

- Clears the specified IRDA pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __FLAG__: specifies the flag to check. This parameter can be any combination of the following values:
  - IRDA_FLAG_TC: Transmission Complete flag.
  - IRDA_FLAG_RXNE: Receive data register not empty flag.

Return value:

- None

Notes:

- PE (Parity error), FE (Framing error), NE (Noise error), ORE (OverRun error) and IDLE (Idle line detected) flags are cleared by software sequence: a read operation to USART_SR register followed by a read operation to USART_DR register. RXNE flag can be also cleared by a read to the USART_DR register. TC flag can be also cleared by software sequence: a read operation to USART_SR register followed by a write operation to USART_DR register. TXE flag is cleared only by a write to the USART_DR register.

__HAL_IRDA_CLEAR_PEFLAG

Description:

- Clear the IRDA PE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_CLEAR_FEFLAG

Description:

- Clear the IRDA FE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_CLEAR_NEFLAG

Description:

- Clear the IRDA NE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_CLEAR_OREFLAG

Description:

- Clear the IRDA ORE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_CLEAR_IDLEFLAG

Description:

- Clear the IRDA IDLE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_IRDA_ENABLE_IT

Description:

- Enables or disables the specified IRDA interrupt.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __INTERRUPT__: specifies the IRDA interrupt source to check. This parameter can be one of the following values:
  - IRDA_IT_TXE: Transmit Data Register empty interrupt
  - IRDA_IT_TC: Transmission complete interrupt
  - IRDA_IT_RXNE: Receive Data register not empty interrupt
  - IRDA_IT_IDLE: Idle line detection interrupt
  - IRDA_IT_PE: Parity Error interrupt
  - IRDA_IT_ERR: Error interrupt(Frame error, noise error, overrun error)

Return value:

- None

__HAL_IRDA_DISABLE_IT

__HAL_IRDA_GET_IT_SOURCE

Description:

- Checks whether the specified IRDA interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __IT__: specifies the IRDA interrupt source to check. This parameter can be one of the following values:
  - IRDA_IT_TXE: Transmit Data Register empty interrupt
  - IRDA_IT_TC: Transmission complete interrupt
  - IRDA_IT_RXNE: Receive Data register not empty interrupt
  - IRDA_IT_IDLE: Idle line detection interrupt

– USART_IT_ERR: Error interrupt
– IRDA_IT_PE: Parity Error interrupt

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_IRDA_ONE_BIT_SAMPLE_ENABLE

Description:

- Macro to enable the IRDA's one bit sample method.

Parameters:

- __HANDLE__: specifies the IRDA Handle.

Return value:

- None

__HAL_IRDA_ONE_BIT_SAMPLE_DISABLE

Description:

- Macro to disable the IRDA's one bit sample method.

Parameters:

- __HANDLE__: specifies the IRDA Handle.

Return value:

- None

__HAL_IRDA_ENABLE

Description:

- Enable UART/USART associated to IRDA Handle.

Parameters:

- __HANDLE__: specifies the IRDA Handle. IRDA Handle selects the USARTx or UARTy peripheral (USART,UART availability and x,y values depending on device).

Return value:

- None

__HAL_IRDA_DISABLE

Description:

- Disable UART/USART associated to IRDA Handle.

Parameters:

- __HANDLE__: specifies the IRDA Handle. IRDA Handle selects the USARTx or UARTy peripheral (USART,UART availability and x,y values depending on device).

Return value:

- None

IRDA Flags

IRDA_FLAG_TXE

IRDA_FLAG_TC

IRDA_FLAG_RXNE

IRDA_FLAG_IDLE

IRDA_FLAG_ORE

IRDA_FLAG_NE

IRDA_FLAG_FE

IRDA_FLAG_PE

IRDA Interrupt Definitions

IRDA_IT_PE

IRDA_IT_TXE

IRDA_IT_TC

IRDA_IT_RXNE

IRDA_IT_IDLE

IRDA_IT_LBD

IRDA_IT_CTS

IRDA_IT_ERR

IRDA Low Power

IRDA_POWERMODE_LOWPOWER

IRDA_POWERMODE_NORMAL

IRDA Transfer Mode

IRDA_MODE_RX

IRDA_MODE_TX

IRDA_MODE_TX_RX

IRDA Parity

IRDA_PARITY_NONE

IRDA_PARITY_EVEN

IRDA_PARITY_ODD

IRDA Word Length

IRDA_WORDLENGTH_8B

IRDA_WORDLENGTH_9B

37 HAL IWDG Generic Driver

37.1 IWDG Firmware driver registers structures

37.1.1 IWDG_InitTypeDef

Data Fields

- ***uint32_t Prescaler***
- ***uint32_t Reload***

Field Documentation

- ***uint32_t IWDG_InitTypeDef::Prescaler***
  Select the prescaler of the IWDG. This parameter can be a value of ***IWDG_Prescaler***
- ***uint32_t IWDG_InitTypeDef::Reload***
  Specifies the IWDG down-counter reload value. This parameter must be a number between Min_Data = 0 and Max_Data = 0x0FFF

37.1.2 IWDG_HandleTypeDef

Data Fields

- ***IWDG_TypeDef * Instance***
- ***IWDG_InitTypeDef Init***

Field Documentation

- ***IWDG_TypeDef* IWDG_HandleTypeDef::Instance***
  Register base address
- ***IWDG_InitTypeDef IWDG_HandleTypeDef::Init***
  IWDG required parameters

37.2 IWDG Firmware driver API description

37.2.1 IWDG Generic features

- The IWDG can be started by either software or hardware (configurable through option byte).
- The IWDG is clocked by Low-Speed clock (LSI) and thus stays active even if the main clock fails.
- Once the IWDG is started, the LSI is forced ON and both can not be disabled. The counter starts counting down from the reset value (0xFFF). When it reaches the end of count value (0x000) a reset signal is generated (IWDG reset).

- Whenever the key value 0x0000 AAAA is written in the IWDG_KR register, the IWDG_RLR value is reloaded in the counter and the watchdog reset is prevented.
- The IWDG is implemented in the VDD voltage domain that is still functional in STOP and STANDBY mode (IWDG reset can wake-up from STANDBY). IWDGRST flag in RCC_CSR register can be used to inform when an IWDG reset occurs.
- Debug mode : When the microcontroller enters debug mode (core halted), the IWDG counter either continues to work normally or stops, depending on DBG_IWDG_STOP configuration bit in DBG module, accessible through __HAL_DBGMCU_FREEZE_IWDG() and __HAL_DBGMCU_UNFREEZE_IWDG() macros

Min-max timeout value @32KHz (LSI): ~125us / ~32.7s The IWDG timeout may vary due to LSI frequency dispersion. STM32F4xx devices provide the capability to measure the LSI frequency (LSI clock connected internally to TIM5 CH4 input capture). The measured value can be used to have an IWDG timeout with an acceptable accuracy.

37.2.2 How to use this driver

1. Use IWDG using HAL_IWDG_Init() function to :
   – Enable instance by writing Start keyword in IWDG_KEY register. LSI clock is forced ON and IWDG counter starts downcounting.
   – Enable write access to configuration register: IWDG_PR & IWDG_RLR.
   – Configure the IWDG prescaler and counter reload value. This reload value will be loaded in the IWDG counter each time the watchdog is reloaded, then the IWDG will start counting down from this value.
   – wait for status flags to be reset"
2. Then the application program must refresh the IWDG counter at regular intervals during normal operation to prevent an MCU reset, using HAL_IWDG_Refresh() function.

IWDG HAL driver macros list

Below the list of most used macros in IWDG HAL driver:

- __HAL_IWDG_START: Enable the IWDG peripheral
- __HAL_IWDG_RELOAD_COUNTER: Reloads IWDG counter with value defined in the reload register

37.2.3 Initialization and Start functions

This section provides functions allowing to:

- Initialize the IWDG according to the specified parameters in the IWDG_InitTypeDef of associated handle.
- Once initialization is performed in HAL_IWDG_Init function, Watchdog is reloaded in order to exit function with correct time base.

This section contains the following APIs:

- ***HAL_IWDG_Init()***

37.2.4 IO operation functions

This section provides functions allowing to:

- Refresh the IWDG.

This section contains the following APIs:

- ***HAL_IWDG_Refresh()***

37.2.5 Detailed description of functions

HAL_IWDG_Init

Function Name	**HAL_StatusTypeDef HAL_IWDG_Init (IWDG_HandleTypeDef * hiwdg)**
Function Description	Initialize the IWDG according to the specified parameters in the IWDG_InitTypeDef and start watchdog.
Parameters	• **hiwdg:** pointer to a IWDG_HandleTypeDef structure that contains the configuration information for the specified IWDG module.
Return values	• **HAL:** status

HAL_IWDG_Refresh

Function Name	**HAL_StatusTypeDef HAL_IWDG_Refresh (IWDG_HandleTypeDef * hiwdg)**
Function Description	Refresh the IWDG.
Parameters	• **hiwdg:** pointer to a IWDG_HandleTypeDef structure that contains the configuration information for the specified IWDG module.
Return values	• **HAL:** status

37.3 IWDG Firmware driver defines

37.3.1 IWDG

IWDG Exported Macros

__HAL_IWDG_START

Description:

- Enable the IWDG peripheral.

Parameters:

- __HANDLE__: IWDG handle

Return value:

- None

__HAL_IWDG_RELOAD_COUNTER

Description:

- Reload IWDG counter with value defined in the reload register (write access to IWDG_PR & IWDG_RLR registers disabled).

Parameters:

- __HANDLE__: IWDG handle

Return value:

- None

IWDG Prescaler

IWDG_PRESCALER_4	IWDG prescaler set to 4
IWDG_PRESCALER_8	IWDG prescaler set to 8
IWDG_PRESCALER_16	IWDG prescaler set to 16
IWDG_PRESCALER_32	IWDG prescaler set to 32
IWDG_PRESCALER_64	IWDG prescaler set to 64
IWDG_PRESCALER_128	IWDG prescaler set to 128
IWDG_PRESCALER_256	IWDG prescaler set to 256

38 HAL LPTIM Generic Driver

38.1 LPTIM Firmware driver registers structures

38.1.1 LPTIM_ClockConfigTypeDef

Data Fields

- ***uint32_t Source***
- ***uint32_t Prescaler***

Field Documentation

- ***uint32_t LPTIM_ClockConfigTypeDef::Source***
  Selects the clock source. This parameter can be a value of ***LPTIM_Clock_Source***
- ***uint32_t LPTIM_ClockConfigTypeDef::Prescaler***
  Specifies the counter clock Prescaler. This parameter can be a value of ***LPTIM_Clock_Prescaler***

38.1.2 LPTIM_ULPClockConfigTypeDef

Data Fields

- ***uint32_t Polarity***
- ***uint32_t SampleTime***

Field Documentation

- ***uint32_t LPTIM_ULPClockConfigTypeDef::Polarity***
  Selects the polarity of the active edge for the counter unit if the ULPTIM input is selected. Note: This parameter is used only when Ultra low power clock source is used. Note: If the polarity is configured on 'both edges', an auxiliary clock (one of the Low power oscillator) must be active. This parameter can be a value of ***LPTIM_Clock_Polarity***
- ***uint32_t LPTIM_ULPClockConfigTypeDef::SampleTime***
  Selects the clock sampling time to configure the clock glitch filter. Note: This parameter is used only when Ultra low power clock source is used. This parameter can be a value of ***LPTIM_Clock_Sample_Time***

38.1.3 LPTIM_TriggerConfigTypeDef

Data Fields

- ***uint32_t Source***
- ***uint32_t ActiveEdge***
- ***uint32_t SampleTime***

Field Documentation

- ***uint32_t LPTIM_TriggerConfigTypeDef::Source***
  Selects the Trigger source. This parameter can be a value of ***LPTIM_Trigger_Source***
- ***uint32_t LPTIM_TriggerConfigTypeDef::ActiveEdge***
  Selects the Trigger active edge. Note: This parameter is used only when an external trigger is used. This parameter can be a value of ***LPTIM_External_Trigger_Polarity***
- ***uint32_t LPTIM_TriggerConfigTypeDef::SampleTime***
  Selects the trigger sampling time to configure the clock glitch filter. Note: This parameter is used only when an external trigger is used. This parameter can be a value of ***LPTIM_Trigger_Sample_Time***

38.1.4 LPTIM_InitTypeDef

Data Fields

- ***LPTIM_ClockConfigTypeDef Clock***
- ***LPTIM_ULPClockConfigTypeDef UltraLowPowerClock***
- ***LPTIM_TriggerConfigTypeDef Trigger***
- ***uint32_t OutputPolarity***
- ***uint32_t UpdateMode***
- ***uint32_t CounterSource***

Field Documentation

- ***LPTIM_ClockConfigTypeDef LPTIM_InitTypeDef::Clock***
  Specifies the clock parameters
- ***LPTIM_ULPClockConfigTypeDef LPTIM_InitTypeDef::UltraLowPowerClock***
  Specifies the Ultra Low Power clock parameters
- ***LPTIM_TriggerConfigTypeDef LPTIM_InitTypeDef::Trigger***
  Specifies the Trigger parameters
- ***uint32_t LPTIM_InitTypeDef::OutputPolarity***
  Specifies the Output polarity. This parameter can be a value of ***LPTIM_Output_Polarity***
- ***uint32_t LPTIM_InitTypeDef::UpdateMode***
  Specifies whether the update of the autorelaod and the compare values is done immediately or after the end of current period. This parameter can be a value of ***LPTIM_Updating_Mode***
- ***uint32_t LPTIM_InitTypeDef::CounterSource***
  Specifies whether the counter is incremented each internal event or each external event. This parameter can be a value of ***LPTIM_Counter_Source***

38.1.5 LPTIM_HandleTypeDef

Data Fields

- ***LPTIM_TypeDef * Instance***
- ***LPTIM_InitTypeDef Init***

- ***HAL_StatusTypeDef Status***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_LPTIM_StateTypeDef State***

Field Documentation

- ***LPTIM_TypeDef* LPTIM_HandleTypeDef::Instance***
  Register base address
- ***LPTIM_InitTypeDef LPTIM_HandleTypeDef::Init***
  LPTIM required parameters
- ***HAL_StatusTypeDef LPTIM_HandleTypeDef::Status***
  LPTIM peripheral status
- ***HAL_LockTypeDef LPTIM_HandleTypeDef::Lock***
  LPTIM locking object
- ***__IO HAL_LPTIM_StateTypeDef LPTIM_HandleTypeDef::State***
  LPTIM peripheral state

38.2 LPTIM Firmware driver API description

38.2.1 How to use this driver

The LPTIM HAL driver can be used as follows:

1. Initialize the LPTIM low level resources by implementing the HAL_LPTIM_MspInit():
   a. Enable the LPTIM interface clock using __LPTIMx_CLK_ENABLE().
   b. In case of using interrupts (e.g. HAL_LPTIM_PWM_Start_IT()):
      – Configure the LPTIM interrupt priority using HAL_NVIC_SetPriority().
      – Enable the LPTIM IRQ handler using HAL_NVIC_EnableIRQ().
      – In LPTIM IRQ handler, call HAL_LPTIM_IRQHandler().
2. Initialize the LPTIM HAL using HAL_LPTIM_Init(). This function configures mainly:
   a. The instance: LPTIM1.
   b. Clock: the counter clock.
      – Source : it can be either the ULPTIM input (IN1) or one of the internal clock; (APB, LSE or LSI).
      – Prescaler: select the clock divider.
   c. UltraLowPowerClock : To be used only if the ULPTIM is selected as counter clock source.
      – Polarity: polarity of the active edge for the counter unit if the ULPTIM input is selected.
      – SampleTime: clock sampling time to configure the clock glitch filter.
   d. Trigger: How the counter start.
      – Source: trigger can be software or one of the hardware triggers.
      – ActiveEdge : only for hardware trigger.
      – SampleTime : trigger sampling time to configure the trigger glitch filter.
   e. OutputPolarity : 2 opposite polarities are possibles.
   f. UpdateMode: specifies whether the update of the autoreload and the compare values is done immediately or after the end of current period.
3. Six modes are available:
   a. PWM Mode: To generate a PWM signal with specified period and pulse, call HAL_LPTIM_PWM_Start() or HAL_LPTIM_PWM_Start_IT() for interruption mode.

b. One Pulse Mode: To generate pulse with specified width in response to a stimulus, call HAL_LPTIM_OnePulse_Start() or HAL_LPTIM_OnePulse_Start_IT() for interruption mode.
c. Set once Mode: In this mode, the output changes the level (from low level to high level if the output polarity is configured high, else the opposite) when a compare match occurs. To start this mode, call HAL_LPTIM_SetOnce_Start() or HAL_LPTIM_SetOnce_Start_IT() for interruption mode.
d. Encoder Mode: To use the encoder interface call HAL_LPTIM_Encoder_Start() or HAL_LPTIM_Encoder_Start_IT() for interruption mode.
e. Time out Mode: an active edge on one selected trigger input rests the counter. The first trigger event will start the timer, any successive trigger event will reset the counter and the timer will restart. To start this mode call HAL_LPTIM_TimeOut_Start_IT() or HAL_LPTIM_TimeOut_Start_IT() for interruption mode.
f. Counter Mode: counter can be used to count external events on the LPTIM Input1 or it can be used to count internal clock cycles. To start this mode, call HAL_LPTIM_Counter_Start() or HAL_LPTIM_Counter_Start_IT() for interruption mode.

4. User can stop any process by calling the corresponding API: HAL_LPTIM_Xxx_Stop() or HAL_LPTIM_Xxx_Stop_IT() if the process is already started in interruption mode.
5. Call HAL_LPTIM_DeInit() to deinitialize the LPTIM peripheral.

38.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the LPTIM according to the specified parameters in the LPTIM_InitTypeDef and creates the associated handle.
- DeInitialize the LPTIM peripheral.
- Initialize the LPTIM MSP.
- DeInitialize LPTIM MSP.

This section contains the following APIs:

- ***HAL_LPTIM_Init()***
- ***HAL_LPTIM_DeInit()***
- ***HAL_LPTIM_MspInit()***
- ***HAL_LPTIM_MspDeInit()***

38.2.3 LPTIM Start Stop operation functions

This section provides functions allowing to:

- Start the PWM mode.
- Stop the PWM mode.
- Start the One pulse mode.
- Stop the One pulse mode.
- Start the Set once mode.
- Stop the Set once mode.
- Start the Encoder mode.
- Stop the Encoder mode.
- Start the Timeout mode.
- Stop the Timeout mode.
- Start the Counter mode.
- Stop the Counter mode.

This section contains the following APIs:

- ***HAL_LPTIM_PWM_Start()***
- ***HAL_LPTIM_PWM_Stop()***
- ***HAL_LPTIM_PWM_Start_IT()***
- ***HAL_LPTIM_PWM_Stop_IT()***
- ***HAL_LPTIM_OnePulse_Start()***
- ***HAL_LPTIM_OnePulse_Stop()***
- ***HAL_LPTIM_OnePulse_Start_IT()***
- ***HAL_LPTIM_OnePulse_Stop_IT()***
- ***HAL_LPTIM_SetOnce_Start()***
- ***HAL_LPTIM_SetOnce_Stop()***
- ***HAL_LPTIM_SetOnce_Start_IT()***
- ***HAL_LPTIM_SetOnce_Stop_IT()***
- ***HAL_LPTIM_Encoder_Start()***
- ***HAL_LPTIM_Encoder_Stop()***
- ***HAL_LPTIM_Encoder_Start_IT()***
- ***HAL_LPTIM_Encoder_Stop_IT()***
- ***HAL_LPTIM_TimeOut_Start()***
- ***HAL_LPTIM_TimeOut_Stop()***
- ***HAL_LPTIM_TimeOut_Start_IT()***
- ***HAL_LPTIM_TimeOut_Stop_IT()***
- ***HAL_LPTIM_Counter_Start()***
- ***HAL_LPTIM_Counter_Stop()***
- ***HAL_LPTIM_Counter_Start_IT()***
- ***HAL_LPTIM_Counter_Stop_IT()***

38.2.4 LPTIM Read operation functions

This section provides LPTIM Reading functions.

- Read the counter value.
- Read the period (Auto-reload) value.
- Read the pulse (Compare)value.

This section contains the following APIs:

- ***HAL_LPTIM_ReadCounter()***
- ***HAL_LPTIM_ReadAutoReload()***
- ***HAL_LPTIM_ReadCompare()***

38.2.5 LPTIM IRQ handler

This section provides LPTIM IRQ handler function.

This section contains the following APIs:

- ***HAL_LPTIM_IRQHandler()***
- ***HAL_LPTIM_CompareMatchCallback()***
- ***HAL_LPTIM_AutoReloadMatchCallback()***
- ***HAL_LPTIM_TriggerCallback()***
- ***HAL_LPTIM_CompareWriteCallback()***
- ***HAL_LPTIM_AutoReloadWriteCallback()***
- ***HAL_LPTIM_DirectionUpCallback()***
- ***HAL_LPTIM_DirectionDownCallback()***

38.2.6 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral.

This section contains the following APIs:

- ***HAL_LPTIM_GetState()***

38.2.7 Detailed description of functions

HAL_LPTIM_Init

Function Name	**HAL_StatusTypeDef HAL_LPTIM_Init (LPTIM_HandleTypeDef * hlptim)**
Function Description	Initializes the LPTIM according to the specified parameters in the LPTIM_InitTypeDef and creates the associated handle.
Parameters	• **hlptim:** LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_DeInit

Function Name	**HAL_StatusTypeDef HAL_LPTIM_DeInit (LPTIM_HandleTypeDef * hlptim)**
Function Description	DeInitializes the LPTIM peripheral.
Parameters	• **hlptim:** LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_MspInit

Function Name	**void HAL_LPTIM_MspInit (LPTIM_HandleTypeDef * hlptim)**
Function Description	Initializes the LPTIM MSP.
Parameters	• **hlptim:** LPTIM handle
Return values	• **None**

HAL_LPTIM_MspDeInit

Function Name	**void HAL_LPTIM_MspDeInit (LPTIM_HandleTypeDef * hlptim)**
Function Description	DeInitializes LPTIM MSP.
Parameters	• **hlptim:** LPTIM handle
Return values	• **None**

HAL_LPTIM_PWM_Start

Function Name	**HAL_StatusTypeDef HAL_LPTIM_PWM_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**
Function Description	Starts the LPTIM PWM generation.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_PWM_Stop

Function Name **HAL_StatusTypeDef HAL_LPTIM_PWM_Stop (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the LPTIM PWM generation.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_PWM_Start_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_PWM_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**

Function Description Starts the LPTIM PWM generation in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF
- **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF

Return values
- **HAL:** status

HAL_LPTIM_PWM_Stop_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_PWM_Stop_IT (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the LPTIM PWM generation in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_OnePulse_Start

Function Name **HAL_StatusTypeDef HAL_LPTIM_OnePulse_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**

Function Description Starts the LPTIM One pulse generation.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_OnePulse_Stop

Function Name **HAL_StatusTypeDef HAL_LPTIM_OnePulse_Stop (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the LPTIM One pulse generation.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_OnePulse_Start_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_OnePulse_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**

Function Description Starts the LPTIM One pulse generation in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_OnePulse_Stop_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_OnePulse_Stop_IT (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the LPTIM One pulse generation in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_SetOnce_Start

Function Name **HAL_StatusTypeDef HAL_LPTIM_SetOnce_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**

Function Description Starts the LPTIM in Set once mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_SetOnce_Stop

Function Name	**HAL_StatusTypeDef HAL_LPTIM_SetOnce_Stop (LPTIM_HandleTypeDef * hlptim)**
Function Description	Stops the LPTIM Set once mode.
Parameters	• **hlptim:** : LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_SetOnce_Start_IT

Function Name	**HAL_StatusTypeDef HAL_LPTIM_SetOnce_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Pulse)**
Function Description	Starts the LPTIM Set once mode in interrupt mode.
Parameters	• **hlptim:** : LPTIM handle • **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF. • **Pulse:** : Specifies the compare value. This parameter must be a value between 0x0000 and 0xFFFF.
Return values	• **HAL:** status

HAL_LPTIM_SetOnce_Stop_IT

Function Name	**HAL_StatusTypeDef HAL_LPTIM_SetOnce_Stop_IT (LPTIM_HandleTypeDef * hlptim)**
Function Description	Stops the LPTIM Set once mode in interrupt mode.
Parameters	• **hlptim:** : LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_Encoder_Start

Function Name	**HAL_StatusTypeDef HAL_LPTIM_Encoder_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period)**
Function Description	Starts the Encoder interface.
Parameters	• **hlptim:** : LPTIM handle • **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
Return values	• **HAL:** status

HAL_LPTIM_Encoder_Stop

Function Name	**HAL_StatusTypeDef HAL_LPTIM_Encoder_Stop (LPTIM_HandleTypeDef * hlptim)**
Function Description	Stops the Encoder interface.
Parameters	• **hlptim:** : LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_Encoder_Start_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_Encoder_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period)**

Function Description Starts the Encoder interface in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_Encoder_Stop_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_Encoder_Stop_IT (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the Encoder interface in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_TimeOut_Start

Function Name **HAL_StatusTypeDef HAL_LPTIM_TimeOut_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Timeout)**

Function Description Starts the Timeout function.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Timeout:** : Specifies the TimeOut value to rest the counter. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_TimeOut_Stop

Function Name **HAL_StatusTypeDef HAL_LPTIM_TimeOut_Stop (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the Timeout function.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_TimeOut_Start_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_TimeOut_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period, uint32_t Timeout)**

Function Description Starts the Timeout function in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle

- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.
- **Timeout:** : Specifies the TimeOut value to rest the counter. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_TimeOut_Stop_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_TimeOut_Stop_IT (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the Timeout function in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_Counter_Start

Function Name **HAL_StatusTypeDef HAL_LPTIM_Counter_Start (LPTIM_HandleTypeDef * hlptim, uint32_t Period)**

Function Description Starts the Counter mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_Counter_Stop

Function Name **HAL_StatusTypeDef HAL_LPTIM_Counter_Stop (LPTIM_HandleTypeDef * hlptim)**

Function Description Stops the Counter mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **HAL:** status

HAL_LPTIM_Counter_Start_IT

Function Name **HAL_StatusTypeDef HAL_LPTIM_Counter_Start_IT (LPTIM_HandleTypeDef * hlptim, uint32_t Period)**

Function Description Starts the Counter mode in interrupt mode.

Parameters
- **hlptim:** : LPTIM handle
- **Period:** : Specifies the Autoreload value. This parameter must be a value between 0x0000 and 0xFFFF.

Return values
- **HAL:** status

HAL_LPTIM_Counter_Stop_IT

Function Name	**HAL_StatusTypeDef HAL_LPTIM_Counter_Stop_IT (LPTIM_HandleTypeDef * hlptim)**
Function Description	Stops the Counter mode in interrupt mode.
Parameters	• **hlptim:** : LPTIM handle
Return values	• **HAL:** status

HAL_LPTIM_ReadCounter

Function Name	**uint32_t HAL_LPTIM_ReadCounter (LPTIM_HandleTypeDef * hlptim)**
Function Description	This function returns the current counter value.
Parameters	• **hlptim:** LPTIM handle
Return values	• **Counter:** value.

HAL_LPTIM_ReadAutoReload

Function Name	**uint32_t HAL_LPTIM_ReadAutoReload (LPTIM_HandleTypeDef * hlptim)**
Function Description	This function return the current Autoreload (Period) value.
Parameters	• **hlptim:** LPTIM handle
Return values	• **Autoreload:** value.

HAL_LPTIM_ReadCompare

Function Name	**uint32_t HAL_LPTIM_ReadCompare (LPTIM_HandleTypeDef * hlptim)**
Function Description	This function return the current Compare (Pulse) value.
Parameters	• **hlptim:** LPTIM handle
Return values	• **Compare:** value.

HAL_LPTIM_IRQHandler

Function Name	**void HAL_LPTIM_IRQHandler (LPTIM_HandleTypeDef * hlptim)**
Function Description	This function handles LPTIM interrupt request.
Parameters	• **hlptim:** LPTIM handle
Return values	• **None**

HAL_LPTIM_CompareMatchCallback

Function Name	**void HAL_LPTIM_CompareMatchCallback (LPTIM_HandleTypeDef * hlptim)**
Function Description	Compare match callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_AutoReloadMatchCallback

Function Name **void HAL_LPTIM_AutoReloadMatchCallback (LPTIM_HandleTypeDef * hlptim)**

Function Description Autoreload match callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_TriggerCallback

Function Name **void HAL_LPTIM_TriggerCallback (LPTIM_HandleTypeDef * hlptim)**

Function Description Trigger detected callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_CompareWriteCallback

Function Name **void HAL_LPTIM_CompareWriteCallback (LPTIM_HandleTypeDef * hlptim)**

Function Description Compare write callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_AutoReloadWriteCallback

Function Name **void HAL_LPTIM_AutoReloadWriteCallback (LPTIM_HandleTypeDef * hlptim)**

Function Description Autoreload write callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_DirectionUpCallback

Function Name **void HAL_LPTIM_DirectionUpCallback (LPTIM_HandleTypeDef * hlptim)**

Function Description Direction counter changed from Down to Up callback in non blocking mode.

Parameters
- **hlptim:** : LPTIM handle

Return values
- **None**

HAL_LPTIM_DirectionDownCallback

Function Name	**void HAL_LPTIM_DirectionDownCallback (LPTIM_HandleTypeDef * hlptim)**
Function Description	Direction counter changed from Up to Down callback in non blocking mode.
Parameters	• **hlptim:** : LPTIM handle
Return values	• **None**

HAL_LPTIM_GetState

Function Name	**HAL_LPTIM_StateTypeDef HAL_LPTIM_GetState (LPTIM_HandleTypeDef * hlptim)**
Function Description	Returns the LPTIM state.
Parameters	• **hlptim:** LPTIM handle
Return values	• **HAL:** state

38.3 LPTIM Firmware driver defines

38.3.1 LPTIM

LPTIM Clock Polarity

LPTIM_CLOCKPOLARITY_RISING

LPTIM_CLOCKPOLARITY_FALLING

LPTIM_CLOCKPOLARITY_RISING_FALLING

LPTIM Clock Prescaler

LPTIM_PRESCALER_DIV1

LPTIM_PRESCALER_DIV2

LPTIM_PRESCALER_DIV4

LPTIM_PRESCALER_DIV8

LPTIM_PRESCALER_DIV16

LPTIM_PRESCALER_DIV32

LPTIM_PRESCALER_DIV64

LPTIM_PRESCALER_DIV128

LPTIM Clock Sample Time

LPTIM_CLOCKSAMPLETIME_DIRECTTRANSITION

LPTIM_CLOCKSAMPLETIME_2TRANSITIONS

LPTIM_CLOCKSAMPLETIME_4TRANSITIONS

LPTIM_CLOCKSAMPLETIME_8TRANSITIONS

LPTIM Clock Source

LPTIM_CLOCKSOURCE_APBCLOCK_LPOSC

LPTIM_CLOCKSOURCE_ULPTIM

LPTIM Counter Source

LPTIM_COUNTERSOURCE_INTERNAL

LPTIM_COUNTERSOURCE_EXTERNAL

LPTIM Exported Macros

__HAL_LPTIM_RESET_HANDLE_STATE

Description:

- Reset LPTIM handle state.

Parameters:

- __HANDLE__: LPTIM handle

Return value:

- None

__HAL_LPTIM_ENABLE

Description:

- Enable/Disable the LPTIM peripheral.

Parameters:

- __HANDLE__: LPTIM handle

Return value:

- None

__HAL_LPTIM_DISABLE

__HAL_LPTIM_START_CONTINUOUS

Description:

- Starts the LPTIM peripheral in Continuous or in single mode.

Parameters:

- __HANDLE__: DMA handle

Return value:

- None

__HAL_LPTIM_START_SINGLE

__HAL_LPTIM_AUTORELOAD_SET

Description:

- Writes the passed parameter in the Autoreload register.

Parameters:

- __HANDLE__: LPTIM handle
- __VALUE__: : Autoreload value

Return value:

- None

__HAL_LPTIM_COMPARE_SET

Description:

- Writes the passed parameter in the Compare register.

Parameters:

- __HANDLE__: LPTIM handle
- __VALUE__: : Compare value

Return value:

- None

__HAL_LPTIM_GET_FLAG

Description:

- Checks whether the specified LPTIM flag is set or not.

Parameters:

- __HANDLE__: LPTIM handle
- __FLAG__: : LPTIM flag to check This parameter can be a value of:
  - LPTIM_FLAG_DOWN : Counter direction change up Flag.
  - LPTIM_FLAG_UP : Counter direction change down to up Flag.
  - LPTIM_FLAG_ARROK : Autoreload register update OK Flag.
  - LPTIM_FLAG_CMPOK : Compare register update OK Flag.
  - LPTIM_FLAG_EXTTRIG : External trigger edge event Flag.
  - LPTIM_FLAG_ARR

M : Autoreload match Flag.
  - LPTIM_FLAG_CMPM : Compare match Flag.

Return value:

- The: state of the specified flag (SET or RESET).

__HAL_LPTIM_CLEAR_FLAG

Description:

- Clears the specified LPTIM flag.

Parameters:

- __HANDLE__: LPTIM handle.
- __FLAG__: : LPTIM flag to clear. This parameter can be a value of:
  - LPTIM_FLAG_DOWN : Counter direction change up Flag.
  - LPTIM_FLAG_UP : Counter direction change down to up Flag.
  - LPTIM_FLAG_ARROK : Autoreload register update OK Flag.
  - LPTIM_FLAG_CMPOK : Compare register update OK Flag.
  - LPTIM_FLAG_EXTTRIG : External trigger edge event Flag.
  - LPTIM_FLAG_ARRM : Autoreload match Flag.
  - LPTIM_FLAG_CMPM : Compare match Flag.

Return value:

- None.

__HAL_LPTIM_ENABLE_IT

Description:

- Enable the specified LPTIM interrupt.

Parameters:

- __HANDLE__: : LPTIM handle.
- __INTERRUPT__: : LPTIM interrupt to set. This parameter can be a value of:
  - LPTIM_IT_DOWN : Counter direction change up Interrupt.
  - LPTIM_IT_UP : Counter direction change down to up Interrupt.
  - LPTIM_IT_ARROK : Autoreload register update OK Interrupt.
  - LPTIM_IT_CMPOK : Compare register update OK Interrupt.
  - LPTIM_IT_EXTTRIG : External trigger edge event Interrupt.
  - LPTIM_IT_ARRM : Autoreload match Interrupt.
  - LPTIM_IT_CMPM : Compare match Interrupt.

Return value:

- None.

__HAL_LPTIM_DISABLE_IT

Description:

- Disable the specified LPTIM interrupt.

Parameters:

- __HANDLE__: : LPTIM handle.
- __INTERRUPT__: : LPTIM interrupt to set. This parameter can be a value of:
  - LPTIM_IT_DOWN : Counter direction change up Interrupt.
  - LPTIM_IT_UP : Counter direction change down to up Interrupt.

– LPTIM_IT_ARROK : Autoreload register update OK Interrupt.
– LPTIM_IT_CMPOK : Compare register update OK Interrupt.
– LPTIM_IT_EXTTRIG : External trigger edge event Interrupt.
– LPTIM_IT_ARRM : Autoreload match Interrupt.
– LPTIM_IT_CMPM : Compare match Interrupt.

Return value:

- None.

__HAL_LPTIM_GET_IT_SOURCE

Description:

- Checks whether the specified LPTIM interrupt is set or not.

Parameters:

- __HANDLE__: : LPTIM handle.
- __INTERRUPT__: : LPTIM interrupt to check. This parameter can be a value of:
  – LPTIM_IT_DOWN : Counter direction change up Interrupt.
  – LPTIM_IT_UP : Counter direction change down to up Interrupt.
  – LPTIM_IT_ARROK : Autoreload register update OK Interrupt.
  – LPTIM_IT_CMPOK : Compare register update OK Interrupt.
  – LPTIM_IT_EXTTRIG : External trigger edge event Interrupt.
  – LPTIM_IT_ARRM :

Autoreload match Interrupt.
  - LPTIM_IT_CMPM : Compare match Interrupt.

Return value:

- Interrupt: status.

__HAL_LPTIM_OPTR_CONFIG

Description:

- LPTIM Option Register.

Parameters:

- __HANDLE__: LPTIM handle
- __VALUE__: This parameter can be a value of :
  - LPTIM_OP_PAD_AF
  - LPTIM_OP_PAD_PA4
  - LPTIM_OP_PAD_PB9
  - LPTIM_OP_TIM_DAC

Return value:

- None

__HAL_LPTIM_WAKEUPTIMER_EXTI_ENABLE_IT

Description:

- Enable interrupt on the LPTIM Wake-up Timer associated Exti line.

Return value:

- None

__HAL_LPTIM_WAKEUPTIMER_EXTI_DISABLE_IT

Description:

- Disable interrupt on the LPTIM Wake-up Timer associated Exti line.

Return value:

- None

__HAL_LPTIM_WAKEUPTIMER_EXTI_ENABLE_EVENT

Description:

- Enable event on the LPTIM Wake-up Timer associated Exti line.

Return value:

- None.

__HAL_LPTIM_WAKEUPTIMER_EXTI_DISABLE_EVENT	**Description:** • Disable event on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_ENABLE_FALLING_EDGE	**Description:** • Enable falling edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_DISABLE_FALLING_EDGE	**Description:** • Disable falling edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_ENABLE_RISING_EDGE	**Description:** • Enable rising edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_DISABLE_RISING_EDGE	**Description:** • Disable rising edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_ENABLE_RISING_FALLING_EDGE	**Description:** • Enable rising & falling edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_DISABLE_RISIN	**Description:**

G_FALLING_EDGE	• Disable rising & falling edge trigger on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_GET_FLAG	**Description:** • Check whether the LPTIM Wake-up Timer associated Exti line interrupt flag is set or not. **Return value:** • Line: Status.
__HAL_LPTIM_WAKEUPTIMER_EXTI_CLEAR_FLAG	**Description:** • Clear the LPTIM Wake-up Timer associated Exti line flag. **Return value:** • None.
__HAL_LPTIM_WAKEUPTIMER_EXTI_GENERATE_SWIT	**Description:** • Generate a Software interrupt on the LPTIM Wake-up Timer associated Exti line. **Return value:** • None.

LPTIM External Trigger Polarity

LPTIM_ACTIVEEDGE_RISING

LPTIM_ACTIVEEDGE_FALLING

LPTIM_ACTIVEEDGE_RISING_FALLING

LPTIM Flag Definition

LPTIM_FLAG_DOWN

LPTIM_FLAG_UP

LPTIM_FLAG_ARROK

LPTIM_FLAG_CMPOK

LPTIM_FLAG_EXTTRIG

LPTIM_FLAG_ARRM

LPTIM_FLAG_CMPM

LPTIM Interrupts Definition

LPTIM_IT_DOWN

LPTIM_IT_UP

LPTIM_IT_ARROK

LPTIM_IT_CMPOK

LPTIM_IT_EXTTRIG

LPTIM_IT_ARRM

LPTIM_IT_CMPM

Register Definition

LPTIM_OP_PAD_AF

LPTIM_OP_PAD_PA4

LPTIM_OP_PAD_PB9

LPTIM_OP_TIM_DAC

LPTIM Output Polarity

LPTIM_OUTPUTPOLARITY_HIGH

LPTIM_OUTPUTPOLARITY_LOW

LPTIM Trigger Sample Time

LPTIM_TRIGSAMPLETIME_DIRECTTRANSITION

LPTIM_TRIGSAMPLETIME_2TRANSITIONS

LPTIM_TRIGSAMPLETIME_4TRANSITIONS

LPTIM_TRIGSAMPLETIME_8TRANSITIONS

LPTIM Trigger Source

LPTIM_TRIGSOURCE_SOFTWARE

LPTIM_TRIGSOURCE_0

LPTIM_TRIGSOURCE_1

LPTIM_TRIGSOURCE_2

LPTIM_TRIGSOURCE_3

LPTIM_TRIGSOURCE_4

LPTIM_TRIGSOURCE_5

LPTIM Updating Mode

LPTIM_UPDATE_IMMEDIATE

LPTIM_UPDATE_ENDOFPERIOD

LPTIM WAKEUP Timer EXTI Line

LPTIM_EXTI_LINE_WAKEUPTIMER_EVENT	External interrupt line 23 Connected to the LPTIM EXTI Line

39 HAL LTDC Generic Driver

39.1 LTDC Firmware driver registers structures

39.1.1 LTDC_ColorTypeDef

Data Fields

- ***uint8_t Blue***
- ***uint8_t Green***
- ***uint8_t Red***
- ***uint8_t Reserved***

Field Documentation

- ***uint8_t LTDC_ColorTypeDef::Blue***
  Configures the blue value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint8_t LTDC_ColorTypeDef::Green***
  Configures the green value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint8_t LTDC_ColorTypeDef::Red***
  Configures the red value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint8_t LTDC_ColorTypeDef::Reserved***
  Reserved 0xFF

39.1.2 LTDC_InitTypeDef

Data Fields

- ***uint32_t HSPolarity***
- ***uint32_t VSPolarity***
- ***uint32_t DEPolarity***
- ***uint32_t PCPolarity***
- ***uint32_t HorizontalSync***
- ***uint32_t VerticalSync***
- ***uint32_t AccumulatedHBP***
- ***uint32_t AccumulatedVBP***
- ***uint32_t AccumulatedActiveW***
- ***uint32_t AccumulatedActiveH***
- ***uint32_t TotalWidth***
- ***uint32_t TotalHeigh***
- ***LTDC_ColorTypeDef Backcolor***

Field Documentation

- ***uint32_t LTDC_InitTypeDef::HSPolarity***
  configures the horizontal synchronization polarity. This parameter can be one value of ***LTDC_HS_POLARITY***
- ***uint32_t LTDC_InitTypeDef::VSPolarity***
  configures the vertical synchronization polarity. This parameter can be one value of ***LTDC_VS_POLARITY***
- ***uint32_t LTDC_InitTypeDef::DEPolarity***
  configures the data enable polarity. This parameter can be one of value of ***LTDC_DE_POLARITY***
- ***uint32_t LTDC_InitTypeDef::PCPolarity***
  configures the pixel clock polarity. This parameter can be one of value of ***LTDC_PC_POLARITY***
- ***uint32_t LTDC_InitTypeDef::HorizontalSync***
  configures the number of Horizontal synchronization width. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0xFFF.
- ***uint32_t LTDC_InitTypeDef::VerticalSync***
  configures the number of Vertical synchronization height. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0x7FF.
- ***uint32_t LTDC_InitTypeDef::AccumulatedHBP***
  configures the accumulated horizontal back porch width. This parameter must be a number between Min_Data = LTDC_HorizontalSync and Max_Data = 0xFFF.
- ***uint32_t LTDC_InitTypeDef::AccumulatedVBP***
  configures the accumulated vertical back porch height. This parameter must be a number between Min_Data = LTDC_VerticalSync and Max_Data = 0x7FF.
- ***uint32_t LTDC_InitTypeDef::AccumulatedActiveW***
  configures the accumulated active width. This parameter must be a number between Min_Data = LTDC_AccumulatedHBP and Max_Data = 0xFFF.
- ***uint32_t LTDC_InitTypeDef::AccumulatedActiveH***
  configures the accumulated active height. This parameter must be a number between Min_Data = LTDC_AccumulatedVBP and Max_Data = 0x7FF.
- ***uint32_t LTDC_InitTypeDef::TotalWidth***
  configures the total width. This parameter must be a number between Min_Data = LTDC_AccumulatedActiveW and Max_Data = 0xFFF.
- ***uint32_t LTDC_InitTypeDef::TotalHeigh***
  configures the total height. This parameter must be a number between Min_Data = LTDC_AccumulatedActiveH and Max_Data = 0x7FF.
- ***LTDC_ColorTypeDef LTDC_InitTypeDef::Backcolor***
  Configures the background color.

39.1.3 LTDC_LayerCfgTypeDef

Data Fields

- ***uint32_t WindowX0***
- ***uint32_t WindowX1***
- ***uint32_t WindowY0***
- ***uint32_t WindowY1***
- ***uint32_t PixelFormat***
- ***uint32_t Alpha***

- ***uint32_t Alpha0***
- ***uint32_t BlendingFactor1***
- ***uint32_t BlendingFactor2***
- ***uint32_t FBStartAdress***
- ***uint32_t ImageWidth***
- ***uint32_t ImageHeight***
- ***LTDC_ColorTypeDef Backcolor***

Field Documentation

- ***uint32_t LTDC_LayerCfgTypeDef::WindowX0***
  Configures the Window Horizontal Start Position. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0xFFF.
- ***uint32_t LTDC_LayerCfgTypeDef::WindowX1***
  Configures the Window Horizontal Stop Position. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0xFFF.
- ***uint32_t LTDC_LayerCfgTypeDef::WindowY0***
  Configures the Window vertical Start Position. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0x7FF.
- ***uint32_t LTDC_LayerCfgTypeDef::WindowY1***
  Configures the Window vertical Stop Position. This parameter must be a number between Min_Data = 0x0000 and Max_Data = 0x7FF.
- ***uint32_t LTDC_LayerCfgTypeDef::PixelFormat***
  Specifies the pixel format. This parameter can be one of value of ***LTDC_Pixelformat***
- ***uint32_t LTDC_LayerCfgTypeDef::Alpha***
  Specifies the constant alpha used for blending. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint32_t LTDC_LayerCfgTypeDef::Alpha0***
  Configures the default alpha value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
- ***uint32_t LTDC_LayerCfgTypeDef::BlendingFactor1***
  Select the blending factor 1. This parameter can be one of value of ***LTDC_BlendingFactor1***
- ***uint32_t LTDC_LayerCfgTypeDef::BlendingFactor2***
  Select the blending factor 2. This parameter can be one of value of ***LTDC_BlendingFactor2***
- ***uint32_t LTDC_LayerCfgTypeDef::FBStartAdress***
  Configures the color frame buffer address
- ***uint32_t LTDC_LayerCfgTypeDef::ImageWidth***
  Configures the color frame buffer line length. This parameter must be a number between Min_Data = 0x0000U and Max_Data = 0x1FFF.
- ***uint32_t LTDC_LayerCfgTypeDef::ImageHeight***
  Specifies the number of line in frame buffer. This parameter must be a number between Min_Data = 0x000 and Max_Data = 0x7FF.
- ***LTDC_ColorTypeDef LTDC_LayerCfgTypeDef::Backcolor***
  Configures the layer background color.

39.1.4 LTDC_HandleTypeDef

Data Fields

- ***LTDC_TypeDef * Instance***
- ***LTDC_InitTypeDef Init***
- ***LTDC_LayerCfgTypeDef LayerCfg***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_LTDC_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***LTDC_TypeDef* LTDC_HandleTypeDef::Instance***
  LTDC Register base address
- ***LTDC_InitTypeDef LTDC_HandleTypeDef::Init***
  LTDC parameters
- ***LTDC_LayerCfgTypeDef LTDC_HandleTypeDef::LayerCfg[MAX_LAYER]***
  LTDC Layers parameters
- ***HAL_LockTypeDef LTDC_HandleTypeDef::Lock***
  LTDC Lock
- ***__IO HAL_LTDC_StateTypeDef LTDC_HandleTypeDef::State***
  LTDC state
- ***__IO uint32_t LTDC_HandleTypeDef::ErrorCode***
  LTDC Error code

39.2 LTDC Firmware driver API description

39.2.1 How to use this driver

1. Program the required configuration through the following parameters: the LTDC timing, the horizontal and vertical polarity, the pixel clock polarity, Data Enable polarity and the LTDC background color value using HAL_LTDC_Init() function
2. Program the required configuration through the following parameters: the pixel format, the blending factors, input alpha value, the window size and the image size using HAL_LTDC_ConfigLayer() function for foreground or/and background layer.
3. Optionally, configure and enable the CLUT using HAL_LTDC_ConfigCLUT() and HAL_LTDC_EnableCLUT functions.
4. Optionally, enable the Dither using HAL_LTDC_EnableDither().
5. Optionally, configure and enable the Color keying using HAL_LTDC_ConfigColorKeying() and HAL_LTDC_EnableColorKeying functions.
6. Optionally, configure LineInterrupt using HAL_LTDC_ProgramLineEvent() function
7. If needed, reconfigure and change the pixel format value, the alpha value value, the window size, the window position and the layer start address for foreground or/and background layer using respectively the following functions: HAL_LTDC_SetPixelFormat(), HAL_LTDC_SetAlpha(), HAL_LTDC_SetWindowSize(), HAL_LTDC_SetWindowPosition(), HAL_LTDC_SetAddress.

8. Variant functions with "_NoReload" post fix allows to set the LTDC configuration/settings without immediate reload. This is useful in case when the program requires to modify serval LTDC settings (on one or both layers) then applying(reload) these settings in one shot by calling the function "HAL_LTDC_Relaod" After calling the "_NoReload" functions to set different color/format/layer settings, the program can call the function "HAL_LTDC_Relaod" To apply(Reload) these settings. Function "HAL_LTDC_Relaod" can be called with the parameter "ReloadType" set to LTDC_RELOAD_IMMEDIATE if an immediate reload is required. Function "HAL_LTDC_Relaod" can be called with the parameter "ReloadType" set to LTDC_RELOAD_VERTICAL_BLANKING if the reload should be done in the next vertical blanking period, this option allows to avoid display flicker by applying the new settings during the vertical blanking period.
9. To control LTDC state you can use the following function: HAL_LTDC_GetState()

LTDC HAL driver macros list

Below the list of most used macros in LTDC HAL driver.

- __HAL_LTDC_ENABLE: Enable the LTDC.
- __HAL_LTDC_DISABLE: Disable the LTDC.
- __HAL_LTDC_LAYER_ENABLE: Enable the LTDC Layer.
- __HAL_LTDC_LAYER_DISABLE: Disable the LTDC Layer.
- __HAL_LTDC_RELOAD_CONFIG: Reload Layer Configuration.
- __HAL_LTDC_GET_FLAG: Get the LTDC pending flags.
- __HAL_LTDC_CLEAR_FLAG: Clear the LTDC pending flags.
- __HAL_LTDC_ENABLE_IT: Enable the specified LTDC interrupts.
- __HAL_LTDC_DISABLE_IT: Disable the specified LTDC interrupts.
- __HAL_LTDC_GET_IT_SOURCE: Check whether the specified LTDC interrupt has occurred or not.

You can refer to the LTDC HAL driver header file for more useful macros

39.2.2 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and configure the LTDC
- De-initialize the LTDC

This section contains the following APIs:

- ***HAL_LTDC_Init()***
- ***HAL_LTDC_DeInit()***
- ***HAL_LTDC_MspInit()***
- ***HAL_LTDC_MspDeInit()***
- ***HAL_LTDC_ErrorCallback()***
- ***HAL_LTDC_LineEventCallback()***
- ***HAL_LTDC_ReloadEventCallback()***

39.2.3 IO operation functions

This section provides function allowing to:

- Handle LTDC interrupt request

This section contains the following APIs:

- ***HAL_LTDC_IRQHandler()***
- ***HAL_LTDC_ErrorCallback()***
- ***HAL_LTDC_LineEventCallback()***
- ***HAL_LTDC_ReloadEventCallback()***

39.2.4 Peripheral Control functions

This section provides functions allowing to:

- Configure the LTDC foreground or/and background parameters.
- Set the active layer.
- Configure the color keying.
- Configure the C-LUT.
- Enable / Disable the color keying.
- Enable / Disable the C-LUT.
- Update the layer position.
- Update the layer size.
- Update pixel format on the fly.
- Update transparency on the fly.
- Update address on the fly.

This section contains the following APIs:

- ***HAL_LTDC_ConfigLayer()***
- ***HAL_LTDC_ConfigColorKeying()***
- ***HAL_LTDC_ConfigCLUT()***
- ***HAL_LTDC_EnableColorKeying()***
- ***HAL_LTDC_DisableColorKeying()***
- ***HAL_LTDC_EnableCLUT()***
- ***HAL_LTDC_DisableCLUT()***
- ***HAL_LTDC_EnableDither()***
- ***HAL_LTDC_DisableDither()***
- ***HAL_LTDC_SetWindowSize()***
- ***HAL_LTDC_SetWindowPosition()***
- ***HAL_LTDC_SetPixelFormat()***
- ***HAL_LTDC_SetAlpha()***
- ***HAL_LTDC_SetAddress()***
- ***HAL_LTDC_SetPitch()***
- ***HAL_LTDC_ProgramLineEvent()***
- ***HAL_LTDC_Relaod()***
- ***HAL_LTDC_ConfigLayer_NoReload()***
- ***HAL_LTDC_SetWindowSize_NoReload()***
- ***HAL_LTDC_SetWindowPosition_NoReload()***
- ***HAL_LTDC_SetPixelFormat_NoReload()***
- ***HAL_LTDC_SetAlpha_NoReload()***
- ***HAL_LTDC_SetAddress_NoReload()***
- ***HAL_LTDC_SetPitch_NoReload()***
- ***HAL_LTDC_ConfigColorKeying_NoReload()***
- ***HAL_LTDC_EnableColorKeying_NoReload()***
- ***HAL_LTDC_DisableColorKeying_NoReload()***
- ***HAL_LTDC_EnableCLUT_NoReload()***
- ***HAL_LTDC_DisableCLUT_NoReload()***

39.2.5 Peripheral State and Errors functions

This subsection provides functions allowing to

- Check the LTDC state.
- Get error code.

This section contains the following APIs:

- ***HAL_LTDC_GetState()***
- ***HAL_LTDC_GetError()***

39.2.6 Detailed description of functions

HAL_LTDC_Init

Function Name	**HAL_StatusTypeDef HAL_LTDC_Init (LTDC_HandleTypeDef * hltdc)**
Function Description	Initializes the LTDC according to the specified parameters in the LTDC_InitTypeDef and create the associated handle.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **HAL:** status

HAL_LTDC_DeInit

Function Name	**HAL_StatusTypeDef HAL_LTDC_DeInit (LTDC_HandleTypeDef * hltdc)**
Function Description	Deinitializes the LTDC peripheral registers to their default reset values.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_MspInit

Function Name	**void HAL_LTDC_MspInit (LTDC_HandleTypeDef * hltdc)**
Function Description	Initializes the LTDC MSP.
Parameters	• **hltdc:** : pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_MspDeInit

Function Name	**void HAL_LTDC_MspDeInit (LTDC_HandleTypeDef * hltdc)**
Function Description	DeInitializes the LTDC MSP.
Parameters	• **hltdc:** : pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_ErrorCallback

Function Name	**void HAL_LTDC_ErrorCallback (LTDC_HandleTypeDef * hltdc)**
Function Description	Error LTDC callback.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_LineEventCallback

Function Name	**void HAL_LTDC_LineEventCallback (LTDC_HandleTypeDef * hltdc)**
Function Description	Line Event callback.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_ReloadEventCallback

Function Name	**void HAL_LTDC_ReloadEventCallback (LTDC_HandleTypeDef * hltdc)**
Function Description	Reload Event callback.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **None**

HAL_LTDC_IRQHandler

Function Name	**void HAL_LTDC_IRQHandler (LTDC_HandleTypeDef * hltdc)**
Function Description	Handles LTDC interrupt request.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
Return values	• **HAL:** status

HAL_LTDC_ConfigLayer

Function Name	**HAL_StatusTypeDef HAL_LTDC_ConfigLayer (LTDC_HandleTypeDef * hltdc, LTDC_LayerCfgTypeDef * pLayerCfg, uint32_t LayerIdx)**
Function Description	Configure the LTDC Layer according to the specified parameters in the LTDC_InitTypeDef and create the associated handle.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **pLayerCfg:** pointer to a LTDC_LayerCfgTypeDef structure that contains the configuration information for the Layer. • **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetWindowSize

Function Name **HAL_StatusTypeDef HAL_LTDC_SetWindowSize (LTDC_HandleTypeDef * hltdc, uint32_t XSize, uint32_t YSize, uint32_t LayerIdx)**

Function Description Set the LTDC window size.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **XSize:** LTDC Pixel per line
- **YSize:** LTDC Line number
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetWindowPosition

Function Name **HAL_StatusTypeDef HAL_LTDC_SetWindowPosition (LTDC_HandleTypeDef * hltdc, uint32_t X0, uint32_t Y0, uint32_t LayerIdx)**

Function Description Set the LTDC window position.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **X0:** LTDC window X offset
- **Y0:** LTDC window Y offset
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetPixelFormat

Function Name **HAL_StatusTypeDef HAL_LTDC_SetPixelFormat (LTDC_HandleTypeDef * hltdc, uint32_t Pixelformat, uint32_t LayerIdx)**

Function Description Reconfigure the pixel format.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **Pixelformat:** new pixel format value.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1.

Return values
- **HAL:** status

HAL_LTDC_SetAlpha

Function Name **HAL_StatusTypeDef HAL_LTDC_SetAlpha (LTDC_HandleTypeDef * hltdc, uint32_t Alpha, uint32_t LayerIdx)**

Function Description	Reconfigure the layer alpha value.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **Alpha:** new alpha value. • **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1
Return values	• **HAL:** status

HAL_LTDC_SetAddress

Function Name	**HAL_StatusTypeDef HAL_LTDC_SetAddress (LTDC_HandleTypeDef * hltdc, uint32_t Address, uint32_t LayerIdx)**
Function Description	Reconfigure the frame buffer Address.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **Address:** new address value. • **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1.
Return values	• **HAL:** status

HAL_LTDC_SetPitch

Function Name	**HAL_StatusTypeDef HAL_LTDC_SetPitch (LTDC_HandleTypeDef * hltdc, uint32_t LinePitchInPixels, uint32_t LayerIdx)**
Function Description	Function used to reconfigure the pitch for specific cases where the attached LayerIdx buffer have a width that is larger than the one intended to be displayed on screen.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **LinePitchInPixels:** New line pitch in pixels to configure for LTDC layer 'LayerIdx'. • **LayerIdx:** LTDC layer index concerned by the modification of line pitch.
Return values	• **HAL:** status

HAL_LTDC_ConfigColorKeying

Function Name	**HAL_StatusTypeDef HAL_LTDC_ConfigColorKeying (LTDC_HandleTypeDef * hltdc, uint32_t RGBValue, uint32_t LayerIdx)**
Function Description	Configure the color keying.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **RGBValue:** the color key value • **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_ConfigCLUT

Function Name **HAL_StatusTypeDef HAL_LTDC_ConfigCLUT (LTDC_HandleTypeDef * hltdc, uint32_t * pCLUT, uint32_t CLUTSize, uint32_t LayerIdx)**

Function Description Load the color lookup table.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **pCLUT:** pointer to the color lookup table address.
- **CLUTSize:** the color lookup table size.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_EnableColorKeying

Function Name **HAL_StatusTypeDef HAL_LTDC_EnableColorKeying (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Enable the color keying.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_DisableColorKeying

Function Name **HAL_StatusTypeDef HAL_LTDC_DisableColorKeying (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Disable the color keying.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_EnableCLUT

Function Name **HAL_StatusTypeDef HAL_LTDC_EnableCLUT (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Enable the color lookup table.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_DisableCLUT

Function Name **HAL_StatusTypeDef HAL_LTDC_DisableCLUT (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Disable the color lookup table.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_ProgramLineEvent

Function Name **HAL_StatusTypeDef HAL_LTDC_ProgramLineEvent (LTDC_HandleTypeDef * hltdc, uint32_t Line)**

Function Description Define the position of the line interrupt .

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **Line:** Line Interrupt Position.

Return values
- **HAL:** status

HAL_LTDC_EnableDither

Function Name **HAL_StatusTypeDef HAL_LTDC_EnableDither (LTDC_HandleTypeDef * hltdc)**

Function Description Enables Dither.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.

Return values
- **HAL:** status

HAL_LTDC_DisableDither

Function Name **HAL_StatusTypeDef HAL_LTDC_DisableDither (LTDC_HandleTypeDef * hltdc)**

Function Description Disables Dither.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.

Return values
- **HAL:** status

HAL_LTDC_Relaod

Function Name **HAL_StatusTypeDef HAL_LTDC_Relaod (LTDC_HandleTypeDef * hltdc, uint32_t ReloadType)**

Function Description LTDC configuration relaod .

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **ReloadType:** This parameter can be one of the following values : LTDC_RELOAD_IMMEDIATE : Immediate Reload LTDC_SRCR_VBR : Reload in the next Vertical Blanking

Return values
- **HAL:** status

HAL_LTDC_ConfigLayer_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_ConfigLayer_NoReload (LTDC_HandleTypeDef * hltdc, LTDC_LayerCfgTypeDef * pLayerCfg, uint32_t LayerIdx)**

Function Description Configure the LTDC Layer according to the specified without reloading parameters in the LTDC_InitTypeDef and create the associated handle.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **pLayerCfg:** pointer to a LTDC_LayerCfgTypeDef structure that contains the configuration information for the Layer.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetWindowSize_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetWindowSize_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t XSize, uint32_t YSize, uint32_t LayerIdx)**

Function Description Set the LTDC window size without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **XSize:** LTDC Pixel per line
- **YSize:** LTDC Line number
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetWindowPosition_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetWindowPosition_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t X0, uint32_t Y0, uint32_t LayerIdx)**

Function Description Set the LTDC window position without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **X0:** LTDC window X offset
- **Y0:** LTDC window Y offset
- **LayerIdx:** LTDC Layer index. This parameter can be one of

the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetPixelFormat_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetPixelFormat_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t Pixelformat, uint32_t LayerIdx)**

Function Description Reconfigure the pixel format without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDfef structure that contains the configuration information for the LTDC.
- **Pixelformat:** new pixel format value.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1.

Return values
- **HAL:** status

HAL_LTDC_SetAlpha_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetAlpha_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t Alpha, uint32_t LayerIdx)**

Function Description Reconfigure the layer alpha value without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **Alpha:** new alpha value.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_SetAddress_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetAddress_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t Address, uint32_t LayerIdx)**

Function Description Reconfigure the frame buffer Address without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **Address:** new address value.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1.

Return values
- **HAL:** status

HAL_LTDC_SetPitch_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_SetPitch_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t LinePitchInPixels, uint32_t LayerIdx)**

Function Description Function used to reconfigure the pitch for specific cases where the

attached LayerIdx buffer have a width that is larger than the one intended to be displayed on screen.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LinePitchInPixels:** New line pitch in pixels to configure for LTDC layer 'LayerIdx'.
- **LayerIdx:** LTDC layer index concerned by the modification of line pitch.

Return values
- **HAL:** status

HAL_LTDC_ConfigColorKeying_NoReload

Function Name
HAL_StatusTypeDef HAL_LTDC_ConfigColorKeying_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t RGBValue, uint32_t LayerIdx)

Function Description
Configure the color keying without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **RGBValue:** the color key value
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_EnableColorKeying_NoReload

Function Name
HAL_StatusTypeDef HAL_LTDC_EnableColorKeying_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)

Function Description
Enable the color keying without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_DisableColorKeying_NoReload

Function Name
HAL_StatusTypeDef HAL_LTDC_DisableColorKeying_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)

Function Description
Disable the color keying without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_EnableCLUT_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_EnableCLUT_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Enable the color lookup table without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_DisableCLUT_NoReload

Function Name **HAL_StatusTypeDef HAL_LTDC_DisableCLUT_NoReload (LTDC_HandleTypeDef * hltdc, uint32_t LayerIdx)**

Function Description Disable the color lookup table without reloading.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.
- **LayerIdx:** LTDC Layer index. This parameter can be one of the following values: 0 or 1

Return values
- **HAL:** status

HAL_LTDC_GetState

Function Name **HAL_LTDC_StateTypeDef HAL_LTDC_GetState (LTDC_HandleTypeDef * hltdc)**

Function Description Return the LTDC state.

Parameters
- **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.

Return values
- **HAL:** state

HAL_LTDC_GetError

Function Name **uint32_t HAL_LTDC_GetError (LTDC_HandleTypeDef * hltdc)**

Function Description Return the LTDC error code.

Parameters
- **hltdc:** : pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC.

Return values
- **LTDC:** Error Code

39.3 LTDC Firmware driver defines

39.3.1 LTDC

LTDC Alpha

LTDC_ALPHA LTDC Cte Alpha mask

LTDC BACK COLOR

LTDC_COLOR Color mask

LTDC Blending Factor1

LTDC_BLENDING_FACTOR1_CA Blending factor : Cte Alpha

LTDC_BLENDING_FACTOR1_PAxCA Blending factor : Cte Alpha x Pixel Alpha

LTDC Blending Factor2

LTDC_BLENDING_FACTOR2_CA Blending factor : Cte Alpha

LTDC_BLENDING_FACTOR2_PAxCA Blending factor : Cte Alpha x Pixel Alpha

LTDC DE POLARITY

LTDC_DEPOLARITY_AL Data Enable, is active low.

LTDC_DEPOLARITY_AH Data Enable, is active high.

LTDC Error Code

HAL_LTDC_ERROR_NONE LTDC No error

HAL_LTDC_ERROR_TE LTDC Transfer error

HAL_LTDC_ERROR_FU LTDC FIFO Underrun

HAL_LTDC_ERROR_TIMEOUT LTDC Timeout error

LTDC Exported Macros

__HAL_LTDC_RESET_HANDLE_STATE **Description:**

- Reset LTDC handle state.

Parameters:

- __HANDLE__: specifies the LTDC handle.

Return value:

- None

__HAL_LTDC_ENABLE **Description:**

- Enable the LTDC.

Parameters:

- __HANDLE__: LTDC handle

Return value:

- None.

__HAL_LTDC_DISABLE **Description:**

- Disable the LTDC.

Parameters:

- __HANDLE__: LTDC handle

Return value:

- None.

__HAL_LTDC_LAYER_ENABLE

Description:

- Enable the LTDC Layer.

Parameters:

- __HANDLE__: LTDC handle
- __LAYER__: Specify the layer to be enabled This parameter can be 0 or 1

Return value:

- None.

__HAL_LTDC_LAYER_DISABLE

Description:

- Disable the LTDC Layer.

Parameters:

- __HANDLE__: LTDC handle
- __LAYER__: Specify the layer to be disabled This parameter can be 0 or 1

Return value:

- None.

__HAL_LTDC_RELOAD_CONFIG

Description:

- Reload Layer Configuration.

Parameters:

- __HANDLE__: LTDC handle

Return value:

- None.

__HAL_LTDC_GET_FLAG

Description:

- Get the LTDC pending flags.

Parameters:

- __HANDLE__: LTDC handle
- __FLAG__: Get the specified flag. This parameter can be any combination of the following values:
  - LTDC_FLAG_LI: Line Interrupt flag
  - LTDC_FLAG_FU: FIFO Underrun Interrupt flag
  - LTDC_FLAG_TE: Transfer Error interrupt flag
  - LTDC_FLAG_RR: Register Reload Interrupt Flag

Return value:

- The: state of FLAG (SET or RESET).

__HAL_LTDC_CLEAR_FLAG

Description:

- Clears the LTDC pending flags.

Parameters:

- __HANDLE__: LTDC handle
- __FLAG__: specifies the flag to clear. This parameter can be any combination of the following values:
  - LTDC_FLAG_LI: Line Interrupt flag
  - LTDC_FLAG_FU: FIFO Underrun Interrupt flag
  - LTDC_FLAG_TE: Transfer Error interrupt flag
  - LTDC_FLAG_RR: Register Reload Interrupt Flag

Return value:

- None

__HAL_LTDC_ENABLE_IT

Description:

- Enables the specified LTDC interrupts.

Parameters:

- __HANDLE__: LTDC handle
- __INTERRUPT__: specifies the LTDC interrupt sources to be enabled. This parameter can be any combination of the following values:
  - LTDC_IT_LI: Line Interrupt flag
  - LTDC_IT_FU: FIFO Underrun Interrupt flag
  - LTDC_IT_TE: Transfer Error interrupt flag
  - LTDC_IT_RR: Register Reload Interrupt Flag

Return value:

- None

__HAL_LTDC_DISABLE_IT

Description:

- Disables the specified LTDC interrupts.

Parameters:

- __HANDLE__: LTDC handle
- __INTERRUPT__: specifies the LTDC interrupt sources to be disabled. This parameter can be any combination of the following values:
  - LTDC_IT_LI: Line Interrupt flag
  - LTDC_IT_FU: FIFO Underrun

Interrupt flag
- LTDC_IT_TE: Transfer Error interrupt flag
- LTDC_IT_RR: Register Reload Interrupt Flag

Return value:

- None

__HAL_LTDC_GET_IT_SOURCE

Description:

- Checks whether the specified LTDC interrupt has occurred or not.

Parameters:

- __HANDLE__: LTDC handle
- __INTERRUPT__: specifies the LTDC interrupt source to check. This parameter can be one of the following values:
  - LTDC_IT_LI: Line Interrupt flag
  - LTDC_IT_FU: FIFO Underrun Interrupt flag
  - LTDC_IT_TE: Transfer Error interrupt flag
  - LTDC_IT_RR: Register Reload Interrupt Flag

Return value:

- The: state of INTERRUPT (SET or RESET).

LTDC Exported Types

MAX_LAYER

LTDC Flag

LTDC_FLAG_LI

LTDC_FLAG_FU

LTDC_FLAG_TE

LTDC_FLAG_RR

LTDC HS POLARITY

LTDC_HSPOLARITY_AL	Horizontal Synchronization is active low.
LTDC_HSPOLARITY_AH	Horizontal Synchronization is active high.

LTDC Interrupts

LTDC_IT_LI

LTDC_IT_FU

LTDC_IT_TE

LTDC_IT_RR

LTDC LAYER Config

LTDC_STOPPOSITION	LTDC Layer stop position
LTDC_STARTPOSITION	LTDC Layer start position
LTDC_COLOR_FRAME_BUFFER	LTDC Layer Line length
LTDC_LINE_NUMBER	LTDC Layer Line number

LTDC PC POLARITY

LTDC_PCPOLARITY_IPC	input pixel clock.
LTDC_PCPOLARITY_IIPC	inverted input pixel clock.

LTDC Pixel format

LTDC_PIXEL_FORMAT_ARGB8888	ARGB8888 LTDC pixel format
LTDC_PIXEL_FORMAT_RGB888	RGB888 LTDC pixel format
LTDC_PIXEL_FORMAT_RGB565	RGB565 LTDC pixel format
LTDC_PIXEL_FORMAT_ARGB1555	ARGB1555 LTDC pixel format
LTDC_PIXEL_FORMAT_ARGB4444	ARGB4444 LTDC pixel format
LTDC_PIXEL_FORMAT_L8	L8 LTDC pixel format
LTDC_PIXEL_FORMAT_AL44	AL44 LTDC pixel format
LTDC_PIXEL_FORMAT_AL88	AL88 LTDC pixel format

LTDC Reload Type

LTDC_RELOAD_IMMEDIATE	Immediate Reload
LTDC_RELOAD_VERTICAL_BLANKING	Vertical Blanking Reload

LTDC SYNC

LTDC_HORIZONTALSYNC	Horizontal synchronization width.
LTDC_VERTICALSYNC	Vertical synchronization height.

LTDC VS POLARITY

LTDC_VSPOLARITY_AL	Vertical Synchronization is active low.
LTDC_VSPOLARITY_AH	Vertical Synchronization is active high.

40 HAL LTDC Extension Driver

40.1 LTDCEx Firmware driver API description

40.1.1 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and configure the LTDC

This section contains the following APIs:

- ***HAL_LTDC_StructInitFromVideoConfig()***
- ***HAL_LTDC_StructInitFromAdaptedCommandConfig()***

40.1.2 Detailed description of functions

HAL_LTDC_StructInitFromVideoConfig

Function Name	**HAL_StatusTypeDef HAL_LTDC_StructInitFromVideoConfig (LTDC_HandleTypeDef * hltdc, DSI_VidCfgTypeDef * VidCfg)**
Function Description	Retrieve common parameters from DSI Video mode configuration structure.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **VidCfg:** pointer to a DSI_VidCfgTypeDef structure that contains the DSI video mode configuration parameters
Return values	• **HAL:** status
Notes	• The implementation of this function is taking into account the LTDC polarities inversion as described in the current LTDC specification

HAL_LTDC_StructInitFromAdaptedCommandConfig

Function Name	**HAL_StatusTypeDef HAL_LTDC_StructInitFromAdaptedCommandConfig (LTDC_HandleTypeDef * hltdc, DSI_CmdCfgTypeDef * CmdCfg)**
Function Description	Retrieve common parameters from DSI Adapted command mode configuration structure.
Parameters	• **hltdc:** pointer to a LTDC_HandleTypeDef structure that contains the configuration information for the LTDC. • **CmdCfg:** pointer to a DSI_CmdCfgTypeDef structure that contains the DSI command mode configuration parameters
Return values	• **HAL:** status
Notes	• The implementation of this function is taking into account the LTDC polarities inversion as described in the current LTDC specification

41 HAL NAND Generic Driver

41.1 NAND Firmware driver registers structures

41.1.1 NAND_IDTypeDef

Data Fields

- ***uint8_t Maker_Id***
- ***uint8_t Device_Id***
- ***uint8_t Third_Id***
- ***uint8_t Fourth_Id***

Field Documentation

- ***uint8_t NAND_IDTypeDef::Maker_Id***
- ***uint8_t NAND_IDTypeDef::Device_Id***
- ***uint8_t NAND_IDTypeDef::Third_Id***
- ***uint8_t NAND_IDTypeDef::Fourth_Id***

41.1.2 NAND_AddressTypeDef

Data Fields

- ***uint16_t Page***
- ***uint16_t Zone***
- ***uint16_t Block***

Field Documentation

- ***uint16_t NAND_AddressTypeDef::Page***
  NAND memory Page address
- ***uint16_t NAND_AddressTypeDef::Zone***
  NAND memory Zone address
- ***uint16_t NAND_AddressTypeDef::Block***
  NAND memory Block address

41.1.3 NAND_InfoTypeDef

Data Fields

- ***uint32_t PageSize***
- ***uint32_t SpareAreaSize***
- ***uint32_t BlockSize***
- ***uint32_t BlockNbr***

- ***uint32_t ZoneSize***

Field Documentation

- ***uint32_t NAND_InfoTypeDef::PageSize***
  NAND memory page (without spare area) size measured in K. bytes
- ***uint32_t NAND_InfoTypeDef::SpareAreaSize***
  NAND memory spare area size measured in K. bytes
- ***uint32_t NAND_InfoTypeDef::BlockSize***
  NAND memory block size number of pages
- ***uint32_t NAND_InfoTypeDef::BlockNbr***
  NAND memory number of blocks
- ***uint32_t NAND_InfoTypeDef::ZoneSize***
  NAND memory zone size measured in number of blocks

41.1.4 NAND_HandleTypeDef

Data Fields

- ***FMC_NAND_TypeDef * Instance***
- ***FMC_NAND_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_NAND_StateTypeDef State***
- ***NAND_InfoTypeDef Info***

Field Documentation

- ***FMC_NAND_TypeDef* NAND_HandleTypeDef::Instance***
  Register base address
- ***FMC_NAND_InitTypeDef NAND_HandleTypeDef::Init***
  NAND device control configuration parameters
- ***HAL_LockTypeDef NAND_HandleTypeDef::Lock***
  NAND locking object
- ***__IO HAL_NAND_StateTypeDef NAND_HandleTypeDef::State***
  NAND device access state
- ***NAND_InfoTypeDef NAND_HandleTypeDef::Info***
  NAND characteristic information structure

41.2 NAND Firmware driver API description

41.2.1 How to use this driver

This driver is a generic layered driver which contains a set of APIs used to control NAND flash memories. It uses the FMC/FSMC layer functions to interface with NAND devices. This driver is used as follows:

- NAND flash memory configuration sequence using the function HAL_NAND_Init() with control and timing parameters for both common and attribute spaces.

- Read NAND flash memory maker and device IDs using the function HAL_NAND_Read_ID(). The read information is stored in the NAND_ID_TypeDef structure declared by the function caller.
- Access NAND flash memory by read/write operations using the functions HAL_NAND_Read_Page()/HAL_NAND_Read_SpareArea(), HAL_NAND_Write_Page()/HAL_NAND_Write_SpareArea() to read/write page(s)/spare area(s). These functions use specific device information (Block, page size..) predefined by the user in the HAL_NAND_Info_TypeDef structure. The read/write address information is contained by the Nand_Address_Typedef structure passed as parameter.
- Perform NAND flash Reset chip operation using the function HAL_NAND_Reset().
- Perform NAND flash erase block operation using the function HAL_NAND_Erase_Block(). The erase block address information is contained in the Nand_Address_Typedef structure passed as parameter.
- Read the NAND flash status operation using the function HAL_NAND_Read_Status().
- You can also control the NAND device by calling the control APIs HAL_NAND_ECC_Enable()/ HAL_NAND_ECC_Disable() to respectively enable/disable the ECC code correction feature or the function HAL_NAND_GetECC() to get the ECC correction code.
- You can monitor the NAND device HAL state by calling the function HAL_NAND_GetState()

This driver is a set of generic APIs which handle standard NAND flash operations. If a NAND flash device contains different operations and/or implementations, it should be implemented separately.

41.2.2 NAND Initialization and de-initialization functions

This section provides functions allowing to initialize/de-initialize the NAND memory

This section contains the following APIs:

- ***HAL_NAND_Init()***
- ***HAL_NAND_DeInit()***
- ***HAL_NAND_MspInit()***
- ***HAL_NAND_MspDeInit()***
- ***HAL_NAND_IRQHandler()***
- ***HAL_NAND_ITCallback()***

41.2.3 NAND Input and Output functions

This section provides functions allowing to use and control the NAND memory

This section contains the following APIs:

- ***HAL_NAND_Read_ID()***
- ***HAL_NAND_Reset()***
- ***HAL_NAND_Read_Page()***
- ***HAL_NAND_Write_Page()***
- ***HAL_NAND_Read_SpareArea()***
- ***HAL_NAND_Write_SpareArea()***
- ***HAL_NAND_Erase_Block()***
- ***HAL_NAND_Read_Status()***
- ***HAL_NAND_Address_Inc()***

41.2.4 NAND Control functions

This subsection provides a set of functions allowing to control dynamically the NAND interface.

This section contains the following APIs:

- ***HAL_NAND_ECC_Enable()***
- ***HAL_NAND_ECC_Disable()***
- ***HAL_NAND_GetECC()***

41.2.5 NAND State functions

This subsection permits to get in run-time the status of the NAND controller and the data flow.

This section contains the following APIs:

- ***HAL_NAND_GetState()***
- ***HAL_NAND_Read_Status()***

41.2.6 Detailed description of functions

HAL_NAND_Init

Function Name	**HAL_StatusTypeDef HAL_NAND_Init (NAND_HandleTypeDef * hnand, FMC_NAND_PCC_TimingTypeDef * ComSpace_Timing, FMC_NAND_PCC_TimingTypeDef * AttSpace_Timing)**
Function Description	Perform NAND memory Initialization sequence.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **ComSpace_Timing:** pointer to Common space timing structure • **AttSpace_Timing:** pointer to Attribute space timing structure
Return values	• **HAL:** status

HAL_NAND_DeInit

Function Name	**HAL_StatusTypeDef HAL_NAND_DeInit (NAND_HandleTypeDef * hnand)**
Function Description	Perform NAND memory De-Initialization sequence.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
Return values	• **HAL:** status

HAL_NAND_MspInit

Function Name	**void HAL_NAND_MspInit (NAND_HandleTypeDef * hnand)**
Function Description	NAND MSP Init.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **None**

HAL_NAND_MspDeInit

Function Name **void HAL_NAND_MspDeInit (NAND_HandleTypeDef * hnand)**

Function Description NAND MSP DeInit.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **None**

HAL_NAND_IRQHandler

Function Name **void HAL_NAND_IRQHandler (NAND_HandleTypeDef * hnand)**

Function Description This function handles NAND device interrupt request.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **HAL:** status

HAL_NAND_ITCallback

Function Name **void HAL_NAND_ITCallback (NAND_HandleTypeDef * hnand)**

Function Description NAND interrupt feature callback.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **None**

HAL_NAND_Read_ID

Function Name **HAL_StatusTypeDef HAL_NAND_Read_ID (NAND_HandleTypeDef * hnand, NAND_IDTypeDef * pNAND_ID)**

Function Description Read the NAND memory electronic signature.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
- **pNAND_ID:** NAND ID structure

Return values
- **HAL:** status

HAL_NAND_Reset

Function Name **HAL_StatusTypeDef HAL_NAND_Reset (NAND_HandleTypeDef * hnand)**

Function Description NAND memory reset.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **HAL:** status

HAL_NAND_Read_Page

Function Name	**HAL_StatusTypeDef HAL_NAND_Read_Page (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress, uint8_t * pBuffer, uint32_t NumPageToRead)**
Function Description	Read Page(s) from NAND memory block.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **pAddress:** : pointer to NAND address structure • **pBuffer:** : pointer to destination read buffer • **NumPageToRead:** : number of pages to read from block
Return values	• **HAL:** status

HAL_NAND_Write_Page

Function Name	**HAL_StatusTypeDef HAL_NAND_Write_Page (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress, uint8_t * pBuffer, uint32_t NumPageToWrite)**
Function Description	Write Page(s) to NAND memory block.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **pAddress:** : pointer to NAND address structure • **pBuffer:** : pointer to source buffer to write • **NumPageToWrite:** : number of pages to write to block
Return values	• **HAL:** status

HAL_NAND_Read_SpareArea

Function Name	**HAL_StatusTypeDef HAL_NAND_Read_SpareArea (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress, uint8_t * pBuffer, uint32_t NumSpareAreaToRead)**
Function Description	Read Spare area(s) from NAND memory.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **pAddress:** : pointer to NAND address structure • **pBuffer:** pointer to source buffer to write • **NumSpareAreaToRead:** Number of spare area to read
Return values	• **HAL:** status

HAL_NAND_Write_SpareArea

Function Name	**HAL_StatusTypeDef HAL_NAND_Write_SpareArea (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress, uint8_t * pBuffer, uint32_t NumSpareAreaTowrite)**
Function Description	Write Spare area(s) to NAND memory.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **pAddress:** : pointer to NAND address structure • **pBuffer:** : pointer to source buffer to write

- **NumSpareAreaTowrite:** : number of spare areas to write to block

Return values
- **HAL:** status

HAL_NAND_Erase_Block

Function Name **HAL_StatusTypeDef HAL_NAND_Erase_Block (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress)**

Function Description NAND memory Block erase.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
- **pAddress:** : pointer to NAND address structure

Return values
- **HAL:** status

HAL_NAND_Read_Status

Function Name **uint32_t HAL_NAND_Read_Status (NAND_HandleTypeDef * hnand)**

Function Description NAND memory read status.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **NAND:** status

HAL_NAND_Address_Inc

Function Name **uint32_t HAL_NAND_Address_Inc (NAND_HandleTypeDef * hnand, NAND_AddressTypeDef * pAddress)**

Function Description Increment the NAND memory address.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
- **pAddress:** pointer to NAND address structure

Return values
- **The:** new status of the increment address operation. It can be:
  - NAND_VALID_ADDRESS: When the new address is valid address
  - NAND_INVALID_ADDRESS: When the new address is invalid address

HAL_NAND_ECC_Enable

Function Name **HAL_StatusTypeDef HAL_NAND_ECC_Enable (NAND_HandleTypeDef * hnand)**

Function Description Enables dynamically NAND ECC feature.

Parameters
- **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.

Return values
- **HAL:** status

HAL_NAND_ECC_Disable

Function Name	**HAL_StatusTypeDef HAL_NAND_ECC_Disable (NAND_HandleTypeDef * hnand)**
Function Description	Disables dynamically FMC_NAND ECC feature.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
Return values	• **HAL:** status

HAL_NAND_GetECC

Function Name	**HAL_StatusTypeDef HAL_NAND_GetECC (NAND_HandleTypeDef * hnand, uint32_t * ECCval, uint32_t Timeout)**
Function Description	Disables dynamically NAND ECC feature.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module. • **ECCval:** pointer to ECC value • **Timeout:** maximum timeout to wait
Return values	• **HAL:** status

HAL_NAND_GetState

Function Name	**HAL_NAND_StateTypeDef HAL_NAND_GetState (NAND_HandleTypeDef * hnand)**
Function Description	return the NAND state
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
Return values	• **HAL:** state

HAL_NAND_Read_Status

Function Name	**uint32_t HAL_NAND_Read_Status (NAND_HandleTypeDef * hnand)**
Function Description	NAND memory read status.
Parameters	• **hnand:** pointer to a NAND_HandleTypeDef structure that contains the configuration information for NAND module.
Return values	• **NAND:** status

41.3 NAND Firmware driver defines

41.3.1 NAND

NAND Exported Macros

__HAL_NAND_RESET_HANDLE_STATE **Description:**

- Reset NAND handle state.

Parameters:

- __HANDLE__: specifies the NAND handle.

Return value:

- None

42 HAL NOR Generic Driver

42.1 NOR Firmware driver registers structures

42.1.1 NOR_IDTypeDef

Data Fields

- ***uint16_t Manufacturer_Code***
- ***uint16_t Device_Code1***
- ***uint16_t Device_Code2***
- ***uint16_t Device_Code3***

Field Documentation

- ***uint16_t NOR_IDTypeDef::Manufacturer_Code***
  Defines the device's manufacturer code used to identify the memory
- ***uint16_t NOR_IDTypeDef::Device_Code1***
- ***uint16_t NOR_IDTypeDef::Device_Code2***
- ***uint16_t NOR_IDTypeDef::Device_Code3***
  Defines the device's codes used to identify the memory. These codes can be accessed by performing read operations with specific control signals and addresses set.They can also be accessed by issuing an Auto Select command

42.1.2 NOR_CFITypeDef

Data Fields

- ***uint16_t CFI_1***
- ***uint16_t CFI_2***
- ***uint16_t CFI_3***
- ***uint16_t CFI_4***

Field Documentation

- ***uint16_t NOR_CFITypeDef::CFI_1***
  < Defines the information stored in the memory's Common flash interface which contains a description of various electrical and timing parameters, density information and functions supported by the memory
- ***uint16_t NOR_CFITypeDef::CFI_2***
- ***uint16_t NOR_CFITypeDef::CFI_3***
- ***uint16_t NOR_CFITypeDef::CFI_4***

42.1.3 NOR_HandleTypeDef

Data Fields

- ***FMC_NORSRAM_TypeDef * Instance***
- ***FMC_NORSRAM_EXTENDED_TypeDef * Extended***
- ***FMC_NORSRAM_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_NOR_StateTypeDef State***

Field Documentation

- ***FMC_NORSRAM_TypeDef* NOR_HandleTypeDef::Instance***
  Register base address
- ***FMC_NORSRAM_EXTENDED_TypeDef* NOR_HandleTypeDef::Extended***
  Extended mode register base address
- ***FMC_NORSRAM_InitTypeDef NOR_HandleTypeDef::Init***
  NOR device control configuration parameters
- ***HAL_LockTypeDef NOR_HandleTypeDef::Lock***
  NOR locking object
- ***__IO HAL_NOR_StateTypeDef NOR_HandleTypeDef::State***
  NOR device access state

42.2 NOR Firmware driver API description

42.2.1 How to use this driver

This driver is a generic layered driver which contains a set of APIs used to control NOR flash memories. It uses the FMC/FSMC layer functions to interface with NOR devices. This driver is used as follows:

- NOR flash memory configuration sequence using the function HAL_NOR_Init() with control and timing parameters for both normal and extended mode.
- Read NOR flash memory manufacturer code and device IDs using the function HAL_NOR_Read_ID(). The read information is stored in the NOR_ID_TypeDef structure declared by the function caller.
- Access NOR flash memory by read/write data unit operations using the functions HAL_NOR_Read(), HAL_NOR_Program().
- Perform NOR flash erase block/chip operations using the functions HAL_NOR_Erase_Block() and HAL_NOR_Erase_Chip().
- Read the NOR flash CFI (common flash interface) IDs using the function HAL_NOR_Read_CFI(). The read information is stored in the NOR_CFI_TypeDef structure declared by the function caller.
- You can also control the NOR device by calling the control APIs HAL_NOR_WriteOperation_Enable()/ HAL_NOR_WriteOperation_Disable() to respectively enable/disable the NOR write operation
- You can monitor the NOR device HAL state by calling the function HAL_NOR_GetState()

This driver is a set of generic APIs which handle standard NOR flash operations. If a NOR flash device contains different operations and/or implementations, it should be implemented separately.

NOR HAL driver macros list

Below the list of most used macros in NOR HAL driver.

- NOR_WRITE : NOR memory write data to specified address

42.2.2 NOR Initialization and de_initialization functions

This section provides functions allowing to initialize/de-initialize the NOR memory

This section contains the following APIs:

- ***HAL_NOR_Init()***
- ***HAL_NOR_DeInit()***
- ***HAL_NOR_MspInit()***
- ***HAL_NOR_MspDeInit()***
- ***HAL_NOR_MspWait()***

42.2.3 NOR Input and Output functions

This section provides functions allowing to use and control the NOR memory

This section contains the following APIs:

- ***HAL_NOR_Read_ID()***
- ***HAL_NOR_ReturnToReadMode()***
- ***HAL_NOR_Read()***
- ***HAL_NOR_Program()***
- ***HAL_NOR_ReadBuffer()***
- ***HAL_NOR_ProgramBuffer()***
- ***HAL_NOR_Erase_Block()***
- ***HAL_NOR_Erase_Chip()***
- ***HAL_NOR_Read_CFI()***

42.2.4 NOR Control functions

This subsection provides a set of functions allowing to control dynamically the NOR interface.

This section contains the following APIs:

- ***HAL_NOR_WriteOperation_Enable()***
- ***HAL_NOR_WriteOperation_Disable()***

42.2.5 NOR State functions

This subsection permits to get in run-time the status of the NOR controller and the data flow.

This section contains the following APIs:

- ***HAL_NOR_GetState()***
- ***HAL_NOR_GetStatus()***

42.2.6 Detailed description of functions

HAL_NOR_Init

Function Name	**HAL_StatusTypeDef HAL_NOR_Init (NOR_HandleTypeDef * hnor, FMC_NORSRAM_TimingTypeDef * Timing, FMC_NORSRAM_TimingTypeDef * ExtTiming)**
Function Description	Perform the NOR memory Initialization sequence.
Parameters	• **hnor:** pointer to the NOR handle • **Timing:** pointer to NOR control timing structure • **ExtTiming:** pointer to NOR extended mode timing structure
Return values	• **HAL:** status

HAL_NOR_DeInit

Function Name	**HAL_StatusTypeDef HAL_NOR_DeInit (NOR_HandleTypeDef * hnor)**
Function Description	Perform NOR memory De-Initialization sequence.
Parameters	• **hnor:** pointer to a NOR_HandleTypeDef structure that contains the configuration information for NOR module.
Return values	• **HAL:** status

HAL_NOR_MspInit

Function Name	**void HAL_NOR_MspInit (NOR_HandleTypeDef * hnor)**
Function Description	NOR MSP Init.
Parameters	• **hnor:** pointer to a NOR_HandleTypeDef structure that contains the configuration information for NOR module.
Return values	• **None**

HAL_NOR_MspDeInit

Function Name	**void HAL_NOR_MspDeInit (NOR_HandleTypeDef * hnor)**
Function Description	NOR MSP DeInit.
Parameters	• **hnor:** pointer to a NOR_HandleTypeDef structure that contains the configuration information for NOR module.
Return values	• **None**

HAL_NOR_MspWait

Function Name	**void HAL_NOR_MspWait (NOR_HandleTypeDef * hnor, uint32_t Timeout)**
Function Description	NOR BSP Wait for Ready/Busy signal.
Parameters	• **hnor:** pointer to a NOR_HandleTypeDef structure that contains the configuration information for NOR module. • **Timeout:** Maximum timeout value

Return values
- **None**

HAL_NOR_Read_ID

Function Name **HAL_StatusTypeDef HAL_NOR_Read_ID (NOR_HandleTypeDef * hnor, NOR_IDTypeDef * pNOR_ID)**

Function Description Read NOR flash IDs.

Parameters
- **hnor:** pointer to the NOR handle
- **pNOR_ID:** : pointer to NOR ID structure

Return values
- **HAL:** status

HAL_NOR_ReturnToReadMode

Function Name **HAL_StatusTypeDef HAL_NOR_ReturnToReadMode (NOR_HandleTypeDef * hnor)**

Function Description Returns the NOR memory to Read mode.

Parameters
- **hnor:** pointer to the NOR handle

Return values
- **HAL:** status

HAL_NOR_Read

Function Name **HAL_StatusTypeDef HAL_NOR_Read (NOR_HandleTypeDef * hnor, uint32_t * pAddress, uint16_t * pData)**

Function Description Read data from NOR memory.

Parameters
- **hnor:** pointer to the NOR handle
- **pAddress:** pointer to Device address
- **pData:** : pointer to read data

Return values
- **HAL:** status

HAL_NOR_Program

Function Name **HAL_StatusTypeDef HAL_NOR_Program (NOR_HandleTypeDef * hnor, uint32_t * pAddress, uint16_t * pData)**

Function Description Program data to NOR memory.

Parameters
- **hnor:** pointer to the NOR handle
- **pAddress:** Device address
- **pData:** : pointer to the data to write

Return values
- **HAL:** status

HAL_NOR_ReadBuffer

Function Name **HAL_StatusTypeDef HAL_NOR_ReadBuffer (NOR_HandleTypeDef * hnor, uint32_t uwAddress, uint16_t * pData, uint32_t uwBufferSize)**

Function Description Reads a half-word buffer from the NOR memory.

Parameters
- **hnor:** pointer to the NOR handle
- **uwAddress:** NOR memory internal address to read from.
- **pData:** pointer to the buffer that receives the data read from the NOR memory.
- **uwBufferSize:** : number of Half word to read.

Return values
- **HAL:** status

HAL_NOR_ProgramBuffer

Function Name **HAL_StatusTypeDef HAL_NOR_ProgramBuffer (NOR_HandleTypeDef * hnor, uint32_t uwAddress, uint16_t * pData, uint32_t uwBufferSize)**

Function Description Writes a half-word buffer to the NOR memory.

Parameters
- **hnor:** pointer to the NOR handle
- **uwAddress:** NOR memory internal start write address
- **pData:** pointer to source data buffer.
- **uwBufferSize:** Size of the buffer to write

Return values
- **HAL:** status

HAL_NOR_Erase_Block

Function Name **HAL_StatusTypeDef HAL_NOR_Erase_Block (NOR_HandleTypeDef * hnor, uint32_t BlockAddress, uint32_t Address)**

Function Description Erase the specified block of the NOR memory.

Parameters
- **hnor:** pointer to the NOR handle
- **BlockAddress:** : Block to erase address
- **Address:** Device address

Return values
- **HAL:** status

HAL_NOR_Erase_Chip

Function Name **HAL_StatusTypeDef HAL_NOR_Erase_Chip (NOR_HandleTypeDef * hnor, uint32_t Address)**

Function Description Erase the entire NOR chip.

Parameters
- **hnor:** pointer to the NOR handle
- **Address:** : Device address

Return values
- **HAL:** status

HAL_NOR_Read_CFI

Function Name **HAL_StatusTypeDef HAL_NOR_Read_CFI (NOR_HandleTypeDef * hnor, NOR_CFITypeDef * pNOR_CFI)**

Function Description Read NOR flash CFI IDs.

Parameters
- **hnor:** pointer to the NOR handle
- **pNOR_CFI:** : pointer to NOR CFI IDs structure

Return values
- **HAL:** status

HAL_NOR_WriteOperation_Enable

Function Name **HAL_StatusTypeDef HAL_NOR_WriteOperation_Enable (NOR_HandleTypeDef * hnor)**

Function Description Enables dynamically NOR write operation.

Parameters
- **hnor:** pointer to the NOR handle

Return values
- **HAL:** status

HAL_NOR_WriteOperation_Disable

Function Name **HAL_StatusTypeDef HAL_NOR_WriteOperation_Disable (NOR_HandleTypeDef * hnor)**

Function Description Disables dynamically NOR write operation.

Parameters
- **hnor:** pointer to the NOR handle

Return values
- **HAL:** status

HAL_NOR_GetState

Function Name **HAL_NOR_StateTypeDef HAL_NOR_GetState (NOR_HandleTypeDef * hnor)**

Function Description return the NOR controller state

Parameters
- **hnor:** pointer to the NOR handle

Return values
- **NOR:** controller state

HAL_NOR_GetStatus

Function Name **HAL_NOR_StatusTypeDef HAL_NOR_GetStatus (NOR_HandleTypeDef * hnor, uint32_t Address, uint32_t Timeout)**

Function Description Returns the NOR operation status.

Parameters
- **hnor:** pointer to the NOR handle
- **Address:** Device address
- **Timeout:** NOR programming Timeout

Return values

- **NOR_Status:** The returned value can be: HAL_NOR_STATUS_SUCCESS, HAL_NOR_STATUS_ERROR or HAL_NOR_STATUS_TIMEOUT

42.3 NOR Firmware driver defines

42.3.1 NOR

NOR Exported Macros

__HAL_NOR_RESET_HANDLE_STATE

Description:

- Reset NOR handle state.

Parameters:

- __HANDLE__: specifies the NOR handle.

Return value:

- None

43 HAL PCCARD Generic Driver

43.1 PCCARD Firmware driver registers structures

43.1.1 PCCARD_HandleTypeDef

Data Fields

- ***FMC_PCCARD_TypeDef * Instance***
- ***FMC_PCCARD_InitTypeDef Init***
- ***__IO HAL_PCCARD_StateTypeDef State***
- ***HAL_LockTypeDef Lock***

Field Documentation

- ***FMC_PCCARD_TypeDef* PCCARD_HandleTypeDef::Instance***
  Register base address for PCCARD device
- ***FMC_PCCARD_InitTypeDef PCCARD_HandleTypeDef::Init***
  PCCARD device control configuration parameters
- ***__IO HAL_PCCARD_StateTypeDef PCCARD_HandleTypeDef::State***
  PCCARD device access state
- ***HAL_LockTypeDef PCCARD_HandleTypeDef::Lock***
  PCCARD Lock

43.2 PCCARD Firmware driver API description

43.2.1 How to use this driver

This driver is a generic layered driver which contains a set of APIs used to control PCCARD/compact flash memories. It uses the FMC/FSMC layer functions to interface with PCCARD devices. This driver is used for:

- PCCARD/Compact Flash memory configuration sequence using the function HAL_PCCARD_Init()/HAL_CF_Init() with control and timing parameters for both common and attribute spaces.
- Read PCCARD/Compact Flash memory maker and device IDs using the function HAL_PCCARD_Read_ID()/HAL_CF_Read_ID(). The read information is stored in the CompactFlash_ID structure declared by the function caller.
- Access PCCARD/Compact Flash memory by read/write operations using the functions HAL_PCCARD_Read_Sector()/ HAL_PCCARD_Write_Sector() - HAL_CF_Read_Sector()/HAL_CF_Write_Sector(), to read/write sector.
- Perform PCCARD/Compact Flash Reset chip operation using the function HAL_PCCARD_Reset()/HAL_CF_Reset.
- Perform PCCARD/Compact Flash erase sector operation using the function HAL_PCCARD_Erase_Sector()/HAL_CF_Erase_Sector.
- Read the PCCARD/Compact Flash status operation using the function HAL_PCCARD_ReadStatus()/HAL_CF_ReadStatus().
- You can monitor the PCCARD/Compact Flash device HAL state by calling the function HAL_PCCARD_GetState()/HAL_CF_GetState()

This driver is a set of generic APIs which handle standard PCCARD/compact flash operations. If a PCCARD/Compact Flash device contains different operations and/or implementations, it should be implemented separately.

43.2.2 PCCARD Initialization and de-initialization functions

This section provides functions allowing to initialize/de-initialize the PCCARD memory

This section contains the following APIs:

- ***HAL_PCCARD_Init()***
- ***HAL_PCCARD_DeInit()***
- ***HAL_PCCARD_MspInit()***
- ***HAL_PCCARD_MspDeInit()***

43.2.3 PCCARD Input and Output functions

This section provides functions allowing to use and control the PCCARD memory

This section contains the following APIs:

- ***HAL_PCCARD_Read_ID()***
- ***HAL_PCCARD_Read_Sector()***
- ***HAL_PCCARD_Write_Sector()***
- ***HAL_PCCARD_Erase_Sector()***
- ***HAL_PCCARD_Reset()***
- ***HAL_PCCARD_IRQHandler()***
- ***HAL_PCCARD_ITCallback()***

43.2.4 PCCARD State functions

This subsection permits to get in run-time the status of the PCCARD controller and the data flow.

This section contains the following APIs:

- ***HAL_PCCARD_GetState()***
- ***HAL_PCCARD_GetStatus()***
- ***HAL_PCCARD_ReadStatus()***

43.2.5 Detailed description of functions

HAL_PCCARD_Init

Function Name	**HAL_StatusTypeDef HAL_PCCARD_Init (PCCARD_HandleTypeDef * hpccard, FMC_NAND_PCC_TimingTypeDef * ComSpaceTiming, FMC_NAND_PCC_TimingTypeDef * AttSpaceTiming, FMC_NAND_PCC_TimingTypeDef * IOSpaceTiming)**
Function Description	Perform the PCCARD memory Initialization sequence.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module. • **ComSpaceTiming:** Common space timing structure • **AttSpaceTiming:** Attribute space timing structure

- **IOSpaceTiming:** IO space timing structure

Return values
- **HAL:** status

HAL_PCCARD_DeInit

Function Name **HAL_StatusTypeDef HAL_PCCARD_DeInit (PCCARD_HandleTypeDef * hpccard)**

Function Description Perform the PCCARD memory De-initialization sequence.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **HAL:** status

HAL_PCCARD_MspInit

Function Name **void HAL_PCCARD_MspInit (PCCARD_HandleTypeDef * hpccard)**

Function Description PCCARD MSP Init.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **None**

HAL_PCCARD_MspDeInit

Function Name **void HAL_PCCARD_MspDeInit (PCCARD_HandleTypeDef * hpccard)**

Function Description PCCARD MSP DeInit.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **None**

HAL_PCCARD_Read_ID

Function Name **HAL_StatusTypeDef HAL_PCCARD_Read_ID (PCCARD_HandleTypeDef * hpccard, uint8_t CompactFlash_ID, uint8_t * pStatus)**

Function Description Read Compact Flash's ID.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.
- **CompactFlash_ID:** Compact flash ID structure.
- **pStatus:** pointer to compact flash status

Return values
- **HAL:** status

HAL_PCCARD_Write_Sector

Function Name	**HAL_StatusTypeDef HAL_PCCARD_Write_Sector (PCCARD_HandleTypeDef * hpccard, uint16_t * pBuffer, uint16_t SectorAddress, uint8_t * pStatus)**
Function Description	Write sector to PCCARD memory.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module. • **pBuffer:** pointer to source write buffer • **SectorAddress:** Sector address to write • **pStatus:** pointer to PCCARD status
Return values	• **HAL:** status

HAL_PCCARD_Read_Sector

Function Name	**HAL_StatusTypeDef HAL_PCCARD_Read_Sector (PCCARD_HandleTypeDef * hpccard, uint16_t * pBuffer, uint16_t SectorAddress, uint8_t * pStatus)**
Function Description	Read sector from PCCARD memory.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module. • **pBuffer:** pointer to destination read buffer • **SectorAddress:** Sector address to read • **pStatus:** pointer to PCCARD status
Return values	• **HAL:** status

HAL_PCCARD_Erase_Sector

Function Name	**HAL_StatusTypeDef HAL_PCCARD_Erase_Sector (PCCARD_HandleTypeDef * hpccard, uint16_t SectorAddress, uint8_t * pStatus)**
Function Description	Erase sector from PCCARD memory.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module. • **SectorAddress:** Sector address to erase • **pStatus:** pointer to PCCARD status
Return values	• **HAL:** status

HAL_PCCARD_Reset

Function Name	**HAL_StatusTypeDef HAL_PCCARD_Reset (PCCARD_HandleTypeDef * hpccard)**
Function Description	Reset the PCCARD memory.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD

module.

Return values
- **HAL:** status

HAL_PCCARD_IRQHandler

Function Name **void HAL_PCCARD_IRQHandler (PCCARD_HandleTypeDef * hpccard)**

Function Description This function handles PCCARD device interrupt request.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **HAL:** status

HAL_PCCARD_ITCallback

Function Name **void HAL_PCCARD_ITCallback (PCCARD_HandleTypeDef * hpccard)**

Function Description PCCARD interrupt feature callback.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **None**

HAL_PCCARD_GetState

Function Name **HAL_PCCARD_StateTypeDef HAL_PCCARD_GetState (PCCARD_HandleTypeDef * hpccard)**

Function Description return the PCCARD controller state

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **HAL:** state

HAL_PCCARD_GetStatus

Function Name **HAL_PCCARD_StatusTypeDef HAL_PCCARD_GetStatus (PCCARD_HandleTypeDef * hpccard)**

Function Description Get the compact flash memory status.

Parameters
- **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.

Return values
- **New:** status of the PCCARD operation. This parameter can be:
  - CompactFlash_TIMEOUT_ERROR: when the previous operation generate a Timeout error
  - CompactFlash_READY: when memory is ready for the next operation

HAL_PCCARD_ReadStatus

Function Name	**HAL_PCCARD_StatusTypeDef HAL_PCCARD_ReadStatus (PCCARD_HandleTypeDef * hpccard)**
Function Description	Reads the Compact Flash memory status using the Read status command.
Parameters	• **hpccard:** pointer to a PCCARD_HandleTypeDef structure that contains the configuration information for PCCARD module.
Return values	• **The:** status of the Compact Flash memory. This parameter can be: – CompactFlash_BUSY: when memory is busy – CompactFlash_READY: when memory is ready for the next operation – CompactFlash_ERROR: when the previous operation generates error

43.3 PCCARD Firmware driver defines

43.3.1 PCCARD

PCCARD Exported Macros

__HAL_PCCARD_RESET_HANDLE_STATE

Description:

- Reset PCCARD handle state.

Parameters:

- __HANDLE__: specifies the PCCARD handle.

Return value:

- None

44 HAL PCD Generic Driver

44.1 PCD Firmware driver registers structures

44.1.1 PCD_HandleTypeDef

Data Fields

- ***PCD_TypeDef * Instance***
- ***PCD_InitTypeDef Init***
- ***PCD_EPTypeDef IN_ep***
- ***PCD_EPTypeDef OUT_ep***
- ***HAL_LockTypeDef Lock***
- ***__IO PCD_StateTypeDef State***
- ***uint32_t Setup***
- ***PCD_LPM_StateTypeDef LPM_State***
- ***uint32_t BESL***
- ***uint32_t lpm_active***
- ***void * pData***

Field Documentation

- ***PCD_TypeDef* PCD_HandleTypeDef::Instance***
  Register base address
- ***PCD_InitTypeDef PCD_HandleTypeDef::Init***
  PCD required parameters
- ***PCD_EPTypeDef PCD_HandleTypeDef::IN_ep[15]***
  IN endpoint parameters
- ***PCD_EPTypeDef PCD_HandleTypeDef::OUT_ep[15]***
  OUT endpoint parameters
- ***HAL_LockTypeDef PCD_HandleTypeDef::Lock***
  PCD peripheral status
- ***__IO PCD_StateTypeDef PCD_HandleTypeDef::State***
  PCD communication state
- ***uint32_t PCD_HandleTypeDef::Setup[12]***
  Setup packet buffer
- ***PCD_LPM_StateTypeDef PCD_HandleTypeDef::LPM_State***
  LPM State
- ***uint32_t PCD_HandleTypeDef::BESL***
- ***uint32_t PCD_HandleTypeDef::lpm_active***
  Enable or disable the Link Power Management . This parameter can be set to ENABLE or DISABLE
- ***void* PCD_HandleTypeDef::pData***
  Pointer to upper stack Handler

44.2 PCD Firmware driver API description

44.2.1 How to use this driver

The PCD HAL driver can be used as follows:

1. Declare a PCD_HandleTypeDef handle structure, for example: PCD_HandleTypeDef hpcd;
2. Fill parameters of Init structure in HCD handle
3. Call HAL_PCD_Init() API to initialize the PCD peripheral (Core, Device core, ...)
4. Initialize the PCD low level resources through the HAL_PCD_MspInit() API:
   a. Enable the PCD/USB Low Level interface clock using
      – __HAL_RCC_USB_OTG_FS_CLK_ENABLE();
      – __HAL_RCC_USB_OTG_HS_CLK_ENABLE(); (For High Speed Mode)
   b. Initialize the related GPIO clocks
   c. Configure PCD pin-out
   d. Configure PCD NVIC interrupt
5. Associate the Upper USB device stack to the HAL PCD Driver:
   a. hpcd.pData = pdev;
6. Enable PCD transmission and reception:
   a. HAL_PCD_Start();

44.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

This section contains the following APIs:

- ***HAL_PCD_Init()***
- ***HAL_PCD_DeInit()***
- ***HAL_PCD_MspInit()***
- ***HAL_PCD_MspDeInit()***

44.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the PCD data transfers.

This section contains the following APIs:

- ***HAL_PCD_Start()***
- ***HAL_PCD_Stop()***
- ***HAL_PCD_IRQHandler()***
- ***HAL_PCD_DataOutStageCallback()***
- ***HAL_PCD_DataInStageCallback()***
- ***HAL_PCD_SetupStageCallback()***
- ***HAL_PCD_SOFCallback()***
- ***HAL_PCD_ResetCallback()***
- ***HAL_PCD_SuspendCallback()***
- ***HAL_PCD_ResumeCallback()***
- ***HAL_PCD_ISOOUTIncompleteCallback()***
- ***HAL_PCD_ISOINIncompleteCallback()***
- ***HAL_PCD_ConnectCallback()***
- ***HAL_PCD_DisconnectCallback()***

44.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the PCD data transfers.

This section contains the following APIs:

- ***HAL_PCD_DevConnect()***
- ***HAL_PCD_DevDisconnect()***
- ***HAL_PCD_SetAddress()***
- ***HAL_PCD_EP_Open()***
- ***HAL_PCD_EP_Close()***
- ***HAL_PCD_EP_Receive()***
- ***HAL_PCD_EP_GetRxCount()***
- ***HAL_PCD_EP_Transmit()***
- ***HAL_PCD_EP_SetStall()***
- ***HAL_PCD_EP_ClrStall()***
- ***HAL_PCD_EP_Flush()***
- ***HAL_PCD_ActivateRemoteWakeup()***
- ***HAL_PCD_DeActivateRemoteWakeup()***

44.2.5 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_PCD_GetState()***

44.2.6 Detailed description of functions

HAL_PCD_Init

Function Name	**HAL_StatusTypeDef HAL_PCD_Init (PCD_HandleTypeDef * hpcd)**
Function Description	Initializes the PCD according to the specified parameters in the PCD_InitTypeDef and initialize the associated handle.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_DeInit

Function Name	**HAL_StatusTypeDef HAL_PCD_DeInit (PCD_HandleTypeDef * hpcd)**
Function Description	DeInitializes the PCD peripheral.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_MspInit

Function Name	**void HAL_PCD_MspInit (PCD_HandleTypeDef * hpcd)**
Function Description	Initializes the PCD MSP.
Parameters	• **hpcd:** PCD handle
Return values	• **None**

HAL_PCD_MspDeInit

Function Name	**void HAL_PCD_MspDeInit (PCD_HandleTypeDef * hpcd)**
Function Description	DeInitializes PCD MSP.
Parameters	• **hpcd:** PCD handle
Return values	• **None**

HAL_PCD_Start

Function Name	**HAL_StatusTypeDef HAL_PCD_Start (PCD_HandleTypeDef * hpcd)**
Function Description	Start The USB OTG Device.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_Stop

Function Name	**HAL_StatusTypeDef HAL_PCD_Stop (PCD_HandleTypeDef * hpcd)**
Function Description	Stop The USB OTG Device.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_IRQHandler

Function Name	**void HAL_PCD_IRQHandler (PCD_HandleTypeDef * hpcd)**
Function Description	Handles PCD interrupt request.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_DataOutStageCallback

Function Name	**void HAL_PCD_DataOutStageCallback (PCD_HandleTypeDef * hpcd, uint8_t epnum)**
Function Description	Data OUT stage callback.
Parameters	• **hpcd:** PCD handle • **epnum:** endpoint number

Return values
- **None**

HAL_PCD_DataInStageCallback

Function Name **void HAL_PCD_DataInStageCallback (PCD_HandleTypeDef * hpcd, uint8_t epnum)**

Function Description Data IN stage callback.

Parameters
- **hpcd:** PCD handle
- **epnum:** endpoint number

Return values
- **None**

HAL_PCD_SetupStageCallback

Function Name **void HAL_PCD_SetupStageCallback (PCD_HandleTypeDef * hpcd)**

Function Description Setup stage callback.

Parameters
- **hpcd:** PCD handle

Return values
- **None**

HAL_PCD_SOFCallback

Function Name **void HAL_PCD_SOFCallback (PCD_HandleTypeDef * hpcd)**

Function Description USB Start Of Frame callback.

Parameters
- **hpcd:** PCD handle

Return values
- **None**

HAL_PCD_ResetCallback

Function Name **void HAL_PCD_ResetCallback (PCD_HandleTypeDef * hpcd)**

Function Description USB Reset callback.

Parameters
- **hpcd:** PCD handle

Return values
- **None**

HAL_PCD_SuspendCallback

Function Name **void HAL_PCD_SuspendCallback (PCD_HandleTypeDef * hpcd)**

Function Description Suspend event callback.

Parameters
- **hpcd:** PCD handle

Return values
- **None**

HAL_PCD_ResumeCallback

Function Name	**void HAL_PCD_ResumeCallback (PCD_HandleTypeDef * hpcd)**
Function Description	Resume event callback.
Parameters	• **hpcd:** PCD handle
Return values	• **None**

HAL_PCD_ISOOUTIncompleteCallback

Function Name	**void HAL_PCD_ISOOUTIncompleteCallback (PCD_HandleTypeDef * hpcd, uint8_t epnum)**
Function Description	Incomplete ISO OUT callback.
Parameters	• **hpcd:** PCD handle • **epnum:** endpoint number
Return values	• **None**

HAL_PCD_ISOINIncompleteCallback

Function Name	**void HAL_PCD_ISOINIncompleteCallback (PCD_HandleTypeDef * hpcd, uint8_t epnum)**
Function Description	Incomplete ISO IN callback.
Parameters	• **hpcd:** PCD handle • **epnum:** endpoint number
Return values	• **None**

HAL_PCD_ConnectCallback

Function Name	**void HAL_PCD_ConnectCallback (PCD_HandleTypeDef * hpcd)**
Function Description	Connection event callback.
Parameters	• **hpcd:** PCD handle
Return values	• **None**

HAL_PCD_DisconnectCallback

Function Name	**void HAL_PCD_DisconnectCallback (PCD_HandleTypeDef * hpcd)**
Function Description	Disconnection event callback.
Parameters	• **hpcd:** PCD handle
Return values	• **None**

HAL_PCD_DevConnect

Function Name	**HAL_StatusTypeDef HAL_PCD_DevConnect (PCD_HandleTypeDef * hpcd)**
Function Description	Connect the USB device.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_DevDisconnect

Function Name	**HAL_StatusTypeDef HAL_PCD_DevDisconnect (PCD_HandleTypeDef * hpcd)**
Function Description	Disconnect the USB device.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCD_SetAddress

Function Name	**HAL_StatusTypeDef HAL_PCD_SetAddress (PCD_HandleTypeDef * hpcd, uint8_t address)**
Function Description	Set the USB Device address.
Parameters	• **hpcd:** PCD handle • **address:** new device address
Return values	• **HAL:** status

HAL_PCD_EP_Open

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_Open (PCD_HandleTypeDef * hpcd, uint8_t ep_addr, uint16_t ep_mps, uint8_t ep_type)**
Function Description	Open and configure an endpoint.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address • **ep_mps:** endpoint max packet size • **ep_type:** endpoint type
Return values	• **HAL:** status

HAL_PCD_EP_Close

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_Close (PCD_HandleTypeDef * hpcd, uint8_t ep_addr)**
Function Description	Deactivate an endpoint.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address
Return values	• **HAL:** status

HAL_PCD_EP_Receive

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_Receive (PCD_HandleTypeDef * hpcd, uint8_t ep_addr, uint8_t * pBuf, uint32_t len)**
Function Description	Receive an amount of data.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address • **pBuf:** pointer to the reception buffer • **len:** amount of data to be received
Return values	• **HAL:** status

HAL_PCD_EP_Transmit

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_Transmit (PCD_HandleTypeDef * hpcd, uint8_t ep_addr, uint8_t * pBuf, uint32_t len)**
Function Description	Send an amount of data.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address • **pBuf:** pointer to the transmission buffer • **len:** amount of data to be sent
Return values	• **HAL:** status

HAL_PCD_EP_GetRxCount

Function Name	**uint16_t HAL_PCD_EP_GetRxCount (PCD_HandleTypeDef * hpcd, uint8_t ep_addr)**
Function Description	Get Received Data Size.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address
Return values	• **Data:** Size

HAL_PCD_EP_SetStall

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_SetStall (PCD_HandleTypeDef * hpcd, uint8_t ep_addr)**
Function Description	Set a STALL condition over an endpoint.
Parameters	• **hpcd:** PCD handle • **ep_addr:** endpoint address
Return values	• **HAL:** status

HAL_PCD_EP_ClrStall

Function Name	**HAL_StatusTypeDef HAL_PCD_EP_ClrStall (PCD_HandleTypeDef * hpcd, uint8_t ep_addr)**
Function Description	Clear a STALL condition over in an endpoint.

Parameters
- **hpcd:** PCD handle
- **ep_addr:** endpoint address

Return values
- **HAL:** status

HAL_PCD_EP_Flush

Function Name **HAL_StatusTypeDef HAL_PCD_EP_Flush (PCD_HandleTypeDef * hpcd, uint8_t ep_addr)**

Function Description Flush an endpoint.

Parameters
- **hpcd:** PCD handle
- **ep_addr:** endpoint address

Return values
- **HAL:** status

HAL_PCD_ActivateRemoteWakeup

Function Name **HAL_StatusTypeDef HAL_PCD_ActivateRemoteWakeup (PCD_HandleTypeDef * hpcd)**

Function Description Activate remote wakeup signalling.

Parameters
- **hpcd:** PCD handle

Return values
- **HAL:** status

HAL_PCD_DeActivateRemoteWakeup

Function Name **HAL_StatusTypeDef HAL_PCD_DeActivateRemoteWakeup (PCD_HandleTypeDef * hpcd)**

Function Description De-activate remote wakeup signalling.

Parameters
- **hpcd:** PCD handle

Return values
- **HAL:** status

HAL_PCD_GetState

Function Name **PCD_StateTypeDef HAL_PCD_GetState (PCD_HandleTypeDef * hpcd)**

Function Description Return the PCD handle state.

Parameters
- **hpcd:** PCD handle

Return values
- **HAL:** state

44.3 PCD Firmware driver defines

44.3.1 PCD

PCD Exported Macros

__HAL_PCD_ENABLE

__HAL_PCD_DISABLE

__HAL_PCD_GET_FLAG

__HAL_PCD_CLEAR_FLAG	
__HAL_PCD_IS_INVALID_INTERRUPT	
__HAL_PCD_UNGATE_PHYCLOCK	
__HAL_PCD_GATE_PHYCLOCK	
__HAL_PCD_IS_PHY_SUSPENDED	
USB_OTG_FS_WAKEUP_EXTI_RISING_EDGE	
USB_OTG_FS_WAKEUP_EXTI_FALLING_EDGE	
USB_OTG_FS_WAKEUP_EXTI_RISING_FALLING_EDGE	
USB_OTG_HS_WAKEUP_EXTI_RISING_EDGE	
USB_OTG_HS_WAKEUP_EXTI_FALLING_EDGE	
USB_OTG_HS_WAKEUP_EXTI_RISING_FALLING_EDGE	
USB_OTG_HS_WAKEUP_EXTI_LINE	External interrupt line 20 Connected to the USB HS EXTI Line
USB_OTG_FS_WAKEUP_EXTI_LINE	External interrupt line 18 Connected to the USB FS EXTI Line
__HAL_USB_OTG_HS_WAKEUP_EXTI_ENABLE_IT	
__HAL_USB_OTG_HS_WAKEUP_EXTI_DISABLE_IT	
__HAL_USB_OTG_HS_WAKEUP_EXTI_GET_FLAG	
__HAL_USB_OTG_HS_WAKEUP_EXTI_CLEAR_FLAG	
__HAL_USB_OTG_HS_WAKEUP_EXTI_ENABLE_RISING_EDGE	
__HAL_USB_OTG_HS_WAKEUP_EXTI_ENABLE_FALLING_EDGE	
__HAL_USB_OTG_HS_WAKEUP_EXTI_ENABLE_RISING_FALLING_EDGE	
__HAL_USB_OTG_HS_WAKEUP_EXTI_GENERATE_SWIT	
__HAL_USB_OTG_FS_WAKEUP_EXTI_ENABLE_IT	
__HAL_USB_OTG_FS_WAKEUP_EXTI_DISABLE_IT	
__HAL_USB_OTG_FS_WAKEUP_EXTI_GET_FLAG	
__HAL_USB_OTG_FS_WAKEUP_EXTI_CLEAR_FLAG	
__HAL_USB_OTG_FS_WAKEUP_EXTI_ENABLE_RISING_EDGE	
__HAL_USB_OTG_FS_WAKEUP_EXTI_ENABLE_FALLING_EDGE	
__HAL_USB_OTG_FS_WAKEUP_EXTI_ENABLE_RISING_FALLING_EDGE	

__HAL_USB_OTG_FS_WAKEUP_EXTI_GENERATE_SWIT

PCD PHY Module

PCD_PHY_ULPI

PCD_PHY_EMBEDDED

PCD Speed

PCD_SPEED_HIGH

PCD_SPEED_HIGH_IN_FULL

PCD_SPEED_FULL

Turnaround Timeout Value

USBD_HS_TRDT_VALUE

USBD_FS_TRDT_VALUE

45 HAL PCD Extension Driver

45.1 PCDEx Firmware driver API description

45.1.1 Extended features functions

This section provides functions allowing to:

- Update FIFO configuration

This section contains the following APIs:

- ***HAL_PCDEx_SetTxFiFo()***
- ***HAL_PCDEx_SetRxFiFo()***
- ***HAL_PCDEx_ActivateLPM()***
- ***HAL_PCDEx_DeActivateLPM()***
- ***HAL_PCDEx_LPM_Callback()***

45.1.2 Detailed description of functions

HAL_PCDEx_SetTxFiFo

Function Name	**HAL_StatusTypeDef HAL_PCDEx_SetTxFiFo (PCD_HandleTypeDef * hpcd, uint8_t fifo, uint16_t size)**
Function Description	Set Tx FIFO.
Parameters	• **hpcd:** PCD handle • **fifo:** The number of Tx fifo • **size:** Fifo size
Return values	• **HAL:** status

HAL_PCDEx_SetRxFiFo

Function Name	**HAL_StatusTypeDef HAL_PCDEx_SetRxFiFo (PCD_HandleTypeDef * hpcd, uint16_t size)**
Function Description	Set Rx FIFO.
Parameters	• **hpcd:** PCD handle • **size:** Size of Rx fifo
Return values	• **HAL:** status

HAL_PCDEx_ActivateLPM

Function Name	**HAL_StatusTypeDef HAL_PCDEx_ActivateLPM (PCD_HandleTypeDef * hpcd)**
Function Description	Activate LPM feature.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCDEx_DeActivateLPM

Function Name	**HAL_StatusTypeDef HAL_PCDEx_DeActivateLPM (PCD_HandleTypeDef * hpcd)**
Function Description	Deactivate LPM feature.
Parameters	• **hpcd:** PCD handle
Return values	• **HAL:** status

HAL_PCDEx_LPM_Callback

Function Name	**void HAL_PCDEx_LPM_Callback (PCD_HandleTypeDef * hpcd, PCD_LPM_MsgTypeDef msg)**
Function Description	Send LPM message to user layer callback.
Parameters	• **hpcd:** PCD handle • **msg:** LPM message
Return values	• **HAL:** status

46 HAL PWR Generic Driver

46.1 PWR Firmware driver registers structures

46.1.1 PWR_PVDTypeDef

Data Fields

- ***uint32_t PVDLevel***
- ***uint32_t Mode***

Field Documentation

- ***uint32_t PWR_PVDTypeDef::PVDLevel***
  PVDLevel: Specifies the PVD detection level. This parameter can be a value of ***PWR_PVD_detection_level***
- ***uint32_t PWR_PVDTypeDef::Mode***
  Mode: Specifies the operating mode for the selected pins. This parameter can be a value of ***PWR_PVD_Mode***

46.2 PWR Firmware driver API description

46.2.1 Initialization and de-initialization functions

After reset, the backup domain (RTC registers, RTC backup data registers and backup SRAM) is protected against possible unwanted write accesses. To enable access to the RTC Domain and RTC registers, proceed as follows:

- Enable the Power Controller (PWR) APB1 interface clock using the __HAL_RCC_PWR_CLK_ENABLE() macro.
- Enable access to RTC domain using the HAL_PWR_EnableBkUpAccess() function.

This section contains the following APIs:

- ***HAL_PWR_DeInit()***
- ***HAL_PWR_EnableBkUpAccess()***
- ***HAL_PWR_DisableBkUpAccess()***

46.2.2 Peripheral Control functions

PVD configuration

- The PVD is used to monitor the VDD power supply by comparing it to a threshold selected by the PVD Level (PLS[2:0] bits in the PWR_CR).
- A PVDO flag is available to indicate if VDD/VDDA is higher or lower than the PVD threshold. This event is internally connected to the EXTI line16 and can generate an interrupt if enabled. This is done through __HAL_PWR_PVD_EXTI_ENABLE_IT() macro.
- The PVD is stopped in Standby mode.

Wake-up pin configuration

- Wake-up pin is used to wake up the system from Standby mode. This pin is forced in input pull-down configuration and is active on rising edges.
- There is one Wake-up pin: Wake-up Pin 1 on PA.00.
  - For STM32F446xx there are two Wake-Up pins: Pin1 on PA.00 and Pin2 on PC.13
  - For STM32F410xx/STM32F412Zx/STM32F412Rx/STM32F412Vx/STM32F412Cx there are three Wake-Up pins: Pin1 on PA.00, Pin2 on PC.00 and Pin3 on PC.01

Low Power modes configuration

The devices feature 3 low-power modes:

- Sleep mode: Cortex-M4 core stopped, peripherals kept running.
- Stop mode: all clocks are stopped, regulator running, regulator in low power mode
- Standby mode: 1.2V domain powered off.

Sleep mode

- Entry: The Sleep mode is entered by using the HAL_PWR_EnterSLEEPMode(PWR_MAINREGULATOR_ON, PWR_SLEEPENTRY_WFI) functions with
  - PWR_SLEEPENTRY_WFI: enter SLEEP mode with WFI instruction
  - PWR_SLEEPENTRY_WFE: enter SLEEP mode with WFE instruction The Regulator parameter is not used for the STM32F4 family and is kept as parameter just to maintain compatibility with the lower power families (STM32L).
- Exit: Any peripheral interrupt acknowledged by the nested vectored interrupt controller (NVIC) can wake up the device from Sleep mode.

Stop mode

In Stop mode, all clocks in the 1.2V domain are stopped, the PLL, the HSI, and the HSE RC oscillators are disabled. Internal SRAM and register contents are preserved. The voltage regulator can be configured either in normal or low-power mode. To minimize the consumption In Stop mode, FLASH can be powered off before entering the Stop mode using the HAL_PWREx_EnableFlashPowerDown() function. It can be switched on again by software after exiting the Stop mode using the HAL_PWREx_DisableFlashPowerDown() function.

- Entry: The Stop mode is entered using the HAL_PWR_EnterSTOPMode(PWR_MAINREGULATOR_ON) function with:
  - Main regulator ON.
  - Low Power regulator ON.
- Exit: Any EXTI Line (Internal or External) configured in Interrupt/Event mode.

Standby mode

- The Standby mode allows to achieve the lowest power consumption. It is based on the Cortex-M4 deep sleep mode, with the voltage regulator disabled. The 1.2V domain is consequently powered off. The PLL, the HSI oscillator and the HSE oscillator are also switched off. SRAM and register contents are lost except for the RTC registers,

RTC backup registers, backup SRAM and Standby circuitry. The voltage regulator is OFF.

– Entry:
  – The Standby mode is entered using the HAL_PWR_EnterSTANDBYMode() function.
– Exit:
  – WKUP pin rising edge, RTC alarm (Alarm A and Alarm B), RTC wake-up, tamper event, time-stamp event, external reset in NRST pin, IWDG reset.

Auto-wake-up (AWU) from low-power mode

- The MCU can be woken up from low-power mode by an RTC Alarm event, an RTC Wake-up event, a tamper event or a time-stamp event, without depending on an external interrupt (Auto-wake-up mode).
- RTC auto-wake-up (AWU) from the Stop and Standby modes
  – To wake up from the Stop mode with an RTC alarm event, it is necessary to configure the RTC to generate the RTC alarm using the HAL_RTC_SetAlarm_IT() function.
  – To wake up from the Stop mode with an RTC Tamper or time stamp event, it is necessary to configure the RTC to detect the tamper or time stamp event using the HAL_RTCEx_SetTimeStamp_IT() or HAL_RTCEx_SetTamper_IT() functions.
  – To wake up from the Stop mode with an RTC Wake-up event, it is necessary to configure the RTC to generate the RTC Wake-up event using the HAL_RTCEx_SetWakeUpTimer_IT() function.

This section contains the following APIs:

- ***HAL_PWR_ConfigPVD()***
- ***HAL_PWR_EnablePVD()***
- ***HAL_PWR_DisablePVD()***
- ***HAL_PWR_EnableWakeUpPin()***
- ***HAL_PWR_DisableWakeUpPin()***
- ***HAL_PWR_EnterSLEEPMode()***
- ***HAL_PWR_EnterSTOPMode()***
- ***HAL_PWR_EnterSTANDBYMode()***
- ***HAL_PWR_PVD_IRQHandler()***
- ***HAL_PWR_PVDCallback()***
- ***HAL_PWR_EnableSleepOnExit()***
- ***HAL_PWR_DisableSleepOnExit()***
- ***HAL_PWR_EnableSEVOnPend()***
- ***HAL_PWR_DisableSEVOnPend()***

46.2.3 Detailed description of functions

HAL_PWR_DeInit

Function Name	**void HAL_PWR_DeInit (void)**
Function Description	Deinitializes the HAL PWR peripheral registers to their default reset values.
Return values	• **None**

HAL_PWR_EnableBkUpAccess

Function Name	**void HAL_PWR_EnableBkUpAccess (void)**
Function Description	Enables access to the backup domain (RTC registers, RTC backup data registers and backup SRAM).
Return values	• **None**
Notes	• If the HSE divided by 2, 3, ..31 is used as the RTC clock, the Backup Domain Access should be kept enabled.

HAL_PWR_DisableBkUpAccess

Function Name	**void HAL_PWR_DisableBkUpAccess (void)**
Function Description	Disables access to the backup domain (RTC registers, RTC backup data registers and backup SRAM).
Return values	• **None**
Notes	• If the HSE divided by 2, 3, ..31 is used as the RTC clock, the Backup Domain Access should be kept enabled.

HAL_PWR_ConfigPVD

Function Name	**void HAL_PWR_ConfigPVD (PWR_PVDTypeDef * sConfigPVD)**
Function Description	Configures the voltage threshold detected by the Power Voltage Detector(PVD).
Parameters	• **sConfigPVD:** pointer to an PWR_PVDTypeDef structure that contains the configuration information for the PVD.
Return values	• **None**
Notes	• Refer to the electrical characteristics of your device datasheet for more details about the voltage threshold corresponding to each detection level.

HAL_PWR_EnablePVD

Function Name	**void HAL_PWR_EnablePVD (void)**
Function Description	Enables the Power Voltage Detector(PVD).
Return values	• **None**

HAL_PWR_DisablePVD

Function Name	**void HAL_PWR_DisablePVD (void)**
Function Description	Disables the Power Voltage Detector(PVD).
Return values	• **None**

HAL_PWR_EnableWakeUpPin

Function Name	**void HAL_PWR_EnableWakeUpPin (uint32_t WakeUpPinx)**
Function Description	Enables the Wake-up PINx functionality.
Parameters	• **WakeUpPinx:** Specifies the Power Wake-Up pin to enable. This parameter can be one of the following values: – PWR_WAKEUP_PIN1 – PWR_WAKEUP_PIN2 available only on STM32F410xx/STM32F446xx/STM32F412Zx/STM32F412Rx/STM32F412Vx/STM32F412Cx devices – PWR_WAKEUP_PIN3 available only on STM32F410xx/STM32F412xx devices
Return values	• **None**

HAL_PWR_DisableWakeUpPin

Function Name	**void HAL_PWR_DisableWakeUpPin (uint32_t WakeUpPinx)**
Function Description	Disables the Wake-up PINx functionality.
Parameters	• **WakeUpPinx:** Specifies the Power Wake-Up pin to disable. This parameter can be one of the following values: – PWR_WAKEUP_PIN1 – PWR_WAKEUP_PIN2 available only on STM32F410xx/STM32F446xx/STM32F412Zx/STM32F412Rx/STM32F412Vx/STM32F412Cx devices – PWR_WAKEUP_PIN3 available only on STM32F410xx/STM32F412Zx/STM32F412Rx/STM32F412Vx/STM32F412Cx devices
Return values	• **None**

HAL_PWR_EnterSTOPMode

Function Name	**void HAL_PWR_EnterSTOPMode (uint32_t Regulator, uint8_t STOPEntry)**
Function Description	Enters Stop mode.
Parameters	• **Regulator:** Specifies the regulator state in Stop mode. This parameter can be one of the following values: – PWR_MAINREGULATOR_ON: Stop mode with regulator ON – PWR_LOWPOWERREGULATOR_ON: Stop mode with low power regulator ON • **STOPEntry:** Specifies if Stop mode in entered with WFI or WFE instruction. This parameter can be one of the following values:

  - PWR_STOPENTRY_WFI: Enter Stop mode with WFI instruction
  - PWR_STOPENTRY_WFE: Enter Stop mode with WFE instruction

Return values
- **None**

Notes
- In Stop mode, all I/O pins keep the same state as in Run mode.
- When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock.
- When the voltage regulator operates in low power mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

HAL_PWR_EnterSLEEPMode

Function Name **void HAL_PWR_EnterSLEEPMode (uint32_t Regulator, uint8_t SLEEPEntry)**

Function Description Enters Sleep mode.

Parameters
- **Regulator:** Specifies the regulator state in SLEEP mode. This parameter can be one of the following values:
  - PWR_MAINREGULATOR_ON: SLEEP mode with regulator ON
  - PWR_LOWPOWERREGULATOR_ON: SLEEP mode with low power regulator ON
- **SLEEPEntry:** Specifies if SLEEP mode in entered with WFI or WFE instruction. This parameter can be one of the following values:
  - PWR_SLEEPENTRY_WFI: enter SLEEP mode with WFI instruction
  - PWR_SLEEPENTRY_WFE: enter SLEEP mode with WFE instruction

Return values
- **None**

Notes
- In Sleep mode, all I/O pins keep the same state as in Run mode.
- In Sleep mode, the systick is stopped to avoid exit from this mode with systick interrupt when used as time base for Timeout
- This parameter is not used for the STM32F4 family and is kept as parameter just to maintain compatibility with the lower power families.

HAL_PWR_EnterSTANDBYMode

Function Name **void HAL_PWR_EnterSTANDBYMode (void)**

Function Description Enters Standby mode.

Return values
- **None**

Notes
- In Standby mode, all I/O pins are high impedance except for:

Reset pad (still available)RTC_AF1 pin (PC13) if configured for tamper, time-stamp, RTC Alarm out, or RTC clock calibration out.RTC_AF2 pin (PI8) if configured for tamper or time-stamp.WKUP pin 1 (PA0) if enabled.

HAL_PWR_PVD_IRQHandler

Function Name	**void HAL_PWR_PVD_IRQHandler (void)**
Function Description	This function handles the PWR PVD interrupt request.
Return values	• **None**
Notes	• This API should be called under the PVD_IRQHandler().

HAL_PWR_PVDCallback

Function Name	**void HAL_PWR_PVDCallback (void)**
Function Description	PWR PVD interrupt callback.
Return values	• **None**

HAL_PWR_EnableSleepOnExit

Function Name	**void HAL_PWR_EnableSleepOnExit (void)**
Function Description	Indicates Sleep-On-Exit when returning from Handler mode to Thread mode.
Return values	• **None**
Notes	• Set SLEEPONEXIT bit of SCR register. When this bit is set, the processor re-enters SLEEP mode when an interruption handling is over. Setting this bit is useful when the processor is expected to run only on interruptions handling.

HAL_PWR_DisableSleepOnExit

Function Name	**void HAL_PWR_DisableSleepOnExit (void)**
Function Description	Disables Sleep-On-Exit feature when returning from Handler mode to Thread mode.
Return values	• **None**
Notes	• Clears SLEEPONEXIT bit of SCR register. When this bit is set, the processor re-enters SLEEP mode when an interruption handling is over.

HAL_PWR_EnableSEVOnPend

Function Name	**void HAL_PWR_EnableSEVOnPend (void)**
Function Description	Enables CORTEX M4 SEVONPEND bit.
Return values	• **None**
Notes	• Sets SEVONPEND bit of SCR register. When this bit is set, this causes WFE to wake up when an interrupt moves from

inactive to pended.

HAL_PWR_DisableSEVOnPend

Function Name	**void HAL_PWR_DisableSEVOnPend (void)**
Function Description	Disables CORTEX M4 SEVONPEND bit.
Return values	• **None**
Notes	• Clears SEVONPEND bit of SCR register. When this bit is set, this causes WFE to wake up when an interrupt moves from inactive to pended.

46.3 PWR Firmware driver defines

46.3.1 PWR

PWR CR Register alias address

DBP_BIT_NUMBER

CR_DBP_BB

PVDE_BIT_NUMBER

CR_PVDE_BB

PMODE_BIT_NUMBER

CR_PMODE_BB

PWR CSR Register alias address

EWUP_BIT_NUMBER

CSR_EWUP_BB

PWR Exported Macro

__HAL_PWR_GET_FLAG

Description:

- Check PWR flag is set or not.

Parameters:

- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - PWR_FLAG_WU: Wake Up flag. This flag indicates that a wakeup event was received from the WKUP pin or from the RTC alarm (Alarm A or Alarm B), RTC Tamper event, RTC TimeStamp event or RTC Wakeup. An additional wakeup event is detected if the

WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.
- PWR_FLAG_SB: StandBy flag. This flag indicates that the system was resumed from StandBy mode.
- PWR_FLAG_PVDO: PVD Output. This flag is valid only if PVD is enabled by the HAL_PWR_EnablePVD() function. The PVD is stopped by Standby mode For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.
- PWR_FLAG_BRR: Backup regulator ready flag. This bit is not reset when the device wakes up from Standby mode or by a system reset or power reset.
- PWR_FLAG_VOSRDY: This flag indicates that the Regulator voltage scaling output selection is ready.

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_PWR_CLEAR_FLAG

Description:

- Clear the PWR's pending flags.

Parameters:

- __FLAG__: specifies the flag to clear. This parameter can be one of the following values:
  - PWR_FLAG_WU: Wake Up flag
  - PWR_FLAG_SB: StandBy flag

__HAL_PWR_PVD_EXTI_ENABLE_IT

Description:

- Enable the PVD Exti Line 16.

Return value:

- None.

__HAL_PWR_PVD_EXTI_DISABLE_IT

Description:

- Disable the PVD EXTI Line 16.

Return value:

- None.

__HAL_PWR_PVD_EXTI_ENABLE_EVENT

Description:

- Enable event on PVD Exti Line 16.

Return value:

- None.

__HAL_PWR_PVD_EXTI_DISABLE_EVENT

Description:

- Disable event on PVD Exti Line 16.

Return value:

- None.

__HAL_PWR_PVD_EXTI_ENABLE_RISING_EDGE

Description:

- Enable the PVD Extended Interrupt Rising Trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_DISABLE_RISING_EDGE

Description:

- Disable the PVD Extended Interrupt Rising Trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_ENABLE_FALLING_EDGE

Description:

- Enable the PVD Extended Interrupt Falling Trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_DISABLE_FALLING_EDGE

Description:

- Disable the PVD Extended Interrupt Falling Trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_ENABLE_RISING_FALLING_EDGE

Description:

- PVD EXTI line configuration: set rising & falling edge trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_DISABLE_RISING_FALLING_EDGE

Description:

- Disable the PVD Extended Interrupt Rising & Falling Trigger.

Return value:

- None.

__HAL_PWR_PVD_EXTI_GET_FLAG

Description:

- checks whether the specified PVD Exti interrupt flag is set or not.

Return value:

- EXTI: PVD Line Status.

__HAL_PWR_PVD_EXTI_CLEAR_FLAG

Description:

- Clear the PVD Exti flag.

Return value:

- None.

__HAL_PWR_PVD_EXTI_GENERATE_SWIT

Description:

- Generates a Software interrupt on PVD EXTI line.

Return value:

- None

PWR Flag

PWR_FLAG_WU

PWR_FLAG_SB

PWR_FLAG_PVDO

PWR_FLAG_BRR

PWR_FLAG_VOSRDY

PWR Private macros to check input parameters

IS_PWR_PVD_LEVEL

IS_PWR_PVD_MODE

IS_PWR_REGULATOR

IS_PWR_SLEEP_ENTRY

IS_PWR_STOP_ENTRY

PWR PVD detection level

PWR_PVDLEVEL_0

PWR_PVDLEVEL_1

PWR_PVDLEVEL_2

PWR_PVDLEVEL_3

PWR_PVDLEVEL_4

PWR_PVDLEVEL_5

PWR_PVDLEVEL_6

PWR_PVDLEVEL_7

PWR PVD EXTI Line

PWR_EXTI_LINE_PVD External interrupt line 16 Connected to the PVD EXTI Line

PWR PVD Mode

PWR_PVD_MODE_NORMAL	basic mode is used
PWR_PVD_MODE_IT_RISING	External Interrupt Mode with Rising edge trigger detection
PWR_PVD_MODE_IT_FALLING	External Interrupt Mode with Falling edge trigger detection
PWR_PVD_MODE_IT_RISING_FALLING	External Interrupt Mode with Rising/Falling edge trigger detection
PWR_PVD_MODE_EVENT_RISING	Event Mode with Rising edge trigger detection
PWR_PVD_MODE_EVENT_FALLING	Event Mode with Falling edge trigger detection
PWR_PVD_MODE_EVENT_RISING_FALLING	Event Mode with Rising/Falling edge trigger detection

PWR PVD Mode Mask

PVD_MODE_IT

PVD_MODE_EVT

PVD_RISING_EDGE

PVD_FALLING_EDGE

PWR Register alias address

PWR_OFFSET

PWR_CR_OFFSET

PWR_CSR_OFFSET

PWR_CR_OFFSET_BB

PWR_CSR_OFFSET_BB

PWR Regulator state in SLEEP/STOP mode

PWR_MAINREGULATOR_ON

PWR_LOWPOWERREGULATOR_ON

PWR SLEEP mode entry

PWR_SLEEPENTRY_WFI

PWR_SLEEPENTRY_WFE

PWR STOP mode entry

PWR_STOPENTRY_WFI

PWR_STOPENTRY_WFE

PWR WakeUp Pins

PWR_WAKEUP_PIN1

47 HAL PWR Extension Driver

47.1 PWREx Firmware driver API description

47.1.1 Peripheral extended features functions

Main and Backup Regulators configuration

- The backup domain includes 4 Kbytes of backup SRAM accessible only from the CPU, and address in 32-bit, 16-bit or 8-bit mode. Its content is retained even in Standby or VBAT mode when the low power backup regulator is enabled. It can be considered as an internal EEPROM when VBAT is always present. You can use the HAL_PWREx_EnableBkUpReg() function to enable the low power backup regulator.
- When the backup domain is supplied by VDD (analog switch connected to VDD) the backup SRAM is powered from VDD which replaces the VBAT power supply to save battery life.
- The backup SRAM is not mass erased by a tamper event. It is read protected to prevent confidential data, such as cryptographic private key, from being accessed. The backup SRAM can be erased only through the Flash interface when a protection level change from level 1 to level 0 is requested. Refer to the description of Read protection (RDP) in the Flash programming manual.
- The main internal regulator can be configured to have a tradeoff between performance and power consumption when the device does not operate at the maximum frequency. This is done through __HAL_PWR_MAINREGULATORMODE_CONFIG() macro which configure VOS bit in PWR_CR register Refer to the product datasheets for more details.

FLASH Power Down configuration

- By setting the FPDS bit in the PWR_CR register by using the HAL_PWREx_EnableFlashPowerDown() function, the Flash memory also enters power down mode when the device enters Stop mode. When the Flash memory is in power down mode, an additional startup delay is incurred when waking up from Stop mode.
- For STM32F42xxx/43xxx/446xx/469xx/479xx Devices, the scale can be modified only when the PLL is OFF and the HSI or HSE clock source is selected as system clock. The new value programmed is active only when the PLL is ON. When the PLL is OFF, the voltage scale 3 is automatically selected. Refer to the datasheets for more details.

Over-Drive and Under-Drive configuration

- For STM32F42xxx/43xxx/446xx/469xx/479xx Devices, in Run mode: the main regulator has 2 operating modes available:
  - Normal mode: The CPU and core logic operate at maximum frequency at a given voltage scaling (scale 1, scale 2 or scale 3)
  - Over-drive mode: This mode allows the CPU and the core logic to operate at a higher frequency than the normal mode for a given voltage scaling (scale 1, scale 2 or scale 3). This mode is enabled through HAL_PWREx_EnableOverDrive()

function and disabled by HAL_PWREx_DisableOverDrive() function, to enter or exit from Over-drive mode please follow the sequence described in Reference manual.

- For STM32F42xxx/43xxx/446xx/469xx/479xx Devices, in Stop mode: the main regulator or low power regulator supplies a low power voltage to the 1.2V domain, thus preserving the content of registers and internal SRAM. 2 operating modes are available:
  - Normal mode: the 1.2V domain is preserved in nominal leakage mode. This mode is only available when the main regulator or the low power regulator is used in Scale 3 or low voltage mode.
  - Under-drive mode: the 1.2V domain is preserved in reduced leakage mode. This mode is only available when the main regulator or the low power regulator is in low voltage mode.

This section contains the following APIs:

- ***HAL_PWREx_EnableBkUpReg()***
- ***HAL_PWREx_DisableBkUpReg()***
- ***HAL_PWREx_EnableFlashPowerDown()***
- ***HAL_PWREx_DisableFlashPowerDown()***
- ***HAL_PWREx_GetVoltageRange()***
- ***HAL_PWREx_ControlVoltageScaling()***
- ***HAL_PWREx_EnableWakeUpPinPolarityRisingEdge()***
- ***HAL_PWREx_EnableWakeUpPinPolarityFallingEdge()***
- ***HAL_PWREx_EnableOverDrive()***
- ***HAL_PWREx_DisableOverDrive()***
- ***HAL_PWREx_EnterUnderDriveSTOPMode()***

47.1.2 Detailed description of functions

HAL_PWREx_EnableFlashPowerDown

Function Name	**void HAL_PWREx_EnableFlashPowerDown (void)**
Function Description	Enables the Flash Power Down in Stop mode.
Return values	• **None**

HAL_PWREx_DisableFlashPowerDown

Function Name	**void HAL_PWREx_DisableFlashPowerDown (void)**
Function Description	Disables the Flash Power Down in Stop mode.
Return values	• **None**

HAL_PWREx_EnableBkUpReg

Function Name	**HAL_StatusTypeDef HAL_PWREx_EnableBkUpReg (void)**
Function Description	Enables the Backup Regulator.
Return values	• **HAL:** status

HAL_PWREx_DisableBkUpReg

Function Name	**HAL_StatusTypeDef HAL_PWREx_DisableBkUpReg (void)**

Function Description Disables the Backup Regulator.

Return values
- **HAL:** status

HAL_PWREx_GetVoltageRange

Function Name **uint32_t HAL_PWREx_GetVoltageRange (void)**

Function Description Return Voltage Scaling Range.

Return values
- **The:** configured scale for the regulator voltage(VOS bit field). The returned value can be one of the following:
  - PWR_REGULATOR_VOLTAGE_SCALE1: Regulator voltage output Scale 1 mode
  - PWR_REGULATOR_VOLTAGE_SCALE2: Regulator voltage output Scale 2 mode
  - PWR_REGULATOR_VOLTAGE_SCALE3: Regulator voltage output Scale 3 mode

HAL_PWREx_ControlVoltageScaling

Function Name **HAL_StatusTypeDef HAL_PWREx_ControlVoltageScaling (uint32_t VoltageScaling)**

Function Description Configures the main internal regulator output voltage.

Parameters
- **VoltageScaling:** specifies the regulator output voltage to achieve a tradeoff between performance and power consumption. This parameter can be one of the following values:
  - PWR_REGULATOR_VOLTAGE_SCALE1: Regulator voltage output range 1 mode, the maximum value of fHCLK is 168 MHz. It can be extended to 180 MHz by activating the over-drive mode.
  - PWR_REGULATOR_VOLTAGE_SCALE2: Regulator voltage output range 2 mode, the maximum value of fHCLK is 144 MHz. It can be extended to, 168 MHz by activating the over-drive mode.
  - PWR_REGULATOR_VOLTAGE_SCALE3: Regulator voltage output range 3 mode, the maximum value of fHCLK is 120 MHz.

Return values
- **HAL:** Status

Notes
- To update the system clock frequency(SYSCLK): Set the HSI or HSE as system clock frequency using the HAL_RCC_ClockConfig().Call the HAL_RCC_OscConfig() to configure the PLL.Call HAL_PWREx_ConfigVoltageScaling() API to adjust the voltage scale.Set the new system clock frequency using the HAL_RCC_ClockConfig().
- The scale can be modified only when the HSI or HSE clock source is selected as system clock source, otherwise the API returns HAL_ERROR.
- When the PLL is OFF, the voltage scale 3 is automatically selected and the VOS bits value in the PWR_CR1 register are not taken in account.
- This API forces the PLL state ON to allow the possibility to configure the voltage scale 1 or 2.

- The new voltage scale is active only when the PLL is ON.

HAL_PWREx_EnableWakeUpPinPolarityRisingEdge

Function Name	**void HAL_PWREx_EnableWakeUpPinPolarityRisingEdge (void)**
Function Description	Enables Wakeup Pin Detection on high level (rising edge).
Return values	• **None**

HAL_PWREx_EnableWakeUpPinPolarityFallingEdge

Function Name	**void HAL_PWREx_EnableWakeUpPinPolarityFallingEdge (void)**
Function Description	Enables Wakeup Pin Detection on low level (falling edge).
Return values	• **None**

HAL_PWREx_EnableOverDrive

Function Name	**HAL_StatusTypeDef HAL_PWREx_EnableOverDrive (void)**
Function Description	Activates the Over-Drive mode.
Return values	• **HAL:** status
Notes	• This function can be used only for STM32F42xx/STM32F43xx/STM32F446xx/STM32F469xx/STM32F479xx devices. This mode allows the CPU and the core logic to operate at a higher frequency than the normal mode for a given voltage scaling (scale 1, scale 2 or scale 3). • It is recommended to enter or exit Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE. During the Over-drive switch activation, no peripheral clocks should be enabled. The peripheral clocks must be enabled once the Over-drive mode is activated.

HAL_PWREx_DisableOverDrive

Function Name	**HAL_StatusTypeDef HAL_PWREx_DisableOverDrive (void)**
Function Description	Deactivates the Over-Drive mode.
Return values	• **HAL:** status
Notes	• This function can be used only for STM32F42xx/STM32F43xx/STM32F446xx/STM32F469xx/STM32F479xx devices. This mode allows the CPU and the core logic to operate at a higher frequency than the normal mode for a given voltage scaling (scale 1, scale 2 or scale 3). • It is recommended to enter or exit Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE. During the Over-drive switch activation, no peripheral clocks should be enabled. The peripheral clocks must be enabled once the Over-drive mode is activated.

HAL_PWREx_EnterUnderDriveSTOPMode

Function Name	**HAL_StatusTypeDef HAL_PWREx_EnterUnderDriveSTOPMode (uint32_t Regulator, uint8_t STOPEntry)**
Function Description	Enters in Under-Drive STOP mode.
Parameters	• **Regulator:** specifies the regulator state in STOP mode. This parameter can be one of the following values: – PWR_MAINREGULATOR_UNDERDRIVE_ON: Main Regulator in under-drive mode and Flash memory in power-down when the device is in Stop under-drive mode – PWR_LOWPOWERREGULATOR_UNDERDRIVE_ON: Low Power Regulator in under-drive mode and Flash memory in power-down when the device is in Stop under-drive mode • **STOPEntry:** specifies if STOP mode in entered with WFI or WFE instruction. This parameter can be one of the following values: – PWR_SLEEPENTRY_WFI: enter STOP mode with WFI instruction – PWR_SLEEPENTRY_WFE: enter STOP mode with WFE instruction
Return values	• **None**
Notes	• This mode is only available for STM32F42xxx/STM324F3xxx/STM32F446xx/STM32F469xx/STM32F479xx devices. • This mode can be selected only when the Under-Drive is already active • This mode is enabled only with STOP low power mode. In this mode, the 1.2V domain is preserved in reduced leakage mode. This mode is only available when the main regulator or the low power regulator is in low voltage mode • If the Under-drive mode was enabled, it is automatically disabled after exiting Stop mode. When the voltage regulator operates in Under-drive mode, an additional startup delay is induced when waking up from Stop mode. • In Stop mode, all I/O pins keep the same state as in Run mode. • When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock. • When the voltage regulator operates in low power mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

47.2 PWREx Firmware driver defines

47.2.1 PWREx

PWRx CSR Register alias address

BRE_BIT_NUMBER

CSR_BRE_BB

WUPP_BIT_NUMBER

CSR_WUPP_BB

PWREx Exported Constants

__HAL_PWR_VOLTAGESCALING_CONFIG

Description:

- macros configure the main internal regulator output voltage.

Parameters:

- __REGULATOR__: specifies the regulator output voltage to achieve a tradeoff between performance and power consumption when the device does not operate at the maximum frequency (refer to the datasheets for more details). This parameter can be one of the following values:
  - PWR_REGULATOR_VOLTAGE_SCALE1: Regulator voltage output Scale 1 mode
  - PWR_REGULATOR_VOLTAGE_SCALE2: Regulator voltage output Scale 2 mode
  - PWR_REGULATOR_VOLTAGE_SCALE3: Regulator voltage output Scale 3 mode

Return value:

- None

__HAL_PWR_OVERDRIVE_ENABLE

Notes:

- These macros can be used only for STM32F42xx/STM3243xx devices.

__HAL_PWR_OVERDRIVE_DISABLE

__HAL_PWR_OVERDRIVESWITCHING_ENABLE

Notes:

- These macros can be used only for STM32F42xx/STM3243xx devices.

__HAL_PWR_OVERDRIVESWITCHING_DISABLE

__HAL_PWR_UNDERDRIVE_ENABLE

Notes:

- This mode is enabled only with STOP low power mode. In this mode, the 1.2V domain is preserved in reduced leakage mode. This mode is only available when the main regulator or the low power regulator is in low voltage mode. If the Under-drive mode was enabled, it is automatically disabled after exiting Stop mode. When the voltage regulator operates in Under-drive mode, an additional startup delay is induced when

waking up from Stop mode.

__HAL_PWR_UNDERDRIVE_DISABLE

__HAL_PWR_GET_ODRUDR_FLAG

Description:

- Check PWR flag is set or not.

Parameters:

- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - PWR_FLAG_ODRDY: This flag indicates that the Over-drive mode is ready
  - PWR_FLAG_ODSWRDY: This flag indicates that the Over-drive mode switching is ready
  - PWR_FLAG_UDRDY: This flag indicates that the Under-drive mode is enabled in Stop mode

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

Notes:

- These macros can be used only for STM32F42xx/STM3243xx devices.

__HAL_PWR_CLEAR_ODRUDR_FLAG

Notes:

- These macros can be used only for STM32F42xx/STM3243xx devices.

PWREx Private macros to check input parameters

IS_PWR_REGULATOR_UNDERDRIVE

IS_PWR_VOLTAGE_SCALING_RANGE

IS_PWR_WAKEUP_PIN

PWREx Over Under Drive Flag

PWR_FLAG_ODRDY

PWR_FLAG_ODSWRDY

PWR_FLAG_UDRDY

PWREx Register alias address

FPDS_BIT_NUMBER

CR_FPDS_BB

ODEN_BIT_NUMBER

CR_ODEN_BB

ODSWEN_BIT_NUMBER

CR_ODSWEN_BB

MRLVDS_BIT_NUMBER

CR_MRLVDS_BB

LPLVDS_BIT_NUMBER

CR_LPLVDS_BB

PWREx Regulator state in UnderDrive mode

PWR_MAINREGULATOR_UNDERDRIVE_ON

PWR_LOWPOWERREGULATOR_UNDERDRIVE_ON

PWREx Regulator Voltage Scale

PWR_REGULATOR_VOLTAGE_SCALE1

PWR_REGULATOR_VOLTAGE_SCALE2

PWR_REGULATOR_VOLTAGE_SCALE3

48 HAL QSPI Generic Driver

48.1 QSPI Firmware driver registers structures

48.1.1 QSPI_InitTypeDef

Data Fields

- ***uint32_t ClockPrescaler***
- ***uint32_t FifoThreshold***
- ***uint32_t SampleShifting***
- ***uint32_t FlashSize***
- ***uint32_t ChipSelectHighTime***
- ***uint32_t ClockMode***
- ***uint32_t FlashID***
- ***uint32_t DualFlash***

Field Documentation

- ***uint32_t QSPI_InitTypeDef::ClockPrescaler***
- ***uint32_t QSPI_InitTypeDef::FifoThreshold***
- ***uint32_t QSPI_InitTypeDef::SampleShifting***
- ***uint32_t QSPI_InitTypeDef::FlashSize***
- ***uint32_t QSPI_InitTypeDef::ChipSelectHighTime***
- ***uint32_t QSPI_InitTypeDef::ClockMode***
- ***uint32_t QSPI_InitTypeDef::FlashID***
- ***uint32_t QSPI_InitTypeDef::DualFlash***

48.1.2 QSPI_HandleTypeDef

Data Fields

- ***QUADSPI_TypeDef * Instance***
- ***QSPI_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***__IO uint16_t TxXferSize***
- ***__IO uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***__IO uint16_t RxXferSize***
- ***__IO uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdma***
- ***__IO HAL_LockTypeDef Lock***
- ***__IO HAL_QSPI_StateTypeDef State***
- ***__IO uint32_t ErrorCode***
- ***uint32_t Timeout***

Field Documentation

- ***QUADSPI_TypeDef* QSPI_HandleTypeDef::Instance***
- ***QSPI_InitTypeDef QSPI_HandleTypeDef::Init***
- ***uint8_t* QSPI_HandleTypeDef::pTxBuffPtr***
- ***__IO uint16_t QSPI_HandleTypeDef::TxXferSize***
- ***__IO uint16_t QSPI_HandleTypeDef::TxXferCount***
- ***uint8_t* QSPI_HandleTypeDef::pRxBuffPtr***
- ***__IO uint16_t QSPI_HandleTypeDef::RxXferSize***
- ***__IO uint16_t QSPI_HandleTypeDef::RxXferCount***
- ***DMA_HandleTypeDef* QSPI_HandleTypeDef::hdma***
- ***__IO HAL_LockTypeDef QSPI_HandleTypeDef::Lock***
- ***__IO HAL_QSPI_StateTypeDef QSPI_HandleTypeDef::State***
- ***__IO uint32_t QSPI_HandleTypeDef::ErrorCode***
- ***uint32_t QSPI_HandleTypeDef::Timeout***

48.1.3 QSPI_CommandTypeDef

Data Fields

- ***uint32_t Instruction***
- ***uint32_t Address***
- ***uint32_t AlternateBytes***
- ***uint32_t AddressSize***
- ***uint32_t AlternateBytesSize***
- ***uint32_t DummyCycles***
- ***uint32_t InstructionMode***
- ***uint32_t AddressMode***
- ***uint32_t AlternateByteMode***
- ***uint32_t DataMode***
- ***uint32_t NbData***
- ***uint32_t DdrMode***
- ***uint32_t DdrHoldHalfCycle***
- ***uint32_t SIOOMode***

Field Documentation

- ***uint32_t QSPI_CommandTypeDef::Instruction***
- ***uint32_t QSPI_CommandTypeDef::Address***
- ***uint32_t QSPI_CommandTypeDef::AlternateBytes***
- ***uint32_t QSPI_CommandTypeDef::AddressSize***
- ***uint32_t QSPI_CommandTypeDef::AlternateBytesSize***
- ***uint32_t QSPI_CommandTypeDef::DummyCycles***
- ***uint32_t QSPI_CommandTypeDef::InstructionMode***
- ***uint32_t QSPI_CommandTypeDef::AddressMode***
- ***uint32_t QSPI_CommandTypeDef::AlternateByteMode***
- ***uint32_t QSPI_CommandTypeDef::DataMode***
- ***uint32_t QSPI_CommandTypeDef::NbData***
- ***uint32_t QSPI_CommandTypeDef::DdrMode***
- ***uint32_t QSPI_CommandTypeDef::DdrHoldHalfCycle***
- ***uint32_t QSPI_CommandTypeDef::SIOOMode***

48.1.4 QSPI_AutoPollingTypeDef

Data Fields

- ***uint32_t Match***
- ***uint32_t Mask***
- ***uint32_t Interval***
- ***uint32_t StatusBytesSize***
- ***uint32_t MatchMode***
- ***uint32_t AutomaticStop***

Field Documentation

- ***uint32_t QSPI_AutoPollingTypeDef::Match***
- ***uint32_t QSPI_AutoPollingTypeDef::Mask***
- ***uint32_t QSPI_AutoPollingTypeDef::Interval***
- ***uint32_t QSPI_AutoPollingTypeDef::StatusBytesSize***
- ***uint32_t QSPI_AutoPollingTypeDef::MatchMode***
- ***uint32_t QSPI_AutoPollingTypeDef::AutomaticStop***

48.1.5 QSPI_MemoryMappedTypeDef

Data Fields

- ***uint32_t TimeOutPeriod***
- ***uint32_t TimeOutActivation***

Field Documentation

- ***uint32_t QSPI_MemoryMappedTypeDef::TimeOutPeriod***
- ***uint32_t QSPI_MemoryMappedTypeDef::TimeOutActivation***

48.2 QSPI Firmware driver API description

48.2.1 How to use this driver

Initialization

1. As prerequisite, fill in the HAL_QSPI_MspInit() :
   – Enable QuadSPI clock interface with __HAL_RCC_QSPI_CLK_ENABLE().
   – Reset QuadSPI IP with __HAL_RCC_QSPI_FORCE_RESET() and __HAL_RCC_QSPI_RELEASE_RESET().
   – Enable the clocks for the QuadSPI GPIOS with __HAL_RCC_GPIOx_CLK_ENABLE().

- Configure these QuadSPI pins in alternate mode using HAL_GPIO_Init().
- If interrupt mode is used, enable and configure QuadSPI global interrupt with HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ().
- If DMA mode is used, enable the clocks for the QuadSPI DMA channel with __HAL_RCC_DMAx_CLK_ENABLE(), configure DMA with HAL_DMA_Init(), link it with QuadSPI handle using __HAL_LINKDMA(), enable and configure DMA channel global interrupt with HAL_NVIC_SetPriority() and HAL_NVIC_EnableIRQ().

2. Configure the flash size, the clock prescaler, the fifo threshold, the clock mode, the sample shifting and the CS high time using the HAL_QSPI_Init() function.

Indirect functional mode

1. Configure the command sequence using the HAL_QSPI_Command() or HAL_QSPI_Command_IT() functions :
   - Instruction phase : the mode used and if present the instruction opcode.
   - Address phase : the mode used and if present the size and the address value.
   - Alternate-bytes phase : the mode used and if present the size and the alternate bytes values.
   - Dummy-cycles phase : the number of dummy cycles (mode used is same as data phase).
   - Data phase : the mode used and if present the number of bytes.
   - Double Data Rate (DDR) mode : the activation (or not) of this mode and the delay if activated.
   - Sending Instruction Only Once (SIOO) mode : the activation (or not) of this mode.
2. If no data is required for the command, it is sent directly to the memory :
   - In polling mode, the output of the function is done when the transfer is complete.
   - In interrupt mode, HAL_QSPI_CmdCpltCallback() will be called when the transfer is complete.
3. For the indirect write mode, use HAL_QSPI_Transmit(), HAL_QSPI_Transmit_DMA() or HAL_QSPI_Transmit_IT() after the command configuration :
   - In polling mode, the output of the function is done when the transfer is complete.
   - In interrupt mode, HAL_QSPI_FifoThresholdCallback() will be called when the fifo threshold is reached and HAL_QSPI_TxCpltCallback() will be called when the transfer is complete.
   - In DMA mode, HAL_QSPI_TxHalfCpltCallback() will be called at the half transfer and HAL_QSPI_TxCpltCallback() will be called when the transfer is complete.
4. For the indirect read mode, use HAL_QSPI_Receive(), HAL_QSPI_Receive_DMA() or HAL_QSPI_Receive_IT() after the command configuration :
   - In polling mode, the output of the function is done when the transfer is complete.
   - In interrupt mode, HAL_QSPI_FifoThresholdCallback() will be called when the fifo threshold is reached and HAL_QSPI_RxCpltCallback() will be called when the transfer is complete.
   - In DMA mode, HAL_QSPI_RxHalfCpltCallback() will be called at the half transfer and HAL_QSPI_RxCpltCallback() will be called when the transfer is complete.

Auto-polling functional mode

1. Configure the command sequence and the auto-polling functional mode using the HAL_QSPI_AutoPolling() or HAL_QSPI_AutoPolling_IT() functions :
   - Instruction phase : the mode used and if present the instruction opcode.

- Address phase : the mode used and if present the size and the address value.
- Alternate-bytes phase : the mode used and if present the size and the alternate bytes values.
- Dummy-cycles phase : the number of dummy cycles (mode used is same as data phase).
- Data phase : the mode used.
- Double Data Rate (DDR) mode : the activation (or not) of this mode and the delay if activated.
- Sending Instruction Only Once (SIOO) mode : the activation (or not) of this mode.
- The size of the status bytes, the match value, the mask used, the match mode (OR/AND), the polling interval and the automatic stop activation.

2. After the configuration :
   - In polling mode, the output of the function is done when the status match is reached. The automatic stop is activated to avoid an infinite loop.
   - In interrupt mode, HAL_QSPI_StatusMatchCallback() will be called each time the status match is reached.

Memory-mapped functional mode

1. Configure the command sequence and the memory-mapped functional mode using the HAL_QSPI_MemoryMapped() functions :
   - Instruction phase : the mode used and if present the instruction opcode.
   - Address phase : the mode used and the size.
   - Alternate-bytes phase : the mode used and if present the size and the alternate bytes values.
   - Dummy-cycles phase : the number of dummy cycles (mode used is same as data phase).
   - Data phase : the mode used.
   - Double Data Rate (DDR) mode : the activation (or not) of this mode and the delay if activated.
   - Sending Instruction Only Once (SIOO) mode : the activation (or not) of this mode.
   - The timeout activation and the timeout period.
2. After the configuration, the QuadSPI will be used as soon as an access on the AHB is done on the address range. HAL_QSPI_TimeOutCallback() will be called when the timeout expires.

Errors management and abort functionality

1. HAL_QSPI_GetError() function gives the error raised during the last operation.
2. HAL_QSPI_Abort() and HAL_QSPI_AbortIT() functions aborts any on-going operation and flushes the fifo :
   - In polling mode, the output of the function is done when the transfer complete bit is set and the busy bit cleared.
   - In interrupt mode, HAL_QSPI_AbortCpltCallback() will be called when the transfer complete bi is set.

Control functions

1. HAL_QSPI_GetState() function gives the current state of the HAL QuadSPI driver.
2. HAL_QSPI_SetTimeout() function configures the timeout value used in the driver.
3. HAL_QSPI_SetFifoThreshold() function configures the threshold on the Fifo of the QSPI IP.
4. HAL_QSPI_GetFifoThreshold() function gives the current of the Fifo's threshold

Workarounds linked to Silicon Limitation

1. Workarounds Implemented inside HAL Driver
   – Extra data written in the FIFO at the end of a read transfer

48.2.2 Initialization and Configuration functions

This subsection provides a set of functions allowing to :

- Initialize the QuadSPI.
- De-initialize the QuadSPI.

This section contains the following APIs:

- ***HAL_QSPI_Init()***
- ***HAL_QSPI_DeInit()***
- ***HAL_QSPI_MspInit()***
- ***HAL_QSPI_MspDeInit()***

48.2.3 IO operation functions

This subsection provides a set of functions allowing to :

- Handle the interrupts.
- Handle the command sequence.
- Transmit data in blocking, interrupt or DMA mode.
- Receive data in blocking, interrupt or DMA mode.
- Manage the auto-polling functional mode.
- Manage the memory-mapped functional mode.

This section contains the following APIs:

- ***HAL_QSPI_IRQHandler()***
- ***HAL_QSPI_Command()***
- ***HAL_QSPI_Command_IT()***
- ***HAL_QSPI_Transmit()***
- ***HAL_QSPI_Receive()***
- ***HAL_QSPI_Transmit_IT()***
- ***HAL_QSPI_Receive_IT()***
- ***HAL_QSPI_Transmit_DMA()***
- ***HAL_QSPI_Receive_DMA()***
- ***HAL_QSPI_AutoPolling()***
- ***HAL_QSPI_AutoPolling_IT()***
- ***HAL_QSPI_MemoryMapped()***
- ***HAL_QSPI_ErrorCallback()***

- ***HAL_QSPI_AbortCpltCallback()***
- ***HAL_QSPI_CmdCpltCallback()***
- ***HAL_QSPI_RxCpltCallback()***
- ***HAL_QSPI_TxCpltCallback()***
- ***HAL_QSPI_RxHalfCpltCallback()***
- ***HAL_QSPI_TxHalfCpltCallback()***
- ***HAL_QSPI_FifoThresholdCallback()***
- ***HAL_QSPI_StatusMatchCallback()***
- ***HAL_QSPI_TimeOutCallback()***

48.2.4 Peripheral Control and State functions

This subsection provides a set of functions allowing to :

- Check in run-time the state of the driver.
- Check the error code set during last operation.
- Abort any operation.

This section contains the following APIs:

- ***HAL_QSPI_GetState()***
- ***HAL_QSPI_GetError()***
- ***HAL_QSPI_Abort()***
- ***HAL_QSPI_Abort_IT()***
- ***HAL_QSPI_SetTimeout()***
- ***HAL_QSPI_SetFifoThreshold()***
- ***HAL_QSPI_GetFifoThreshold()***
- ***HAL_QSPI_ErrorCallback()***
- ***HAL_QSPI_AbortCpltCallback()***
- ***HAL_QSPI_FifoThresholdCallback()***
- ***HAL_QSPI_CmdCpltCallback()***
- ***HAL_QSPI_RxCpltCallback()***
- ***HAL_QSPI_TxCpltCallback()***
- ***HAL_QSPI_RxHalfCpltCallback()***
- ***HAL_QSPI_TxHalfCpltCallback()***
- ***HAL_QSPI_StatusMatchCallback()***
- ***HAL_QSPI_TimeOutCallback()***

48.2.5 Detailed description of functions

HAL_QSPI_Init

Function Name	**HAL_StatusTypeDef HAL_QSPI_Init (QSPI_HandleTypeDef * hqspi)**
Function Description	Initializes the QSPI mode according to the specified parameters in the QSPI_InitTypeDef and creates the associated handle.
Parameters	• **hqspi:** qspi handle
Return values	• **HAL:** status

HAL_QSPI_DeInit

Function Name	**HAL_StatusTypeDef HAL_QSPI_DeInit (QSPI_HandleTypeDef * hqspi)**
Function Description	DeInitializes the QSPI peripheral.
Parameters	• **hqspi:** qspi handle
Return values	• **HAL:** status

HAL_QSPI_MspInit

Function Name	**void HAL_QSPI_MspInit (QSPI_HandleTypeDef * hqspi)**
Function Description	QSPI MSP Init.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_MspDeInit

Function Name	**void HAL_QSPI_MspDeInit (QSPI_HandleTypeDef * hqspi)**
Function Description	QSPI MSP DeInit.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_IRQHandler

Function Name	**void HAL_QSPI_IRQHandler (QSPI_HandleTypeDef * hqspi)**
Function Description	This function handles QSPI interrupt request.
Parameters	• **hqspi:** QSPI handle
Return values	• **None.:**

HAL_QSPI_Command

Function Name	**HAL_StatusTypeDef HAL_QSPI_Command (QSPI_HandleTypeDef * hqspi, QSPI_CommandTypeDef * cmd, uint32_t Timeout)**
Function Description	Sets the command configuration.
Parameters	• **hqspi:** QSPI handle • **cmd:** : structure that contains the command configuration information • **Timeout:** : Time out duration
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Read or Write Modes

HAL_QSPI_Transmit

Function Name	**HAL_StatusTypeDef HAL_QSPI_Transmit (QSPI_HandleTypeDef * hqspi, uint8_t * pData, uint32_t Timeout)**
Function Description	Transmit an amount of data in blocking mode.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer • **Timeout:** : Time out duration
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Write Mode

HAL_QSPI_Receive

Function Name	**HAL_StatusTypeDef HAL_QSPI_Receive (QSPI_HandleTypeDef * hqspi, uint8_t * pData, uint32_t Timeout)**
Function Description	Receive an amount of data in blocking mode.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer • **Timeout:** : Time out duration
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Read Mode

HAL_QSPI_Command_IT

Function Name	**HAL_StatusTypeDef HAL_QSPI_Command_IT (QSPI_HandleTypeDef * hqspi, QSPI_CommandTypeDef * cmd)**
Function Description	Sets the command configuration in interrupt mode.
Parameters	• **hqspi:** QSPI handle • **cmd:** : structure that contains the command configuration information
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Read or Write Modes

HAL_QSPI_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_QSPI_Transmit_IT (QSPI_HandleTypeDef * hqspi, uint8_t * pData)**
Function Description	Send an amount of data in interrupt mode.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Write Mode

HAL_QSPI_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_QSPI_Receive_IT (QSPI_HandleTypeDef * hqspi, uint8_t * pData)**
Function Description	Receive an amount of data in no-blocking mode with Interrupt.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Read Mode

HAL_QSPI_Transmit_DMA

Function Name	**HAL_StatusTypeDef HAL_QSPI_Transmit_DMA (QSPI_HandleTypeDef * hqspi, uint8_t * pData)**
Function Description	Sends an amount of data in non blocking mode with DMA.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Write Mode • If DMA peripheral access is configured as halfword, the number of data and the fifo threshold should be aligned on halfword • If DMA peripheral access is configured as word, the number of data and the fifo threshold should be aligned on word

HAL_QSPI_Receive_DMA

Function Name	**HAL_StatusTypeDef HAL_QSPI_Receive_DMA (QSPI_HandleTypeDef * hqspi, uint8_t * pData)**
Function Description	Receives an amount of data in non blocking mode with DMA.
Parameters	• **hqspi:** QSPI handle • **pData:** pointer to data buffer.
Return values	• **HAL:** status
Notes	• This function is used only in Indirect Read Mode • If DMA peripheral access is configured as halfword, the number of data and the fifo threshold should be aligned on halfword • If DMA peripheral access is configured as word, the number of data and the fifo threshold should be aligned on word

HAL_QSPI_AutoPolling

Function Name	**HAL_StatusTypeDef HAL_QSPI_AutoPolling (QSPI_HandleTypeDef * hqspi, QSPI_CommandTypeDef * cmd, QSPI_AutoPollingTypeDef * cfg, uint32_t Timeout)**
Function Description	Configure the QSPI Automatic Polling Mode in blocking mode.
Parameters	• **hqspi:** QSPI handle

- **cmd:** structure that contains the command configuration information.
- **cfg:** structure that contains the polling configuration information.
- **Timeout:** : Time out duration

Return values
- **HAL:** status

Notes
- This function is used only in Automatic Polling Mode

HAL_QSPI_AutoPolling_IT

Function Name **HAL_StatusTypeDef HAL_QSPI_AutoPolling_IT (QSPI_HandleTypeDef * hqspi, QSPI_CommandTypeDef * cmd, QSPI_AutoPollingTypeDef * cfg)**

Function Description Configure the QSPI Automatic Polling Mode in non-blocking mode.

Parameters
- **hqspi:** QSPI handle
- **cmd:** structure that contains the command configuration information.
- **cfg:** structure that contains the polling configuration information.

Return values
- **HAL:** status

Notes
- This function is used only in Automatic Polling Mode

HAL_QSPI_MemoryMapped

Function Name **HAL_StatusTypeDef HAL_QSPI_MemoryMapped (QSPI_HandleTypeDef * hqspi, QSPI_CommandTypeDef * cmd, QSPI_MemoryMappedTypeDef * cfg)**

Function Description Configure the Memory Mapped mode.

Parameters
- **hqspi:** QSPI handle
- **cmd:** structure that contains the command configuration information.
- **cfg:** structure that contains the memory mapped configuration information.

Return values
- **HAL:** status

Notes
- This function is used only in Memory mapped Mode

HAL_QSPI_ErrorCallback

Function Name **void HAL_QSPI_ErrorCallback (QSPI_HandleTypeDef * hqspi)**

Function Description Transfer Error callbacks.

Parameters
- **hqspi:** QSPI handle

Return values
- **None**

HAL_QSPI_AbortCpltCallback

Function Name	**void HAL_QSPI_AbortCpltCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	Abort completed callback.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_FifoThresholdCallback

Function Name	**void HAL_QSPI_FifoThresholdCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	FIFO Threshold callbacks.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_CmdCpltCallback

Function Name	**void HAL_QSPI_CmdCpltCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	Command completed callback.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_RxCpltCallback

Function Name	**void HAL_QSPI_RxCpltCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	Rx Transfer completed callbacks.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_TxCpltCallback

Function Name	**void HAL_QSPI_TxCpltCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	Tx Transfer completed callbacks.
Parameters	• **hqspi:** QSPI handle
Return values	• **None**

HAL_QSPI_RxHalfCpltCallback

Function Name	**void HAL_QSPI_RxHalfCpltCallback (QSPI_HandleTypeDef * hqspi)**
Function Description	Rx Half Transfer completed callbacks.

Parameters
- **hqspi:** QSPI handle

Return values
- **None**

HAL_QSPI_TxHalfCpltCallback

Function Name **void HAL_QSPI_TxHalfCpltCallback (QSPI_HandleTypeDef * hqspi)**

Function Description Tx Half Transfer completed callbacks.

Parameters
- **hqspi:** QSPI handle

Return values
- **None**

HAL_QSPI_StatusMatchCallback

Function Name **void HAL_QSPI_StatusMatchCallback (QSPI_HandleTypeDef * hqspi)**

Function Description Status Match callbacks.

Parameters
- **hqspi:** QSPI handle

Return values
- **None**

HAL_QSPI_TimeOutCallback

Function Name **void HAL_QSPI_TimeOutCallback (QSPI_HandleTypeDef * hqspi)**

Function Description Timeout callbacks.

Parameters
- **hqspi:** QSPI handle

Return values
- **None**

HAL_QSPI_GetState

Function Name **HAL_QSPI_StateTypeDef HAL_QSPI_GetState (QSPI_HandleTypeDef * hqspi)**

Function Description Return the QSPI handle state.

Parameters
- **hqspi:** QSPI handle

Return values
- **HAL:** state

HAL_QSPI_GetError

Function Name **uint32_t HAL_QSPI_GetError (QSPI_HandleTypeDef * hqspi)**

Function Description Return the QSPI error code.

Parameters
- **hqspi:** QSPI handle

Return values
- **QSPI:** Error Code

HAL_QSPI_Abort

Function Name	**HAL_StatusTypeDef HAL_QSPI_Abort (QSPI_HandleTypeDef * hqspi)**
Function Description	Abort the current transmission.
Parameters	• **hqspi:** QSPI handle
Return values	• **HAL:** status

HAL_QSPI_Abort_IT

Function Name	**HAL_StatusTypeDef HAL_QSPI_Abort_IT (QSPI_HandleTypeDef * hqspi)**
Function Description	Abort the current transmission (non-blocking function)
Parameters	• **hqspi:** QSPI handle
Return values	• **HAL:** status

HAL_QSPI_SetTimeout

Function Name	**void HAL_QSPI_SetTimeout (QSPI_HandleTypeDef * hqspi, uint32_t Timeout)**
Function Description	Set QSPI timeout.
Parameters	• **hqspi:** QSPI handle. • **Timeout:** Timeout for the QSPI memory access.
Return values	• **None**

HAL_QSPI_SetFifoThreshold

Function Name	**HAL_StatusTypeDef HAL_QSPI_SetFifoThreshold (QSPI_HandleTypeDef * hqspi, uint32_t Threshold)**
Function Description	Set QSPI Fifo threshold.
Parameters	• **hqspi:** QSPI handle. • **Threshold:** Threshold of the Fifo (value between 1 and 16).
Return values	• **HAL:** status

HAL_QSPI_GetFifoThreshold

Function Name	**uint32_t HAL_QSPI_GetFifoThreshold (QSPI_HandleTypeDef * hqspi)**
Function Description	Get QSPI Fifo threshold.
Parameters	• **hqspi:** QSPI handle.
Return values	• **Fifo:** threshold (value between 1 and 16)

48.3 QSPI Firmware driver defines

48.3.1 QSPI

QSPI Address Mode

QSPI_ADDRESS_NONE	No address
QSPI_ADDRESS_1_LINE	Address on a single line
QSPI_ADDRESS_2_LINES	Address on two lines
QSPI_ADDRESS_4_LINES	Address on four lines

QSPI Address Size

QSPI_ADDRESS_8_BITS	8-bit address
QSPI_ADDRESS_16_BITS	16-bit address
QSPI_ADDRESS_24_BITS	24-bit address
QSPI_ADDRESS_32_BITS	32-bit address

QSPI Alternate Bytes Mode

QSPI_ALTERNATE_BYTES_NONE	No alternate bytes
QSPI_ALTERNATE_BYTES_1_LINE	Alternate bytes on a single line
QSPI_ALTERNATE_BYTES_2_LINES	Alternate bytes on two lines
QSPI_ALTERNATE_BYTES_4_LINES	Alternate bytes on four lines

QSPI Alternate Bytes Size

QSPI_ALTERNATE_BYTES_8_BITS	8-bit alternate bytes
QSPI_ALTERNATE_BYTES_16_BITS	16-bit alternate bytes
QSPI_ALTERNATE_BYTES_24_BITS	24-bit alternate bytes
QSPI_ALTERNATE_BYTES_32_BITS	32-bit alternate bytes

QSPI Automatic Stop

QSPI_AUTOMATIC_STOP_DISABLE	AutoPolling stops only with abort or QSPI disabling
QSPI_AUTOMATIC_STOP_ENABLE	AutoPolling stops as soon as there is a match

QSPI Chip Select High Time

QSPI_CS_HIGH_TIME_1_CYCLE	nCS stay high for at least 1 clock cycle between commands
QSPI_CS_HIGH_TIME_2_CYCLE	nCS stay high for at least 2 clock cycles between commands
QSPI_CS_HIGH_TIME_3_CYCLE	nCS stay high for at least 3 clock cycles between commands
QSPI_CS_HIGH_TIME_4_CYCLE	nCS stay high for at least 4 clock cycles between commands
QSPI_CS_HIGH_TIME_5_CYCLE	nCS stay high for at least 5 clock cycles between commands
QSPI_CS_HIGH_TIME_6_CYCLE	nCS stay high for at least 6 clock cycles between

	commands
QSPI_CS_HIGH_TIME_7_CYCLE	nCS stay high for at least 7 clock cycles between commands
QSPI_CS_HIGH_TIME_8_CYCLE	nCS stay high for at least 8 clock cycles between commands

QSPI Clock Mode

QSPI_CLOCK_MODE_0	Clk stays low while nCS is released
QSPI_CLOCK_MODE_3	Clk goes high while nCS is released

QSPI Clock Prescaler

IS_QSPI_CLOCK_PRESCALER

QSPI Data Mode

QSPI_DATA_NONE	No data
QSPI_DATA_1_LINE	Data on a single line
QSPI_DATA_2_LINES	Data on two lines
QSPI_DATA_4_LINES	Data on four lines

QSPI Ddr HoldHalfCycle

QSPI_DDR_HHC_ANALOG_DELAY	Delay the data output using analog delay in DDR mode
QSPI_DDR_HHC_HALF_CLK_DELAY	Delay the data output by 1/2 clock cycle in DDR mode

QSPI Ddr Mode

QSPI_DDR_MODE_DISABLE	Double data rate mode disabled
QSPI_DDR_MODE_ENABLE	Double data rate mode enabled

QSPI Dual Flash Mode

QSPI_DUALFLASH_ENABLE

QSPI_DUALFLASH_DISABLE

QSPI Dummy Cycles

IS_QSPI_DUMMY_CYCLES

QSPI Error Code

HAL_QSPI_ERROR_NONE	No error
HAL_QSPI_ERROR_TIMEOUT	Timeout error
HAL_QSPI_ERROR_TRANSFER	Transfer error
HAL_QSPI_ERROR_DMA	DMA transfer error
HAL_QSPI_ERROR_INVALID_PARAM	Invalid parameters error

QSPI Exported Macros

__HAL_QSPI_RESET_HANDLE_STATE **Description:**

- Reset QSPI handle state.

Parameters:

- __HANDLE__: QSPI handle.

Return value:

- None

__HAL_QSPI_ENABLE

Description:

- Enable QSPI.

Parameters:

- __HANDLE__: specifies the QSPI Handle.

Return value:

- None

__HAL_QSPI_DISABLE

Description:

- Disable QSPI.

Parameters:

- __HANDLE__: specifies the QSPI Handle.

Return value:

- None

__HAL_QSPI_ENABLE_IT

Description:

- Enables the specified QSPI interrupt.

Parameters:

- __HANDLE__: specifies the QSPI Handle.
- __INTERRUPT__: specifies the QSPI interrupt source to enable. This parameter can be one of the following values:
  - QSPI_IT_TO: QSPI Time out interrupt
  - QSPI_IT_SM: QSPI Status match interrupt
  - QSPI_IT_FT: QSPI FIFO threshold interrupt
  - QSPI_IT_TC: QSPI Transfer complete interrupt
  - QSPI_IT_TE: QSPI Transfer error interrupt

Return value:

- None

__HAL_QSPI_DISABLE_IT

Description:

- Disables the specified QSPI interrupt.

Parameters:

- __HANDLE__: specifies the QSPI Handle.
- __INTERRUPT__: specifies the QSPI interrupt source to disable. This parameter can be one of the following values:
  - QSPI_IT_TO: QSPI Timeout interrupt

- QSPI_IT_SM: QSPI Status match interrupt
- QSPI_IT_FT: QSPI FIFO threshold interrupt
- QSPI_IT_TC: QSPI Transfer complete interrupt
- QSPI_IT_TE: QSPI Transfer error interrupt

Return value:

- None

__HAL_QSPI_GET_IT_SOURCE

Description:

- Checks whether the specified QSPI interrupt source is enabled.

Parameters:

- __HANDLE__: specifies the QSPI Handle.
- __INTERRUPT__: specifies the QSPI interrupt source to check. This parameter can be one of the following values:
  - QSPI_IT_TO: QSPI Time out interrupt
  - QSPI_IT_SM: QSPI Status match interrupt
  - QSPI_IT_FT: QSPI FIFO threshold interrupt
  - QSPI_IT_TC: QSPI Transfer complete interrupt
  - QSPI_IT_TE: QSPI Transfer error interrupt

Return value:

- The: new state of __INTERRUPT__ (TRUE or FALSE).

__HAL_QSPI_GET_FLAG

Description:

- Get the selected QSPI's flag status.

Parameters:

- __HANDLE__: specifies the QSPI Handle.
- __FLAG__: specifies the QSPI flag to check. This parameter can be one of the following values:
  - QSPI_FLAG_BUSY: QSPI Busy flag
  - QSPI_FLAG_TO: QSPI Time out flag
  - QSPI_FLAG_SM: QSPI Status match flag
  - QSPI_FLAG_FT: QSPI FIFO threshold flag
  - QSPI_FLAG_TC: QSPI Transfer complete flag
  - QSPI_FLAG_TE: QSPI Transfer error

flag

Return value:

- None

__HAL_QSPI_CLEAR_FLAG

Description:

- Clears the specified QSPI's flag status.

Parameters:

- __HANDLE__: specifies the QSPI Handle.
- __FLAG__: specifies the QSPI clear register flag that needs to be set This parameter can be one of the following values:
  - QSPI_FLAG_TO: QSPI Time out flag
  - QSPI_FLAG_SM: QSPI Status match flag
  - QSPI_FLAG_TC: QSPI Transfer complete flag
  - QSPI_FLAG_TE: QSPI Transfer error flag

Return value:

- None

QSPI Fifo Threshold

IS_QSPI_FIFO_THRESHOLD

QSPI Flags

QSPI_FLAG_BUSY	Busy flag: operation is ongoing
QSPI_FLAG_TO	Timeout flag: timeout occurs in memory-mapped mode
QSPI_FLAG_SM	Status match flag: received data matches in autopolling mode
QSPI_FLAG_FT	Fifo threshold flag: Fifo threshold reached or data left after read from memory is complete
QSPI_FLAG_TC	Transfer complete flag: programmed number of data have been transferred or the transfer has been aborted
QSPI_FLAG_TE	Transfer error flag: invalid address is being accessed

QSPI Flash Size

IS_QSPI_FLASH_SIZE

QSPI Flash Select

QSPI_FLASH_ID_1

QSPI_FLASH_ID_2

QSPI Instruction

IS_QSPI_INSTRUCTION

QSPI Instruction Mode

QSPI_INSTRUCTION_NONE	No instruction

QSPI_INSTRUCTION_1_LINE	Instruction on a single line
QSPI_INSTRUCTION_2_LINES	Instruction on two lines
QSPI_INSTRUCTION_4_LINES	Instruction on four lines

QSPI Interrupts

QSPI_IT_TO	Interrupt on the timeout flag
QSPI_IT_SM	Interrupt on the status match flag
QSPI_IT_FT	Interrupt on the fifo threshold flag
QSPI_IT_TC	Interrupt on the transfer complete flag
QSPI_IT_TE	Interrupt on the transfer error flag

QSPI Interval

IS_QSPI_INTERVAL

QSPI Match Mode

QSPI_MATCH_MODE_AND	AND match mode between unmasked bits
QSPI_MATCH_MODE_OR	OR match mode between unmasked bits

QSPI Sample Shifting

QSPI_SAMPLE_SHIFTING_NONE	No clock cycle shift to sample data
QSPI_SAMPLE_SHIFTING_HALFCYCLE	1/2 clock cycle shift to sample data

QSPI SIOO Mode

QSPI_SIOO_INST_EVERY_CMD	Send instruction on every transaction
QSPI_SIOO_INST_ONLY_FIRST_CMD	Send instruction only for the first command

QSPI Status Bytes Size

IS_QSPI_STATUS_BYTES_SIZE

QSPI TimeOut Activation

QSPI_TIMEOUT_COUNTER_DISABLE	Timeout counter disabled, nCS remains active
QSPI_TIMEOUT_COUNTER_ENABLE	Timeout counter enabled, nCS released when timeout expires

QSPI TimeOut Period

IS_QSPI_TIMEOUT_PERIOD

QSPI Timeout definition

HAL_QPSI_TIMEOUT_DEFAULT_VALUE

49 HAL RCC Generic Driver

49.1 RCC Firmware driver registers structures

49.1.1 RCC_OscInitTypeDef

Data Fields

- ***uint32_t OscillatorType***
- ***uint32_t HSEState***
- ***uint32_t LSEState***
- ***uint32_t HSIState***
- ***uint32_t HSICalibrationValue***
- ***uint32_t LSIState***
- ***RCC_PLLInitTypeDef PLL***

Field Documentation

- ***uint32_t RCC_OscInitTypeDef::OscillatorType***
  The oscillators to be configured. This parameter can be a value of ***RCC_Oscillator_Type***
- ***uint32_t RCC_OscInitTypeDef::HSEState***
  The new state of the HSE. This parameter can be a value of ***RCC_HSE_Config***
- ***uint32_t RCC_OscInitTypeDef::LSEState***
  The new state of the LSE. This parameter can be a value of ***RCC_LSE_Config***
- ***uint32_t RCC_OscInitTypeDef::HSIState***
  The new state of the HSI. This parameter can be a value of ***RCC_HSI_Config***
- ***uint32_t RCC_OscInitTypeDef::HSICalibrationValue***
  The HSI calibration trimming value (default is RCC_HSICALIBRATION_DEFAULT). This parameter must be a number between Min_Data = 0x00 and Max_Data = 0x1F
- ***uint32_t RCC_OscInitTypeDef::LSIState***
  The new state of the LSI. This parameter can be a value of ***RCC_LSI_Config***
- ***RCC_PLLInitTypeDef RCC_OscInitTypeDef::PLL***
  PLL structure parameters

49.1.2 RCC_ClkInitTypeDef

Data Fields

- ***uint32_t ClockType***
- ***uint32_t SYSCLKSource***
- ***uint32_t AHBCLKDivider***
- ***uint32_t APB1CLKDivider***
- ***uint32_t APB2CLKDivider***

Field Documentation

- ***uint32_t RCC_ClkInitTypeDef::ClockType***
  The clock to be configured. This parameter can be a value of ***RCC_System_Clock_Type***
- ***uint32_t RCC_ClkInitTypeDef::SYSCLKSource***
  The clock source (SYSCLKS) used as system clock. This parameter can be a value of ***RCC_System_Clock_Source***
- ***uint32_t RCC_ClkInitTypeDef::AHBCLKDivider***
  The AHB clock (HCLK) divider. This clock is derived from the system clock (SYSCLK). This parameter can be a value of ***RCC_AHB_Clock_Source***
- ***uint32_t RCC_ClkInitTypeDef::APB1CLKDivider***
  The APB1 clock (PCLK1) divider. This clock is derived from the AHB clock (HCLK). This parameter can be a value of ***RCC_APB1_APB2_Clock_Source***
- ***uint32_t RCC_ClkInitTypeDef::APB2CLKDivider***
  The APB2 clock (PCLK2) divider. This clock is derived from the AHB clock (HCLK). This parameter can be a value of ***RCC_APB1_APB2_Clock_Source***

49.2 RCC Firmware driver API description

49.2.1 RCC specific features

After reset the device is running from Internal High Speed oscillator (HSI 16MHz) with Flash 0 wait state, Flash prefetch buffer, D-Cache and I-Cache are disabled, and all peripherals are off except internal SRAM, Flash and JTAG.

- There is no prescaler on High speed (AHB) and Low speed (APB) busses; all peripherals mapped on these busses are running at HSI speed.
- The clock for all peripherals is switched off, except the SRAM and FLASH.
- All GPIOs are in input floating state, except the JTAG pins which are assigned to be used for debug purpose.

Once the device started from reset, the user application has to:

- Configure the clock source to be used to drive the System clock (if the application needs higher frequency/performance)
- Configure the System clock frequency and Flash settings
- Configure the AHB and APB busses prescalers
- Enable the clock for the peripheral(s) to be used
- Configure the clock source(s) for peripherals which clocks are not derived from the System clock (I2S, RTC, ADC, USB OTG FS/SDIO/RNG)

49.2.2 RCC Limitations

A delay between an RCC peripheral clock enable and the effective peripheral enabling should be taken into account in order to manage the peripheral read/write from/to registers.

- This delay depends on the peripheral mapping.
- If peripheral is mapped on AHB: the delay is 2 AHB clock cycle after the clock enable bit is set on the hardware register
- If peripheral is mapped on APB: the delay is 2 APB clock cycle after the clock enable bit is set on the hardware register

Implemented Workaround:

- For AHB & APB peripherals, a dummy read to the peripheral register has been inserted in each __HAL_RCC_PPP_CLK_ENABLE() macro.

49.2.3 Initialization and de-initialization functions

This section provides functions allowing to configure the internal/external oscillators (HSE, HSI, LSE, LSI, PLL, CSS and MCO) and the System busses clocks (SYSCLK, AHB, APB1 and APB2).

Internal/external clock and PLL configuration

1. HSI (high-speed internal), 16 MHz factory-trimmed RC used directly or through the PLL as System clock source.
2. LSI (low-speed internal), 32 KHz low consumption RC used as IWDG and/or RTC clock source.
3. HSE (high-speed external), 4 to 26 MHz crystal oscillator used directly or through the PLL as System clock source. Can be used also as RTC clock source.
4. LSE (low-speed external), 32 KHz oscillator used as RTC clock source.
5. PLL (clocked by HSI or HSE), featuring two different output clocks:
   – The first output is used to generate the high speed system clock (up to 168 MHz)
   – The second output is used to generate the clock for the USB OTG FS (48 MHz), the random analog generator (<=48 MHz) and the SDIO (<= 48 MHz).
6. CSS (Clock security system), once enable using the macro __HAL_RCC_CSS_ENABLE() and if a HSE clock failure occurs(HSE used directly or through PLL as System clock source), the System clocks automatically switched to HSI and an interrupt is generated if enabled. The interrupt is linked to the Cortex-M4 NMI (Non-Maskable Interrupt) exception vector.
7. MCO1 (microcontroller clock output), used to output HSI, LSE, HSE or PLL clock (through a configurable prescaler) on PA8 pin.
8. MCO2 (microcontroller clock output), used to output HSE, PLL, SYSCLK or PLLI2S clock (through a configurable prescaler) on PC9 pin.

System, AHB and APB busses clocks configuration

1. Several clock sources can be used to drive the System clock (SYSCLK): HSI, HSE and PLL. The AHB clock (HCLK) is derived from System clock through configurable prescaler and used to clock the CPU, memory and peripherals mapped on AHB bus (DMA, GPIO...). APB1 (PCLK1) and APB2 (PCLK2) clocks are derived from AHB clock through configurable prescalers and used to clock the peripherals mapped on these busses. You can use "HAL_RCC_GetSysClockFreq()" function to retrieve the frequencies of these clocks.
2. For the STM32F405xx/07xx and STM32F415xx/17xx devices, the maximum frequency of the SYSCLK and HCLK is 168 MHz, PCLK2 84 MHz and PCLK1 42 MHz. Depending on the device voltage range, the maximum frequency should be adapted accordingly (refer to the product datasheets for more details).
3. For the STM32F42xxx, STM32F43xxx, STM32F446xx, STM32F469xx and STM32F479xx devices, the maximum frequency of the SYSCLK and HCLK is 180 MHz, PCLK2 90 MHz and PCLK1 45 MHz. Depending on the device voltage range, the maximum frequency should be adapted accordingly (refer to the product datasheets for more details).
4. For the STM32F401xx, the maximum frequency of the SYSCLK and HCLK is 84 MHz, PCLK2 84 MHz and PCLK1 42 MHz. Depending on the device voltage range, the maximum frequency should be adapted accordingly (refer to the product datasheets for more details).
5. For the STM32F41xxx, the maximum frequency of the SYSCLK and HCLK is 100 MHz, PCLK2 100 MHz and PCLK1 50 MHz. Depending on the device voltage range, the maximum frequency should be adapted accordingly (refer to the product datasheets for more details).

This section contains the following APIs:

- *HAL_RCC_DeInit()*
- *HAL_RCC_OscConfig()*
- *HAL_RCC_ClockConfig()*

49.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the RCC Clocks frequencies.

This section contains the following APIs:

- *HAL_RCC_MCOConfig()*
- *HAL_RCC_EnableCSS()*
- *HAL_RCC_DisableCSS()*
- *HAL_RCC_GetSysClockFreq()*
- *HAL_RCC_GetHCLKFreq()*
- *HAL_RCC_GetPCLK1Freq()*
- *HAL_RCC_GetPCLK2Freq()*
- *HAL_RCC_GetOscConfig()*
- *HAL_RCC_GetClockConfig()*
- *HAL_RCC_NMI_IRQHandler()*
- *HAL_RCC_CSSCallback()*

49.2.5 Detailed description of functions

HAL_RCC_DeInit

Function Name	**void HAL_RCC_DeInit (void)**
Function Description	Resets the RCC clock configuration to the default reset state.
Return values	• **None**
Notes	• The default reset state of the clock configuration is given below: HSI ON and used as system clock sourceHSE and PLL OFFAHB, APB1 and APB2 prescaler set to 1.CSS, MCO1 and MCO2 OFFAll interrupts disabled • This function doesn't modify the configuration of the Peripheral clocksLSI, LSE and RTC clocks

HAL_RCC_OscConfig

Function Name	**HAL_StatusTypeDef HAL_RCC_OscConfig (RCC_OscInitTypeDef * RCC_OscInitStruct)**
Function Description	Initializes the RCC Oscillators according to the specified parameters in the RCC_OscInitTypeDef.
Parameters	• **RCC_OscInitStruct:** pointer to an RCC_OscInitTypeDef structure that contains the configuration information for the RCC Oscillators.
Return values	• **HAL:** status
Notes	• The PLL is not disabled when used as system clock. • Transitions LSE Bypass to LSE On and LSE On to LSE Bypass are not supported by this API. User should request a transition to LSE Off first and then LSE On or LSE Bypass.

- Transition HSE Bypass to HSE On and HSE On to HSE Bypass are not supported by this API. User should request a transition to HSE Off first and then HSE On or HSE Bypass.

HAL_RCC_ClockConfig

Function Name **HAL_StatusTypeDef HAL_RCC_ClockConfig (RCC_ClkInitTypeDef * RCC_ClkInitStruct, uint32_t FLatency)**

Function Description Initializes the CPU, AHB and APB busses clocks according to the specified parameters in the RCC_ClkInitStruct.

Parameters
- **RCC_ClkInitStruct:** pointer to an RCC_OscInitTypeDef structure that contains the configuration information for the RCC peripheral.
- **FLatency:** FLASH Latency, this parameter depend on device selected

Return values
- **None**

Notes
- The SystemCoreClock CMSIS variable is used to store System Clock Frequency and updated by HAL_RCC_GetHCLKFreq() function called within this function
- The HSI is used (enabled by hardware) as system clock source after startup from Reset, wake-up from STOP and STANDBY mode, or in case of failure of the HSE used directly or indirectly as system clock (if the Clock Security System CSS is enabled).
- A switch from one clock source to another occurs only if the target clock source is ready (clock stable after startup delay or PLL locked). If a clock source which is not yet ready is selected, the switch will occur when the clock source will be ready.
- Depending on the device voltage range, the software has to set correctly HPRE[3:0] bits to ensure that HCLK not exceed the maximum allowed frequency (for more details refer to section above "Initialization/de-initialization functions")

HAL_RCC_MCOConfig

Function Name **void HAL_RCC_MCOConfig (uint32_t RCC_MCOx, uint32_t RCC_MCOSource, uint32_t RCC_MCODiv)**

Function Description Selects the clock source to output on MCO1 pin(PA8) or on MCO2 pin(PC9).

Parameters
- **RCC_MCOx:** specifies the output direction for the clock source. This parameter can be one of the following values:
  - RCC_MCO1: Clock source to output on MCO1 pin(PA8).
  - RCC_MCO2: Clock source to output on MCO2 pin(PC9).
- **RCC_MCOSource:** specifies the clock source to output. This parameter can be one of the following values:
  - RCC_MCO1SOURCE_HSI: HSI clock selected as MCO1 source
  - RCC_MCO1SOURCE_LSE: LSE clock selected as MCO1 source
  - RCC_MCO1SOURCE_HSE: HSE clock selected as

MCO1 source
  - RCC_MCO1SOURCE_PLLCLK: main PLL clock selected as MCO1 source
  - RCC_MCO2SOURCE_SYSCLK: System clock (SYSCLK) selected as MCO2 source
  - RCC_MCO2SOURCE_PLLI2SCLK: PLLI2S clock selected as MCO2 source, available for all STM32F4 devices except STM32F410xx
  - RCC_MCO2SOURCE_I2SCLK: I2SCLK clock selected as MCO2 source, available only for STM32F410Rx devices
  - RCC_MCO2SOURCE_HSE: HSE clock selected as MCO2 source
  - RCC_MCO2SOURCE_PLLCLK: main PLL clock selected as MCO2 source
- **RCC_MCODiv:** specifies the MCOx prescaler. This parameter can be one of the following values:
  - RCC_MCODIV_1: no division applied to MCOx clock
  - RCC_MCODIV_2: division by 2 applied to MCOx clock
  - RCC_MCODIV_3: division by 3 applied to MCOx clock
  - RCC_MCODIV_4: division by 4 applied to MCOx clock
  - RCC_MCODIV_5: division by 5 applied to MCOx clock

Return values
- **None**

Notes
- PA8/PC9 should be configured in alternate function mode.
- For STM32F410Rx devices to output I2SCLK clock on MCO2 you should have at last one of the SPI clocks enabled (SPI1, SPI2 or SPI5).

HAL_RCC_EnableCSS

Function Name **void HAL_RCC_EnableCSS (void)**

Function Description Enables the Clock Security System.

Return values
- **None**

Notes
- If a failure is detected on the HSE oscillator clock, this oscillator is automatically disabled and an interrupt is generated to inform the software about the failure (Clock Security System Interrupt, CSSI), allowing the MCU to perform rescue operations. The CSSI is linked to the Cortex-M4 NMI (Non-Maskable Interrupt) exception vector.

HAL_RCC_DisableCSS

Function Name **void HAL_RCC_DisableCSS (void)**

Function Description Disables the Clock Security System.

Return values
- **None**

HAL_RCC_GetSysClockFreq

Function Name	**uint32_t HAL_RCC_GetSysClockFreq (void)**
Function Description	Returns the SYSCLK frequency.
Return values	• **SYSCLK:** frequency
Notes	• The system frequency computed by this function is not the real frequency in the chip. It is calculated based on the predefined constant and the selected clock source: • If SYSCLK source is HSI, function returns values based on HSI_VALUE(*) • If SYSCLK source is HSE, function returns values based on HSE_VALUE(**) • If SYSCLK source is PLL, function returns values based on HSE_VALUE(**) or HSI_VALUE(*) multiplied/divided by the PLL factors. • (*) HSI_VALUE is a constant defined in stm32f4xx_hal_conf.h file (default value 16 MHz) but the real value may vary depending on the variations in voltage and temperature. • (**) HSE_VALUE is a constant defined in stm32f4xx_hal_conf.h file (default value 25 MHz), user has to ensure that HSE_VALUE is same as the real frequency of the crystal used. Otherwise, this function may have wrong result. • The result of this function could be not correct when using fractional value for HSE crystal. • This function can be used by the user application to compute the baudrate for the communication peripherals or configure other parameters. • Each time SYSCLK changes, this function must be called to update the right SYSCLK value. Otherwise, any configuration based on this function will be incorrect.

HAL_RCC_GetHCLKFreq

Function Name	**uint32_t HAL_RCC_GetHCLKFreq (void)**
Function Description	Returns the HCLK frequency.
Return values	• **HCLK:** frequency
Notes	• Each time HCLK changes, this function must be called to update the right HCLK value. Otherwise, any configuration based on this function will be incorrect. • The SystemCoreClock CMSIS variable is used to store System Clock Frequency and updated within this function

HAL_RCC_GetPCLK1Freq

Function Name	**uint32_t HAL_RCC_GetPCLK1Freq (void)**
Function Description	Returns the PCLK1 frequency.
Return values	• **PCLK1:** frequency
Notes	• Each time PCLK1 changes, this function must be called to update the right PCLK1 value. Otherwise, any configuration

based on this function will be incorrect.

HAL_RCC_GetPCLK2Freq

Function Name	**uint32_t HAL_RCC_GetPCLK2Freq (void)**
Function Description	Returns the PCLK2 frequency.
Return values	• **PCLK2:** frequency
Notes	• Each time PCLK2 changes, this function must be called to update the right PCLK2 value. Otherwise, any configuration based on this function will be incorrect.

HAL_RCC_GetOscConfig

Function Name	**void HAL_RCC_GetOscConfig (RCC_OscInitTypeDef * RCC_OscInitStruct)**
Function Description	Configures the RCC_OscInitStruct according to the internal RCC configuration registers.
Parameters	• **RCC_OscInitStruct:** pointer to an RCC_OscInitTypeDef structure that will be configured.
Return values	• **None**

HAL_RCC_GetClockConfig

Function Name	**void HAL_RCC_GetClockConfig (RCC_ClkInitTypeDef * RCC_ClkInitStruct, uint32_t * pFLatency)**
Function Description	Configures the RCC_ClkInitStruct according to the internal RCC configuration registers.
Parameters	• **RCC_ClkInitStruct:** pointer to an RCC_ClkInitTypeDef structure that will be configured. • **pFLatency:** Pointer on the Flash Latency.
Return values	• **None**

HAL_RCC_NMI_IRQHandler

Function Name	**void HAL_RCC_NMI_IRQHandler (void)**
Function Description	This function handles the RCC CSS interrupt request.
Return values	• **None**
Notes	• This API should be called under the NMI_Handler().

HAL_RCC_CSSCallback

Function Name	**void HAL_RCC_CSSCallback (void)**
Function Description	RCC Clock Security System interrupt callback.
Return values	• **None**

49.3 RCC Firmware driver defines

49.3.1 RCC

AHB1 Peripheral Clock Enable Disable

__HAL_RCC_GPIOA_CLK_ENABLE

__HAL_RCC_GPIOB_CLK_ENABLE

__HAL_RCC_GPIOC_CLK_ENABLE

__HAL_RCC_GPIOH_CLK_ENABLE

__HAL_RCC_DMA1_CLK_ENABLE

__HAL_RCC_DMA2_CLK_ENABLE

__HAL_RCC_GPIOA_CLK_DISABLE

__HAL_RCC_GPIOB_CLK_DISABLE

__HAL_RCC_GPIOC_CLK_DISABLE

__HAL_RCC_GPIOH_CLK_DISABLE

__HAL_RCC_DMA1_CLK_DISABLE

__HAL_RCC_DMA2_CLK_DISABLE

AHB1 Force Release Reset

__HAL_RCC_AHB1_FORCE_RESET

__HAL_RCC_GPIOA_FORCE_RESET

__HAL_RCC_GPIOB_FORCE_RESET

__HAL_RCC_GPIOC_FORCE_RESET

__HAL_RCC_GPIOH_FORCE_RESET

__HAL_RCC_DMA1_FORCE_RESET

__HAL_RCC_DMA2_FORCE_RESET

__HAL_RCC_AHB1_RELEASE_RESET

__HAL_RCC_GPIOA_RELEASE_RESET

__HAL_RCC_GPIOB_RELEASE_RESET

__HAL_RCC_GPIOC_RELEASE_RESET

__HAL_RCC_GPIOH_RELEASE_RESET

__HAL_RCC_DMA1_RELEASE_RESET

__HAL_RCC_DMA2_RELEASE_RESET

AHB1 Peripheral Low Power Enable Disable

__HAL_RCC_GPIOA_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOB_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOC_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOH_CLK_SLEEP_ENABLE

__HAL_RCC_DMA1_CLK_SLEEP_ENABLE

__HAL_RCC_DMA2_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOA_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOB_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOC_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOH_CLK_SLEEP_DISABLE

__HAL_RCC_DMA1_CLK_SLEEP_DISABLE

__HAL_RCC_DMA2_CLK_SLEEP_DISABLE

AHB1 Peripheral Clock Enable Disable Status

__HAL_RCC_GPIOA_IS_CLK_ENABLED

__HAL_RCC_GPIOB_IS_CLK_ENABLED

__HAL_RCC_GPIOC_IS_CLK_ENABLED

__HAL_RCC_GPIOH_IS_CLK_ENABLED

__HAL_RCC_DMA1_IS_CLK_ENABLED

__HAL_RCC_DMA2_IS_CLK_ENABLED

__HAL_RCC_GPIOA_IS_CLK_DISABLED

__HAL_RCC_GPIOB_IS_CLK_DISABLED

__HAL_RCC_GPIOC_IS_CLK_DISABLED

__HAL_RCC_GPIOH_IS_CLK_DISABLED

__HAL_RCC_DMA1_IS_CLK_DISABLED

__HAL_RCC_DMA2_IS_CLK_DISABLED

AHB Clock Source

RCC_SYSCLK_DIV1

RCC_SYSCLK_DIV2

RCC_SYSCLK_DIV4

RCC_SYSCLK_DIV8

RCC_SYSCLK_DIV16

RCC_SYSCLK_DIV64

RCC_SYSCLK_DIV128

RCC_SYSCLK_DIV256

RCC_SYSCLK_DIV512

APB1/APB2 Clock Source

RCC_HCLK_DIV1

RCC_HCLK_DIV2

RCC_HCLK_DIV4

RCC_HCLK_DIV8

RCC_HCLK_DIV16

APB1 Peripheral Clock Enable Disable

__HAL_RCC_TIM5_CLK_ENABLE

__HAL_RCC_WWDG_CLK_ENABLE

__HAL_RCC_SPI2_CLK_ENABLE

__HAL_RCC_USART2_CLK_ENABLE

__HAL_RCC_I2C1_CLK_ENABLE

__HAL_RCC_I2C2_CLK_ENABLE

__HAL_RCC_PWR_CLK_ENABLE

__HAL_RCC_TIM5_CLK_DISABLE

__HAL_RCC_WWDG_CLK_DISABLE

__HAL_RCC_SPI2_CLK_DISABLE

__HAL_RCC_USART2_CLK_DISABLE

__HAL_RCC_I2C1_CLK_DISABLE

__HAL_RCC_I2C2_CLK_DISABLE

__HAL_RCC_PWR_CLK_DISABLE

APB1 Force Release Reset

__HAL_RCC_APB1_FORCE_RESET

__HAL_RCC_TIM5_FORCE_RESET

__HAL_RCC_WWDG_FORCE_RESET

__HAL_RCC_SPI2_FORCE_RESET

__HAL_RCC_USART2_FORCE_RESET

__HAL_RCC_I2C1_FORCE_RESET

__HAL_RCC_I2C2_FORCE_RESET

__HAL_RCC_PWR_FORCE_RESET

__HAL_RCC_APB1_RELEASE_RESET

__HAL_RCC_TIM5_RELEASE_RESET

__HAL_RCC_WWDG_RELEASE_RESET

__HAL_RCC_SPI2_RELEASE_RESET

__HAL_RCC_USART2_RELEASE_RESET

__HAL_RCC_I2C1_RELEASE_RESET

__HAL_RCC_I2C2_RELEASE_RESET

__HAL_RCC_PWR_RELEASE_RESET

APB1 Peripheral Low Power Enable Disable

__HAL_RCC_TIM5_CLK_SLEEP_ENABLE

__HAL_RCC_WWDG_CLK_SLEEP_ENABLE

__HAL_RCC_SPI2_CLK_SLEEP_ENABLE

__HAL_RCC_USART2_CLK_SLEEP_ENABLE

__HAL_RCC_I2C1_CLK_SLEEP_ENABLE

__HAL_RCC_I2C2_CLK_SLEEP_ENABLE

__HAL_RCC_PWR_CLK_SLEEP_ENABLE

__HAL_RCC_TIM5_CLK_SLEEP_DISABLE

__HAL_RCC_WWDG_CLK_SLEEP_DISABLE

__HAL_RCC_SPI2_CLK_SLEEP_DISABLE

__HAL_RCC_USART2_CLK_SLEEP_DISABLE

__HAL_RCC_I2C1_CLK_SLEEP_DISABLE

__HAL_RCC_I2C2_CLK_SLEEP_DISABLE

__HAL_RCC_PWR_CLK_SLEEP_DISABLE

APB1 Peripheral Clock Enable Disable Status

__HAL_RCC_TIM5_IS_CLK_ENABLED

__HAL_RCC_WWDG_IS_CLK_ENABLED

__HAL_RCC_SPI2_IS_CLK_ENABLED

__HAL_RCC_USART2_IS_CLK_ENABLED

__HAL_RCC_I2C1_IS_CLK_ENABLED

__HAL_RCC_I2C2_IS_CLK_ENABLED

__HAL_RCC_PWR_IS_CLK_ENABLED

__HAL_RCC_TIM5_IS_CLK_DISABLED

__HAL_RCC_WWDG_IS_CLK_DISABLED

__HAL_RCC_SPI2_IS_CLK_DISABLED

__HAL_RCC_USART2_IS_CLK_DISABLED

__HAL_RCC_I2C1_IS_CLK_DISABLED

__HAL_RCC_I2C2_IS_CLK_DISABLED

__HAL_RCC_PWR_IS_CLK_DISABLED

APB2 Peripheral Clock Enable Disable

__HAL_RCC_TIM1_CLK_ENABLE

__HAL_RCC_USART1_CLK_ENABLE

__HAL_RCC_USART6_CLK_ENABLE

__HAL_RCC_ADC1_CLK_ENABLE

__HAL_RCC_SPI1_CLK_ENABLE

__HAL_RCC_SYSCFG_CLK_ENABLE

__HAL_RCC_TIM9_CLK_ENABLE

__HAL_RCC_TIM11_CLK_ENABLE

__HAL_RCC_TIM1_CLK_DISABLE

__HAL_RCC_USART1_CLK_DISABLE

__HAL_RCC_USART6_CLK_DISABLE

__HAL_RCC_ADC1_CLK_DISABLE

__HAL_RCC_SPI1_CLK_DISABLE

__HAL_RCC_SYSCFG_CLK_DISABLE

__HAL_RCC_TIM9_CLK_DISABLE

__HAL_RCC_TIM11_CLK_DISABLE

APB2 Force Release Reset

__HAL_RCC_APB2_FORCE_RESET

__HAL_RCC_TIM1_FORCE_RESET

__HAL_RCC_USART1_FORCE_RESET

__HAL_RCC_USART6_FORCE_RESET

__HAL_RCC_ADC_FORCE_RESET

__HAL_RCC_SPI1_FORCE_RESET

__HAL_RCC_SYSCFG_FORCE_RESET

__HAL_RCC_TIM9_FORCE_RESET

__HAL_RCC_TIM11_FORCE_RESET

__HAL_RCC_APB2_RELEASE_RESET

__HAL_RCC_TIM1_RELEASE_RESET

__HAL_RCC_USART1_RELEASE_RESET

__HAL_RCC_USART6_RELEASE_RESET

__HAL_RCC_ADC_RELEASE_RESET

__HAL_RCC_SPI1_RELEASE_RESET

__HAL_RCC_SYSCFG_RELEASE_RESET

__HAL_RCC_TIM9_RELEASE_RESET

__HAL_RCC_TIM11_RELEASE_RESET

APB2 Peripheral Low Power Enable Disable

__HAL_RCC_TIM1_CLK_SLEEP_ENABLE

__HAL_RCC_USART1_CLK_SLEEP_ENABLE

__HAL_RCC_USART6_CLK_SLEEP_ENABLE

__HAL_RCC_ADC1_CLK_SLEEP_ENABLE

__HAL_RCC_SPI1_CLK_SLEEP_ENABLE

__HAL_RCC_SYSCFG_CLK_SLEEP_ENABLE

__HAL_RCC_TIM9_CLK_SLEEP_ENABLE

__HAL_RCC_TIM11_CLK_SLEEP_ENABLE

__HAL_RCC_TIM1_CLK_SLEEP_DISABLE

__HAL_RCC_USART1_CLK_SLEEP_DISABLE

__HAL_RCC_USART6_CLK_SLEEP_DISABLE

__HAL_RCC_ADC1_CLK_SLEEP_DISABLE

__HAL_RCC_SPI1_CLK_SLEEP_DISABLE

__HAL_RCC_SYSCFG_CLK_SLEEP_DISABLE

__HAL_RCC_TIM9_CLK_SLEEP_DISABLE

__HAL_RCC_TIM11_CLK_SLEEP_DISABLE

APB2 Peripheral Clock Enable Disable Status

__HAL_RCC_TIM1_IS_CLK_ENABLED

__HAL_RCC_USART1_IS_CLK_ENABLED

__HAL_RCC_USART6_IS_CLK_ENABLED

__HAL_RCC_ADC1_IS_CLK_ENABLED

__HAL_RCC_SPI1_IS_CLK_ENABLED

__HAL_RCC_SYSCFG_IS_CLK_ENABLED

__HAL_RCC_TIM9_IS_CLK_ENABLED

__HAL_RCC_TIM11_IS_CLK_ENABLED

__HAL_RCC_TIM1_IS_CLK_DISABLED

__HAL_RCC_USART1_IS_CLK_DISABLED

__HAL_RCC_USART6_IS_CLK_DISABLED

__HAL_RCC_ADC1_IS_CLK_DISABLED

__HAL_RCC_SPI1_IS_CLK_DISABLED

__HAL_RCC_SYSCFG_IS_CLK_DISABLED

__HAL_RCC_TIM9_IS_CLK_DISABLED

__HAL_RCC_TIM11_IS_CLK_DISABLED

RCC BitAddress AliasRegion

RCC_OFFSET

RCC_CR_OFFSET

RCC_HSION_BIT_NUMBER

RCC_CR_HSION_BB

RCC_CSSON_BIT_NUMBER

RCC_CR_CSSON_BB

RCC_PLLON_BIT_NUMBER

RCC_CR_PLLON_BB

RCC_BDCR_OFFSET

RCC_RTCEN_BIT_NUMBER

RCC_BDCR_RTCEN_BB

RCC_BDRST_BIT_NUMBER

RCC_BDCR_BDRST_BB

RCC_CSR_OFFSET

RCC_LSION_BIT_NUMBER

RCC_CSR_LSION_BB

RCC_CR_BYTE2_ADDRESS

RCC_CIR_BYTE1_ADDRESS

RCC_CIR_BYTE2_ADDRESS

RCC_BDCR_BYTE0_ADDRESS

RCC_DBP_TIMEOUT_VALUE

RCC_LSE_TIMEOUT_VALUE

HSE_TIMEOUT_VALUE

HSI_TIMEOUT_VALUE

LSI_TIMEOUT_VALUE

Flags

RCC_FLAG_HSIRDY

RCC_FLAG_HSERDY

RCC_FLAG_PLLRDY

RCC_FLAG_PLLI2SRDY

RCC_FLAG_LSERDY

RCC_FLAG_LSIRDY

RCC_FLAG_BORRST

RCC_FLAG_PINRST

RCC_FLAG_PORRST

RCC_FLAG_SFTRST

RCC_FLAG_IWDGRST

RCC_FLAG_WWDGRST

RCC_FLAG_LPWRRST

Flags Interrupts Management

__HAL_RCC_ENABLE_IT

Description:

- Enable RCC interrupt (Perform Byte access to RCC_CIR[14:8] bits to enable the selected interrupts).

Parameters:

- __INTERRUPT__: specifies the RCC interrupt sources to be enabled. This parameter can be any combination of the following values:
  – RCC_IT_LSIRDY: LSI ready interrupt.

- RCC_IT_LSERDY: LSE ready interrupt.
- RCC_IT_HSIRDY: HSI ready interrupt.
- RCC_IT_HSERDY: HSE ready interrupt.
- RCC_IT_PLLRDY: Main PLL ready interrupt.
- RCC_IT_PLLI2SRDY: PLLI2S ready interrupt.

__HAL_RCC_DISABLE_IT

Description:

- Disable RCC interrupt (Perform Byte access to RCC_CIR[14:8] bits to disable the selected interrupts).

Parameters:

- __INTERRUPT__: specifies the RCC interrupt sources to be disabled. This parameter can be any combination of the following values:
  - RCC_IT_LSIRDY: LSI ready interrupt.
  - RCC_IT_LSERDY: LSE ready interrupt.
  - RCC_IT_HSIRDY: HSI ready interrupt.
  - RCC_IT_HSERDY: HSE ready interrupt.
  - RCC_IT_PLLRDY: Main PLL ready interrupt.
  - RCC_IT_PLLI2SRDY: PLLI2S ready interrupt.

__HAL_RCC_CLEAR_IT

Description:

- Clear the RCC's interrupt pending bits (Perform Byte access to RCC_CIR[23:16] bits to clear the selected interrupt pending bits.

Parameters:

- __INTERRUPT__: specifies the interrupt pending bit to clear. This parameter can be any combination of the following values:
  - RCC_IT_LSIRDY: LSI ready interrupt.
  - RCC_IT_LSERDY: LSE ready interrupt.
  - RCC_IT_HSIRDY: HSI ready interrupt.
  - RCC_IT_HSERDY: HSE ready interrupt.
  - RCC_IT_PLLRDY: Main PLL ready interrupt.
  - RCC_IT_PLLI2SRDY: PLLI2S ready interrupt.
  - RCC_IT_CSS: Clock Security System interrupt

__HAL_RCC_GET_IT

Description:

- Check the RCC's interrupt has occurred or

not.

Parameters:

- __INTERRUPT__: specifies the RCC interrupt source to check. This parameter can be one of the following values:
  - RCC_IT_LSIRDY: LSI ready interrupt.
  - RCC_IT_LSERDY: LSE ready interrupt.
  - RCC_IT_HSIRDY: HSI ready interrupt.
  - RCC_IT_HSERDY: HSE ready interrupt.
  - RCC_IT_PLLRDY: Main PLL ready interrupt.
  - RCC_IT_PLLI2SRDY: PLLI2S ready interrupt.
  - RCC_IT_CSS: Clock Security System interrupt

Return value:

- The: new state of __INTERRUPT__ (TRUE or FALSE).

__HAL_RCC_CLEAR_RESET_FLAGS

RCC_FLAG_MASK

Description:

- Check RCC flag is set or not.

Parameters:

- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - RCC_FLAG_HSIRDY: HSI oscillator clock ready.
  - RCC_FLAG_HSERDY: HSE oscillator clock ready.
  - RCC_FLAG_PLLRDY: Main PLL clock ready.
  - RCC_FLAG_PLLI2SRDY: PLLI2S clock ready.
  - RCC_FLAG_LSERDY: LSE oscillator clock ready.
  - RCC_FLAG_LSIRDY: LSI oscillator clock ready.
  - RCC_FLAG_BORRST: POR/PDR or BOR reset.
  - RCC_FLAG_PINRST: Pin reset.
  - RCC_FLAG_PORRST: POR/PDR reset.
  - RCC_FLAG_SFTRST: Software reset.
  - RCC_FLAG_IWDGRST: Independent Watchdog reset.
  - RCC_FLAG_WWDGRST: Window Watchdog reset.
  - RCC_FLAG_LPWRRST: Low Power

reset.

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_RCC_GET_FLAG

Get Clock source

__HAL_RCC_SYSCLK_CONFIG

Description:

- Macro to configure the system clock source.

Parameters:

- __RCC_SYSCLKSOURCE__: specifies the system clock source. This parameter can be one of the following values:
  - RCC_SYSCLKSOURCE_HSI: HSI oscillator is used as system clock source.
  - RCC_SYSCLKSOURCE_HSE: HSE oscillator is used as system clock source.
  - RCC_SYSCLKSOURCE_PLLCLK: PLL output is used as system clock source.
  - RCC_SYSCLKSOURCE_PLLRCLK: PLLR output is used as system clock source.

__HAL_RCC_GET_SYSCLK_SOURCE

Description:

- Macro to get the clock source used as system clock.

Return value:

- The: clock source used as system clock. The returned value can be one of the following:
  - RCC_SYSCLKSOURCE_STATUS_HSI: HSI used as system clock.
  - RCC_SYSCLKSOURCE_STATUS_HSE: HSE used as system clock.
  - RCC_SYSCLKSOURCE_STATUS_PLLCLK: PLL used as system clock.
  - RCC_SYSCLKSOURCE_STATUS_PLLRCLK: PLLR used as system clock.

__HAL_RCC_GET_PLL_OSCSOURCE

Description:

- Macro to get the oscillator used as PLL clock source.

Return value:

- The: oscillator used as PLL clock source. The returned value can be one of the following:
  - RCC_PLLSOURCE_HSI: HSI oscillator is used as PLL clock source.
  - RCC_PLLSOURCE_HSE: HSE oscillator is used as PLL clock source.

HSE Config

RCC_HSE_OFF

RCC_HSE_ON

RCC_HSE_BYPASS

HSE Configuration

__HAL_RCC_HSE_CONFIG **Description:**

- Macro to configure the External High Speed oscillator (HSE).

Parameters:

- __STATE__: specifies the new state of the HSE. This parameter can be one of the following values:
  - RCC_HSE_OFF: turn OFF the HSE oscillator, HSERDY flag goes low after 6 HSE oscillator clock cycles.
  - RCC_HSE_ON: turn ON the HSE oscillator.
  - RCC_HSE_BYPASS: HSE oscillator bypassed with external clock.

Notes:

- Transition HSE Bypass to HSE On and HSE On to HSE Bypass are not supported by this macro. User should request a transition to HSE Off first and then HSE On or HSE Bypass. After enabling the HSE (RCC_HSE_ON or RCC_HSE_Bypass), the application software should wait on HSERDY flag to be set indicating that HSE clock is stable and can be used to clock the PLL and/or system clock. HSE state can not be changed if it is used directly or through the PLL as system clock. In this case, you have to select another source of the system clock then change the HSE state (ex. disable it). The HSE is stopped by hardware when entering STOP and STANDBY modes. This function reset the CSSON bit, so if the clock security system(CSS) was previously enabled you have to enable it again after calling this function.

HSI Config

RCC_HSI_OFF

RCC_HSI_ON

RCC_HSICALIBRATION_DEFAULT

HSI Configuration

__HAL_RCC_HSI_ENABLE **Notes:**

- The HSI is stopped by hardware when entering STOP and STANDBY modes. It is used (enabled by hardware) as system clock source after startup from Reset, wake-up from STOP and STANDBY mode, or in case of failure of the

HSE used directly or indirectly as system clock (if the Clock Security System CSS is enabled). HSI can not be stopped if it is used as system clock source. In this case, you have to select another source of the system clock then stop the HSI. After enabling the HSI, the application software should wait on HSIRDY flag to be set indicating that HSI clock is stable and can be used as system clock source. This parameter can be: ENABLE or DISABLE. When the HSI is stopped, HSIRDY flag goes low after 6 HSI oscillator clock cycles.

__HAL_RCC_HSI_DISABLE

__HAL_RCC_HSI_CALIBRATIONVALUE_ADJUST

Description:

- Macro to adjust the Internal High Speed oscillator (HSI) calibration value.

Parameters:

- __HSICalibrationValue__: specifies the calibration trimming value. (default is RCC_HSICALIBRATION_DEFAULT). This parameter must be a number between 0 and 0x1F.

Notes:

- The calibration is used to compensate for the variations in voltage and temperature that influence the frequency of the internal HSI RC.

RTC Clock Configuration

__HAL_RCC_RTC_ENABLE

Notes:

- These macros must be used only after the RTC clock source was selected.

__HAL_RCC_RTC_DISABLE

__HAL_RCC_RTC_CLKPRESCALER

Description:

- Macros to configure the RTC clock (RTCCLK).

Parameters:

- __RTCCLKSource__: specifies the RTC

clock source. This parameter can be one of the following values:
- RCC_RTCCLKSOURCE_LSE: LSE selected as RTC clock.
- RCC_RTCCLKSOURCE_LSI: LSI selected as RTC clock.
- RCC_RTCCLKSOURCE_HSE_DIVx: HSE clock divided by x selected as RTC clock, where x:[2,31]

Notes:

- As the RTC clock configuration bits are in the Backup domain and write access is denied to this domain after reset, you have to enable write access using the Power Backup Access macro before to configure the RTC clock source (to be done once after reset). Once the RTC clock is configured it can't be changed unless the Backup domain is reset using __HAL_RCC_BackupReset_RELEASE() macro, or by a Power On Reset (POR).
- If the LSE or LSI is used as RTC clock source, the RTC continues to work in STOP and STANDBY modes, and can be used as wake-up source. However, when the HSE clock is used as RTC clock source, the RTC cannot be used in STOP and STANDBY modes. The maximum input clock frequency for RTC is 1MHz (when using HSE as RTC clock source).

__HAL_RCC_RTC_CONFIG

__HAL_RCC_BACKUPRESET_FORCE

Notes:

- This function resets the RTC peripheral (including the backup registers) and the RTC clock source selection in RCC_CSR register. The BKPSRAM is not affected by this reset.

__HAL_RCC_BACKUPRESET_RELEASE

Interrupts

RCC_IT_LSIRDY

RCC_IT_LSERDY

RCC_IT_HSIRDY

RCC_IT_HSERDY

RCC_IT_PLLRDY

RCC_IT_PLLI2SRDY

RCC_IT_CSS

RCC Private macros to check input parameters

IS_RCC_OSCILLATORTYPE

IS_RCC_HSE

IS_RCC_LSE

IS_RCC_HSI

IS_RCC_LSI

IS_RCC_PLL

IS_RCC_PLLSOURCE

IS_RCC_SYSCLKSOURCE

IS_RCC_RTCCLKSOURCE

IS_RCC_PLLM_VALUE

IS_RCC_PLLP_VALUE

IS_RCC_PLLQ_VALUE

IS_RCC_HCLK

IS_RCC_CLOCKTYPE

IS_RCC_PCLK

IS_RCC_MCO

IS_RCC_MCO1SOURCE

IS_RCC_MCODIV

IS_RCC_CALIBRATION_VALUE

LSE Config

RCC_LSE_OFF

RCC_LSE_ON

RCC_LSE_BYPASS

LSE Configuration

__HAL_RCC_LSE_CONFIG

Description:

- Macro to configure the External Low Speed oscillator (LSE).

Parameters:

- __STATE__: specifies the new state of the LSE. This parameter can be one of the following values:
  - RCC_LSE_OFF: turn OFF the LSE oscillator, LSERDY flag goes low after 6 LSE oscillator clock cycles.
  - RCC_LSE_ON: turn ON the LSE oscillator.
  - RCC_LSE_BYPASS: LSE oscillator bypassed with external clock.

Notes:

- Transition LSE Bypass to LSE On and LSE On to LSE Bypass are not supported by this macro. User should

request a transition to LSE Off first and then LSE On or LSE Bypass. As the LSE is in the Backup domain and write access is denied to this domain after reset, you have to enable write access using HAL_PWR_EnableBkUpAccess() function before to configure the LSE (to be done once after reset). After enabling the LSE (RCC_LSE_ON or RCC_LSE_BYPASS), the application software should wait on LSERDY flag to be set indicating that LSE clock is stable and can be used to clock the RTC.

LSI Config

RCC_LSI_OFF

RCC_LSI_ON

LSI Configuration

__HAL_RCC_LSI_ENABLE

Notes:

- After enabling the LSI, the application software should wait on LSIRDY flag to be set indicating that LSI clock is stable and can be used to clock the IWDG and/or the RTC. LSI can not be disabled if the IWDG is running. When the LSI is stopped, LSIRDY flag goes low after 6 LSI oscillator clock cycles.

__HAL_RCC_LSI_DISABLE

MCO1 Clock Source

RCC_MCO1SOURCE_HSI

RCC_MCO1SOURCE_LSE

RCC_MCO1SOURCE_HSE

RCC_MCO1SOURCE_PLLCLK

MCO2 Clock Source

RCC_MCO2SOURCE_SYSCLK

RCC_MCO2SOURCE_PLLI2SCLK

RCC_MCO2SOURCE_HSE

RCC_MCO2SOURCE_PLLCLK

MCOx Clock Prescaler

RCC_MCODIV_1

RCC_MCODIV_2

RCC_MCODIV_3

RCC_MCODIV_4

RCC_MCODIV_5

MCO Index

RCC_MCO1

RCC_MCO2

Oscillator Type

RCC_OSCILLATORTYPE_NONE

RCC_OSCILLATORTYPE_HSE

RCC_OSCILLATORTYPE_HSI

RCC_OSCILLATORTYPE_LSE

RCC_OSCILLATORTYPE_LSI

PLLP Clock Divider

RCC_PLLP_DIV2

RCC_PLLP_DIV4

RCC_PLLP_DIV6

RCC_PLLP_DIV8

PLL Clock Source

RCC_PLLSOURCE_HSI

RCC_PLLSOURCE_HSE

PLL Config

RCC_PLL_NONE

RCC_PLL_OFF

RCC_PLL_ON

PLL Configuration

__HAL_RCC_PLL_ENABLE

Notes:

- After enabling the main PLL, the application software should wait on PLLRDY flag to be set indicating that PLL clock is stable and can be used as system clock source. The main PLL can not be disabled if it is used as system clock source The main PLL is disabled by hardware when entering STOP and STANDBY modes.

__HAL_RCC_PLL_DISABLE

__HAL_RCC_PLL_PLLSOURCE_CONFIG

Description:

- Macro to configure the PLL clock source.

Parameters:

- __PLLSOURCE__: specifies the PLL entry clock source. This parameter can be one of the following values:
  - RCC_PLLSOURCE_HSI: HSI oscillator clock selected as PLL clock entry
  - RCC_PLLSOURCE_HSE: HSE oscillator clock selected as PLL

clock entry

Notes:

- This function must be used only when the main PLL is disabled.

__HAL_RCC_PLL_PLLM_CONFIG

Description:

- Macro to configure the PLL multiplication factor.

Parameters:

- __PLLM__: specifies the division factor for PLL VCO input clock This parameter must be a number between Min_Data = 2 and Max_Data = 63.

Notes:

- This function must be used only when the main PLL is disabled.
- You have to set the PLLM parameter correctly to ensure that the VCO input frequency ranges from 1 to 2 MHz. It is recommended to select a frequency of 2 MHz to limit PLL jitter.

RTC Clock Source

RCC_RTCCLKSOURCE_LSE

RCC_RTCCLKSOURCE_LSI

RCC_RTCCLKSOURCE_HSE_DIV2

RCC_RTCCLKSOURCE_HSE_DIV3

RCC_RTCCLKSOURCE_HSE_DIV4

RCC_RTCCLKSOURCE_HSE_DIV5

RCC_RTCCLKSOURCE_HSE_DIV6

RCC_RTCCLKSOURCE_HSE_DIV7

RCC_RTCCLKSOURCE_HSE_DIV8

RCC_RTCCLKSOURCE_HSE_DIV9

RCC_RTCCLKSOURCE_HSE_DIV10

RCC_RTCCLKSOURCE_HSE_DIV11

RCC_RTCCLKSOURCE_HSE_DIV12

RCC_RTCCLKSOURCE_HSE_DIV13

RCC_RTCCLKSOURCE_HSE_DIV14

RCC_RTCCLKSOURCE_HSE_DIV15

RCC_RTCCLKSOURCE_HSE_DIV16

RCC_RTCCLKSOURCE_HSE_DIV17

RCC_RTCCLKSOURCE_HSE_DIV18

RCC_RTCCLKSOURCE_HSE_DIV19
RCC_RTCCLKSOURCE_HSE_DIV20
RCC_RTCCLKSOURCE_HSE_DIV21
RCC_RTCCLKSOURCE_HSE_DIV22
RCC_RTCCLKSOURCE_HSE_DIV23
RCC_RTCCLKSOURCE_HSE_DIV24
RCC_RTCCLKSOURCE_HSE_DIV25
RCC_RTCCLKSOURCE_HSE_DIV26
RCC_RTCCLKSOURCE_HSE_DIV27
RCC_RTCCLKSOURCE_HSE_DIV28
RCC_RTCCLKSOURCE_HSE_DIV29
RCC_RTCCLKSOURCE_HSE_DIV30
RCC_RTCCLKSOURCE_HSE_DIV31

System Clock Source

RCC_SYSCLKSOURCE_HSI
RCC_SYSCLKSOURCE_HSE
RCC_SYSCLKSOURCE_PLLCLK
RCC_SYSCLKSOURCE_PLLRCLK

System Clock Source Status

RCC_SYSCLKSOURCE_STATUS_HSI	HSI used as system clock
RCC_SYSCLKSOURCE_STATUS_HSE	HSE used as system clock
RCC_SYSCLKSOURCE_STATUS_PLLCLK	PLL used as system clock
RCC_SYSCLKSOURCE_STATUS_PLLRCLK	PLLR used as system clock

System Clock Type

RCC_CLOCKTYPE_SYSCLK
RCC_CLOCKTYPE_HCLK
RCC_CLOCKTYPE_PCLK1
RCC_CLOCKTYPE_PCLK2

50 HAL RCC Extension Driver

50.1 RCCEx Firmware driver registers structures

50.1.1 RCC_PLLInitTypeDef

Data Fields

- ***uint32_t PLLState***
- ***uint32_t PLLSource***
- ***uint32_t PLLM***
- ***uint32_t PLLN***
- ***uint32_t PLLP***
- ***uint32_t PLLQ***
- ***uint32_t PLLR***

Field Documentation

- ***uint32_t RCC_PLLInitTypeDef::PLLState***
  The new state of the PLL. This parameter can be a value of ***RCC_PLL_Config***
- ***uint32_t RCC_PLLInitTypeDef::PLLSource***
  RCC_PLLSource: PLL entry clock source. This parameter must be a value of ***RCC_PLL_Clock_Source***
- ***uint32_t RCC_PLLInitTypeDef::PLLM***
  PLLM: Division factor for PLL VCO input clock. This parameter must be a number between Min_Data = 0 and Max_Data = 63
- ***uint32_t RCC_PLLInitTypeDef::PLLN***
  PLLN: Multiplication factor for PLL VCO output clock. This parameter must be a number between Min_Data = 50 and Max_Data = 432 except for STM32F411xE devices where the Min_Data = 192
- ***uint32_t RCC_PLLInitTypeDef::PLLP***
  PLLP: Division factor for main system clock (SYSCLK). This parameter must be a value of ***RCC_PLLP_Clock_Divider***
- ***uint32_t RCC_PLLInitTypeDef::PLLQ***
  PLLQ: Division factor for OTG FS, SDIO and RNG clocks. This parameter must be a number between Min_Data = 4 and Max_Data = 15
- ***uint32_t RCC_PLLInitTypeDef::PLLR***
  PLLR: PLL division factor for I2S, SAI, SYSTEM, SPDIFRX clocks. This parameter is only available in STM32F410xx/STM32F446xx/STM32F469xx/STM32F479xx and STM32F412Zx/STM32F412Vx/STM32F412Rx/STM32F412Cx devices. This parameter must be a number between Min_Data = 2 and Max_Data = 7

50.1.2 RCC_PLLI2SInitTypeDef

Data Fields

- ***uint32_t PLLI2SN***
- ***uint32_t PLLI2SR***

- ***uint32_t PLLI2SQ***

Field Documentation

- ***uint32_t RCC_PLLI2SInitTypeDef::PLLI2SN***
  Specifies the multiplication factor for PLLI2S VCO output clock. This parameter must be a number between Min_Data = 50 and Max_Data = 432. This parameter will be used only when PLLI2S is selected as Clock Source I2S or SAI
- ***uint32_t RCC_PLLI2SInitTypeDef::PLLI2SR***
  Specifies the division factor for I2S clock. This parameter must be a number between Min_Data = 2 and Max_Data = 7. This parameter will be used only when PLLI2S is selected as Clock Source I2S or SAI
- ***uint32_t RCC_PLLI2SInitTypeDef::PLLI2SQ***
  Specifies the division factor for SAI1 clock. This parameter must be a number between Min_Data = 2 and Max_Data = 15. This parameter will be used only when PLLI2S is selected as Clock Source SAI

50.1.3 RCC_PLLSAIInitTypeDef

Data Fields

- ***uint32_t PLLSAIN***
- ***uint32_t PLLSAIP***
- ***uint32_t PLLSAIQ***
- ***uint32_t PLLSAIR***

Field Documentation

- ***uint32_t RCC_PLLSAIInitTypeDef::PLLSAIN***
  Specifies the multiplication factor for PLLI2S VCO output clock. This parameter must be a number between Min_Data = 50 and Max_Data = 432. This parameter will be used only when PLLSAI is selected as Clock Source SAI or LTDC
- ***uint32_t RCC_PLLSAIInitTypeDef::PLLSAIP***
  Specifies division factor for OTG FS and SDIO clocks. This parameter is only available in STM32F469xx/STM32F479xx devices. This parameter must be a value of ***RCCEx_PLLSAIP_Clock_Divider***
- ***uint32_t RCC_PLLSAIInitTypeDef::PLLSAIQ***
  Specifies the division factor for SAI1 clock. This parameter must be a number between Min_Data = 2 and Max_Data = 15. This parameter will be used only when PLLSAI is selected as Clock Source SAI or LTDC
- ***uint32_t RCC_PLLSAIInitTypeDef::PLLSAIR***
  specifies the division factor for LTDC clock This parameter must be a number between Min_Data = 2 and Max_Data = 7. This parameter will be used only when PLLSAI is selected as Clock Source LTDC

50.1.4 RCC_PeriphCLKInitTypeDef

Data Fields

- ***uint32_t PeriphClockSelection***
- ***RCC_PLLI2SInitTypeDef PLLI2S***
- ***RCC_PLLSAIInitTypeDef PLLSAI***
- ***uint32_t PLLI2SDivQ***
- ***uint32_t PLLSAIDivQ***
- ***uint32_t PLLSAIDivR***
- ***uint32_t RTCClockSelection***
- ***uint8_t TIMPresSelection***
- ***uint32_t Clk48ClockSelection***
- ***uint32_t SdioClockSelection***

Field Documentation

- ***uint32_t RCC_PeriphCLKInitTypeDef::PeriphClockSelection***
  The Extended Clock to be configured. This parameter can be a value of ***RCCEx_Periph_Clock_Selection***
- ***RCC_PLLI2SInitTypeDef RCC_PeriphCLKInitTypeDef::PLLI2S***
  PLL I2S structure parameters. This parameter will be used only when PLLI2S is selected as Clock Source I2S or SAI
- ***RCC_PLLSAIInitTypeDef RCC_PeriphCLKInitTypeDef::PLLSAI***
  PLL SAI structure parameters. This parameter will be used only when PLLI2S is selected as Clock Source SAI or LTDC
- ***uint32_t RCC_PeriphCLKInitTypeDef::PLLI2SDivQ***
  Specifies the PLLI2S division factor for SAI1 clock. This parameter must be a number between Min_Data = 1 and Max_Data = 32 This parameter will be used only when PLLI2S is selected as Clock Source SAI
- ***uint32_t RCC_PeriphCLKInitTypeDef::PLLSAIDivQ***
  Specifies the PLLI2S division factor for SAI1 clock. This parameter must be a number between Min_Data = 1 and Max_Data = 32 This parameter will be used only when PLLSAI is selected as Clock Source SAI
- ***uint32_t RCC_PeriphCLKInitTypeDef::PLLSAIDivR***
  Specifies the PLLSAI division factor for LTDC clock. This parameter must be one value of ***RCCEx_PLLSAI_DIVR***
- ***uint32_t RCC_PeriphCLKInitTypeDef::RTCClockSelection***
  Specifies RTC Clock Prescalers Selection. This parameter can be a value of ***RCC_RTC_Clock_Source***
- ***uint8_t RCC_PeriphCLKInitTypeDef::TIMPresSelection***
  Specifies TIM Clock Prescalers Selection. This parameter can be a value of ***RCCEx_TIM_PRescaler_Selection***
- ***uint32_t RCC_PeriphCLKInitTypeDef::Clk48ClockSelection***
  Specifies CLK48 Clock Selection this clock used OTG FS, SDIO and RNG clocks. This parameter can be a value of ***RCCEx_CLK48_Clock_Source***
- ***uint32_t RCC_PeriphCLKInitTypeDef::SdioClockSelection***
  Specifies SDIO Clock Source Selection. This parameter can be a value of ***RCCEx_SDIO_Clock_Source***

50.2 RCCEx Firmware driver API description

50.2.1 Extended Peripheral Control functions

This subsection provides a set of functions allowing to control the RCC Clocks frequencies.

Important note: Care must be taken when HAL_RCCEx_PeriphCLKConfig() is used to select the RTC clock source; in this case the Backup domain will be reset in order to modify the RTC Clock source, as consequence RTC registers (including the backup registers) and RCC_BDCR register are set to their reset values.

This section contains the following APIs:

- ***HAL_RCC_DeInit()***
- ***HAL_RCCEx_PeriphCLKConfig()***
- ***HAL_RCCEx_GetPeriphCLKConfig()***
- ***HAL_RCC_OscConfig()***
- ***HAL_RCC_GetOscConfig()***
- ***HAL_RCCEx_SelectLSEMode()***

50.2.2 Detailed description of functions

HAL_RCCEx_PeriphCLKConfig

Function Name	**HAL_StatusTypeDef HAL_RCCEx_PeriphCLKConfig (RCC_PeriphCLKInitTypeDef * PeriphClkInit)**
Function Description	Initializes the RCC extended peripherals clocks according to the specified parameters in the RCC_PeriphCLKInitTypeDef.
Parameters	• **PeriphClkInit:** pointer to an RCC_PeriphCLKInitTypeDef structure that contains the configuration information for the Extended Peripherals clocks(I2S, SAI, LTDC, RTC and TIM).
Return values	• **HAL:** status
Notes	• Care must be taken when HAL_RCCEx_PeriphCLKConfig() is used to select the RTC clock source; in this case the Backup domain will be reset in order to modify the RTC Clock source, as consequence RTC registers (including the backup registers) and RCC_BDCR register are set to their reset values.

HAL_RCCEx_GetPeriphCLKConfig

Function Name	**void HAL_RCCEx_GetPeriphCLKConfig (RCC_PeriphCLKInitTypeDef * PeriphClkInit)**
Function Description	Configures the RCC_PeriphCLKInitTypeDef according to the internal RCC configuration registers.
Parameters	• **PeriphClkInit:** pointer to an RCC_PeriphCLKInitTypeDef structure that will be configured.
Return values	• **None**

HAL_RCCEx_SelectLSEMode

Function Name	**void HAL_RCCEx_SelectLSEMode (uint8_t Mode)**
Function Description	Select LSE mode.
Parameters	• **Mode:** specifies the LSE mode. This parameter can be one of the following values: – RCC_LSE_LOWPOWER_MODE: LSE oscillator in low power mode selection – RCC_LSE_HIGHDRIVE_MODE: LSE oscillator in High Drive mode selection
Return values	• **None**
Notes	• This mode is only available for STM32F410xx/STM32F411xx/STM32F446xx/STM32F469xx/STM32F479xx/ STM32F412Zx/STM32F412Vx/STM32F412Rx/STM32F412Cx devices.

50.3 RCCEx Firmware driver defines

50.3.1 RCCEx

AHB1 Peripheral Clock Enable Disable

__HAL_RCC_BKPSRAM_CLK_ENABLE

__HAL_RCC_CCMDATARAMEN_CLK_ENABLE

__HAL_RCC_CRC_CLK_ENABLE

__HAL_RCC_GPIOD_CLK_ENABLE

__HAL_RCC_GPIOE_CLK_ENABLE

__HAL_RCC_GPIOI_CLK_ENABLE

__HAL_RCC_GPIOF_CLK_ENABLE

__HAL_RCC_GPIOG_CLK_ENABLE

__HAL_RCC_GPIOJ_CLK_ENABLE

__HAL_RCC_GPIOK_CLK_ENABLE

__HAL_RCC_DMA2D_CLK_ENABLE

__HAL_RCC_ETHMAC_CLK_ENABLE

__HAL_RCC_ETHMACTX_CLK_ENABLE

__HAL_RCC_ETHMACRX_CLK_ENABLE

__HAL_RCC_ETHMACPTP_CLK_ENABLE

__HAL_RCC_USB_OTG_HS_CLK_ENABLE

__HAL_RCC_USB_OTG_HS_ULPI_CLK_ENABLE

__HAL_RCC_GPIOD_CLK_DISABLE
__HAL_RCC_GPIOE_CLK_DISABLE
__HAL_RCC_GPIOF_CLK_DISABLE
__HAL_RCC_GPIOG_CLK_DISABLE
__HAL_RCC_GPIOI_CLK_DISABLE
__HAL_RCC_GPIOJ_CLK_DISABLE
__HAL_RCC_GPIOK_CLK_DISABLE
__HAL_RCC_DMA2D_CLK_DISABLE
__HAL_RCC_ETHMAC_CLK_DISABLE
__HAL_RCC_ETHMACTX_CLK_DISABLE
__HAL_RCC_ETHMACRX_CLK_DISABLE
__HAL_RCC_ETHMACPTP_CLK_DISABLE
__HAL_RCC_USB_OTG_HS_CLK_DISABLE
__HAL_RCC_USB_OTG_HS_ULPI_CLK_DISABLE
__HAL_RCC_BKPSRAM_CLK_DISABLE
__HAL_RCC_CCMDATARAMEN_CLK_DISABLE
__HAL_RCC_CRC_CLK_DISABLE
__HAL_RCC_ETH_CLK_ENABLE
__HAL_RCC_ETH_CLK_DISABLE

AHB1 Force Release Reset

__HAL_RCC_GPIOD_FORCE_RESET
__HAL_RCC_GPIOE_FORCE_RESET
__HAL_RCC_GPIOF_FORCE_RESET
__HAL_RCC_GPIOG_FORCE_RESET
__HAL_RCC_GPIOI_FORCE_RESET
__HAL_RCC_ETHMAC_FORCE_RESET
__HAL_RCC_USB_OTG_HS_FORCE_RESET
__HAL_RCC_GPIOJ_FORCE_RESET
__HAL_RCC_GPIOK_FORCE_RESET
__HAL_RCC_DMA2D_FORCE_RESET
__HAL_RCC_CRC_FORCE_RESET
__HAL_RCC_GPIOD_RELEASE_RESET
__HAL_RCC_GPIOE_RELEASE_RESET
__HAL_RCC_GPIOF_RELEASE_RESET
__HAL_RCC_GPIOG_RELEASE_RESET
__HAL_RCC_GPIOI_RELEASE_RESET

__HAL_RCC_ETHMAC_RELEASE_RESET

__HAL_RCC_USB_OTG_HS_RELEASE_RESET

__HAL_RCC_GPIOJ_RELEASE_RESET

__HAL_RCC_GPIOK_RELEASE_RESET

__HAL_RCC_DMA2D_RELEASE_RESET

__HAL_RCC_CRC_RELEASE_RESET

AHB1 Peripheral Low Power Enable Disable

__HAL_RCC_GPIOD_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOE_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOF_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOG_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOI_CLK_SLEEP_ENABLE

__HAL_RCC_SRAM2_CLK_SLEEP_ENABLE

__HAL_RCC_ETHMAC_CLK_SLEEP_ENABLE

__HAL_RCC_ETHMACTX_CLK_SLEEP_ENABLE

__HAL_RCC_ETHMACRX_CLK_SLEEP_ENABLE

__HAL_RCC_ETHMACPTP_CLK_SLEEP_ENABLE

__HAL_RCC_USB_OTG_HS_CLK_SLEEP_ENABLE

__HAL_RCC_USB_OTG_HS_ULPI_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOJ_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOK_CLK_SLEEP_ENABLE

__HAL_RCC_SRAM3_CLK_SLEEP_ENABLE

__HAL_RCC_DMA2D_CLK_SLEEP_ENABLE

__HAL_RCC_CRC_CLK_SLEEP_ENABLE

__HAL_RCC_FLITF_CLK_SLEEP_ENABLE

__HAL_RCC_SRAM1_CLK_SLEEP_ENABLE

__HAL_RCC_BKPSRAM_CLK_SLEEP_ENABLE

__HAL_RCC_GPIOD_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOE_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOF_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOG_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOI_CLK_SLEEP_DISABLE

__HAL_RCC_SRAM2_CLK_SLEEP_DISABLE

__HAL_RCC_ETHMAC_CLK_SLEEP_DISABLE

__HAL_RCC_ETHMACTX_CLK_SLEEP_DISABLE

__HAL_RCC_ETHMACRX_CLK_SLEEP_DISABLE

__HAL_RCC_ETHMACPTP_CLK_SLEEP_DISABLE

__HAL_RCC_USB_OTG_HS_CLK_SLEEP_DISABLE

__HAL_RCC_USB_OTG_HS_ULPI_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOJ_CLK_SLEEP_DISABLE

__HAL_RCC_GPIOK_CLK_SLEEP_DISABLE

__HAL_RCC_DMA2D_CLK_SLEEP_DISABLE

__HAL_RCC_CRC_CLK_SLEEP_DISABLE

__HAL_RCC_FLITF_CLK_SLEEP_DISABLE

__HAL_RCC_SRAM1_CLK_SLEEP_DISABLE

__HAL_RCC_BKPSRAM_CLK_SLEEP_DISABLE

AHB1 Peripheral Clock Enable Disable Status

__HAL_RCC_GPIOD_IS_CLK_ENABLED

__HAL_RCC_GPIOE_IS_CLK_ENABLED

__HAL_RCC_GPIOF_IS_CLK_ENABLED

__HAL_RCC_GPIOG_IS_CLK_ENABLED

__HAL_RCC_GPIOI_IS_CLK_ENABLED

__HAL_RCC_GPIOJ_IS_CLK_ENABLED

__HAL_RCC_GPIOK_IS_CLK_ENABLED

__HAL_RCC_DMA2D_IS_CLK_ENABLED

__HAL_RCC_ETHMAC_IS_CLK_ENABLED

__HAL_RCC_ETHMACTX_IS_CLK_ENABLED

__HAL_RCC_ETHMACRX_IS_CLK_ENABLED

__HAL_RCC_ETHMACPTP_IS_CLK_ENABLED

__HAL_RCC_USB_OTG_HS_IS_CLK_ENABLED

__HAL_RCC_USB_OTG_HS_ULPI_IS_CLK_ENABLED

__HAL_RCC_BKPSRAM_IS_CLK_ENABLED

__HAL_RCC_CCMDATARAMEN_IS_CLK_ENABLED

__HAL_RCC_CRC_IS_CLK_ENABLED

__HAL_RCC_ETH_IS_CLK_ENABLED

__HAL_RCC_GPIOD_IS_CLK_DISABLED

__HAL_RCC_GPIOE_IS_CLK_DISABLED

__HAL_RCC_GPIOF_IS_CLK_DISABLED

__HAL_RCC_GPIOG_IS_CLK_DISABLED

__HAL_RCC_GPIOI_IS_CLK_DISABLED

__HAL_RCC_GPIOJ_IS_CLK_DISABLED

__HAL_RCC_GPIOK_IS_CLK_DISABLED

__HAL_RCC_DMA2D_IS_CLK_DISABLED

__HAL_RCC_ETHMAC_IS_CLK_DISABLED

__HAL_RCC_ETHMACTX_IS_CLK_DISABLED

__HAL_RCC_ETHMACRX_IS_CLK_DISABLED

__HAL_RCC_ETHMACPTP_IS_CLK_DISABLED

__HAL_RCC_USB_OTG_HS_IS_CLK_DISABLED

__HAL_RCC_USB_OTG_HS_ULPI_IS_CLK_DISABLED

__HAL_RCC_BKPSRAM_IS_CLK_DISABLED

__HAL_RCC_CCMDATARAMEN_IS_CLK_DISABLED

__HAL_RCC_CRC_IS_CLK_DISABLED

__HAL_RCC_ETH_IS_CLK_DISABLED

AHB2 Peripheral Clock Enable Disable

__HAL_RCC_DCMI_CLK_ENABLE

__HAL_RCC_DCMI_CLK_DISABLE

__HAL_RCC_CRYP_CLK_ENABLE

__HAL_RCC_HASH_CLK_ENABLE

__HAL_RCC_CRYP_CLK_DISABLE

__HAL_RCC_HASH_CLK_DISABLE

__HAL_RCC_USB_OTG_FS_CLK_ENABLE

__HAL_RCC_USB_OTG_FS_CLK_DISABLE

__HAL_RCC_RNG_CLK_ENABLE

__HAL_RCC_RNG_CLK_DISABLE

AHB2 Force Release Reset

__HAL_RCC_AHB2_FORCE_RESET

__HAL_RCC_USB_OTG_FS_FORCE_RESET

__HAL_RCC_RNG_FORCE_RESET

__HAL_RCC_DCMI_FORCE_RESET

__HAL_RCC_AHB2_RELEASE_RESET

__HAL_RCC_USB_OTG_FS_RELEASE_RESET

__HAL_RCC_RNG_RELEASE_RESET

__HAL_RCC_DCMI_RELEASE_RESET

__HAL_RCC_CRYP_FORCE_RESET

__HAL_RCC_HASH_FORCE_RESET

__HAL_RCC_CRYP_RELEASE_RESET

__HAL_RCC_HASH_RELEASE_RESET

AHB2 Peripheral Low Power Enable Disable

__HAL_RCC_USB_OTG_FS_CLK_SLEEP_ENABLE

__HAL_RCC_USB_OTG_FS_CLK_SLEEP_DISABLE

__HAL_RCC_RNG_CLK_SLEEP_ENABLE

__HAL_RCC_RNG_CLK_SLEEP_DISABLE

__HAL_RCC_DCMI_CLK_SLEEP_ENABLE

__HAL_RCC_DCMI_CLK_SLEEP_DISABLE

__HAL_RCC_CRYP_CLK_SLEEP_ENABLE

__HAL_RCC_HASH_CLK_SLEEP_ENABLE

__HAL_RCC_CRYP_CLK_SLEEP_DISABLE

__HAL_RCC_HASH_CLK_SLEEP_DISABLE

AHB2 Peripheral Clock Enable Disable Status

__HAL_RCC_DCMI_IS_CLK_ENABLED

__HAL_RCC_DCMI_IS_CLK_DISABLED

__HAL_RCC_CRYP_IS_CLK_ENABLED

__HAL_RCC_CRYP_IS_CLK_DISABLED

__HAL_RCC_HASH_IS_CLK_ENABLED

__HAL_RCC_HASH_IS_CLK_DISABLED

__HAL_RCC_USB_OTG_FS_IS_CLK_ENABLED

__HAL_RCC_USB_OTG_FS_IS_CLK_DISABLED

__HAL_RCC_RNG_IS_CLK_ENABLED

__HAL_RCC_RNG_IS_CLK_DISABLED

AHB3 Peripheral Clock Enable Disable

__HAL_RCC_FMC_CLK_ENABLE

__HAL_RCC_FMC_CLK_DISABLE

__HAL_RCC_QSPI_CLK_ENABLE

__HAL_RCC_QSPI_CLK_DISABLE

AHB3 Force Release Reset

__HAL_RCC_AHB3_FORCE_RESET

__HAL_RCC_AHB3_RELEASE_RESET

__HAL_RCC_FMC_FORCE_RESET

__HAL_RCC_FMC_RELEASE_RESET

__HAL_RCC_QSPI_FORCE_RESET

__HAL_RCC_QSPI_RELEASE_RESET

AHB3 Peripheral Low Power Enable Disable

__HAL_RCC_FMC_CLK_SLEEP_ENABLE

__HAL_RCC_FMC_CLK_SLEEP_DISABLE

__HAL_RCC_QSPI_CLK_SLEEP_ENABLE

__HAL_RCC_QSPI_CLK_SLEEP_DISABLE

AHB3 Peripheral Clock Enable Disable Status

__HAL_RCC_FMC_IS_CLK_ENABLED

__HAL_RCC_FMC_IS_CLK_DISABLED

__HAL_RCC_QSPI_IS_CLK_ENABLED

__HAL_RCC_QSPI_IS_CLK_DISABLED

APB1 Peripheral Clock Enable Disable

__HAL_RCC_TIM6_CLK_ENABLE

__HAL_RCC_TIM7_CLK_ENABLE

__HAL_RCC_TIM12_CLK_ENABLE

__HAL_RCC_TIM13_CLK_ENABLE

__HAL_RCC_TIM14_CLK_ENABLE

__HAL_RCC_TIM14_CLK_ENABLE

__HAL_RCC_USART3_CLK_ENABLE

__HAL_RCC_UART4_CLK_ENABLE

__HAL_RCC_UART5_CLK_ENABLE

__HAL_RCC_CAN1_CLK_ENABLE

__HAL_RCC_CAN2_CLK_ENABLE

__HAL_RCC_DAC_CLK_ENABLE

__HAL_RCC_UART7_CLK_ENABLE

__HAL_RCC_UART8_CLK_ENABLE

__HAL_RCC_TIM2_CLK_ENABLE

__HAL_RCC_TIM3_CLK_ENABLE

__HAL_RCC_TIM4_CLK_ENABLE

__HAL_RCC_SPI3_CLK_ENABLE

__HAL_RCC_I2C3_CLK_ENABLE

__HAL_RCC_TIM2_CLK_DISABLE

__HAL_RCC_TIM3_CLK_DISABLE

__HAL_RCC_TIM4_CLK_DISABLE

__HAL_RCC_SPI3_CLK_DISABLE

__HAL_RCC_I2C3_CLK_DISABLE

__HAL_RCC_TIM6_CLK_DISABLE

__HAL_RCC_TIM7_CLK_DISABLE

__HAL_RCC_TIM12_CLK_DISABLE

__HAL_RCC_TIM13_CLK_DISABLE

__HAL_RCC_TIM14_CLK_DISABLE
__HAL_RCC_USART3_CLK_DISABLE
__HAL_RCC_UART4_CLK_DISABLE
__HAL_RCC_UART5_CLK_DISABLE
__HAL_RCC_CAN1_CLK_DISABLE
__HAL_RCC_CAN2_CLK_DISABLE
__HAL_RCC_DAC_CLK_DISABLE
__HAL_RCC_UART7_CLK_DISABLE
__HAL_RCC_UART8_CLK_DISABLE

APB1 Force Release Reset

__HAL_RCC_TIM6_FORCE_RESET
__HAL_RCC_TIM7_FORCE_RESET
__HAL_RCC_TIM12_FORCE_RESET
__HAL_RCC_TIM13_FORCE_RESET
__HAL_RCC_TIM14_FORCE_RESET
__HAL_RCC_USART3_FORCE_RESET
__HAL_RCC_UART4_FORCE_RESET
__HAL_RCC_UART5_FORCE_RESET
__HAL_RCC_CAN1_FORCE_RESET
__HAL_RCC_CAN2_FORCE_RESET
__HAL_RCC_DAC_FORCE_RESET
__HAL_RCC_UART7_FORCE_RESET
__HAL_RCC_UART8_FORCE_RESET
__HAL_RCC_TIM2_FORCE_RESET
__HAL_RCC_TIM3_FORCE_RESET
__HAL_RCC_TIM4_FORCE_RESET
__HAL_RCC_SPI3_FORCE_RESET
__HAL_RCC_I2C3_FORCE_RESET
__HAL_RCC_TIM2_RELEASE_RESET
__HAL_RCC_TIM3_RELEASE_RESET
__HAL_RCC_TIM4_RELEASE_RESET
__HAL_RCC_SPI3_RELEASE_RESET
__HAL_RCC_I2C3_RELEASE_RESET
__HAL_RCC_TIM6_RELEASE_RESET
__HAL_RCC_TIM7_RELEASE_RESET
__HAL_RCC_TIM12_RELEASE_RESET

__HAL_RCC_TIM13_RELEASE_RESET

__HAL_RCC_TIM14_RELEASE_RESET

__HAL_RCC_USART3_RELEASE_RESET

__HAL_RCC_UART4_RELEASE_RESET

__HAL_RCC_UART5_RELEASE_RESET

__HAL_RCC_CAN1_RELEASE_RESET

__HAL_RCC_CAN2_RELEASE_RESET

__HAL_RCC_DAC_RELEASE_RESET

__HAL_RCC_UART7_RELEASE_RESET

__HAL_RCC_UART8_RELEASE_RESET

APB1 Peripheral Low Power Enable Disable

__HAL_RCC_TIM6_CLK_SLEEP_ENABLE

__HAL_RCC_TIM7_CLK_SLEEP_ENABLE

__HAL_RCC_TIM12_CLK_SLEEP_ENABLE

__HAL_RCC_TIM13_CLK_SLEEP_ENABLE

__HAL_RCC_TIM14_CLK_SLEEP_ENABLE

__HAL_RCC_USART3_CLK_SLEEP_ENABLE

__HAL_RCC_UART4_CLK_SLEEP_ENABLE

__HAL_RCC_UART5_CLK_SLEEP_ENABLE

__HAL_RCC_CAN1_CLK_SLEEP_ENABLE

__HAL_RCC_CAN2_CLK_SLEEP_ENABLE

__HAL_RCC_DAC_CLK_SLEEP_ENABLE

__HAL_RCC_UART7_CLK_SLEEP_ENABLE

__HAL_RCC_UART8_CLK_SLEEP_ENABLE

__HAL_RCC_TIM2_CLK_SLEEP_ENABLE

__HAL_RCC_TIM3_CLK_SLEEP_ENABLE

__HAL_RCC_TIM4_CLK_SLEEP_ENABLE

__HAL_RCC_SPI3_CLK_SLEEP_ENABLE

__HAL_RCC_I2C3_CLK_SLEEP_ENABLE

__HAL_RCC_TIM2_CLK_SLEEP_DISABLE

__HAL_RCC_TIM3_CLK_SLEEP_DISABLE

__HAL_RCC_TIM4_CLK_SLEEP_DISABLE

__HAL_RCC_SPI3_CLK_SLEEP_DISABLE

__HAL_RCC_I2C3_CLK_SLEEP_DISABLE

__HAL_RCC_TIM6_CLK_SLEEP_DISABLE

__HAL_RCC_TIM7_CLK_SLEEP_DISABLE

__HAL_RCC_TIM12_CLK_SLEEP_DISABLE

__HAL_RCC_TIM13_CLK_SLEEP_DISABLE

__HAL_RCC_TIM14_CLK_SLEEP_DISABLE

__HAL_RCC_USART3_CLK_SLEEP_DISABLE

__HAL_RCC_UART4_CLK_SLEEP_DISABLE

__HAL_RCC_UART5_CLK_SLEEP_DISABLE

__HAL_RCC_CAN1_CLK_SLEEP_DISABLE

__HAL_RCC_CAN2_CLK_SLEEP_DISABLE

__HAL_RCC_DAC_CLK_SLEEP_DISABLE

__HAL_RCC_UART7_CLK_SLEEP_DISABLE

__HAL_RCC_UART8_CLK_SLEEP_DISABLE

APB1 Peripheral Clock Enable Disable Status

__HAL_RCC_TIM2_IS_CLK_ENABLED

__HAL_RCC_TIM3_IS_CLK_ENABLED

__HAL_RCC_TIM4_IS_CLK_ENABLED

__HAL_RCC_SPI3_IS_CLK_ENABLED

__HAL_RCC_I2C3_IS_CLK_ENABLED

__HAL_RCC_TIM6_IS_CLK_ENABLED

__HAL_RCC_TIM7_IS_CLK_ENABLED

__HAL_RCC_TIM12_IS_CLK_ENABLED

__HAL_RCC_TIM13_IS_CLK_ENABLED

__HAL_RCC_TIM14_IS_CLK_ENABLED

__HAL_RCC_USART3_IS_CLK_ENABLED

__HAL_RCC_UART4_IS_CLK_ENABLED

__HAL_RCC_UART5_IS_CLK_ENABLED

__HAL_RCC_CAN1_IS_CLK_ENABLED

__HAL_RCC_CAN2_IS_CLK_ENABLED

__HAL_RCC_DAC_IS_CLK_ENABLED

__HAL_RCC_UART7_IS_CLK_ENABLED

__HAL_RCC_UART8_IS_CLK_ENABLED

__HAL_RCC_TIM2_IS_CLK_DISABLED

__HAL_RCC_TIM3_IS_CLK_DISABLED

__HAL_RCC_TIM4_IS_CLK_DISABLED

__HAL_RCC_SPI3_IS_CLK_DISABLED

__HAL_RCC_I2C3_IS_CLK_DISABLED

__HAL_RCC_TIM6_IS_CLK_DISABLED

__HAL_RCC_TIM7_IS_CLK_DISABLED

__HAL_RCC_TIM12_IS_CLK_DISABLED

__HAL_RCC_TIM13_IS_CLK_DISABLED

__HAL_RCC_TIM14_IS_CLK_DISABLED

__HAL_RCC_USART3_IS_CLK_DISABLED

__HAL_RCC_UART4_IS_CLK_DISABLED

__HAL_RCC_UART5_IS_CLK_DISABLED

__HAL_RCC_CAN1_IS_CLK_DISABLED

__HAL_RCC_CAN2_IS_CLK_DISABLED

__HAL_RCC_DAC_IS_CLK_DISABLED

__HAL_RCC_UART7_IS_CLK_DISABLED

__HAL_RCC_UART8_IS_CLK_DISABLED

APB2 Peripheral Clock Enable Disable

__HAL_RCC_TIM8_CLK_ENABLE

__HAL_RCC_ADC2_CLK_ENABLE

__HAL_RCC_ADC3_CLK_ENABLE

__HAL_RCC_SPI5_CLK_ENABLE

__HAL_RCC_SPI6_CLK_ENABLE

__HAL_RCC_SAI1_CLK_ENABLE

__HAL_RCC_SDIO_CLK_ENABLE

__HAL_RCC_SPI4_CLK_ENABLE

__HAL_RCC_TIM10_CLK_ENABLE

__HAL_RCC_SDIO_CLK_DISABLE

__HAL_RCC_SPI4_CLK_DISABLE

__HAL_RCC_TIM10_CLK_DISABLE

__HAL_RCC_TIM8_CLK_DISABLE

__HAL_RCC_ADC2_CLK_DISABLE

__HAL_RCC_ADC3_CLK_DISABLE

__HAL_RCC_SPI5_CLK_DISABLE

__HAL_RCC_SPI6_CLK_DISABLE

__HAL_RCC_SAI1_CLK_DISABLE

__HAL_RCC_LTDC_CLK_ENABLE

__HAL_RCC_LTDC_CLK_DISABLE

__HAL_RCC_DSI_CLK_ENABLE

__HAL_RCC_DSI_CLK_DISABLE

APB2 Force Release Reset

__HAL_RCC_TIM8_FORCE_RESET

__HAL_RCC_SPI5_FORCE_RESET

__HAL_RCC_SPI6_FORCE_RESET

__HAL_RCC_SAI1_FORCE_RESET

__HAL_RCC_SDIO_FORCE_RESET

__HAL_RCC_SPI4_FORCE_RESET

__HAL_RCC_TIM10_FORCE_RESET

__HAL_RCC_SDIO_RELEASE_RESET

__HAL_RCC_SPI4_RELEASE_RESET

__HAL_RCC_TIM10_RELEASE_RESET

__HAL_RCC_TIM8_RELEASE_RESET

__HAL_RCC_SPI5_RELEASE_RESET

__HAL_RCC_SPI6_RELEASE_RESET

__HAL_RCC_SAI1_RELEASE_RESET

__HAL_RCC_LTDC_FORCE_RESET

__HAL_RCC_LTDC_RELEASE_RESET

__HAL_RCC_DSI_FORCE_RESET

__HAL_RCC_DSI_RELEASE_RESET

APB2 Peripheral Low Power Enable Disable

__HAL_RCC_TIM8_CLK_SLEEP_ENABLE

__HAL_RCC_ADC2_CLK_SLEEP_ENABLE

__HAL_RCC_ADC3_CLK_SLEEP_ENABLE

__HAL_RCC_SPI5_CLK_SLEEP_ENABLE

__HAL_RCC_SPI6_CLK_SLEEP_ENABLE

__HAL_RCC_SAI1_CLK_SLEEP_ENABLE

__HAL_RCC_SDIO_CLK_SLEEP_ENABLE

__HAL_RCC_SPI4_CLK_SLEEP_ENABLE

__HAL_RCC_TIM10_CLK_SLEEP_ENABLE

__HAL_RCC_SDIO_CLK_SLEEP_DISABLE

__HAL_RCC_SPI4_CLK_SLEEP_DISABLE

__HAL_RCC_TIM10_CLK_SLEEP_DISABLE

__HAL_RCC_TIM8_CLK_SLEEP_DISABLE

__HAL_RCC_ADC2_CLK_SLEEP_DISABLE

__HAL_RCC_ADC3_CLK_SLEEP_DISABLE

__HAL_RCC_SPI5_CLK_SLEEP_DISABLE

__HAL_RCC_SPI6_CLK_SLEEP_DISABLE

__HAL_RCC_SAI1_CLK_SLEEP_DISABLE

__HAL_RCC_LTDC_CLK_SLEEP_ENABLE

__HAL_RCC_LTDC_CLK_SLEEP_DISABLE

__HAL_RCC_DSI_CLK_SLEEP_ENABLE

__HAL_RCC_DSI_CLK_SLEEP_DISABLE

APB2 Peripheral Clock Enable Disable Status

__HAL_RCC_TIM8_IS_CLK_ENABLED

__HAL_RCC_ADC2_IS_CLK_ENABLED

__HAL_RCC_ADC3_IS_CLK_ENABLED

__HAL_RCC_SPI5_IS_CLK_ENABLED

__HAL_RCC_SPI6_IS_CLK_ENABLED

__HAL_RCC_SAI1_IS_CLK_ENABLED

__HAL_RCC_SDIO_IS_CLK_ENABLED

__HAL_RCC_SPI4_IS_CLK_ENABLED

__HAL_RCC_TIM10_IS_CLK_ENABLED

__HAL_RCC_SDIO_IS_CLK_DISABLED

__HAL_RCC_SPI4_IS_CLK_DISABLED

__HAL_RCC_TIM10_IS_CLK_DISABLED

__HAL_RCC_TIM8_IS_CLK_DISABLED

__HAL_RCC_ADC2_IS_CLK_DISABLED

__HAL_RCC_ADC3_IS_CLK_DISABLED

__HAL_RCC_SPI5_IS_CLK_DISABLED

__HAL_RCC_SPI6_IS_CLK_DISABLED

__HAL_RCC_SAI1_IS_CLK_DISABLED

__HAL_RCC_LTDC_IS_CLK_ENABLED

__HAL_RCC_LTDC_IS_CLK_DISABLED

__HAL_RCC_DSI_IS_CLK_ENABLED

__HAL_RCC_DSI_IS_CLK_DISABLED

RCC BitAddress AliasRegion

RCC_PLLSAION_BIT_NUMBER

RCC_CR_PLLSAION_BB

PLLSAI_TIMEOUT_VALUE

RCC_PLLI2SON_BIT_NUMBER

RCC_CR_PLLI2SON_BB

RCC_DCKCFGR_OFFSET

RCC_TIMPRE_BIT_NUMBER

RCC_DCKCFGR_TIMPRE_BB

RCC_CFGR_OFFSET

RCC_I2SSRC_BIT_NUMBER

RCC_CFGR_I2SSRC_BB

PLLI2S_TIMEOUT_VALUE

PLL_TIMEOUT_VALUE

RCC CLK48 Clock Source

RCC_CLK48CLKSOURCE_PLLQ

RCC_CLK48CLKSOURCE_PLLSAIP

RCC DSI Clock Source

RCC_DSICLKSOURCE_DSIPHY

RCC_DSICLKSOURCE_PLLR

RCCEx Exported Macros

__HAL_RCC_PLL_CONFIG

Description:

- Macro to configure the main PLL clock source, multiplication and division factors.

Parameters:

- __RCC_PLLSource__: specifies the PLL entry clock source. This parameter can be one of the following values:
  - RCC_PLLSOURCE_HSI: HSI oscillator clock selected as PLL clock entry
  - RCC_PLLSOURCE_HSE: HSE oscillator clock selected as PLL clock entry
- __PLLM__: specifies the division factor for PLL VCO input clock This parameter must be a number between Min_Data = 2 and Max_Data = 63.
- __PLLN__: specifies the multiplication factor for PLL VCO output clock This parameter must be a number between Min_Data = 50 and Max_Data = 432.
- __PLLP__: specifies the division factor for main system clock (SYSCLK) This parameter must be a number in the range {2, 4, 6, or 8}.
- __PLLQ__: specifies the division factor for OTG FS, SDIO and RNG clocks This parameter must be a number between Min_Data = 2 and Max_Data = 15.
- __PLLR__: PLL division factor for I2S, SAI, SYSTEM, SPDIFRX clocks. This parameter must be a number between

Min_Data = 2 and Max_Data = 7.

Notes:

- This function must be used only when the main PLL is disabled.
- This clock source (RCC_PLLSource) is common for the main PLL and PLLI2S.
- You have to set the PLLM parameter correctly to ensure that the VCO input frequency ranges from 1 to 2 MHz. It is recommended to select a frequency of 2 MHz to limit PLL jitter.
- You have to set the PLLN parameter correctly to ensure that the VCO output frequency is between 100 and 432 MHz.
- If the USB OTG FS is used in your application, you have to set the PLLQ parameter correctly to have 48 MHz clock for the USB. However, the SDIO and RNG need a frequency lower than or equal to 48 MHz to work correctly.
- This parameter is only available in STM32F446xx/STM32F469xx/STM32F479xx/ STM32F412Zx/STM32F412Vx/STM32F412Rx/STM32F412Cx devices.

__HAL_RCC_PLLI2S_ENABLE

Notes:

- The PLLI2S is disabled by hardware when entering STOP and STANDBY modes.

__HAL_RCC_PLLI2S_DISABLE

__HAL_RCC_PLLI2S_CONFIG

Description:

- Macro to configure the PLLI2S clock multiplication and division factors .

Parameters:

- __PLLI2SN__: specifies the multiplication factor for PLLI2S VCO output clock This parameter must be a number between Min_Data = 50 and Max_Data = 432.
- __PLLI2SR__: specifies the division factor for I2S clock This parameter must be a number between Min_Data = 2 and Max_Data = 7.

Notes:

- This macro must be used only when the PLLI2S is disabled. PLLI2S clock source is common with the main PLL (configured in HAL_RCC_ClockConfig() API).
- You have to set the PLLI2SN parameter correctly to ensure that the VCO output frequency is between Min_Data = 100 and

Max_Data = 432 MHz.
- You have to set the PLLI2SR parameter correctly to not exceed 192 MHz on the I2S clock frequency.

__HAL_RCC_PLLI2S_SAICLK_CONFIG

Description:

- Macro used by the SAI HAL driver to configure the PLLI2S clock multiplication and division factors.

Parameters:

- __PLLI2SN__: specifies the multiplication factor for PLLI2S VCO output clock. This parameter must be a number between Min_Data = 50 and Max_Data = 432.
- __PLLI2SQ__: specifies the division factor for SAI1 clock. This parameter must be a number between Min_Data = 2 and Max_Data = 15.
- __PLLI2SR__: specifies the division factor for I2S clock This parameter must be a number between Min_Data = 2 and Max_Data = 7.

Notes:

- This macro must be used only when the PLLI2S is disabled. PLLI2S clock source is common with the main PLL (configured in HAL_RCC_ClockConfig() API)
- You have to set the PLLI2SN parameter correctly to ensure that the VCO output frequency is between Min_Data = 100 and Max_Data = 432 MHz.
- the PLLI2SQ parameter is only available with STM32F427xx/437xx/429xx/439xx/469xx/479xx Devices and can be configured using the __HAL_RCC_PLLI2S_PLLSAICLK_CONFIG() macro
- You have to set the PLLI2SR parameter correctly to not exceed 192 MHz on the I2S clock frequency.

__HAL_RCC_PLLSAI_ENABLE

Notes:

- The PLLSAI is only available with STM32F429x/439x Devices. The PLLSAI is disabled by hardware when entering STOP and STANDBY modes.

__HAL_RCC_PLLSAI_DISABLE

__HAL_RCC_PLLSAI_CONFIG

Description:

- Macro to configure the PLLSAI clock multiplication and division factors.

Parameters:

- __PLLSAIN__: specifies the multiplication factor for PLLSAI VCO output clock. This parameter must be a number between Min_Data = 50 and Max_Data = 432.
- __PLLSAIP__: specifies division factor for SDIO and CLK48 clocks. This parameter must be a number in the range {2, 4, 6, or 8}.
- __PLLSAIQ__: specifies the division factor for SAI clock This parameter must be a number between Min_Data = 2 and Max_Data = 15.
- __PLLSAIR__: specifies the division factor for LTDC clock This parameter must be a number between Min_Data = 2 and Max_Data = 7.

Notes:

- You have to set the PLLSAIN parameter correctly to ensure that the VCO output frequency is between Min_Data = 100 and Max_Data = 432 MHz.

__HAL_RCC_PLLI2S_PLLSAICLKDIVQ_CONFIG

Description:

- Macro to configure the SAI clock Divider coming from PLLI2S.

Parameters:

- __PLLI2SDivQ__: specifies the PLLI2S division factor for SAI1 clock. This parameter must be a number between 1 and 32. SAI1 clock frequency = f(PLLI2SQ) / __PLLI2SDivQ__

Notes:

- This function must be called before enabling the PLLI2S.

__HAL_RCC_PLLSAI_PLLSAICLKDIVQ_CONFIG

Description:

- Macro to configure the SAI clock Divider coming from PLLSAI.

Parameters:

- __PLLSAIDivQ__: specifies the PLLSAI division factor for SAI1 clock . This parameter must be a number between Min_Data = 1 and Max_Data = 32. SAI1 clock frequency = f(PLLSAIQ) / __PLLSAIDivQ__

Notes:

- This function must be called before enabling the PLLSAI.

__HAL_RCC_PLLSAI_PLLSAICLKDIVR_CONFIG

Description:

- Macro to configure the LTDC clock Divider coming from PLLSAI.

Parameters:

- __PLLSAIDivR__: specifies the PLLSAI division factor for LTDC clock . This parameter must be a number between Min_Data = 2 and Max_Data = 16. LTDC clock frequency = f(PLLSAIR) / __PLLSAIDivR__

Notes:

- The LTDC peripheral is only available with STM32F427/437/429/439/469/479xx Devices. This function must be called before enabling the PLLSAI.

__HAL_RCC_I2S_CONFIG

Description:

- Macro to configure the I2S clock source (I2SCLK).

Parameters:

- __SOURCE__: specifies the I2S clock source. This parameter can be one of the following values:
  - RCC_I2SCLKSOURCE_PLLI2S: PLLI2S clock used as I2S clock source.
  - RCC_I2SCLKSOURCE_EXT: External clock mapped on the I2S_CKIN pin used as I2S clock source.

Notes:

- This function must be called before enabling the I2S APB clock.

__HAL_RCC_SAI_BLOCKACLKSOURCE_CONFIG

Description:

- Macro to configure SAI1BlockA clock source selection.

Parameters:

- __SOURCE__: specifies the SAI Block A clock source. This parameter can be one of the following values:
  - RCC_SAIACLKSOURCE_PLLI2S: PLLI2S_Q clock divided by PLLI2SDIVQ used as SAI1 Block A clock.
  - RCC_SAIACLKSOURCE_PLLSAI: PLLISAI_Q clock divided by PLLSAIDIVQ used as SAI1 Block A clock.

– RCC_SAIACLKSOURCE_Ext: External clock mapped on the I2S_CKIN pin used as SAI1 Block A clock.

Notes:

- The SAI peripheral is only available with STM32F427/437/429/439/469/479xx Devices. This function must be called before enabling PLLSAI, PLLI2S and the SAI clock.

__HAL_RCC_SAI_BLOCKBCLKSOURCE_CONFIG

Description:

- Macro to configure SAI1BlockB clock source selection.

Parameters:

- __SOURCE__: specifies the SAI Block B clock source. This parameter can be one of the following values:
  – RCC_SAIBCLKSOURCE_PLLI2S: PLLI2S_Q clock divided by PLLI2SDIVQ used as SAI1 Block B clock.
  – RCC_SAIBCLKSOURCE_PLLSAI: PLLISAI_Q clock divided by PLLSAIDIVQ used as SAI1 Block B clock.
  – RCC_SAIBCLKSOURCE_Ext: External clock mapped on the I2S_CKIN pin used as SAI1 Block B clock.

Notes:

- The SAI peripheral is only available with STM32F427/437/429/439/469/479xx Devices. This function must be called before enabling PLLSAI, PLLI2S and the SAI clock.

__HAL_RCC_CLK48_CONFIG

Description:

- Macro to configure the CLK48 clock.

Parameters:

- __SOURCE__: specifies the CLK48 clock source. This parameter can be one of the following values:
  – RCC_CLK48CLKSOURCE_PLLQ: PLL VCO Output divided by PLLQ used as CLK48 clock.
  – RCC_CLK48CLKSOURCE_PLLSAIP: PLLSAI VCO Output divided by PLLSAIP used as CLK48 clock.

__HAL_RCC_GET_CLK48_SOURCE

Description:

- Macro to Get the CLK48 clock.

Return value:

- The: clock source can be one of the following values:
  - RCC_CLK48CLKSOURCE_PLLQ: PLL VCO Output divided by PLLQ used as CLK48 clock.
  - RCC_CLK48CLKSOURCE_PLLSAIP: PLLSAI VCO Output divided by PLLSAIP used as CLK48 clock.

__HAL_RCC_SDIO_CONFIG

Description:

- Macro to configure the SDIO clock.

Parameters:

- __SOURCE__: specifies the SDIO clock source. This parameter can be one of the following values:
  - RCC_SDIOCLKSOURCE_CLK48: CLK48 output used as SDIO clock.
  - RCC_SDIOCLKSOURCE_SYSCLK: System clock output used as SDIO clock.

__HAL_RCC_GET_SDIO_SOURCE

Description:

- Macro to Get the SDIO clock.

Return value:

- The: clock source can be one of the following values:
  - RCC_SDIOCLKSOURCE_CLK48: CLK48 output used as SDIO clock.
  - RCC_SDIOCLKSOURCE_SYSCLK: System clock output used as SDIO clock.

__HAL_RCC_DSI_CONFIG

Description:

- Macro to configure the DSI clock.

Parameters:

- __SOURCE__: specifies the DSI clock source. This parameter can be one of the following values:
  - RCC_DSICLKSOURCE_PLLR: PLLR output used as DSI clock.
  - RCC_DSICLKSOURCE_DSIPHY: DSI-PHY output used as DSI clock.

__HAL_RCC_GET_DSI_SOURCE

Description:

- Macro to Get the DSI clock.

Return value:

- The: clock source can be one of the following values:
  - RCC_DSICLKSOURCE_PLLR: PLLR output used as DSI clock.
  - RCC_DSICLKSOURCE_DSIPHY: DSI-PHY output used as DSI clock.

__HAL_RCC_TIMCLKPRESCALER

Description:

- Macro to configure the Timers clocks prescalers.

Parameters:

- __PRESC__: : specifies the Timers clocks prescalers selection This parameter can be one of the following values:
  - RCC_TIMPRES_DESACTIVATED: The Timers kernels clocks prescaler is equal to HPRE if PPREx is corresponding to division by 1 or 2, else it is equal to [(HPRE * PPREx) / 2] if PPREx is corresponding to division by 4 or more.
  - RCC_TIMPRES_ACTIVATED: The Timers kernels clocks prescaler is equal to HPRE if PPREx is corresponding to division by 1, 2 or 4, else it is equal to [(HPRE * PPREx) / 4] if PPREx is corresponding to division by 8 or more.

Notes:

- This feature is only available with STM32F429x/439x Devices.

__HAL_RCC_PLLSAI_ENABLE_IT

__HAL_RCC_PLLSAI_DISABLE_IT

__HAL_RCC_PLLSAI_CLEAR_IT

__HAL_RCC_PLLSAI_GET_IT

Description:

- Check the PLLSAI RDY interrupt has occurred or not.

Return value:

- The: new state (TRUE or FALSE).

__HAL_RCC_PLLSAI_GET_FLAG

Description:

- Check PLLSAI RDY flag is set or not.

Return value:

- The: new state (TRUE or FALSE).

I2S Clock Source

RCC_I2SCLKSOURCE_PLLI2S

RCC_I2SCLKSOURCE_EXT

RCC Private macros to check input parameters

IS_RCC_PLLN_VALUE

IS_RCC_PLLI2SN_VALUE

IS_RCC_PERIPHCLOCK

IS_RCC_PLLI2SR_VALUE

IS_RCC_PLLI2SQ_VALUE

IS_RCC_PLLSAIN_VALUE

IS_RCC_PLLSAIQ_VALUE

IS_RCC_PLLSAIR_VALUE

IS_RCC_PLLSAI_DIVQ_VALUE

IS_RCC_PLLI2S_DIVQ_VALUE

IS_RCC_PLLSAI_DIVR_VALUE

IS_RCC_PLLR_VALUE

IS_RCC_PLLSAIP_VALUE

IS_RCC_CLK48CLKSOURCE

IS_RCC_SDIOCLKSOURCE

IS_RCC_DSIBYTELANECLKSOURCE

IS_RCC_LSE_MODE

IS_RCC_MCO2SOURCE

RCC LSE Dual Mode Selection

RCC_LSE_LOWPOWER_MODE

RCC_LSE_HIGHDRIVE_MODE

RCC Extended MCOx Clock Config

__HAL_RCC_MCO1_CONFIG **Description:**

- Macro to configure the MCO1 clock.

Parameters:

- __MCOCLKSOURCE__: specifies the MCO clock source. This parameter can be one of the following values:
  - RCC_MCO1SOURCE_HSI: HSI clock selected as MCO1 source
  - RCC_MCO1SOURCE_LSE: LSE clock selected as MCO1 source
  - RCC_MCO1SOURCE_HSE: HSE clock selected as MCO1 source
  - RCC_MCO1SOURCE_PLLCLK: main PLL clock selected as MCO1 source

- __MCODIV__: specifies the MCO clock prescaler. This parameter can be one of the following values:
  - RCC_MCODIV_1: no division applied to MCOx clock
  - RCC_MCODIV_2: division by 2 applied to MCOx clock
  - RCC_MCODIV_3: division by 3 applied to MCOx clock
  - RCC_MCODIV_4: division by 4 applied to MCOx clock
  - RCC_MCODIV_5: division by 5 applied to MCOx clock

__HAL_RCC_MCO2_CONFIG

Description:

- Macro to configure the MCO2 clock.

Parameters:

- __MCOCLKSOURCE__: specifies the MCO clock source. This parameter can be one of the following values:
  - RCC_MCO2SOURCE_SYSCLK: System clock (SYSCLK) selected as MCO2 source
  - RCC_MCO2SOURCE_PLLI2SCLK: PLLI2S clock selected as MCO2 source, available for all STM32F4 devices except STM32F410xx
  - RCC_MCO2SOURCE_I2SCLK: I2SCLK clock selected as MCO2 source, available only for STM32F410Rx devices
  - RCC_MCO2SOURCE_HSE: HSE clock selected as MCO2 source
  - RCC_MCO2SOURCE_PLLCLK: main PLL clock selected as MCO2 source
- __MCODIV__: specifies the MCO clock prescaler. This parameter can be one of the following values:
  - RCC_MCODIV_1: no division applied to MCOx clock
  - RCC_MCODIV_2: division by 2 applied to MCOx clock
  - RCC_MCODIV_3: division by 3 applied to MCOx clock
  - RCC_MCODIV_4: division by 4 applied to MCOx clock
  - RCC_MCODIV_5: division by 5 applied to MCOx clock

Notes:

- For STM32F410Rx devices, to output I2SCLK clock on MCO2, you should have at least one of the SPI clocks enabled (SPI1, SPI2 or SPI5).

RCC Periph Clock Selection

RCC_PERIPHCLK_I2S

RCC_PERIPHCLK_SAI_PLLI2S

RCC_PERIPHCLK_SAI_PLLSAI

RCC_PERIPHCLK_LTDC

RCC_PERIPHCLK_TIM

RCC_PERIPHCLK_RTC

RCC_PERIPHCLK_PLLI2S

RCC_PERIPHCLK_CLK48

RCC_PERIPHCLK_SDIO

RCC PLLSAIP Clock Divider

RCC_PLLSAIP_DIV2

RCC_PLLSAIP_DIV4

RCC_PLLSAIP_DIV6

RCC_PLLSAIP_DIV8

RCC PLLSAI DIVR

RCC_PLLSAIDIVR_2

RCC_PLLSAIDIVR_4

RCC_PLLSAIDIVR_8

RCC_PLLSAIDIVR_16

RCC SAI BlockA Clock Source

RCC_SAIACLKSOURCE_PLLSAI

RCC_SAIACLKSOURCE_PLLI2S

RCC_SAIACLKSOURCE_EXT

RCC SAI BlockB Clock Source

RCC_SAIBCLKSOURCE_PLLSAI

RCC_SAIBCLKSOURCE_PLLI2S

RCC_SAIBCLKSOURCE_EXT

RCC SDIO Clock Source

RCC_SDIOCLKSOURCE_CLK48

RCC_SDIOCLKSOURCE_SYSCLK

RCC TIM PRescaler Selection

RCC_TIMPRES_DESACTIVATED

RCC_TIMPRES_ACTIVATED

51 HAL RNG Generic Driver

51.1 RNG Firmware driver registers structures

51.1.1 RNG_HandleTypeDef

Data Fields

- ***RNG_TypeDef * Instance***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_RNG_StateTypeDef State***
- ***uint32_t RandomNumber***

Field Documentation

- ***RNG_TypeDef* RNG_HandleTypeDef::Instance***
  Register base address
- ***HAL_LockTypeDef RNG_HandleTypeDef::Lock***
  RNG locking object
- ***__IO HAL_RNG_StateTypeDef RNG_HandleTypeDef::State***
  RNG communication state
- ***uint32_t RNG_HandleTypeDef::RandomNumber***
  Last Generated RNG Data

51.2 RNG Firmware driver API description

51.2.1 How to use this driver

The RNG HAL driver can be used as follows:

1. Enable the RNG controller clock using __HAL_RCC_RNG_CLK_ENABLE() macro in HAL_RNG_MspInit().
2. Activate the RNG peripheral using HAL_RNG_Init() function.
3. Wait until the 32 bit Random Number Generator contains a valid random data using (polling/interrupt) mode.
4. Get the 32 bit random number using HAL_RNG_GenerateRandomNumber() function.

51.2.2 Initialization and de-initialization functions

This section provides functions allowing to:

- Initialize the RNG according to the specified parameters in the RNG_InitTypeDef and create the associated handle
- DeInitialize the RNG peripheral
- Initialize the RNG MSP
- DeInitialize RNG MSP

This section contains the following APIs:

- ***HAL_RNG_Init()***
- ***HAL_RNG_DeInit()***

- ***HAL_RNG_MspInit()***
- ***HAL_RNG_MspDeInit()***

51.2.3 Peripheral Control functions

This section provides functions allowing to:

- Get the 32 bit Random number
- Get the 32 bit Random number with interrupt enabled
- Handle RNG interrupt request

This section contains the following APIs:

- ***HAL_RNG_GenerateRandomNumber()***
- ***HAL_RNG_GenerateRandomNumber_IT()***
- ***HAL_RNG_IRQHandler()***
- ***HAL_RNG_GetRandomNumber()***
- ***HAL_RNG_GetRandomNumber_IT()***
- ***HAL_RNG_ReadLastRandomNumber()***
- ***HAL_RNG_ReadyDataCallback()***
- ***HAL_RNG_ErrorCallback()***

51.2.4 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_RNG_GetState()***

51.2.5 Detailed description of functions

HAL_RNG_Init

Function Name	**HAL_StatusTypeDef HAL_RNG_Init (RNG_HandleTypeDef * hrng)**
Function Description	Initializes the RNG peripheral and creates the associated handle.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **HAL:** status

HAL_RNG_DeInit

Function Name	**HAL_StatusTypeDef HAL_RNG_DeInit (RNG_HandleTypeDef * hrng)**
Function Description	DeInitializes the RNG peripheral.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **HAL:** status

HAL_RNG_MspInit

Function Name	**void HAL_RNG_MspInit (RNG_HandleTypeDef * hrng)**

Function Description	Initializes the RNG MSP.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **None**

HAL_RNG_MspDeInit

Function Name	**void HAL_RNG_MspDeInit (RNG_HandleTypeDef * hrng)**
Function Description	DeInitializes the RNG MSP.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **None**

HAL_RNG_GetRandomNumber

Function Name	**uint32_t HAL_RNG_GetRandomNumber (RNG_HandleTypeDef * hrng)**
Function Description	Returns generated random number in polling mode (Obsolete) Use HAL_RNG_GenerateRandomNumber() API instead.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **Random:** value

HAL_RNG_GetRandomNumber_IT

Function Name	**uint32_t HAL_RNG_GetRandomNumber_IT (RNG_HandleTypeDef * hrng)**
Function Description	Returns a 32-bit random number with interrupt enabled (Obsolete), Use HAL_RNG_GenerateRandomNumber_IT() API instead.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **32-bit:** random number

HAL_RNG_GenerateRandomNumber

Function Name	**HAL_StatusTypeDef HAL_RNG_GenerateRandomNumber (RNG_HandleTypeDef * hrng, uint32_t * random32bit)**
Function Description	Generates a 32-bit random number.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG. • **random32bit:** pointer to generated random number variable if successful.
Return values	• **HAL:** status
Notes	• Each time the random number data is read the RNG_FLAG_DRDY flag is automatically cleared.

HAL_RNG_GenerateRandomNumber_IT

Function Name	**HAL_StatusTypeDef HAL_RNG_GenerateRandomNumber_IT (RNG_HandleTypeDef * hrng)**
Function Description	Generates a 32-bit random number in interrupt mode.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **HAL:** status

HAL_RNG_ReadLastRandomNumber

Function Name	**uint32_t HAL_RNG_ReadLastRandomNumber (RNG_HandleTypeDef * hrng)**
Function Description	Read latest generated random number.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **random:** value

HAL_RNG_IRQHandler

Function Name	**void HAL_RNG_IRQHandler (RNG_HandleTypeDef * hrng)**
Function Description	Handles RNG interrupt request.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
Return values	• **None**
Notes	• In the case of a clock error, the RNG is no more able to generate random numbers because the PLL48CLK clock is not correct. User has to check that the clock controller is correctly configured to provide the RNG clock and clear the CEIS bit using __HAL_RNG_CLEAR_IT(). The clock error has no impact on the previously generated random numbers, and the RNG_DR register contents can be used. • In the case of a seed error, the generation of random numbers is interrupted as long as the SECS bit is '1'. If a number is available in the RNG_DR register, it must not be used because it may not have enough entropy. In this case, it is recommended to clear the SEIS bit using __HAL_RNG_CLEAR_IT(), then disable and enable the RNG peripheral to reinitialize and restart the RNG. • User-written HAL_RNG_ErrorCallback() API is called once whether SEIS or CEIS are set.

HAL_RNG_ErrorCallback

Function Name	**void HAL_RNG_ErrorCallback (RNG_HandleTypeDef * hrng)**
Function Description	RNG error callbacks.
Parameters	• **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.

Return values
- **None**

HAL_RNG_ReadyDataCallback

Function Name **void HAL_RNG_ReadyDataCallback (RNG_HandleTypeDef * hrng, uint32_t random32bit)**

Function Description Data Ready callback in non-blocking mode.

Parameters
- **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.
- **random32bit:** generated random number.

Return values
- **None**

HAL_RNG_GetState

Function Name **HAL_RNG_StateTypeDef HAL_RNG_GetState (RNG_HandleTypeDef * hrng)**

Function Description Returns the RNG state.

Parameters
- **hrng:** pointer to a RNG_HandleTypeDef structure that contains the configuration information for RNG.

Return values
- **HAL:** state

51.3 RNG Firmware driver defines

51.3.1 RNG

RNG Interrupt definition

RNG_IT_DRDY Data Ready interrupt

RNG_IT_CEI Clock error interrupt

RNG_IT_SEI Seed error interrupt

RNG Flag definition

RNG_FLAG_DRDY Data ready

RNG_FLAG_CECS Clock error current status

RNG_FLAG_SECS Seed error current status

RNG Exported Macros

__HAL_RNG_RESET_HANDLE_STATE **Description:**
- Reset RNG handle state.

Parameters:
- __HANDLE__: RNG Handle

Return value:
- None

__HAL_RNG_ENABLE **Description:**
- Enables the RNG peripheral.

Parameters:

- __HANDLE__: RNG Handle

Return value:

- None

__HAL_RNG_DISABLE

Description:

- Disables the RNG peripheral.

Parameters:

- __HANDLE__: RNG Handle

Return value:

- None

__HAL_RNG_GET_FLAG

Description:

- Check the selected RNG flag status.

Parameters:

- __HANDLE__: RNG Handle
- __FLAG__: RNG flag This parameter can be one of the following values:
  - RNG_FLAG_DRDY: Data ready
  - RNG_FLAG_CECS: Clock error current status
  - RNG_FLAG_SECS: Seed error current status

Return value:

- The: new state of __FLAG__ (SET or RESET).

__HAL_RNG_CLEAR_FLAG

Description:

- Clears the selected RNG flag status.

Parameters:

- __HANDLE__: RNG handle
- __FLAG__: RNG flag to clear

Return value:

- None

Notes:

- WARNING: This is a dummy macro for HAL code alignment, flags RNG_FLAG_DRDY, RNG_FLAG_CECS and RNG_FLAG_SECS are read-only.

__HAL_RNG_ENABLE_IT

Description:

- Enables the RNG interrupts.

Parameters:

- __HANDLE__: RNG Handle

Return value:

- None

__HAL_RNG_DISABLE_IT

Description:

- Disables the RNG interrupts.

Parameters:

- __HANDLE__: RNG Handle

Return value:

- None

__HAL_RNG_GET_IT

Description:

- Checks whether the specified RNG interrupt has occurred or not.

Parameters:

- __HANDLE__: RNG Handle
- __INTERRUPT__: specifies the RNG interrupt status flag to check. This parameter can be one of the following values:
  - RNG_IT_DRDY: Data ready interrupt
  - RNG_IT_CEI: Clock error interrupt
  - RNG_IT_SEI: Seed error interrupt

Return value:

- The: new state of __INTERRUPT__ (SET or RESET).

__HAL_RNG_CLEAR_IT

Description:

- Clear the RNG interrupt status flags.

Parameters:

- __HANDLE__: RNG Handle
- __INTERRUPT__: specifies the RNG interrupt status flag to clear. This parameter can be one of the following values:
  - RNG_IT_CEI: Clock error interrupt
  - RNG_IT_SEI: Seed error interrupt

Return value:

- None

Notes:

- RNG_IT_DRDY flag is read-only, reading RNG_DR register automatically clears RNG_IT_DRDY.

52 HAL RTC Generic Driver

52.1 RTC Firmware driver registers structures

52.1.1 RTC_InitTypeDef

Data Fields

- ***uint32_t HourFormat***
- ***uint32_t AsynchPrediv***
- ***uint32_t SynchPrediv***
- ***uint32_t OutPut***
- ***uint32_t OutPutPolarity***
- ***uint32_t OutPutType***

Field Documentation

- ***uint32_t RTC_InitTypeDef::HourFormat***
  Specifies the RTC Hour Format. This parameter can be a value of ***RTC_Hour_Formats***
- ***uint32_t RTC_InitTypeDef::AsynchPrediv***
  Specifies the RTC Asynchronous Predivider value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0x7F
- ***uint32_t RTC_InitTypeDef::SynchPrediv***
  Specifies the RTC Synchronous Predivider value. This parameter must be a number between Min_Data = 0x00 and Max_Data = 0x7FFFU
- ***uint32_t RTC_InitTypeDef::OutPut***
  Specifies which signal will be routed to the RTC output. This parameter can be a value of ***RTC_Output_selection_Definitions***
- ***uint32_t RTC_InitTypeDef::OutPutPolarity***
  Specifies the polarity of the output signal. This parameter can be a value of ***RTC_Output_Polarity_Definitions***
- ***uint32_t RTC_InitTypeDef::OutPutType***
  Specifies the RTC Output Pin mode. This parameter can be a value of ***RTC_Output_Type_ALARM_OUT***

52.1.2 RTC_TimeTypeDef

Data Fields

- ***uint8_t Hours***
- ***uint8_t Minutes***
- ***uint8_t Seconds***
- ***uint8_t TimeFormat***
- ***uint32_t SubSeconds***
- ***uint32_t SecondFraction***
- ***uint32_t DayLightSaving***
- ***uint32_t StoreOperation***

Field Documentation

- ***uint8_t RTC_TimeTypeDef::Hours***
  Specifies the RTC Time Hour. This parameter must be a number between Min_Data = 0 and Max_Data = 12 if the RTC_HourFormat_12 is selected. This parameter must be a number between Min_Data = 0 and Max_Data = 23 if the RTC_HourFormat_24 is selected
- ***uint8_t RTC_TimeTypeDef::Minutes***
  Specifies the RTC Time Minutes. This parameter must be a number between Min_Data = 0 and Max_Data = 59
- ***uint8_t RTC_TimeTypeDef::Seconds***
  Specifies the RTC Time Seconds. This parameter must be a number between Min_Data = 0 and Max_Data = 59
- ***uint8_t RTC_TimeTypeDef::TimeFormat***
  Specifies the RTC AM/PM Time. This parameter can be a value of ***RTC_AM_PM_Definitions***
- ***uint32_t RTC_TimeTypeDef::SubSeconds***
  Specifies the RTC_SSR RTC Sub Second register content. This parameter corresponds to a time unit range between [0-1] Second with [1 Sec / SecondFraction +1] granularity
- ***uint32_t RTC_TimeTypeDef::SecondFraction***
  Specifies the range or granularity of Sub Second register content corresponding to Synchronous pre-scaler factor value (PREDIV_S) This parameter corresponds to a time unit range between [0-1] Second with [1 Sec / SecondFraction +1] granularity. This field will be used only by HAL_RTC_GetTime function
- ***uint32_t RTC_TimeTypeDef::DayLightSaving***
  Specifies DayLight Save Operation. This parameter can be a value of ***RTC_DayLightSaving_Definitions***
- ***uint32_t RTC_TimeTypeDef::StoreOperation***
  Specifies RTC_StoreOperation value to be written in the BCK bit in CR register to store the operation. This parameter can be a value of ***RTC_StoreOperation_Definitions***

52.1.3 RTC_DateTypeDef

Data Fields

- ***uint8_t WeekDay***
- ***uint8_t Month***
- ***uint8_t Date***
- ***uint8_t Year***

Field Documentation

- ***uint8_t RTC_DateTypeDef::WeekDay***
  Specifies the RTC Date WeekDay. This parameter can be a value of ***RTC_WeekDay_Definitions***
- ***uint8_t RTC_DateTypeDef::Month***
  Specifies the RTC Date Month (in BCD format). This parameter can be a value of ***RTC_Month_Date_Definitions***

- ***uint8_t RTC_DateTypeDef::Date***
  Specifies the RTC Date. This parameter must be a number between Min_Data = 1 and Max_Data = 31
- ***uint8_t RTC_DateTypeDef::Year***
  Specifies the RTC Date Year. This parameter must be a number between Min_Data = 0 and Max_Data = 99

52.1.4 RTC_AlarmTypeDef

Data Fields

- ***RTC_TimeTypeDef AlarmTime***
- ***uint32_t AlarmMask***
- ***uint32_t AlarmSubSecondMask***
- ***uint32_t AlarmDateWeekDaySel***
- ***uint8_t AlarmDateWeekDay***
- ***uint32_t Alarm***

Field Documentation

- ***RTC_TimeTypeDef RTC_AlarmTypeDef::AlarmTime***
  Specifies the RTC Alarm Time members
- ***uint32_t RTC_AlarmTypeDef::AlarmMask***
  Specifies the RTC Alarm Masks. This parameter can be a value of ***RTC_AlarmMask_Definitions***
- ***uint32_t RTC_AlarmTypeDef::AlarmSubSecondMask***
  Specifies the RTC Alarm SubSeconds Masks. This parameter can be a value of ***RTC_Alarm_Sub_Seconds_Masks_Definitions***
- ***uint32_t RTC_AlarmTypeDef::AlarmDateWeekDaySel***
  Specifies the RTC Alarm is on Date or WeekDay. This parameter can be a value of ***RTC_AlarmDateWeekDay_Definitions***
- ***uint8_t RTC_AlarmTypeDef::AlarmDateWeekDay***
  Specifies the RTC Alarm Date/WeekDay. If the Alarm Date is selected, this parameter must be set to a value in the 1-31 range. If the Alarm WeekDay is selected, this parameter can be a value of ***RTC_WeekDay_Definitions***
- ***uint32_t RTC_AlarmTypeDef::Alarm***
  Specifies the alarm . This parameter can be a value of ***RTC_Alarms_Definitions***

52.1.5 RTC_HandleTypeDef

Data Fields

- ***RTC_TypeDef * Instance***
- ***RTC_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_RTCStateTypeDef State***

Field Documentation

- ***RTC_TypeDef* RTC_HandleTypeDef::Instance***
  Register base address
- ***RTC_InitTypeDef RTC_HandleTypeDef::Init***
  RTC required parameters
- ***HAL_LockTypeDef RTC_HandleTypeDef::Lock***
  RTC locking object
- ***__IO HAL_RTCStateTypeDef RTC_HandleTypeDef::State***
  Time communication state

52.2 RTC Firmware driver API description

52.2.1 Backup Domain Operating Condition

The real-time clock (RTC), the RTC backup registers, and the backup SRAM (BKP SRAM) can be powered from the VBAT voltage when the main VDD supply is powered off. To retain the content of the RTC backup registers, backup SRAM, and supply the RTC when VDD is turned off, VBAT pin can be connected to an optional standby voltage supplied by a battery or by another source.

To allow the RTC operating even when the main digital supply (VDD) is turned off, the VBAT pin powers the following blocks:

1. The RTC
2. The LSE oscillator
3. The backup SRAM when the low power backup regulator is enabled
4. PC13 to PC15 I/Os, plus PI8 I/O (when available)

When the backup domain is supplied by VDD (analog switch connected to VDD), the following pins are available:

1. PC14 and PC15 can be used as either GPIO or LSE pins
2. PC13 can be used as a GPIO or as the RTC_AF1 pin
3. PI8 can be used as a GPIO or as the RTC_AF2 pin

When the backup domain is supplied by VBAT (analog switch connected to VBAT because VDD is not present), the following pins are available:

1. PC14 and PC15 can be used as LSE pins only
2. PC13 can be used as the RTC_AF1 pin
3. PI8 can be used as the RTC_AF2 pin

52.2.2 Backup Domain Reset

The backup domain reset sets all RTC registers and the RCC_BDCR register to their reset values. The BKPSRAM is not affected by this reset. The only way to reset the BKPSRAM is through the Flash interface by requesting a protection level change from 1 to 0.

A backup domain reset is generated when one of the following events occurs:

1. Software reset, triggered by setting the BDRST bit in the RCC Backup domain control register (RCC_BDCR).
2. VDD or VBAT power on, if both supplies have previously been powered off.

52.2.3 Backup Domain Access

After reset, the backup domain (RTC registers, RTC backup data registers and backup SRAM) is protected against possible unwanted write accesses.

To enable access to the RTC Domain and RTC registers, proceed as follows:

- Enable the Power Controller (PWR) APB1 interface clock using the __HAL_RCC_PWR_CLK_ENABLE() function.
- Enable access to RTC domain using the HAL_PWR_EnableBkUpAccess() function.
- Select the RTC clock source using the __HAL_RCC_RTC_CONFIG() function.
- Enable RTC Clock using the __HAL_RCC_RTC_ENABLE() function.

52.2.4 How to use this driver

- Enable the RTC domain access (see description in the section above).
- Configure the RTC Prescaler (Asynchronous and Synchronous) and RTC hour format using the HAL_RTC_Init() function.

Time and Date configuration

- To configure the RTC Calendar (Time and Date) use the HAL_RTC_SetTime() and HAL_RTC_SetDate() functions.
- To read the RTC Calendar, use the HAL_RTC_GetTime() and HAL_RTC_GetDate() functions.

Alarm configuration

- To configure the RTC Alarm use the HAL_RTC_SetAlarm() function. You can also configure the RTC Alarm with interrupt mode using the HAL_RTC_SetAlarm_IT() function.
- To read the RTC Alarm, use the HAL_RTC_GetAlarm() function.

52.2.5 RTC and low power modes

The MCU can be woken up from a low power mode by an RTC alternate function.

The RTC alternate functions are the RTC alarms (Alarm A and Alarm B), RTC wake-up, RTC tamper event detection and RTC time stamp event detection. These RTC alternate functions can wake up the system from the Stop and Standby low power modes.

The system can also wake up from low power modes without depending on an external interrupt (Auto-wake-up mode), by using the RTC alarm or the RTC wake-up events.

The RTC provides a programmable time base for waking up from the Stop or Standby mode at regular intervals. Wake-up from STOP and STANDBY modes is possible only when the RTC clock source is LSE or LSI.

52.2.6 Initialization and de-initialization functions

This section provides functions allowing to initialize and configure the RTC Prescaler (Synchronous and Asynchronous), RTC Hour format, disable RTC registers Write protection, enter and exit the RTC initialization mode, RTC registers synchronization check and reference clock detection enable.

1. The RTC Prescaler is programmed to generate the RTC 1Hz time base. It is split into 2 programmable prescalers to minimize power consumption.
   – A 7-bit asynchronous prescaler and a 13-bit synchronous prescaler.
   – When both prescalers are used, it is recommended to configure the asynchronous prescaler to a high value to minimize power consumption.
2. All RTC registers are Write protected. Writing to the RTC registers is enabled by writing a key into the Write Protection register, RTC_WPR.
3. To configure the RTC Calendar, user application should enter initialization mode. In this mode, the calendar counter is stopped and its value can be updated. When the initialization sequence is complete, the calendar restarts counting after 4 RTCCLK cycles.
4. To read the calendar through the shadow registers after Calendar initialization, calendar update or after wake-up from low power modes the software must first clear the RSF flag. The software must then wait until it is set again before reading the calendar, which means that the calendar registers have been correctly copied into the RTC_TR and RTC_DR shadow registers.The HAL_RTC_WaitForSynchro() function implements the above software sequence (RSF clear and RSF check).

This section contains the following APIs:

- ***HAL_RTC_Init()***
- ***HAL_RTC_DeInit()***
- ***HAL_RTC_MspInit()***
- ***HAL_RTC_MspDeInit()***

52.2.7 RTC Time and Date functions

This section provides functions allowing to configure Time and Date features

This section contains the following APIs:

- ***HAL_RTC_SetTime()***
- ***HAL_RTC_GetTime()***
- ***HAL_RTC_SetDate()***
- ***HAL_RTC_GetDate()***

52.2.8 RTC Alarm functions

This section provides functions allowing to configure Alarm feature

This section contains the following APIs:

- ***HAL_RTC_SetAlarm()***
- ***HAL_RTC_SetAlarm_IT()***
- ***HAL_RTC_DeactivateAlarm()***
- ***HAL_RTC_GetAlarm()***
- ***HAL_RTC_AlarmIRQHandler()***
- ***HAL_RTC_AlarmAEventCallback()***
- ***HAL_RTC_PollForAlarmAEvent()***

52.2.9 Peripheral Control functions

This subsection provides functions allowing to

- Wait for RTC Time and Date Synchronization

This section contains the following APIs:

- ***HAL_RTC_WaitForSynchro()***

52.2.10 Peripheral State functions

This subsection provides functions allowing to

- Get RTC state

This section contains the following APIs:

- ***HAL_RTC_GetState()***

52.2.11 Detailed description of functions

HAL_RTC_Init

Function Name	**HAL_StatusTypeDef HAL_RTC_Init (RTC_HandleTypeDef * hrtc)**
Function Description	Initializes the RTC peripheral.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status

HAL_RTC_DeInit

Function Name	**HAL_StatusTypeDef HAL_RTC_DeInit (RTC_HandleTypeDef * hrtc)**
Function Description	DeInitializes the RTC peripheral.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status
Notes	• This function doesn't reset the RTC Backup Data registers.

HAL_RTC_MspInit

Function Name	**void HAL_RTC_MspInit (RTC_HandleTypeDef * hrtc)**
Function Description	Initializes the RTC MSP.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **None**

HAL_RTC_MspDeInit

Function Name	**void HAL_RTC_MspDeInit (RTC_HandleTypeDef * hrtc)**
Function Description	DeInitializes the RTC MSP.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **None**

HAL_RTC_SetTime

Function Name **HAL_StatusTypeDef HAL_RTC_SetTime (RTC_HandleTypeDef * hrtc, RTC_TimeTypeDef * sTime, uint32_t Format)**

Function Description Sets RTC current time.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sTime:** Pointer to Time structure
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:
  - RTC_FORMAT_BIN: Binary data format
  - RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTC_GetTime

Function Name **HAL_StatusTypeDef HAL_RTC_GetTime (RTC_HandleTypeDef * hrtc, RTC_TimeTypeDef * sTime, uint32_t Format)**

Function Description Gets RTC current time.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sTime:** Pointer to Time structure
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:
  - RTC_FORMAT_BIN: Binary data format
  - RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

Notes
- You can use SubSeconds and SecondFraction (sTime structure fields returned) to convert SubSeconds value in second fraction ratio with time unit following generic formula: Second fraction ratio * time_unit= [(SecondFraction-SubSeconds)/(SecondFraction+1)] * time_unit This conversion can be performed only if no shift operation is pending (ie. SHFP=0) when PREDIV_S >= SS
- You must call HAL_RTC_GetDate() after HAL_RTC_GetTime() to unlock the values in the higher-order calendar shadow registers to ensure consistency between the time and date values. Reading RTC current time locks the values in calendar shadow registers until current date is read.

HAL_RTC_SetDate

Function Name **HAL_StatusTypeDef HAL_RTC_SetDate (RTC_HandleTypeDef * hrtc, RTC_DateTypeDef * sDate, uint32_t Format)**

Function Description Sets RTC current date.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sDate:** Pointer to date structure
- **Format:** specifies the format of the entered parameters. This parameter can be one of the following values:

– RTC_FORMAT_BIN: Binary data format
– RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTC_GetDate

Function Name **HAL_StatusTypeDef HAL_RTC_GetDate (RTC_HandleTypeDef * hrtc, RTC_DateTypeDef * sDate, uint32_t Format)**

Function Description Gets RTC current date.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sDate:** Pointer to Date structure
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:
  – RTC_FORMAT_BIN: Binary data format
  – RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

Notes
- You must call HAL_RTC_GetDate() after HAL_RTC_GetTime() to unlock the values in the higher-order calendar shadow registers to ensure consistency between the time and date values. Reading RTC current time locks the values in calendar shadow registers until Current date is read.

HAL_RTC_SetAlarm

Function Name **HAL_StatusTypeDef HAL_RTC_SetAlarm (RTC_HandleTypeDef * hrtc, RTC_AlarmTypeDef * sAlarm, uint32_t Format)**

Function Description Sets the specified RTC Alarm.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sAlarm:** Pointer to Alarm structure
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:
  – RTC_FORMAT_BIN: Binary data format
  – RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTC_SetAlarm_IT

Function Name **HAL_StatusTypeDef HAL_RTC_SetAlarm_IT (RTC_HandleTypeDef * hrtc, RTC_AlarmTypeDef * sAlarm, uint32_t Format)**

Function Description Sets the specified RTC Alarm with Interrupt.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sAlarm:** Pointer to Alarm structure
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:

  - RTC_FORMAT_BIN: Binary data format
  - RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTC_DeactivateAlarm

Function Name **HAL_StatusTypeDef HAL_RTC_DeactivateAlarm (RTC_HandleTypeDef * hrtc, uint32_t Alarm)**

Function Description Deactivate the specified RTC Alarm.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **Alarm:** Specifies the Alarm. This parameter can be one of the following values:
  - RTC_ALARM_A: AlarmA
  - RTC_ALARM_B: AlarmB

Return values
- **HAL:** status

HAL_RTC_GetAlarm

Function Name **HAL_StatusTypeDef HAL_RTC_GetAlarm (RTC_HandleTypeDef * hrtc, RTC_AlarmTypeDef * sAlarm, uint32_t Alarm, uint32_t Format)**

Function Description Gets the RTC Alarm value and masks.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sAlarm:** Pointer to Date structure
- **Alarm:** Specifies the Alarm. This parameter can be one of the following values:
  - RTC_ALARM_A: AlarmA
  - RTC_ALARM_B: AlarmB
- **Format:** Specifies the format of the entered parameters. This parameter can be one of the following values:
  - RTC_FORMAT_BIN: Binary data format
  - RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTC_AlarmIRQHandler

Function Name **void HAL_RTC_AlarmIRQHandler (RTC_HandleTypeDef * hrtc)**

Function Description This function handles Alarm interrupt request.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTC_PollForAlarmAEvent

Function Name **HAL_StatusTypeDef HAL_RTC_PollForAlarmAEvent (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**

Function Description	This function handles AlarmA Polling request.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC. • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_RTC_AlarmAEventCallback

Function Name	**void HAL_RTC_AlarmAEventCallback (RTC_HandleTypeDef * hrtc)**
Function Description	Alarm A callback.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **None**

HAL_RTC_WaitForSynchro

Function Name	**HAL_StatusTypeDef HAL_RTC_WaitForSynchro (RTC_HandleTypeDef * hrtc)**
Function Description	Waits until the RTC Time and Date registers (RTC_TR and RTC_DR) are synchronized with RTC APB clock.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status
Notes	• The RTC Resynchronization mode is write protected, use the __HAL_RTC_WRITEPROTECTION_DISABLE() before calling this function. • To read the calendar through the shadow registers after Calendar initialization, calendar update or after wake-up from low power modes the software must first clear the RSF flag. The software must then wait until it is set again before reading the calendar, which means that the calendar registers have been correctly copied into the RTC_TR and RTC_DR shadow registers.

HAL_RTC_GetState

Function Name	**HAL_RTCStateTypeDef HAL_RTC_GetState (RTC_HandleTypeDef * hrtc)**
Function Description	Returns the RTC state.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** state

RTC_EnterInitMode

Function Name	**HAL_StatusTypeDef RTC_EnterInitMode (RTC_HandleTypeDef * hrtc)**
Function Description	Enters the RTC Initialization mode.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status
Notes	• The RTC Initialization mode is write protected, use the __HAL_RTC_WRITEPROTECTION_DISABLE() before calling this function.

RTC_ByteToBcd2

Function Name	**uint8_t RTC_ByteToBcd2 (uint8_t Value)**
Function Description	Converts a 2 digit decimal to BCD format.
Parameters	• **Value:** Byte to be converted
Return values	• **Converted:** byte

RTC_Bcd2ToByte

Function Name	**uint8_t RTC_Bcd2ToByte (uint8_t Value)**
Function Description	Converts from 2 digit BCD to Binary.
Parameters	• **Value:** BCD value to be converted
Return values	• **Converted:** word

52.3 RTC Firmware driver defines

52.3.1 RTC

RTC Alarm Date WeekDay Definitions

RTC_ALARMDATEWEEKDAYSEL_DATE

RTC_ALARMDATEWEEKDAYSEL_WEEKDAY

RTC Alarm Mask Definitions

RTC_ALARMMASK_NONE

RTC_ALARMMASK_DATEWEEKDAY

RTC_ALARMMASK_HOURS

RTC_ALARMMASK_MINUTES

RTC_ALARMMASK_SECONDS

RTC_ALARMMASK_ALL

RTC Alarms Definitions

RTC_ALARM_A

RTC_ALARM_B

RTC Alarm Sub Seconds Masks Definitions

RTC_ALARMSUBSECONDMASK_ALL	All Alarm SS fields are masked. There is no comparison on sub seconds for Alarm
RTC_ALARMSUBSECONDMASK_SS14_1	SS[14:1] are don't care in Alarm comparison. Only SS[0] is compared.
RTC_ALARMSUBSECONDMASK_SS14_2	SS[14:2] are don't care in Alarm comparison. Only SS[1:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_3	SS[14:3] are don't care in Alarm comparison. Only SS[2:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_4	SS[14:4] are don't care in Alarm comparison. Only SS[3:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_5	SS[14:5] are don't care in Alarm comparison. Only SS[4:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_6	SS[14:6] are don't care in Alarm comparison. Only SS[5:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_7	SS[14:7] are don't care in Alarm comparison. Only SS[6:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_8	SS[14:8] are don't care in Alarm comparison. Only SS[7:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_9	SS[14:9] are don't care in Alarm comparison. Only SS[8:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_10	SS[14:10] are don't care in Alarm comparison. Only SS[9:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_11	SS[14:11] are don't care in Alarm comparison. Only SS[10:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_12	SS[14:12] are don't care in Alarm comparison.Only SS[11:0] are compared
RTC_ALARMSUBSECONDMASK_SS14_13	SS[14:13] are don't care in Alarm comparison. Only SS[12:0] are compared
RTC_ALARMSUBSECONDMASK_SS14	SS[14] is don't care in Alarm comparison.Only SS[13:0] are compared
RTC_ALARMSUBSECONDMASK_NONE	SS[14:0] are compared and must match to activate alarm.

RTC AM PM Definitions

RTC_HOURFORMAT12_AM

RTC_HOURFORMAT12_PM

RTC DayLight Saving Definitions

RTC_DAYLIGHTSAVING_SUB1H

RTC_DAYLIGHTSAVING_ADD1H

RTC_DAYLIGHTSAVING_NONE

RTC Exported Macros

__HAL_RTC_RESET_HANDLE_STATE

Description:

- Reset RTC handle state.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_WRITEPROTECTION_DISABLE

Description:

- Disable the write protection for RTC registers.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_WRITEPROTECTION_ENABLE

Description:

- Enable the write protection for RTC registers.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_ALARMA_ENABLE

Description:

- Enable the RTC ALARMA peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_ALARMA_DISABLE

Description:

- Disable the RTC ALARMA peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_ALARMB_ENABLE

Description:

- Enable the RTC ALARMB peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_ALARMB_DISABLE

Description:

- Disable the RTC ALARMB peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_ALARM_ENABLE_IT

Description:

- Enable the RTC Alarm interrupt.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Alarm interrupt sources to be enabled or disabled. This parameter can be any combination of the following values:
  - RTC_IT_ALRA: Alarm A interrupt
  - RTC_IT_ALRB: Alarm B interrupt

Return value:

- None

__HAL_RTC_ALARM_DISABLE_IT

Description:

- Disable the RTC Alarm interrupt.

Parameters:

- __HANDLE__: specifies the RTC handle.

- __INTERRUPT__: specifies the RTC Alarm interrupt sources to be enabled or disabled. This parameter can be any combination of the following values:
  - RTC_IT_ALRA: Alarm A interrupt
  - RTC_IT_ALRB: Alarm B interrupt

Return value:

- None

__HAL_RTC_ALARM_GET_IT

Description:

- Check whether the specified RTC Alarm interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Alarm interrupt to check. This parameter can be:
  - RTC_IT_ALRA: Alarm A interrupt
  - RTC_IT_ALRB: Alarm B interrupt

Return value:

- None

__HAL_RTC_ALARM_GET_FLAG

Description:

- Get the selected RTC Alarm's flag status.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Alarm Flag to check. This parameter can be:
  - RTC_FLAG_ALRAF
  - RTC_FLAG_ALRBF
  - RTC_FLAG_ALRAWF
  - RTC_FLAG_ALRBWF

Return value:

- None

__HAL_RTC_ALARM_CLEAR_FLAG

Description:

- Clear the RTC Alarm's pending flags.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Alarm Flag sources to be enabled or disabled. This parameter can be:
  - RTC_FLAG_ALRAF
  - RTC_FLAG_ALRBF

Return value:

- None

__HAL_RTC_ALARM_GET_IT_SOURCE

Description:

- Check whether the specified RTC Alarm interrupt has been enabled or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Alarm interrupt sources to check. This parameter can be:
  - RTC_IT_ALRA: Alarm A interrupt
  - RTC_IT_ALRB: Alarm B interrupt

Return value:

- None

__HAL_RTC_ALARM_EXTI_ENABLE_IT

Description:

- Enable interrupt on the RTC Alarm associated Exti line.

Return value:

- None

__HAL_RTC_ALARM_EXTI_DISABLE_IT

Description:

- Disable interrupt on the RTC Alarm associated Exti line.

Return value:

- None

__HAL_RTC_ALARM_EXTI_ENABLE_EVENT

Description:

- Enable event on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_DISABLE_EVENT

Description:

- Disable event on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_ENABLE_FALLING_EDGE

Description:

- Enable falling edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_DISABLE_FALLING_EDGE

Description:

- Disable falling edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_ENABLE_RISING_EDGE

Description:

- Enable rising edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_DISABLE_RISING_EDGE

Description:

- Disable rising edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_ENABLE_RISING_FALLING_EDGE

Description:

- Enable rising & falling edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_DISABLE_RISING_FALLING_EDGE

Description:

- Disable rising & falling edge trigger on the RTC Alarm associated Exti line.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_GET_FLAG

Description:

- Check whether the RTC Alarm associated Exti line interrupt flag is set or not.

Return value:

- Line: Status.

__HAL_RTC_ALARM_EXTI_CLEAR_FLAG

Description:

- Clear the RTC Alarm associated Exti line flag.

Return value:

- None.

__HAL_RTC_ALARM_EXTI_GENERATE_SWIT

Description:

- Generate a Software interrupt on RTC Alarm associated Exti line.

Return value:

- None.

RTC Flags Definitions

RTC_FLAG_RECALPF

RTC_FLAG_TAMP2F

RTC_FLAG_TAMP1F

RTC_FLAG_TSOVF

RTC_FLAG_TSF

RTC_FLAG_WUTF

RTC_FLAG_ALRBF

RTC_FLAG_ALRAF

RTC_FLAG_INITF

RTC_FLAG_RSF

RTC_FLAG_INITS

RTC_FLAG_SHPF

RTC_FLAG_WUTWF

RTC_FLAG_ALRBWF

RTC_FLAG_ALRAWF

RTC Hour Formats

RTC_HOURFORMAT_24

RTC_HOURFORMAT_12

RTC Input Parameter Format Definitions

RTC_FORMAT_BIN

RTC_FORMAT_BCD

RTC Interrupts Definitions

RTC_IT_TS

RTC_IT_WUT

RTC_IT_ALRB

RTC_IT_ALRA

RTC_IT_TAMP

RTC_IT_TAMP1

RTC_IT_TAMP2

RTC Private macros to check input parameters

IS_RTC_HOUR_FORMAT

IS_RTC_OUTPUT

IS_RTC_OUTPUT_POL

IS_RTC_OUTPUT_TYPE

IS_RTC_HOUR12

IS_RTC_HOUR24

IS_RTC_ASYNCH_PREDIV

IS_RTC_SYNCH_PREDIV

IS_RTC_MINUTES

IS_RTC_SECONDS

IS_RTC_HOURFORMAT12

IS_RTC_DAYLIGHT_SAVING

IS_RTC_STORE_OPERATION

IS_RTC_FORMAT

IS_RTC_YEAR

IS_RTC_MONTH

IS_RTC_DATE

IS_RTC_WEEKDAY

IS_RTC_ALARM_DATE_WEEKDAY_DATE

IS_RTC_ALARM_DATE_WEEKDAY_WEEKDAY

IS_RTC_ALARM_DATE_WEEKDAY_SEL

IS_RTC_ALARM_MASK

IS_RTC_ALARM

IS_RTC_ALARM_SUB_SECOND_VALUE

IS_RTC_ALARM_SUB_SECOND_MASK

RTC Month Date Definitions

RTC_MONTH_JANUARY

RTC_MONTH_FEBRUARY

RTC_MONTH_MARCH

RTC_MONTH_APRIL

RTC_MONTH_MAY

RTC_MONTH_JUNE

RTC_MONTH_JULY

RTC_MONTH_AUGUST

RTC_MONTH_SEPTEMBER

RTC_MONTH_OCTOBER

RTC_MONTH_NOVEMBER

RTC_MONTH_DECEMBER

RTC Output Polarity Definitions

RTC_OUTPUT_POLARITY_HIGH

RTC_OUTPUT_POLARITY_LOW

RTC Output Selection Definitions

RTC_OUTPUT_DISABLE

RTC_OUTPUT_ALARMA

RTC_OUTPUT_ALARMB

RTC_OUTPUT_WAKEUP

RTC Output Type ALARM OUT

RTC_OUTPUT_TYPE_OPENDRAIN

RTC_OUTPUT_TYPE_PUSHPULL

RTC Store Operation Definitions

RTC_STOREOPERATION_RESET

RTC_STOREOPERATION_SET

RTC WeekDay Definitions

RTC_WEEKDAY_MONDAY

RTC_WEEKDAY_TUESDAY

RTC_WEEKDAY_WEDNESDAY

RTC_WEEKDAY_THURSDAY

RTC_WEEKDAY_FRIDAY

RTC_WEEKDAY_SATURDAY

RTC_WEEKDAY_SUNDAY

53 HAL RTC Extension Driver

53.1 RTCEx Firmware driver registers structures

53.1.1 RTC_TamperTypeDef

Data Fields

- ***uint32_t Tamper***
- ***uint32_t PinSelection***
- ***uint32_t Trigger***
- ***uint32_t Filter***
- ***uint32_t SamplingFrequency***
- ***uint32_t PrechargeDuration***
- ***uint32_t TamperPullUp***
- ***uint32_t TimeStampOnTamperDetection***

Field Documentation

- ***uint32_t RTC_TamperTypeDef::Tamper***
  Specifies the Tamper Pin. This parameter can be a value of ***RTCEx_Tamper_Pins_Definitions***
- ***uint32_t RTC_TamperTypeDef::PinSelection***
  Specifies the Tamper Pin. This parameter can be a value of ***RTCEx_Tamper_Pins_Selection***
- ***uint32_t RTC_TamperTypeDef::Trigger***
  Specifies the Tamper Trigger. This parameter can be a value of ***RTCEx_Tamper_Trigger_Definitions***
- ***uint32_t RTC_TamperTypeDef::Filter***
  Specifies the RTC Filter Tamper. This parameter can be a value of ***RTCEx_Tamper_Filter_Definitions***
- ***uint32_t RTC_TamperTypeDef::SamplingFrequency***
  Specifies the sampling frequency. This parameter can be a value of ***RTCEx_Tamper_Sampling_Frequencies_Definitions***
- ***uint32_t RTC_TamperTypeDef::PrechargeDuration***
  Specifies the Precharge Duration . This parameter can be a value of ***RTCEx_Tamper_Pin_Precharge_Duration_Definitions***
- ***uint32_t RTC_TamperTypeDef::TamperPullUp***
  Specifies the Tamper PullUp . This parameter can be a value of ***RTCEx_Tamper_Pull_UP_Definitions***
- ***uint32_t RTC_TamperTypeDef::TimeStampOnTamperDetection***
  Specifies the TimeStampOnTamperDetection. This parameter can be a value of ***RTCEx_Tamper_TimeStampOnTamperDetection_Definitions***

53.2 RTCEx Firmware driver API description

53.2.1 How to use this driver

- Enable the RTC domain access.
- Configure the RTC Prescaler (Asynchronous and Synchronous) and RTC hour format using the HAL_RTC_Init() function.

RTC Wake-up configuration

- To configure the RTC Wake-up Clock source and Counter use the HAL_RTC_SetWakeUpTimer() function. You can also configure the RTC Wake-up timer in interrupt mode using the HAL_RTC_SetWakeUpTimer_IT() function.
- To read the RTC Wake-up Counter register, use the HAL_RTC_GetWakeUpTimer() function.

TimeStamp configuration

- Configure the RTC_AFx trigger and enable the RTC TimeStamp using the HAL_RTC_SetTimeStamp() function. You can also configure the RTC TimeStamp with interrupt mode using the HAL_RTC_SetTimeStamp_IT() function.
- To read the RTC TimeStamp Time and Date register, use the HAL_RTC_GetTimeStamp() function.
- The TIMESTAMP alternate function can be mapped either to RTC_AF1 (PC13) or RTC_AF2 (PI8 or PA0 only for STM32F446xx devices) depending on the value of TSINSEL bit in RTC_TAFCR register. The corresponding pin is also selected by HAL_RTC_SetTimeStamp() or HAL_RTC_SetTimeStamp_IT() function.

Tamper configuration

- Enable the RTC Tamper and configure the Tamper filter count, trigger Edge or Level according to the Tamper filter (if equal to 0 Edge else Level) value, sampling frequency, precharge or discharge and Pull-UP using the HAL_RTC_SetTamper() function. You can configure RTC Tamper in interrupt mode using HAL_RTC_SetTamper_IT() function.
- The TAMPER1 alternate function can be mapped either to RTC_AF1 (PC13) or RTC_AF2 (PI8 or PA0 only for STM32F446xx devices) depending on the value of TAMP1INSEL bit in RTC_TAFCR register. The corresponding pin is also selected by HAL_RTC_SetTamper() or HAL_RTC_SetTamper_IT() function.

Backup Data Registers configuration

- To write to the RTC Backup Data registers, use the HAL_RTC_BKUPWrite() function.
- To read the RTC Backup Data registers, use the HAL_RTC_BKUPRead() function.

53.2.2 RTC TimeStamp and Tamper functions

This section provides functions allowing to configure TimeStamp feature

This section contains the following APIs:

- ***HAL_RTCEx_SetTimeStamp()***
- ***HAL_RTCEx_SetTimeStamp_IT()***
- ***HAL_RTCEx_DeactivateTimeStamp()***
- ***HAL_RTCEx_GetTimeStamp()***
- ***HAL_RTCEx_SetTamper()***
- ***HAL_RTCEx_SetTamper_IT()***
- ***HAL_RTCEx_DeactivateTamper()***
- ***HAL_RTCEx_TamperTimeStampIRQHandler()***
- ***HAL_RTCEx_TimeStampEventCallback()***
- ***HAL_RTCEx_Tamper1EventCallback()***
- ***HAL_RTCEx_Tamper2EventCallback()***
- ***HAL_RTCEx_PollForTimeStampEvent()***
- ***HAL_RTCEx_PollForTamper1Event()***
- ***HAL_RTCEx_PollForTamper2Event()***

53.2.3 RTC Wake-up functions

This section provides functions allowing to configure Wake-up feature

This section contains the following APIs:

- ***HAL_RTCEx_SetWakeUpTimer()***
- ***HAL_RTCEx_SetWakeUpTimer_IT()***
- ***HAL_RTCEx_DeactivateWakeUpTimer()***
- ***HAL_RTCEx_GetWakeUpTimer()***
- ***HAL_RTCEx_WakeUpTimerIRQHandler()***
- ***HAL_RTCEx_WakeUpTimerEventCallback()***
- ***HAL_RTCEx_PollForWakeUpTimerEvent()***

53.2.4 Extension Peripheral Control functions

This subsection provides functions allowing to

- Write a data in a specified RTC Backup data register
- Read a data in a specified RTC Backup data register
- Set the Coarse calibration parameters.
- Deactivate the Coarse calibration parameters
- Set the Smooth calibration parameters.
- Configure the Synchronization Shift Control Settings.
- Configure the Calibration Pinout (RTC_CALIB) Selection (1Hz or 512Hz).
- Deactivate the Calibration Pinout (RTC_CALIB) Selection (1Hz or 512Hz).
- Enable the RTC reference clock detection.
- Disable the RTC reference clock detection.
- Enable the Bypass Shadow feature.
- Disable the Bypass Shadow feature.

This section contains the following APIs:

- ***HAL_RTCEx_BKUPWrite()***
- ***HAL_RTCEx_BKUPRead()***
- ***HAL_RTCEx_SetCoarseCalib()***
- ***HAL_RTCEx_DeactivateCoarseCalib()***

- ***HAL_RTCEx_SetSmoothCalib()***
- ***HAL_RTCEx_SetSynchroShift()***
- ***HAL_RTCEx_SetCalibrationOutPut()***
- ***HAL_RTCEx_DeactivateCalibrationOutPut()***
- ***HAL_RTCEx_SetRefClock()***
- ***HAL_RTCEx_DeactivateRefClock()***
- ***HAL_RTCEx_EnableBypassShadow()***
- ***HAL_RTCEx_DisableBypassShadow()***

53.2.5 Extended features functions

This section provides functions allowing to:

- RTC Alarm B callback
- RTC Poll for Alarm B request

This section contains the following APIs:

- ***HAL_RTCEx_AlarmBEventCallback()***
- ***HAL_RTCEx_PollForAlarmBEvent()***

53.2.6 Detailed description of functions

HAL_RTCEx_SetTimeStamp

Function Name — **HAL_StatusTypeDef HAL_RTCEx_SetTimeStamp (RTC_HandleTypeDef * hrtc, uint32_t TimeStampEdge, uint32_t RTC_TimeStampPin)**

Function Description — Sets TimeStamp.

Parameters

- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **TimeStampEdge:** Specifies the pin edge on which the TimeStamp is activated. This parameter can be one of the following values:
  - RTC_TIMESTAMPEDGE_RISING: the Time stamp event occurs on the rising edge of the related pin.
  - RTC_TIMESTAMPEDGE_FALLING: the Time stamp event occurs on the falling edge of the related pin.
- **RTC_TimeStampPin:** specifies the RTC TimeStamp Pin. This parameter can be one of the following values:
  - RTC_TIMESTAMPPIN_DEFAULT: PC13 is selected as RTC TimeStamp Pin.
  - RTC_TIMESTAMPPIN_POS1: PI8/PA0 is selected as RTC TimeStamp Pin. (PI8 for all STM32 devices except for STM32F446xx devices the PA0 is used)
  - RTC_TIMESTAMPPIN_PA0: PA0 is selected as RTC TimeStamp Pin only for STM32F446xx devices

Return values

- **HAL:** status

Notes

- This API must be called before enabling the TimeStamp feature.

HAL_RTCEx_SetTimeStamp_IT

Function Name	**HAL_StatusTypeDef HAL_RTCEx_SetTimeStamp_IT (RTC_HandleTypeDef * hrtc, uint32_t TimeStampEdge, uint32_t RTC_TimeStampPin)**
Function Description	Sets TimeStamp with Interrupt.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC. • **TimeStampEdge:** Specifies the pin edge on which the TimeStamp is activated. This parameter can be one of the following values: – RTC_TIMESTAMPEDGE_RISING: the Time stamp event occurs on the rising edge of the related pin. – RTC_TIMESTAMPEDGE_FALLING: the Time stamp event occurs on the falling edge of the related pin. • **RTC_TimeStampPin:** Specifies the RTC TimeStamp Pin. This parameter can be one of the following values: – RTC_TIMESTAMPPIN_DEFAULT: PC13 is selected as RTC TimeStamp Pin. – RTC_TIMESTAMPPIN_PI8: PI8 is selected as RTC TimeStamp Pin. (not applicable in the case of STM32F446xx devices) – RTC_TIMESTAMPPIN_PA0: PA0 is selected as RTC TimeStamp Pin only for STM32F446xx devices
Return values	• **HAL:** status
Notes	• This API must be called before enabling the TimeStamp feature.

HAL_RTCEx_DeactivateTimeStamp

Function Name	**HAL_StatusTypeDef HAL_RTCEx_DeactivateTimeStamp (RTC_HandleTypeDef * hrtc)**
Function Description	Deactivates TimeStamp.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status

HAL_RTCEx_GetTimeStamp

Function Name	**HAL_StatusTypeDef HAL_RTCEx_GetTimeStamp (RTC_HandleTypeDef * hrtc, RTC_TimeTypeDef * sTimeStamp, RTC_DateTypeDef * sTimeStampDate, uint32_t Format)**
Function Description	Gets the RTC TimeStamp value.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC. • **sTimeStamp:** Pointer to Time structure • **sTimeStampDate:** Pointer to Date structure • **Format:** specifies the format of the entered parameters. This parameter can be one of the following values:

RTC_FORMAT_BIN: Binary data format
RTC_FORMAT_BCD: BCD data format

Return values
- **HAL:** status

HAL_RTCEx_SetTamper

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetTamper (RTC_HandleTypeDef * hrtc, RTC_TamperTypeDef * sTamper)**

Function Description Sets Tamper.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sTamper:** Pointer to Tamper Structure.

Return values
- **HAL:** status

Notes
- By calling this API we disable the tamper interrupt for all tampers.

HAL_RTCEx_SetTamper_IT

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetTamper_IT (RTC_HandleTypeDef * hrtc, RTC_TamperTypeDef * sTamper)**

Function Description Sets Tamper with interrupt.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **sTamper:** Pointer to RTC Tamper.

Return values
- **HAL:** status

Notes
- By calling this API we force the tamper interrupt for all tampers.

HAL_RTCEx_DeactivateTamper

Function Name **HAL_StatusTypeDef HAL_RTCEx_DeactivateTamper (RTC_HandleTypeDef * hrtc, uint32_t Tamper)**

Function Description Deactivates Tamper.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **Tamper:** Selected tamper pin. This parameter can be RTC_Tamper_1 and/or RTC_TAMPER_2.

Return values
- **HAL:** status

HAL_RTCEx_TamperTimeStampIRQHandler

Function Name **void HAL_RTCEx_TamperTimeStampIRQHandler (RTC_HandleTypeDef * hrtc)**

Function Description This function handles TimeStamp interrupt request.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_Tamper1EventCallback

Function Name **void HAL_RTCEx_Tamper1EventCallback (RTC_HandleTypeDef * hrtc)**

Function Description Tamper 1 callback.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_Tamper2EventCallback

Function Name **void HAL_RTCEx_Tamper2EventCallback (RTC_HandleTypeDef * hrtc)**

Function Description Tamper 2 callback.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_TimeStampEventCallback

Function Name **void HAL_RTCEx_TimeStampEventCallback (RTC_HandleTypeDef * hrtc)**

Function Description TimeStamp callback.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_PollForTimeStampEvent

Function Name **HAL_StatusTypeDef HAL_RTCEx_PollForTimeStampEvent (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**

Function Description This function handles TimeStamp polling request.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_RTCEx_PollForTamper1Event

Function Name **HAL_StatusTypeDef HAL_RTCEx_PollForTamper1Event (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**

Function Description This function handles Tamper1 Polling.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_RTCEx_PollForTamper2Event

Function Name **HAL_StatusTypeDef HAL_RTCEx_PollForTamper2Event (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**

Function Description This function handles Tamper2 Polling.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_RTCEx_SetWakeUpTimer

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetWakeUpTimer (RTC_HandleTypeDef * hrtc, uint32_t WakeUpCounter, uint32_t WakeUpClock)**

Function Description Sets wake up timer.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **WakeUpCounter:** Wake up counter
- **WakeUpClock:** Wake up clock

Return values
- **HAL:** status

HAL_RTCEx_SetWakeUpTimer_IT

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetWakeUpTimer_IT (RTC_HandleTypeDef * hrtc, uint32_t WakeUpCounter, uint32_t WakeUpClock)**

Function Description Sets wake up timer with interrupt.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **WakeUpCounter:** Wake up counter
- **WakeUpClock:** Wake up clock

Return values
- **HAL:** status

HAL_RTCEx_DeactivateWakeUpTimer

Function Name **uint32_t HAL_RTCEx_DeactivateWakeUpTimer (RTC_HandleTypeDef * hrtc)**

Function Description Deactivates wake up timer counter.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **HAL:** status

HAL_RTCEx_GetWakeUpTimer

Function Name **uint32_t HAL_RTCEx_GetWakeUpTimer (RTC_HandleTypeDef * hrtc)**

Function Description Gets wake up timer counter.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **Counter:** value

HAL_RTCEx_WakeUpTimerIRQHandler

Function Name **void HAL_RTCEx_WakeUpTimerIRQHandler (RTC_HandleTypeDef * hrtc)**

Function Description This function handles Wake Up Timer interrupt request.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_WakeUpTimerEventCallback

Function Name **void HAL_RTCEx_WakeUpTimerEventCallback (RTC_HandleTypeDef * hrtc)**

Function Description Wake Up Timer callback.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **None**

HAL_RTCEx_PollForWakeUpTimerEvent

Function Name **HAL_StatusTypeDef HAL_RTCEx_PollForWakeUpTimerEvent (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**

Function Description This function handles Wake Up Timer Polling.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_RTCEx_BKUPWrite

Function Name **void HAL_RTCEx_BKUPWrite (RTC_HandleTypeDef * hrtc, uint32_t BackupRegister, uint32_t Data)**

Function Description Writes a data in a specified RTC Backup data register.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **BackupRegister:** RTC Backup data Register number. This parameter can be: RTC_BKP_DRx where x can be from 0 to 19 to specify the register.

- **Data:** Data to be written in the specified RTC Backup data register.

Return values
- **None**

HAL_RTCEx_BKUPRead

Function Name **uint32_t HAL_RTCEx_BKUPRead (RTC_HandleTypeDef * hrtc, uint32_t BackupRegister)**

Function Description Reads data from the specified RTC Backup data Register.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **BackupRegister:** RTC Backup data Register number. This parameter can be: RTC_BKP_DRx where x can be from 0 to 19 to specify the register.

Return values
- **Read:** value

HAL_RTCEx_SetCoarseCalib

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetCoarseCalib (RTC_HandleTypeDef * hrtc, uint32_t CalibSign, uint32_t Value)**

Function Description Sets the Coarse calibration parameters.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **CalibSign:** Specifies the sign of the coarse calibration value. This parameter can be one of the following values :
  - RTC_CALIBSIGN_POSITIVE: The value sign is positive
  - RTC_CALIBSIGN_NEGATIVE: The value sign is negative
- **Value:** value of coarse calibration expressed in ppm (coded on 5 bits).

Return values
- **HAL:** status

Notes
- This Calibration value should be between 0 and 63 when using negative sign with a 2-ppm step.
- This Calibration value should be between 0 and 126 when using positive sign with a 4-ppm step.

HAL_RTCEx_DeactivateCoarseCalib

Function Name **HAL_StatusTypeDef HAL_RTCEx_DeactivateCoarseCalib (RTC_HandleTypeDef * hrtc)**

Function Description Deactivates the Coarse calibration parameters.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.

Return values
- **HAL:** status

HAL_RTCEx_SetSmoothCalib

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetSmoothCalib (RTC_HandleTypeDef * hrtc, uint32_t SmoothCalibPeriod, uint32_t SmoothCalibPlusPulses, uint32_t SmouthCalibMinusPulsesValue)**

Function Description Sets the Smooth calibration parameters.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **SmoothCalibPeriod:** Select the Smooth Calibration Period. This parameter can be can be one of the following values :
  - RTC_SMOOTHCALIB_PERIOD_32SEC: The smooth calibration period is 32s.
  - RTC_SMOOTHCALIB_PERIOD_16SEC: The smooth calibration period is 16s.
  - RTC_SMOOTHCALIB_PERIOD_8SEC: The smooth calibration period is 8s.
- **SmoothCalibPlusPulses:** Select to Set or reset the CALP bit. This parameter can be one of the following values:
  - RTC_SMOOTHCALIB_PLUSPULSES_SET: Add one RTCCLK pulse every 2*11 pulses.
  - RTC_SMOOTHCALIB_PLUSPULSES_RESET: No RTCCLK pulses are added.
- **SmouthCalibMinusPulsesValue:** Select the value of CALM[8:0] bits. This parameter can be one any value from 0 to 0x000001FF.

Return values
- **HAL:** status

Notes
- To deactivate the smooth calibration, the field SmoothCalibPlusPulses must be equal to SMOOTHCALIB_PLUSPULSES_RESET and the field SmouthCalibMinusPulsesValue must be equal to 0.

HAL_RTCEx_SetSynchroShift

Function Name **HAL_StatusTypeDef HAL_RTCEx_SetSynchroShift (RTC_HandleTypeDef * hrtc, uint32_t ShiftAdd1S, uint32_t ShiftSubFS)**

Function Description Configures the Synchronization Shift Control Settings.

Parameters
- **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
- **ShiftAdd1S:** Select to add or not 1 second to the time calendar. This parameter can be one of the following values :
  - RTC_SHIFTADD1S_SET: Add one second to the clock calendar.
  - RTC_SHIFTADD1S_RESET: No effect.
- **ShiftSubFS:** Select the number of Second Fractions to substitute. This parameter can be one any value from 0 to 0x7FFF.

Return values
- **HAL:** status

Notes
- When REFCKON is set, firmware must not write to Shift

control register.

HAL_RTCEx_SetCalibrationOutPut

Function Name	**HAL_StatusTypeDef HAL_RTCEx_SetCalibrationOutPut (RTC_HandleTypeDef * hrtc, uint32_t CalibOutput)**
Function Description	Configures the Calibration Pinout (RTC_CALIB) Selection (1Hz or 512Hz).
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC. • **CalibOutput:** Select the Calibration output Selection . This parameter can be one of the following values: – RTC_CALIBOUTPUT_512HZ: A signal has a regular waveform at 512Hz. – RTC_CALIBOUTPUT_1HZ: A signal has a regular waveform at 1Hz.
Return values	• **HAL:** status

HAL_RTCEx_DeactivateCalibrationOutPut

Function Name	**HAL_StatusTypeDef HAL_RTCEx_DeactivateCalibrationOutPut (RTC_HandleTypeDef * hrtc)**
Function Description	Deactivates the Calibration Pinout (RTC_CALIB) Selection (1Hz or 512Hz).
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status

HAL_RTCEx_SetRefClock

Function Name	**HAL_StatusTypeDef HAL_RTCEx_SetRefClock (RTC_HandleTypeDef * hrtc)**
Function Description	Enables the RTC reference clock detection.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status

HAL_RTCEx_DeactivateRefClock

Function Name	**HAL_StatusTypeDef HAL_RTCEx_DeactivateRefClock (RTC_HandleTypeDef * hrtc)**
Function Description	Disable the RTC reference clock detection.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status

HAL_RTCEx_EnableBypassShadow

Function Name	**HAL_StatusTypeDef HAL_RTCEx_EnableBypassShadow (RTC_HandleTypeDef * hrtc)**
Function Description	Enables the Bypass Shadow feature.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status
Notes	• When the Bypass Shadow is enabled the calendar value are taken directly from the Calendar counter.

HAL_RTCEx_DisableBypassShadow

Function Name	**HAL_StatusTypeDef HAL_RTCEx_DisableBypassShadow (RTC_HandleTypeDef * hrtc)**
Function Description	Disables the Bypass Shadow feature.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **HAL:** status
Notes	• When the Bypass Shadow is enabled the calendar value are taken directly from the Calendar counter.

HAL_RTCEx_AlarmBEventCallback

Function Name	**void HAL_RTCEx_AlarmBEventCallback (RTC_HandleTypeDef * hrtc)**
Function Description	Alarm B callback.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC.
Return values	• **None**

HAL_RTCEx_PollForAlarmBEvent

Function Name	**HAL_StatusTypeDef HAL_RTCEx_PollForAlarmBEvent (RTC_HandleTypeDef * hrtc, uint32_t Timeout)**
Function Description	This function handles AlarmB Polling request.
Parameters	• **hrtc:** pointer to a RTC_HandleTypeDef structure that contains the configuration information for RTC. • **Timeout:** Timeout duration
Return values	• **HAL:** status

53.3 RTCEx Firmware driver defines

53.3.1 RTCEx

RTC Add 1 Second Parameter Definitions

RTC_SHIFTADD1S_RESET

RTC_SHIFTADD1S_SET

RTC Backup Registers Definitions

RTC_BKP_DR0

RTC_BKP_DR1

RTC_BKP_DR2

RTC_BKP_DR3

RTC_BKP_DR4

RTC_BKP_DR5

RTC_BKP_DR6

RTC_BKP_DR7

RTC_BKP_DR8

RTC_BKP_DR9

RTC_BKP_DR10

RTC_BKP_DR11

RTC_BKP_DR12

RTC_BKP_DR13

RTC_BKP_DR14

RTC_BKP_DR15

RTC_BKP_DR16

RTC_BKP_DR17

RTC_BKP_DR18

RTC_BKP_DR19

RTC Calibration

__HAL_RTC_COARSE_CALIB_ENABLE

Description:

- Enable the Coarse calibration process.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_COARSE_CALIB_DISABLE

Description:

- Disable the Coarse calibration

process.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_CALIBRATION_OUTPUT_ENABLE

Description:

- Enable the RTC calibration output.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_CALIBRATION_OUTPUT_DISABLE

Description:

- Disable the calibration output.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_CLOCKREF_DETECTION_ENABLE

Description:

- Enable the clock reference detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_CLOCKREF_DETECTION_DISABLE

Description:

- Disable the clock reference detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_SHIFT_GET_FLAG

Description:

- Get the selected RTC shift

operation's flag status.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC shift operation Flag is pending or not. This parameter can be:
  – RTC_FLAG_SHPF

Return value:

- None

RTC Calib Output Selection Definitions

RTC_CALIBOUTPUT_512HZ

RTC_CALIBOUTPUT_1HZ

RTC Digital Calib Definitions

RTC_CALIBSIGN_POSITIVE

RTC_CALIBSIGN_NEGATIVE

Private macros to check input parameters

IS_RTC_BKP

IS_TIMESTAMP_EDGE

IS_RTC_TAMPER

IS_RTC_TAMPER_PIN

IS_RTC_TIMESTAMP_PIN

IS_RTC_TAMPER_TRIGGER

IS_RTC_TAMPER_FILTER

IS_RTC_TAMPER_SAMPLING_FREQ

IS_RTC_TAMPER_PRECHARGE_DURATION

IS_RTC_TAMPER_TIMESTAMPONTAMPER_DETECTION

IS_RTC_TAMPER_PULLUP_STATE

IS_RTC_WAKEUP_CLOCK

IS_RTC_WAKEUP_COUNTER

IS_RTC_CALIB_SIGN

IS_RTC_CALIB_VALUE

IS_RTC_SMOOTH_CALIB_PERIOD

IS_RTC_SMOOTH_CALIB_PLUS

IS_RTC_SMOOTH_CALIB_MINUS

IS_RTC_SHIFT_ADD1S

IS_RTC_SHIFT_SUBFS

IS_RTC_CALIB_OUTPUT

RTC Smooth Calib Period Definitions

RTC_SMOOTHCALIB_PERIOD_32SEC	If RTCCLK = 32768 Hz, Smooth calibration period is 32s, else 2exp20 RTCCLK seconds
RTC_SMOOTHCALIB_PERIOD_16SEC	If RTCCLK = 32768 Hz, Smooth calibration period is 16s, else 2exp19 RTCCLK seconds
RTC_SMOOTHCALIB_PERIOD_8SEC	If RTCCLK = 32768 Hz, Smooth calibration period is 8s, else 2exp18 RTCCLK seconds

RTC Smooth Calib Plus Pulses Definitions

RTC_SMOOTHCALIB_PLUSPULSES_SET	The number of RTCCLK pulses added during a X -second window = Y - CALM[8:0] with Y = 512, 256, 128 when X = 32, 16, 8
RTC_SMOOTHCALIB_PLUSPULSES_RESET	The number of RTCCLK pulses subbstited during a 32-second window = CALM[8:0]

RTC Tamper

__HAL_RTC_TAMPER1_ENABLE

Description:

- Enable the RTC Tamper1 input detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TAMPER1_DISABLE

Description:

- Disable the RTC Tamper1 input detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TAMPER2_ENABLE

Description:

- Enable the RTC Tamper2 input detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TAMPER2_DISABLE

Description:

- Disable the RTC Tamper2 input detection.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TAMPER_GET_IT

Description:

- Check whether the specified RTC Tamper interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Tamper interrupt to check. This parameter can be:
  - RTC_IT_TAMP1
  - RTC_IT_TAMP2

Return value:

- None

__HAL_RTC_TAMPER_GET_IT_SOURCE

Description:

- Check whether the specified RTC Tamper interrupt has been enabled or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Tamper interrupt source to check. This parameter can be:
  - RTC_IT_TAMP: Tamper interrupt

Return value:

- None

__HAL_RTC_TAMPER_GET_FLAG

Description:

- Get the selected RTC Tamper's flag status.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Tamper Flag sources to be enabled or disabled. This parameter can be:
  - RTC_FLAG_TAMP1F

– RTC_FLAG_TAMP2F

Return value:

- None

__HAL_RTC_TAMPER_CLEAR_FLAG

Description:

- Clear the RTC Tamper's pending flags.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Tamper Flag to clear. This parameter can be:
  – RTC_FLAG_TAMP1F
  – RTC_FLAG_TAMP2F

Return value:

- None

RTC Tamper Filter Definitions

RTC_TAMPERFILTER_DISABLE	Tamper filter is disabled
RTC_TAMPERFILTER_2SAMPLE	Tamper is activated after 2 consecutive samples at the active level
RTC_TAMPERFILTER_4SAMPLE	Tamper is activated after 4 consecutive samples at the active level
RTC_TAMPERFILTER_8SAMPLE	Tamper is activated after 8 consecutive samples at the active level.

RTC Tamper Pins Definitions

RTC_TAMPER_1

RTC_TAMPER_2

RTC tamper Pins Selection

RTC_TAMPERPIN_DEFAULT

RTC_TAMPERPIN_POS1

RTC Tamper Pin Precharge Duration Definitions

RTC_TAMPERPRECHARGEDURATION_1RTCCLK	Tamper pins are pre-charged before sampling during 1 RTCCLK cycle
RTC_TAMPERPRECHARGEDURATION_2RTCCLK	Tamper pins are pre-charged before sampling during 2 RTCCLK cycles
RTC_TAMPERPRECHARGEDURATION_4RTCCLK	Tamper pins are pre-charged before sampling during 4 RTCCLK cycles
RTC_TAMPERPRECHARGEDURATION_8RTCCLK	Tamper pins are pre-charged before sampling during 8 RTCCLK cycles

RTC Tamper Pull Up Definitions

RTC_TAMPER_PULLUP_ENABLE	TimeStamp on Tamper Detection event saved
RTC_TAMPER_PULLUP_DISABLE	TimeStamp on Tamper Detection event is not saved

RTC Tamper Sampling Frequencies Definitions

RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV32768	Each of the tamper inputs are sampled with a frequency = RTCCLK / 32768
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV16384	Each of the tamper inputs are sampled with a frequency = RTCCLK / 16384
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV8192	Each of the tamper inputs are sampled with a frequency = RTCCLK / 8192
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV4096	Each of the tamper inputs are sampled with a frequency = RTCCLK / 4096
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV2048	Each of the tamper inputs are sampled with a frequency = RTCCLK / 2048
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV1024	Each of the tamper inputs are sampled with a frequency = RTCCLK / 1024
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV512	Each of the tamper inputs are sampled with a frequency = RTCCLK / 512
RTC_TAMPERSAMPLINGFREQ_RTCCLK_DIV256	Each of the tamper inputs are sampled with a frequency = RTCCLK / 256

EXTI RTC Tamper Timestamp EXTI

__HAL_RTC_TAMPER_TIMESTAMP_EXTI_ENABLE_IT	**Description:** • Enable interrupt on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_DISABLE_IT	**Description:** • Disable interrupt on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_ENABLE_	**Description:**

EVENT	• Enable event on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_DISABLE_EVENT	**Description:** • Disable event on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_ENABLE_FALLING_EDGE	**Description:** • Enable falling edge trigger on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_DISABLE_FALLING_EDGE	**Description:** • Disable falling edge trigger on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_ENABLE_RISING_EDGE	**Description:** • Enable rising edge trigger on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_DISABLE_RISING_EDGE	**Description:** • Disable rising edge trigger on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_ENABLE_RISING_FALLING_EDGE	**Description:** • Enable rising & falling edge trigger on the RTC

	Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_DISABLE_RISING_FALLING_EDGE	**Description:** • Disable rising & falling edge trigger on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_GET_FLAG	**Description:** • Check whether the RTC Tamper and Timestamp associated Exti line interrupt flag is set or not. **Return value:** • Line: Status.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_CLEAR_FLAG	**Description:** • Clear the RTC Tamper and Timestamp associated Exti line flag. **Return value:** • None.
__HAL_RTC_TAMPER_TIMESTAMP_EXTI_GENERATE_SWIT	**Description:** • Generate a Software interrupt on the RTC Tamper and Timestamp associated Exti line. **Return value:** • None.

RTC Tamper TimeStamp On Tamper Detection Definitions

RTC_TIMESTAMPONTAMPERDETECTION_ENABLE	TimeStamp on Tamper Detection event saved
RTC_TIMESTAMPONTAMPERDETECTION_DISABLE	TimeStamp on Tamper Detection event is not saved

RTC Tamper Triggers Definitions

RTC_TAMPERTRIGGER_RISINGEDGE

RTC_TAMPERTRIGGER_FALLINGEDGE

RTC_TAMPERTRIGGER_LOWLEVEL

RTC_TAMPERTRIGGER_HIGHLEVEL

RTC Timestamp

__HAL_RTC_TIMESTAMP_ENABLE

Description:

- Enable the RTC TimeStamp peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TIMESTAMP_DISABLE

Description:

- Disable the RTC TimeStamp peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_TIMESTAMP_ENABLE_IT

Description:

- Enable the RTC TimeStamp interrupt.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC TimeStamp interrupt sources to be enabled or disabled. This parameter can be:
  - RTC_IT_TS: TimeStamp interrupt

Return value:

- None

__HAL_RTC_TIMESTAMP_DISABLE_IT

Description:

- Disable the RTC TimeStamp interrupt.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC TimeStamp interrupt sources to be enabled or disabled. This parameter can be:
  - RTC_IT_TS: TimeStamp interrupt

Return value:

- None

__HAL_RTC_TIMESTAMP_GET_IT

Description:

- Check whether the specified RTC TimeStamp interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC TimeStamp interrupt to check. This parameter can be:
  - RTC_IT_TS: TimeStamp interrupt

Return value:

- None

__HAL_RTC_TIMESTAMP_GET_IT_SOURCE

Description:

- Check whether the specified RTC Time Stamp interrupt has been enabled or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC Time Stamp interrupt source to check. This parameter can be:
  - RTC_IT_TS: TimeStamp interrupt

Return value:

- None

__HAL_RTC_TIMESTAMP_GET_FLAG

Description:

- Get the selected RTC TimeStamp's flag status.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC TimeStamp flag to check. This parameter can be:
  - RTC_FLAG_TSF
  - RTC_FLAG_TSOVF

Return value:

- None

__HAL_RTC_TIMESTAMP_CLEAR_FLAG

Description:

- Clear the RTC Time Stamp's pending flags.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Alarm Flag sources to be enabled or disabled. This parameter can be:
  – RTC_FLAG_TSF

Return value:

- None

RTC TimeStamp Pins Selection

RTC_TIMESTAMPPIN_DEFAULT

RTC_TIMESTAMPPIN_POS1

RTC TimeStamp Edges Definitions

RTC_TIMESTAMPEDGE_RISING

RTC_TIMESTAMPEDGE_FALLING

RTC WakeUp Timer

__HAL_RTC_WAKEUPTIMER_ENABLE

Description:

- Enable the RTC WakeUp Timer peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_WAKEUPTIMER_DISABLE

Description:

- Disable the RTC Wake-up Timer peripheral.

Parameters:

- __HANDLE__: specifies the RTC handle.

Return value:

- None

__HAL_RTC_WAKEUPTIMER_ENABLE_IT

Description:

- Enable the RTC WakeUpTimer interrupt.

Parameters:

- __HANDLE__: specifies the RTC

handle.
- __INTERRUPT__: specifies the RTC WakeUpTimer interrupt sources to be enabled or disabled. This parameter can be:
  - RTC_IT_WUT: WakeUpTimer A interrupt

Return value:

- None

__HAL_RTC_WAKEUPTIMER_DISABLE_IT

Description:

- Disable the RTC WakeUpTimer interrupt.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC WakeUpTimer interrupt sources to be enabled or disabled. This parameter can be:
  - RTC_IT_WUT: WakeUpTimer A interrupt

Return value:

- None

__HAL_RTC_WAKEUPTIMER_GET_IT

Description:

- Check whether the specified RTC WakeUpTimer interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the RTC WakeUpTimer interrupt to check. This parameter can be:
  - RTC_IT_WUT: WakeUpTimer A interrupt

Return value:

- None

__HAL_RTC_WAKEUPTIMER_GET_IT_SOURCE

Description:

- Check whether the specified RTC Wake Up timer interrupt has been enabled or not.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __INTERRUPT__: specifies the

RTC Wake Up timer interrupt sources to check. This parameter can be:
  – RTC_IT_WUT: WakeUpTimer interrupt

Return value:

- None

__HAL_RTC_WAKEUPTIMER_GET_FLAG

Description:

- Get the selected RTC WakeUpTimer's flag status.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC WakeUpTimer Flag to check. This parameter can be:
  – RTC_FLAG_WUTF
  – RTC_FLAG_WUTWF

Return value:

- None

__HAL_RTC_WAKEUPTIMER_CLEAR_FLAG

Description:

- Clear the RTC Wake Up timer's pending flags.

Parameters:

- __HANDLE__: specifies the RTC handle.
- __FLAG__: specifies the RTC Tamper Flag sources to be enabled or disabled. This parameter can be:
  – RTC_FLAG_WUTF

Return value:

- None

__HAL_RTC_WAKEUPTIMER_EXTI_ENABLE_IT

Description:

- Enable interrupt on the RTC Wake-up Timer associated Exti line.

Return value:

- None

__HAL_RTC_WAKEUPTIMER_EXTI_DISABLE_IT

Description:

- Disable interrupt on the RTC Wake-up Timer associated Exti line.

Return value:

- None

__HAL_RTC_WAKEUPTIMER_EXTI_ENABLE_EVENT	**Description:** • Enable event on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_DISABLE_EVENT	**Description:** • Disable event on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_ENABLE_FALLING_EDGE	**Description:** • Enable falling edge trigger on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_DISABLE_FALLING_EDGE	**Description:** • Disable falling edge trigger on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_ENABLE_RISING_EDGE	**Description:** • Enable rising edge trigger on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_DISABLE_RISING_EDGE	**Description:** • Disable rising edge trigger on the RTC Wake-up Timer associated Exti line. **Return value:** • None.
__HAL_RTC_WAKEUPTIMER_EXTI_ENABLE_RISING_FALLING_EDGE	**Description:** • Enable rising & falling edge trigger on the RTC Wake-up Timer associated Exti line. **Return value:** • None.

__HAL_RTC_WAKEUPTIMER_EXTI_DISABLE_RISING_FALLING_EDGE

Description:

- Disable rising & falling edge trigger on the RTC Wake-up Timer associated Exti line.

Return value:

- None.

__HAL_RTC_WAKEUPTIMER_EXTI_GET_FLAG

Description:

- Check whether the RTC Wake-up Timer associated Exti line interrupt flag is set or not.

Return value:

- Line: Status.

__HAL_RTC_WAKEUPTIMER_EXTI_CLEAR_FLAG

Description:

- Clear the RTC Wake-up Timer associated Exti line flag.

Return value:

- None.

__HAL_RTC_WAKEUPTIMER_EXTI_GENERATE_SWIT

Description:

- Generate a Software interrupt on the RTC Wake-up Timer associated Exti line.

Return value:

- None.

RTC Wake-up Timer Definitions

RTC_WAKEUPCLOCK_RTCCLK_DIV16

RTC_WAKEUPCLOCK_RTCCLK_DIV8

RTC_WAKEUPCLOCK_RTCCLK_DIV4

RTC_WAKEUPCLOCK_RTCCLK_DIV2

RTC_WAKEUPCLOCK_CK_SPRE_16BITS

RTC_WAKEUPCLOCK_CK_SPRE_17BITS

54 HAL SAI Generic Driver

54.1 SAI Firmware driver registers structures

54.1.1 SAI_InitTypeDef

Data Fields

- ***uint32_t AudioMode***
- ***uint32_t Synchro***
- ***uint32_t SynchroExt***
- ***uint32_t OutputDrive***
- ***uint32_t NoDivider***
- ***uint32_t FIFOThreshold***
- ***uint32_t ClockSource***
- ***uint32_t AudioFrequency***
- ***uint32_t Mckdiv***
- ***uint32_t MonoStereoMode***
- ***uint32_t CompandingMode***
- ***uint32_t TriState***
- ***uint32_t Protocol***
- ***uint32_t DataSize***
- ***uint32_t FirstBit***
- ***uint32_t ClockStrobing***

Field Documentation

- ***uint32_t SAI_InitTypeDef::AudioMode***
  Specifies the SAI Block audio Mode. This parameter can be a value of ***SAI_Block_Mode***
- ***uint32_t SAI_InitTypeDef::Synchro***
  Specifies SAI Block synchronization This parameter can be a value of ***SAI_Block_Synchronization***
- ***uint32_t SAI_InitTypeDef::SynchroExt***
  Specifies SAI external output synchronization, this setup is common for BlockA and BlockB This parameter can be a value of ***SAI_Block_SyncExt***
  Note:: If both audio blocks of same SAI are used, this parameter has to be set to the same value for each audio block
- ***uint32_t SAI_InitTypeDef::OutputDrive***
  Specifies when SAI Block outputs are driven. This parameter can be a value of ***SAI_Block_Output_Drive***
  Note:this value has to be set before enabling the audio block but after the audio block configuration.
- ***uint32_t SAI_InitTypeDef::NoDivider***
  Specifies whether master clock will be divided or not. This parameter can be a value of ***SAI_Block_NoDivider***
  Note: If bit NODIV in the SAI_xCR1 register is cleared, the frame length should be aligned to a number equal to a power of 2, from 8 to 256. If bit NODIV in the SAI_xCR1 register is set, the frame length can take any of the values without

constraint since the input clock of the audio block should be equal to the bit clock. There is no MCLK_x clock which can be output.

- ***uint32_t SAI_InitTypeDef::FIFOThreshold***
  Specifies SAI Block FIFO threshold. This parameter can be a value of ***SAI_Block_Fifo_Threshold***
- ***uint32_t SAI_InitTypeDef::ClockSource***
  Specifies the SAI Block x Clock source. This parameter is not used for STM32F446xx devices.
- ***uint32_t SAI_InitTypeDef::AudioFrequency***
  Specifies the audio frequency sampling. This parameter can be a value of ***SAI_Audio_Frequency***
- ***uint32_t SAI_InitTypeDef::Mckdiv***
  Specifies the master clock divider, the parameter will be used if for AudioFrequency the user choice This parameter must be a number between Min_Data = 0 and Max_Data = 15
- ***uint32_t SAI_InitTypeDef::MonoStereoMode***
  Specifies if the mono or stereo mode is selected. This parameter can be a value of ***SAI_Mono_Stereo_Mode***
- ***uint32_t SAI_InitTypeDef::CompandingMode***
  Specifies the companding mode type. This parameter can be a value of ***SAI_Block_Companding_Mode***
- ***uint32_t SAI_InitTypeDef::TriState***
  Specifies the companding mode type. This parameter can be a value of ***SAI_TRIState_Management***
- ***uint32_t SAI_InitTypeDef::Protocol***
  Specifies the SAI Block protocol. This parameter can be a value of ***SAI_Block_Protocol***
- ***uint32_t SAI_InitTypeDef::DataSize***
  Specifies the SAI Block data size. This parameter can be a value of ***SAI_Block_Data_Size***
- ***uint32_t SAI_InitTypeDef::FirstBit***
  Specifies whether data transfers start from MSB or LSB bit. This parameter can be a value of ***SAI_Block_MSB_LSB_transmission***
- ***uint32_t SAI_InitTypeDef::ClockStrobing***
  Specifies the SAI Block clock strobing edge sensitivity. This parameter can be a value of ***SAI_Block_Clock_Strobing***

54.1.2 SAI_FrameInitTypeDef

Data Fields

- ***uint32_t FrameLength***
- ***uint32_t ActiveFrameLength***
- ***uint32_t FSDefinition***
- ***uint32_t FSPolarity***
- ***uint32_t FSOffset***

Field Documentation

- ***uint32_t SAI_FrameInitTypeDef::FrameLength***
  Specifies the Frame length, the number of SCK clocks for each audio frame. This parameter must be a number between Min_Data = 8 and Max_Data = 256.

Note: If master clock MCLK_x pin is declared as an output, the frame length should be aligned to a number equal to power of 2 in order to keep in an audio frame, an integer number of MCLK pulses by bit Clock.

- ***uint32_t SAI_FrameInitTypeDef::ActiveFrameLength***
  Specifies the Frame synchronization active level length. This Parameter specifies the length in number of bit clock (SCK + 1) of the active level of FS signal in audio frame. This parameter must be a number between Min_Data = 1 and Max_Data = 128
- ***uint32_t SAI_FrameInitTypeDef::FSDefinition***
  Specifies the Frame synchronization definition. This parameter can be a value of ***SAI_Block_FS_Definition***
- ***uint32_t SAI_FrameInitTypeDef::FSPolarity***
  Specifies the Frame synchronization Polarity. This parameter can be a value of ***SAI_Block_FS_Polarity***
- ***uint32_t SAI_FrameInitTypeDef::FSOffset***
  Specifies the Frame synchronization Offset. This parameter can be a value of ***SAI_Block_FS_Offset***

54.1.3 SAI_SlotInitTypeDef

Data Fields

- ***uint32_t FirstBitOffset***
- ***uint32_t SlotSize***
- ***uint32_t SlotNumber***
- ***uint32_t SlotActive***

Field Documentation

- ***uint32_t SAI_SlotInitTypeDef::FirstBitOffset***
  Specifies the position of first data transfer bit in the slot. This parameter must be a number between Min_Data = 0 and Max_Data = 24
- ***uint32_t SAI_SlotInitTypeDef::SlotSize***
  Specifies the Slot Size. This parameter can be a value of ***SAI_Block_Slot_Size***
- ***uint32_t SAI_SlotInitTypeDef::SlotNumber***
  Specifies the number of slot in the audio frame. This parameter must be a number between Min_Data = 1 and Max_Data = 16
- ***uint32_t SAI_SlotInitTypeDef::SlotActive***
  Specifies the slots in audio frame that will be activated. This parameter can be a value of ***SAI_Block_Slot_Active***

54.1.4 __SAI_HandleTypeDef

Data Fields

- ***SAI_Block_TypeDef * Instance***
- ***SAI_InitTypeDef Init***
- ***SAI_FrameInitTypeDef FrameInit***
- ***SAI_SlotInitTypeDef SlotInit***
- ***uint8_t * pBuffPtr***
- ***uint16_t XferSize***

- ***uint16_t XferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***SAIcallback mutecallback***
- ***void(* InterruptServiceRoutine***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_SAI_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***SAI_Block_TypeDef* __SAI_HandleTypeDef::Instance***
  SAI Blockx registers base address
- ***SAI_InitTypeDef __SAI_HandleTypeDef::Init***
  SAI communication parameters
- ***SAI_FrameInitTypeDef __SAI_HandleTypeDef::FrameInit***
  SAI Frame configuration parameters
- ***SAI_SlotInitTypeDef __SAI_HandleTypeDef::SlotInit***
  SAI Slot configuration parameters
- ***uint8_t* __SAI_HandleTypeDef::pBuffPtr***
  Pointer to SAI transfer Buffer
- ***uint16_t __SAI_HandleTypeDef::XferSize***
  SAI transfer size
- ***uint16_t __SAI_HandleTypeDef::XferCount***
  SAI transfer counter
- ***DMA_HandleTypeDef* __SAI_HandleTypeDef::hdmatx***
  SAI Tx DMA handle parameters
- ***DMA_HandleTypeDef* __SAI_HandleTypeDef::hdmarx***
  SAI Rx DMA handle parameters
- ***SAIcallback __SAI_HandleTypeDef::mutecallback***
  SAI mute callback
- ***void(* __SAI_HandleTypeDef::InterruptServiceRoutine)(struct __SAI_HandleTypeDef *hsai)***
- ***HAL_LockTypeDef __SAI_HandleTypeDef::Lock***
  SAI locking object
- ***__IO HAL_SAI_StateTypeDef __SAI_HandleTypeDef::State***
  SAI communication state
- ***__IO uint32_t __SAI_HandleTypeDef::ErrorCode***
  SAI Error code

54.2 SAI Firmware driver API description

54.2.1 How to use this driver

The SAI HAL driver can be used as follows:

1. Declare a SAI_HandleTypeDef handle structure (eg. SAI_HandleTypeDef hsai).
2. Initialize the SAI low level resources by implementing the HAL_SAI_MspInit() API:
   a. Enable the SAI interface clock.
   b. SAI pins configuration:
      – Enable the clock for the SAI GPIOs.
      – Configure these SAI pins as alternate function pull-up.

c. NVIC configuration if you need to use interrupt process (HAL_SAI_Transmit_IT() and HAL_SAI_Receive_IT() APIs):
   – Configure the SAI interrupt priority.
   – Enable the NVIC SAI IRQ handle.

d. DMA Configuration if you need to use DMA process (HAL_SAI_Transmit_DMA() and HAL_SAI_Receive_DMA() APIs):
   – Declare a DMA handle structure for the Tx/Rx stream.
   – Enable the DMAx interface clock.
   – Configure the declared DMA handle structure with the required Tx/Rx parameters.
   – Configure the DMA Tx/Rx Stream.
   – Associate the initialized DMA handle to the SAI DMA Tx/Rx handle.
   – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.

3. The initialization can be done by two ways
   a. Expert mode : Initialize the structures Init, FrameInit and SlotInit and call HAL_SAI_Init().
   b. Simplified mode : Initialize the high part of Init Structure and call HAL_SAI_InitProtocol().

The specific SAI interrupts (FIFO request and Overrun underrun interrupt) will be managed using the macros __HAL_SAI_ENABLE_IT() and __HAL_SAI_DISABLE_IT() inside the transmit and receive process.

SAI Clock Source configuration is managed differently depending on the selected STM32F4 devices :

- For STM32F446xx devices, the configuration is managed through RCCEx_PeriphCLKConfig() function in the HAL RCC drivers
- For STM32F439xx/STM32F437xx/STM32F429xx/STM32F427xx devices, the configuration is managed within HAL SAI drivers through HAL_SAI_Init() function using ClockSource field of SAI_InitTypeDef structure.

Make sure that either:

- I2S PLL is configured or
- SAI PLL is configured or
- External clock source is configured after setting correctly the define constant EXTERNAL_CLOCK_VALUE in the stm32f4xx_hal_conf.h file.

In master Tx mode: enabling the audio block immediately generates the bit clock for the external slaves even if there is no data in the FIFO, However FS signal generation is conditioned by the presence of data in the FIFO.

In master Rx mode: enabling the audio block immediately generates the bit clock and FS signal for the external slaves.

It is mandatory to respect the following conditions in order to avoid bad SAI behavior:

- First bit Offset <= (SLOT size - Data size)
- Data size <= SLOT size
- Number of SLOT x SLOT size = Frame length
- The number of slots should be even when SAI_FS_CHANNEL_IDENTIFICATION is selected.

Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_SAI_Transmit()
- Receive an amount of data in blocking mode using HAL_SAI_Receive()

Interrupt mode IO operation

- Send an amount of data in non-blocking mode using HAL_SAI_Transmit_IT()
- At transmission end of transfer HAL_SAI_TxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_SAI_TxCpltCallback()
- Receive an amount of data in non-blocking mode using HAL_SAI_Receive_IT()
- At reception end of transfer HAL_SAI_RxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_SAI_RxCpltCallback()
- In case of flag error, HAL_SAI_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SAI_ErrorCallback()

DMA mode IO operation

- Send an amount of data in non-blocking mode (DMA) using HAL_SAI_Transmit_DMA()
- At transmission end of transfer HAL_SAI_TxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_SAI_TxCpltCallback()
- Receive an amount of data in non-blocking mode (DMA) using HAL_SAI_Receive_DMA()
- At reception end of transfer HAL_SAI_RxCpltCallback() is executed and user can add his own code by customization of function pointer HAL_SAI_RxCpltCallback()
- In case of flag error, HAL_SAI_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SAI_ErrorCallback()
- Pause the DMA Transfer using HAL_SAI_DMAPause()
- Resume the DMA Transfer using HAL_SAI_DMAResume()
- Stop the DMA Transfer using HAL_SAI_DMAStop()

SAI HAL driver additional function list

Below the list the others API available SAI HAL driver :

- HAL_SAI_EnableTxMuteMode(): Enable the mute in tx mode
- HAL_SAI_DisableTxMuteMode(): Disable the mute in tx mode
- HAL_SAI_EnableRxMuteMode(): Enable the mute in Rx mode
- HAL_SAI_DisableRxMuteMode(): Disable the mute in Rx mode

- HAL_SAI_FlushRxFifo(): Flush the rx fifo.
- HAL_SAI_Abort(): Abort the current transfer

SAI HAL driver macros list

Below the list of most used macros in SAI HAL driver :

- __HAL_SAI_ENABLE(): Enable the SAI peripheral
- __HAL_SAI_DISABLE(): Disable the SAI peripheral
- __HAL_SAI_ENABLE_IT(): Enable the specified SAI interrupts
- __HAL_SAI_DISABLE_IT(): Disable the specified SAI interrupts
- __HAL_SAI_GET_IT_SOURCE(): Check if the specified SAI interrupt source is enabled or disabled
- __HAL_SAI_GET_FLAG(): Check whether the specified SAI flag is set or not

54.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and de-initialize the SAIx peripheral:

- User must implement HAL_SAI_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).
- Call the function HAL_SAI_Init() to configure the selected device with the selected configuration:
  - Mode (Master/slave TX/RX)
  - Protocol
  - Data Size
  - MCLK Output
  - Audio frequency
  - FIFO Threshold
  - Frame Config
  - Slot Config
- Call the function HAL_SAI_DeInit() to restore the default configuration of the selected SAI peripheral.

This section contains the following APIs:

- ***HAL_SAI_InitProtocol()***
- ***HAL_SAI_Init()***
- ***HAL_SAI_DeInit()***
- ***HAL_SAI_MspInit()***
- ***HAL_SAI_MspDeInit()***

54.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the SAI data transfers.

- There are two modes of transfer:
  - Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
  - No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer startup. The end of the data processing will be indicated through the dedicated SAI IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
- Blocking mode functions are :
  - HAL_SAI_Transmit()

  - HAL_SAI_Receive()
  - HAL_SAI_TransmitReceive()
- Non Blocking mode functions with Interrupt are :
  - HAL_SAI_Transmit_IT()
  - HAL_SAI_Receive_IT()
  - HAL_SAI_TransmitReceive_IT()
- Non Blocking mode functions with DMA are :
  - HAL_SAI_Transmit_DMA()
  - HAL_SAI_Receive_DMA()
  - HAL_SAI_TransmitReceive_DMA()
- A set of Transfer Complete Callbacks are provided in non Blocking mode:
  - HAL_SAI_TxCpltCallback()
  - HAL_SAI_RxCpltCallback()
  - HAL_SAI_ErrorCallback()

This section contains the following APIs:

- ***HAL_SAI_Transmit()***
- ***HAL_SAI_Receive()***
- ***HAL_SAI_Transmit_IT()***
- ***HAL_SAI_Receive_IT()***
- ***HAL_SAI_DMAPause()***
- ***HAL_SAI_DMAResume()***
- ***HAL_SAI_DMAStop()***
- ***HAL_SAI_Abort()***
- ***HAL_SAI_Transmit_DMA()***
- ***HAL_SAI_Receive_DMA()***
- ***HAL_SAI_EnableTxMuteMode()***
- ***HAL_SAI_DisableTxMuteMode()***
- ***HAL_SAI_EnableRxMuteMode()***
- ***HAL_SAI_DisableRxMuteMode()***
- ***HAL_SAI_IRQHandler()***
- ***HAL_SAI_TxCpltCallback()***
- ***HAL_SAI_TxHalfCpltCallback()***
- ***HAL_SAI_RxCpltCallback()***
- ***HAL_SAI_RxHalfCpltCallback()***
- ***HAL_SAI_ErrorCallback()***

54.2.4 Peripheral State and Errors functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_SAI_GetState()***
- ***HAL_SAI_GetError()***

54.2.5 Detailed description of functions

HAL_SAI_InitProtocol

Function Name	**HAL_StatusTypeDef HAL_SAI_InitProtocol (SAI_HandleTypeDef * hsai, uint32_t protocol, uint32_t datasize, uint32_t nbslot)**
Function Description	Initialize the structure FrameInit, SlotInit and the low part of Init

according to the specified parameters and call the function HAL_SAI_Init to initialize the SAI block.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
- **protocol:** one of the supported protocol SAI Supported protocol
- **datasize:** one of the supported datasize SAI protocol data size the configuration information for SAI module.
- **nbslot:** Number of slot.

Return values
- **HAL:** status

HAL_SAI_Init

Function Name **HAL_StatusTypeDef HAL_SAI_Init (SAI_HandleTypeDef * hsai)**

Function Description Initialize the SAI according to the specified parameters.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_DeInit

Function Name **HAL_StatusTypeDef HAL_SAI_DeInit (SAI_HandleTypeDef * hsai)**

Function Description DeInitialize the SAI peripheral.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_MspInit

Function Name **void HAL_SAI_MspInit (SAI_HandleTypeDef * hsai)**

Function Description Initialize the SAI MSP.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_MspDeInit

Function Name **void HAL_SAI_MspDeInit (SAI_HandleTypeDef * hsai)**

Function Description DeInitialize the SAI MSP.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_Transmit

Function Name	**HAL_StatusTypeDef HAL_SAI_Transmit (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Transmit an amount of data in blocking mode.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_SAI_Receive

Function Name	**HAL_StatusTypeDef HAL_SAI_Receive (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size, uint32_t Timeout)**
Function Description	Receive an amount of data in blocking mode.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_SAI_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_SAI_Transmit_IT (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size)**
Function Description	Transmit an amount of data in non-blocking mode with Interrupt.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_SAI_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_SAI_Receive_IT (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size)**
Function Description	Receive an amount of data in non-blocking mode with Interrupt.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status

HAL_SAI_Transmit_DMA

Function Name	**HAL_StatusTypeDef HAL_SAI_Transmit_DMA (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size)**
Function Description	Transmit an amount of data in non-blocking mode with DMA.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_SAI_Receive_DMA

Function Name	**HAL_StatusTypeDef HAL_SAI_Receive_DMA (SAI_HandleTypeDef * hsai, uint8_t * pData, uint16_t Size)**
Function Description	Receive an amount of data in non-blocking mode with DMA.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status

HAL_SAI_DMAPause

Function Name	**HAL_StatusTypeDef HAL_SAI_DMAPause (SAI_HandleTypeDef * hsai)**
Function Description	Pause the audio stream playing from the Media.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **HAL:** status

HAL_SAI_DMAResume

Function Name	**HAL_StatusTypeDef HAL_SAI_DMAResume (SAI_HandleTypeDef * hsai)**
Function Description	Resume the audio stream playing from the Media.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **HAL:** status

HAL_SAI_DMAStop

Function Name	**HAL_StatusTypeDef HAL_SAI_DMAStop (SAI_HandleTypeDef * hsai)**
Function Description	Stop the audio stream playing from the Media.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_Abort

Function Name **HAL_StatusTypeDef HAL_SAI_Abort (SAI_HandleTypeDef * hsai)**

Function Description Abort the current transfer and disable the SAI.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_EnableTxMuteMode

Function Name **HAL_StatusTypeDef HAL_SAI_EnableTxMuteMode (SAI_HandleTypeDef * hsai, uint16_t val)**

Function Description Enable the Tx mute mode.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
- **val:** value sent during the mute SAI Block Mute Value

Return values
- **HAL:** status

HAL_SAI_DisableTxMuteMode

Function Name **HAL_StatusTypeDef HAL_SAI_DisableTxMuteMode (SAI_HandleTypeDef * hsai)**

Function Description Disable the Tx mute mode.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_EnableRxMuteMode

Function Name **HAL_StatusTypeDef HAL_SAI_EnableRxMuteMode (SAI_HandleTypeDef * hsai, SAIcallback callback, uint16_t counter)**

Function Description Enable the Rx mute detection.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
- **callback:** function called when the mute is detected.
- **counter:** number a data before mute detection max 63.

Return values
- **HAL:** status

HAL_SAI_DisableRxMuteMode

Function Name **HAL_StatusTypeDef HAL_SAI_DisableRxMuteMode (SAI_HandleTypeDef * hsai)**

Function Description Disable the Rx mute detection.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **HAL:** status

HAL_SAI_IRQHandler

Function Name **void HAL_SAI_IRQHandler (SAI_HandleTypeDef * hsai)**

Function Description Handle SAI interrupt request.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_TxHalfCpltCallback

Function Name **void HAL_SAI_TxHalfCpltCallback (SAI_HandleTypeDef * hsai)**

Function Description Tx Transfer Half completed callback.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_TxCpltCallback

Function Name **void HAL_SAI_TxCpltCallback (SAI_HandleTypeDef * hsai)**

Function Description Tx Transfer completed callback.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_RxHalfCpltCallback

Function Name **void HAL_SAI_RxHalfCpltCallback (SAI_HandleTypeDef * hsai)**

Function Description Rx Transfer half completed callback.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_RxCpltCallback

Function Name **void HAL_SAI_RxCpltCallback (SAI_HandleTypeDef * hsai)**

Function Description Rx Transfer completed callback.

Parameters
- **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.

Return values
- **None**

HAL_SAI_ErrorCallback

Function Name	**void HAL_SAI_ErrorCallback (SAI_HandleTypeDef * hsai)**
Function Description	SAI error callback.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **None**

HAL_SAI_GetState

Function Name	**HAL_SAI_StateTypeDef HAL_SAI_GetState (SAI_HandleTypeDef * hsai)**
Function Description	Return the SAI handle state.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **HAL:** state

HAL_SAI_GetError

Function Name	**uint32_t HAL_SAI_GetError (SAI_HandleTypeDef * hsai)**
Function Description	Return the SAI error code.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for the specified SAI Block.
Return values	• **SAI:** Error Code

54.3 SAI Firmware driver defines

54.3.1 SAI

SAI Audio Frequency

SAI_AUDIO_FREQUENCY_192K

SAI_AUDIO_FREQUENCY_96K

SAI_AUDIO_FREQUENCY_48K

SAI_AUDIO_FREQUENCY_44K

SAI_AUDIO_FREQUENCY_32K

SAI_AUDIO_FREQUENCY_22K

SAI_AUDIO_FREQUENCY_16K

SAI_AUDIO_FREQUENCY_11K

SAI_AUDIO_FREQUENCY_8K

SAI_AUDIO_FREQUENCY_MCKDIV

SAI Block Clock Strobing

SAI_CLOCKSTROBING_FALLINGEDGE

SAI_CLOCKSTROBING_RISINGEDGE

SAI Block Companding Mode

SAI_NOCOMPANDING

SAI_ULAW_1CPL_COMPANDING

SAI_ALAW_1CPL_COMPANDING

SAI_ULAW_2CPL_COMPANDING

SAI_ALAW_2CPL_COMPANDING

SAI Block Data Size

SAI_DATASIZE_8

SAI_DATASIZE_10

SAI_DATASIZE_16

SAI_DATASIZE_20

SAI_DATASIZE_24

SAI_DATASIZE_32

SAI Block Fifo Status Level

SAI_FIFOSTATUS_EMPTY

SAI_FIFOSTATUS_LESS1QUARTERFULL

SAI_FIFOSTATUS_1QUARTERFULL

SAI_FIFOSTATUS_HALFFULL

SAI_FIFOSTATUS_3QUARTERFULL

SAI_FIFOSTATUS_FULL

SAI Block Fifo Threshold

SAI_FIFOTHRESHOLD_EMPTY

SAI_FIFOTHRESHOLD_1QF

SAI_FIFOTHRESHOLD_HF

SAI_FIFOTHRESHOLD_3QF

SAI_FIFOTHRESHOLD_FULL

SAI Block Flags Definition

SAI_FLAG_OVRUDR

SAI_FLAG_MUTEDET

SAI_FLAG_WCKCFG

SAI_FLAG_FREQ

SAI_FLAG_CNRDY

SAI_FLAG_AFSDET

SAI_FLAG_LFSDET

SAI Block FS Definition

SAI_FS_STARTFRAME

SAI_FS_CHANNEL_IDENTIFICATION

SAI Block FS Offset

SAI_FS_FIRSTBIT

SAI_FS_BEFOREFIRSTBIT

SAI Block FS Polarity

SAI_FS_ACTIVE_LOW

SAI_FS_ACTIVE_HIGH

SAI Block Interrupts Definition

SAI_IT_OVRUDR

SAI_IT_MUTEDET

SAI_IT_WCKCFG

SAI_IT_FREQ

SAI_IT_CNRDY

SAI_IT_AFSDET

SAI_IT_LFSDET

SAI Block Mode

SAI_MODEMASTER_TX

SAI_MODEMASTER_RX

SAI_MODESLAVE_TX

SAI_MODESLAVE_RX

SAI Block MSB LSB transmission

SAI_FIRSTBIT_MSB

SAI_FIRSTBIT_LSB

SAI Block Mute Value

SAI_ZERO_VALUE

SAI_LAST_SENT_VALUE

SAI Block NoDivider

SAI_MASTERDIVIDER_ENABLE

SAI_MASTERDIVIDER_DISABLE

SAI Block Output Drive

SAI_OUTPUTDRIVE_DISABLE

SAI_OUTPUTDRIVE_ENABLE

SAI Block Protocol

SAI_FREE_PROTOCOL

SAI_SPDIF_PROTOCOL

SAI_AC97_PROTOCOL

SAI Block Slot Active

SAI_SLOT_NOTACTIVE
SAI_SLOTACTIVE_0
SAI_SLOTACTIVE_1
SAI_SLOTACTIVE_2
SAI_SLOTACTIVE_3
SAI_SLOTACTIVE_4
SAI_SLOTACTIVE_5
SAI_SLOTACTIVE_6
SAI_SLOTACTIVE_7
SAI_SLOTACTIVE_8
SAI_SLOTACTIVE_9
SAI_SLOTACTIVE_10
SAI_SLOTACTIVE_11
SAI_SLOTACTIVE_12
SAI_SLOTACTIVE_13
SAI_SLOTACTIVE_14
SAI_SLOTACTIVE_15
SAI_SLOTACTIVE_ALL

SAI Block Slot Size

SAI_SLOTSIZE_DATASIZE
SAI_SLOTSIZE_16B
SAI_SLOTSIZE_32B

SAI External synchronisation

SAI_SYNCEXT_DISABLE
SAI_SYNCEXT_OUTBLOCKA_ENABLE
SAI_SYNCEXT_OUTBLOCKB_ENABLE

SAI Block Synchronization

SAI_ASYNCHRONOUS	Asynchronous
SAI_SYNCHRONOUS	Synchronous with other block of same SAI
SAI_SYNCHRONOUS_EXT_SAI1	Synchronous with other SAI, SAI1
SAI_SYNCHRONOUS_EXT_SAI2	Synchronous with other SAI, SAI2

SAI Clock Source

SAI_CLKSOURCE_PLLSAI
SAI_CLKSOURCE_PLLI2S
SAI_CLKSOURCE_EXT
SAI_CLKSOURCE_NA No applicable for STM32F446xx

SAI Error Code

HAL_SAI_ERROR_NONE	No error
HAL_SAI_ERROR_OVR	Overrun Error
HAL_SAI_ERROR_UDR	Underrun error
HAL_SAI_ERROR_AFSDET	Anticipated Frame synchronisation detection
HAL_SAI_ERROR_LFSDET	Late Frame synchronisation detection
HAL_SAI_ERROR_CNREADY	codec not ready
HAL_SAI_ERROR_WCKCFG	Wrong clock configuration
HAL_SAI_ERROR_TIMEOUT	Timeout error
HAL_SAI_ERROR_DMA	DMA error

SAI Exported Macros

__HAL_SAI_RESET_HANDLE_STATE

Description:

- Reset SAI handle state.

Parameters:

- __HANDLE__: specifies the SAI Handle.

Return value:

- NoneS

__HAL_SAI_ENABLE_IT

Description:

- Enable or disable the specified SAI interrupts.

Parameters:

- __HANDLE__: specifies the SAI Handle.
- __INTERRUPT__: specifies the interrupt source to enable or disable. This parameter can be one of the following values:
  - SAI_IT_OVRUDR: Overrun underrun interrupt enable
  - SAI_IT_MUTEDET: Mute detection interrupt enable
  - SAI_IT_WCKCFG: Wrong Clock Configuration interrupt enable
  - SAI_IT_FREQ: FIFO request interrupt enable
  - SAI_IT_CNRDY: Codec not ready interrupt enable
  - SAI_IT_AFSDET: Anticipated frame synchronization detection interrupt enable
  - SAI_IT_LFSDET: Late frame synchronization detection interrupt enable

Return value:

- None

__HAL_SAI_DISABLE_IT

__HAL_SAI_GET_IT_SOURCE

Description:

- Check if the specified SAI interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: specifies the SAI Handle. This parameter can be SAI where x: 1, 2, or 3 to select the SAI peripheral.
- __INTERRUPT__: specifies the SAI interrupt source to check. This parameter can be one of the following values:
  - SAI_IT_OVRUDR: Overrun underrun interrupt enable
  - SAI_IT_MUTEDET: Mute detection interrupt enable
  - SAI_IT_WCKCFG: Wrong Clock Configuration interrupt enable
  - SAI_IT_FREQ: FIFO request interrupt enable
  - SAI_IT_CNRDY: Codec not ready interrupt enable
  - SAI_IT_AFSDET: Anticipated frame synchronization detection interrupt enable
  - SAI_IT_LFSDET: Late frame synchronization detection interrupt enable

Return value:

- The: new state of __INTERRUPT__ (TRUE or FALSE).

__HAL_SAI_GET_FLAG

Description:

- Check whether the specified SAI flag is set or not.

Parameters:

- __HANDLE__: specifies the SAI Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - SAI_FLAG_OVRUDR: Overrun underrun flag.
  - SAI_FLAG_MUTEDET: Mute detection flag.
  - SAI_FLAG_WCKCFG: Wrong Clock Configuration flag.
  - SAI_FLAG_FREQ: FIFO request flag.
  - SAI_FLAG_CNRDY: Codec not ready flag.
  - SAI_FLAG_AFSDET: Anticipated frame synchronization detection flag.

– SAI_FLAG_LFSDET: Late frame synchronization detection flag.

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_SAI_CLEAR_FLAG

Description:

- Clear the specified SAI pending flag.

Parameters:

- __HANDLE__: specifies the SAI Handle.
- __FLAG__: specifies the flag to check. This parameter can be any combination of the following values:
  – SAI_FLAG_OVRUDR: Clear Overrun underrun
  – SAI_FLAG_MUTEDET: Clear Mute detection
  – SAI_FLAG_WCKCFG: Clear Wrong Clock Configuration
  – SAI_FLAG_FREQ: Clear FIFO request
  – SAI_FLAG_CNRDY: Clear Codec not ready
  – SAI_FLAG_AFSDET: Clear Anticipated frame synchronization detection
  – SAI_FLAG_LFSDET: Clear Late frame synchronization detection

Return value:

- None

__HAL_SAI_ENABLE

Description:

- Enable SAI.

Parameters:

- __HANDLE__: specifies the SAI Handle.

Return value:

- None

__HAL_SAI_DISABLE

Description:

- Disable SAI.

Parameters:

- __HANDLE__: specifies the SAI Handle.

Return value:

- None

SAI Mono Stereo Mode

SAI_STEREOMODE

SAI_MONOMODE

SAI Supported protocol

SAI_I2S_STANDARD

SAI_I2S_MSBJUSTIFIED

SAI_I2S_LSBJUSTIFIED

SAI_PCM_LONG

SAI_PCM_SHORT

SAI protocol data size

SAI_PROTOCOL_DATASIZE_16BIT

SAI_PROTOCOL_DATASIZE_16BITEXTENDED

SAI_PROTOCOL_DATASIZE_24BIT

SAI_PROTOCOL_DATASIZE_32BIT

SAI TRIState Management

SAI_OUTPUT_NOTRELEASED

SAI_OUTPUT_RELEASED

55 HAL SAI Extension Driver

55.1 SAIEx Firmware driver API description

55.1.1 SAI peripheral extension features

Comparing to other previous devices, the SAI interface for STM32F446xx devices contains the following additional features :

- Possibility to be clocked from PLLR

55.1.2 How to use this driver

This driver provides functions to manage several sources to clock SAI

55.1.3 Extension features Functions

This subsection provides a set of functions allowing to manage the possible SAI clock sources.

This section contains the following APIs:

- ***SAI_BlockSynchroConfig()***
- ***SAI_GetInputClock()***

55.1.4 Detailed description of functions

SAI_BlockSynchroConfig

Function Name	**void SAI_BlockSynchroConfig (SAI_HandleTypeDef * hsai)**
Function Description	Configure SAI Block synchronization mode.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **SAI:** Clock Input

SAI_GetInputClock

Function Name	**uint32_t SAI_GetInputClock (SAI_HandleTypeDef * hsai)**
Function Description	Get SAI Input Clock based on SAI source clock selection.
Parameters	• **hsai:** pointer to a SAI_HandleTypeDef structure that contains the configuration information for SAI module.
Return values	• **SAI:** Clock Input

56 HAL SDRAM Generic Driver

56.1 SDRAM Firmware driver registers structures

56.1.1 SDRAM_HandleTypeDef

Data Fields

- ***FMC_SDRAM_TypeDef * Instance***
- ***FMC_SDRAM_InitTypeDef Init***
- ***__IO HAL_SDRAM_StateTypeDef State***
- ***HAL_LockTypeDef Lock***
- ***DMA_HandleTypeDef * hdma***

Field Documentation

- ***FMC_SDRAM_TypeDef* SDRAM_HandleTypeDef::Instance***
  Register base address
- ***FMC_SDRAM_InitTypeDef SDRAM_HandleTypeDef::Init***
  SDRAM device configuration parameters
- ***__IO HAL_SDRAM_StateTypeDef SDRAM_HandleTypeDef::State***
  SDRAM access state
- ***HAL_LockTypeDef SDRAM_HandleTypeDef::Lock***
  SDRAM locking object
- ***DMA_HandleTypeDef* SDRAM_HandleTypeDef::hdma***
  Pointer DMA handler

56.2 SDRAM Firmware driver API description

56.2.1 How to use this driver

This driver is a generic layered driver which contains a set of APIs used to control SDRAM memories. It uses the FMC layer functions to interface with SDRAM devices. The following sequence should be followed to configure the FMC to interface with SDRAM memories:

1. Declare a SDRAM_HandleTypeDef handle structure, for example: SDRAM_HandleTypeDef hdsram
   – Fill the SDRAM_HandleTypeDef handle "Init" field with the allowed values of the structure member.
   – Fill the SDRAM_HandleTypeDef handle "Instance" field with a predefined base register instance for NOR or SDRAM device
2. Declare a FMC_SDRAM_TimingTypeDef structure; for example: FMC_SDRAM_TimingTypeDef Timing; and fill its fields with the allowed values of the structure member.
3. Initialize the SDRAM Controller by calling the function HAL_SDRAM_Init(). This function performs the following sequence:
   a. MSP hardware layer configuration using the function HAL_SDRAM_MspInit()
   b. Control register configuration using the FMC SDRAM interface function FMC_SDRAM_Init()

c. Timing register configuration using the FMC SDRAM interface function FMC_SDRAM_Timing_Init()
d. Program the SDRAM external device by applying its initialization sequence according to the device plugged in your hardware. This step is mandatory for accessing the SDRAM device.

4. At this stage you can perform read/write accesses from/to the memory connected to the SDRAM Bank. You can perform either polling or DMA transfer using the following APIs:
   – HAL_SDRAM_Read()/HAL_SDRAM_Write() for polling read/write access
   – HAL_SDRAM_Read_DMA()/HAL_SDRAM_Write_DMA() for DMA read/write transfer
5. You can also control the SDRAM device by calling the control APIs HAL_SDRAM_WriteOperation_Enable()/ HAL_SDRAM_WriteOperation_Disable() to respectively enable/disable the SDRAM write operation or the function HAL_SDRAM_SendCommand() to send a specified command to the SDRAM device. The command to be sent must be configured with the FMC_SDRAM_CommandTypeDef structure.
6. You can continuously monitor the SDRAM device HAL state by calling the function HAL_SDRAM_GetState()

56.2.2 SDRAM Initialization and de_initialization functions

This section provides functions allowing to initialize/de-initialize the SDRAM memory

This section contains the following APIs:

- ***HAL_SDRAM_Init()***
- ***HAL_SDRAM_DeInit()***
- ***HAL_SDRAM_MspInit()***
- ***HAL_SDRAM_MspDeInit()***
- ***HAL_SDRAM_IRQHandler()***
- ***HAL_SDRAM_RefreshErrorCallback()***
- ***HAL_SDRAM_DMA_XferCpltCallback()***
- ***HAL_SDRAM_DMA_XferErrorCallback()***

56.2.3 SDRAM Input and Output functions

This section provides functions allowing to use and control the SDRAM memory

This section contains the following APIs:

- ***HAL_SDRAM_Read_8b()***
- ***HAL_SDRAM_Write_8b()***
- ***HAL_SDRAM_Read_16b()***
- ***HAL_SDRAM_Write_16b()***
- ***HAL_SDRAM_Read_32b()***
- ***HAL_SDRAM_Write_32b()***
- ***HAL_SDRAM_Read_DMA()***
- ***HAL_SDRAM_Write_DMA()***

56.2.4 SDRAM Control functions

This subsection provides a set of functions allowing to control dynamically the SDRAM interface.

This section contains the following APIs:

- ***HAL_SDRAM_WriteProtection_Enable()***

- ***HAL_SDRAM_WriteProtection_Disable()***
- ***HAL_SDRAM_SendCommand()***
- ***HAL_SDRAM_ProgramRefreshRate()***
- ***HAL_SDRAM_SetAutoRefreshNumber()***
- ***HAL_SDRAM_GetModeStatus()***

56.2.5 SDRAM State functions

This subsection permits to get in run-time the status of the SDRAM controller and the data flow.

This section contains the following APIs:

- ***HAL_SDRAM_GetState()***

56.2.6 Detailed description of functions

HAL_SDRAM_Init

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Init (SDRAM_HandleTypeDef * hsdram, FMC_SDRAM_TimingTypeDef * Timing)**
Function Description	Performs the SDRAM device initialization sequence.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **Timing:** Pointer to SDRAM control timing structure
Return values	• **HAL:** status

HAL_SDRAM_DeInit

Function Name	**HAL_StatusTypeDef HAL_SDRAM_DeInit (SDRAM_HandleTypeDef * hsdram)**
Function Description	Perform the SDRAM device initialization sequence.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **HAL:** status

HAL_SDRAM_MspInit

Function Name	**void HAL_SDRAM_MspInit (SDRAM_HandleTypeDef * hsdram)**
Function Description	SDRAM MSP Init.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **None**

HAL_SDRAM_MspDeInit

Function Name	**void HAL_SDRAM_MspDeInit (SDRAM_HandleTypeDef * hsdram)**

Function Description	SDRAM MSP DeInit.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **None**

HAL_SDRAM_IRQHandler

Function Name	**void HAL_SDRAM_IRQHandler (SDRAM_HandleTypeDef * hsdram)**
Function Description	This function handles SDRAM refresh error interrupt request.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **HAL:** status

HAL_SDRAM_RefreshErrorCallback

Function Name	**void HAL_SDRAM_RefreshErrorCallback (SDRAM_HandleTypeDef * hsdram)**
Function Description	SDRAM Refresh error callback.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **None**

HAL_SDRAM_DMA_XferCpltCallback

Function Name	**void HAL_SDRAM_DMA_XferCpltCallback (DMA_HandleTypeDef * hdma)**
Function Description	DMA transfer complete callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

HAL_SDRAM_DMA_XferErrorCallback

Function Name	**void HAL_SDRAM_DMA_XferErrorCallback (DMA_HandleTypeDef * hdma)**
Function Description	DMA transfer complete error callback.
Parameters	• **hdma:** DMA handle
Return values	• **None**

HAL_SDRAM_Read_8b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Read_8b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint8_t * pDstBuffer, uint32_t BufferSize)**

Function Description	Reads 8-bit data buffer from the SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to read start address • **pDstBuffer:** Pointer to destination buffer • **BufferSize:** Size of the buffer to read from memory
Return values	• **HAL:** status

HAL_SDRAM_Write_8b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Write_8b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint8_t * pSrcBuffer, uint32_t BufferSize)**
Function Description	Writes 8-bit data buffer to SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to write start address • **pSrcBuffer:** Pointer to source buffer to write • **BufferSize:** Size of the buffer to write to memory
Return values	• **HAL:** status

HAL_SDRAM_Read_16b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Read_16b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint16_t * pDstBuffer, uint32_t BufferSize)**
Function Description	Reads 16-bit data buffer from the SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to read start address • **pDstBuffer:** Pointer to destination buffer • **BufferSize:** Size of the buffer to read from memory
Return values	• **HAL:** status

HAL_SDRAM_Write_16b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Write_16b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint16_t * pSrcBuffer, uint32_t BufferSize)**
Function Description	Writes 16-bit data buffer to SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to write start address • **pSrcBuffer:** Pointer to source buffer to write • **BufferSize:** Size of the buffer to write to memory
Return values	• **HAL:** status

HAL_SDRAM_Read_32b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Read_32b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint32_t * pDstBuffer, uint32_t BufferSize)**
Function Description	Reads 32-bit data buffer from the SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to read start address • **pDstBuffer:** Pointer to destination buffer • **BufferSize:** Size of the buffer to read from memory
Return values	• **HAL:** status

HAL_SDRAM_Write_32b

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Write_32b (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint32_t * pSrcBuffer, uint32_t BufferSize)**
Function Description	Writes 32-bit data buffer to SDRAM memory.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to write start address • **pSrcBuffer:** Pointer to source buffer to write • **BufferSize:** Size of the buffer to write to memory
Return values	• **HAL:** status

HAL_SDRAM_Read_DMA

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Read_DMA (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint32_t * pDstBuffer, uint32_t BufferSize)**
Function Description	Reads a Words data from the SDRAM memory using DMA transfer.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **pAddress:** Pointer to read start address • **pDstBuffer:** Pointer to destination buffer • **BufferSize:** Size of the buffer to read from memory
Return values	• **HAL:** status

HAL_SDRAM_Write_DMA

Function Name	**HAL_StatusTypeDef HAL_SDRAM_Write_DMA (SDRAM_HandleTypeDef * hsdram, uint32_t * pAddress, uint32_t * pSrcBuffer, uint32_t BufferSize)**
Function Description	Writes a Words data buffer to SDRAM memory using DMA transfer.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.

- **pAddress:** Pointer to write start address
- **pSrcBuffer:** Pointer to source buffer to write
- **BufferSize:** Size of the buffer to write to memory

Return values
- **HAL:** status

HAL_SDRAM_WriteProtection_Enable

Function Name **HAL_StatusTypeDef HAL_SDRAM_WriteProtection_Enable (SDRAM_HandleTypeDef * hsdram)**

Function Description Enables dynamically SDRAM write protection.

Parameters
- **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.

Return values
- **HAL:** status

HAL_SDRAM_WriteProtection_Disable

Function Name **HAL_StatusTypeDef HAL_SDRAM_WriteProtection_Disable (SDRAM_HandleTypeDef * hsdram)**

Function Description Disables dynamically SDRAM write protection.

Parameters
- **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.

Return values
- **HAL:** status

HAL_SDRAM_SendCommand

Function Name **HAL_StatusTypeDef HAL_SDRAM_SendCommand (SDRAM_HandleTypeDef * hsdram, FMC_SDRAM_CommandTypeDef * Command, uint32_t Timeout)**

Function Description Sends Command to the SDRAM bank.

Parameters
- **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
- **Command:** SDRAM command structure
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SDRAM_ProgramRefreshRate

Function Name **HAL_StatusTypeDef HAL_SDRAM_ProgramRefreshRate (SDRAM_HandleTypeDef * hsdram, uint32_t RefreshRate)**

Function Description Programs the SDRAM Memory Refresh rate.

Parameters
- **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
- **RefreshRate:** The SDRAM refresh rate value

Return values
- **HAL:** status

HAL_SDRAM_SetAutoRefreshNumber

Function Name	**HAL_StatusTypeDef HAL_SDRAM_SetAutoRefreshNumber (SDRAM_HandleTypeDef * hsdram, uint32_t AutoRefreshNumber)**
Function Description	Sets the Number of consecutive SDRAM Memory auto Refresh commands.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module. • **AutoRefreshNumber:** The SDRAM auto Refresh number
Return values	• **HAL:** status

HAL_SDRAM_GetModeStatus

Function Name	**uint32_t HAL_SDRAM_GetModeStatus (SDRAM_HandleTypeDef * hsdram)**
Function Description	Returns the SDRAM memory current mode.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **The:** SDRAM memory mode.

HAL_SDRAM_GetState

Function Name	**HAL_SDRAM_StateTypeDef HAL_SDRAM_GetState (SDRAM_HandleTypeDef * hsdram)**
Function Description	Returns the SDRAM state.
Parameters	• **hsdram:** pointer to a SDRAM_HandleTypeDef structure that contains the configuration information for SDRAM module.
Return values	• **HAL:** state

56.3 SDRAM Firmware driver defines

56.3.1 SDRAM

SDRAM Exported Macros

__HAL_SDRAM_RESET_HANDLE_STATE

Description:

- Reset SDRAM handle state.

Parameters:

- __HANDLE__: specifies the SDRAM handle.

Return value:

- None

57 HAL SD Generic Driver

57.1 SD Firmware driver registers structures

57.1.1 SD_HandleTypeDef

Data Fields

- ***SD_TypeDef * Instance***
- ***SD_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***uint32_t CardType***
- ***uint32_t RCA***
- ***uint32_t CSD***
- ***uint32_t CID***
- ***__IO uint32_t SdTransferCplt***
- ***__IO uint32_t SdTransferErr***
- ***__IO uint32_t DmaTransferCplt***
- ***__IO uint32_t SdOperation***
- ***DMA_HandleTypeDef * hdmarx***
- ***DMA_HandleTypeDef * hdmatx***

Field Documentation

- ***SD_TypeDef* SD_HandleTypeDef::Instance***
  SDIO register base address
- ***SD_InitTypeDef SD_HandleTypeDef::Init***
  SD required parameters
- ***HAL_LockTypeDef SD_HandleTypeDef::Lock***
  SD locking object
- ***uint32_t SD_HandleTypeDef::CardType***
  SD card type
- ***uint32_t SD_HandleTypeDef::RCA***
  SD relative card address
- ***uint32_t SD_HandleTypeDef::CSD[4]***
  SD card specific data table
- ***uint32_t SD_HandleTypeDef::CID[4]***
  SD card identification number table
- ***__IO uint32_t SD_HandleTypeDef::SdTransferCplt***
  SD transfer complete flag in non blocking mode
- ***__IO uint32_t SD_HandleTypeDef::SdTransferErr***
  SD transfer error flag in non blocking mode
- ***__IO uint32_t SD_HandleTypeDef::DmaTransferCplt***
  SD DMA transfer complete flag
- ***__IO uint32_t SD_HandleTypeDef::SdOperation***
  SD transfer operation (read/write)
- ***DMA_HandleTypeDef* SD_HandleTypeDef::hdmarx***
  SD Rx DMA handle parameters
- ***DMA_HandleTypeDef* SD_HandleTypeDef::hdmatx***
  SD Tx DMA handle parameters

57.1.2 HAL_SD_CSDTypedef

Data Fields

- ***__IO uint8_t CSDStruct***
- ***__IO uint8_t SysSpecVersion***
- ***__IO uint8_t Reserved1***
- ***__IO uint8_t TAAC***
- ***__IO uint8_t NSAC***
- ***__IO uint8_t MaxBusClkFrec***
- ***__IO uint16_t CardComdClasses***
- ***__IO uint8_t RdBlockLen***
- ***__IO uint8_t PartBlockRead***
- ***__IO uint8_t WrBlockMisalign***
- ***__IO uint8_t RdBlockMisalign***
- ***__IO uint8_t DSRImpl***
- ***__IO uint8_t Reserved2***
- ***__IO uint32_t DeviceSize***
- ***__IO uint8_t MaxRdCurrentVDDMin***
- ***__IO uint8_t MaxRdCurrentVDDMax***
- ***__IO uint8_t MaxWrCurrentVDDMin***
- ***__IO uint8_t MaxWrCurrentVDDMax***
- ***__IO uint8_t DeviceSizeMul***
- ***__IO uint8_t EraseGrSize***
- ***__IO uint8_t EraseGrMul***
- ***__IO uint8_t WrProtectGrSize***
- ***__IO uint8_t WrProtectGrEnable***
- ***__IO uint8_t ManDeflECC***
- ***__IO uint8_t WrSpeedFact***
- ***__IO uint8_t MaxWrBlockLen***
- ***__IO uint8_t WriteBlockPaPartial***
- ***__IO uint8_t Reserved3***
- ***__IO uint8_t ContentProtectAppli***
- ***__IO uint8_t FileFormatGrouop***
- ***__IO uint8_t CopyFlag***
- ***__IO uint8_t PermWrProtect***
- ***__IO uint8_t TempWrProtect***
- ***__IO uint8_t FileFormat***
- ***__IO uint8_t ECC***
- ***__IO uint8_t CSD_CRC***
- ***__IO uint8_t Reserved4***

Field Documentation

- ***__IO uint8_t HAL_SD_CSDTypedef::CSDStruct***
  CSD structure
- ***__IO uint8_t HAL_SD_CSDTypedef::SysSpecVersion***
  System specification version
- ***__IO uint8_t HAL_SD_CSDTypedef::Reserved1***
  Reserved

- ***__IO uint8_t HAL_SD_CSDTypedef::TAAC***
  Data read access time 1
- ***__IO uint8_t HAL_SD_CSDTypedef::NSAC***
  Data read access time 2 in CLK cycles
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxBusClkFrec***
  Max. bus clock frequency
- ***__IO uint16_t HAL_SD_CSDTypedef::CardComdClasses***
  Card command classes
- ***__IO uint8_t HAL_SD_CSDTypedef::RdBlockLen***
  Max. read data block length
- ***__IO uint8_t HAL_SD_CSDTypedef::PartBlockRead***
  Partial blocks for read allowed
- ***__IO uint8_t HAL_SD_CSDTypedef::WrBlockMisalign***
  Write block misalignment
- ***__IO uint8_t HAL_SD_CSDTypedef::RdBlockMisalign***
  Read block misalignment
- ***__IO uint8_t HAL_SD_CSDTypedef::DSRImpl***
  DSR implemented
- ***__IO uint8_t HAL_SD_CSDTypedef::Reserved2***
  Reserved
- ***__IO uint32_t HAL_SD_CSDTypedef::DeviceSize***
  Device Size
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxRdCurrentVDDMin***
  Max. read current @ VDD min
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxRdCurrentVDDMax***
  Max. read current @ VDD max
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxWrCurrentVDDMin***
  Max. write current @ VDD min
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxWrCurrentVDDMax***
  Max. write current @ VDD max
- ***__IO uint8_t HAL_SD_CSDTypedef::DeviceSizeMul***
  Device size multiplier
- ***__IO uint8_t HAL_SD_CSDTypedef::EraseGrSize***
  Erase group size
- ***__IO uint8_t HAL_SD_CSDTypedef::EraseGrMul***
  Erase group size multiplier
- ***__IO uint8_t HAL_SD_CSDTypedef::WrProtectGrSize***
  Write protect group size
- ***__IO uint8_t HAL_SD_CSDTypedef::WrProtectGrEnable***
  Write protect group enable
- ***__IO uint8_t HAL_SD_CSDTypedef::ManDeflECC***
  Manufacturer default ECC
- ***__IO uint8_t HAL_SD_CSDTypedef::WrSpeedFact***
  Write speed factor
- ***__IO uint8_t HAL_SD_CSDTypedef::MaxWrBlockLen***
  Max. write data block length
- ***__IO uint8_t HAL_SD_CSDTypedef::WriteBlockPaPartial***
  Partial blocks for write allowed
- ***__IO uint8_t HAL_SD_CSDTypedef::Reserved3***
  Reserved
- ***__IO uint8_t HAL_SD_CSDTypedef::ContentProtectAppli***
  Content protection application
- ***__IO uint8_t HAL_SD_CSDTypedef::FileFormatGrouop***
  File format group

- ***__IO uint8_t HAL_SD_CSDTypedef::CopyFlag***
  Copy flag (OTP)
- ***__IO uint8_t HAL_SD_CSDTypedef::PermWrProtect***
  Permanent write protection
- ***__IO uint8_t HAL_SD_CSDTypedef::TempWrProtect***
  Temporary write protection
- ***__IO uint8_t HAL_SD_CSDTypedef::FileFormat***
  File format
- ***__IO uint8_t HAL_SD_CSDTypedef::ECC***
  ECC code
- ***__IO uint8_t HAL_SD_CSDTypedef::CSD_CRC***
  CSD CRC
- ***__IO uint8_t HAL_SD_CSDTypedef::Reserved4***
  Always 1

57.1.3 HAL_SD_CIDTypedef

Data Fields

- ***__IO uint8_t ManufacturerID***
- ***__IO uint16_t OEM_AppliID***
- ***__IO uint32_t ProdName1***
- ***__IO uint8_t ProdName2***
- ***__IO uint8_t ProdRev***
- ***__IO uint32_t ProdSN***
- ***__IO uint8_t Reserved1***
- ***__IO uint16_t ManufactDate***
- ***__IO uint8_t CID_CRC***
- ***__IO uint8_t Reserved2***

Field Documentation

- ***__IO uint8_t HAL_SD_CIDTypedef::ManufacturerID***
  Manufacturer ID
- ***__IO uint16_t HAL_SD_CIDTypedef::OEM_AppliID***
  OEM/Application ID
- ***__IO uint32_t HAL_SD_CIDTypedef::ProdName1***
  Product Name part1
- ***__IO uint8_t HAL_SD_CIDTypedef::ProdName2***
  Product Name part2
- ***__IO uint8_t HAL_SD_CIDTypedef::ProdRev***
  Product Revision
- ***__IO uint32_t HAL_SD_CIDTypedef::ProdSN***
  Product Serial Number
- ***__IO uint8_t HAL_SD_CIDTypedef::Reserved1***
  Reserved1
- ***__IO uint16_t HAL_SD_CIDTypedef::ManufactDate***
  Manufacturing Date
- ***__IO uint8_t HAL_SD_CIDTypedef::CID_CRC***
  CID CRC

- ***__IO uint8_t HAL_SD_CIDTypedef::Reserved2***
  Always 1

57.1.4 HAL_SD_CardStatusTypedef

Data Fields

- ***__IO uint8_t DAT_BUS_WIDTH***
- ***__IO uint8_t SECURED_MODE***
- ***__IO uint16_t SD_CARD_TYPE***
- ***__IO uint32_t SIZE_OF_PROTECTED_AREA***
- ***__IO uint8_t SPEED_CLASS***
- ***__IO uint8_t PERFORMANCE_MOVE***
- ***__IO uint8_t AU_SIZE***
- ***__IO uint16_t ERASE_SIZE***
- ***__IO uint8_t ERASE_TIMEOUT***
- ***__IO uint8_t ERASE_OFFSET***

Field Documentation

- ***__IO uint8_t HAL_SD_CardStatusTypedef::DAT_BUS_WIDTH***
  Shows the currently defined data bus width
- ***__IO uint8_t HAL_SD_CardStatusTypedef::SECURED_MODE***
  Card is in secured mode of operation
- ***__IO uint16_t HAL_SD_CardStatusTypedef::SD_CARD_TYPE***
  Carries information about card type
- ***__IO uint32_t HAL_SD_CardStatusTypedef::SIZE_OF_PROTECTED_AREA***
  Carries information about the capacity of protected area
- ***__IO uint8_t HAL_SD_CardStatusTypedef::SPEED_CLASS***
  Carries information about the speed class of the card
- ***__IO uint8_t HAL_SD_CardStatusTypedef::PERFORMANCE_MOVE***
  Carries information about the card's performance move
- ***__IO uint8_t HAL_SD_CardStatusTypedef::AU_SIZE***
  Carries information about the card's allocation unit size
- ***__IO uint16_t HAL_SD_CardStatusTypedef::ERASE_SIZE***
  Determines the number of AUs to be erased in one operation
- ***__IO uint8_t HAL_SD_CardStatusTypedef::ERASE_TIMEOUT***
  Determines the timeout for any number of AU erase
- ***__IO uint8_t HAL_SD_CardStatusTypedef::ERASE_OFFSET***
  Carries information about the erase offset

57.1.5 HAL_SD_CardInfoTypedef

Data Fields

- ***HAL_SD_CSDTypedef SD_csd***
- ***HAL_SD_CIDTypedef SD_cid***
- ***uint64_t CardCapacity***
- ***uint32_t CardBlockSize***

- ***uint16_t RCA***
- ***uint8_t CardType***

Field Documentation

- ***HAL_SD_CSDTypedef HAL_SD_CardInfoTypedef::SD_csd***
  SD card specific data register
- ***HAL_SD_CIDTypedef HAL_SD_CardInfoTypedef::SD_cid***
  SD card identification number register
- ***uint64_t HAL_SD_CardInfoTypedef::CardCapacity***
  Card capacity
- ***uint32_t HAL_SD_CardInfoTypedef::CardBlockSize***
  Card block size
- ***uint16_t HAL_SD_CardInfoTypedef::RCA***
  SD relative card address
- ***uint8_t HAL_SD_CardInfoTypedef::CardType***
  SD card type

57.2 SD Firmware driver API description

57.2.1 How to use this driver

This driver implements a high level communication layer for read and write from/to this memory. The needed STM32 hardware resources (SDIO and GPIO) are performed by the user in HAL_SD_MspInit() function (MSP layer). Basically, the MSP layer configuration should be the same as we provide in the examples. You can easily tailor this configuration according to hardware resources.

This driver is a generic layered driver for SDIO memories which uses the HAL SDIO driver functions to interface with SD and uSD cards devices. It is used as follows:

1. Initialize the SDIO low level resources by implement the HAL_SD_MspInit() API:
   a. Enable the SDIO interface clock using __HAL_RCC_SDIO_CLK_ENABLE();
   b. SDIO pins configuration for SD card
      – Enable the clock for the SDIO GPIOs using the functions __HAL_RCC_GPIOx_CLK_ENABLE();
      – Configure these SDIO pins as alternate function pull-up using HAL_GPIO_Init() and according to your pin assignment;
   c. DMA Configuration if you need to use DMA process (HAL_SD_ReadBlocks_DMA() and HAL_SD_WriteBlocks_DMA() APIs).
      – Enable the DMAx interface clock using __HAL_RCC_DMAx_CLK_ENABLE();
      – Configure the DMA using the function HAL_DMA_Init() with predeclared and filled.
   d. NVIC configuration if you need to use interrupt process when using DMA transfer.
      – Configure the SDIO and DMA interrupt priorities using functions HAL_NVIC_SetPriority(); DMA priority is superior to SDIO's priority
      – Enable the NVIC DMA and SDIO IRQs using function HAL_NVIC_EnableIRQ()
      – SDIO interrupts are managed using the macros __HAL_SD_SDIO_ENABLE_IT() and __HAL_SD_SDIO_DISABLE_IT() inside the communication process.

  - SDIO interrupts pending bits are managed using the macros __HAL_SD_SDIO_GET_IT() and __HAL_SD_SDIO_CLEAR_IT()

2. At this stage, you can perform SD read/write/erase operations after SD card initialization

SD Card Initialization and configuration

To initialize the SD Card, use the HAL_SD_Init() function. It Initializes the SD Card and put it into Standby State (Ready for data transfer). This function provide the following operations:

1. Apply the SD Card initialization process at 400KHz and check the SD Card type (Standard Capacity or High Capacity). You can change or adapt this frequency by adjusting the "ClockDiv" field. The SD Card frequency (SDIO_CK) is computed as follows: SDIO_CK = SDIOCLK / (ClockDiv + 2) In initialization mode and according to the SD Card standard, make sure that the SDIO_CK frequency doesn't exceed 400KHz.
2. Get the SD CID and CSD data. All these information are managed by the SDCardInfo structure. This structure provide also ready computed SD Card capacity and Block size. These information are stored in SD handle structure in case of future use.
3. Configure the SD Card Data transfer frequency. By Default, the card transfer frequency is set to 24MHz. You can change or adapt this frequency by adjusting the "ClockDiv" field. In transfer mode and according to the SD Card standard, make sure that the SDIO_CK frequency doesn't exceed 25MHz and 50MHz in High-speed mode switch. To be able to use a frequency higher than 24MHz, you should use the SDIO peripheral in bypass mode. Refer to the corresponding reference manual for more details.
4. Select the corresponding SD Card according to the address read with the step 2.
5. Configure the SD Card in wide bus mode: 4-bits data.

SD Card Read operation

- You can read from SD card in polling mode by using function HAL_SD_ReadBlocks(). This function support only 512-bytes block length (the block size should be chosen as 512 bytes). You can choose either one block read operation or multiple block read operation by adjusting the "NumberOfBlocks" parameter.
- You can read from SD card in DMA mode by using function HAL_SD_ReadBlocks_DMA(). This function support only 512-bytes block length (the block size should be chosen as 512 bytes). You can choose either one block read operation or multiple block read operation by adjusting the "NumberOfBlocks" parameter. After this, you have to call the function HAL_SD_CheckReadOperation(), to insure that the read transfer is done correctly in both DMA and SD sides.

SD Card Write operation

- You can write to SD card in polling mode by using function HAL_SD_WriteBlocks(). This function support only 512-bytes block length (the block size should be chosen as 512 bytes). You can choose either one block read operation or multiple block read operation by adjusting the "NumberOfBlocks" parameter.
- You can write to SD card in DMA mode by using function HAL_SD_WriteBlocks_DMA(). This function support only 512-bytes block length (the block size should be chosen as 512 byte). You can choose either one block read operation or multiple block read operation by adjusting the "NumberOfBlocks"

parameter. After this, you have to call the function HAL_SD_CheckWriteOperation(), to insure that the write transfer is done correctly in both DMA and SD sides.

SD card status

- At any time, you can check the SD Card status and get the SD card state by using the HAL_SD_GetStatus() function. This function checks first if the SD card is still connected and then get the internal SD Card transfer state.
- You can also get the SD card SD Status register by using the HAL_SD_SendSDStatus() function.

SD HAL driver macros list

You can refer to the SD HAL driver header file for more useful macros

57.2.2 Initialization and de-initialization functions

This section provides functions allowing to initialize/de-initialize the SD card device to be ready for use.

This section contains the following APIs:

- ***HAL_SD_Init()***
- ***HAL_SD_DeInit()***
- ***HAL_SD_MspInit()***
- ***HAL_SD_MspDeInit()***

57.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the data transfer from/to SD card.

This section contains the following APIs:

- ***HAL_SD_ReadBlocks()***
- ***HAL_SD_WriteBlocks()***
- ***HAL_SD_ReadBlocks_DMA()***
- ***HAL_SD_WriteBlocks_DMA()***
- ***HAL_SD_CheckReadOperation()***
- ***HAL_SD_CheckWriteOperation()***
- ***HAL_SD_Erase()***
- ***HAL_SD_IRQHandler()***
- ***HAL_SD_XferCpltCallback()***
- ***HAL_SD_XferErrorCallback()***
- ***HAL_SD_DMA_RxCpltCallback()***
- ***HAL_SD_DMA_RxErrorCallback()***
- ***HAL_SD_DMA_TxCpltCallback()***
- ***HAL_SD_DMA_TxErrorCallback()***

57.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the SD card operations.

This section contains the following APIs:

- ***HAL_SD_Get_CardInfo()***
- ***HAL_SD_WideBusOperation_Config()***
- ***HAL_SD_StopTransfer()***
- ***HAL_SD_HighSpeed()***

57.2.5 Peripheral State functions

This subsection permits to get in runtime the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_SD_SendSDStatus()***
- ***HAL_SD_GetStatus()***
- ***HAL_SD_GetCardStatus()***

57.2.6 Detailed description of functions

HAL_SD_Init

Function Name	**HAL_SD_ErrorTypedef HAL_SD_Init (SD_HandleTypeDef * hsd, HAL_SD_CardInfoTypedef * SDCardInfo)**
Function Description	Initializes the SD card according to the specified parameters in the SD_HandleTypeDef and create the associated handle.
Parameters	• **hsd:** SD handle • **SDCardInfo:** HAL_SD_CardInfoTypedef structure for SD card information
Return values	• **HAL:** SD error state

HAL_SD_DeInit

Function Name	**HAL_StatusTypeDef HAL_SD_DeInit (SD_HandleTypeDef * hsd)**
Function Description	De-Initializes the SD card.
Parameters	• **hsd:** SD handle
Return values	• **HAL:** status

HAL_SD_MspInit

Function Name	**void HAL_SD_MspInit (SD_HandleTypeDef * hsd)**
Function Description	Initializes the SD MSP.
Parameters	• **hsd:** SD handle
Return values	• **None**

HAL_SD_MspDeInit

Function Name	**void HAL_SD_MspDeInit (SD_HandleTypeDef * hsd)**
Function Description	De-Initialize SD MSP.
Parameters	• **hsd:** SD handle
Return values	• **None**

HAL_SD_ReadBlocks

Function Name	**HAL_SD_ErrorTypedef HAL_SD_ReadBlocks (SD_HandleTypeDef * hsd, uint32_t * pReadBuffer, uint64_t ReadAddr, uint32_t BlockSize, uint32_t NumberOfBlocks)**
Function Description	Reads block(s) from a specified address in a card.
Parameters	• **hsd:** SD handle • **pReadBuffer:** pointer to the buffer that will contain the received data • **ReadAddr:** Address from where data is to be read • **BlockSize:** SD card Data block size • **NumberOfBlocks:** Number of SD blocks to read
Return values	• **SD:** Card error state
Notes	• BlockSize must be 512 bytes.

HAL_SD_WriteBlocks

Function Name	**HAL_SD_ErrorTypedef HAL_SD_WriteBlocks (SD_HandleTypeDef * hsd, uint32_t * pWriteBuffer, uint64_t WriteAddr, uint32_t BlockSize, uint32_t NumberOfBlocks)**
Function Description	Allows to write block(s) to a specified address in a card.
Parameters	• **hsd:** SD handle • **pWriteBuffer:** pointer to the buffer that will contain the data to transmit • **WriteAddr:** Address from where data is to be written • **BlockSize:** SD card Data block size • **NumberOfBlocks:** Number of SD blocks to write
Return values	• **SD:** Card error state
Notes	• BlockSize must be 512 bytes.

HAL_SD_Erase

Function Name	**HAL_SD_ErrorTypedef HAL_SD_Erase (SD_HandleTypeDef * hsd, uint64_t startaddr, uint64_t endaddr)**
Function Description	Erases the specified memory area of the given SD card.
Parameters	• **hsd:** SD handle • **startaddr:** Start byte address • **endaddr:** End byte address
Return values	• **SD:** Card error state

HAL_SD_IRQHandler

Function Name	**void HAL_SD_IRQHandler (SD_HandleTypeDef * hsd)**
Function Description	This function handles SD card interrupt request.
Parameters	• **hsd:** SD handle
Return values	• **None**

HAL_SD_DMA_RxCpltCallback

Function Name	**void HAL_SD_DMA_RxCpltCallback (DMA_HandleTypeDef * hdma)**
Function Description	SD Transfer complete Rx callback in non blocking mode.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

HAL_SD_DMA_RxErrorCallback

Function Name	**void HAL_SD_DMA_RxErrorCallback (DMA_HandleTypeDef * hdma)**
Function Description	SD DMA transfer complete Rx error callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

HAL_SD_DMA_TxCpltCallback

Function Name	**void HAL_SD_DMA_TxCpltCallback (DMA_HandleTypeDef * hdma)**
Function Description	SD Transfer complete Tx callback in non blocking mode.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

HAL_SD_DMA_TxErrorCallback

Function Name	**void HAL_SD_DMA_TxErrorCallback (DMA_HandleTypeDef * hdma)**
Function Description	SD DMA transfer complete error Tx callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

HAL_SD_XferCpltCallback

Function Name	**void HAL_SD_XferCpltCallback (SD_HandleTypeDef * hsd)**
Function Description	SD end of transfer callback.
Parameters	• **hsd:** SD handle
Return values	• **None**

HAL_SD_XferErrorCallback

Function Name	**void HAL_SD_XferErrorCallback (SD_HandleTypeDef * hsd)**
Function Description	SD Transfer Error callback.
Parameters	• **hsd:** SD handle
Return values	• **None**

HAL_SD_ReadBlocks_DMA

Function Name	**HAL_SD_ErrorTypedef HAL_SD_ReadBlocks_DMA (SD_HandleTypeDef * hsd, uint32_t * pReadBuffer, uint64_t ReadAddr, uint32_t BlockSize, uint32_t NumberOfBlocks)**
Function Description	Reads block(s) from a specified address in a card.
Parameters	• **hsd:** SD handle • **pReadBuffer:** Pointer to the buffer that will contain the received data • **ReadAddr:** Address from where data is to be read • **BlockSize:** SD card Data block size • **NumberOfBlocks:** Number of blocks to read.
Return values	• **SD:** Card error state
Notes	• This API should be followed by the function HAL_SD_CheckReadOperation() to check the completion of the read process • BlockSize must be 512 bytes.

HAL_SD_WriteBlocks_DMA

Function Name	**HAL_SD_ErrorTypedef HAL_SD_WriteBlocks_DMA (SD_HandleTypeDef * hsd, uint32_t * pWriteBuffer, uint64_t WriteAddr, uint32_t BlockSize, uint32_t NumberOfBlocks)**
Function Description	Writes block(s) to a specified address in a card.
Parameters	• **hsd:** SD handle • **pWriteBuffer:** pointer to the buffer that will contain the data to transmit • **WriteAddr:** Address from where data is to be read • **BlockSize:** the SD card Data block size • **NumberOfBlocks:** Number of blocks to write
Return values	• **SD:** Card error state
Notes	• This API should be followed by the function HAL_SD_CheckWriteOperation() to check the completion of

the write process (by SD current status polling).
- BlockSize must be 512 bytes.

HAL_SD_CheckWriteOperation

Function Name	**HAL_SD_ErrorTypedef HAL_SD_CheckWriteOperation (SD_HandleTypeDef * hsd, uint32_t Timeout)**
Function Description	This function waits until the SD DMA data write transfer is finished.
Parameters	• **hsd:** SD handle • **Timeout:** Timeout duration
Return values	• **SD:** Card error state

HAL_SD_CheckReadOperation

Function Name	**HAL_SD_ErrorTypedef HAL_SD_CheckReadOperation (SD_HandleTypeDef * hsd, uint32_t Timeout)**
Function Description	This function waits until the SD DMA data read transfer is finished.
Parameters	• **hsd:** SD handle • **Timeout:** Timeout duration
Return values	• **SD:** Card error state

HAL_SD_Get_CardInfo

Function Name	**HAL_SD_ErrorTypedef HAL_SD_Get_CardInfo (SD_HandleTypeDef * hsd, HAL_SD_CardInfoTypedef * pCardInfo)**
Function Description	Returns information about specific card.
Parameters	• **hsd:** SD handle • **pCardInfo:** Pointer to a HAL_SD_CardInfoTypedef structure that contains all SD cardinformation
Return values	• **SD:** Card error state

HAL_SD_WideBusOperation_Config

Function Name	**HAL_SD_ErrorTypedef HAL_SD_WideBusOperation_Config (SD_HandleTypeDef * hsd, uint32_t WideMode)**
Function Description	Enables wide bus operation for the requested card if supported by card.
Parameters	• **hsd:** SD handle • **WideMode:** Specifies the SD card wide bus mode This parameter can be one of the following values: – SDIO_BUS_WIDE_8B: 8-bit data transfer (Only for MMC) – SDIO_BUS_WIDE_4B: 4-bit data transfer – SDIO_BUS_WIDE_1B: 1-bit data transfer
Return values	• **SD:** Card error state

HAL_SD_StopTransfer

Function Name	**HAL_SD_ErrorTypedef HAL_SD_StopTransfer (SD_HandleTypeDef * hsd)**
Function Description	Aborts an ongoing data transfer.
Parameters	• **hsd:** SD handle
Return values	• **SD:** Card error state

HAL_SD_HighSpeed

Function Name	**HAL_SD_ErrorTypedef HAL_SD_HighSpeed (SD_HandleTypeDef * hsd)**
Function Description	Switches the SD card to High Speed mode.
Parameters	• **hsd:** SD handle
Return values	• **SD:** Card error state
Notes	• This operation should be followed by the configuration of PLL to have SDIOCK clock between 67 and 75 MHz

HAL_SD_SendSDStatus

Function Name	**HAL_SD_ErrorTypedef HAL_SD_SendSDStatus (SD_HandleTypeDef * hsd, uint32_t * pSDstatus)**
Function Description	Returns the current SD card's status.
Parameters	• **hsd:** SD handle • **pSDstatus:** Pointer to the buffer that will contain the SD card status SD Status register)
Return values	• **SD:** Card error state

HAL_SD_GetCardStatus

Function Name	**HAL_SD_ErrorTypedef HAL_SD_GetCardStatus (SD_HandleTypeDef * hsd, HAL_SD_CardStatusTypedef * pCardStatus)**
Function Description	Gets the SD card status.
Parameters	• **hsd:** SD handle • **pCardStatus:** Pointer to the HAL_SD_CardStatusTypedef structure that will contain the SD card status information
Return values	• **SD:** Card error state

HAL_SD_GetStatus

Function Name	**HAL_SD_TransferStateTypedef HAL_SD_GetStatus (SD_HandleTypeDef * hsd)**
Function Description	Gets the current sd card data status.
Parameters	• **hsd:** SD handle

Return values
- **Data:** Transfer state

57.3 SD Firmware driver defines

57.3.1 SD

SD Exported Constants

SD_CMD_GO_IDLE_STATE	Resets the SD memory card.
SD_CMD_SEND_OP_COND	Sends host capacity support information and activates the card's initialization process.
SD_CMD_ALL_SEND_CID	Asks any card connected to the host to send the CID numbers on the CMD line.
SD_CMD_SET_REL_ADDR	Asks the card to publish a new relative address (RCA).
SD_CMD_SET_DSR	Programs the DSR of all cards.
SD_CMD_SDIO_SEN_OP_COND	Sends host capacity support information (HCS) and asks the accessed card to send its operating condition register (OCR) content in the response on the CMD line.
SD_CMD_HS_SWITCH	Checks switchable function (mode 0) and switch card function (mode 1).
SD_CMD_SEL_DESEL_CARD	Selects the card by its own relative address and gets deselected by any other address
SD_CMD_HS_SEND_EXT_CSD	Sends SD Memory Card interface condition, which includes host supply voltage information and asks the card whether card supports voltage.
SD_CMD_SEND_CSD	Addressed card sends its card specific data (CSD) on the CMD line.
SD_CMD_SEND_CID	Addressed card sends its card identification (CID) on the CMD line.
SD_CMD_READ_DAT_UNTIL_STOP	SD card doesn't support it.
SD_CMD_STOP_TRANSMISSION	Forces the card to stop

	transmission.
SD_CMD_SEND_STATUS	Addressed card sends its status register.
SD_CMD_HS_BUSTEST_READ	
SD_CMD_GO_INACTIVE_STATE	Sends an addressed card into the inactive state.
SD_CMD_SET_BLOCKLEN	Sets the block length (in bytes for SDSC) for all following block commands (read, write, lock). Default block length is fixed to 512 Bytes. Not effective for SDHS and SDXC.
SD_CMD_READ_SINGLE_BLOCK	Reads single block of size selected by SET_BLOCKLEN in case of SDSC, and a block of fixed 512 bytes in case of SDHC and SDXC.
SD_CMD_READ_MULT_BLOCK	Continuously transfers data blocks from card to host until interrupted by STOP_TRANSMISSION command.
SD_CMD_HS_BUSTEST_WRITE	64 bytes tuning pattern is sent for SDR50 and SDR104.
SD_CMD_WRITE_DAT_UNTIL_STOP	Speed class control command.
SD_CMD_SET_BLOCK_COUNT	Specify block count for CMD18 and CMD25.
SD_CMD_WRITE_SINGLE_BLOCK	Writes single block of size selected by SET_BLOCKLEN in case of SDSC, and a block of fixed 512 bytes in case of SDHC and SDXC.
SD_CMD_WRITE_MULT_BLOCK	Continuously writes blocks of data until a STOP_TRANSMISSION follows.
SD_CMD_PROG_CID	Reserved for manufacturers.
SD_CMD_PROG_CSD	Programming of the programmable bits of the CSD.
SD_CMD_SET_WRITE_PROT	Sets the write protection bit of the addressed group.

SD_CMD_CLR_WRITE_PROT	Clears the write protection bit of the addressed group.
SD_CMD_SEND_WRITE_PROT	Asks the card to send the status of the write protection bits.
SD_CMD_SD_ERASE_GRP_START	Sets the address of the first write block to be erased. (For SD card only).
SD_CMD_SD_ERASE_GRP_END	Sets the address of the last write block of the continuous range to be erased.
SD_CMD_ERASE_GRP_START	Sets the address of the first write block to be erased. Reserved for each command system set by switch function command (CMD6).
SD_CMD_ERASE_GRP_END	Sets the address of the last write block of the continuous range to be erased. Reserved for each command system set by switch function command (CMD6).
SD_CMD_ERASE	Reserved for SD security applications.
SD_CMD_FAST_IO	SD card doesn't support it (Reserved).
SD_CMD_GO_IRQ_STATE	SD card doesn't support it (Reserved).
SD_CMD_LOCK_UNLOCK	Sets/resets the password or lock/unlock the card. The size of the data block is set by the SET_BLOCK_LEN command.
SD_CMD_APP_CMD	Indicates to the card that the next command is an application specific command rather than a standard command.
SD_CMD_GEN_CMD	Used either to transfer a data block to the card or to get a data block from the card for general purpose/application specific commands.
SD_CMD_NO_CMD	
SD_CMD_APP_SD_SET_BUSWIDTH	SDIO_APP_CMD should be sent before sending these

	commands. (ACMD6) Defines the data bus width to be used for data transfer. The allowed data bus widths are given in SCR register.
SD_CMD_SD_APP_STATUS	(ACMD13) Sends the SD status.
SD_CMD_SD_APP_SEND_NUM_WRITE_BLOCKS	(ACMD22) Sends the number of the written (without errors) write blocks. Responds with 32bit+CRC data block.
SD_CMD_SD_APP_OP_COND	(ACMD41) Sends host capacity support information (HCS) and asks the accessed card to send its operating condition register (OCR) content in the response on the CMD line.
SD_CMD_SD_APP_SET_CLR_CARD_DETECT	(ACMD42) Connects/Disconnects the 50 KOhm pull-up resistor on CD/DAT3 (pin 1) of the card.
SD_CMD_SD_APP_SEND_SCR	Reads the SD Configuration Register (SCR).
SD_CMD_SDIO_RW_DIRECT	For SD I/O card only, reserved for security specification.
SD_CMD_SDIO_RW_EXTENDED	For SD I/O card only, reserved for security specification.
SD_CMD_SD_APP_GET_MKB	SD_CMD_APP_CMD should be sent before sending these commands. For SD card only
SD_CMD_SD_APP_GET_MID	For SD card only
SD_CMD_SD_APP_SET_CER_RN1	For SD card only
SD_CMD_SD_APP_GET_CER_RN2	For SD card only
SD_CMD_SD_APP_SET_CER_RES2	For SD card only
SD_CMD_SD_APP_GET_CER_RES1	For SD card only
SD_CMD_SD_APP_SECURE_READ_MULTIPLE_BLOCK	For SD card only
SD_CMD_SD_APP_SECURE_WRITE_MULTIPLE_BLOCK	For SD card only
SD_CMD_SD_APP_SECURE_ERASE	For SD card only
SD_CMD_SD_APP_CHANGE_SECURE_AREA	For SD card only
SD_CMD_SD_APP_SECURE_WRITE_MKB	For SD card only

STD_CAPACITY_SD_CARD_V1_1

STD_CAPACITY_SD_CARD_V2_0

HIGH_CAPACITY_SD_CARD

MULTIMEDIA_CARD

SECURE_DIGITAL_IO_CARD

HIGH_SPEED_MULTIMEDIA_CARD

SECURE_DIGITAL_IO_COMBO_CARD

HIGH_CAPACITY_MMC_CARD

SD Exported Macros

__HAL_SD_SDIO_ENABLE

Description:

- Enable the SD device.

Return value:

- None

__HAL_SD_SDIO_DISABLE

Description:

- Disable the SD device.

Return value:

- None

__HAL_SD_SDIO_DMA_ENABLE

Description:

- Enable the SDIO DMA transfer.

Return value:

- None

__HAL_SD_SDIO_DMA_DISABLE

Description:

- Disable the SDIO DMA transfer.

Return value:

- None

__HAL_SD_SDIO_ENABLE_IT

Description:

- Enable the SD device interrupt.

Parameters:

- __HANDLE__: SD Handle
- __INTERRUPT__: specifies the SDIO interrupt sources to be enabled. This parameter can be one or a combination of the following values:
  - SDIO_IT_CCRCFAIL: Command response received (CRC check failed) interrupt
  - SDIO_IT_DCRCFAIL: Data block sent/received (CRC check failed) interrupt
  - SDIO_IT_CTIMEOUT: Command response timeout interrupt
  - SDIO_IT_DTIMEOUT: Data timeout interrupt

- SDIO_IT_TXUNDERR: Transmit FIFO underrun error interrupt
- SDIO_IT_RXOVERR: Received FIFO overrun error interrupt
- SDIO_IT_CMDREND: Command response received (CRC check passed) interrupt
- SDIO_IT_CMDSENT: Command sent (no response required) interrupt
- SDIO_IT_DATAEND: Data end (data counter, SDIDCOUNT, is zero) interrupt
- SDIO_IT_STBITERR: Start bit not detected on all data signals in wide bus mode interrupt
- SDIO_IT_DBCKEND: Data block sent/received (CRC check passed) interrupt
- SDIO_IT_CMDACT: Command transfer in progress interrupt
- SDIO_IT_TXACT: Data transmit in progress interrupt
- SDIO_IT_RXACT: Data receive in progress interrupt
- SDIO_IT_TXFIFOHE: Transmit FIFO Half Empty interrupt
- SDIO_IT_RXFIFOHF: Receive FIFO Half Full interrupt
- SDIO_IT_TXFIFOF: Transmit FIFO full interrupt
- SDIO_IT_RXFIFOF: Receive FIFO full interrupt
- SDIO_IT_TXFIFOE: Transmit FIFO empty interrupt
- SDIO_IT_RXFIFOE: Receive FIFO empty interrupt
- SDIO_IT_TXDAVL: Data available in transmit FIFO interrupt
- SDIO_IT_RXDAVL: Data available in receive FIFO interrupt
- SDIO_IT_SDIOIT: SD I/O interrupt received interrupt
- SDIO_IT_CEATAEND: CE-ATA command completion signal received for CMD61 interrupt

Return value:

- None

__HAL_SD_SDIO_DISABLE_IT

Description:

- Disable the SD device interrupt.

Parameters:

- __HANDLE__: SD Handle
- __INTERRUPT__: specifies the SDIO interrupt sources to be disabled. This parameter can be one or a combination of the following values:

  - SDIO_IT_CCRCFAIL: Command response received (CRC check failed) interrupt
  - SDIO_IT_DCRCFAIL: Data block sent/received (CRC check failed) interrupt
  - SDIO_IT_CTIMEOUT: Command response timeout interrupt
  - SDIO_IT_DTIMEOUT: Data timeout interrupt
  - SDIO_IT_TXUNDERR: Transmit FIFO underrun error interrupt
  - SDIO_IT_RXOVERR: Received FIFO overrun error interrupt
  - SDIO_IT_CMDREND: Command response received (CRC check passed) interrupt
  - SDIO_IT_CMDSENT: Command sent (no response required) interrupt
  - SDIO_IT_DATAEND: Data end (data counter, SDIDCOUNT, is zero) interrupt
  - SDIO_IT_STBITERR: Start bit not detected on all data signals in wide bus mode interrupt
  - SDIO_IT_DBCKEND: Data block sent/received (CRC check passed) interrupt
  - SDIO_IT_CMDACT: Command transfer in progress interrupt
  - SDIO_IT_TXACT: Data transmit in progress interrupt
  - SDIO_IT_RXACT: Data receive in progress interrupt
  - SDIO_IT_TXFIFOHE: Transmit FIFO Half Empty interrupt
  - SDIO_IT_RXFIFOHF: Receive FIFO Half Full interrupt
  - SDIO_IT_TXFIFOF: Transmit FIFO full interrupt
  - SDIO_IT_RXFIFOF: Receive FIFO full interrupt
  - SDIO_IT_TXFIFOE: Transmit FIFO empty interrupt
  - SDIO_IT_RXFIFOE: Receive FIFO empty interrupt
  - SDIO_IT_TXDAVL: Data available in transmit FIFO interrupt
  - SDIO_IT_RXDAVL: Data available in receive FIFO interrupt
  - SDIO_IT_SDIOIT: SD I/O interrupt received interrupt
  - SDIO_IT_CEATAEND: CE-ATA command completion signal received for CMD61 interrupt

Return value:

- None

__HAL_SD_SDIO_GET_FLAG

Description:

- Check whether the specified SD flag is set or not.

Parameters:

- __HANDLE__: SD Handle
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - SDIO_FLAG_CCRCFAIL: Command response received (CRC check failed)
  - SDIO_FLAG_DCRCFAIL: Data block sent/received (CRC check failed)
  - SDIO_FLAG_CTIMEOUT: Command response timeout
  - SDIO_FLAG_DTIMEOUT: Data timeout
  - SDIO_FLAG_TXUNDERR: Transmit FIFO underrun error
  - SDIO_FLAG_RXOVERR: Received FIFO overrun error
  - SDIO_FLAG_CMDREND: Command response received (CRC check passed)
  - SDIO_FLAG_CMDSENT: Command sent (no response required)
  - SDIO_FLAG_DATAEND: Data end (data counter, SDIDCOUNT, is zero)
  - SDIO_FLAG_STBITERR: Start bit not detected on all data signals in wide bus mode.
  - SDIO_FLAG_DBCKEND: Data block sent/received (CRC check passed)
  - SDIO_FLAG_CMDACT: Command transfer in progress
  - SDIO_FLAG_TXACT: Data transmit in progress
  - SDIO_FLAG_RXACT: Data receive in progress
  - SDIO_FLAG_TXFIFOHE: Transmit FIFO Half Empty
  - SDIO_FLAG_RXFIFOHF: Receive FIFO Half Full
  - SDIO_FLAG_TXFIFOF: Transmit FIFO full
  - SDIO_FLAG_RXFIFOF: Receive FIFO full
  - SDIO_FLAG_TXFIFOE: Transmit FIFO empty
  - SDIO_FLAG_RXFIFOE: Receive FIFO empty
  - SDIO_FLAG_TXDAVL: Data available in transmit FIFO
  - SDIO_FLAG_RXDAVL: Data available in receive FIFO
  - SDIO_FLAG_SDIOIT: SD I/O interrupt received
  - SDIO_FLAG_CEATAEND: CE-ATA

command completion signal received for CMD61

Return value:

- The: new state of SD FLAG (SET or RESET).

__HAL_SD_SDIO_CLEAR_FLAG

Description:

- Clear the SD's pending flags.

Parameters:

- __HANDLE__: SD Handle
- __FLAG__: specifies the flag to clear. This parameter can be one or a combination of the following values:
  - SDIO_FLAG_CCRCFAIL: Command response received (CRC check failed)
  - SDIO_FLAG_DCRCFAIL: Data block sent/received (CRC check failed)
  - SDIO_FLAG_CTIMEOUT: Command response timeout
  - SDIO_FLAG_DTIMEOUT: Data timeout
  - SDIO_FLAG_TXUNDERR: Transmit FIFO underrun error
  - SDIO_FLAG_RXOVERR: Received FIFO overrun error
  - SDIO_FLAG_CMDREND: Command response received (CRC check passed)
  - SDIO_FLAG_CMDSENT: Command sent (no response required)
  - SDIO_FLAG_DATAEND: Data end (data counter, SDIDCOUNT, is zero)
  - SDIO_FLAG_STBITERR: Start bit not detected on all data signals in wide bus mode
  - SDIO_FLAG_DBCKEND: Data block sent/received (CRC check passed)
  - SDIO_FLAG_SDIOIT: SD I/O interrupt received
  - SDIO_FLAG_CEATAEND: CE-ATA command completion signal received for CMD61

Return value:

- None

__HAL_SD_SDIO_GET_IT

Description:

- Check whether the specified SD interrupt has occurred or not.

Parameters:

- __HANDLE__: SD Handle
- __INTERRUPT__: specifies the SDIO interrupt source to check. This parameter can be one of the following values:

  - SDIO_IT_CCRCFAIL: Command response received (CRC check failed) interrupt
  - SDIO_IT_DCRCFAIL: Data block sent/received (CRC check failed) interrupt
  - SDIO_IT_CTIMEOUT: Command response timeout interrupt
  - SDIO_IT_DTIMEOUT: Data timeout interrupt
  - SDIO_IT_TXUNDERR: Transmit FIFO underrun error interrupt
  - SDIO_IT_RXOVERR: Received FIFO overrun error interrupt
  - SDIO_IT_CMDREND: Command response received (CRC check passed) interrupt
  - SDIO_IT_CMDSENT: Command sent (no response required) interrupt
  - SDIO_IT_DATAEND: Data end (data counter, SDIDCOUNT, is zero) interrupt
  - SDIO_IT_STBITERR: Start bit not detected on all data signals in wide bus mode interrupt
  - SDIO_IT_DBCKEND: Data block sent/received (CRC check passed) interrupt
  - SDIO_IT_CMDACT: Command transfer in progress interrupt
  - SDIO_IT_TXACT: Data transmit in progress interrupt
  - SDIO_IT_RXACT: Data receive in progress interrupt
  - SDIO_IT_TXFIFOHE: Transmit FIFO Half Empty interrupt
  - SDIO_IT_RXFIFOHF: Receive FIFO Half Full interrupt
  - SDIO_IT_TXFIFOF: Transmit FIFO full interrupt
  - SDIO_IT_RXFIFOF: Receive FIFO full interrupt
  - SDIO_IT_TXFIFOE: Transmit FIFO empty interrupt
  - SDIO_IT_RXFIFOE: Receive FIFO empty interrupt
  - SDIO_IT_TXDAVL: Data available in transmit FIFO interrupt
  - SDIO_IT_RXDAVL: Data available in receive FIFO interrupt
  - SDIO_IT_SDIOIT: SD I/O interrupt received interrupt
  - SDIO_IT_CEATAEND: CE-ATA command completion signal received for CMD61 interrupt

Return value:

- The: new state of SD IT (SET or RESET).

__HAL_SD_SDIO_CLEAR_IT

Description:

- Clear the SD's interrupt pending bits.

Parameters:

- __HANDLE__: SD Handle
- __INTERRUPT__: specifies the interrupt pending bit to clear. This parameter can be one or a combination of the following values:
  - SDIO_IT_CCRCFAIL: Command response received (CRC check failed) interrupt
  - SDIO_IT_DCRCFAIL: Data block sent/received (CRC check failed) interrupt
  - SDIO_IT_CTIMEOUT: Command response timeout interrupt
  - SDIO_IT_DTIMEOUT: Data timeout interrupt
  - SDIO_IT_TXUNDERR: Transmit FIFO underrun error interrupt
  - SDIO_IT_RXOVERR: Received FIFO overrun error interrupt
  - SDIO_IT_CMDREND: Command response received (CRC check passed) interrupt
  - SDIO_IT_CMDSENT: Command sent (no response required) interrupt
  - SDIO_IT_DATAEND: Data end (data counter, SDIO_DCOUNT, is zero) interrupt
  - SDIO_IT_STBITERR: Start bit not detected on all data signals in wide bus mode interrupt
  - SDIO_IT_SDIOIT: SD I/O interrupt received interrupt
  - SDIO_IT_CEATAEND: CE-ATA command completion signal received for CMD61

Return value:

- None

SD Handle Structure definition

SD_InitTypeDef

SD_TypeDef

58 HAL SMARTCARD Generic Driver

58.1 SMARTCARD Firmware driver registers structures

58.1.1 SMARTCARD_InitTypeDef

Data Fields

- ***uint32_t BaudRate***
- ***uint32_t WordLength***
- ***uint32_t StopBits***
- ***uint32_t Parity***
- ***uint32_t Mode***
- ***uint32_t CLKPolarity***
- ***uint32_t CLKPhase***
- ***uint32_t CLKLastBit***
- ***uint32_t Prescaler***
- ***uint32_t GuardTime***
- ***uint32_t NACKState***

Field Documentation

- ***uint32_t SMARTCARD_InitTypeDef::BaudRate***
  This member configures the SmartCard communication baud rate. The baud rate is computed using the following formula:IntegerDivider = ((PCLKx) / (8 * (hirda->Init.BaudRate)))FractionalDivider = ((IntegerDivider - ((uint32_t) IntegerDivider)) * 8) + 0.5
- ***uint32_t SMARTCARD_InitTypeDef::WordLength***
  Specifies the number of data bits transmitted or received in a frame. This parameter can be a value of ***SMARTCARD_Word_Length***
- ***uint32_t SMARTCARD_InitTypeDef::StopBits***
  Specifies the number of stop bits transmitted. This parameter can be a value of ***SMARTCARD_Stop_Bits***
- ***uint32_t SMARTCARD_InitTypeDef::Parity***
  Specifies the parity mode. This parameter can be a value of ***SMARTCARD_Parity*** **Note:**When parity is enabled, the computed parity is inserted at the MSB position of the transmitted data (9th bit when the word length is set to 9 data bits; 8th bit when the word length is set to 8 data bits).
- ***uint32_t SMARTCARD_InitTypeDef::Mode***
  Specifies whether the Receive or Transmit mode is enabled or disabled. This parameter can be a value of ***SMARTCARD_Mode***
- ***uint32_t SMARTCARD_InitTypeDef::CLKPolarity***
  Specifies the steady state of the serial clock. This parameter can be a value of ***SMARTCARD_Clock_Polarity***
- ***uint32_t SMARTCARD_InitTypeDef::CLKPhase***
  Specifies the clock transition on which the bit capture is made. This parameter can be a value of ***SMARTCARD_Clock_Phase***
- ***uint32_t SMARTCARD_InitTypeDef::CLKLastBit***
  Specifies whether the clock pulse corresponding to the last transmitted data bit (MSB)

has to be output on the SCLK pin in synchronous mode. This parameter can be a value of ***SMARTCARD_Last_Bit***

- ***uint32_t SMARTCARD_InitTypeDef::Prescaler***
  Specifies the SmartCard Prescaler value used for dividing the system clock to provide the smartcard clock. The value given in the register (5 significant bits) is multiplied by 2 to give the division factor of the source clock frequency. This parameter can be a value of ***SMARTCARD_Prescaler***
- ***uint32_t SMARTCARD_InitTypeDef::GuardTime***
  Specifies the SmartCard Guard Time value in terms of number of baud clocks
- ***uint32_t SMARTCARD_InitTypeDef::NACKState***
  Specifies the SmartCard NACK Transmission state. This parameter can be a value of ***SMARTCARD_NACK_State***

58.1.2 SMARTCARD_HandleTypeDef

Data Fields

- ***USART_TypeDef * Instance***
- ***SMARTCARD_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferSize***
- ***uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***uint16_t RxXferSize***
- ***uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_SMARTCARD_StateTypeDef gState***
- ***__IO HAL_SMARTCARD_StateTypeDef RxState***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***USART_TypeDef* SMARTCARD_HandleTypeDef::Instance***
- ***SMARTCARD_InitTypeDef SMARTCARD_HandleTypeDef::Init***
- ***uint8_t* SMARTCARD_HandleTypeDef::pTxBuffPtr***
- ***uint16_t SMARTCARD_HandleTypeDef::TxXferSize***
- ***uint16_t SMARTCARD_HandleTypeDef::TxXferCount***
- ***uint8_t* SMARTCARD_HandleTypeDef::pRxBuffPtr***
- ***uint16_t SMARTCARD_HandleTypeDef::RxXferSize***
- ***uint16_t SMARTCARD_HandleTypeDef::RxXferCount***
- ***DMA_HandleTypeDef* SMARTCARD_HandleTypeDef::hdmatx***
- ***DMA_HandleTypeDef* SMARTCARD_HandleTypeDef::hdmarx***
- ***HAL_LockTypeDef SMARTCARD_HandleTypeDef::Lock***
- ***__IO HAL_SMARTCARD_StateTypeDef SMARTCARD_HandleTypeDef::gState***
- ***__IO HAL_SMARTCARD_StateTypeDef SMARTCARD_HandleTypeDef::RxState***
- ***__IO uint32_t SMARTCARD_HandleTypeDef::ErrorCode***

58.2 SMARTCARD Firmware driver API description

58.2.1 How to use this driver

The SMARTCARD HAL driver can be used as follows:

1. Declare a SMARTCARD_HandleTypeDef handle structure.
2. Initialize the SMARTCARD low level resources by implementing the HAL_SMARTCARD_MspInit() API:
   a. Enable the USARTx interface clock.
   b. SMARTCARD pins configuration:
      – Enable the clock for the SMARTCARD GPIOs.
      – Configure these SMARTCARD pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_SMARTCARD_Transmit_IT() and HAL_SMARTCARD_Receive_IT() APIs):
      – Configure the USARTx interrupt priority.
      – Enable the NVIC USART IRQ handle.
   d. DMA Configuration if you need to use DMA process (HAL_SMARTCARD_Transmit_DMA() and HAL_SMARTCARD_Receive_DMA() APIs):
      – Declare a DMA handle structure for the Tx/Rx stream.
      – Enable the DMAx interface clock.
      – Configure the declared DMA handle structure with the required Tx/Rx parameters.
      – Configure the DMA Tx/Rx Stream.
      – Associate the initialized DMA handle to the SMARTCARD DMA Tx/Rx handle.
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.
3. Program the Baud Rate, Word Length , Stop Bit, Parity, Hardware flow control and Mode(Receiver/Transmitter) in the SMARTCARD Init structure.
4. Initialize the SMARTCARD registers by calling the HAL_SMARTCARD_Init() API:
   – These APIs configure also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customized HAL_SMARTCARD_MspInit() API.

The specific SMARTCARD interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_SMARTCARD_ENABLE_IT() and __HAL_SMARTCARD_DISABLE_IT() inside the transmit and receive process.

Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_SMARTCARD_Transmit()
- Receive an amount of data in blocking mode using HAL_SMARTCARD_Receive()

Interrupt mode IO operation

- Send an amount of data in non blocking mode using HAL_SMARTCARD_Transmit_IT()

- At transmission end of transfer HAL_SMARTCARD_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_TxCpltCallback
- Receive an amount of data in non blocking mode using HAL_SMARTCARD_Receive_IT()
- At reception end of transfer HAL_SMARTCARD_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_RxCpltCallback
- In case of transfer Error, HAL_SMARTCARD_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_ErrorCallback

DMA mode IO operation

- Send an amount of data in non blocking mode (DMA) using HAL_SMARTCARD_Transmit_DMA()
- At transmission end of transfer HAL_SMARTCARD_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_TxCpltCallback
- Receive an amount of data in non blocking mode (DMA) using HAL_SMARTCARD_Receive_DMA()
- At reception end of transfer HAL_SMARTCARD_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_RxCpltCallback
- In case of transfer Error, HAL_SMARTCARD_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SMARTCARD_ErrorCallback

SMARTCARD HAL driver macros list

Below the list of most used macros in SMARTCARD HAL driver.

- __HAL_SMARTCARD_ENABLE: Enable the SMARTCARD peripheral
- __HAL_SMARTCARD_DISABLE: Disable the SMARTCARD peripheral
- __HAL_SMARTCARD_GET_FLAG : Check whether the specified SMARTCARD flag is set or not
- __HAL_SMARTCARD_CLEAR_FLAG : Clear the specified SMARTCARD pending flag
- __HAL_SMARTCARD_ENABLE_IT: Enable the specified SMARTCARD interrupt
- __HAL_SMARTCARD_DISABLE_IT: Disable the specified SMARTCARD interrupt

You can refer to the SMARTCARD HAL driver header file for more useful macros

58.2.2 Initialization and Configuration functions

This subsection provides a set of functions allowing to initialize the USART in Smartcard mode.

The Smartcard interface is designed to support asynchronous protocol Smartcards as defined in the ISO 7816-3 standard.

The USART can provide a clock to the smartcard through the SCLK output. In smartcard mode, SCLK is not associated to the communication but is simply derived from the internal peripheral input clock through a 5-bit prescaler.

- For the Smartcard mode only these parameters can be configured:
  - Baud Rate
  - Word Length => Should be 9 bits (8 bits + parity)
  - Stop Bit
  - Parity: => Should be enabled
  - USART polarity
  - USART phase
  - USART LastBit
  - Receiver/transmitter modes
  - Prescaler
  - GuardTime
  - NACKState: The Smartcard NACK state
- Recommended SmartCard interface configuration to get the Answer to Reset from the Card:
  - Word Length = 9 Bits
  - 1.5 Stop Bit
  - Even parity
  - BaudRate = 12096 baud
  - Tx and Rx enabled

Please refer to the ISO 7816-3 specification for more details.

It is also possible to choose 0.5 stop bit for receiving but it is recommended to use 1.5 stop bits for both transmitting and receiving to avoid switching between the two configurations.

The HAL_SMARTCARD_Init() function follows the USART SmartCard configuration procedure (details for the procedure are available in reference manual (RM0329)).

This section contains the following APIs:

- ***HAL_SMARTCARD_Init()***
- ***HAL_SMARTCARD_DeInit()***
- ***HAL_SMARTCARD_MspInit()***
- ***HAL_SMARTCARD_MspDeInit()***
- ***HAL_SMARTCARD_ReInit()***

58.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the SMARTCARD data transfers. Smartcard is a single wire half duplex communication protocol. The Smartcard interface is designed to support asynchronous protocol Smartcards as defined in the ISO 7816-3 standard. The USART should be configured as: (+) 8 bits plus parity: where M=1 and PCE=1 in the USART_CR1 register (+) 1.5 stop bits when transmitting and receiving: where STOP=11 in the USART_CR2 register.

1. There are two modes of transfer:
   - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.

- Non Blocking mode: The communication is performed using Interrupts or DMA, These APIs return the HAL status. The end of the data processing will be indicated through the dedicated SMARTCARD IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_SMARTCARD_TxCpltCallback(), HAL_SMARTCARD_RxCpltCallback() user callbacks will be executed respectively at the end of the Transmit or Receive process The HAL_SMARTCARD_ErrorCallback() user callback will be executed when a communication error is detected

2. Blocking mode APIs are :
   - HAL_SMARTCARD_Transmit()
   - HAL_SMARTCARD_Receive()
3. Non Blocking mode APIs with Interrupt are :
   - HAL_SMARTCARD_Transmit_IT()
   - HAL_SMARTCARD_Receive_IT()
   - HAL_SMARTCARD_IRQHandler()
4. Non Blocking mode functions with DMA are :
   - HAL_SMARTCARD_Transmit_DMA()
   - HAL_SMARTCARD_Receive_DMA()
5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   - HAL_SMARTCARD_TxCpltCallback()
   - HAL_SMARTCARD_RxCpltCallback()
   - HAL_SMARTCARD_ErrorCallback()

This subsection provides a set of functions allowing to manage the SMARTCARD data transfers. Smartcard is a single wire half duplex communication protocol. The Smartcard interface is designed to support asynchronous protocol Smartcards as defined in the ISO 7816-3 standard. The USART should be configured as:

- 8 bits plus parity: where M=1 and PCE=1 in the USART_CR1 register
- 1.5 stop bits when transmitting and receiving: where STOP=11 in the USART_CR2 register. (#) There are two modes of transfer:
  - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.
  - Non Blocking mode: The communication is performed using Interrupts or DMA, These APIs return the HAL status. The end of the data processing will be indicated through the dedicated SMARTCARD IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_SMARTCARD_TxCpltCallback(), HAL_SMARTCARD_RxCpltCallback() user callbacks will be executed respectively at the end of the Transmit or Receive process The HAL_SMARTCARD_ErrorCallback() user callback will be executed when a communication error is detected (#) Blocking mode APIs are :
  - HAL_SMARTCARD_Transmit()
  - HAL_SMARTCARD_Receive() (#) Non Blocking mode APIs with Interrupt are :
  - HAL_SMARTCARD_Transmit_IT()
  - HAL_SMARTCARD_Receive_IT()
  - HAL_SMARTCARD_IRQHandler() (#) Non Blocking mode functions with DMA are :
  - HAL_SMARTCARD_Transmit_DMA()
  - HAL_SMARTCARD_Receive_DMA() (#) A set of Transfer Complete Callbacks are provided in non Blocking mode:
  - HAL_SMARTCARD_TxCpltCallback()
  - HAL_SMARTCARD_RxCpltCallback()
  - HAL_SMARTCARD_ErrorCallback()

This section contains the following APIs:

- ***HAL_SMARTCARD_Transmit()***
- ***HAL_SMARTCARD_Receive()***
- ***HAL_SMARTCARD_Transmit_IT()***
- ***HAL_SMARTCARD_Receive_IT()***
- ***HAL_SMARTCARD_Transmit_DMA()***
- ***HAL_SMARTCARD_Receive_DMA()***
- ***HAL_SMARTCARD_IRQHandler()***
- ***HAL_SMARTCARD_TxCpltCallback()***
- ***HAL_SMARTCARD_RxCpltCallback()***
- ***HAL_SMARTCARD_ErrorCallback()***

58.2.4 Peripheral State and Errors functions

This subsection provides a set of functions allowing to control the SmartCard.

- HAL_SMARTCARD_GetState() API can be helpful to check in run-time the state of the SmartCard peripheral.
- HAL_SMARTCARD_GetError() check in run-time errors that could be occurred during communication.

This section contains the following APIs:

- ***HAL_SMARTCARD_GetState()***
- ***HAL_SMARTCARD_GetError()***

58.2.5 Detailed description of functions

HAL_SMARTCARD_Init

Function Name	**HAL_StatusTypeDef HAL_SMARTCARD_Init (SMARTCARD_HandleTypeDef * hsc)**
Function Description	Initializes the SmartCard mode according to the specified parameters in the SMARTCARD_InitTypeDef and create the associated handle .
Parameters	• **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
Return values	• **HAL:** status

HAL_SMARTCARD_ReInit

Function Name	**HAL_StatusTypeDef HAL_SMARTCARD_ReInit (SMARTCARD_HandleTypeDef * hsc)**
Function Description	

HAL_SMARTCARD_DeInit

Function Name	**HAL_StatusTypeDef HAL_SMARTCARD_DeInit (SMARTCARD_HandleTypeDef * hsc)**
Function Description	DeInitializes the USART SmartCard peripheral.
Parameters	• **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD

module.

Return values
- **HAL:** status

HAL_SMARTCARD_MspInit

Function Name **void HAL_SMARTCARD_MspInit (SMARTCARD_HandleTypeDef * hsc)**

Function Description SMARTCARD MSP Init.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_MspDeInit

Function Name **void HAL_SMARTCARD_MspDeInit (SMARTCARD_HandleTypeDef * hsc)**

Function Description SMARTCARD MSP DeInit.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_Transmit

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Transmit (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Send an amount of data in blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SMARTCARD_Receive

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Receive (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receive an amount of data in blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be received

- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SMARTCARD_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Transmit_IT (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size)**

Function Description Send an amount of data in non blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent

Return values
- **HAL:** status

HAL_SMARTCARD_Receive_IT

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Receive_IT (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size)**

Function Description Receive an amount of data in non blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be received

Return values
- **HAL:** status

HAL_SMARTCARD_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Transmit_DMA (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size)**

Function Description Send an amount of data in non blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent

Return values
- **HAL:** status

HAL_SMARTCARD_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_SMARTCARD_Receive_DMA (SMARTCARD_HandleTypeDef * hsc, uint8_t * pData, uint16_t Size)**

Function Description Receive an amount of data in non blocking mode.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be received

Return values
- **HAL:** status

Notes
- When the SMARTCARD parity is enabled (PCE = 1) the data received contain the parity bit.s

HAL_SMARTCARD_IRQHandler

Function Name **void HAL_SMARTCARD_IRQHandler (SMARTCARD_HandleTypeDef * hsc)**

Function Description This function handles SMARTCARD interrupt request.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_TxCpltCallback

Function Name **void HAL_SMARTCARD_TxCpltCallback (SMARTCARD_HandleTypeDef * hsc)**

Function Description Tx Transfer completed callbacks.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_RxCpltCallback

Function Name **void HAL_SMARTCARD_RxCpltCallback (SMARTCARD_HandleTypeDef * hsc)**

Function Description Rx Transfer completed callbacks.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_ErrorCallback

Function Name **void HAL_SMARTCARD_ErrorCallback (SMARTCARD_HandleTypeDef * hsc)**

Function Description SMARTCARD error callbacks.

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **None**

HAL_SMARTCARD_GetState

Function Name **HAL_SMARTCARD_StateTypeDef HAL_SMARTCARD_GetState (SMARTCARD_HandleTypeDef * hsc)**

Function Description return the SMARTCARD state

Parameters
- **hsc:** pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for SMARTCARD module.

Return values
- **HAL:** state

HAL_SMARTCARD_GetError

Function Name **uint32_t HAL_SMARTCARD_GetError (SMARTCARD_HandleTypeDef * hsc)**

Function Description Return the SMARTCARD error code.

Parameters
- **hsc:** : pointer to a SMARTCARD_HandleTypeDef structure that contains the configuration information for the specified SMARTCARD.

Return values
- **SMARTCARD:** Error Code

58.3 SMARTCARD Firmware driver defines

58.3.1 SMARTCARD

SMARTCARD Clock Phase

SMARTCARD_PHASE_1EDGE

SMARTCARD_PHASE_2EDGE

SMARTCARD Clock Polarity

SMARTCARD_POLARITY_LOW

SMARTCARD_POLARITY_HIGH

SMARTCARD DMA requests

SMARTCARD_DMAREQ_TX

SMARTCARD_DMAREQ_RX

SMARTCARD Error Code

HAL_SMARTCARD_ERROR_NONE	No error
HAL_SMARTCARD_ERROR_PE	Parity error
HAL_SMARTCARD_ERROR_NE	Noise error
HAL_SMARTCARD_ERROR_FE	Frame error
HAL_SMARTCARD_ERROR_ORE	Overrun error

HAL_SMARTCARD_ERROR_DMA DMA transfer error

SMARTCARD Exported Macros

__HAL_SMARTCARD_RESET_HANDLE_STATE

Description:

- Reset SMARTCARD handle gstate & RxState.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.

Return value:

- None

__HAL_SMARTCARD_FLUSH_DRREGISTER

Description:

- Flushs the Smartcard DR register.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.

__HAL_SMARTCARD_GET_FLAG

Description:

- Checks whether the specified Smartcard flag is set or not.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - SMARTCARD_FLAG_TXE: Transmit data register empty flag
  - SMARTCARD_FLAG_TC: Transmission Complete flag
  - SMARTCARD_FLAG_RXNE: Receive data register not empty flag
  - SMARTCARD_FLAG_IDLE: Idle Line detection flag
  - SMARTCARD_FLAG_ORE: Overrun Error flag
  - SMARTCARD_FLAG_NE: Noise Error flag
  - SMARTCARD_FLAG_FE: Framing Error flag
  - SMARTCARD_FLAG_PE: Parity Error flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_SMARTCARD_CLEAR_FLAG

Description:

- Clears the specified Smartcard pending flags.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.
- __FLAG__: specifies the flag to check. This parameter can be any combination of the following values:
  - SMARTCARD_FLAG_TC: Transmission Complete flag.
  - SMARTCARD_FLAG_RXNE: Receive data register not empty flag.

Notes:

- PE (Parity error), FE (Framing error), NE (Noise error) and ORE (Overrun error) flags are cleared by software sequence: a read operation to USART_SR register followed by a read operation to USART_DR register. RXNE flag can be also cleared by a read to the USART_DR register. TC flag can be also cleared by software sequence: a read operation to USART_SR register followed by a write operation to USART_DR register. TXE flag is cleared only by a write to the USART_DR register.

__HAL_SMARTCARD_CLEAR_PEFLAG

Description:

- Clear the SMARTCARD PE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or SMARTCARD peripheral.

Return value:

- None

__HAL_SMARTCARD_CLEAR_FEFLAG

Description:

- Clear the SMARTCARD FE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or SMARTCARD peripheral.

Return value:

- None

__HAL_SMARTCARD_CLEAR_NEFLAG

Description:

- Clear the SMARTCARD NE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or SMARTCARD peripheral.

Return value:

- None

__HAL_SMARTCARD_CLEAR_OREFLAG

Description:

- Clear the SMARTCARD ORE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or SMARTCARD peripheral.

Return value:

- None

__HAL_SMARTCARD_CLEAR_IDLEFLAG

Description:

- Clear the SMARTCARD IDLE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or SMARTCARD peripheral.

Return value:

- None

__HAL_SMARTCARD_ENABLE_IT

Description:

- Enables or disables the specified SmartCard interrupts.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.
- __INTERRUPT__: specifies the SMARTCARD interrupt source to check. This parameter can be one of the following values:
  - SMARTCARD_IT_TXE: Transmit Data Register empty interrupt
  - SMARTCARD_IT_TC: Transmission complete interrupt
  - SMARTCARD_IT_RXNE: Receive Data register not empty interrupt
  - SMARTCARD_IT_IDLE: Idle line detection interrupt
  - SMARTCARD_IT_PE: Parity Error interrupt
  - SMARTCARD_IT_ERR: Error interrupt(Frame error, noise error, overrun error)

__HAL_SMARTCARD_DISABLE_IT

__HAL_SMARTCARD_GET_IT_SOURCE

Description:

- Checks whether the specified SmartCard interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the SmartCard Handle.
- __IT__: specifies the SMARTCARD interrupt source to check. This parameter can be one of the following values:
  - SMARTCARD_IT_TXE: Transmit Data Register empty interrupt
  - SMARTCARD_IT_TC: Transmission complete interrupt
  - SMARTCARD_IT_RXNE: Receive Data register not empty interrupt
  - SMARTCARD_IT_IDLE: Idle line detection interrupt
  - SMARTCARD_IT_ERR: Error interrupt
  - SMARTCARD_IT_PE: Parity Error interrupt

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_SMARTCARD_ONE_BIT_SAMPLE_ENABLE

Description:

- Macro to enable the SMARTCARD's one bit sample method.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.

Return value:

- None

__HAL_SMARTCARD_ONE_BIT_SAMPLE_DISABLE

Description:

- Macro to disable the SMARTCARD's one bit sample method.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle.

Return value:

- None

__HAL_SMARTCARD_ENABLE

Description:

- Enable the USART associated to the SMARTCARD Handle.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle. SMARTCARD Handle selects the USARTx peripheral (USART availability and x value depending on device).

Return value:

- None

__HAL_SMARTCARD_DISABLE

Description:

- Disable the USART associated to the SMARTCARD Handle.

Parameters:

- __HANDLE__: specifies the SMARTCARD Handle. SMARTCARD Handle selects the USARTx peripheral (USART availability and x value depending

on device).

Return value:

- None

__HAL_SMARTCARD_DMA_REQUEST_ENABLE

Description:

- Macros to enable or disable the SmartCard DMA request.

Parameters:

- __HANDLE__: specifies the SmartCard Handle.
- __REQUEST__: specifies the SmartCard DMA request. This parameter can be one of the following values:
  - SMARTCARD_DMAREQ_TX: SmartCard DMA transmit request
  - SMARTCARD_DMAREQ_RX : SmartCard DMA receive request

__HAL_SMARTCARD_DMA_REQUEST_DISABLE

SMARTCARD Last Bit

SMARTCARD_LASTBIT_DISABLE

SMARTCARD_LASTBIT_ENABLE

SMARTCARD Mode

SMARTCARD_MODE_RX

SMARTCARD_MODE_TX

SMARTCARD_MODE_TX_RX

SMARTCARD NACK State

SMARTCARD_NACK_ENABLE

SMARTCARD_NACK_DISABLE

SMARTCARD Parity

SMARTCARD_PARITY_EVEN

SMARTCARD_PARITY_ODD

SMARTCARD Prescaler

SMARTCARD_PRESCALER_SYSCLK_DIV2	SYSCLK divided by 2
SMARTCARD_PRESCALER_SYSCLK_DIV4	SYSCLK divided by 4
SMARTCARD_PRESCALER_SYSCLK_DIV6	SYSCLK divided by 6
SMARTCARD_PRESCALER_SYSCLK_DIV8	SYSCLK divided by 8
SMARTCARD_PRESCALER_SYSCLK_DIV10	SYSCLK divided by 10
SMARTCARD_PRESCALER_SYSCLK_DIV12	SYSCLK divided by 12

SMARTCARD_PRESCALER_SYSCLK_DIV14	SYSCLK divided by 14
SMARTCARD_PRESCALER_SYSCLK_DIV16	SYSCLK divided by 16
SMARTCARD_PRESCALER_SYSCLK_DIV18	SYSCLK divided by 18
SMARTCARD_PRESCALER_SYSCLK_DIV20	SYSCLK divided by 20
SMARTCARD_PRESCALER_SYSCLK_DIV22	SYSCLK divided by 22
SMARTCARD_PRESCALER_SYSCLK_DIV24	SYSCLK divided by 24
SMARTCARD_PRESCALER_SYSCLK_DIV26	SYSCLK divided by 26
SMARTCARD_PRESCALER_SYSCLK_DIV28	SYSCLK divided by 28
SMARTCARD_PRESCALER_SYSCLK_DIV30	SYSCLK divided by 30
SMARTCARD_PRESCALER_SYSCLK_DIV32	SYSCLK divided by 32
SMARTCARD_PRESCALER_SYSCLK_DIV34	SYSCLK divided by 34
SMARTCARD_PRESCALER_SYSCLK_DIV36	SYSCLK divided by 36
SMARTCARD_PRESCALER_SYSCLK_DIV38	SYSCLK divided by 38
SMARTCARD_PRESCALER_SYSCLK_DIV40	SYSCLK divided by 40
SMARTCARD_PRESCALER_SYSCLK_DIV42	SYSCLK divided by 42
SMARTCARD_PRESCALER_SYSCLK_DIV44	SYSCLK divided by 44
SMARTCARD_PRESCALER_SYSCLK_DIV46	SYSCLK divided by 46
SMARTCARD_PRESCALER_SYSCLK_DIV48	SYSCLK divided by 48
SMARTCARD_PRESCALER_SYSCLK_DIV50	SYSCLK divided by 50
SMARTCARD_PRESCALER_SYSCLK_DIV52	SYSCLK divided by 52
SMARTCARD_PRESCALER_SYSCLK_DIV54	SYSCLK divided by 54
SMARTCARD_PRESCALER_SYSCLK_DIV56	SYSCLK divided by 56
SMARTCARD_PRESCALER_SYSCLK_DIV58	SYSCLK divided by 58
SMARTCARD_PRESCALER_SYSCLK_DIV60	SYSCLK divided by 60
SMARTCARD_PRESCALER_SYSCLK_DIV62	SYSCLK divided by 62

SMARTCARD Number of Stop Bits

SMARTCARD_STOPBITS_0_5

SMARTCARD_STOPBITS_1_5

SMARTCARD Word Length

SMARTCARD_WORDLENGTH_9B

59 HAL SPDIFRX Generic Driver

59.1 SPDIFRX Firmware driver registers structures

59.1.1 SPDIFRX_InitTypeDef

Data Fields

- ***uint32_t InputSelection***
- ***uint32_t Retries***
- ***uint32_t WaitForActivity***
- ***uint32_t ChannelSelection***
- ***uint32_t DataFormat***
- ***uint32_t StereoMode***
- ***uint32_t PreambleTypeMask***
- ***uint32_t ChannelStatusMask***
- ***uint32_t ValidityBitMask***
- ***uint32_t ParityErrorMask***

Field Documentation

- ***uint32_t SPDIFRX_InitTypeDef::InputSelection***
  Specifies the SPDIF input selection. This parameter can be a value of ***SPDIFRX_Input_Selection***
- ***uint32_t SPDIFRX_InitTypeDef::Retries***
  Specifies the Maximum allowed re-tries during synchronization phase. This parameter can be a value of ***SPDIFRX_Max_Retries***
- ***uint32_t SPDIFRX_InitTypeDef::WaitForActivity***
  Specifies the wait for activity on SPDIF selected input. This parameter can be a value of ***SPDIFRX_Wait_For_Activity***.
- ***uint32_t SPDIFRX_InitTypeDef::ChannelSelection***
  Specifies whether the control flow will take the channel status from channel A or B. This parameter can be a value of ***SPDIFRX_Channel_Selection***
- ***uint32_t SPDIFRX_InitTypeDef::DataFormat***
  Specifies the Data samples format (LSB, MSB, ...). This parameter can be a value of ***SPDIFRX_Data_Format***
- ***uint32_t SPDIFRX_InitTypeDef::StereoMode***
  Specifies whether the peripheral is in stereo or mono mode. This parameter can be a value of ***SPDIFRX_Stereo_Mode***
- ***uint32_t SPDIFRX_InitTypeDef::PreambleTypeMask***
  Specifies whether The preamble type bits are copied or not into the received frame. This parameter can be a value of ***SPDIFRX_PT_Mask***
- ***uint32_t SPDIFRX_InitTypeDef::ChannelStatusMask***
  Specifies whether the channel status and user bits are copied or not into the received frame. This parameter can be a value of ***SPDIFRX_ChannelStatus_Mask***
- ***uint32_t SPDIFRX_InitTypeDef::ValidityBitMask***
  Specifies whether the validity bit is copied or not into the received frame. This parameter can be a value of ***SPDIFRX_V_Mask***

- ***uint32_t SPDIFRX_InitTypeDef::ParityErrorMask***
  Specifies whether the parity error bit is copied or not into the received frame. This parameter can be a value of ***SPDIFRX_PE_Mask***

59.1.2 SPDIFRX_SetDataFormatTypeDef

Data Fields

- ***uint32_t DataFormat***
- ***uint32_t StereoMode***
- ***uint32_t PreambleTypeMask***
- ***uint32_t ChannelStatusMask***
- ***uint32_t ValidityBitMask***
- ***uint32_t ParityErrorMask***

Field Documentation

- ***uint32_t SPDIFRX_SetDataFormatTypeDef::DataFormat***
  Specifies the Data samples format (LSB, MSB, ...). This parameter can be a value of ***SPDIFRX_Data_Format***
- ***uint32_t SPDIFRX_SetDataFormatTypeDef::StereoMode***
  Specifies whether the peripheral is in stereo or mono mode. This parameter can be a value of ***SPDIFRX_Stereo_Mode***
- ***uint32_t SPDIFRX_SetDataFormatTypeDef::PreambleTypeMask***
  Specifies whether The preamble type bits are copied or not into the received frame. This parameter can be a value of ***SPDIFRX_PT_Mask***
- ***uint32_t SPDIFRX_SetDataFormatTypeDef::ChannelStatusMask***
  Specifies whether the channel status and user bits are copied or not into the received frame. This parameter can be a value of ***SPDIFRX_ChannelStatus_Mask***
- ***uint32_t SPDIFRX_SetDataFormatTypeDef::ValidityBitMask***
  Specifies whether the validity bit is copied or not into the received frame. This parameter can be a value of ***SPDIFRX_V_Mask***
- ***uint32_t SPDIFRX_SetDataFormatTypeDef::ParityErrorMask***
  Specifies whether the parity error bit is copied or not into the received frame. This parameter can be a value of ***SPDIFRX_PE_Mask***

59.1.3 SPDIFRX_HandleTypeDef

Data Fields

- ***SPDIFRX_TypeDef * Instance***
- ***SPDIFRX_InitTypeDef Init***
- ***uint32_t * pRxBuffPtr***
- ***uint32_t * pCsBuffPtr***
- ***__IO uint16_t RxXferSize***
- ***__IO uint16_t RxXferCount***
- ***__IO uint16_t CsXferSize***
- ***__IO uint16_t CsXferCount***
- ***DMA_HandleTypeDef * hdmaCsRx***

- ***DMA_HandleTypeDef * hdmaDrRx***
- ***__IO HAL_LockTypeDef Lock***
- ***__IO HAL_SPDIFRX_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***SPDIFRX_TypeDef* SPDIFRX_HandleTypeDef::Instance***
- ***SPDIFRX_InitTypeDef SPDIFRX_HandleTypeDef::Init***
- ***uint32_t* SPDIFRX_HandleTypeDef::pRxBuffPtr***
- ***uint32_t* SPDIFRX_HandleTypeDef::pCsBuffPtr***
- ***__IO uint16_t SPDIFRX_HandleTypeDef::RxXferSize***
- ***__IO uint16_t SPDIFRX_HandleTypeDef::RxXferCount***
- ***__IO uint16_t SPDIFRX_HandleTypeDef::CsXferSize***
- ***__IO uint16_t SPDIFRX_HandleTypeDef::CsXferCount***
- ***DMA_HandleTypeDef* SPDIFRX_HandleTypeDef::hdmaCsRx***
- ***DMA_HandleTypeDef* SPDIFRX_HandleTypeDef::hdmaDrRx***
- ***__IO HAL_LockTypeDef SPDIFRX_HandleTypeDef::Lock***
- ***__IO HAL_SPDIFRX_StateTypeDef SPDIFRX_HandleTypeDef::State***
- ***__IO uint32_t SPDIFRX_HandleTypeDef::ErrorCode***

59.2 SPDIFRX Firmware driver API description

59.2.1 How to use this driver

The SPDIFRX HAL driver can be used as follow:

1. Declare SPDIFRX_HandleTypeDef handle structure.
2. Initialize the SPDIFRX low level resources by implement the HAL_SPDIFRX_MspInit() API:
   a. Enable the SPDIFRX interface clock.
   b. SPDIFRX pins configuration:
      – Enable the clock for the SPDIFRX GPIOs.
      – Configure these SPDIFRX pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_SPDIFRX_ReceiveControlFlow_IT() and HAL_SPDIFRX_ReceiveDataFlow_IT() API's).
      – Configure the SPDIFRX interrupt priority.
      – Enable the NVIC SPDIFRX IRQ handle.
   d. DMA Configuration if you need to use DMA process (HAL_SPDIFRX_ReceiveDataFlow_DMA() and HAL_SPDIFRX_ReceiveControlFlow_DMA() API's).
      – Declare a DMA handle structure for the reception of the Data Flow channel.
      – Declare a DMA handle structure for the reception of the Control Flow channel.
      – Enable the DMAx interface clock.
      – Configure the declared DMA handle structure CtrlRx/DataRx with the required parameters.
      – Configure the DMA Channel.
      – Associate the initialized DMA handle to the SPDIFRX DMA CtrlRx/DataRx handle.

  - Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA CtrlRx/DataRx channel.

3. Program the input selection, re-tries number, wait for activity, channel status selection, data format, stereo mode and masking of user bits using HAL_SPDIFRX_Init() function. The specific SPDIFRX interrupts (RXNE/CSRNE and Error Interrupts) will be managed using the macros __SPDIFRX_ENABLE_IT() and __SPDIFRX_DISABLE_IT() inside the receive process. Make sure that ck_spdif clock is configured.
4. Three operation modes are available within this driver :

Polling mode for reception operation (for debug purpose)

- Receive data flow in blocking mode using HAL_SPDIFRX_ReceiveDataFlow()
- Receive control flow of data in blocking mode using HAL_SPDIFRX_ReceiveControlFlow()

Interrupt mode for reception operation

- Receive an amount of data (Data Flow) in non blocking mode using HAL_SPDIFRX_ReceiveDataFlow_IT()
- Receive an amount of data (Control Flow) in non blocking mode using HAL_SPDIFRX_ReceiveControlFlow_IT()
- At reception end of half transfer HAL_SPDIFRX_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_RxHalfCpltCallback
- At reception end of transfer HAL_SPDIFRX_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_RxCpltCallback
- In case of transfer Error, HAL_SPDIFRX_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_ErrorCallback

DMA mode for reception operation

- Receive an amount of data (Data Flow) in non blocking mode (DMA) using HAL_SPDIFRX_ReceiveDataFlow_DMA()
- Receive an amount of data (Control Flow) in non blocking mode (DMA) using HAL_SPDIFRX_ReceiveControlFlow_DMA()
- At reception end of half transfer HAL_SPDIFRX_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_RxHalfCpltCallback
- At reception end of transfer HAL_SPDIFRX_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_RxCpltCallback
- In case of transfer Error, HAL_SPDIFRX_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_SPDIFRX_ErrorCallback
- Stop the DMA Transfer using HAL_SPDIFRX_DMAStop()

SPDIFRX HAL driver macros list

Below the list of most used macros in SPDIFRX HAL driver.

- __HAL_SPDIFRX_IDLE: Disable the specified SPDIFRX peripheral (IDEL State)

- __HAL_SPDIFRX_SYNC: Enable the synchronization state of the specified SPDIFRX peripheral (SYNC State)
- __HAL_SPDIFRX_RCV: Enable the receive state of the specified SPDIFRX peripheral (RCV State)
- __HAL_SPDIFRX_ENABLE_IT : Enable the specified SPDIFRX interrupts
- __HAL_SPDIFRX_DISABLE_IT : Disable the specified SPDIFRX interrupts
- __HAL_SPDIFRX_GET_FLAG: Check whether the specified SPDIFRX flag is set or not.

You can refer to the SPDIFRX HAL driver header file for more useful macros

59.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and de-initialize the SPDIFRX peripheral:

- User must Implement HAL_SPDIFRX_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).
- Call the function HAL_SPDIFRX_Init() to configure the SPDIFRX peripheral with the selected configuration:
  - Input Selection (IN0, IN1,...)
  - Maximum allowed re-tries during synchronization phase
  - Wait for activity on SPDIF selected input
  - Channel status selection (from channel A or B)
  - Data format (LSB, MSB, ...)
  - Stereo mode
  - User bits masking (PT,C,U,V,...)
- Call the function HAL_SPDIFRX_DeInit() to restore the default configuration of the selected SPDIFRXx peripheral.

This section contains the following APIs:

- ***HAL_SPDIFRX_Init()***
- ***HAL_SPDIFRX_DeInit()***
- ***HAL_SPDIFRX_MspInit()***
- ***HAL_SPDIFRX_MspDeInit()***
- ***HAL_SPDIFRX_SetDataFormat()***

59.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the SPDIFRX data transfers.

1. There is two mode of transfer:
   - Blocking mode : The communication is performed in the polling mode. The status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode : The communication is performed using Interrupts or DMA. These functions return the status of the transfer start-up. The end of the data processing will be indicated through the dedicated SPDIFRX IRQ when using Interrupt mode or the DMA IRQ when using DMA mode.
2. Blocking mode functions are :
   - HAL_SPDIFRX_ReceiveDataFlow()

– HAL_SPDIFRX_ReceiveControlFlow() (+@) Do not use blocking mode to receive both control and data flow at the same time.

3. No-Blocking mode functions with Interrupt are :
   – HAL_SPDIFRX_ReceiveControlFlow_IT()
   – HAL_SPDIFRX_ReceiveDataFlow_IT()
4. No-Blocking mode functions with DMA are :
   – HAL_SPDIFRX_ReceiveControlFlow_DMA()
   – HAL_SPDIFRX_ReceiveDataFlow_DMA()
5. A set of Transfer Complete Callbacks are provided in No_Blocking mode:
   – HAL_SPDIFRX_RxCpltCallback()
   – HAL_SPDIFRX_ErrorCallback()

This section contains the following APIs:

- ***HAL_SPDIFRX_ReceiveDataFlow()***
- ***HAL_SPDIFRX_ReceiveControlFlow()***
- ***HAL_SPDIFRX_ReceiveDataFlow_IT()***
- ***HAL_SPDIFRX_ReceiveControlFlow_IT()***
- ***HAL_SPDIFRX_ReceiveDataFlow_DMA()***
- ***HAL_SPDIFRX_ReceiveControlFlow_DMA()***
- ***HAL_SPDIFRX_DMAStop()***
- ***HAL_SPDIFRX_IRQHandler()***
- ***HAL_SPDIFRX_RxHalfCpltCallback()***
- ***HAL_SPDIFRX_RxCpltCallback()***
- ***HAL_SPDIFRX_CxHalfCpltCallback()***
- ***HAL_SPDIFRX_CxCpltCallback()***
- ***HAL_SPDIFRX_ErrorCallback()***

59.2.4 Peripheral State and Errors functions

This subsection permit to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_SPDIFRX_GetState()***
- ***HAL_SPDIFRX_GetError()***

59.2.5 Detailed description of functions

HAL_SPDIFRX_Init

Function Name	**HAL_StatusTypeDef HAL_SPDIFRX_Init (SPDIFRX_HandleTypeDef * hspdif)**
Function Description	Initializes the SPDIFRX according to the specified parameters in the SPDIFRX_InitTypeDef and create the associated handle.
Parameters	• **hspdif:** SPDIFRX handle
Return values	• **HAL:** status

HAL_SPDIFRX_DeInit

Function Name	**HAL_StatusTypeDef HAL_SPDIFRX_DeInit (SPDIFRX_HandleTypeDef * hspdif)**
Function Description	DeInitializes the SPDIFRX peripheral.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **HAL:** status

HAL_SPDIFRX_MspInit

Function Name **void HAL_SPDIFRX_MspInit (SPDIFRX_HandleTypeDef * hspdif)**

Function Description SPDIFRX MSP Init.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_MspDeInit

Function Name **void HAL_SPDIFRX_MspDeInit (SPDIFRX_HandleTypeDef * hspdif)**

Function Description SPDIFRX MSP DeInit.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_SetDataFormat

Function Name **HAL_StatusTypeDef HAL_SPDIFRX_SetDataFormat (SPDIFRX_HandleTypeDef * hspdif, SPDIFRX_SetDataFormatTypeDef sDataFormat)**

Function Description Sets the SPDIFRX dtat format according to the specified parameters in the SPDIFRX_InitTypeDef.

Parameters
- **hspdif:** SPDIFRX handle
- **sDataFormat:** SPDIFRX data format

Return values
- **HAL:** status

HAL_SPDIFRX_ReceiveDataFlow

Function Name **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveDataFlow (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receives an amount of data (Data Flow) in blocking mode.

Parameters
- **hspdif:** pointer to SPDIFRX_HandleTypeDef structure that contains the configuration information for SPDIFRX module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be received
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SPDIFRX_ReceiveControlFlow

Function Name **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveControlFlow (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t**

Size, uint32_t Timeout)

Function Description: Receives an amount of data (Control Flow) in blocking mode.

Parameters:
- **hspdif:** pointer to a SPDIFRX_HandleTypeDef structure that contains the configuration information for SPDIFRX module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be received
- **Timeout:** Timeout duration

Return values:
- **HAL:** status

HAL_SPDIFRX_ReceiveControlFlow_IT

Function Name: **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveControlFlow_IT (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t Size)**

Function Description: Receive an amount of data (Control Flow) with Interrupt.

Parameters:
- **hspdif:** SPDIFRX handle
- **pData:** a 32-bit pointer to the Receive data buffer.
- **Size:** number of data sample (Control Flow) to be received :

Return values:
- **HAL:** status

HAL_SPDIFRX_ReceiveDataFlow_IT

Function Name: **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveDataFlow_IT (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t Size)**

Function Description: Receive an amount of data (Data Flow) in non-blocking mode with Interrupt.

Parameters:
- **hspdif:** SPDIFRX handle
- **pData:** a 32-bit pointer to the Receive data buffer.
- **Size:** number of data sample to be received .

Return values:
- **HAL:** status

HAL_SPDIFRX_IRQHandler

Function Name: **void HAL_SPDIFRX_IRQHandler (SPDIFRX_HandleTypeDef * hspdif)**

Function Description: This function handles SPDIFRX interrupt request.

Parameters:
- **hspdif:** SPDIFRX handle

Return values:
- **HAL:** status

HAL_SPDIFRX_ReceiveControlFlow_DMA

Function Name: **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveControlFlow_DMA (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t Size)**

Function Description: Receive an amount of data (Control Flow) with DMA.

Parameters
- **hspdif:** SPDIFRX handle
- **pData:** a 32-bit pointer to the Receive data buffer.
- **Size:** number of data (Control Flow) sample to be received :

Return values
- **HAL:** status

HAL_SPDIFRX_ReceiveDataFlow_DMA

Function Name **HAL_StatusTypeDef HAL_SPDIFRX_ReceiveDataFlow_DMA (SPDIFRX_HandleTypeDef * hspdif, uint32_t * pData, uint16_t Size)**

Function Description Receive an amount of data (Data Flow) mode with DMA.

Parameters
- **hspdif:** SPDIFRX handle
- **pData:** a 32-bit pointer to the Receive data buffer.
- **Size:** number of data sample to be received :

Return values
- **HAL:** status

HAL_SPDIFRX_DMAStop

Function Name **HAL_StatusTypeDef HAL_SPDIFRX_DMAStop (SPDIFRX_HandleTypeDef * hspdif)**

Function Description stop the audio stream receive from the Media.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_RxHalfCpltCallback

Function Name **void HAL_SPDIFRX_RxHalfCpltCallback (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Rx Transfer (Data flow) half completed callbacks.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_RxCpltCallback

Function Name **void HAL_SPDIFRX_RxCpltCallback (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Rx Transfer (Data flow) completed callbacks.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_ErrorCallback

Function Name **void HAL_SPDIFRX_ErrorCallback (SPDIFRX_HandleTypeDef * hspdif)**

Function Description SPDIFRX error callbacks.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_CxHalfCpltCallback

Function Name **void HAL_SPDIFRX_CxHalfCpltCallback (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Rx (Control flow) Transfer half completed callbacks.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_CxCpltCallback

Function Name **void HAL_SPDIFRX_CxCpltCallback (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Rx Transfer (Control flow) completed callbacks.

Parameters
- **hspdif:** SPDIFRX handle

Return values
- **None**

HAL_SPDIFRX_GetState

Function Name **HAL_SPDIFRX_StateTypeDef HAL_SPDIFRX_GetState (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Return the SPDIFRX state.

Parameters
- **hspdif:** : SPDIFRX handle

Return values
- **HAL:** state

HAL_SPDIFRX_GetError

Function Name **uint32_t HAL_SPDIFRX_GetError (SPDIFRX_HandleTypeDef * hspdif)**

Function Description Return the SPDIFRX error code.

Parameters
- **hspdif:** : SPDIFRX handle

Return values
- **SPDIFRX:** Error Code

59.3 SPDIFRX Firmware driver defines

59.3.1 SPDIFRX

SPDIFRX Channel Status Mask

SPDIFRX_CHANNELSTATUS_OFF

SPDIFRX_CHANNELSTATUS_ON

SPDIFRX Channel Selection

SPDIFRX_CHANNEL_A

SPDIFRX_CHANNEL_B

SPDIFRX Data Format

SPDIFRX_DATAFORMAT_LSB

SPDIFRX_DATAFORMAT_MSB

SPDIFRX_DATAFORMAT_32BITS

SPDIFRX Error Code

HAL_SPDIFRX_ERROR_NONE	No error
HAL_SPDIFRX_ERROR_TIMEOUT	Timeout error
HAL_SPDIFRX_ERROR_OVR	OVR error
HAL_SPDIFRX_ERROR_PE	Parity error
HAL_SPDIFRX_ERROR_DMA	DMA transfer error
HAL_SPDIFRX_ERROR_UNKNOWN	Unknown Error error

SPDIFRX Exported Macros

__HAL_SPDIFRX_RESET_HANDLE_STATE

Description:

- Reset SPDIFRX handle state.

Parameters:

- __HANDLE__: SPDIFRX handle.

Return value:

- None

__HAL_SPDIFRX_IDLE

Description:

- Disable the specified SPDIFRX peripheral (IDLE State).

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.

Return value:

- None

__HAL_SPDIFRX_SYNC

Description:

- Enable the specified SPDIFRX peripheral (SYNC State).

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.

Return value:

- None

__HAL_SPDIFRX_RCV

Description:

- Enable the specified SPDIFRX peripheral (RCV State).

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.

Return value:

- None

__HAL_SPDIFRX_ENABLE_IT

Description:

- Enable or disable the specified SPDIFRX interrupts.

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.
- __INTERRUPT__: specifies the interrupt source to enable or disable. This parameter can be one of the following values:
  - SPDIFRX_IT_RXNE
  - SPDIFRX_IT_CSRNE
  - SPDIFRX_IT_PERRIE
  - SPDIFRX_IT_OVRIE
  - SPDIFRX_IT_SBLKIE
  - SPDIFRX_IT_SYNCDIE
  - SPDIFRX_IT_IFEIE

Return value:

- None

__HAL_SPDIFRX_DISABLE_IT

__HAL_SPDIFRX_GET_IT_SOURCE

Description:

- Checks if the specified SPDIFRX interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.
- __INTERRUPT__: specifies the SPDIFRX interrupt source to check. This parameter can be one of the following values:
  - SPDIFRX_IT_RXNE
  - SPDIFRX_IT_CSRNE
  - SPDIFRX_IT_PERRIE
  - SPDIFRX_IT_OVRIE
  - SPDIFRX_IT_SBLKIE
  - SPDIFRX_IT_SYNCDIE
  - SPDIFRX_IT_IFEIE

Return value:

- The: new state of __IT__ (TRUE or

FALSE).

__HAL_SPDIFRX_GET_FLAG

Description:

- Checks whether the specified SPDIFRX flag is set or not.

Parameters:

- __HANDLE__: specifies the SPDIFRX Handle.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - SPDIFRX_FLAG_RXNE
  - SPDIFRX_FLAG_CSRNE
  - SPDIFRX_FLAG_PERR
  - SPDIFRX_FLAG_OVR
  - SPDIFRX_FLAG_SBD
  - SPDIFRX_FLAG_SYNCD
  - SPDIFRX_FLAG_FERR
  - SPDIFRX_FLAG_SERR
  - SPDIFRX_FLAG_TERR

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_SPDIFRX_CLEAR_IT

Description:

- Clears the specified SPDIFRX SR flag, in setting the proper IFCR register bit.

Parameters:

- __HANDLE__: specifies the USART Handle.
- __IT_CLEAR__: specifies the interrupt clear register flag that needs to be set to clear the corresponding interrupt This parameter can be one of the following values:
  - SPDIFRX_FLAG_PERR
  - SPDIFRX_FLAG_OVR
  - SPDIFRX_SR_SBD
  - SPDIFRX_SR_SYNCD

Return value:

- None

SPDIFRX Flags Definition

SPDIFRX_FLAG_RXNE

SPDIFRX_FLAG_CSRNE

SPDIFRX_FLAG_PERR

SPDIFRX_FLAG_OVR

SPDIFRX_FLAG_SBD

SPDIFRX_FLAG_SYNCD

SPDIFRX_FLAG_FERR

SPDIFRX_FLAG_SERR

SPDIFRX_FLAG_TERR

SPDIFRX Input Selection

SPDIFRX_INPUT_IN0

SPDIFRX_INPUT_IN1

SPDIFRX_INPUT_IN2

SPDIFRX_INPUT_IN3

SPDIFRX Interrupts Definition

SPDIFRX_IT_RXNE

SPDIFRX_IT_CSRNE

SPDIFRX_IT_PERRIE

SPDIFRX_IT_OVRIE

SPDIFRX_IT_SBLKIE

SPDIFRX_IT_SYNCDIE

SPDIFRX_IT_IFEIE

SPDIFRX Maximum Retries

SPDIFRX_MAXRETRIES_NONE

SPDIFRX_MAXRETRIES_3

SPDIFRX_MAXRETRIES_15

SPDIFRX_MAXRETRIES_63

SPDIFRX Parity Error Mask

SPDIFRX_PARITYERRORMASK_OFF

SPDIFRX_PARITYERRORMASK_ON

SPDIFRX Preamble Type Mask

SPDIFRX_PREAMBLETYPEMASK_OFF

SPDIFRX_PREAMBLETYPEMASK_ON

SPDIFRX State

SPDIFRX_STATE_IDLE

SPDIFRX_STATE_SYNC

SPDIFRX_STATE_RCV

SPDIFRX Stereo Mode

SPDIFRX_STEREOMODE_DISABLE

SPDIFRX_STEREOMODE_ENABLE

SPDIFRX Validity Mask

SPDIFRX_VALIDITYMASK_OFF

SPDIFRX_VALIDITYMASK_ON

SPDIFRX Wait For Activity

SPDIFRX_WAITFORACTIVITY_OFF

SPDIFRX_WAITFORACTIVITY_ON

60 HAL SPI Generic Driver

60.1 SPI Firmware driver registers structures

60.1.1 SPI_InitTypeDef

Data Fields

- ***uint32_t Mode***
- ***uint32_t Direction***
- ***uint32_t DataSize***
- ***uint32_t CLKPolarity***
- ***uint32_t CLKPhase***
- ***uint32_t NSS***
- ***uint32_t BaudRatePrescaler***
- ***uint32_t FirstBit***
- ***uint32_t TIMode***
- ***uint32_t CRCCalculation***
- ***uint32_t CRCPolynomial***

Field Documentation

- ***uint32_t SPI_InitTypeDef::Mode***
  Specifies the SPI operating mode. This parameter can be a value of ***SPI_Mode***
- ***uint32_t SPI_InitTypeDef::Direction***
  Specifies the SPI bidirectional mode state. This parameter can be a value of ***SPI_Direction***
- ***uint32_t SPI_InitTypeDef::DataSize***
  Specifies the SPI data size. This parameter can be a value of ***SPI_Data_Size***
- ***uint32_t SPI_InitTypeDef::CLKPolarity***
  Specifies the serial clock steady state. This parameter can be a value of ***SPI_Clock_Polarity***
- ***uint32_t SPI_InitTypeDef::CLKPhase***
  Specifies the clock active edge for the bit capture. This parameter can be a value of ***SPI_Clock_Phase***
- ***uint32_t SPI_InitTypeDef::NSS***
  Specifies whether the NSS signal is managed by hardware (NSS pin) or by software using the SSI bit. This parameter can be a value of ***SPI_Slave_Select_management***
- ***uint32_t SPI_InitTypeDef::BaudRatePrescaler***
  Specifies the Baud Rate prescaler value which will be used to configure the transmit and receive SCK clock. This parameter can be a value of ***SPI_BaudRate_Prescaler***
  Note: The communication clock is derived from the master clock. The slave clock does not need to be set.
- ***uint32_t SPI_InitTypeDef::FirstBit***
  Specifies whether data transfers start from MSB or LSB bit. This parameter can be a value of ***SPI_MSB_LSB_transmission***
- ***uint32_t SPI_InitTypeDef::TIMode***
  Specifies if the TI mode is enabled or not. This parameter can be a value of ***SPI_TI_mode***

- ***uint32_t SPI_InitTypeDef::CRCCalculation***
  Specifies if the CRC calculation is enabled or not. This parameter can be a value of ***SPI_CRC_Calculation***
- ***uint32_t SPI_InitTypeDef::CRCPolynomial***
  Specifies the polynomial used for the CRC calculation. This parameter must be a number between Min_Data = 0 and Max_Data = 65535

60.1.2 __SPI_HandleTypeDef

Data Fields

- ***SPI_TypeDef * Instance***
- ***SPI_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferSize***
- ***__IO uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***uint16_t RxXferSize***
- ***__IO uint16_t RxXferCount***
- ***void(* RxISR***
- ***void(* TxISR***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_SPI_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***SPI_TypeDef* __SPI_HandleTypeDef::Instance***
- ***SPI_InitTypeDef __SPI_HandleTypeDef::Init***
- ***uint8_t* __SPI_HandleTypeDef::pTxBuffPtr***
- ***uint16_t __SPI_HandleTypeDef::TxXferSize***
- ***__IO uint16_t __SPI_HandleTypeDef::TxXferCount***
- ***uint8_t* __SPI_HandleTypeDef::pRxBuffPtr***
- ***uint16_t __SPI_HandleTypeDef::RxXferSize***
- ***__IO uint16_t __SPI_HandleTypeDef::RxXferCount***
- ***void(* __SPI_HandleTypeDef::RxISR)(struct __SPI_HandleTypeDef *hspi)***
- ***void(* __SPI_HandleTypeDef::TxISR)(struct __SPI_HandleTypeDef *hspi)***
- ***DMA_HandleTypeDef* __SPI_HandleTypeDef::hdmatx***
- ***DMA_HandleTypeDef* __SPI_HandleTypeDef::hdmarx***
- ***HAL_LockTypeDef __SPI_HandleTypeDef::Lock***
- ***__IO HAL_SPI_StateTypeDef __SPI_HandleTypeDef::State***
- ***__IO uint32_t __SPI_HandleTypeDef::ErrorCode***

60.2 SPI Firmware driver API description

60.2.1 How to use this driver

The SPI HAL driver can be used as follows:

1. Declare a SPI_HandleTypeDef handle structure, for example: SPI_HandleTypeDef hspi;
2. Initialize the SPI low level resources by implementing the HAL_SPI_MspInit() API:
   a. Enable the SPIx interface clock
   b. SPI pins configuration
      – Enable the clock for the SPI GPIOs
      – Configure these SPI pins as alternate function push-pull
   c. NVIC configuration if you need to use interrupt process
      – Configure the SPIx interrupt priority
      – Enable the NVIC SPI IRQ handle
   d. DMA Configuration if you need to use DMA process
      – Declare a DMA_HandleTypeDef handle structure for the transmit or receive stream
      – Enable the DMAx clock
      – Configure the DMA handle parameters
      – Configure the DMA Tx or Rx stream
      – Associate the initialized hdma_tx handle to the hspi DMA Tx or Rx handle
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx or Rx stream
3. Program the Mode, BidirectionalMode , Data size, Baudrate Prescaler, NSS management, Clock polarity and phase, FirstBit and CRC configuration in the hspi Init structure.
4. Initialize the SPI registers by calling the HAL_SPI_Init() API:
   – This API configures also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customized HAL_SPI_MspInit() API.

Circular mode restriction:

1. The DMA circular mode cannot be used when the SPI is configured in these modes:
   a. Master 2Lines RxOnly
   b. Master 1Line Rx
2. The CRC feature is not managed when the DMA circular mode is enabled
3. When the SPI DMA Pause/Stop features are used, we must use the following APIs the HAL_SPI_DMAPause()/ HAL_SPI_DMAStop() only under the SPI callbacks

Master Receive mode restriction:

1. In Master unidirectional receive-only mode (MSTR =1, BIDIMODE=0, RXONLY=0) or bidirectional receive mode (MSTR=1, BIDIMODE=1, BIDIOE=0), to ensure that the SPI does not initiate a new transfer the following procedure has to be respected:
   a. HAL_SPI_DeInit()
   b. HAL_SPI_Init()

The max SPI frequency depend on SPI data size (8bits, 16bits), SPI mode(2 Lines fullduplex, 2 lines RxOnly, 1 line TX/RX) and Process mode (Polling, IT, DMA).

- TX/RX processes are HAL_SPI_TransmitReceive(), HAL_SPI_TransmitReceive_IT() and HAL_SPI_TransmitReceive_DMA()
- RX processes are HAL_SPI_Receive(), HAL_SPI_Receive_IT() and HAL_SPI_Receive_DMA()
- TX processes are HAL_SPI_Transmit(), HAL_SPI_Transmit_IT() and HAL_SPI_Transmit_DMA()

60.2.2 Initialization and de-initialization functions

This subsection provides a set of functions allowing to initialize and de-initialize the SPIx peripheral:

- User must implement HAL_SPI_MspInit() function in which he configures all related peripherals resources (CLOCK, GPIO, DMA, IT and NVIC).
- Call the function HAL_SPI_Init() to configure the selected device with the selected configuration:
  - Mode
  - Direction
  - Data Size
  - Clock Polarity and Phase
  - NSS Management
  - BaudRate Prescaler
  - FirstBit
  - TIMode
  - CRC Calculation
  - CRC Polynomial if CRC enabled
- Call the function HAL_SPI_DeInit() to restore the default configuration of the selected SPIx peripheral.

This section contains the following APIs:

- ***HAL_SPI_Init()***
- ***HAL_SPI_DeInit()***
- ***HAL_SPI_MspInit()***
- ***HAL_SPI_MspDeInit()***

60.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the SPI data transfers.

The SPI supports master and slave mode :

1. There are two modes of transfer:
   - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode: The communication is performed using Interrupts or DMA, These APIs return the HAL status. The end of the data processing will be indicated through the dedicated SPI IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_SPI_TxCpltCallback(), HAL_SPI_RxCpltCallback() and HAL_SPI_TxRxCpltCallback() user callbacks will be executed respectively at the end of the transmit or Receive process The HAL_SPI_ErrorCallback()user callback will be executed when a communication error is detected
2. APIs provided for these 2 transfer modes (Blocking mode or Non blocking mode using either Interrupt or DMA) exist for 1Line (simplex) and 2Lines (full duplex) modes.

This section contains the following APIs:

- ***HAL_SPI_Transmit()***
- ***HAL_SPI_Receive()***
- ***HAL_SPI_TransmitReceive()***
- ***HAL_SPI_Transmit_IT()***
- ***HAL_SPI_Receive_IT()***
- ***HAL_SPI_TransmitReceive_IT()***
- ***HAL_SPI_Transmit_DMA()***
- ***HAL_SPI_Receive_DMA()***
- ***HAL_SPI_TransmitReceive_DMA()***
- ***HAL_SPI_DMAPause()***
- ***HAL_SPI_DMAResume()***
- ***HAL_SPI_DMAStop()***
- ***HAL_SPI_IRQHandler()***
- ***HAL_SPI_TxCpltCallback()***
- ***HAL_SPI_RxCpltCallback()***
- ***HAL_SPI_TxRxCpltCallback()***
- ***HAL_SPI_TxHalfCpltCallback()***
- ***HAL_SPI_RxHalfCpltCallback()***
- ***HAL_SPI_TxRxHalfCpltCallback()***
- ***HAL_SPI_ErrorCallback()***

60.2.4 Peripheral State and Errors functions

This subsection provides a set of functions allowing to control the SPI.

- HAL_SPI_GetState() API can be helpful to check in run-time the state of the SPI peripheral
- HAL_SPI_GetError() check in run-time Errors occurring during communication

This section contains the following APIs:

- ***HAL_SPI_GetState()***
- ***HAL_SPI_GetError()***

60.2.5 Detailed description of functions

HAL_SPI_Init

Function Name	**HAL_StatusTypeDef HAL_SPI_Init (SPI_HandleTypeDef * hspi)**
Function Description	Initialize the SPI according to the specified parameters in the SPI_InitTypeDef and initialize the associated handle.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **HAL:** status

HAL_SPI_DeInit

Function Name	**HAL_StatusTypeDef HAL_SPI_DeInit (SPI_HandleTypeDef * hspi)**
Function Description	De Initialize the SPI peripheral.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains

the configuration information for SPI module.

Return values
- **HAL:** status

HAL_SPI_MspInit

Function Name **void HAL_SPI_MspInit (SPI_HandleTypeDef * hspi)**

Function Description Initialize the SPI MSP.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.

Return values
- **None**

HAL_SPI_MspDeInit

Function Name **void HAL_SPI_MspDeInit (SPI_HandleTypeDef * hspi)**

Function Description De-Initialize the SPI MSP.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.

Return values
- **None**

HAL_SPI_Transmit

Function Name **HAL_StatusTypeDef HAL_SPI_Transmit (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Transmit an amount of data in blocking mode.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SPI_Receive

Function Name **HAL_StatusTypeDef HAL_SPI_Receive (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receive an amount of data in blocking mode.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be received
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_SPI_TransmitReceive

Function Name	**HAL_StatusTypeDef HAL_SPI_TransmitReceive (SPI_HandleTypeDef * hspi, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size, uint32_t Timeout)**
Function Description	Transmit and Receive an amount of data in blocking mode.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module. • **pTxData:** pointer to transmission data buffer • **pRxData:** pointer to reception data buffer • **Size:** amount of data to be sent and received • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_SPI_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_SPI_Transmit_IT (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size)**
Function Description	Transmit an amount of data in non-blocking mode with Interrupt.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module. • **pData:** pointer to data buffer • **Size:** amount of data to be sent
Return values	• **HAL:** status

HAL_SPI_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_SPI_Receive_IT (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size)**
Function Description	Receive an amount of data in non-blocking mode with Interrupt.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module. • **pData:** pointer to data buffer • **Size:** amount of data to be sent
Return values	• **HAL:** status

HAL_SPI_TransmitReceive_IT

Function Name	**HAL_StatusTypeDef HAL_SPI_TransmitReceive_IT (SPI_HandleTypeDef * hspi, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size)**
Function Description	Transmit and Receive an amount of data in non-blocking mode with Interrupt.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module. • **pTxData:** pointer to transmission data buffer • **pRxData:** pointer to reception data buffer • **Size:** amount of data to be sent and received

Return values
- **HAL:** status

HAL_SPI_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_SPI_Transmit_DMA (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size)**

Function Description Transmit an amount of data in non-blocking mode with DMA.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent

Return values
- **HAL:** status

HAL_SPI_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_SPI_Receive_DMA (SPI_HandleTypeDef * hspi, uint8_t * pData, uint16_t Size)**

Function Description Receive an amount of data in non-blocking mode with DMA.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
- **pData:** pointer to data buffer
- **Size:** amount of data to be sent

Return values
- **HAL:** status

Notes
- When the CRC feature is enabled the pData Length must be Size + 1.

HAL_SPI_TransmitReceive_DMA

Function Name **HAL_StatusTypeDef HAL_SPI_TransmitReceive_DMA (SPI_HandleTypeDef * hspi, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size)**

Function Description Transmit and Receive an amount of data in non-blocking mode with DMA.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
- **pTxData:** pointer to transmission data buffer
- **pRxData:** pointer to reception data buffer
- **Size:** amount of data to be sent

Return values
- **HAL:** status

Notes
- When the CRC feature is enabled the pRxData Length must be Size + 1

HAL_SPI_DMAPause

Function Name **HAL_StatusTypeDef HAL_SPI_DMAPause (SPI_HandleTypeDef * hspi)**

Function Description Pause the DMA Transfer.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for the specified SPI module.

Return values
- **HAL:** status

HAL_SPI_DMAResume

Function Name **HAL_StatusTypeDef HAL_SPI_DMAResume (SPI_HandleTypeDef * hspi)**

Function Description Resume the DMA Transfer.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for the specified SPI module.

Return values
- **HAL:** status

HAL_SPI_DMAStop

Function Name **HAL_StatusTypeDef HAL_SPI_DMAStop (SPI_HandleTypeDef * hspi)**

Function Description Stop the DMA Transfer.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for the specified SPI module.

Return values
- **HAL:** status

HAL_SPI_IRQHandler

Function Name **void HAL_SPI_IRQHandler (SPI_HandleTypeDef * hspi)**

Function Description Handle SPI interrupt request.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for the specified SPI module.

Return values
- **None**

HAL_SPI_TxCpltCallback

Function Name **void HAL_SPI_TxCpltCallback (SPI_HandleTypeDef * hspi)**

Function Description Tx Transfer completed callback.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.

Return values
- **None**

HAL_SPI_RxCpltCallback

Function Name **void HAL_SPI_RxCpltCallback (SPI_HandleTypeDef * hspi)**

Function Description Rx Transfer completed callback.

Parameters
- **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.

Return values
- **None**

HAL_SPI_TxRxCpltCallback

Function Name	**void HAL_SPI_TxRxCpltCallback (SPI_HandleTypeDef * hspi)**
Function Description	Tx and Rx Transfer completed callback.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **None**

HAL_SPI_TxHalfCpltCallback

Function Name	**void HAL_SPI_TxHalfCpltCallback (SPI_HandleTypeDef * hspi)**
Function Description	Tx Half Transfer completed callback.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **None**

HAL_SPI_RxHalfCpltCallback

Function Name	**void HAL_SPI_RxHalfCpltCallback (SPI_HandleTypeDef * hspi)**
Function Description	Rx Half Transfer completed callback.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **None**

HAL_SPI_TxRxHalfCpltCallback

Function Name	**void HAL_SPI_TxRxHalfCpltCallback (SPI_HandleTypeDef * hspi)**
Function Description	Tx and Rx Half Transfer callback.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **None**

HAL_SPI_ErrorCallback

Function Name	**void HAL_SPI_ErrorCallback (SPI_HandleTypeDef * hspi)**
Function Description	SPI error callback.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **None**

HAL_SPI_GetState

Function Name	**HAL_SPI_StateTypeDef HAL_SPI_GetState (SPI_HandleTypeDef * hspi)**

Function Description	Return the SPI handle state.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **SPI:** state

HAL_SPI_GetError

Function Name	**uint32_t HAL_SPI_GetError (SPI_HandleTypeDef * hspi)**
Function Description	Return the SPI error code.
Parameters	• **hspi:** pointer to a SPI_HandleTypeDef structure that contains the configuration information for SPI module.
Return values	• **SPI:** error code in bitmap format

60.3 SPI Firmware driver defines

60.3.1 SPI

SPI BaudRate Prescaler

SPI_BAUDRATEPRESCALER_2

SPI_BAUDRATEPRESCALER_4

SPI_BAUDRATEPRESCALER_8

SPI_BAUDRATEPRESCALER_16

SPI_BAUDRATEPRESCALER_32

SPI_BAUDRATEPRESCALER_64

SPI_BAUDRATEPRESCALER_128

SPI_BAUDRATEPRESCALER_256

SPI Clock Phase

SPI_PHASE_1EDGE

SPI_PHASE_2EDGE

SPI Clock Polarity

SPI_POLARITY_LOW

SPI_POLARITY_HIGH

SPI CRC Calculation

SPI_CRCCALCULATION_DISABLE

SPI_CRCCALCULATION_ENABLE

SPI Data Size

SPI_DATASIZE_8BIT

SPI_DATASIZE_16BIT

SPI Direction Mode

SPI_DIRECTION_2LINES

SPI_DIRECTION_2LINES_RXONLY

SPI_DIRECTION_1LINE

SPI Error Code

HAL_SPI_ERROR_NONE	No error
HAL_SPI_ERROR_MODF	MODF error
HAL_SPI_ERROR_CRC	CRC error
HAL_SPI_ERROR_OVR	OVR error
HAL_SPI_ERROR_FRE	FRE error
HAL_SPI_ERROR_DMA	DMA transfer error
HAL_SPI_ERROR_FLAG	Flag: RXNE,TXE, BSY

SPI Exported Macros

__HAL_SPI_RESET_HANDLE_STATE

Description:

- Reset SPI handle state.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_ENABLE_IT

Description:

- Enable or disable the specified SPI interrupts.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.
- __INTERRUPT__: specifies the interrupt source to enable or disable. This parameter can be one of the following values:
  - SPI_IT_TXE: Tx buffer empty interrupt enable
  - SPI_IT_RXNE: RX buffer not empty interrupt enable
  - SPI_IT_ERR: Error interrupt enable

Return value:

- None

__HAL_SPI_DISABLE_IT

__HAL_SPI_GET_IT_SOURCE

Description:

- Check whether the specified SPI interrupt source is enabled or not.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.
- __INTERRUPT__: specifies the SPI interrupt source to check. This parameter can be one of the following values:
  - SPI_IT_TXE: Tx buffer empty interrupt enable
  - SPI_IT_RXNE: RX buffer not empty interrupt enable
  - SPI_IT_ERR: Error interrupt enable

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_SPI_GET_FLAG

Description:

- Check whether the specified SPI flag is set or not.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - SPI_FLAG_RXNE: Receive buffer not empty flag
  - SPI_FLAG_TXE: Transmit buffer empty flag
  - SPI_FLAG_CRCERR: CRC error flag
  - SPI_FLAG_MODF: Mode fault flag
  - SPI_FLAG_OVR: Overrun flag
  - SPI_FLAG_BSY: Busy flag
  - SPI_FLAG_FRE: Frame format error flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_SPI_CLEAR_CRCERRFLAG

Description:

- Clear the SPI CRCERR pending flag.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_CLEAR_MODFFLAG

Description:

- Clear the SPI MODF pending flag.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_CLEAR_OVRFLAG

Description:

- Clear the SPI OVR pending flag.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_CLEAR_FREFLAG

Description:

- Clear the SPI FRE pending flag.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_ENABLE

Description:

- Enable the SPI peripheral.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

__HAL_SPI_DISABLE

Description:

- Disable the SPI peripheral.

Parameters:

- __HANDLE__: specifies the SPI Handle. This parameter can be SPI where x: 1, 2, or 3 to select the SPI peripheral.

Return value:

- None

SPI Flags Definition

SPI_FLAG_RXNE

SPI_FLAG_TXE

SPI_FLAG_BSY

SPI_FLAG_CRCERR

SPI_FLAG_MODF

SPI_FLAG_OVR

SPI_FLAG_FRE

SPI Interrupt Definition

SPI_IT_TXE

SPI_IT_RXNE

SPI_IT_ERR

SPI Mode

SPI_MODE_SLAVE

SPI_MODE_MASTER

SPI MSB LSB Transmission

SPI_FIRSTBIT_MSB

SPI_FIRSTBIT_LSB

SPI Slave Select Management

SPI_NSS_SOFT

SPI_NSS_HARD_INPUT

SPI_NSS_HARD_OUTPUT

SPI TI Mode

SPI_TIMODE_DISABLE

SPI_TIMODE_ENABLE

61 HAL SRAM Generic Driver

61.1 SRAM Firmware driver registers structures

61.1.1 SRAM_HandleTypeDef

Data Fields

- ***FMC_NORSRAM_TypeDef * Instance***
- ***FMC_NORSRAM_EXTENDED_TypeDef * Extended***
- ***FMC_NORSRAM_InitTypeDef Init***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_SRAM_StateTypeDef State***
- ***DMA_HandleTypeDef * hdma***

Field Documentation

- ***FMC_NORSRAM_TypeDef* SRAM_HandleTypeDef::Instance***
  Register base address
- ***FMC_NORSRAM_EXTENDED_TypeDef* SRAM_HandleTypeDef::Extended***
  Extended mode register base address
- ***FMC_NORSRAM_InitTypeDef SRAM_HandleTypeDef::Init***
  SRAM device control configuration parameters
- ***HAL_LockTypeDef SRAM_HandleTypeDef::Lock***
  SRAM locking object
- ***__IO HAL_SRAM_StateTypeDef SRAM_HandleTypeDef::State***
  SRAM device access state
- ***DMA_HandleTypeDef* SRAM_HandleTypeDef::hdma***
  Pointer DMA handler

61.2 SRAM Firmware driver API description

61.2.1 How to use this driver

This driver is a generic layered driver which contains a set of APIs used to control SRAM memories. It uses the FMC layer functions to interface with SRAM devices. The following sequence should be followed to configure the FMC/FSMC to interface with SRAM/PSRAM memories:

1. Declare a SRAM_HandleTypeDef handle structure, for example: SRAM_HandleTypeDef hsram; and:
   – Fill the SRAM_HandleTypeDef handle "Init" field with the allowed values of the structure member.
   – Fill the SRAM_HandleTypeDef handle "Instance" field with a predefined base register instance for NOR or SRAM device
   – Fill the SRAM_HandleTypeDef handle "Extended" field with a predefined base register instance for NOR or SRAM extended mode
2. Declare two FMC_NORSRAM_TimingTypeDef structures, for both normal and extended mode timings; for example: FMC_NORSRAM_TimingTypeDef Timing and

FMC_NORSRAM_TimingTypeDef ExTiming; and fill its fields with the allowed values of the structure member.

3. Initialize the SRAM Controller by calling the function HAL_SRAM_Init(). This function performs the following sequence:
   a. MSP hardware layer configuration using the function HAL_SRAM_MspInit()
   b. Control register configuration using the FMC NORSRAM interface function FMC_NORSRAM_Init()
   c. Timing register configuration using the FMC NORSRAM interface function FMC_NORSRAM_Timing_Init()
   d. Extended mode Timing register configuration using the FMC NORSRAM interface function FMC_NORSRAM_Extended_Timing_Init()
   e. Enable the SRAM device using the macro __FMC_NORSRAM_ENABLE()
4. At this stage you can perform read/write accesses from/to the memory connected to the NOR/SRAM Bank. You can perform either polling or DMA transfer using the following APIs:
   – HAL_SRAM_Read()/HAL_SRAM_Write() for polling read/write access
   – HAL_SRAM_Read_DMA()/HAL_SRAM_Write_DMA() for DMA read/write transfer
5. You can also control the SRAM device by calling the control APIs HAL_SRAM_WriteOperation_Enable()/ HAL_SRAM_WriteOperation_Disable() to respectively enable/disable the SRAM write operation
6. You can continuously monitor the SRAM device HAL state by calling the function HAL_SRAM_GetState()

61.2.2 SRAM Initialization and de_initialization functions

This section provides functions allowing to initialize/de-initialize the SRAM memory

This section contains the following APIs:

- ***HAL_SRAM_Init()***
- ***HAL_SRAM_DeInit()***
- ***HAL_SRAM_MspInit()***
- ***HAL_SRAM_MspDeInit()***
- ***HAL_SRAM_DMA_XferCpltCallback()***
- ***HAL_SRAM_DMA_XferErrorCallback()***

61.2.3 SRAM Input and Output functions

This section provides functions allowing to use and control the SRAM memory

This section contains the following APIs:

- ***HAL_SRAM_Read_8b()***
- ***HAL_SRAM_Write_8b()***
- ***HAL_SRAM_Read_16b()***
- ***HAL_SRAM_Write_16b()***
- ***HAL_SRAM_Read_32b()***
- ***HAL_SRAM_Write_32b()***
- ***HAL_SRAM_Read_DMA()***
- ***HAL_SRAM_Write_DMA()***

61.2.4 SRAM Control functions

This subsection provides a set of functions allowing to control dynamically the SRAM interface.

This section contains the following APIs:

- *HAL_SRAM_WriteOperation_Enable()*
- *HAL_SRAM_WriteOperation_Disable()*

61.2.5 SRAM State functions

This subsection permits to get in run-time the status of the SRAM controller and the data flow.

This section contains the following APIs:

- *HAL_SRAM_GetState()*

61.2.6 Detailed description of functions

HAL_SRAM_Init

Function Name	**HAL_StatusTypeDef HAL_SRAM_Init (SRAM_HandleTypeDef * hsram, FMC_NORSRAM_TimingTypeDef * Timing, FMC_NORSRAM_TimingTypeDef * ExtTiming)**
Function Description	Performs the SRAM device initialization sequence.
Parameters	• **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module. • **Timing:** Pointer to SRAM control timing structure • **ExtTiming:** Pointer to SRAM extended mode timing structure
Return values	• **HAL:** status

HAL_SRAM_DeInit

Function Name	**HAL_StatusTypeDef HAL_SRAM_DeInit (SRAM_HandleTypeDef * hsram)**
Function Description	Performs the SRAM device De-initialization sequence.
Parameters	• **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
Return values	• **HAL:** status

HAL_SRAM_MspInit

Function Name	**void HAL_SRAM_MspInit (SRAM_HandleTypeDef * hsram)**
Function Description	SRAM MSP Init.
Parameters	• **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
Return values	• **None**

HAL_SRAM_MspDeInit

Function Name	**void HAL_SRAM_MspDeInit (SRAM_HandleTypeDef * hsram)**
Function Description	SRAM MSP DeInit.
Parameters	• **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.

Return values
- **None**

HAL_SRAM_DMA_XferCpltCallback

Function Name **void HAL_SRAM_DMA_XferCpltCallback (DMA_HandleTypeDef * hdma)**

Function Description DMA transfer complete callback.

Parameters
- **hdma:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.

Return values
- **None**

HAL_SRAM_DMA_XferErrorCallback

Function Name **void HAL_SRAM_DMA_XferErrorCallback (DMA_HandleTypeDef * hdma)**

Function Description DMA transfer complete error callback.

Parameters
- **hdma:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.

Return values
- **None**

HAL_SRAM_Read_8b

Function Name **HAL_StatusTypeDef HAL_SRAM_Read_8b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint8_t * pDstBuffer, uint32_t BufferSize)**

Function Description Reads 8-bit buffer from SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to read start address
- **pDstBuffer:** Pointer to destination buffer
- **BufferSize:** Size of the buffer to read from memory

Return values
- **HAL:** status

HAL_SRAM_Write_8b

Function Name **HAL_StatusTypeDef HAL_SRAM_Write_8b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint8_t * pSrcBuffer, uint32_t BufferSize)**

Function Description Writes 8-bit buffer to SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to write start address
- **pSrcBuffer:** Pointer to source buffer to write
- **BufferSize:** Size of the buffer to write to memory

Return values
- **HAL:** status

HAL_SRAM_Read_16b

Function Name **HAL_StatusTypeDef HAL_SRAM_Read_16b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint16_t * pDstBuffer, uint32_t BufferSize)**

Function Description Reads 16-bit buffer from SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to read start address
- **pDstBuffer:** Pointer to destination buffer
- **BufferSize:** Size of the buffer to read from memory

Return values
- **HAL:** status

HAL_SRAM_Write_16b

Function Name **HAL_StatusTypeDef HAL_SRAM_Write_16b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint16_t * pSrcBuffer, uint32_t BufferSize)**

Function Description Writes 16-bit buffer to SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to write start address
- **pSrcBuffer:** Pointer to source buffer to write
- **BufferSize:** Size of the buffer to write to memory

Return values
- **HAL:** status

HAL_SRAM_Read_32b

Function Name **HAL_StatusTypeDef HAL_SRAM_Read_32b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint32_t * pDstBuffer, uint32_t BufferSize)**

Function Description Reads 32-bit buffer from SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to read start address
- **pDstBuffer:** Pointer to destination buffer
- **BufferSize:** Size of the buffer to read from memory

Return values
- **HAL:** status

HAL_SRAM_Write_32b

Function Name **HAL_StatusTypeDef HAL_SRAM_Write_32b (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint32_t * pSrcBuffer, uint32_t BufferSize)**

Function Description Writes 32-bit buffer to SRAM memory.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to write start address
- **pSrcBuffer:** Pointer to source buffer to write

- **BufferSize:** Size of the buffer to write to memory

Return values
- **HAL:** status

HAL_SRAM_Read_DMA

Function Name **HAL_StatusTypeDef HAL_SRAM_Read_DMA (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint32_t * pDstBuffer, uint32_t BufferSize)**

Function Description Reads a Words data from the SRAM memory using DMA transfer.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to read start address
- **pDstBuffer:** Pointer to destination buffer
- **BufferSize:** Size of the buffer to read from memory

Return values
- **HAL:** status

HAL_SRAM_Write_DMA

Function Name **HAL_StatusTypeDef HAL_SRAM_Write_DMA (SRAM_HandleTypeDef * hsram, uint32_t * pAddress, uint32_t * pSrcBuffer, uint32_t BufferSize)**

Function Description Writes a Words data buffer to SRAM memory using DMA transfer.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
- **pAddress:** Pointer to write start address
- **pSrcBuffer:** Pointer to source buffer to write
- **BufferSize:** Size of the buffer to write to memory

Return values
- **HAL:** status

HAL_SRAM_WriteOperation_Enable

Function Name **HAL_StatusTypeDef HAL_SRAM_WriteOperation_Enable (SRAM_HandleTypeDef * hsram)**

Function Description Enables dynamically SRAM write operation.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.

Return values
- **HAL:** status

HAL_SRAM_WriteOperation_Disable

Function Name **HAL_StatusTypeDef HAL_SRAM_WriteOperation_Disable (SRAM_HandleTypeDef * hsram)**

Function Description Disables dynamically SRAM write operation.

Parameters
- **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.

Return values
- **HAL:** status

HAL_SRAM_GetState

Function Name	**HAL_SRAM_StateTypeDef HAL_SRAM_GetState (SRAM_HandleTypeDef * hsram)**
Function Description	Returns the SRAM controller state.
Parameters	• **hsram:** pointer to a SRAM_HandleTypeDef structure that contains the configuration information for SRAM module.
Return values	• **HAL:** state

61.3 SRAM Firmware driver defines

61.3.1 SRAM

SRAM Exported Macros

__HAL_SRAM_RESET_HANDLE_STATE

Description:

- Reset SRAM handle state.

Parameters:

- __HANDLE__: SRAM handle

Return value:

- None

62 HAL TIM Generic Driver

62.1 TIM Firmware driver registers structures

62.1.1 TIM_Base_InitTypeDef

Data Fields

- ***uint32_t Prescaler***
- ***uint32_t CounterMode***
- ***uint32_t Period***
- ***uint32_t ClockDivision***
- ***uint32_t RepetitionCounter***

Field Documentation

- ***uint32_t TIM_Base_InitTypeDef::Prescaler***
  Specifies the prescaler value used to divide the TIM clock. This parameter can be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t TIM_Base_InitTypeDef::CounterMode***
  Specifies the counter mode. This parameter can be a value of ***TIM_Counter_Mode***
- ***uint32_t TIM_Base_InitTypeDef::Period***
  Specifies the period value to be loaded into the active Auto-Reload Register at the next update event. This parameter can be a number between Min_Data = 0x0000U and Max_Data = 0xFFFF.
- ***uint32_t TIM_Base_InitTypeDef::ClockDivision***
  Specifies the clock division. This parameter can be a value of ***TIM_ClockDivision***
- ***uint32_t TIM_Base_InitTypeDef::RepetitionCounter***
  Specifies the repetition counter value. Each time the RCR downcounter reaches zero, an update event is generated and counting restarts from the RCR value (N). This means in PWM mode that (N+1) corresponds to:the number of PWM periods in edge-aligned modethe number of half PWM period in center-aligned mode This parameter must be a number between Min_Data = 0x00 and Max_Data = 0xFF.
  Note: This parameter is valid only for TIM1 and TIM8.

62.1.2 TIM_OC_InitTypeDef

Data Fields

- ***uint32_t OCMode***
- ***uint32_t Pulse***
- ***uint32_t OCPolarity***
- ***uint32_t OCNPolarity***
- ***uint32_t OCFastMode***
- ***uint32_t OCIdleState***
- ***uint32_t OCNIdleState***

Field Documentation

- ***uint32_t TIM_OC_InitTypeDef::OCMode***
  Specifies the TIM mode. This parameter can be a value of ***TIM_Output_Compare_and_PWM_modes***
- ***uint32_t TIM_OC_InitTypeDef::Pulse***
  Specifies the pulse value to be loaded into the Capture Compare Register. This parameter can be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU
- ***uint32_t TIM_OC_InitTypeDef::OCPolarity***
  Specifies the output polarity. This parameter can be a value of ***TIM_Output_Compare_Polarity***
- ***uint32_t TIM_OC_InitTypeDef::OCNPolarity***
  Specifies the complementary output polarity. This parameter can be a value of ***TIM_Output_Compare_N_Polarity***
  Note: This parameter is valid only for TIM1 and TIM8.
- ***uint32_t TIM_OC_InitTypeDef::OCFastMode***
  Specifies the Fast mode state. This parameter can be a value of ***TIM_Output_Fast_State***
  Note: This parameter is valid only in PWM1 and PWM2 mode.
- ***uint32_t TIM_OC_InitTypeDef::OCIdleState***
  Specifies the TIM Output Compare pin state during Idle state. This parameter can be a value of ***TIM_Output_Compare_Idle_State***
  Note: This parameter is valid only for TIM1 and TIM8.
- ***uint32_t TIM_OC_InitTypeDef::OCNIdleState***
  Specifies the TIM Output Compare pin state during Idle state. This parameter can be a value of ***TIM_Output_Compare_N_Idle_State***
  Note: This parameter is valid only for TIM1 and TIM8.

62.1.3 TIM_OnePulse_InitTypeDef

Data Fields

- ***uint32_t OCMode***
- ***uint32_t Pulse***
- ***uint32_t OCPolarity***
- ***uint32_t OCNPolarity***
- ***uint32_t OCIdleState***
- ***uint32_t OCNIdleState***
- ***uint32_t ICPolarity***
- ***uint32_t ICSelection***
- ***uint32_t ICFilter***

Field Documentation

- ***uint32_t TIM_OnePulse_InitTypeDef::OCMode***
  Specifies the TIM mode. This parameter can be a value of ***TIM_Output_Compare_and_PWM_modes***
- ***uint32_t TIM_OnePulse_InitTypeDef::Pulse***
  Specifies the pulse value to be loaded into the Capture Compare Register. This parameter can be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU

- ***uint32_t TIM_OnePulse_InitTypeDef::OCPolarity***
  Specifies the output polarity. This parameter can be a value of ***TIM_Output_Compare_Polarity***
- ***uint32_t TIM_OnePulse_InitTypeDef::OCNPolarity***
  Specifies the complementary output polarity. This parameter can be a value of ***TIM_Output_Compare_N_Polarity***
  Note: This parameter is valid only for TIM1 and TIM8.
- ***uint32_t TIM_OnePulse_InitTypeDef::OCIdleState***
  Specifies the TIM Output Compare pin state during Idle state. This parameter can be a value of ***TIM_Output_Compare_Idle_State***
  Note: This parameter is valid only for TIM1 and TIM8.
- ***uint32_t TIM_OnePulse_InitTypeDef::OCNIdleState***
  Specifies the TIM Output Compare pin state during Idle state. This parameter can be a value of ***TIM_Output_Compare_N_Idle_State***
  Note: This parameter is valid only for TIM1 and TIM8.
- ***uint32_t TIM_OnePulse_InitTypeDef::ICPolarity***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Input_Capture_Polarity***
- ***uint32_t TIM_OnePulse_InitTypeDef::ICSelection***
  Specifies the input. This parameter can be a value of ***TIM_Input_Capture_Selection***
- ***uint32_t TIM_OnePulse_InitTypeDef::ICFilter***
  Specifies the input capture filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.4 TIM_IC_InitTypeDef

Data Fields

- ***uint32_t ICPolarity***
- ***uint32_t ICSelection***
- ***uint32_t ICPrescaler***
- ***uint32_t ICFilter***

Field Documentation

- ***uint32_t TIM_IC_InitTypeDef::ICPolarity***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Input_Capture_Polarity***
- ***uint32_t TIM_IC_InitTypeDef::ICSelection***
  Specifies the input. This parameter can be a value of ***TIM_Input_Capture_Selection***
- ***uint32_t TIM_IC_InitTypeDef::ICPrescaler***
  Specifies the Input Capture Prescaler. This parameter can be a value of ***TIM_Input_Capture_Prescaler***
- ***uint32_t TIM_IC_InitTypeDef::ICFilter***
  Specifies the input capture filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.5 TIM_Encoder_InitTypeDef

Data Fields

- ***uint32_t EncoderMode***
- ***uint32_t IC1Polarity***
- ***uint32_t IC1Selection***
- ***uint32_t IC1Prescaler***
- ***uint32_t IC1Filter***
- ***uint32_t IC2Polarity***
- ***uint32_t IC2Selection***
- ***uint32_t IC2Prescaler***
- ***uint32_t IC2Filter***

Field Documentation

- ***uint32_t TIM_Encoder_InitTypeDef::EncoderMode***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Encoder_Mode***
- ***uint32_t TIM_Encoder_InitTypeDef::IC1Polarity***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Input_Capture_Polarity***
- ***uint32_t TIM_Encoder_InitTypeDef::IC1Selection***
  Specifies the input. This parameter can be a value of ***TIM_Input_Capture_Selection***
- ***uint32_t TIM_Encoder_InitTypeDef::IC1Prescaler***
  Specifies the Input Capture Prescaler. This parameter can be a value of ***TIM_Input_Capture_Prescaler***
- ***uint32_t TIM_Encoder_InitTypeDef::IC1Filter***
  Specifies the input capture filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF
- ***uint32_t TIM_Encoder_InitTypeDef::IC2Polarity***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Input_Capture_Polarity***
- ***uint32_t TIM_Encoder_InitTypeDef::IC2Selection***
  Specifies the input. This parameter can be a value of ***TIM_Input_Capture_Selection***
- ***uint32_t TIM_Encoder_InitTypeDef::IC2Prescaler***
  Specifies the Input Capture Prescaler. This parameter can be a value of ***TIM_Input_Capture_Prescaler***
- ***uint32_t TIM_Encoder_InitTypeDef::IC2Filter***
  Specifies the input capture filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.6 TIM_ClockConfigTypeDef

Data Fields

- ***uint32_t ClockSource***
- ***uint32_t ClockPolarity***
- ***uint32_t ClockPrescaler***

- ***uint32_t ClockFilter***

Field Documentation

- ***uint32_t TIM_ClockConfigTypeDef::ClockSource***
  TIM clock sources. This parameter can be a value of ***TIM_Clock_Source***
- ***uint32_t TIM_ClockConfigTypeDef::ClockPolarity***
  TIM clock polarity. This parameter can be a value of ***TIM_Clock_Polarity***
- ***uint32_t TIM_ClockConfigTypeDef::ClockPrescaler***
  TIM clock prescaler. This parameter can be a value of ***TIM_Clock_Prescaler***
- ***uint32_t TIM_ClockConfigTypeDef::ClockFilter***
  TIM clock filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.7 TIM_ClearInputConfigTypeDef

Data Fields

- ***uint32_t ClearInputState***
- ***uint32_t ClearInputSource***
- ***uint32_t ClearInputPolarity***
- ***uint32_t ClearInputPrescaler***
- ***uint32_t ClearInputFilter***

Field Documentation

- ***uint32_t TIM_ClearInputConfigTypeDef::ClearInputState***
  TIM clear Input state. This parameter can be ENABLE or DISABLE
- ***uint32_t TIM_ClearInputConfigTypeDef::ClearInputSource***
  TIM clear Input sources. This parameter can be a value of ***TIM_ClearInput_Source***
- ***uint32_t TIM_ClearInputConfigTypeDef::ClearInputPolarity***
  TIM Clear Input polarity. This parameter can be a value of ***TIM_ClearInput_Polarity***
- ***uint32_t TIM_ClearInputConfigTypeDef::ClearInputPrescaler***
  TIM Clear Input prescaler. This parameter can be a value of ***TIM_ClearInput_Prescaler***
- ***uint32_t TIM_ClearInputConfigTypeDef::ClearInputFilter***
  TIM Clear Input filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.8 TIM_SlaveConfigTypeDef

Data Fields

- ***uint32_t SlaveMode***
- ***uint32_t InputTrigger***
- ***uint32_t TriggerPolarity***
- ***uint32_t TriggerPrescaler***
- ***uint32_t TriggerFilter***

Field Documentation

- ***uint32_t TIM_SlaveConfigTypeDef::SlaveMode***
  Slave mode selection This parameter can be a value of ***TIM_Slave_Mode***
- ***uint32_t TIM_SlaveConfigTypeDef::InputTrigger***
  Input Trigger source This parameter can be a value of ***TIM_Trigger_Selection***
- ***uint32_t TIM_SlaveConfigTypeDef::TriggerPolarity***
  Input Trigger polarity This parameter can be a value of ***TIM_Trigger_Polarity***
- ***uint32_t TIM_SlaveConfigTypeDef::TriggerPrescaler***
  Input trigger prescaler This parameter can be a value of ***TIM_Trigger_Prescaler***
- ***uint32_t TIM_SlaveConfigTypeDef::TriggerFilter***
  Input trigger filter This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF

62.1.9 TIM_HandleTypeDef

Data Fields

- ***TIM_TypeDef * Instance***
- ***TIM_Base_InitTypeDef Init***
- ***HAL_TIM_ActiveChannel Channel***
- ***DMA_HandleTypeDef * hdma***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_TIM_StateTypeDef State***

Field Documentation

- ***TIM_TypeDef* TIM_HandleTypeDef::Instance***
  Register base address
- ***TIM_Base_InitTypeDef TIM_HandleTypeDef::Init***
  TIM Time Base required parameters
- ***HAL_TIM_ActiveChannel TIM_HandleTypeDef::Channel***
  Active channel
- ***DMA_HandleTypeDef* TIM_HandleTypeDef::hdma[7]***
  DMA Handlers array This array is accessed by a ***DMA_Handle_index***
- ***HAL_LockTypeDef TIM_HandleTypeDef::Lock***
  Locking object
- ***__IO HAL_TIM_StateTypeDef TIM_HandleTypeDef::State***
  TIM operation state

62.2 TIM Firmware driver API description

62.2.1 TIMER Generic features

The Timer features include:

1. 16-bit up, down, up/down auto-reload counter.
2. 16-bit programmable prescaler allowing dividing (also on the fly) the counter clock frequency either by any factor between 1 and 65536.
3. Up to 4 independent channels for:

- Input Capture
- Output Compare
- PWM generation (Edge and Center-aligned Mode)
- One-pulse mode output

62.2.2 How to use this driver

1. Initialize the TIM low level resources by implementing the following functions depending from feature used :
   - Time Base : HAL_TIM_Base_MspInit()
   - Input Capture : HAL_TIM_IC_MspInit()
   - Output Compare : HAL_TIM_OC_MspInit()
   - PWM generation : HAL_TIM_PWM_MspInit()
   - One-pulse mode output : HAL_TIM_OnePulse_MspInit()
   - Encoder mode output : HAL_TIM_Encoder_MspInit()
2. Initialize the TIM low level resources :
   a. Enable the TIM interface clock using __TIMx_CLK_ENABLE();
   b. TIM pins configuration
      - Enable the clock for the TIM GPIOs using the following function: __GPIOx_CLK_ENABLE();
      - Configure these TIM pins in Alternate function mode using HAL_GPIO_Init();
3. The external Clock can be configured, if needed (the default clock is the internal clock from the APBx), using the following function: HAL_TIM_ConfigClockSource, the clock configuration should be done before any start function.
4. Configure the TIM in the desired functioning mode using one of the initialization function of this driver:
   - HAL_TIM_Base_Init: to use the Timer to generate a simple time base
   - HAL_TIM_OC_Init and HAL_TIM_OC_ConfigChannel: to use the Timer to generate an Output Compare signal.
   - HAL_TIM_PWM_Init and HAL_TIM_PWM_ConfigChannel: to use the Timer to generate a PWM signal.
   - HAL_TIM_IC_Init and HAL_TIM_IC_ConfigChannel: to use the Timer to measure an external signal.
   - HAL_TIM_OnePulse_Init and HAL_TIM_OnePulse_ConfigChannel: to use the Timer in One Pulse Mode.
   - HAL_TIM_Encoder_Init: to use the Timer Encoder Interface.
5. Activate the TIM peripheral using one of the start functions depending from the feature used:
   - Time Base : HAL_TIM_Base_Start(), HAL_TIM_Base_Start_DMA(), HAL_TIM_Base_Start_IT()
   - Input Capture : HAL_TIM_IC_Start(), HAL_TIM_IC_Start_DMA(), HAL_TIM_IC_Start_IT()
   - Output Compare : HAL_TIM_OC_Start(), HAL_TIM_OC_Start_DMA(), HAL_TIM_OC_Start_IT()
   - PWM generation : HAL_TIM_PWM_Start(), HAL_TIM_PWM_Start_DMA(), HAL_TIM_PWM_Start_IT()
   - One-pulse mode output : HAL_TIM_OnePulse_Start(), HAL_TIM_OnePulse_Start_IT()
   - Encoder mode output : HAL_TIM_Encoder_Start(), HAL_TIM_Encoder_Start_DMA(), HAL_TIM_Encoder_Start_IT().
6. The DMA Burst is managed with the two following functions: HAL_TIM_DMABurst_WriteStart() HAL_TIM_DMABurst_ReadStart()

62.2.3 Time Base functions

This section provides functions allowing to:

- Initialize and configure the TIM base.
- De-initialize the TIM base.
- Start the Time Base.
- Stop the Time Base.
- Start the Time Base and enable interrupt.
- Stop the Time Base and disable interrupt.
- Start the Time Base and enable DMA transfer.
- Stop the Time Base and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_Base_Init()***
- ***HAL_TIM_Base_DeInit()***
- ***HAL_TIM_Base_MspInit()***
- ***HAL_TIM_Base_MspDeInit()***
- ***HAL_TIM_Base_Start()***
- ***HAL_TIM_Base_Stop()***
- ***HAL_TIM_Base_Start_IT()***
- ***HAL_TIM_Base_Stop_IT()***
- ***HAL_TIM_Base_Start_DMA()***
- ***HAL_TIM_Base_Stop_DMA()***

62.2.4 Time Output Compare functions

This section provides functions allowing to:

- Initialize and configure the TIM Output Compare.
- De-initialize the TIM Output Compare.
- Start the Time Output Compare.
- Stop the Time Output Compare.
- Start the Time Output Compare and enable interrupt.
- Stop the Time Output Compare and disable interrupt.
- Start the Time Output Compare and enable DMA transfer.
- Stop the Time Output Compare and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_OC_Init()***
- ***HAL_TIM_OC_DeInit()***
- ***HAL_TIM_OC_MspInit()***
- ***HAL_TIM_OC_MspDeInit()***
- ***HAL_TIM_OC_Start()***
- ***HAL_TIM_OC_Stop()***
- ***HAL_TIM_OC_Start_IT()***
- ***HAL_TIM_OC_Stop_IT()***
- ***HAL_TIM_OC_Start_DMA()***
- ***HAL_TIM_OC_Stop_DMA()***

62.2.5 Time PWM functions

This section provides functions allowing to:

- Initialize and configure the TIM OPWM.
- De-initialize the TIM PWM.

- Start the Time PWM.
- Stop the Time PWM.
- Start the Time PWM and enable interrupt.
- Stop the Time PWM and disable interrupt.
- Start the Time PWM and enable DMA transfer.
- Stop the Time PWM and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_PWM_Init()***
- ***HAL_TIM_PWM_DeInit()***
- ***HAL_TIM_PWM_MspInit()***
- ***HAL_TIM_PWM_MspDeInit()***
- ***HAL_TIM_PWM_Start()***
- ***HAL_TIM_PWM_Stop()***
- ***HAL_TIM_PWM_Start_IT()***
- ***HAL_TIM_PWM_Stop_IT()***
- ***HAL_TIM_PWM_Start_DMA()***
- ***HAL_TIM_PWM_Stop_DMA()***

62.2.6 Time Input Capture functions

This section provides functions allowing to:

- Initialize and configure the TIM Input Capture.
- De-initialize the TIM Input Capture.
- Start the Time Input Capture.
- Stop the Time Input Capture.
- Start the Time Input Capture and enable interrupt.
- Stop the Time Input Capture and disable interrupt.
- Start the Time Input Capture and enable DMA transfer.
- Stop the Time Input Capture and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_IC_Init()***
- ***HAL_TIM_IC_DeInit()***
- ***HAL_TIM_IC_MspInit()***
- ***HAL_TIM_IC_MspDeInit()***
- ***HAL_TIM_IC_Start()***
- ***HAL_TIM_IC_Stop()***
- ***HAL_TIM_IC_Start_IT()***
- ***HAL_TIM_IC_Stop_IT()***
- ***HAL_TIM_IC_Start_DMA()***
- ***HAL_TIM_IC_Stop_DMA()***

62.2.7 Time One Pulse functions

This section provides functions allowing to:

- Initialize and configure the TIM One Pulse.
- De-initialize the TIM One Pulse.
- Start the Time One Pulse.
- Stop the Time One Pulse.
- Start the Time One Pulse and enable interrupt.
- Stop the Time One Pulse and disable interrupt.
- Start the Time One Pulse and enable DMA transfer.

- Stop the Time One Pulse and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_OnePulse_Init()***
- ***HAL_TIM_OnePulse_DeInit()***
- ***HAL_TIM_OnePulse_MspInit()***
- ***HAL_TIM_OnePulse_MspDeInit()***
- ***HAL_TIM_OnePulse_Start()***
- ***HAL_TIM_OnePulse_Stop()***
- ***HAL_TIM_OnePulse_Start_IT()***
- ***HAL_TIM_OnePulse_Stop_IT()***

62.2.8 Time Encoder functions

This section provides functions allowing to:

- Initialize and configure the TIM Encoder.
- De-initialize the TIM Encoder.
- Start the Time Encoder.
- Stop the Time Encoder.
- Start the Time Encoder and enable interrupt.
- Stop the Time Encoder and disable interrupt.
- Start the Time Encoder and enable DMA transfer.
- Stop the Time Encoder and disable DMA transfer.

This section contains the following APIs:

- ***HAL_TIM_Encoder_Init()***
- ***HAL_TIM_Encoder_DeInit()***
- ***HAL_TIM_Encoder_MspInit()***
- ***HAL_TIM_Encoder_MspDeInit()***
- ***HAL_TIM_Encoder_Start()***
- ***HAL_TIM_Encoder_Stop()***
- ***HAL_TIM_Encoder_Start_IT()***
- ***HAL_TIM_Encoder_Stop_IT()***
- ***HAL_TIM_Encoder_Start_DMA()***
- ***HAL_TIM_Encoder_Stop_DMA()***

62.2.9 IRQ handler management

This section provides Timer IRQ handler function.

This section contains the following APIs:

- ***HAL_TIM_IRQHandler()***

62.2.10 Peripheral Control functions

This section provides functions allowing to:

- Configure The Input Output channels for OC, PWM, IC or One Pulse mode.
- Configure External Clock source.
- Configure Complementary channels, break features and dead time.
- Configure Master and the Slave synchronization.
- Configure the DMA Burst Mode.

This section contains the following APIs:

- ***HAL_TIM_OC_ConfigChannel()***
- ***HAL_TIM_IC_ConfigChannel()***
- ***HAL_TIM_PWM_ConfigChannel()***
- ***HAL_TIM_OnePulse_ConfigChannel()***
- ***HAL_TIM_DMABurst_WriteStart()***
- ***HAL_TIM_DMABurst_WriteStop()***
- ***HAL_TIM_DMABurst_ReadStart()***
- ***HAL_TIM_DMABurst_ReadStop()***
- ***HAL_TIM_GenerateEvent()***
- ***HAL_TIM_ConfigOCrefClear()***
- ***HAL_TIM_ConfigClockSource()***
- ***HAL_TIM_ConfigTI1Input()***
- ***HAL_TIM_SlaveConfigSynchronization()***
- ***HAL_TIM_SlaveConfigSynchronization_IT()***
- ***HAL_TIM_ReadCapturedValue()***

62.2.11 TIM Callbacks functions

This section provides TIM callback functions:

- Timer Period elapsed callback
- Timer Output Compare callback
- Timer Input capture callback
- Timer Trigger callback
- Timer Error callback

This section contains the following APIs:

- ***HAL_TIM_PeriodElapsedCallback()***
- ***HAL_TIM_OC_DelayElapsedCallback()***
- ***HAL_TIM_IC_CaptureCallback()***
- ***HAL_TIM_PWM_PulseFinishedCallback()***
- ***HAL_TIM_TriggerCallback()***
- ***HAL_TIM_ErrorCallback()***

62.2.12 Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_TIM_Base_GetState()***
- ***HAL_TIM_OC_GetState()***
- ***HAL_TIM_PWM_GetState()***
- ***HAL_TIM_IC_GetState()***
- ***HAL_TIM_OnePulse_GetState()***
- ***HAL_TIM_Encoder_GetState()***

62.2.13 Detailed description of functions

HAL_TIM_Base_Init

Function Name	**HAL_StatusTypeDef HAL_TIM_Base_Init (TIM_HandleTypeDef * htim)**
Function Description	Initializes the TIM Time base Unit according to the specified parameters in the TIM_HandleTypeDef and create the associated

handle.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_DeInit

Function Name **HAL_StatusTypeDef HAL_TIM_Base_DeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes the TIM Base peripheral.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_MspInit

Function Name **void HAL_TIM_Base_MspInit (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM Base MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_Base_MspDeInit

Function Name **void HAL_TIM_Base_MspDeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes TIM Base MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_Base_Start

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Start (TIM_HandleTypeDef * htim)**

Function Description Starts the TIM Base generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Stop (TIM_HandleTypeDef * htim)**

Function Description Stops the TIM Base generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that

contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Start_IT (TIM_HandleTypeDef * htim)**

Function Description Starts the TIM Base generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Stop_IT (TIM_HandleTypeDef * htim)**

Function Description Stops the TIM Base generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Base_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Start_DMA (TIM_HandleTypeDef * htim, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM Base generation in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **pData:** The source Buffer address.
- **Length:** The length of data to be transferred from memory to peripheral.

Return values
- **HAL:** status

HAL_TIM_Base_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_Base_Stop_DMA (TIM_HandleTypeDef * htim)**

Function Description Stops the TIM Base generation in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_OC_Init

Function Name **HAL_StatusTypeDef HAL_TIM_OC_Init (TIM_HandleTypeDef * htim)**

Function Description	Initializes the TIM Output Compare according to the specified parameters in the TIM_HandleTypeDef and create the associated handle.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **HAL:** status

HAL_TIM_OC_DeInit

Function Name	**HAL_StatusTypeDef HAL_TIM_OC_DeInit (TIM_HandleTypeDef * htim)**
Function Description	DeInitializes the TIM peripheral.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **HAL:** status

HAL_TIM_OC_MspInit

Function Name	**void HAL_TIM_OC_MspInit (TIM_HandleTypeDef * htim)**
Function Description	Initializes the TIM Output Compare MSP.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIM_OC_MspDeInit

Function Name	**void HAL_TIM_OC_MspDeInit (TIM_HandleTypeDef * htim)**
Function Description	DeInitializes TIM Output Compare MSP.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIM_OC_Start

Function Name	**HAL_StatusTypeDef HAL_TIM_OC_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**
Function Description	Starts the TIM Output Compare signal generation.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module. • **Channel:** TIM Channel to be enabled. This parameter can be one of the following values: – TIM_CHANNEL_1: TIM Channel 1 selected – TIM_CHANNEL_2: TIM Channel 2 selected – TIM_CHANNEL_3: TIM Channel 3 selected – TIM_CHANNEL_4: TIM Channel 4 selected
Return values	• **HAL:** status

HAL_TIM_OC_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_OC_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Output Compare signal generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_OC_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_OC_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the TIM Output Compare signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_OC_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_OC_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Output Compare signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_OC_Start_DMA

Function Name	**HAL_StatusTypeDef HAL_TIM_OC_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData, uint16_t Length)**
Function Description	Starts the TIM Output Compare signal generation in DMA mode.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module. • **Channel:** TIM Channel to be enabled. This parameter can be one of the following values: – TIM_CHANNEL_1: TIM Channel 1 selected – TIM_CHANNEL_2: TIM Channel 2 selected – TIM_CHANNEL_3: TIM Channel 3 selected – TIM_CHANNEL_4: TIM Channel 4 selected • **pData:** The source Buffer address. • **Length:** The length of data to be transferred from memory to TIM peripheral
Return values	• **HAL:** status

HAL_TIM_OC_Stop_DMA

Function Name	**HAL_StatusTypeDef HAL_TIM_OC_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**
Function Description	Stops the TIM Output Compare signal generation in DMA mode.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module. • **Channel:** TIM Channel to be disabled. This parameter can be one of the following values: – TIM_CHANNEL_1: TIM Channel 1 selected – TIM_CHANNEL_2: TIM Channel 2 selected – TIM_CHANNEL_3: TIM Channel 3 selected – TIM_CHANNEL_4: TIM Channel 4 selected
Return values	• **HAL:** status

HAL_TIM_PWM_Init

Function Name	**HAL_StatusTypeDef HAL_TIM_PWM_Init (TIM_HandleTypeDef * htim)**
Function Description	Initializes the TIM PWM Time Base according to the specified parameters in the TIM_HandleTypeDef and create the associated handle.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **HAL:** status

HAL_TIM_PWM_DeInit

Function Name	**HAL_StatusTypeDef HAL_TIM_PWM_DeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes the TIM peripheral.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_PWM_MspInit

Function Name **void HAL_TIM_PWM_MspInit (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM PWM MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_PWM_MspDeInit

Function Name **void HAL_TIM_PWM_MspDeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes TIM PWM MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_PWM_Start

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the PWM signal generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_PWM_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the PWM signal generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

– TIM_CHANNEL_3: TIM Channel 3 selected
– TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_PWM_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the PWM signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_PWM_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the PWM signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_PWM_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM PWM signal generation in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected
- **pData:** The source Buffer address.

- **Length:** The length of data to be transferred from memory to TIM peripheral

Return values
- **HAL:** status

HAL_TIM_PWM_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM PWM signal generation in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_Init

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Init (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM Input Capture Time base according to the specified parameters in the TIM_HandleTypeDef and create the associated handle.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_IC_DeInit

Function Name **HAL_StatusTypeDef HAL_TIM_IC_DeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes the TIM peripheral.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_IC_MspInit

Function Name **void HAL_TIM_IC_MspInit (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM INput Capture MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_IC_MspDeInit

Function Name **void HAL_TIM_IC_MspDeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes TIM Input Capture MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_IC_Start

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the TIM Input Capture measurement.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Input Capture measurement.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the TIM Input Capture measurement in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected

– TIM_CHANNEL_2: TIM Channel 2 selected
– TIM_CHANNEL_3: TIM Channel 3 selected
– TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Input Capture measurement in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM Input Capture measurement on in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected
- **pData:** The destination Buffer address.
- **Length:** The length of data to be transferred from TIM peripheral to memory.

Return values
- **HAL:** status

HAL_TIM_IC_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_IC_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Input Capture measurement on in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected

  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_OnePulse_Init

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_Init (TIM_HandleTypeDef * htim, uint32_t OnePulseMode)**

Function Description Initializes the TIM One Pulse Time Base according to the specified parameters in the TIM_HandleTypeDef and create the associated handle.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OnePulseMode:** Select the One pulse mode. This parameter can be one of the following values:
  - TIM_OPMODE_SINGLE: Only one pulse will be generated.
  - TIM_OPMODE_REPETITIVE: Repetitive pulses will be generated.

Return values
- **HAL:** status

HAL_TIM_OnePulse_DeInit

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_DeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes the TIM One Pulse.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_OnePulse_MspInit

Function Name **void HAL_TIM_OnePulse_MspInit (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM One Pulse MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_OnePulse_MspDeInit

Function Name **void HAL_TIM_OnePulse_MspDeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes TIM One Pulse MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_OnePulse_Start

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_Start (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Starts the TIM One Pulse signal generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** : TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIM_OnePulse_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_Stop (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Stops the TIM One Pulse signal generation.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** : TIM Channels to be disable. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIM_OnePulse_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_Start_IT (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Starts the TIM One Pulse signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** : TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIM_OnePulse_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_Stop_IT (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Stops the TIM One Pulse signal generation in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** : TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected

– TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIM_Encoder_Init

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Init (TIM_HandleTypeDef * htim, TIM_Encoder_InitTypeDef * sConfig)**

Function Description Initializes the TIM Encoder Interface and create the associated handle.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sConfig:** TIM Encoder Interface configuration structure

Return values
- **HAL:** status

HAL_TIM_Encoder_DeInit

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_DeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes the TIM Encoder interface.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIM_Encoder_MspInit

Function Name **void HAL_TIM_Encoder_MspInit (TIM_HandleTypeDef * htim)**

Function Description Initializes the TIM Encoder Interface MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_Encoder_MspDeInit

Function Name **void HAL_TIM_Encoder_MspDeInit (TIM_HandleTypeDef * htim)**

Function Description DeInitializes TIM Encoder Interface MSP.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_Encoder_Start

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the TIM Encoder Interface.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

Return values
- **HAL:** status

HAL_TIM_Encoder_Stop

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Encoder Interface.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

Return values
- **HAL:** status

HAL_TIM_Encoder_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the TIM Encoder Interface in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

Return values
- **HAL:** status

HAL_TIM_Encoder_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Encoder Interface in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be disabled. This parameter can be one of the following values:

- TIM_CHANNEL_1: TIM Channel 1 selected
- TIM_CHANNEL_2: TIM Channel 2 selected
- TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

Return values
- **HAL:** status

HAL_TIM_Encoder_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData1, uint32_t * pData2, uint16_t Length)**

Function Description Starts the TIM Encoder Interface in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected
- **pData1:** The destination Buffer address for IC1.
- **pData2:** The destination Buffer address for IC2.
- **Length:** The length of data to be transferred from TIM peripheral to memory.

Return values
- **HAL:** status

HAL_TIM_Encoder_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIM_Encoder_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Encoder Interface in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

Return values
- **HAL:** status

HAL_TIM_IRQHandler

Function Name **void HAL_TIM_IRQHandler (TIM_HandleTypeDef * htim)**

Function Description This function handles TIM interrupts requests.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_OC_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_TIM_OC_ConfigChannel (TIM_HandleTypeDef * htim, TIM_OC_InitTypeDef * sConfig, uint32_t Channel)**

Function Description Initializes the TIM Output Compare Channels according to the specified parameters in the TIM_OC_InitTypeDef.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sConfig:** TIM Output Compare configuration structure
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_PWM_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_TIM_PWM_ConfigChannel (TIM_HandleTypeDef * htim, TIM_OC_InitTypeDef * sConfig, uint32_t Channel)**

Function Description Initializes the TIM PWM channels according to the specified parameters in the TIM_OC_InitTypeDef.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sConfig:** TIM PWM configuration structure
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_IC_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_TIM_IC_ConfigChannel (TIM_HandleTypeDef * htim, TIM_IC_InitTypeDef * sConfig, uint32_t Channel)**

Function Description Initializes the TIM Input Capture Channels according to the specified parameters in the TIM_IC_InitTypeDef.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sConfig:** TIM Input Capture configuration structure
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

– TIM_CHANNEL_3: TIM Channel 3 selected
– TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_OnePulse_ConfigChannel

Function Name **HAL_StatusTypeDef HAL_TIM_OnePulse_ConfigChannel (TIM_HandleTypeDef * htim, TIM_OnePulse_InitTypeDef * sConfig, uint32_t OutputChannel, uint32_t InputChannel)**

Function Description Initializes the TIM One Pulse Channels according to the specified parameters in the TIM_OnePulse_InitTypeDef.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sConfig:** TIM One Pulse configuration structure
- **OutputChannel:** TIM Channels to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
- **InputChannel:** TIM Channels to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIM_ConfigOCrefClear

Function Name **HAL_StatusTypeDef HAL_TIM_ConfigOCrefClear (TIM_HandleTypeDef * htim, TIM_ClearInputConfigTypeDef * sClearInputConfig, uint32_t Channel)**

Function Description Configures the OCRef clear feature.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sClearInputConfig:** pointer to a TIM_ClearInputConfigTypeDef structure that contains the OCREF clear feature and parameters for the TIM peripheral.
- **Channel:** specifies the TIM Channel. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIM_ConfigClockSource

Function Name **HAL_StatusTypeDef HAL_TIM_ConfigClockSource (TIM_HandleTypeDef * htim, TIM_ClockConfigTypeDef * sClockSourceConfig)**

Function Description Configures the clock source to be used.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sClockSourceConfig:** pointer to a TIM_ClockConfigTypeDef structure that contains the clock source information for the TIM peripheral.

Return values
- **HAL:** status

HAL_TIM_ConfigTI1Input

Function Name **HAL_StatusTypeDef HAL_TIM_ConfigTI1Input (TIM_HandleTypeDef * htim, uint32_t TI1_Selection)**

Function Description Selects the signal connected to the TI1 input: direct from CH1_input or a XOR combination between CH1_input, CH2_input & CH3_input.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **TI1_Selection:** Indicate whether or not channel 1 is connected to the output of a XOR gate. This parameter can be one of the following values:
  - TIM_TI1SELECTION_CH1: The TIMx_CH1 pin is connected to TI1 input
  - TIM_TI1SELECTION_XORCOMBINATION: The TIMx_CH1, CH2 and CH3 pins are connected to the TI1 input (XOR combination)

Return values
- **HAL:** status

HAL_TIM_SlaveConfigSynchronization

Function Name **HAL_StatusTypeDef HAL_TIM_SlaveConfigSynchronization (TIM_HandleTypeDef * htim, TIM_SlaveConfigTypeDef * sSlaveConfig)**

Function Description Configures the TIM in Slave mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sSlaveConfig:** pointer to a TIM_SlaveConfigTypeDef structure that contains the selected trigger (internal trigger input, filtered timer input or external trigger input) and the) and the Slave mode (Disable, Reset, Gated, Trigger, External clock mode 1).

Return values
- **HAL:** status

HAL_TIM_SlaveConfigSynchronization_IT

Function Name **HAL_StatusTypeDef HAL_TIM_SlaveConfigSynchronization_IT (TIM_HandleTypeDef * htim, TIM_SlaveConfigTypeDef * sSlaveConfig)**

Function Description Configures the TIM in Slave mode in interrupt mode.

Parameters
- **htim:** TIM handle.

- **sSlaveConfig:** pointer to a TIM_SlaveConfigTypeDef structure that contains the selected trigger (internal trigger input, filtered timer input or external trigger input) and the) and the Slave mode (Disable, Reset, Gated, Trigger, External clock mode 1).

Return values

- **HAL:** status

HAL_TIM_DMABurst_WriteStart

Function Name

HAL_StatusTypeDef HAL_TIM_DMABurst_WriteStart (TIM_HandleTypeDef * htim, uint32_t BurstBaseAddress, uint32_t BurstRequestSrc, uint32_t * BurstBuffer, uint32_t BurstLength)

Function Description

Configure the DMA Burst to transfer Data from the memory to the TIM peripheral.

Parameters

- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **BurstBaseAddress:** TIM Base address from when the DMA will starts the Data write. This parameters can be on of the following values:
  - TIM_DMABASE_CR1
  - TIM_DMABASE_CR2
  - TIM_DMABASE_SMCR
  - TIM_DMABASE_DIER
  - TIM_DMABASE_SR
  - TIM_DMABASE_EGR
  - TIM_DMABASE_CCMR1
  - TIM_DMABASE_CCMR2
  - TIM_DMABASE_CCER
  - TIM_DMABASE_CNT
  - TIM_DMABASE_PSC
  - TIM_DMABASE_ARR
  - TIM_DMABASE_RCR
  - TIM_DMABASE_CCR1
  - TIM_DMABASE_CCR2
  - TIM_DMABASE_CCR3
  - TIM_DMABASE_CCR4
  - TIM_DMABASE_BDTR
  - TIM_DMABASE_DCR
- **BurstRequestSrc:** TIM DMA Request sources. This parameters can be on of the following values:
  - TIM_DMA_UPDATE: TIM update Interrupt source
  - TIM_DMA_CC1: TIM Capture Compare 1 DMA source
  - TIM_DMA_CC2: TIM Capture Compare 2 DMA source
  - TIM_DMA_CC3: TIM Capture Compare 3 DMA source
  - TIM_DMA_CC4: TIM Capture Compare 4 DMA source
  - TIM_DMA_COM: TIM Commutation DMA source
  - TIM_DMA_TRIGGER: TIM Trigger DMA source
- **BurstBuffer:** The Buffer address.
- **BurstLength:** DMA Burst length. This parameter can be one value between TIM_DMABURSTLENGTH_1TRANSFER and

TIM_DMABURSTLENGTH_18TRANSFERS.

Return values
- **HAL:** status

HAL_TIM_DMABurst_WriteStop

Function Name **HAL_StatusTypeDef HAL_TIM_DMABurst_WriteStop (TIM_HandleTypeDef * htim, uint32_t BurstRequestSrc)**

Function Description Stops the TIM DMA Burst mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **BurstRequestSrc:** TIM DMA Request sources to disable

Return values
- **HAL:** status

HAL_TIM_DMABurst_ReadStart

Function Name **HAL_StatusTypeDef HAL_TIM_DMABurst_ReadStart (TIM_HandleTypeDef * htim, uint32_t BurstBaseAddress, uint32_t BurstRequestSrc, uint32_t * BurstBuffer, uint32_t BurstLength)**

Function Description Configure the DMA Burst to transfer Data from the TIM peripheral to the memory.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **BurstBaseAddress:** TIM Base address from when the DMA will starts the Data read. This parameters can be on of the following values:
  - TIM_DMABASE_CR1
  - TIM_DMABASE_CR2
  - TIM_DMABASE_SMCR
  - TIM_DMABASE_DIER
  - TIM_DMABASE_SR
  - TIM_DMABASE_EGR
  - TIM_DMABASE_CCMR1
  - TIM_DMABASE_CCMR2
  - TIM_DMABASE_CCER
  - TIM_DMABASE_CNT
  - TIM_DMABASE_PSC
  - TIM_DMABASE_ARR
  - TIM_DMABASE_RCR
  - TIM_DMABASE_CCR1
  - TIM_DMABASE_CCR2
  - TIM_DMABASE_CCR3
  - TIM_DMABASE_CCR4
  - TIM_DMABASE_BDTR
  - TIM_DMABASE_DCR
- **BurstRequestSrc:** TIM DMA Request sources. This parameters can be on of the following values:
  - TIM_DMA_UPDATE: TIM update Interrupt source
  - TIM_DMA_CC1: TIM Capture Compare 1 DMA source
  - TIM_DMA_CC2: TIM Capture Compare 2 DMA source

  - TIM_DMA_CC3: TIM Capture Compare 3 DMA source
  - TIM_DMA_CC4: TIM Capture Compare 4 DMA source
  - TIM_DMA_COM: TIM Commutation DMA source
  - TIM_DMA_TRIGGER: TIM Trigger DMA source
- **BurstBuffer:** The Buffer address.
- **BurstLength:** DMA Burst length. This parameter can be one value between TIM_DMABURSTLENGTH_1TRANSFER and TIM_DMABURSTLENGTH_18TRANSFERS.

Return values
- **HAL:** status

HAL_TIM_DMABurst_ReadStop

Function Name **HAL_StatusTypeDef HAL_TIM_DMABurst_ReadStop (TIM_HandleTypeDef * htim, uint32_t BurstRequestSrc)**

Function Description Stop the DMA burst reading.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **BurstRequestSrc:** TIM DMA Request sources to disable.

Return values
- **HAL:** status

HAL_TIM_GenerateEvent

Function Name **HAL_StatusTypeDef HAL_TIM_GenerateEvent (TIM_HandleTypeDef * htim, uint32_t EventSource)**

Function Description Generate a software event.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **EventSource:** specifies the event source. This parameter can be one of the following values:
  - TIM_EVENTSOURCE_UPDATE: Timer update Event source
  - TIM_EVENTSOURCE_CC1: Timer Capture Compare 1 Event source
  - TIM_EVENTSOURCE_CC2: Timer Capture Compare 2 Event source
  - TIM_EVENTSOURCE_CC3: Timer Capture Compare 3 Event source
  - TIM_EVENTSOURCE_CC4: Timer Capture Compare 4 Event source
  - TIM_EVENTSOURCE_COM: Timer COM event source
  - TIM_EVENTSOURCE_TRIGGER: Timer Trigger Event source
  - TIM_EVENTSOURCE_BREAK: Timer Break event source

Return values
- **HAL:** status

Notes
- TIM6 and TIM7 can only generate an update event.
- TIM_EVENTSOURCE_COM and TIM_EVENTSOURCE_BREAK are used only with TIM1 and TIM8.

HAL_TIM_ReadCapturedValue

Function Name	**uint32_t HAL_TIM_ReadCapturedValue (TIM_HandleTypeDef * htim, uint32_t Channel)**
Function Description	Read the captured value from Capture Compare unit.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module. • **Channel:** TIM Channels to be enabled. This parameter can be one of the following values: – TIM_CHANNEL_1: TIM Channel 1 selected – TIM_CHANNEL_2: TIM Channel 2 selected – TIM_CHANNEL_3: TIM Channel 3 selected – TIM_CHANNEL_4: TIM Channel 4 selected
Return values	• **Captured:** value

HAL_TIM_PeriodElapsedCallback

Function Name	**void HAL_TIM_PeriodElapsedCallback (TIM_HandleTypeDef * htim)**
Function Description	Period elapsed callback in non blocking mode.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIM_OC_DelayElapsedCallback

Function Name	**void HAL_TIM_OC_DelayElapsedCallback (TIM_HandleTypeDef * htim)**
Function Description	Output Compare callback in non blocking mode.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIM_IC_CaptureCallback

Function Name	**void HAL_TIM_IC_CaptureCallback (TIM_HandleTypeDef * htim)**
Function Description	Input Capture callback in non blocking mode.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIM_PWM_PulseFinishedCallback

Function Name	**void HAL_TIM_PWM_PulseFinishedCallback (TIM_HandleTypeDef * htim)**
Function Description	PWM Pulse finished callback in non blocking mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_TriggerCallback

Function Name **void HAL_TIM_TriggerCallback (TIM_HandleTypeDef * htim)**

Function Description Hall Trigger detection callback in non blocking mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_ErrorCallback

Function Name **void HAL_TIM_ErrorCallback (TIM_HandleTypeDef * htim)**

Function Description Timer error callback in non blocking mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **None**

HAL_TIM_Base_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_Base_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM Base state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

HAL_TIM_OC_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_OC_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM OC state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

HAL_TIM_PWM_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_PWM_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM PWM state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

HAL_TIM_IC_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_IC_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM Input Capture state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

HAL_TIM_OnePulse_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_OnePulse_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM One Pulse Mode state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

HAL_TIM_Encoder_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIM_Encoder_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM Encoder Mode state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

TIM_Base_SetConfig

Function Name **void TIM_Base_SetConfig (TIM_TypeDef * TIMx, TIM_Base_InitTypeDef * Structure)**

Function Description Time Base configuration.

Parameters
- **TIMx:** TIM peripheral
- **Structure:** pointer on TIM Time Base required parameters

Return values
- **None**

TIM_TI1_SetConfig

Function Name **void TIM_TI1_SetConfig (TIM_TypeDef * TIMx, uint32_t TIM_ICPolarity, uint32_t TIM_ICSelection, uint32_t TIM_ICFilter)**

Function Description Configure the TI1 as Input.

Parameters
- **TIMx:** to select the TIM peripheral.
- **TIM_ICPolarity:** : The Input Polarity. This parameter can be one of the following values:

  - TIM_ICPolarity_Rising
  - TIM_ICPolarity_Falling
  - TIM_ICPolarity_BothEdge
- **TIM_ICSelection:** specifies the input to be used. This parameter can be one of the following values:
  - TIM_ICSelection_DirectTI: TIM Input 1 is selected to be connected to IC1.
  - TIM_ICSelection_IndirectTI: TIM Input 1 is selected to be connected to IC2.
  - TIM_ICSelection_TRC: TIM Input 1 is selected to be connected to TRC.
- **TIM_ICFilter:** Specifies the Input Capture Filter. This parameter must be a value between 0x00 and 0x0F.

Return values
- **None**

Notes
- TIM_ICFilter and TIM_ICPolarity are not used in INDIRECT mode as TI2FP1 (on channel2 path) is used as the input signal. Therefore CCMR1 must be protected against un-initialized filter and polarity values.

TIM_OC2_SetConfig

Function Name **void TIM_OC2_SetConfig (TIM_TypeDef * TIMx, TIM_OC_InitTypeDef * OC_Config)**

Function Description Time Output Compare 2 configuration.

Parameters
- **TIMx:** to select the TIM peripheral
- **OC_Config:** The output configuration structure

Return values
- **None**

TIM_DMADelayPulseCplt

Function Name **void TIM_DMADelayPulseCplt (DMA_HandleTypeDef * hdma)**

Function Description TIM DMA Delay Pulse complete callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

TIM_DMAError

Function Name **void TIM_DMAError (DMA_HandleTypeDef * hdma)**

Function Description TIM DMA error callback.

Parameters
- **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.

Return values
- **None**

TIM_DMACaptureCplt

Function Name	**void TIM_DMACaptureCplt (DMA_HandleTypeDef * hdma)**
Function Description	TIM DMA Capture complete callback.
Parameters	• **hdma:** pointer to a DMA_HandleTypeDef structure that contains the configuration information for the specified DMA module.
Return values	• **None**

TIM_CCxChannelCmd

Function Name	**void TIM_CCxChannelCmd (TIM_TypeDef * TIMx, uint32_t Channel, uint32_t ChannelState)**
Function Description	Enables or disables the TIM Capture Compare Channel x.
Parameters	• **TIMx:** to select the TIM peripheral • **Channel:** specifies the TIM Channel This parameter can be one of the following values: – TIM_Channel_1: TIM Channel 1 – TIM_Channel_2: TIM Channel 2 – TIM_Channel_3: TIM Channel 3 – TIM_Channel_4: TIM Channel 4 • **ChannelState:** specifies the TIM Channel CCxE bit new state. This parameter can be: TIM_CCx_ENABLE or TIM_CCx_Disable.
Return values	• **None**

62.3 TIM Firmware driver defines

62.3.1 TIM

TIM AOE Bit State

TIM_AUTOMATICOUTPUT_ENABLE

TIM_AUTOMATICOUTPUT_DISABLE

TIM Break Input State

TIM_BREAK_ENABLE

TIM_BREAK_DISABLE

TIM Break Polarity

TIM_BREAKPOLARITY_LOW

TIM_BREAKPOLARITY_HIGH

TIM Channel

TIM_CHANNEL_1

TIM_CHANNEL_2

TIM_CHANNEL_3

TIM_CHANNEL_4

TIM_CHANNEL_ALL

TIM Clear Input Polarity

TIM_CLEARINPUTPOLARITY_INVERTED	Polarity for ETRx pin
TIM_CLEARINPUTPOLARITY_NONINVERTED	Polarity for ETRx pin

TIM Clear Input Prescaler

TIM_CLEARINPUTPRESCALER_DIV1	No prescaler is used
TIM_CLEARINPUTPRESCALER_DIV2	Prescaler for External ETR pin: Capture performed once every 2 events.
TIM_CLEARINPUTPRESCALER_DIV4	Prescaler for External ETR pin: Capture performed once every 4 events.
TIM_CLEARINPUTPRESCALER_DIV8	Prescaler for External ETR pin: Capture performed once every 8 events.

TIM Clear Input Source

TIM_CLEARINPUTSOURCE_ETR

TIM_CLEARINPUTSOURCE_NONE

TIM Clock Division

TIM_CLOCKDIVISION_DIV1

TIM_CLOCKDIVISION_DIV2

TIM_CLOCKDIVISION_DIV4

TIM Clock Polarity

TIM_CLOCKPOLARITY_INVERTED	Polarity for ETRx clock sources
TIM_CLOCKPOLARITY_NONINVERTED	Polarity for ETRx clock sources
TIM_CLOCKPOLARITY_RISING	Polarity for TIx clock sources
TIM_CLOCKPOLARITY_FALLING	Polarity for TIx clock sources
TIM_CLOCKPOLARITY_BOTHEDGE	Polarity for TIx clock sources

TIM Clock Prescaler

TIM_CLOCKPRESCALER_DIV1	No prescaler is used
TIM_CLOCKPRESCALER_DIV2	Prescaler for External ETR Clock: Capture performed once every 2 events.
TIM_CLOCKPRESCALER_DIV4	Prescaler for External ETR Clock: Capture performed once every 4 events.
TIM_CLOCKPRESCALER_DIV8	Prescaler for External ETR Clock: Capture performed once every 8 events.

TIM Clock Source

TIM_CLOCKSOURCE_ETRMODE2

TIM_CLOCKSOURCE_INTERNAL

TIM_CLOCKSOURCE_ITR0

TIM_CLOCKSOURCE_ITR1

TIM_CLOCKSOURCE_ITR2

TIM_CLOCKSOURCE_ITR3

TIM_CLOCKSOURCE_TI1ED

TIM_CLOCKSOURCE_TI1

TIM_CLOCKSOURCE_TI2

TIM_CLOCKSOURCE_ETRMODE1

TIM Commutation Source

TIM_COMMUTATION_TRGI

TIM_COMMUTATION_SOFTWARE

TIM Counter Mode

TIM_COUNTERMODE_UP

TIM_COUNTERMODE_DOWN

TIM_COUNTERMODE_CENTERALIGNED1

TIM_COUNTERMODE_CENTERALIGNED2

TIM_COUNTERMODE_CENTERALIGNED3

TIM DMA Base address

TIM_DMABASE_CR1

TIM_DMABASE_CR2

TIM_DMABASE_SMCR

TIM_DMABASE_DIER

TIM_DMABASE_SR

TIM_DMABASE_EGR

TIM_DMABASE_CCMR1

TIM_DMABASE_CCMR2

TIM_DMABASE_CCER

TIM_DMABASE_CNT

TIM_DMABASE_PSC

TIM_DMABASE_ARR

TIM_DMABASE_RCR

TIM_DMABASE_CCR1

TIM_DMABASE_CCR2

TIM_DMABASE_CCR3

TIM_DMABASE_CCR4

TIM_DMABASE_BDTR

TIM_DMABASE_DCR

TIM_DMABASE_OR

TIM DMA Burst Length

TIM_DMABURSTLENGTH_1TRANSFER

TIM_DMABURSTLENGTH_2TRANSFERS

TIM_DMABURSTLENGTH_3TRANSFERS

TIM_DMABURSTLENGTH_4TRANSFERS

TIM_DMABURSTLENGTH_5TRANSFERS

TIM_DMABURSTLENGTH_6TRANSFERS

TIM_DMABURSTLENGTH_7TRANSFERS

TIM_DMABURSTLENGTH_8TRANSFERS

TIM_DMABURSTLENGTH_9TRANSFERS

TIM_DMABURSTLENGTH_10TRANSFERS

TIM_DMABURSTLENGTH_11TRANSFERS

TIM_DMABURSTLENGTH_12TRANSFERS

TIM_DMABURSTLENGTH_13TRANSFERS

TIM_DMABURSTLENGTH_14TRANSFERS

TIM_DMABURSTLENGTH_15TRANSFERS

TIM_DMABURSTLENGTH_16TRANSFERS

TIM_DMABURSTLENGTH_17TRANSFERS

TIM_DMABURSTLENGTH_18TRANSFERS

TIM DMA sources

TIM_DMA_UPDATE

TIM_DMA_CC1

TIM_DMA_CC2

TIM_DMA_CC3

TIM_DMA_CC4

TIM_DMA_COM

TIM_DMA_TRIGGER

TIM Encoder Mode

TIM_ENCODERMODE_TI1

TIM_ENCODERMODE_TI2

TIM_ENCODERMODE_TI12

TIM ETR Polarity

TIM_ETRPOLARITY_INVERTED Polarity for ETR source

TIM_ETRPOLARITY_NONINVERTED Polarity for ETR source

TIM ETR Prescaler

TIM_ETRPRESCALER_DIV1 No prescaler is used

TIM_ETRPRESCALER_DIV2	ETR input source is divided by 2
TIM_ETRPRESCALER_DIV4	ETR input source is divided by 4
TIM_ETRPRESCALER_DIV8	ETR input source is divided by 8

TIM Event Source

TIM_EVENTSOURCE_UPDATE

TIM_EVENTSOURCE_CC1

TIM_EVENTSOURCE_CC2

TIM_EVENTSOURCE_CC3

TIM_EVENTSOURCE_CC4

TIM_EVENTSOURCE_COM

TIM_EVENTSOURCE_TRIGGER

TIM_EVENTSOURCE_BREAK

TIM Exported Macros

__HAL_TIM_RESET_HANDLE_STATE

Description:

- Reset TIM handle state.

Parameters:

- __HANDLE__: TIM handle

Return value:

- None

__HAL_TIM_ENABLE

Description:

- Enable the TIM peripheral.

Parameters:

- __HANDLE__: TIM handle

Return value:

- None

__HAL_TIM_MOE_ENABLE

Description:

- Enable the TIM main Output.

Parameters:

- __HANDLE__: TIM handle

Return value:

- None

__HAL_TIM_DISABLE

Description:

- Disable the TIM peripheral.

Parameters:

- __HANDLE__: TIM handle

Return value:

- None

__HAL_TIM_MOE_DISABLE

Description:

- Disable the TIM main Output.

Parameters:

- __HANDLE__: TIM handle

Return value:

- None

__HAL_TIM_ENABLE_IT

__HAL_TIM_ENABLE_DMA

__HAL_TIM_DISABLE_IT

__HAL_TIM_DISABLE_DMA

__HAL_TIM_GET_FLAG

__HAL_TIM_CLEAR_FLAG

__HAL_TIM_GET_IT_SOURCE

__HAL_TIM_CLEAR_IT

__HAL_TIM_IS_TIM_COUNTING_DOWN

__HAL_TIM_SET_PRESCALER

TIM_SET_ICPRESCALERVALUE

TIM_RESET_ICPRESCALERVALUE

TIM_SET_CAPTUREPOLARITY

TIM_RESET_CAPTUREPOLARITY

__HAL_TIM_SET_COMPARE

Description:

- Sets the TIM Capture Compare Register value on runtime without calling another time ConfigChannel function.

Parameters:

- __HANDLE__: TIM handle.
- __CHANNEL__: : TIM Channels to be configured. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

- __COMPARE__: specifies the Capture Compare register new value.

Return value:

- None

__HAL_TIM_GET_COMPARE

Description:

- Gets the TIM Capture Compare Register value on runtime.

Parameters:

- __HANDLE__: TIM handle.
- __CHANNEL__: : TIM Channel associated with the capture compare register This parameter can be one of the following values:
  - TIM_CHANNEL_1: get capture/compare 1 register value
  - TIM_CHANNEL_2: get capture/compare 2 register value
  - TIM_CHANNEL_3: get capture/compare 3 register value
  - TIM_CHANNEL_4: get capture/compare 4 register value

Return value:

- None

__HAL_TIM_SET_COUNTER

Description:

- Sets the TIM Counter Register value on runtime.

Parameters:

- __HANDLE__: TIM handle.
- __COUNTER__: specifies the Counter register new value.

Return value:

- None

__HAL_TIM_GET_COUNTER

Description:

- Gets the TIM Counter Register value on runtime.

Parameters:

- __HANDLE__: TIM handle.

Return value:

- None

__HAL_TIM_SET_AUTORELOAD

Description:

- Sets the TIM Autoreload Register value on runtime without calling another time any Init function.

Parameters:

- __HANDLE__: TIM handle.
- __AUTORELOAD__: specifies the Counter register new value.

Return value:

- None

__HAL_TIM_GET_AUTORELOAD

Description:

- Gets the TIM Autoreload Register value on runtime.

Parameters:

- __HANDLE__: TIM handle.

Return value:

- None

__HAL_TIM_SET_CLOCKDIVISION

Description:

- Sets the TIM Clock Division value on runtime without calling another time any Init function.

Parameters:

- __HANDLE__: TIM handle.
- __CKD__: specifies the clock division value. This parameter can be one of the following value:
  - TIM_CLOCKDIVISION_DIV1
  - TIM_CLOCKDIVISION_DIV2
  - TIM_CLOCKDIVISION_DIV4

Return value:

- None

__HAL_TIM_GET_CLOCKDIVISION

Description:

- Gets the TIM Clock Division value on runtime.

Parameters:

- __HANDLE__: TIM handle.

Return value:

- None

__HAL_TIM_SET_ICPRESCALER

Description:

- Sets the TIM Input Capture prescaler on runtime without calling another time

Parameters:

- __HANDLE__: TIM handle.
- __CHANNEL__: : TIM Channels to be configured. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1

selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected
- __ICPSC__: specifies the Input Capture4 prescaler new value. This parameter can be one of the following values:
  - TIM_ICPSC_DIV1: no prescaler
  - TIM_ICPSC_DIV2: capture is done once every 2 events
  - TIM_ICPSC_DIV4: capture is done once every 4 events
  - TIM_ICPSC_DIV8: capture is done once every 8 events

Return value:

- None

__HAL_TIM_GET_ICPRESCALER

Description:

- Gets the TIM Input Capture prescaler on runtime.

Parameters:

- __HANDLE__: TIM handle.
- __CHANNEL__: : TIM Channels to be configured. This parameter can be one of the following values:
  - TIM_CHANNEL_1: get input capture 1 prescaler value
  - TIM_CHANNEL_2: get input capture 2 prescaler value
  - TIM_CHANNEL_3: get input capture 3 prescaler value
  - TIM_CHANNEL_4: get input capture 4 prescaler value

Return value:

- None

__HAL_TIM_URS_ENABLE

Description:

- Set the Update Request Source (URS) bit of the TIMx_CR1 register.

Parameters:

- __HANDLE__: TIM handle.

Return value:

- None

Notes:

- When the USR bit of the TIMx_CR1 register is

set, only counter overflow/underflow generates an update interrupt or DMA request (if enabled)

__HAL_TIM_URS_DISABLE

Description:

- Reset the Update Request Source (URS) bit of the TIMx_CR1 register.

Parameters:

- __HANDLE__: TIM handle.

Return value:

- None

Notes:

- When the USR bit of the TIMx_CR1 register is reset, any of the following events generate an update interrupt or DMA request (if enabled): _ Counter overflow/underflow _ Setting the UG bit _ Update generation through the slave mode controller

__HAL_TIM_SET_CAPTURE POLARITY

Description:

- Sets the TIM Capture x input polarity on runtime.

Parameters:

- __HANDLE__: TIM handle.
- __CHANNEL__: TIM Channels to be configured. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected
- __POLARITY__: Polarity for TIx source
  - TIM_INPUTCHANNELPOLARITY_RISING : Rising Edge
  - TIM_INPUTCHANNELPOLARITY_FALLING: Falling Edge
  - TIM_INPUTCHANNELPOLARITY_BOTHEDGE: Rising and Falling Edge

Return value:

- None

Notes:

- The polarity TIM_INPUTCHANNELPOLARITY_BOTHEDGE is not authorized for TIM Channel 4.

TIM Flag definition

TIM_FLAG_UPDATE

TIM_FLAG_CC1

TIM_FLAG_CC2

TIM_FLAG_CC3

TIM_FLAG_CC4

TIM_FLAG_COM

TIM_FLAG_TRIGGER

TIM_FLAG_BREAK

TIM_FLAG_CC1OF

TIM_FLAG_CC2OF

TIM_FLAG_CC3OF

TIM_FLAG_CC4OF

TIM Input Capture Polarity

TIM_ICPOLARITY_RISING

TIM_ICPOLARITY_FALLING

TIM_ICPOLARITY_BOTHEDGE

TIM Input Capture Prescaler

TIM_ICPSC_DIV1	Capture performed each time an edge is detected on the capture input
TIM_ICPSC_DIV2	Capture performed once every 2 events
TIM_ICPSC_DIV4	Capture performed once every 4 events
TIM_ICPSC_DIV8	Capture performed once every 8 events

TIM Input Capture Selection

TIM_ICSELECTION_DIRECTTI	TIM Input 1, 2, 3 or 4 is selected to be connected to IC1, IC2, IC3 or IC4, respectively
TIM_ICSELECTION_INDIRECTTI	TIM Input 1, 2, 3 or 4 is selected to be connected to IC2, IC1, IC4 or IC3, respectively
TIM_ICSELECTION_TRC	TIM Input 1, 2, 3 or 4 is selected to be connected to TRC

TIM Input Channel Polarity

TIM_INPUTCHANNELPOLARITY_RISING	Polarity for TIx source
TIM_INPUTCHANNELPOLARITY_FALLING	Polarity for TIx source
TIM_INPUTCHANNELPOLARITY_BOTHEDGE	Polarity for TIx source

TIM Interrupt definition

TIM_IT_UPDATE

TIM_IT_CC1

TIM_IT_CC2

TIM_IT_CC3

TIM_IT_CC4

TIM_IT_COM

TIM_IT_TRIGGER

TIM_IT_BREAK

TIM Private macros to check input parameters

IS_TIM_COUNTER_MODE

IS_TIM_CLOCKDIVISION_DIV

IS_TIM_PWM_MODE

IS_TIM_OC_MODE

IS_TIM_FAST_STATE

IS_TIM_OC_POLARITY

IS_TIM_OCN_POLARITY

IS_TIM_OCIDLE_STATE

IS_TIM_OCNIDLE_STATE

IS_TIM_CHANNELS

IS_TIM_OPM_CHANNELS

IS_TIM_COMPLEMENTARY_CHANNELS

IS_TIM_IC_POLARITY

IS_TIM_IC_SELECTION

IS_TIM_IC_PRESCALER

IS_TIM_OPM_MODE

IS_TIM_DMA_SOURCE

IS_TIM_ENCODER_MODE

IS_TIM_EVENT_SOURCE

IS_TIM_CLOCKSOURCE

IS_TIM_CLOCKPOLARITY

IS_TIM_CLOCKPRESCALER

IS_TIM_CLOCKFILTER

IS_TIM_CLEARINPUT_SOURCE

IS_TIM_CLEARINPUT_POLARITY

IS_TIM_CLEARINPUT_PRESCALER

IS_TIM_CLEARINPUT_FILTER

IS_TIM_OSSR_STATE

IS_TIM_OSSI_STATE

IS_TIM_LOCK_LEVEL

IS_TIM_BREAK_STATE

IS_TIM_BREAK_POLARITY

IS_TIM_AUTOMATIC_OUTPUT_STATE

IS_TIM_TRGO_SOURCE

IS_TIM_SLAVE_MODE

IS_TIM_MSM_STATE

IS_TIM_TRIGGER_SELECTION

IS_TIM_INTERNAL_TRIGGEREVENT_SELECTION

IS_TIM_TRIGGERPOLARITY

IS_TIM_TRIGGERPRESCALER

IS_TIM_TRIGGERFILTER

IS_TIM_TI1SELECTION

IS_TIM_DMA_BASE

IS_TIM_DMA_LENGTH

IS_TIM_IC_FILTER

TIM Lock level

TIM_LOCKLEVEL_OFF

TIM_LOCKLEVEL_1

TIM_LOCKLEVEL_2

TIM_LOCKLEVEL_3

TIM Mask Definition

TIM_CCER_CCxE_MASK

TIM_CCER_CCxNE_MASK

TIM Master Mode Selection

TIM_TRGO_RESET

TIM_TRGO_ENABLE

TIM_TRGO_UPDATE

TIM_TRGO_OC1

TIM_TRGO_OC1REF

TIM_TRGO_OC2REF

TIM_TRGO_OC3REF

TIM_TRGO_OC4REF

TIM Master Slave Mode

TIM_MASTERSLAVEMODE_ENABLE

TIM_MASTERSLAVEMODE_DISABLE

TIM One Pulse Mode

TIM_OPMODE_SINGLE

TIM_OPMODE_REPETITIVE

TIM OSSI OffState Selection for Idle mode state

TIM_OSSI_ENABLE

TIM_OSSI_DISABLE

TIM OSSR OffState Selection for Run mode state

TIM_OSSR_ENABLE

TIM_OSSR_DISABLE

TIM Output Compare and PWM modes

TIM_OCMODE_TIMING

TIM_OCMODE_ACTIVE

TIM_OCMODE_INACTIVE

TIM_OCMODE_TOGGLE

TIM_OCMODE_PWM1

TIM_OCMODE_PWM2

TIM_OCMODE_FORCED_ACTIVE

TIM_OCMODE_FORCED_INACTIVE

TIM Output Compare Idle State

TIM_OCIDLESTATE_SET

TIM_OCIDLESTATE_RESET

TIM Output Compare N Idle State

TIM_OCNIDLESTATE_SET

TIM_OCNIDLESTATE_RESET

TIM Output CompareN Polarity

TIM_OCNPOLARITY_HIGH

TIM_OCNPOLARITY_LOW

TIM Output Compare Polarity

TIM_OCPOLARITY_HIGH

TIM_OCPOLARITY_LOW

TIM Output Fast State

TIM_OCFAST_DISABLE

TIM_OCFAST_ENABLE

TIM Slave Mode

TIM_SLAVEMODE_DISABLE

TIM_SLAVEMODE_RESET

TIM_SLAVEMODE_GATED

TIM_SLAVEMODE_TRIGGER

TIM_SLAVEMODE_EXTERNAL1

TIM TI1 Selection

TIM_TI1SELECTION_CH1

TIM_TI1SELECTION_XORCOMBINATION

TIM Trigger Polarity

TIM_TRIGGERPOLARITY_INVERTED	Polarity for ETRx trigger sources
TIM_TRIGGERPOLARITY_NONINVERTED	Polarity for ETRx trigger sources
TIM_TRIGGERPOLARITY_RISING	Polarity for TIxFPx or TI1_ED trigger sources
TIM_TRIGGERPOLARITY_FALLING	Polarity for TIxFPx or TI1_ED trigger sources
TIM_TRIGGERPOLARITY_BOTHEDGE	Polarity for TIxFPx or TI1_ED trigger sources

TIM Trigger Prescaler

TIM_TRIGGERPRESCALER_DIV1	No prescaler is used
TIM_TRIGGERPRESCALER_DIV2	Prescaler for External ETR Trigger: Capture performed once every 2 events.
TIM_TRIGGERPRESCALER_DIV4	Prescaler for External ETR Trigger: Capture performed once every 4 events.
TIM_TRIGGERPRESCALER_DIV8	Prescaler for External ETR Trigger: Capture performed once every 8 events.

TIM Trigger Selection

TIM_TS_ITR0

TIM_TS_ITR1

TIM_TS_ITR2

TIM_TS_ITR3

TIM_TS_TI1F_ED

TIM_TS_TI1FP1

TIM_TS_TI2FP2

TIM_TS_ETRF

TIM_TS_NONE

63 HAL TIM Extension Driver

63.1 TIMEx Firmware driver registers structures

63.1.1 TIM_HallSensor_InitTypeDef

Data Fields

- ***uint32_t IC1Polarity***
- ***uint32_t IC1Prescaler***
- ***uint32_t IC1Filter***
- ***uint32_t Commutation_Delay***

Field Documentation

- ***uint32_t TIM_HallSensor_InitTypeDef::IC1Polarity***
  Specifies the active edge of the input signal. This parameter can be a value of ***TIM_Input_Capture_Polarity***
- ***uint32_t TIM_HallSensor_InitTypeDef::IC1Prescaler***
  Specifies the Input Capture Prescaler. This parameter can be a value of ***TIM_Input_Capture_Prescaler***
- ***uint32_t TIM_HallSensor_InitTypeDef::IC1Filter***
  Specifies the input capture filter. This parameter can be a number between Min_Data = 0x0 and Max_Data = 0xF
- ***uint32_t TIM_HallSensor_InitTypeDef::Commutation_Delay***
  Specifies the pulse value to be loaded into the Capture Compare Register. This parameter can be a number between Min_Data = 0x0000U and Max_Data = 0xFFFFU

63.1.2 TIM_MasterConfigTypeDef

Data Fields

- ***uint32_t MasterOutputTrigger***
- ***uint32_t MasterSlaveMode***

Field Documentation

- ***uint32_t TIM_MasterConfigTypeDef::MasterOutputTrigger***
  Trigger output (TRGO) selection. This parameter can be a value of ***TIM_Master_Mode_Selection***
- ***uint32_t TIM_MasterConfigTypeDef::MasterSlaveMode***
  Master/slave mode selection. This parameter can be a value of ***TIM_Master_Slave_Mode***

63.1.3 TIM_BreakDeadTimeConfigTypeDef

Data Fields

- ***uint32_t OffStateRunMode***
- ***uint32_t OffStateIDLEMode***
- ***uint32_t LockLevel***
- ***uint32_t DeadTime***
- ***uint32_t BreakState***
- ***uint32_t BreakPolarity***
- ***uint32_t AutomaticOutput***

Field Documentation

- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::OffStateRunMode***
  TIM off state in run mode. This parameter can be a value of ***TIM_OSSR_Off_State_Selection_for_Run_mode_state***
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::OffStateIDLEMode***
  TIM off state in IDLE mode. This parameter can be a value of ***TIM_OSSI_Off_State_Selection_for_Idle_mode_state***
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::LockLevel***
  TIM Lock level. This parameter can be a value of ***TIM_Lock_level***
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::DeadTime***
  TIM dead Time. This parameter can be a number between Min_Data = 0x00 and Max_Data = 0xFF
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::BreakState***
  TIM Break State. This parameter can be a value of ***TIM_Break_Input_enable_disable***
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::BreakPolarity***
  TIM Break input polarity. This parameter can be a value of ***TIM_Break_Polarity***
- ***uint32_t TIM_BreakDeadTimeConfigTypeDef::AutomaticOutput***
  TIM Automatic Output Enable state. This parameter can be a value of ***TIM_AOE_Bit_Set_Reset***

63.2 TIMEx Firmware driver API description

63.2.1 TIMER Extended features

The Timer Extension features include:

1. Complementary outputs with programmable dead-time for :
   – Input Capture
   – Output Compare
   – PWM generation (Edge and Center-aligned Mode)
   – One-pulse mode output
2. Synchronization circuit to control the timer with external signals and to interconnect several timers together.
3. Break input to put the timer output signals in reset state or in a known state.
4. Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes

63.2.2 How to use this driver

1. Initialize the TIM low level resources by implementing the following functions depending from feature used :
   - Complementary Output Compare : HAL_TIM_OC_MspInit()
   - Complementary PWM generation : HAL_TIM_PWM_MspInit()
   - Complementary One-pulse mode output : HAL_TIM_OnePulse_MspInit()
   - Hall Sensor output : HAL_TIM_HallSensor_MspInit()
2. Initialize the TIM low level resources :
   a. Enable the TIM interface clock using __TIMx_CLK_ENABLE();
   b. TIM pins configuration
      - Enable the clock for the TIM GPIOs using the following function: __GPIOx_CLK_ENABLE();
      - Configure these TIM pins in Alternate function mode using HAL_GPIO_Init();
3. The external Clock can be configured, if needed (the default clock is the internal clock from the APBx), using the following function: HAL_TIM_ConfigClockSource, the clock configuration should be done before any start function.
4. Configure the TIM in the desired functioning mode using one of the initialization function of this driver:
   - HAL_TIMEx_HallSensor_Init and HAL_TIMEx_ConfigCommutationEvent: to use the Timer Hall Sensor Interface and the commutation event with the corresponding Interrupt and DMA request if needed (Note that One Timer is used to interface with the Hall sensor Interface and another Timer should be used to use the commutation event).
5. Activate the TIM peripheral using one of the start functions:
   - Complementary Output Compare : HAL_TIMEx_OCN_Start(), HAL_TIMEx_OCN_Start_DMA(), HAL_TIMEx_OC_Start_IT()
   - Complementary PWM generation : HAL_TIMEx_PWMN_Start(), HAL_TIMEx_PWMN_Start_DMA(), HAL_TIMEx_PWMN_Start_IT()
   - Complementary One-pulse mode output : HAL_TIMEx_OnePulseN_Start(), HAL_TIMEx_OnePulseN_Start_IT()
   - Hall Sensor output : HAL_TIMEx_HallSensor_Start(), HAL_TIMEx_HallSensor_Start_DMA(), HAL_TIMEx_HallSensor_Start_IT().

63.2.3 Timer Hall Sensor functions

This section provides functions allowing to:

- Initialize and configure TIM HAL Sensor.
- De-initialize TIM HAL Sensor.
- Start the Hall Sensor Interface.
- Stop the Hall Sensor Interface.
- Start the Hall Sensor Interface and enable interrupts.
- Stop the Hall Sensor Interface and disable interrupts.
- Start the Hall Sensor Interface and enable DMA transfers.
- Stop the Hall Sensor Interface and disable DMA transfers.

This section contains the following APIs:

- ***HAL_TIMEx_HallSensor_Init()***
- ***HAL_TIMEx_HallSensor_DeInit()***
- ***HAL_TIMEx_HallSensor_MspInit()***
- ***HAL_TIMEx_HallSensor_MspDeInit()***
- ***HAL_TIMEx_HallSensor_Start()***

- ***HAL_TIMEx_HallSensor_Stop()***
- ***HAL_TIMEx_HallSensor_Start_IT()***
- ***HAL_TIMEx_HallSensor_Stop_IT()***
- ***HAL_TIMEx_HallSensor_Start_DMA()***
- ***HAL_TIMEx_HallSensor_Stop_DMA()***

63.2.4 Timer Complementary Output Compare functions

This section provides functions allowing to:

- Start the Complementary Output Compare/PWM.
- Stop the Complementary Output Compare/PWM.
- Start the Complementary Output Compare/PWM and enable interrupts.
- Stop the Complementary Output Compare/PWM and disable interrupts.
- Start the Complementary Output Compare/PWM and enable DMA transfers.
- Stop the Complementary Output Compare/PWM and disable DMA transfers.

This section contains the following APIs:

- ***HAL_TIMEx_OCN_Start()***
- ***HAL_TIMEx_OCN_Stop()***
- ***HAL_TIMEx_OCN_Start_IT()***
- ***HAL_TIMEx_OCN_Stop_IT()***
- ***HAL_TIMEx_OCN_Start_DMA()***
- ***HAL_TIMEx_OCN_Stop_DMA()***

63.2.5 Timer Complementary PWM functions

This section provides functions allowing to:

- Start the Complementary PWM.
- Stop the Complementary PWM.
- Start the Complementary PWM and enable interrupts.
- Stop the Complementary PWM and disable interrupts.
- Start the Complementary PWM and enable DMA transfers.
- Stop the Complementary PWM and disable DMA transfers.
- Start the Complementary Input Capture measurement.
- Stop the Complementary Input Capture.
- Start the Complementary Input Capture and enable interrupts.
- Stop the Complementary Input Capture and disable interrupts.
- Start the Complementary Input Capture and enable DMA transfers.
- Stop the Complementary Input Capture and disable DMA transfers.
- Start the Complementary One Pulse generation.
- Stop the Complementary One Pulse.
- Start the Complementary One Pulse and enable interrupts.
- Stop the Complementary One Pulse and disable interrupts.

This section contains the following APIs:

- ***HAL_TIMEx_PWMN_Start()***
- ***HAL_TIMEx_PWMN_Stop()***
- ***HAL_TIMEx_PWMN_Start_IT()***
- ***HAL_TIMEx_PWMN_Stop_IT()***
- ***HAL_TIMEx_PWMN_Start_DMA()***
- ***HAL_TIMEx_PWMN_Stop_DMA()***

63.2.6 Timer Complementary One Pulse functions

This section provides functions allowing to:

- Start the Complementary One Pulse generation.
- Stop the Complementary One Pulse.
- Start the Complementary One Pulse and enable interrupts.
- Stop the Complementary One Pulse and disable interrupts.

This section contains the following APIs:

- ***HAL_TIMEx_OnePulseN_Start()***
- ***HAL_TIMEx_OnePulseN_Stop()***
- ***HAL_TIMEx_OnePulseN_Start_IT()***
- ***HAL_TIMEx_OnePulseN_Stop_IT()***

63.2.7 Peripheral Control functions

This section provides functions allowing to:

- Configure The Input Output channels for OC, PWM, IC or One Pulse mode.
- Configure External Clock source.
- Configure Complementary channels, break features and dead time.
- Configure Master and the Slave synchronization.
- Configure the commutation event in case of use of the Hall sensor interface.
- Configure the DMA Burst Mode.

This section contains the following APIs:

- ***HAL_TIMEx_ConfigCommutationEvent()***
- ***HAL_TIMEx_ConfigCommutationEvent_IT()***
- ***HAL_TIMEx_ConfigCommutationEvent_DMA()***
- ***HAL_TIMEx_MasterConfigSynchronization()***
- ***HAL_TIMEx_ConfigBreakDeadTime()***
- ***HAL_TIMEx_RemapConfig()***

63.2.8 Extension Callbacks functions

This section provides Extension TIM callback functions:

- Timer Commutation callback
- Timer Break callback

This section contains the following APIs:

- ***HAL_TIMEx_CommutationCallback()***
- ***HAL_TIMEx_BreakCallback()***
- ***TIMEx_DMACommutationCplt()***

63.2.9 Extension Peripheral State functions

This subsection permits to get in run-time the status of the peripheral and the data flow.

This section contains the following APIs:

- ***HAL_TIMEx_HallSensor_GetState()***

63.2.10 Detailed description of functions

HAL_TIMEx_HallSensor_Init

Function Name	**HAL_StatusTypeDef HAL_TIMEx_HallSensor_Init (TIM_HandleTypeDef * htim, TIM_HallSensor_InitTypeDef * sConfig)**
Function Description	Initializes the TIM Hall Sensor Interface and create the associated handle.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module. • **sConfig:** TIM Hall Sensor configuration structure
Return values	• **HAL:** status

HAL_TIMEx_HallSensor_DeInit

Function Name	**HAL_StatusTypeDef HAL_TIMEx_HallSensor_DeInit (TIM_HandleTypeDef * htim)**
Function Description	DeInitializes the TIM Hall Sensor interface.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **HAL:** status

HAL_TIMEx_HallSensor_MspInit

Function Name	**void HAL_TIMEx_HallSensor_MspInit (TIM_HandleTypeDef * htim)**
Function Description	Initializes the TIM Hall Sensor MSP.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIMEx_HallSensor_MspDeInit

Function Name	**void HAL_TIMEx_HallSensor_MspDeInit (TIM_HandleTypeDef * htim)**
Function Description	DeInitializes TIM Hall Sensor MSP.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
Return values	• **None**

HAL_TIMEx_HallSensor_Start

Function Name	**HAL_StatusTypeDef HAL_TIMEx_HallSensor_Start (TIM_HandleTypeDef * htim)**
Function Description	Starts the TIM Hall Sensor Interface.
Parameters	• **htim:** pointer to a TIM_HandleTypeDef structure that

contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIMEx_HallSensor_Stop

Function Name **HAL_StatusTypeDef HAL_TIMEx_HallSensor_Stop (TIM_HandleTypeDef * htim)**

Function Description Stops the TIM Hall sensor Interface.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIMEx_HallSensor_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_HallSensor_Start_IT (TIM_HandleTypeDef * htim)**

Function Description Starts the TIM Hall Sensor Interface in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIMEx_HallSensor_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_HallSensor_Stop_IT (TIM_HandleTypeDef * htim)**

Function Description Stops the TIM Hall Sensor Interface in interrupt mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIMEx_HallSensor_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_HallSensor_Start_DMA (TIM_HandleTypeDef * htim, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM Hall Sensor Interface in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **pData:** The destination Buffer address.
- **Length:** The length of data to be transferred from TIM peripheral to memory.

Return values
- **HAL:** status

HAL_TIMEx_HallSensor_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_HallSensor_Stop_DMA (TIM_HandleTypeDef * htim)**

Function Description — Stops the TIM Hall Sensor Interface in DMA mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** status

HAL_TIMEx_OCN_Start

Function Name — **HAL_StatusTypeDef HAL_TIMEx_OCN_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description — Starts the TIM Output Compare signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_OCN_Stop

Function Name — **HAL_StatusTypeDef HAL_TIMEx_OCN_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description — Stops the TIM Output Compare signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_OCN_Start_IT

Function Name — **HAL_StatusTypeDef HAL_TIMEx_OCN_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description — Starts the TIM Output Compare signal generation in interrupt mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

– TIM_CHANNEL_3: TIM Channel 3 selected
– TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_OCN_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_OCN_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Output Compare signal generation in interrupt mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_OCN_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_OCN_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM Output Compare signal generation in DMA mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected
- **pData:** The source Buffer address.
- **Length:** The length of data to be transferred from memory to TIM peripheral

Return values
- **HAL:** status

HAL_TIMEx_OCN_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_OCN_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM Output Compare signal generation in DMA mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can

be one of the following values:
- TIM_CHANNEL_1: TIM Channel 1 selected
- TIM_CHANNEL_2: TIM Channel 2 selected
- TIM_CHANNEL_3: TIM Channel 3 selected
- TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Start

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Start (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the PWM signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Stop

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Stop (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the PWM signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected
  - TIM_CHANNEL_3: TIM Channel 3 selected
  - TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Start_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Starts the PWM signal generation in interrupt mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

– TIM_CHANNEL_3: TIM Channel 3 selected
– TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Stop_IT (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the PWM signal generation in interrupt mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Start_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Start_DMA (TIM_HandleTypeDef * htim, uint32_t Channel, uint32_t * pData, uint16_t Length)**

Function Description Starts the TIM PWM signal generation in DMA mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be enabled. This parameter can be one of the following values:
  – TIM_CHANNEL_1: TIM Channel 1 selected
  – TIM_CHANNEL_2: TIM Channel 2 selected
  – TIM_CHANNEL_3: TIM Channel 3 selected
  – TIM_CHANNEL_4: TIM Channel 4 selected
- **pData:** The source Buffer address.
- **Length:** The length of data to be transferred from memory to TIM peripheral

Return values
- **HAL:** status

HAL_TIMEx_PWMN_Stop_DMA

Function Name **HAL_StatusTypeDef HAL_TIMEx_PWMN_Stop_DMA (TIM_HandleTypeDef * htim, uint32_t Channel)**

Function Description Stops the TIM PWM signal generation in DMA mode on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channel to be disabled. This parameter can

be one of the following values:
- TIM_CHANNEL_1: TIM Channel 1 selected
- TIM_CHANNEL_2: TIM Channel 2 selected
- TIM_CHANNEL_3: TIM Channel 3 selected
- TIM_CHANNEL_4: TIM Channel 4 selected

Return values
- **HAL:** status

HAL_TIMEx_OnePulseN_Start

Function Name **HAL_StatusTypeDef HAL_TIMEx_OnePulseN_Start (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Starts the TIM One Pulse signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIMEx_OnePulseN_Stop

Function Name **HAL_StatusTypeDef HAL_TIMEx_OnePulseN_Stop (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Stops the TIM One Pulse signal generation on the complementary output.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIMEx_OnePulseN_Start_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_OnePulseN_Start_IT (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Starts the TIM One Pulse signal generation in interrupt mode on the complementary channel.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** TIM Channel to be enabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIMEx_OnePulseN_Stop_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_OnePulseN_Stop_IT (TIM_HandleTypeDef * htim, uint32_t OutputChannel)**

Function Description Stops the TIM One Pulse signal generation in interrupt mode on the complementary channel.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **OutputChannel:** TIM Channel to be disabled. This parameter can be one of the following values:
  - TIM_CHANNEL_1: TIM Channel 1 selected
  - TIM_CHANNEL_2: TIM Channel 2 selected

Return values
- **HAL:** status

HAL_TIMEx_ConfigCommutationEvent

Function Name **HAL_StatusTypeDef HAL_TIMEx_ConfigCommutationEvent (TIM_HandleTypeDef * htim, uint32_t InputTrigger, uint32_t CommutationSource)**

Function Description Configure the TIM commutation event sequence.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **InputTrigger:** the Internal trigger corresponding to the Timer Interfacing with the Hall sensor. This parameter can be one of the following values:
  - TIM_TS_ITR0: Internal trigger 0 selected
  - TIM_TS_ITR1: Internal trigger 1 selected
  - TIM_TS_ITR2: Internal trigger 2 selected
  - TIM_TS_ITR3: Internal trigger 3 selected
  - TIM_TS_NONE: No trigger is needed
- **CommutationSource:** the Commutation Event source. This parameter can be one of the following values:
  - TIM_COMMUTATION_TRGI: Commutation source is the TRGI of the Interface Timer
  - TIM_COMMUTATION_SOFTWARE: Commutation source is set by software using the COMG bit

Return values
- **HAL:** status

Notes
- This function is mandatory to use the commutation event in order to update the configuration at each commutation detection on the TRGI input of the Timer, the typical use of this feature is with the use of another Timer(interface Timer) configured in Hall sensor interface, this interface Timer will generate the commutation at its TRGO output (connected to Timer used in this function) each time the TI1 of the Interface Timer detect a commutation at its input TI1.

HAL_TIMEx_ConfigCommutationEvent_IT

Function Name **HAL_StatusTypeDef HAL_TIMEx_ConfigCommutationEvent_IT (TIM_HandleTypeDef * htim, uint32_t InputTrigger, uint32_t**

CommutationSource)

Function Description — Configure the TIM commutation event sequence with interrupt.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **InputTrigger:** the Internal trigger corresponding to the Timer Interfacing with the Hall sensor. This parameter can be one of the following values:
  - TIM_TS_ITR0: Internal trigger 0 selected
  - TIM_TS_ITR1: Internal trigger 1 selected
  - TIM_TS_ITR2: Internal trigger 2 selected
  - TIM_TS_ITR3: Internal trigger 3 selected
  - TIM_TS_NONE: No trigger is needed
- **CommutationSource:** the Commutation Event source. This parameter can be one of the following values:
  - TIM_COMMUTATION_TRGI: Commutation source is the TRGI of the Interface Timer
  - TIM_COMMUTATION_SOFTWARE: Commutation source is set by software using the COMG bit

Return values
- **HAL:** status

Notes
- This function is mandatory to use the commutation event in order to update the configuration at each commutation detection on the TRGI input of the Timer, the typical use of this feature is with the use of another Timer(interface Timer) configured in Hall sensor interface, this interface Timer will generate the commutation at its TRGO output (connected to Timer used in this function) each time the TI1 of the Interface Timer detect a commutation at its input TI1.

HAL_TIMEx_ConfigCommutationEvent_DMA

Function Name — **HAL_StatusTypeDef HAL_TIMEx_ConfigCommutationEvent_DMA (TIM_HandleTypeDef * htim, uint32_t InputTrigger, uint32_t CommutationSource)**

Function Description — Configure the TIM commutation event sequence with DMA.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **InputTrigger:** the Internal trigger corresponding to the Timer Interfacing with the Hall sensor. This parameter can be one of the following values:
  - TIM_TS_ITR0: Internal trigger 0 selected
  - TIM_TS_ITR1: Internal trigger 1 selected
  - TIM_TS_ITR2: Internal trigger 2 selected
  - TIM_TS_ITR3: Internal trigger 3 selected
  - TIM_TS_NONE: No trigger is needed
- **CommutationSource:** the Commutation Event source. This parameter can be one of the following values:
  - TIM_COMMUTATION_TRGI: Commutation source is the TRGI of the Interface Timer
  - TIM_COMMUTATION_SOFTWARE: Commutation

source is set by software using the COMG bit

Return values
- **HAL:** status

Notes
- This function is mandatory to use the commutation event in order to update the configuration at each commutation detection on the TRGI input of the Timer, the typical use of this feature is with the use of another Timer(interface Timer) configured in Hall sensor interface, this interface Timer will generate the commutation at its TRGO output (connected to Timer used in this function) each time the TI1 of the Interface Timer detect a commutation at its input TI1.
- : The user should configure the DMA in his own software, in This function only the COMDE bit is set

HAL_TIMEx_MasterConfigSynchronization

Function Name **HAL_StatusTypeDef HAL_TIMEx_MasterConfigSynchronization (TIM_HandleTypeDef * htim, TIM_MasterConfigTypeDef * sMasterConfig)**

Function Description Configures the TIM in master mode.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sMasterConfig:** pointer to a TIM_MasterConfigTypeDef structure that contains the selected trigger output (TRGO) and the Master/Slave mode.

Return values
- **HAL:** status

HAL_TIMEx_ConfigBreakDeadTime

Function Name **HAL_StatusTypeDef HAL_TIMEx_ConfigBreakDeadTime (TIM_HandleTypeDef * htim, TIM_BreakDeadTimeConfigTypeDef * sBreakDeadTimeConfig)**

Function Description Configures the Break feature, dead time, Lock level, OSSI/OSSR State and the AOE(automatic output enable).

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **sBreakDeadTimeConfig:** pointer to a TIM_ConfigBreakDeadConfig_TypeDef structure that contains the BDTR Register configuration information for the TIM peripheral.

Return values
- **HAL:** status

HAL_TIMEx_RemapConfig

Function Name **HAL_StatusTypeDef HAL_TIMEx_RemapConfig (TIM_HandleTypeDef * htim, uint32_t Remap)**

Function Description Configures the TIM2, TIM5 and TIM11 Remapping input capabilities.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that

contains the configuration information for TIM module.

- **Remap:** specifies the TIM input remapping source. This parameter can be one of the following values:
  - TIM_TIM2_TIM8_TRGO: TIM2 ITR1 input is connected to TIM8 Trigger output(default)
  - TIM_TIM2_ETH_PTP: TIM2 ITR1 input is connected to ETH PTP trigger output.
  - TIM_TIM2_USBFS_SOF: TIM2 ITR1 input is connected to USB FS SOF.
  - TIM_TIM2_USBHS_SOF: TIM2 ITR1 input is connected to USB HS SOF.
  - TIM_TIM5_GPIO: TIM5 CH4 input is connected to dedicated Timer pin(default)
  - TIM_TIM5_LSI: TIM5 CH4 input is connected to LSI clock.
  - TIM_TIM5_LSE: TIM5 CH4 input is connected to LSE clock.
  - TIM_TIM5_RTC: TIM5 CH4 input is connected to RTC Output event.
  - TIM_TIM11_GPIO: TIM11 CH4 input is connected to dedicated Timer pin(default)
  - TIM_TIM11_HSE: TIM11 CH4 input is connected to HSE_RTC clock (HSE divided by a programmable prescaler)

Return values

- **HAL:** status

HAL_TIMEx_CommutationCallback

Function Name **void HAL_TIMEx_CommutationCallback (TIM_HandleTypeDef * htim)**

Function Description Hall commutation changed callback in non blocking mode.

Parameters

- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values

- **None**

HAL_TIMEx_BreakCallback

Function Name **void HAL_TIMEx_BreakCallback (TIM_HandleTypeDef * htim)**

Function Description Hall Break detection callback in non blocking mode.

Parameters

- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values

- **None**

TIMEx_DMACommutationCplt

Function Name **void TIMEx_DMACommutationCplt (DMA_HandleTypeDef * hdma)**

Function Description TIM DMA Commutation callback.

Parameters

- **hdma:** pointer to a DMA_HandleTypeDef structure that

contains the configuration information for the specified DMA module.

Return values
- **None**

HAL_TIMEx_HallSensor_GetState

Function Name **HAL_TIM_StateTypeDef HAL_TIMEx_HallSensor_GetState (TIM_HandleTypeDef * htim)**

Function Description Return the TIM Hall Sensor interface state.

Parameters
- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.

Return values
- **HAL:** state

63.3 TIMEx Firmware driver defines

63.3.1 TIMEx

TIM Remap

TIM_TIM2_TIM8_TRGO

TIM_TIM2_ETH_PTP

TIM_TIM2_USBFS_SOF

TIM_TIM2_USBHS_SOF

TIM_TIM5_GPIO

TIM_TIM5_LSI

TIM_TIM5_LSE

TIM_TIM5_RTC

TIM_TIM11_GPIO

TIM_TIM11_HSE

64 HAL UART Generic Driver

64.1 UART Firmware driver registers structures

64.1.1 UART_InitTypeDef

Data Fields

- ***uint32_t BaudRate***
- ***uint32_t WordLength***
- ***uint32_t StopBits***
- ***uint32_t Parity***
- ***uint32_t Mode***
- ***uint32_t HwFlowCtl***
- ***uint32_t OverSampling***

Field Documentation

- ***uint32_t UART_InitTypeDef::BaudRate***
  This member configures the UART communication baud rate. The baud rate is computed using the following formula:IntegerDivider = ((PCLKx) / (8 * (OVR8+1) * (huart->Init.BaudRate)))FractionalDivider = ((IntegerDivider - ((uint32_t) IntegerDivider)) * 8 * (OVR8+1)) + 0.5 Where OVR8 is the "oversampling by 8 mode" configuration bit in the CR1 register.
- ***uint32_t UART_InitTypeDef::WordLength***
  Specifies the number of data bits transmitted or received in a frame. This parameter can be a value of ***UART_Word_Length***
- ***uint32_t UART_InitTypeDef::StopBits***
  Specifies the number of stop bits transmitted. This parameter can be a value of ***UART_Stop_Bits***
- ***uint32_t UART_InitTypeDef::Parity***
  Specifies the parity mode. This parameter can be a value of ***UART_Parity***
  Note: When parity is enabled, the computed parity is inserted at the MSB position of the transmitted data (9th bit when the word length is set to 9 data bits; 8th bit when the word length is set to 8 data bits).
- ***uint32_t UART_InitTypeDef::Mode***
  Specifies whether the Receive or Transmit mode is enabled or disabled. This parameter can be a value of ***UART_Mode***
- ***uint32_t UART_InitTypeDef::HwFlowCtl***
  Specifies whether the hardware flow control mode is enabled or disabled. This parameter can be a value of ***UART_Hardware_Flow_Control***
- ***uint32_t UART_InitTypeDef::OverSampling***
  Specifies whether the Over sampling 8 is enabled or disabled, to achieve higher speed (up to fPCLK/8). This parameter can be a value of ***UART_Over_Sampling***

64.1.2 UART_HandleTypeDef

Data Fields

- ***USART_TypeDef * Instance***
- ***UART_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferSize***
- ***uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***uint16_t RxXferSize***
- ***uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_UART_StateTypeDef gState***
- ***__IO HAL_UART_StateTypeDef RxState***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***USART_TypeDef* UART_HandleTypeDef::Instance***
  UART registers base address
- ***UART_InitTypeDef UART_HandleTypeDef::Init***
  UART communication parameters
- ***uint8_t* UART_HandleTypeDef::pTxBuffPtr***
  Pointer to UART Tx transfer Buffer
- ***uint16_t UART_HandleTypeDef::TxXferSize***
  UART Tx Transfer size
- ***uint16_t UART_HandleTypeDef::TxXferCount***
  UART Tx Transfer Counter
- ***uint8_t* UART_HandleTypeDef::pRxBuffPtr***
  Pointer to UART Rx transfer Buffer
- ***uint16_t UART_HandleTypeDef::RxXferSize***
  UART Rx Transfer size
- ***uint16_t UART_HandleTypeDef::RxXferCount***
  UART Rx Transfer Counter
- ***DMA_HandleTypeDef* UART_HandleTypeDef::hdmatx***
  UART Tx DMA Handle parameters
- ***DMA_HandleTypeDef* UART_HandleTypeDef::hdmarx***
  UART Rx DMA Handle parameters
- ***HAL_LockTypeDef UART_HandleTypeDef::Lock***
  Locking object
- ***__IO HAL_UART_StateTypeDef UART_HandleTypeDef::gState***
  UART state information related to global Handle management and also related to Tx operations. This parameter can be a value of **HAL_UART_StateTypeDef**
- ***__IO HAL_UART_StateTypeDef UART_HandleTypeDef::RxState***
  UART state information related to Rx operations. This parameter can be a value of **HAL_UART_StateTypeDef**
- ***__IO uint32_t UART_HandleTypeDef::ErrorCode***
  UART Error code

64.2 UART Firmware driver API description

64.2.1 How to use this driver

The UART HAL driver can be used as follows:

1. Declare a UART_HandleTypeDef handle structure.
2. Initialize the UART low level resources by implementing the HAL_UART_MspInit() API:
   a. Enable the USARTx interface clock.
   b. UART pins configuration:
      – Enable the clock for the UART GPIOs.
      – Configure these UART pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_UART_Transmit_IT() and HAL_UART_Receive_IT() APIs):
      – Configure the USARTx interrupt priority.
      – Enable the NVIC USART IRQ handle.
   d. DMA Configuration if you need to use DMA process (HAL_UART_Transmit_DMA() and HAL_UART_Receive_DMA() APIs):
      – Declare a DMA handle structure for the Tx/Rx stream.
      – Enable the DMAx interface clock.
      – Configure the declared DMA handle structure with the required Tx/Rx parameters.
      – Configure the DMA Tx/Rx Stream.
      – Associate the initialized DMA handle to the UART DMA Tx/Rx handle.
      – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.
3. Program the Baud Rate, Word Length, Stop Bit, Parity, Hardware flow control and Mode(Receiver/Transmitter) in the Init structure.
4. For the UART asynchronous mode, initialize the UART registers by calling the HAL_UART_Init() API.
5. For the UART Half duplex mode, initialize the UART registers by calling the HAL_HalfDuplex_Init() API.
6. For the LIN mode, initialize the UART registers by calling the HAL_LIN_Init() API.
7. For the Multi-Processor mode, initialize the UART registers by calling the HAL_MultiProcessor_Init() API.

The specific UART interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_UART_ENABLE_IT() and __HAL_UART_DISABLE_IT() inside the transmit and receive process.

These APIs (HAL_UART_Init() and HAL_HalfDuplex_Init()) configure also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customized HAL_UART_MspInit() API.

Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_UART_Transmit()
- Receive an amount of data in blocking mode using HAL_UART_Receive()

Interrupt mode IO operation

- Send an amount of data in non blocking mode using HAL_UART_Transmit_IT()
- At transmission end of transfer HAL_UART_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_TxCpltCallback
- Receive an amount of data in non blocking mode using HAL_UART_Receive_IT()
- At reception end of transfer HAL_UART_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_RxCpltCallback
- In case of transfer Error, HAL_UART_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_UART_ErrorCallback

DMA mode IO operation

- Send an amount of data in non blocking mode (DMA) using HAL_UART_Transmit_DMA()
- At transmission end of half transfer HAL_UART_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_TxHalfCpltCallback
- At transmission end of transfer HAL_UART_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_TxCpltCallback
- Receive an amount of data in non blocking mode (DMA) using HAL_UART_Receive_DMA()
- At reception end of half transfer HAL_UART_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_RxHalfCpltCallback
- At reception end of transfer HAL_UART_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_UART_RxCpltCallback
- In case of transfer Error, HAL_UART_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_UART_ErrorCallback
- Pause the DMA Transfer using HAL_UART_DMAPause()
- Resume the DMA Transfer using HAL_UART_DMAResume()
- Stop the DMA Transfer using HAL_UART_DMAStop()

UART HAL driver macros list

Below the list of most used macros in UART HAL driver.

- __HAL_UART_ENABLE: Enable the UART peripheral
- __HAL_UART_DISABLE: Disable the UART peripheral
- __HAL_UART_GET_FLAG : Check whether the specified UART flag is set or not
- __HAL_UART_CLEAR_FLAG : Clear the specified UART pending flag
- __HAL_UART_ENABLE_IT: Enable the specified UART interrupt
- __HAL_UART_DISABLE_IT: Disable the specified UART interrupt

- __HAL_UART_GET_IT_SOURCE: Check whether the specified UART interrupt has occurred or not

You can refer to the UART HAL driver header file for more useful macros

64.2.2 Initialization and Configuration functions

This subsection provides a set of functions allowing to initialize the USARTx or the UARTy in asynchronous mode.

- For the asynchronous mode only these parameters can be configured:
  - Baud Rate
  - Word Length
  - Stop Bit
  - Parity: If the parity is enabled, then the MSB bit of the data written in the data register is transmitted but is changed by the parity bit. Depending on the frame length defined by the M bit (8-bits or 9-bits), please refer to Reference manual for possible UART frame formats.
  - Hardware flow control
  - Receiver/transmitter modes
  - Over Sampling Method

The HAL_UART_Init(), HAL_HalfDuplex_Init(), HAL_LIN_Init() and HAL_MultiProcessor_Init() APIs follow respectively the UART asynchronous, UART Half duplex, LIN and Multi-Processor configuration procedures (details for the procedures are available in reference manual (RM0329)).

This section contains the following APIs:

- ***HAL_UART_Init()***
- ***HAL_HalfDuplex_Init()***
- ***HAL_LIN_Init()***
- ***HAL_MultiProcessor_Init()***
- ***HAL_UART_DeInit()***
- ***HAL_UART_MspInit()***
- ***HAL_UART_MspDeInit()***

64.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the UART asynchronous and Half duplex data transfers.

1. There are two modes of transfer:
   - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.
   - Non blocking mode: The communication is performed using Interrupts or DMA, these APIs return the HAL status. The end of the data processing will be indicated through the dedicated UART IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_UART_TxCpltCallback(), HAL_UART_RxCpltCallback() user callbacks will be executed respectively at the end of the transmit or receive process. The HAL_UART_ErrorCallback() user callback will be executed when a communication error is detected.
2. Blocking mode APIs are:

  - HAL_UART_Transmit()
  - HAL_UART_Receive()
3. Non Blocking mode APIs with Interrupt are:
  - HAL_UART_Transmit_IT()
  - HAL_UART_Receive_IT()
  - HAL_UART_IRQHandler()
4. Non Blocking mode functions with DMA are:
  - HAL_UART_Transmit_DMA()
  - HAL_UART_Receive_DMA()
5. A set of Transfer Complete Callbacks are provided in non blocking mode:
  - HAL_UART_TxCpltCallback()
  - HAL_UART_RxCpltCallback()
  - HAL_UART_ErrorCallback()

In the Half duplex communication, it is forbidden to run the transmit and receive process in parallel, the UART state HAL_UART_STATE_BUSY_TX_RX can't be useful.

This section contains the following APIs:

- ***HAL_UART_Transmit()***
- ***HAL_UART_Receive()***
- ***HAL_UART_Transmit_IT()***
- ***HAL_UART_Receive_IT()***
- ***HAL_UART_Transmit_DMA()***
- ***HAL_UART_Receive_DMA()***
- ***HAL_UART_DMAPause()***
- ***HAL_UART_DMAResume()***
- ***HAL_UART_DMAStop()***
- ***HAL_UART_IRQHandler()***
- ***HAL_UART_TxCpltCallback()***
- ***HAL_UART_TxHalfCpltCallback()***
- ***HAL_UART_RxCpltCallback()***
- ***HAL_UART_RxHalfCpltCallback()***
- ***HAL_UART_ErrorCallback()***

64.2.4 Peripheral Control functions

This subsection provides a set of functions allowing to control the UART:

- HAL_LIN_SendBreak() API can be helpful to transmit the break character.
- HAL_MultiProcessor_EnterMuteMode() API can be helpful to enter the UART in mute mode.
- HAL_MultiProcessor_ExitMuteMode() API can be helpful to exit the UART mute mode by software.

This section contains the following APIs:

- ***HAL_LIN_SendBreak()***
- ***HAL_MultiProcessor_EnterMuteMode()***
- ***HAL_MultiProcessor_ExitMuteMode()***
- ***HAL_HalfDuplex_EnableTransmitter()***
- ***HAL_HalfDuplex_EnableReceiver()***

64.2.5 Peripheral State and Errors functions

This subsection provides a set of functions allowing to return the State of UART communication process, return Peripheral Errors occurred during communication process

- HAL_UART_GetState() API can be helpful to check in run-time the state of the UART peripheral.
- HAL_UART_GetError() check in run-time errors that could be occurred during communication.

This section contains the following APIs:

- ***HAL_UART_GetState()***
- ***HAL_UART_GetError()***

64.2.6 Detailed description of functions

HAL_UART_Init

Function Name	**HAL_StatusTypeDef HAL_UART_Init (UART_HandleTypeDef * huart)**
Function Description	Initializes the UART mode according to the specified parameters in the UART_InitTypeDef and create the associated handle.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
Return values	• **HAL:** status

HAL_HalfDuplex_Init

Function Name	**HAL_StatusTypeDef HAL_HalfDuplex_Init (UART_HandleTypeDef * huart)**
Function Description	Initializes the half-duplex mode according to the specified parameters in the UART_InitTypeDef and create the associated handle.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
Return values	• **HAL:** status

HAL_LIN_Init

Function Name	**HAL_StatusTypeDef HAL_LIN_Init (UART_HandleTypeDef * huart, uint32_t BreakDetectLength)**
Function Description	Initializes the LIN mode according to the specified parameters in the UART_InitTypeDef and create the associated handle.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module. • **BreakDetectLength:** Specifies the LIN break detection length. This parameter can be one of the following values: – UART_LINBREAKDETECTLENGTH_10B: 10-bit break

detection
– UART_LINBREAKDETECTLENGTH_11B: 11-bit break detection

Return values
- **HAL:** status

HAL_MultiProcessor_Init

Function Name **HAL_StatusTypeDef HAL_MultiProcessor_Init (UART_HandleTypeDef * huart, uint8_t Address, uint32_t WakeUpMethod)**

Function Description Initializes the Multi-Processor mode according to the specified parameters in the UART_InitTypeDef and create the associated handle.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
- **Address:** USART address
- **WakeUpMethod:** specifies the USART wake-up method. This parameter can be one of the following values:
  – UART_WAKEUPMETHOD_IDLELINE: Wake-up by an idle line detection
  – UART_WAKEUPMETHOD_ADDRESSMARK: Wake-up by an address mark

Return values
- **HAL:** status

HAL_UART_DeInit

Function Name **HAL_StatusTypeDef HAL_UART_DeInit (UART_HandleTypeDef * huart)**

Function Description DeInitializes the UART peripheral.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_UART_MspInit

Function Name **void HAL_UART_MspInit (UART_HandleTypeDef * huart)**

Function Description UART MSP Init.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_MspDeInit

Function Name **void HAL_UART_MspDeInit (UART_HandleTypeDef * huart)**

Function Description UART MSP DeInit.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_Transmit

Function Name **HAL_StatusTypeDef HAL_UART_Transmit (UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Sends an amount of data in blocking mode.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_UART_Receive

Function Name **HAL_StatusTypeDef HAL_UART_Receive (UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size, uint32_t Timeout)**

Function Description Receives an amount of data in blocking mode.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be received
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_UART_Transmit_IT

Function Name **HAL_StatusTypeDef HAL_UART_Transmit_IT (UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size)**

Function Description Sends an amount of data in non blocking mode.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
- **pData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_UART_Receive_IT

Function Name **HAL_StatusTypeDef HAL_UART_Receive_IT**

(UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size)

Function Description	Receives an amount of data in non blocking mode.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status

HAL_UART_Transmit_DMA

Function Name	**HAL_StatusTypeDef HAL_UART_Transmit_DMA (UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size)**
Function Description	Sends an amount of data in non blocking mode.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status

HAL_UART_Receive_DMA

Function Name	**HAL_StatusTypeDef HAL_UART_Receive_DMA (UART_HandleTypeDef * huart, uint8_t * pData, uint16_t Size)**
Function Description	Receives an amount of data in non blocking mode.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module. • **pData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status
Notes	• When the UART parity is enabled (PCE = 1) the data received contain the parity bit.

HAL_UART_DMAPause

Function Name	**HAL_StatusTypeDef HAL_UART_DMAPause (UART_HandleTypeDef * huart)**
Function Description	Pauses the DMA Transfer.
Parameters	• **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.
Return values	• **HAL:** status

HAL_UART_DMAResume

Function Name **HAL_StatusTypeDef HAL_UART_DMAResume (UART_HandleTypeDef * huart)**

Function Description Resumes the DMA Transfer.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_UART_DMAStop

Function Name **HAL_StatusTypeDef HAL_UART_DMAStop (UART_HandleTypeDef * huart)**

Function Description Stops the DMA Transfer.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_UART_IRQHandler

Function Name **void HAL_UART_IRQHandler (UART_HandleTypeDef * huart)**

Function Description This function handles UART interrupt request.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_TxCpltCallback

Function Name **void HAL_UART_TxCpltCallback (UART_HandleTypeDef * huart)**

Function Description Tx Transfer completed callbacks.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_TxHalfCpltCallback

Function Name **void HAL_UART_TxHalfCpltCallback (UART_HandleTypeDef * huart)**

Function Description Tx Half Transfer completed callbacks.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_RxCpltCallback

Function Name **void HAL_UART_RxCpltCallback (UART_HandleTypeDef * huart)**

Function Description Rx Transfer completed callbacks.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_RxHalfCpltCallback

Function Name **void HAL_UART_RxHalfCpltCallback (UART_HandleTypeDef * huart)**

Function Description Rx Half Transfer completed callbacks.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_UART_ErrorCallback

Function Name **void HAL_UART_ErrorCallback (UART_HandleTypeDef * huart)**

Function Description UART error callbacks.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **None**

HAL_LIN_SendBreak

Function Name **HAL_StatusTypeDef HAL_LIN_SendBreak (UART_HandleTypeDef * huart)**

Function Description Transmits break characters.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_MultiProcessor_EnterMuteMode

Function Name **HAL_StatusTypeDef HAL_MultiProcessor_EnterMuteMode (UART_HandleTypeDef * huart)**

Function Description Enters the UART in mute mode.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_MultiProcessor_ExitMuteMode

Function Name **HAL_StatusTypeDef HAL_MultiProcessor_ExitMuteMode (UART_HandleTypeDef * huart)**

Function Description Exits the UART mute mode: wake up software.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_HalfDuplex_EnableTransmitter

Function Name **HAL_StatusTypeDef HAL_HalfDuplex_EnableTransmitter (UART_HandleTypeDef * huart)**

Function Description Enables the UART transmitter and disables the UART receiver.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_HalfDuplex_EnableReceiver

Function Name **HAL_StatusTypeDef HAL_HalfDuplex_EnableReceiver (UART_HandleTypeDef * huart)**

Function Description Enables the UART receiver and disables the UART transmitter.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** status

HAL_UART_GetState

Function Name **HAL_UART_StateTypeDef HAL_UART_GetState (UART_HandleTypeDef * huart)**

Function Description Returns the UART state.

Parameters
- **huart:** pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART module.

Return values
- **HAL:** state

HAL_UART_GetError

Function Name	**uint32_t HAL_UART_GetError (UART_HandleTypeDef * huart)**
Function Description	Return the UART error code.
Parameters	• **huart:** : pointer to a UART_HandleTypeDef structure that contains the configuration information for the specified UART.
Return values	• **UART:** Error Code

64.3 UART Firmware driver defines

64.3.1 UART

UART Error Code

HAL_UART_ERROR_NONE	No error
HAL_UART_ERROR_PE	Parity error
HAL_UART_ERROR_NE	Noise error
HAL_UART_ERROR_FE	Frame error
HAL_UART_ERROR_ORE	Overrun error
HAL_UART_ERROR_DMA	DMA transfer error

UART Exported Macros

__HAL_UART_RESET_HANDLE_STATE

Description:

- Reset UART handle gstate & RxState.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_UART_FLUSH_DRREGISTER

Description:

- Flushes the UART DR register.

Parameters:

- __HANDLE__: specifies the UART Handle.

__HAL_UART_GET_FLAG

Description:

- Checks whether the specified UART flag is set or not.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be

UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - UART_FLAG_CTS: CTS Change flag (not available for UART4 and UART5)
  - UART_FLAG_LBD: LIN Break detection flag
  - UART_FLAG_TXE: Transmit data register empty flag
  - UART_FLAG_TC: Transmission Complete flag
  - UART_FLAG_RXNE: Receive data register not empty flag
  - UART_FLAG_IDLE: Idle Line detection flag
  - UART_FLAG_ORE: Overrun Error flag
  - UART_FLAG_NE: Noise Error flag
  - UART_FLAG_FE: Framing Error flag
  - UART_FLAG_PE: Parity Error flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_UART_CLEAR_FLAG

Description:

- Clears the specified UART pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __FLAG__: specifies the flag to check. This parameter can be any combination of the following values:
  - UART_FLAG_CTS: CTS Change flag (not available for UART4 and UART5).
  - UART_FLAG_LBD: LIN Break detection flag.
  - UART_FLAG_TC: Transmission Complete flag.
  - UART_FLAG_RXNE: Receive data register not empty flag.

Return value:

- None

Notes:

- PE (Parity error), FE (Framing error), NE (Noise error), ORE (Overrun error) and IDLE (Idle line detected) flags are cleared by software sequence: a read operation to USART_SR register followed by a read operation to USART_DR register. RXNE flag can be also cleared by a read to the USART_DR register. TC flag can be also cleared by software sequence: a read operation to USART_SR register followed by a write operation to USART_DR register. TXE flag is cleared only by a write to the USART_DR register.

__HAL_UART_CLEAR_PEFLAG

Description:

- Clear the UART PE pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_UART_CLEAR_FEFLAG

Description:

- Clear the UART FE pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_UART_CLEAR_NEFLAG

Description:

- Clear the UART NE pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or

8 to select the USART or UART peripheral.

Return value:

- None

__HAL_UART_CLEAR_OREFLAG

Description:

- Clear the UART ORE pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

__HAL_UART_CLEAR_IDLEFLAG

Description:

- Clear the UART IDLE pending flag.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

Return value:

- None

UART_IT_MASK

Description:

- Enable the specified UART interrupt.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __INTERRUPT__: specifies the UART interrupt source to enable. This parameter can be one of the following values:
  - UART_IT_CTS: CTS change interrupt
  - UART_IT_LBD: LIN Break detection interrupt
  - UART_IT_TXE: Transmit Data Register empty interrupt
  - UART_IT_TC: Transmission complete interrupt
  - UART_IT_RXNE: Receive Data

register not empty interrupt
- UART_IT_IDLE: Idle line detection interrupt
- UART_IT_PE: Parity Error interrupt
- UART_IT_ERR: Error interrupt(Frame error, noise error, overrun error)

Return value:

- None

__HAL_UART_ENABLE_IT

__HAL_UART_DISABLE_IT

Description:

- Disable the specified UART interrupt.

Parameters:

- __HANDLE__: specifies the UART Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.
- __INTERRUPT__: specifies the UART interrupt source to disable. This parameter can be one of the following values:
  - UART_IT_CTS: CTS change interrupt
  - UART_IT_LBD: LIN Break detection interrupt
  - UART_IT_TXE: Transmit Data Register empty interrupt
  - UART_IT_TC: Transmission complete interrupt
  - UART_IT_RXNE: Receive Data register not empty interrupt
  - UART_IT_IDLE: Idle line detection interrupt
  - UART_IT_PE: Parity Error interrupt
  - UART_IT_ERR: Error interrupt(Frame error, noise error, overrun error)

Return value:

- None

__HAL_UART_GET_IT_SOURCE

Description:

- Checks whether the specified UART interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the UART

Handle. This parameter can be UARTx where x: 1, 2, 3, 4, 5, 6, 7 or 8 to select the USART or UART peripheral.

- __IT__: specifies the UART interrupt source to check. This parameter can be one of the following values:
  - UART_IT_CTS: CTS change interrupt (not available for UART4 and UART5)
  - UART_IT_LBD: LIN Break detection interrupt
  - UART_IT_TXE: Transmit Data Register empty interrupt
  - UART_IT_TC: Transmission complete interrupt
  - UART_IT_RXNE: Receive Data register not empty interrupt
  - UART_IT_IDLE: Idle line detection interrupt
  - USART_IT_ERR: Error interrupt

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_UART_HWCONTROL_CTS_ENABLE

Description:

- Enable CTS flow control This macro allows to enable CTS hardware flow control for a given UART instance, without need to call

Parameters:

- __HANDLE__: specifies the UART Handle. The Handle Instance can be USART1, USART2 or LPUART.

Return value:

- None

Notes:

- As macro is expected to be used for modifying CTS Hw flow control feature activation, without need for USART instance Deinit/Init, following conditions for macro call should be fulfilled : UART instance should have already been initialised (through call of HAL_UART_Init())macro could only be called when corresponding UART instance is disabled (i.e __HAL_UART_DISABLE(__HANDLE__)) and should be followed by an Enable macro (i.e __HAL_UART_ENABLE(__HANDLE

__)).

__HAL_UART_HWCONTROL_CTS_DISABLE

Description:

- Disable CTS flow control This macro allows to disable CTS hardware flow control for a given UART instance, without need to call

Parameters:

- __HANDLE__: specifies the UART Handle. The Handle Instance can be USART1, USART2 or LPUART.

Return value:

- None

Notes:

- As macro is expected to be used for modifying CTS Hw flow control feature activation, without need for USART instance Deinit/Init, following conditions for macro call should be fulfilled : UART instance should have already been initialised (through call of HAL_UART_Init())macro could only be called when corresponding UART instance is disabled (i.e __HAL_UART_DISABLE(__HANDLE__)) and should be followed by an Enable macro (i.e __HAL_UART_ENABLE(__HANDLE__)).

__HAL_UART_HWCONTROL_RTS_ENABLE

Description:

- Enable RTS flow control This macro allows to enable RTS hardware flow control for a given UART instance, without need to call

Parameters:

- __HANDLE__: specifies the UART Handle. The Handle Instance can be USART1, USART2 or LPUART.

Return value:

- None

Notes:

- As macro is expected to be used for modifying RTS Hw flow control feature activation, without need for USART instance Deinit/Init, following conditions for macro call should be fulfilled : UART instance should have already been initialised (through call

of HAL_UART_Init())macro could only be called when corresponding UART instance is disabled (i.e __HAL_UART_DISABLE(__HANDLE__)) and should be followed by an Enable macro (i.e __HAL_UART_ENABLE(__HANDLE__)).

__HAL_UART_HWCONTROL_RTS_DISABLE

Description:

- Disable RTS flow control This macro allows to disable RTS hardware flow control for a given UART instance, without need to call

Parameters:

- __HANDLE__: specifies the UART Handle. The Handle Instance can be USART1, USART2 or LPUART.

Return value:

- None

Notes:

- As macro is expected to be used for modifying RTS Hw flow control feature activation, without need for USART instance Deinit/Init, following conditions for macro call should be fulfilled : UART instance should have already been initialised (through call of HAL_UART_Init())macro could only be called when corresponding UART instance is disabled (i.e __HAL_UART_DISABLE(__HANDLE__)) and should be followed by an Enable macro (i.e __HAL_UART_ENABLE(__HANDLE__)).

__HAL_UART_ONE_BIT_SAMPLE_ENABLE

Description:

- macros to enables the UART's one bit sample method

Parameters:

- __HANDLE__: specifies the UART Handle.

Return value:

- None

__HAL_UART_ONE_BIT_SAMPLE_DISABLE

Description:

- macros to disables the UART's one bit sample method

Parameters:

- __HANDLE__: specifies the UART Handle.

Return value:

- None

__HAL_UART_ENABLE

Description:

- Enable UART.

Parameters:

- __HANDLE__: specifies the UART Handle.

Return value:

- None

__HAL_UART_DISABLE

Description:

- Disable UART.

Parameters:

- __HANDLE__: specifies the UART Handle.

Return value:

- None

UART FLags

UART_FLAG_CTS

UART_FLAG_LBD

UART_FLAG_TXE

UART_FLAG_TC

UART_FLAG_RXNE

UART_FLAG_IDLE

UART_FLAG_ORE

UART_FLAG_NE

UART_FLAG_FE

UART_FLAG_PE

UART Hardware Flow Control

UART_HWCONTROL_NONE

UART_HWCONTROL_RTS

UART_HWCONTROL_CTS

UART_HWCONTROL_RTS_CTS

UART Interrupt Definitions

UART_IT_PE

UART_IT_TXE

UART_IT_TC

UART_IT_RXNE

UART_IT_IDLE

UART_IT_LBD

UART_IT_CTS

UART_IT_ERR

UART LIN Break Detection Length

UART_LINBREAKDETECTLENGTH_10B

UART_LINBREAKDETECTLENGTH_11B

UART Transfer Mode

UART_MODE_RX

UART_MODE_TX

UART_MODE_TX_RX

UART Over Sampling

UART_OVERSAMPLING_16

UART_OVERSAMPLING_8

UART Parity

UART_PARITY_NONE

UART_PARITY_EVEN

UART_PARITY_ODD

UART State

UART_STATE_DISABLE

UART_STATE_ENABLE

UART Number of Stop Bits

UART_STOPBITS_1

UART_STOPBITS_2

UART Wakeup Functions

UART_WAKEUPMETHOD_IDLELINE

UART_WAKEUPMETHOD_ADDRESSMARK

UART Word Length

UART_WORDLENGTH_8B

UART_WORDLENGTH_9B

65 HAL USART Generic Driver

65.1 USART Firmware driver registers structures

65.1.1 USART_InitTypeDef

Data Fields

- ***uint32_t BaudRate***
- ***uint32_t WordLength***
- ***uint32_t StopBits***
- ***uint32_t Parity***
- ***uint32_t Mode***
- ***uint32_t CLKPolarity***
- ***uint32_t CLKPhase***
- ***uint32_t CLKLastBit***

Field Documentation

- ***uint32_t USART_InitTypeDef::BaudRate***
  This member configures the Usart communication baud rate. The baud rate is computed using the following formula:IntegerDivider = ((PCLKx) / (8 * (husart->Init.BaudRate)))FractionalDivider = ((IntegerDivider - ((uint32_t) IntegerDivider)) * 8) + 0.5
- ***uint32_t USART_InitTypeDef::WordLength***
  Specifies the number of data bits transmitted or received in a frame. This parameter can be a value of ***USART_Word_Length***
- ***uint32_t USART_InitTypeDef::StopBits***
  Specifies the number of stop bits transmitted. This parameter can be a value of ***USART_Stop_Bits***
- ***uint32_t USART_InitTypeDef::Parity***
  Specifies the parity mode. This parameter can be a value of ***USART_Parity***
  Note: When parity is enabled, the computed parity is inserted at the MSB position of the transmitted data (9th bit when the word length is set to 9 data bits; 8th bit when the word length is set to 8 data bits).
- ***uint32_t USART_InitTypeDef::Mode***
  Specifies whether the Receive or Transmit mode is enabled or disabled. This parameter can be a value of ***USART_Mode***
- ***uint32_t USART_InitTypeDef::CLKPolarity***
  Specifies the steady state of the serial clock. This parameter can be a value of ***USART_Clock_Polarity***
- ***uint32_t USART_InitTypeDef::CLKPhase***
  Specifies the clock transition on which the bit capture is made. This parameter can be a value of ***USART_Clock_Phase***
- ***uint32_t USART_InitTypeDef::CLKLastBit***
  Specifies whether the clock pulse corresponding to the last transmitted data bit (MSB) has to be output on the SCLK pin in synchronous mode. This parameter can be a value of ***USART_Last_Bit***

65.1.2 USART_HandleTypeDef

Data Fields

- ***USART_TypeDef * Instance***
- ***USART_InitTypeDef Init***
- ***uint8_t * pTxBuffPtr***
- ***uint16_t TxXferSize***
- ***__IO uint16_t TxXferCount***
- ***uint8_t * pRxBuffPtr***
- ***uint16_t RxXferSize***
- ***__IO uint16_t RxXferCount***
- ***DMA_HandleTypeDef * hdmatx***
- ***DMA_HandleTypeDef * hdmarx***
- ***HAL_LockTypeDef Lock***
- ***__IO HAL_USART_StateTypeDef State***
- ***__IO uint32_t ErrorCode***

Field Documentation

- ***USART_TypeDef* USART_HandleTypeDef::Instance***
- ***USART_InitTypeDef USART_HandleTypeDef::Init***
- ***uint8_t* USART_HandleTypeDef::pTxBuffPtr***
- ***uint16_t USART_HandleTypeDef::TxXferSize***
- ***__IO uint16_t USART_HandleTypeDef::TxXferCount***
- ***uint8_t* USART_HandleTypeDef::pRxBuffPtr***
- ***uint16_t USART_HandleTypeDef::RxXferSize***
- ***__IO uint16_t USART_HandleTypeDef::RxXferCount***
- ***DMA_HandleTypeDef* USART_HandleTypeDef::hdmatx***
- ***DMA_HandleTypeDef* USART_HandleTypeDef::hdmarx***
- ***HAL_LockTypeDef USART_HandleTypeDef::Lock***
- ***__IO HAL_USART_StateTypeDef USART_HandleTypeDef::State***
- ***__IO uint32_t USART_HandleTypeDef::ErrorCode***

65.2 USART Firmware driver API description

65.2.1 How to use this driver

The USART HAL driver can be used as follows:

1. Declare a USART_HandleTypeDef handle structure.
2. Initialize the USART low level resources by implementing the HAL_USART_MspInit () API:
   a. Enable the USARTx interface clock.
   b. USART pins configuration:
      – Enable the clock for the USART GPIOs.
      – Configure these USART pins as alternate function pull-up.
   c. NVIC configuration if you need to use interrupt process (HAL_USART_Transmit_IT(), HAL_USART_Receive_IT() and HAL_USART_TransmitReceive_IT() APIs):
      – Configure the USARTx interrupt priority.

– Enable the NVIC USART IRQ handle.

d. DMA Configuration if you need to use DMA process (HAL_USART_Transmit_DMA() HAL_USART_Receive_IT() and HAL_USART_TransmitReceive_IT() APIs):
   – Declare a DMA handle structure for the Tx/Rx stream.
   – Enable the DMAx interface clock.
   – Configure the declared DMA handle structure with the required Tx/Rx parameters.
   – Configure the DMA Tx/Rx Stream.
   – Associate the initialized DMA handle to the USART DMA Tx/Rx handle.
   – Configure the priority and enable the NVIC for the transfer complete interrupt on the DMA Tx/Rx Stream.

3. Program the Baud Rate, Word Length, Stop Bit, Parity, Hardware flow control and Mode(Receiver/Transmitter) in the husart Init structure.
4. Initialize the USART registers by calling the HAL_USART_Init() API:
   – These APIs configures also the low level Hardware GPIO, CLOCK, CORTEX...etc) by calling the customized HAL_USART_MspInit(&husart) API. The specific USART interrupts (Transmission complete interrupt, RXNE interrupt and Error Interrupts) will be managed using the macros __HAL_USART_ENABLE_IT() and __HAL_USART_DISABLE_IT() inside the transmit and receive process.
5. Three operation modes are available within this driver :

Polling mode IO operation

- Send an amount of data in blocking mode using HAL_USART_Transmit()
- Receive an amount of data in blocking mode using HAL_USART_Receive()

Interrupt mode IO operation

- Send an amount of data in non blocking mode using HAL_USART_Transmit_IT()
- At transmission end of transfer HAL_USART_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_TxCpltCallback
- Receive an amount of data in non blocking mode using HAL_USART_Receive_IT()
- At reception end of transfer HAL_USART_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_RxCpltCallback
- In case of transfer Error, HAL_USART_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_USART_ErrorCallback

DMA mode IO operation

- Send an amount of data in non blocking mode (DMA) using HAL_USART_Transmit_DMA()
- At transmission end of half transfer HAL_USART_TxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_TxHalfCpltCallback
- At transmission end of transfer HAL_USART_TxCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_TxCpltCallback

- Receive an amount of data in non blocking mode (DMA) using HAL_USART_Receive_DMA()
- At reception end of half transfer HAL_USART_RxHalfCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_RxHalfCpltCallback
- At reception end of transfer HAL_USART_RxCpltCallback is executed and user can add his own code by customization of function pointer HAL_USART_RxCpltCallback
- In case of transfer Error, HAL_USART_ErrorCallback() function is executed and user can add his own code by customization of function pointer HAL_USART_ErrorCallback
- Pause the DMA Transfer using HAL_USART_DMAPause()
- Resume the DMA Transfer using HAL_USART_DMAResume()
- Stop the DMA Transfer using HAL_USART_DMAStop()

USART HAL driver macros list

Below the list of most used macros in USART HAL driver.

- __HAL_USART_ENABLE: Enable the USART peripheral
- __HAL_USART_DISABLE: Disable the USART peripheral
- __HAL_USART_GET_FLAG : Check whether the specified USART flag is set or not
- __HAL_USART_CLEAR_FLAG : Clear the specified USART pending flag
- __HAL_USART_ENABLE_IT: Enable the specified USART interrupt
- __HAL_USART_DISABLE_IT: Disable the specified USART interrupt

You can refer to the USART HAL driver header file for more useful macros

65.2.2 Initialization and Configuration functions

This subsection provides a set of functions allowing to initialize the USART in asynchronous and in synchronous modes.

- For the asynchronous mode only these parameters can be configured:
  - Baud Rate
  - Word Length
  - Stop Bit
  - Parity: If the parity is enabled, then the MSB bit of the data written in the data register is transmitted but is changed by the parity bit. Depending on the frame length defined by the M bit (8-bits or 9-bits), please refer to Reference manual for possible USART frame formats.
  - USART polarity
  - USART phase
  - USART LastBit
  - Receiver/transmitter modes

The HAL_USART_Init() function follows the USART synchronous configuration procedure (details for the procedure are available in reference manual (RM0329)).

This section contains the following APIs:

- ***HAL_USART_Init()***
- ***HAL_USART_DeInit()***
- ***HAL_USART_MspInit()***
- ***HAL_USART_MspDeInit()***

65.2.3 IO operation functions

This subsection provides a set of functions allowing to manage the USART synchronous data transfers.

The USART supports master mode only: it cannot receive or send data related to an input clock (SCLK is always an output).

1. There are two modes of transfer:
   - Blocking mode: The communication is performed in polling mode. The HAL status of all data processing is returned by the same function after finishing transfer.
   - No-Blocking mode: The communication is performed using Interrupts or DMA, These API's return the HAL status. The end of the data processing will be indicated through the dedicated USART IRQ when using Interrupt mode or the DMA IRQ when using DMA mode. The HAL_USART_TxCpltCallback(), HAL_USART_RxCpltCallback() and HAL_USART_TxRxCpltCallback() user callbacks will be executed respectively at the end of the transmit or Receive process The HAL_USART_ErrorCallback() user callback will be executed when a communication error is detected
2. Blocking mode APIs are :
   - HAL_USART_Transmit() in simplex mode
   - HAL_USART_Receive() in full duplex receive only
   - HAL_USART_TransmitReceive() in full duplex mode
3. Non Blocking mode APIs with Interrupt are :
   - HAL_USART_Transmit_IT()in simplex mode
   - HAL_USART_Receive_IT() in full duplex receive only
   - HAL_USART_TransmitReceive_IT() in full duplex mode
   - HAL_USART_IRQHandler()
4. Non Blocking mode functions with DMA are :
   - HAL_USART_Transmit_DMA()in simplex mode
   - HAL_USART_Receive_DMA() in full duplex receive only
   - HAL_USART_TransmitReceie_DMA() in full duplex mode
   - HAL_USART_DMAPause()
   - HAL_USART_DMAResume()
   - HAL_USART_DMAStop()
5. A set of Transfer Complete Callbacks are provided in non Blocking mode:
   - HAL_USART_TxHalfCpltCallback()
   - HAL_USART_TxCpltCallback()
   - HAL_USART_RxHalfCpltCallback()
   - HAL_USART_RxCpltCallback()
   - HAL_USART_ErrorCallback()
   - HAL_USART_TxRxCpltCallback()

This section contains the following APIs:

- ***HAL_USART_Transmit()***
- ***HAL_USART_Receive()***
- ***HAL_USART_TransmitReceive()***
- ***HAL_USART_Transmit_IT()***
- ***HAL_USART_Receive_IT()***
- ***HAL_USART_TransmitReceive_IT()***
- ***HAL_USART_Transmit_DMA()***
- ***HAL_USART_Receive_DMA()***
- ***HAL_USART_TransmitReceive_DMA()***
- ***HAL_USART_DMAPause()***

- ***HAL_USART_DMAResume()***
- ***HAL_USART_DMAStop()***
- ***HAL_USART_IRQHandler()***
- ***HAL_USART_TxCpltCallback()***
- ***HAL_USART_TxHalfCpltCallback()***
- ***HAL_USART_RxCpltCallback()***
- ***HAL_USART_RxHalfCpltCallback()***
- ***HAL_USART_TxRxCpltCallback()***
- ***HAL_USART_ErrorCallback()***

65.2.4 Peripheral State and Errors functions

This subsection provides a set of functions allowing to return the State of USART communication process, return Peripheral Errors occurred during communication process

- HAL_USART_GetState() API can be helpful to check in run-time the state of the USART peripheral.
- HAL_USART_GetError() check in run-time errors that could be occurred during communication.

This section contains the following APIs:

- ***HAL_USART_GetState()***
- ***HAL_USART_GetError()***

65.2.5 Detailed description of functions

HAL_USART_Init

Function Name	**HAL_StatusTypeDef HAL_USART_Init (USART_HandleTypeDef * husart)**
Function Description	Initializes the USART mode according to the specified parameters in the USART_InitTypeDef and create the associated handle.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **HAL:** status

HAL_USART_DeInit

Function Name	**HAL_StatusTypeDef HAL_USART_DeInit (USART_HandleTypeDef * husart)**
Function Description	DeInitializes the USART peripheral.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **HAL:** status

HAL_USART_MspInit

Function Name	**void HAL_USART_MspInit (USART_HandleTypeDef * husart)**
Function Description	USART MSP Init.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **None**

HAL_USART_MspDeInit

Function Name **void HAL_USART_MspDeInit (USART_HandleTypeDef * husart)**

Function Description USART MSP DeInit.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **None**

HAL_USART_Transmit

Function Name **HAL_StatusTypeDef HAL_USART_Transmit (USART_HandleTypeDef * husart, uint8_t * pTxData, uint16_t Size, uint32_t Timeout)**

Function Description Simplex Send an amount of data in blocking mode.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
- **pTxData:** Pointer to data buffer
- **Size:** Amount of data to be sent
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_USART_Receive

Function Name **HAL_StatusTypeDef HAL_USART_Receive (USART_HandleTypeDef * husart, uint8_t * pRxData, uint16_t Size, uint32_t Timeout)**

Function Description Full-Duplex Receive an amount of data in blocking mode.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
- **pRxData:** Pointer to data buffer
- **Size:** Amount of data to be received
- **Timeout:** Timeout duration

Return values
- **HAL:** status

HAL_USART_TransmitReceive

Function Name **HAL_StatusTypeDef HAL_USART_TransmitReceive (USART_HandleTypeDef * husart, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size, uint32_t Timeout)**

Function Description	Full-Duplex Send receive an amount of data in full-duplex mode (blocking mode).
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module. • **pTxData:** Pointer to data transmitted buffer • **pRxData:** Pointer to data received buffer • **Size:** Amount of data to be sent • **Timeout:** Timeout duration
Return values	• **HAL:** status

HAL_USART_Transmit_IT

Function Name	**HAL_StatusTypeDef HAL_USART_Transmit_IT (USART_HandleTypeDef * husart, uint8_t * pTxData, uint16_t Size)**
Function Description	Simplex Send an amount of data in non-blocking mode.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module. • **pTxData:** Pointer to data buffer • **Size:** Amount of data to be sent
Return values	• **HAL:** status
Notes	• The USART errors are not managed to avoid the overrun error.

HAL_USART_Receive_IT

Function Name	**HAL_StatusTypeDef HAL_USART_Receive_IT (USART_HandleTypeDef * husart, uint8_t * pRxData, uint16_t Size)**
Function Description	Simplex Receive an amount of data in non-blocking mode.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module. • **pRxData:** Pointer to data buffer • **Size:** Amount of data to be received
Return values	• **HAL:** status

HAL_USART_TransmitReceive_IT

Function Name	**HAL_StatusTypeDef HAL_USART_TransmitReceive_IT (USART_HandleTypeDef * husart, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size)**
Function Description	Full-Duplex Send receive an amount of data in full-duplex mode (non-blocking).
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified

USART module.
- **pTxData:** Pointer to data transmitted buffer
- **pRxData:** Pointer to data received buffer
- **Size:** Amount of data to be received

Return values
- **HAL:** status

HAL_USART_Transmit_DMA

Function Name **HAL_StatusTypeDef HAL_USART_Transmit_DMA (USART_HandleTypeDef * husart, uint8_t * pTxData, uint16_t Size)**

Function Description Simplex Send an amount of data in non-blocking mode.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
- **pTxData:** Pointer to data buffer
- **Size:** Amount of data to be sent

Return values
- **HAL:** status

HAL_USART_Receive_DMA

Function Name **HAL_StatusTypeDef HAL_USART_Receive_DMA (USART_HandleTypeDef * husart, uint8_t * pRxData, uint16_t Size)**

Function Description Full-Duplex Receive an amount of data in non-blocking mode.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
- **pRxData:** Pointer to data buffer
- **Size:** Amount of data to be received

Return values
- **HAL:** status

Notes
- The USART DMA transmit stream must be configured in order to generate the clock for the slave.
- When the USART parity is enabled (PCE = 1) the data received contain the parity bit.

HAL_USART_TransmitReceive_DMA

Function Name **HAL_StatusTypeDef HAL_USART_TransmitReceive_DMA (USART_HandleTypeDef * husart, uint8_t * pTxData, uint8_t * pRxData, uint16_t Size)**

Function Description Full-Duplex Transmit Receive an amount of data in non-blocking mode.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
- **pTxData:** Pointer to data transmitted buffer
- **pRxData:** Pointer to data received buffer

- **Size:** Amount of data to be received

Return values
- **HAL:** status

Notes
- When the USART parity is enabled (PCE = 1) the data received contain the parity bit.

HAL_USART_DMAPause

Function Name **HAL_StatusTypeDef HAL_USART_DMAPause (USART_HandleTypeDef * husart)**

Function Description Pauses the DMA Transfer.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **HAL:** status

HAL_USART_DMAResume

Function Name **HAL_StatusTypeDef HAL_USART_DMAResume (USART_HandleTypeDef * husart)**

Function Description Resumes the DMA Transfer.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **HAL:** status

HAL_USART_DMAStop

Function Name **HAL_StatusTypeDef HAL_USART_DMAStop (USART_HandleTypeDef * husart)**

Function Description Stops the DMA Transfer.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **HAL:** status

HAL_USART_IRQHandler

Function Name **void HAL_USART_IRQHandler (USART_HandleTypeDef * husart)**

Function Description This function handles USART interrupt request.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **None**

HAL_USART_TxCpltCallback

Function Name	**void HAL_USART_TxCpltCallback (USART_HandleTypeDef * husart)**
Function Description	Tx Transfer completed callbacks.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **None**

HAL_USART_TxHalfCpltCallback

Function Name	**void HAL_USART_TxHalfCpltCallback (USART_HandleTypeDef * husart)**
Function Description	Tx Half Transfer completed callbacks.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **None**

HAL_USART_RxCpltCallback

Function Name	**void HAL_USART_RxCpltCallback (USART_HandleTypeDef * husart)**
Function Description	Rx Transfer completed callbacks.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **None**

HAL_USART_RxHalfCpltCallback

Function Name	**void HAL_USART_RxHalfCpltCallback (USART_HandleTypeDef * husart)**
Function Description	Rx Half Transfer completed callbacks.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.
Return values	• **None**

HAL_USART_TxRxCpltCallback

Function Name	**void HAL_USART_TxRxCpltCallback (USART_HandleTypeDef * husart)**
Function Description	Tx/Rx Transfers completed callback for the non-blocking process.
Parameters	• **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified

USART module.

Return values
- **None**

HAL_USART_ErrorCallback

Function Name **void HAL_USART_ErrorCallback (USART_HandleTypeDef * husart)**

Function Description USART error callbacks.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **None**

HAL_USART_GetState

Function Name **HAL_USART_StateTypeDef HAL_USART_GetState (USART_HandleTypeDef * husart)**

Function Description Returns the USART state.

Parameters
- **husart:** pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART module.

Return values
- **HAL:** state

HAL_USART_GetError

Function Name **uint32_t HAL_USART_GetError (USART_HandleTypeDef * husart)**

Function Description Return the USART error code.

Parameters
- **husart:** : pointer to a USART_HandleTypeDef structure that contains the configuration information for the specified USART.

Return values
- **USART:** Error Code

65.3 USART Firmware driver defines

65.3.1 USART

USART Clock

USART_CLOCK_DISABLE

USART_CLOCK_ENABLE

USART Clock Phase

USART_PHASE_1EDGE

USART_PHASE_2EDGE

USART Clock Polarity

USART_POLARITY_LOW

USART_POLARITY_HIGH

USART Error Code

HAL_USART_ERROR_NONE	No error
HAL_USART_ERROR_PE	Parity error
HAL_USART_ERROR_NE	Noise error
HAL_USART_ERROR_FE	Frame error
HAL_USART_ERROR_ORE	Overrun error
HAL_USART_ERROR_DMA	DMA transfer error

USART Exported Macros

__HAL_USART_RESET_HANDLE_STATE

Description:

- Reset USART handle state.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_GET_FLAG

Description:

- Checks whether the specified Smartcard flag is set or not.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  - USART_FLAG_TXE: Transmit data register empty flag
  - USART_FLAG_TC: Transmission Complete flag
  - USART_FLAG_RXNE: Receive data register not empty flag
  - USART_FLAG_IDLE: Idle Line detection flag
  - USART_FLAG_ORE: Overrun Error flag
  - USART_FLAG_NE: Noise Error flag
  - USART_FLAG_FE: Framing Error flag
  - USART_FLAG_PE: Parity

Error flag

Return value:

- The: new state of __FLAG__ (TRUE or FALSE).

__HAL_USART_CLEAR_FLAG

Description:

- Clears the specified Smartcard pending flags.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.
- __FLAG__: specifies the flag to check. This parameter can be any combination of the following values:
  - USART_FLAG_TC: Transmission Complete flag.
  - USART_FLAG_RXNE: Receive data register not empty flag.

Return value:

- None

Notes:

- PE (Parity error), FE (Framing error), NE (Noise error), ORE (Overrun error) and IDLE (Idle line detected) flags are cleared by software sequence: a read operation to USART_SR register followed by a read operation to USART_DR register. RXNE flag can be also cleared by a read to the USART_DR register. TC flag can be also cleared by software sequence: a read operation to USART_SR register followed by a write operation to USART_DR register. TXE flag is cleared only by a write to the USART_DR register.

__HAL_USART_CLEAR_PEFLAG

Description:

- Clear the USART PE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_CLEAR_FEFLAG

Description:

- Clear the USART FE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_CLEAR_NEFLAG

Description:

- Clear the USART NE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_CLEAR_OREFLAG

Description:

- Clear the UART ORE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_CLEAR_IDLEFLAG

Description:

- Clear the USART IDLE pending flag.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.

Return value:

- None

__HAL_USART_ENABLE_IT

Description:

- Enables or disables the specified USART interrupts.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.
- __INTERRUPT__: specifies the USART interrupt source to check. This parameter can be one of the following values:
  - USART_IT_TXE: Transmit Data Register empty interrupt
  - USART_IT_TC: Transmission complete interrupt
  - USART_IT_RXNE: Receive Data register not empty interrupt
  - USART_IT_IDLE: Idle line detection interrupt
  - USART_IT_PE: Parity Error interrupt
  - USART_IT_ERR: Error interrupt(Frame error, noise error, overrun error) This parameter can be: ENABLE or DISABLE.

Return value:

- None

__HAL_USART_DISABLE_IT

__HAL_USART_GET_IT_SOURCE

Description:

- Checks whether the specified USART interrupt has occurred or not.

Parameters:

- __HANDLE__: specifies the USART Handle. This parameter can be USARTx where x: 1, 2, 3 or 6 to select the USART peripheral.
- __IT__: specifies the USART interrupt source to check. This parameter can be one of the following values:
  - USART_IT_TXE: Transmit Data Register empty interrupt
  - USART_IT_TC: Transmission complete interrupt
  - USART_IT_RXNE: Receive Data register not empty interrupt
  - USART_IT_IDLE: Idle line detection interrupt

- USART_IT_ERR: Error interrupt
- USART_IT_PE: Parity Error interrupt

Return value:

- The: new state of __IT__ (TRUE or FALSE).

__HAL_USART_ONE_BIT_SAMPLE_ENABLE

Description:

- Macro to enable the USART's one bit sample method.

Parameters:

- __HANDLE__: specifies the USART Handle.

Return value:

- None

__HAL_USART_ONE_BIT_SAMPLE_DISABLE

Description:

- Macro to disable the USART's one bit sample method.

Parameters:

- __HANDLE__: specifies the USART Handle.

Return value:

- None

__HAL_USART_ENABLE

Description:

- Enable USART.

Parameters:

- __HANDLE__: specifies the USART Handle. USART Handle selects the USARTx peripheral (USART availability and x value depending on device).

Return value:

- None

__HAL_USART_DISABLE

Description:

- Disable USART.

Parameters:

- __HANDLE__: specifies the USART Handle. USART Handle selects the USARTx peripheral (USART availability and x value depending on device).

Return value:

- None

USART Flags

USART_FLAG_TXE

USART_FLAG_TC

USART_FLAG_RXNE

USART_FLAG_IDLE

USART_FLAG_ORE

USART_FLAG_NE

USART_FLAG_FE

USART_FLAG_PE

USART Interrupts Definition

USART_IT_PE

USART_IT_TXE

USART_IT_TC

USART_IT_RXNE

USART_IT_IDLE

USART_IT_LBD

USART_IT_CTS

USART_IT_ERR

USART Last Bit

USART_LASTBIT_DISABLE

USART_LASTBIT_ENABLE

USART Mode

USART_MODE_RX

USART_MODE_TX

USART_MODE_TX_RX

USART NACK State

USART_NACK_ENABLE

USART_NACK_DISABLE

USART Parity

USART_PARITY_NONE

USART_PARITY_EVEN

USART_PARITY_ODD

USART Number of Stop Bits

USART_STOPBITS_1

USART_STOPBITS_0_5

USART_STOPBITS_2

USART_STOPBITS_1_5

USART Word Length

USART_WORDLENGTH_8B

USART_WORDLENGTH_9B

66 HAL WWDG Generic Driver

66.1 WWDG Firmware driver registers structures

66.1.1 WWDG_InitTypeDef

Data Fields

- ***uint32_t Prescaler***
- ***uint32_t Window***
- ***uint32_t Counter***
- ***uint32_t EWIMode***

Field Documentation

- ***uint32_t WWDG_InitTypeDef::Prescaler***
  Specifies the prescaler value of the WWDG. This parameter can be a value of ***WWDG_Prescaler***
- ***uint32_t WWDG_InitTypeDef::Window***
  Specifies the WWDG window value to be compared to the downcounter. This parameter must be a number Min_Data = 0x40 and Max_Data = 0x7F
- ***uint32_t WWDG_InitTypeDef::Counter***
  Specifies the WWDG free-running downcounter value. This parameter must be a number between Min_Data = 0x40 and Max_Data = 0x7F
- ***uint32_t WWDG_InitTypeDef::EWIMode***
  Specifies if WWDG Early Wakeup Interupt is enable or not. This parameter can be a value of ***WWDG_EWI_Mode***

66.1.2 WWDG_HandleTypeDef

Data Fields

- ***WWDG_TypeDef * Instance***
- ***WWDG_InitTypeDef Init***

Field Documentation

- ***WWDG_TypeDef* WWDG_HandleTypeDef::Instance***
  Register base address
- ***WWDG_InitTypeDef WWDG_HandleTypeDef::Init***
  WWDG required parameters

66.2 WWDG Firmware driver API description

66.2.1 WWDG specific features

Once enabled the WWDG generates a system reset on expiry of a programmed time period, unless the program refreshes the counter (downcounter) before reaching 0x3F value (i.e. a reset is generated when the counter value rolls over from 0x40 to 0x3F).

- An MCU reset is also generated if the counter value is refreshed before the counter has reached the refresh window value. This implies that the counter must be refreshed in a limited window.
- Once enabled the WWDG cannot be disabled except by a system reset.
- WWDGRST flag in RCC_CSR register can be used to inform when a WWDG reset occurs.
- The WWDG counter input clock is derived from the APB clock divided by a programmable prescaler.
- WWDG clock (Hz) = PCLK1 / (4096 * Prescaler)
- WWDG timeout (mS) = 1000 * Counter / WWDG clock
- WWDG Counter refresh is allowed between the following limits :
  - min time (mS) = 1000 * (Counter _ Window) / WWDG clock
  - max time (mS) = 1000 * (Counter _ 0x40) / WWDG clock
- Min-max timeout value at 50 MHz(PCLK1): 81.9 us / 41.9 ms
- The Early Wakeup Interrupt (EWI) can be used if specific safety operations or data logging must be performed before the actual reset is generated. When the downcounter reaches the value 0x40, an EWI interrupt is generated and the corresponding interrupt service routine (ISR) can be used to trigger specific actions (such as communications or data logging), before resetting the device. In some applications, the EWI interrupt can be used to manage a software system check and/or system recovery/graceful degradation, without generating a WWDG reset. In this case, the corresponding interrupt service routine (ISR) should reload the WWDG counter to avoid the WWDG reset, then trigger the required actions. Note:When the EWI interrupt cannot be served, e.g. due to a system lock in a higher priority task, the WWDG reset will eventually be generated.
- Debug mode : When the microcontroller enters debug mode (core halted), the WWDG counter either continues to work normally or stops, depending on DBG_WWDG_STOP configuration bit in DBG module, accessible through __HAL_DBGMCU_FREEZE_WWDG() and __HAL_DBGMCU_UNFREEZE_WWDG() macros

66.2.2 How to use this driver

- Enable WWDG APB1 clock using __HAL_RCC_WWDG_CLK_ENABLE().
- Set the WWDG prescaler, refresh window, counter value and Early Wakeup Interrupt mode using using HAL_WWDG_Init() function. This enables WWDG peripheral and the downcounter starts downcounting from given counter value. Init function can be called again to modify all watchdog parameters, however if EWI mode has been set once, it can't be clear until next reset.
- The application program must refresh the WWDG counter at regular intervals during normal operation to prevent an MCU reset using HAL_WWDG_Refresh() function. This operation must occur only when the counter is lower than the window value already programmed.
- if Early Wakeup Interrupt mode is enable an interrupt is generated when the counter reaches 0x40. User can add his own code in weak function HAL_WWDG_EarlyWakeupCallback().

WWDG HAL driver macros list

Below the list of most used macros in WWDG HAL driver.

- __HAL_WWDG_GET_IT_SOURCE: Check the selected WWDG's interrupt source.
- __HAL_WWDG_GET_FLAG: Get the selected WWDG's flag status.
- __HAL_WWDG_CLEAR_FLAG: Clear the WWDG's pending flags.

66.2.3 Initialization and Configuration functions

This section provides functions allowing to:

- Initialize and start the WWDG according to the specified parameters in the WWDG_InitTypeDef of associated handle.
- Initialize the WWDG MSP.

This section contains the following APIs:

- ***HAL_WWDG_Init()***
- ***HAL_WWDG_MspInit()***

66.2.4 IO operation functions

This section provides functions allowing to:

- Refresh the WWDG.
- Handle WWDG interrupt request and associated function callback.

This section contains the following APIs:

- ***HAL_WWDG_Refresh()***
- ***HAL_WWDG_IRQHandler()***
- ***HAL_WWDG_EarlyWakeupCallback()***

66.2.5 Detailed description of functions

HAL_WWDG_Init

Function Name	**HAL_StatusTypeDef HAL_WWDG_Init (WWDG_HandleTypeDef * hwwdg)**
Function Description	Initialize the WWDG according to the specified.
Parameters	• **hwwdg:** pointer to a WWDG_HandleTypeDef structure that contains the configuration information for the specified WWDG module.
Return values	• **HAL:** status

HAL_WWDG_MspInit

Function Name	**void HAL_WWDG_MspInit (WWDG_HandleTypeDef * hwwdg)**
Function Description	Initialize the WWDG MSP.
Parameters	• **hwwdg:** pointer to a WWDG_HandleTypeDef structure that contains the configuration information for the specified WWDG module.
Return values	• **None**
Notes	• When rewriting this function in user file, mechanism may be

added to avoid multiple initialize when HAL_WWDG_Init function is called again to change parameters.

HAL_WWDG_Refresh

Function Name	**HAL_StatusTypeDef HAL_WWDG_Refresh (WWDG_HandleTypeDef * hwwdg)**
Function Description	Refresh the WWDG.
Parameters	• **hwwdg:** pointer to a WWDG_HandleTypeDef structure that contains the configuration information for the specified WWDG module.
Return values	• **HAL:** status

HAL_WWDG_IRQHandler

Function Name	**void HAL_WWDG_IRQHandler (WWDG_HandleTypeDef * hwwdg)**
Function Description	Handle WWDG interrupt request.
Parameters	• **hwwdg:** pointer to a WWDG_HandleTypeDef structure that contains the configuration information for the specified WWDG module.
Return values	• **None**
Notes	• The Early Wakeup Interrupt (EWI) can be used if specific safety operations or data logging must be performed before the actual reset is generated. The EWI interrupt is enabled by calling HAL_WWDG_Init function with EWIMode set to WWDG_EWI_ENABLE. When the downcounter reaches the value 0x40, and EWI interrupt is generated and the corresponding Interrupt Service Routine (ISR) can be used to trigger specific actions (such as communications or data logging), before resetting the device.

HAL_WWDG_EarlyWakeupCallback

Function Name	**void HAL_WWDG_EarlyWakeupCallback (WWDG_HandleTypeDef * hwwdg)**
Function Description	WWDG Early Wakeup callback.
Parameters	• **hwwdg:** : pointer to a WWDG_HandleTypeDef structure that contains the configuration information for the specified WWDG module.
Return values	• **None**

66.3 WWDG Firmware driver defines

66.3.1 WWDG

WWDG Early Wakeup Interrupt Mode

WWDG_EWI_DISABLE EWI Disable

WWDG_EWI_ENABLE EWI Enable

WWDG Exported Macros

__HAL_WWDG_ENABLE

Description:

- Enables the WWDG peripheral.

Parameters:

- __HANDLE__: WWDG handle

Return value:

- None

__HAL_WWDG_ENABLE_IT

Description:

- Enables the WWDG early wakeup interrupt.

Parameters:

- __HANDLE__: WWDG handle
- __INTERRUPT__: specifies the interrupt to enable. This parameter can be one of the following values:
  - WWDG_IT_EWI: Early wakeup interrupt

Return value:

- None

Notes:

- Once enabled this interrupt cannot be disabled except by a system reset.

__HAL_WWDG_GET_IT

Description:

- Checks whether the selected WWDG interrupt has occurred or not.

Parameters:

- __HANDLE__: WWDG handle
- __INTERRUPT__: specifies the it to check. This parameter can be one of the following values:
  - WWDG_FLAG_EWIF: Early wakeup interrupt IT

Return value:

- The: new state of WWDG_FLAG (SET or RESET).

__HAL_WWDG_CLEAR_IT

Description:

- Clear the WWDG's interrupt pending bits bits to clear the selected interrupt pending bits.

Parameters:

- __HANDLE__: WWDG handle
- __INTERRUPT__: specifies the interrupt pending bit to clear. This parameter can be one of the following values:

– WWDG_FLAG_EWIF: Early wakeup interrupt flag

__HAL_WWDG_GET_FLAG

Description:

- Check whether the specified WWDG flag is set or not.

Parameters:

- __HANDLE__: WWDG handle
- __FLAG__: specifies the flag to check. This parameter can be one of the following values:
  – WWDG_FLAG_EWIF: Early wakeup interrupt flag

Return value:

- The: new state of WWDG_FLAG (SET or RESET).

__HAL_WWDG_CLEAR_FLAG

Description:

- Clears the WWDG's pending flags.

Parameters:

- __HANDLE__: WWDG handle
- __FLAG__: specifies the flag to clear. This parameter can be one of the following values:
  – WWDG_FLAG_EWIF: Early wakeup interrupt flag

Return value:

- None

__HAL_WWDG_GET_IT_SOURCE

Description:

- Checks if the specified WWDG interrupt source is enabled or disabled.

Parameters:

- __HANDLE__: WWDG Handle.
- __INTERRUPT__: specifies the WWDG interrupt source to check. This parameter can be one of the following values:
  – WWDG_IT_EWI: Early Wakeup Interrupt

Return value:

- state: of __INTERRUPT__ (TRUE or FALSE).

WWDG Flag definition

WWDG_FLAG_EWIF Early wakeup interrupt flag

WWDG Interrupt definition

WWDG_IT_EWI Early wakeup interrupt

WWDG Prescaler

WWDG_PRESCALER_1 WWDG counter clock = (PCLK1/4096)/1

WWDG_PRESCALER_2	WWDG counter clock = (PCLK1/4096)/2
WWDG_PRESCALER_4	WWDG counter clock = (PCLK1/4096)/4
WWDG_PRESCALER_8	WWDG counter clock = (PCLK1/4096)/8

67 FAQs

General subjects

Why should I use the HAL drivers?

There are many advantages in using the HAL drivers:

- Ease of use: you can use the HAL drivers to configure and control any peripheral embedded within your STM32 MCU without prior in-depth knowledge of the product.
- HAL drivers provide intuitive and ready-to-use APIs to configure the peripherals and support polling, interrupt and DMA programming model to accommodate all application requirements, thus allowing the end-user to build a complete application by calling a few APIs.
- Higher level of abstraction than a standard peripheral library allowing to transparently manage:
  - Data transfers and processing using blocking mode (polling) or non-blocking mode (interrupt or DMA)
  - Error management through peripheral error detection and timeout mechanism.
- Generic architecture speeding up initialization and porting, thus allowing customers to focus on innovation.
- Generic set of APIs with full compatibility across the STM32 series/lines, to ease the porting task between STM32 MCUs.
- The APIs provided within the HAL drivers are feature-oriented and do not required in-depth knowledge of peripheral operation.
- The APIs provided are modular. They include initialization, IO operation and control functions. The end-user has to call init function, then start the process by calling one IO operation functions (write, read, transmit, receive, …). Most of the peripherals have the same architecture.
- The number of functions required to build a complete and useful application is very reduced. As an example, to build a UART communication process, the user only has to call HAL_UART_Init() then HAL_UART_Transmit() or HAL_UART_Receive().

Which STM32F4 devices are supported by the HAL drivers?

The HAL drivers are developed to support all STM32F4 devices. To ensure compatibility between all devices and portability with others series and lines, the API is split into the generic and the extension APIs . For more details, please refer to *Section 4.4: "Devices supported by HAL drivers"*.

What is the cost of using HAL drivers in term of code size and performance?

Like generic architecture drivers, the HAL drivers may induce firmware overhead.

This is due to the high abstraction level and ready-to-use APIs which allow data transfers, errors management and offloads the user application from implementation details.

Architecture

How many files should I modify to configure the HAL drivers?

Only one file needs to be modified: stm32f4xx_hal_conf.h. You can modify this file by disabling unused modules, or adjusting some parameters (i.e. HSE value, System configuration, Ethernet parameters configuration…)

A template is provided in the HAL drivers folders (stm32f4xx_hal_conf_template.c).

Which header files should I include in my application to use the HAL drivers?

Only stm32f4xx_hal.h file has to be included.

What is the difference between stm32f4xx_hal_ppp.c/.h and stm32f4xx_hal_ppp_ex.c/.h?

The HAL driver architecture supports common features across STM32 series/lines. To support specific features, the drivers are split into two groups.

- The generic APIs (stm32f4xx_hal_ppp.c): It includes the common set of APIs across all the STM32 product lines
- The extension APIs (stm32f4xx_hal_ppp_ex.c): It includes the specific APIs for specific device part number or family.

Is it possible to use the APIs available in stm32f4xx_ll_ppp.c?

These APIs cannot be used directly because they are internal and offer services to upper layer drivers. As an example stm32f4xx_ll_fmc.c/h driver is used by stm32f4xx_hal_sram.c, stm32f4xx_hal_nor.c, stm32f4xx_hal_nand.c and stm32f4xx_hal_sdram.c drivers.

Initialization and I/O operation functions

How do I configure the system clock?

Unlike the standard library, the system clock configuration is not performed in CMSIS drivers file (system_stm32f4xx.c) but in the main user application by calling the two main functions, HAL_RCC_OscConfig() and HAL_RCC_ClockConfig(). It can be modified in any user application section.

What is the purpose of the *PPP_HandleTypeDef *pHandle* structure located in each driver in addition to the Initialization structure

PPP_HandleTypeDef *pHandle is the main structure implemented in the HAL drivers. It handles the peripheral configuration and registers, and embeds all the structures and variables required to follow the peripheral device flow (pointer to buffer, Error code, State,...)

However, this structure is not required to service peripherals such as GPIO, SYSTICK, PWR, and RCC.

What is the purpose of HAL_PPP_MspInit() and HAL_PPP_MspDeInit() functions?

These function are called within HAL_PPP_Init() and HAL_PPP_DeInit(), respectively. They are used to perform the low level Initialization/de-initialization related to the additional hardware resources (RCC, GPIO, NVIC and DMA).

These functions are declared in stm32f4xx_hal_msp.c. A template is provided in the HAL driver folders (stm32f4xx_hal_msp_template.c).

When and how should I use callbacks functions (functions declared with the attribute *__weak*)?

Use callback functions for the I/O operations used in DMA or interrupt mode. The PPP process complete callbacks are called to inform the user about process completion in real-time event mode (interrupts).

The Errors callbacks are called when a processing error occurs in DMA or interrupt mode. These callbacks are customized by the user to add user proprietary code. They can be declared in the application. Note that the same process completion callbacks are used for DMA and interrupt mode.

Is it mandatory to use HAL_Init() function at the beginning of the user application?

It is mandatory to use HAL_Init() function to enable the system configuration (Prefetch, Data instruction cache,…), configure the systTick and the NVIC priority grouping and the hardware low level initialization.

The sysTick configuration shall be adjusted by calling ***HAL_RCC_ClockConfig()*** function, to obtain 1 ms whatever the system clock.

Why do I need to configure the SysTick timer to use the HAL drivers?

The SysTick timer is configured to be used to generate variable increments by calling *HAL_IncTick()* function in Systick ISR and retrieve the value of this variable by calling *HAL_GetTick()* function.

The call HAL_GetTick() function is mandatory when using HAL drivers with Polling Process or when using HAL_Delay().

Why is the SysTick timer configured to have 1 ms?

This is mandatory to ensure correct IO operation in particular for polling mode operation where the 1 ms is required as timebase.

Could HAL_Delay() function block my application under certain conditions?

Care must be taken when using HAL_Delay() since this function provides accurate delay based on a variable incremented in SysTick ISR. This implies that if HAL_Delay() is called from a peripheral ISR process, then the SysTick interrupt must have higher priority (numerically lower) than the peripheral interrupt, otherwise the caller ISR process will be blocked. Use HAL_NVIC_SetPriority() function to change the SysTick interrupt priority.

What programming model sequence should I follow to use HAL drivers ?

Follow the sequence below to use the APIs provided in the HAL drivers:

1. Call HAL_Init() function to initialize the system (data cache, NVIC priority,…).
2. Initialize the system clock by calling HAL_RCC_OscConfig() followed by HAL_RCC_ClockConfig().
3. Add HAL_IncTick() function under SysTick_Handler() ISR function to enable polling process when using HAL_Delay() function
4. Start initializing your peripheral by calling HAL_PPP_Init().
5. Implement the hardware low level initialization (Peripheral clock, GPIO, DMA,..) by calling HAL_PPP_MspInit() in stm32f4xx_hal_msp.c
6. Start your process operation by calling IO operation functions.

What is the purpose of HAL_PPP_IRQHandler() function and when should I use it?

HAL_PPP_IRQHandler() is used to handle interrupt process. It is called under PPP_IRQHandler() function in stm32f4xx_it.c. In this case, the end-user has to implement only the callbacks functions (prefixed by __weak) to perform the appropriate action when an interrupt is detected. Advanced users can implement their own code in PPP_IRQHandler() without calling HAL_PPP_IRQHandler().

Can I use directly the macros defined in stm32f4xx_hal_ppp.h ?

Yes, you can: a set of macros is provided with the APIs. They allow accessing directly some specific features using peripheral flags.

Where must PPP_HandleTypedef structure peripheral handler be declared?

PPP_HandleTypedef structure peripheral handler must be declared as a global variable, so that all the structure fields are set to 0 by default. In this way, the peripheral handler default state are set to HAL_PPP_STATE_RESET, which is the default state for each peripheral after a system reset.

68 Revision history

Table 16: Document revision history

Date	Revision	Changes
05-May-2014	1	Initial release.
03-Apr-2015	2	Added CEC, FMPI2C, QSPI and SPDIFRX in *Table 1: "Acronyms and definitions"*. Added STM32F446xx, cec, dcmi, fmpi2c, fmpi2c_ex, spdifrx and qspi in *Table 5: "List of devices supported by HAL drivers"*. Updated Common macros section in *Section 4.9: "HAL common resources"*. Added *Section 9: "HAL CEC Generic Driver"*, *Section 26: "HAL FLASH__RAMFUNC Generic Driver"Section 27: "HAL FMPI2C Generic Driver"*, *Section 28: "HAL FMPI2C Extension Driver"*, *Section 50: "HAL QSPI Generic Driver"* and *Section 61: "HAL SPDIFRX Generic Driver"*.
15-Sep-2015	3	Added DSI and LPTIM and removed msp_template in *Table 1: "Acronyms and definitions"*. Added STM32F469xx, STM32F479xx, STM32F410xx, dsi, ltdc_ex and lptim in *Table 5: "List of devices supported by HAL drivers"*. Added *Section 22: "HAL DSI Generic Driver"*, *Section 40: "HAL LPTIM Generic Driver"* and *Section 41: "HAL LTDC Generic Driver"*.
08-Sep-2016	4	Added DFSDM in *Table 1: "Acronyms and definitions"*. Added STM32F412Cx, STM32F412Rx, STM32F412Vx, STM32F412Zx and DFSDM in *Table 5: "List of devices supported by HAL drivers"*. Added *Section 16: "HAL DFSDM Generic Driver"*.

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